PHD IN CHILDREN AND ADOLESCENT COUNSELING

CAC.903

Child and Adolescent Assessment Techniques

A complete course volume with original counseling lectures, real-life applications, vector diagrams, doctoral seminar tasks, and attributed open educational readings.

Book 3 of 32 • PDF edition

Educational coursebook. Not an official university publication and not a substitute for supervised clinical training, diagnosis, or emergency care.

How to Use This Book

Purpose and scope

This volume covers CAC.903 Child and Adolescent Assessment Techniques within the supplied PhD curriculum. It teaches multimethod assessment, standardized testing, reliability, validity, interpretation of assessment results, and clinical judgment for children, adolescents, families, schools, and community settings. It is a complete course volume, not a claim to contain the entire discipline.

Two clearly identified layers

Each chapter begins with a newly written counseling lecture, a real-life application, and a doctoral seminar task. An original vector diagram makes a central relationship visible. The chapter then provides attributed foundational readings adapted from CC BY open educational texts. The readings supply broader assessment science; the new lectures provide child-adolescent counseling interpretation and critical cautions.

How to study

Read the lecture first, then the diagram, then the adapted reading. Use the workbook and clinical casebook to turn assessment science into a testable professional task. In supervision, compare at least two formulations for the same case before choosing an assessment plan.

Simple English and real-life application

Technical terms are introduced gradually and defined in the glossary. Fictional cases show how assessment changes the questions a counselor asks. The cases are teaching examples, not testing or treatment protocols.

Doctoral level and safety

The book assumes graduate study in counseling or a related field. It does not authorize unsupervised practice. Crisis, abuse, suicide risk, psychosis, and severe impairment require qualified assessment and local emergency procedures.

Source transparency

The adapted readings are used under Creative Commons Attribution 4.0. They have been selected, reorganized, and typeset for this course, with new lectures and diagrams added. They are not presented as wholly original text. Module-level attribution and pinned source snapshots appear in the attribution register. Inherited references are identified separately from the supplemental sources checked for this edition.

Integrative areas

Diversity, ethics, research, and supervision are integrated throughout. Chapter 3 gives focused attention to cultural and individual diversity; Chapter 20 consolidates ethics, supervision, research, and integrated case formulation.

Contents

Chapter 01 — Assessment as a Decision Process
Chapter 02 — Ethics, Consent, Assent, and Test Security
Chapter 03 — Cultural and Individual Diversity in Assessment
Chapter 04 — Constructs: What Are We Measuring?
Chapter 05 — Scores, Scales, and Norms
Chapter 06 — Reliability and Measurement Error
Chapter 07 — Validity: Evidence for Interpretation and Use
Chapter 08 — Test Bias, Fairness, and Equity
Chapter 09 — The Clinical Interview and Diagnostic Systems
Chapter 10 — Behavioral Observation and Functional Assessment
Chapter 11 — Assessment of Cognition and Thought Processes
Chapter 12 — Intellectual and Adaptive Functioning Assessment
Chapter 13 — Objective Personality and Symptom Measures
Chapter 14 — Projective, Expressive, and Play-Based Methods
Chapter 15 — Clinical Judgment and Algorithmic Prediction
Chapter 16 — Prediction, Base Rates, and Decision Errors
Chapter 17 — Repeated Assessment and Progress Monitoring
Chapter 18 — Generalizability, Technology, and Ambulatory Assessment
Chapter 19 — Report Writing, Feedback, and Communication
Chapter 20 — Integrated Child and Adolescent Case Formulation

Glossary
Applied Workbook
Clinical Casebook
References
Attribution and Production Record

CHAPTER 01 / CAC.903

Assessment as a Decision Process

Assessment is not a pile of tests. It is a disciplined way to make better decisions about a young person's life. A parent asks whether a child's worry is normal. A teacher asks whether a student needs a different reading program. A court asks whether a caregiver can keep a child safe. A physician asks whether attention problems explain school failure. In each case, someone must decide what to do next. Assessment is the process of gathering, organizing, and interpreting information so that the decision is less guesswork and more responsible.

This chapter begins with a simple claim: every assessment should answer a question. If the question is unclear, the assessment will drift. A counselor may give a broad battery because it is available, not because it is needed. The result is a long report that does not help the family. A better approach begins with the referral question, translates it into assessable targets, chooses methods that fit the child and context, and ends with a recommendation that can be acted on.

Learning goals

By the end of this chapter, you should be able to distinguish screening, diagnosis, case formulation, and progress monitoring; write a referral question that can be assessed; and explain why assessment is a hypothesis-testing process. You should also be able to describe at least three sources of information and two risks of overconfidence. These goals matter because doctoral assessment is not only technical skill. It is judgment under uncertainty.

Assessment, testing, and formulation

Testing is one part of assessment. A test is a standardized procedure that produces a score or profile. Assessment is broader. It includes interviews, observation, records, collateral reports, cultural context, and clinical reasoning. Formulation is the organized story that links the information to a plan. A diagnosis names a pattern. A formulation explains how the pattern developed, what maintains it, and what may change it.

Consider ten-year-old Amina, who cries at the school door. A test may show elevated separation anxiety. That score is useful, but it does not tell the whole story. A formulation may note recent grief, a bullying incident, parental accommodation, and a temperament sensitive to novelty. The plan may include gradual exposure, caregiver coaching, school coordination, and grief support. The test score is evidence. The formulation is the map.

The referral question

A good referral question is specific enough to guide method selection and broad enough to avoid tunnel vision. "Is this child ADHD?" is too narrow if the real concern is school failure. A better question is: "What explains this child's classroom inattention and work refusal, and what supports are needed?" That question allows assessment of attention, learning, anxiety, sleep, instruction, and family stress. It also keeps the child from being reduced to one label.

Referral questions often come with hidden assumptions. A teacher may ask, "How do we make this student comply?" The counselor should translate the request into a welfare question: "What is the function of the behavior, and what changes would help the student

succeed safely?" Assessment should not become a tool for adult convenience alone. The child's dignity and development remain central.

Multiple methods and multiple informants

Children's behavior changes across settings. A child may be quiet at home and disruptive at school. A teen may report depression that parents do not see. A parent may report aggression that the child denies. These discrepancies are not automatically lies. They may reflect real context effects, different expectations, or different access to inner experience. A strong assessment uses multiple methods and multiple informants, then explains the pattern rather than forcing agreement.

Methods include clinical interview, standardized rating scales, direct observation, cognitive testing, academic achievement testing, play-based tasks, and record review. Informants include the child, caregivers, teachers, and sometimes physicians or child-welfare workers. Each source has a viewpoint. The counselor's job is to weigh the viewpoints, not to average them blindly.

Screening versus diagnosis

Screening asks whether a concern may be present and whether further assessment is needed. It is usually brief and broad. Diagnosis asks whether a defined pattern meets criteria and causes impairment. Screening tools are not diagnoses. A positive screen should lead to evaluation, not automatic labeling. A negative screen does not rule out a problem if clinical concern remains.

In school settings, screening can identify students who may need support. But false positives and false negatives matter. A false positive may cause unnecessary worry. A false negative may delay help. The counselor should consider base rates, cut scores, and consequences. A screening program should have a clear pathway for follow-up, not just a list of scores.

Evidence-based assessment

Evidence-based assessment means using research and theory to choose targets, methods, and interpretations. It asks whether a measure is reliable, valid, culturally appropriate, and useful for the decision. It also asks whether the assessment improves outcomes. A measure can be psychometrically strong but clinically useless if it does not change the plan. A measure can be popular but weak for the population being assessed.

Doctoral practice requires incremental validity: does this method add useful information beyond what is already known? If a caregiver interview and teacher rating already answer the question, an expensive test may not be needed. If the question is complex, a broader battery may be justified. The counselor should be able to explain why each method is included.

Clinical judgment and humility

Assessment is not mechanical. Counselors make judgments about relevance, credibility, and meaning. Judgment can be improved by structure, feedback, and consultation. It can be distorted by confirmation bias, anchoring, halo effects, and cultural stereotypes. A counselor who likes a family may minimize risk. A counselor who expects defiance may see normal protest as pathology. Humility means treating conclusions as provisional and checking them against evidence.

A practical habit is to write alternative hypotheses. For a child with stomachaches before school, hypotheses may include anxiety, bullying, learning difficulty, sleep loss, family conflict, or medical illness. The assessment plan should test these possibilities. If new information appears, the formulation should change. A report that never changes may be a sign of rigid thinking.

Real-life application

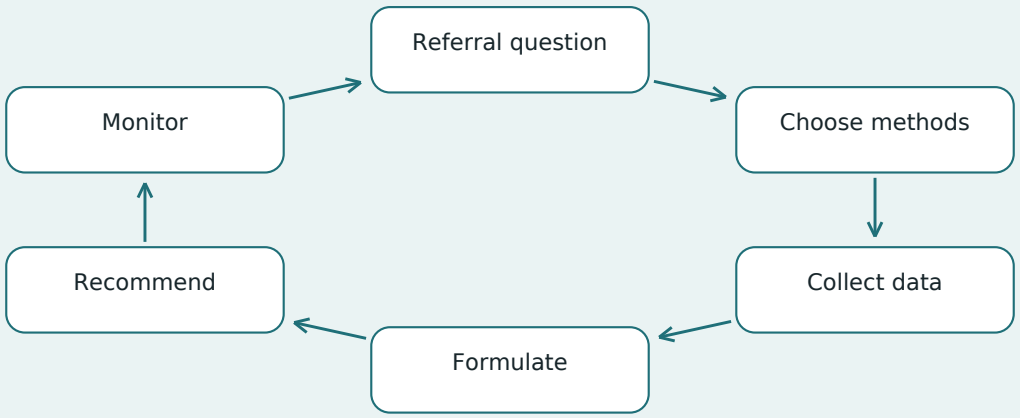
A school counselor receives a referral for thirteen-year-old Daniel, who sleeps in class and argues with teachers. The initial question is oppositional behavior. The counselor broadens the question to include sleep, mood, learning, and home stress. Interview data reveal that Daniel works nights to help his family. Teacher ratings show inattention but not aggression. A sleep log shows short sleep. The formulation shifts from defiance to fatigue and role strain. The recommendation includes schedule adjustment, family support, and monitoring, not only punishment.

This example shows why assessment is a decision process. The same behavior can have different meanings. The counselor's task is to gather enough information to choose a helpful response. The task is not to prove the first hypothesis correct.

Reflection

Write a referral question you have heard or might hear. Rewrite it as an assessable question. List three methods and three informants. Then write one alternative hypothesis that could change your plan. Assessment begins when you allow the possibility that you may be wrong.

Assessment as a decision loop



Assessment begins and ends with a decision. Feedback allows the plan to change.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module intro.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

About This Book

First, let us discuss what this book is not. This book is not a guide on how to assess each psychological construct or disorder. This book is also not a comparative summary of the psychometrics of different measures. There already exist many resources that summarize and compare the reliability and validity of measures in psychology (Buros Center for Testing, 2021). Instead, this book is about the principles of psychological assessment.

This book was originally written for a graduate-level course on psychological assessment. The chapters provide an overview of topics—each of which could have its own class and textbook, such as structural equation modeling, item response theory, generalizability theory, factor analysis, prediction, cognitive assessment, psychophysiological assessment, etc. The book gives readers an overview of the breadth of the field of assessment and various assessment approaches. As a consequence, the book does not cover any one assessment device or method in great depth.

The goal of this book is to help researchers and clinicians learn to think critically about assessments so they can better develop, evaluate, administer, score, integrate, and interpret psychological assessments. Learning important principles of assessment will put you in a better position to learn any assessment device and to develop better ones. This book applies a scientific perspective to the principles of psychological assessment. The assessments used in a given situation—whether in research or practice—should be supported by the strongest available science, or they should be used cautiously while undergoing development and study. In addition to discussing principles, however, analysis scripts in the software R (R-base, n.d.) are provided, so that you are able to apply the principles discussed in this book. Some of the chapters include analysis exercises (at the end of the chapter) for you to test your comprehension.

Why R?

R is free, open source, open platform, and widely used. Unlike proprietary software used for data analysis, R is not a black box. You can examine the code for any function or computation you perform. You can even modify and improve these functions by changing the code, and you can create your own functions. R also has advanced capabilities for data wrangling and has many packages available for advanced statistical analysis and graphing. In addition, there are strong resources available for creating your analyses in R so that they are reproducible by others (Gandrud, 2020).

About the Author

Regarding my background, I am a licensed clinical psychologist. My research examines how children develop behavior problems. I am also a trained clinician, and I supervise training clinicians in assessment and therapy, particularly assessment and treatment of children's disruptive behavior. Given my expertise, many of the examples in the book deal with topics in clinical psychology, but many of the assessment principles discussed are relevant to all areas of psychology—and science more broadly—and are often overlooked in research and practice. As a card-carrying clinical scientist, my perspective is that the scientific epistemology is the strongest approach to knowledge and that assessment should be guided first and foremost by the epistemology of science, regardless of whether one is doing research or practice.

What Is Assessment?

Measurement is the principled assignment of numbers to objects or events (Stevens, 1946). Assessment is a broader process that integrates measurement along with interpretation and decision-making. Assessment involves gathering of information about a person, group, setting, or context. In psychological assessment, we are interested in gathering information about people's psychological functioning, including their thoughts, emotions, and behaviors. Psychological assessment can also consider biological and physiological processes that are linked to people's thoughts, emotions, and behaviors. Many assessment approaches can be used to assess people's thoughts, emotions, and behaviors, including self-report questionnaires, questionnaires reported by others (e.g., spouse, parent, teacher, or friend), interviews, observations, biopsychological assessments (e.g., cortisol, heart rate, brain imaging), performance-based assessments, archival approaches (e.g., chart review), and combinations of these.

Why Should We Care About Assessment (and Science)?

In research, assessments are conducted to advance knowledge, such as improved prediction or understanding. For example, in my research, I use assessments to understand what processes influence children's development of disruptive behavior. In society, assessments are conducted to improve decision-making. For instance, assessments are conducted to determine whether to hire a job candidate or promote an employee. In a clinical context, assessments are conducted to improve treatment and the client's outcomes. As an example, assessments are conducted to determine which treatment would be most effective for a person suffering from depression. Assessments can be valuable for understanding current functioning as well as making predictions. To best answer these questions and address these goals, we need to have confidence that our devices yield accurate answers for these purposes for the assessed individuals. Science is crucial for knowing how much (or how little) confidence we have in a given assessment for a given purpose and population. Effective treatment often depends on accurate assessment. Thus, knowing how to conduct and critically evaluate science will make you more effective at selecting, administering, and interpreting assessments.

Decisions resulting from assessments can have important life-altering consequences. High-stakes decisions based on assessments include decisions about whether a person is hospitalized, whether a child is removed from their abusive home, whether a person is deemed competent to stand trial, whether a prisoner is released on parole, and whether an applicant is admitted to graduate school. These important assessment-related decisions should be made using the best available science.

The problem is that there has been a proliferation of pseudoscience in assessment and treatment. There are widely used psychological assessments and treatments that we know are inaccurate, do not work, or, in some cases, that we know to be harmful. Lists of harmful psychological treatments (Lilienfeld, 2007) and inaccurate assessments (Hunsley et al., 2015) have been published, but these treatments and assessments are still used by professional providers to this day. Practice using such techniques violates the aphorism, "First, do no harm." This would be inconceivable for other applied sciences, such as chemistry, engineering, and medicine. For instance, the prescription of a particular medication for a particular purpose requires approval by the U.S. Food and Drug Administration (FDA). Psychological assessments and treatments do not have the same level of oversight.

The gap between what we know based on science and what is implemented in practice (the science–practice gap) motivated McFall's (McFall, 1991) "Manifesto for a Science of Clinical Psychology," which he later expanded (McFall, 2000). The Manifesto has one cardinal principle and four corollaries:

Cardinal Principle: Scientific clinical psychology is the only legitimate and acceptable form of clinical psychology.

First Corollary: Psychological services should not be administered to the public (except under strict experimental control) until they have satisfied these four minimal criteria:

1. The exact nature of the service must be described clearly.

2. The claimed benefits of the service must be stated explicitly.

3. These claimed benefits must be validated scientifically.

4. Possible negative side effects that might outweigh any benefits must be ruled out empirically.

Second Corollary: The primary and overriding objective of doctoral training programs in clinical psychology must be to produce the most competent clinical scientists possible.

Third Corollary: A scientific epistemology differentiates science from pseudoscience.

Fourth Corollary: The most caring and humane psychological services are those that have been shown empirically to be the most effective, efficient, and safe.

The Manifesto orients you to the scientific perspective from which we will be examining psychological assessment techniques in this book.

Assessment and the Replication Crisis in Science

Assessment is also crucial to advancing knowledge in research, as summarized in the maxim, "What we know depends on how we know it." Findings from studies boil down to the methods that were used to obtain them—thus, everything we know comes down to methods.

Many domains of science, particularly social science, have struggled with a replication crisis, such that a large proportion of findings fail to replicate when independent investigators attempt to replicate the original findings (Duncan et al., 2014; Freese & Peterson, 2017; Larson & Carbine, 2017; Lilienfeld, 2017; Open Science Collaboration, 2015; Shucard et al., 1987; Tackett et al., 2019). There is considerable speculation on what factors account for the replication crisis. For instance, one possible factor is the researcher degrees of freedom, which are unacknowledged choices in how researchers prepare, analyze, and report their data that can lead to detecting significance in the absence of real effects (Loken & Gelman, 2017). This is similar to Gelman & Loken, 2013's description of research as the garden of forking paths, where different decisions along the way can lead to different outcomes (see the source discussion). A second possibility for the replication crisis is that some replication studies have had limited statistical power (e.g., insufficiently large sample sizes). A third possibility may be that there is publication bias such that researchers tend to publish only significant findings, which is known as the file-drawer effect. A fourth possibility is that researchers may engage in ethically questionable research practices, such as multiple testing and selective reporting.

Garden of Forking Paths. (Adapted from https://www.si.umich.edu/about-umsi/news/ditch-stale-pdf-making-research-papers-interactive-and-more-transparent [archived at https://perma.cc/R2V9-CP3F].)

However, difficulties with replication could exist even if researchers have the best of intentions, engage in ethical research practices, and are transparent about all of the methods they used and decisions they made. The replication crisis could owe, in part, to noisy (imprecise and inaccurate) measures. The field has paid insufficient attention to measurement unreliability as a key culprit in the replication crisis. As Loken & Gelman, 2017 demonstrated, when measures are less noisy, measurement error weakens the association between the measures. But when using noisy measures and selecting what to publish based on statistical significance, measurement error can make the association appear stronger than it is. This is what Loken & Gelman, 2017 describe as the statistical significance filter: In a study with noisy measures and a small or moderate sample size,

statistically significant estimates are likely to have a stronger effect size than the actual effect size—the "true" underlying effects could be small or nonexistent. The statistical significance filter exists because, with a small sample size, the effect size will need to be larger in order to detect it as statistically significant due to larger standard errors. That is, when researchers publish a statistically significant effect with a small or moderate sample size and noisy measures, the effect size will necessarily be large enough to detect it (and likely larger than the true effect). However, the effect of noise (measurement error) diminishes as the sample size increases. So, the goal should be to use less noisy measures with larger sample sizes. And, as discussed in the source discussion on ethical considerations in psychological assessment, the use of pre-registration could be useful to control researcher degrees of freedom.

The lack of replicability of findings has the potential to negatively impact the people we study through misinformed assessment, treatment, and policy decisions. Therefore, it is crucial to use assessments with strong psychometric properties and/or to develop better assessments. Psychometrics refer to the reliability and validity of measures. These concepts are described in greater detail in Chapters (the source discussion) and (the source discussion), but for now, think about reliability as consistency of measurement and validity as accuracy of measurement.

Science Versus Pseudoscience in Assessment

Science is the best system of epistemology we have to pursue truth. Science is a process, not a set of facts. It helps us overcome blind spots. The system is revisionary and self-correcting. Science is the epistemology that is the least susceptible to error due to authority, belief, intuition, bias, preference, etc. Clients are in a vulnerable position and deserve to receive services consistent with the strongest available evidence. By providing a client with a service, you are implicitly making a claim and prediction. As a psychologist, you are claiming to have expert knowledge and competence. You are making a prediction that the client will improve because of your services. Ethically, you should be making these predictions based on science and a risk-benefit analysis. It is also important to make sure the client knows when services are unproven so they can provide fully informed consent. Otherwise, because of your position as a psychologist, they may believe that you are using an evidence-based approach when you are not.

We will be examining psychological assessment from a scientific perspective. Here are characteristics of science that distinguish it from pseudoscience:

1. Risky hypotheses are posed that are falsifiable.

The hypotheses can be shown to be wrong.

1. Findings can be replicated independently by different research groups and different methods.

Evidence converges across studies and methods.

1. Potential alternative explanations for findings are specified and examined empirically (with data).

2. Steps are taken to guard against the undue influence of personal beliefs and biases.

3. The strength of claims reflects the strength of evidence.

Findings and the ability to make judgments or predictions are not overstated. For instance, it is important to present the degree of uncertainty from assessments with error bars or confidence intervals.

1. Scientifically supported measurement strategies are used based on their psychometrics, including reliability and validity.

Science does not progress without advances in measurement, including

• more efficient measurement (see Chapters (the source discussion) and (the source discussion))

• more precise measurement (i.e., reliability; see the source discussion)

• more accurate measurement (i.e., validity; see the source discussion)

• more sophisticated modeling (see the source discussion)

• more sophisticated biopsychological (e.g., cognitive neuroscience) techniques, as opposed to self-report and neuropsychological techniques (see the source discussion)

• considerations of cultural and individual diversity (see the source discussion)

• ethical considerations (see the source discussion)

These considerations serve as the focus of this book.

Prerequisites

This book was written using Quarto (Allaire et al., 2025) in R (R-base, n.d.).

Applied examples in R are provided throughout the book. Each chapter that has R examples has a section on "Getting Started," which provides the code to load relevant libraries, load data files, simulate data, add missing data (for realism), perform calculations, and more. The data files used for the examples are available on the Open Science Framework (OSF): https://osf.io/3pwza.

Most of the R packages used in this book can be installed from the Comprehensive R Archive Network (CRAN) using the following command:

Several of the packages are hosted on GitHub repositories, including uroc (Gneiting & Walz, 2021) and dmacs (R-dmacs, n.d.).

You can install the uroc and dmacs packages using the following code:

Many of the R functions used in this book are available from the petersenlab package (Petersen, 2025): https://github.com/DevPsyLab/petersenlab. You can install the latest stable release of the petersenlab package (Petersen, 2025) from CRAN using the following code:

You can install the latest development version of the petersenlab package (Petersen, 2025) from GitHub using the following code:

The code that generates this book is located on GitHub: https://github.com/isaactpetersen/Principles-Psychological-Assessment.

Reproducibility

It is important for work to be reproducible by others. Reproducibility and replicability are key goals of the open science movement (Gandrud, 2020; Open Science Collaboration, 2015; Tackett et al., 2019; Tackett et al., 2019). To that aim, I provide information—in each chapter using R—on my setup that was used to generate the results found in this book. If you run (all of) my code in the same order with the exact same setup, you should get the same results—but let me know if not!

Attribution and critical-reading note

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Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module clinical-assessment.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Historical Perspectives on Clinical Assessment

Historically, assessment has played a key role in clinical psychology. A history of clinical psychology is provided by Benjamin, 2005. Many would consider assessment—especially the assessment of intelligence—the beginning of clinical psychology, and the role that distinguished clinical psychology from other fields. Intelligence testing has been one of the real successes of clinical psychology and assessment. For example, several early measures were developed to assess intelligence, including James Cattell's measure and the Binet-Simon Intelligence scale. Lewis Terman at Stanford extended the Binet-Simon Intelligence scale to study gifted people, which became known as the Stanford-Binet Intelligence Scales.

During World War I, instruments were used to select people for various military occupations (e.g., pilot versus submariner). Intellectual assessment, including the Army Alpha and Beta Tests, was used to decide who was psychologically unfit to serve. Then came the development of personality assessments, such as assessment of "shell shock", which would eventually become known as post-traumatic stress disorder (PTSD). The military conducted personality assessments to identify people who would be susceptible to shell shock, which would be labeled today as a measure of "neuroticism".

Intellectual assessment and personality assessment became the two principal tools for clinical psychologists. Initially, psychologists were primarily psychometricians. They developed assessment devices and helped score and interpret them. Early on, clinical psychologists did not play a large role in making diagnoses or in treatment—instead, psychiatrists did. However, that changed after World War I. Between World War I and World War II, assessment was the dominant role of practicing clinical psychologists, especially

projective personality testing—such as word-association tests, the Rorschach Inkblot Test, and the Thematic Apperception Test—and more objective personality testing. The Rorschach (1921) gave clinical psychologists a "voice at the table" with psychiatrists, even though doubts arose about its accuracy. Projective testing dominated personality assessment through the 1940s. The Minnesota Multiphasic Personality Inventory (MMPI), which became a popular objective personality test, was published in 1943.

During World War II, many of the modern psychology departments, clinical psychology programs, and VA-funded clinical psychology internships were created, not by the psychology community, but by the federal government to address wartime and post-wartime needs. Likewise, many measures were developed as a need to meet wartime needs. One of the origins of the psychometric movement in psychology was spurred by historical pressures to create tests to classify people during wartime, such as with the Army Alpha and Beta Tests.

Contemporary Trends

There has been a general decline in the frequency with which clinical assessment (and psychotherapy) is conducted by clinical psychologists. Assessment and measurement are crucial to research but are declining in treatment. This is in great part due to managed care, in which a primary goal is cost containment of mental health services. Managed care has resulted in greatly reduced patient access to mental health services, a dramatic reduction in insurance funds for reimbursement of services, more clinician time spent on paperwork and less time seeing clients, having less time for assessment because fewer sessions are reimbursed, tendencies to administer less frequent or less comprehensive assessments, and increased provision of psychological services from master's degree-level providers because they are less expensive than doctoral providers.

Nevertheless, there are contexts in which clinical assessment is more common. There is still considerable assessment in the Department of Veterans Affairs (VA) system, including testing and integrated reports. In addition to the VA, there is still considerable neuropsychological and aptitude testing. What is the future of assessment for treatment? My hope is that it will be evidence-based assessment (along with evidence-based treatment).

Terminology

Several terms are relevant for clinical assessment. Prevalence is the proportion of the population that has the condition at a given point in time, whereas incidence is the rate of occurrence of new cases. Incidence is calculated, for a specific time period, as: number of new casesnumber of people who are at risk for getting that condition. Incidence indicates the risk of developing or contracting the condition, whereas prevalence indicates how widespread the condition is. Point prevalence is the proportion of the population that has the condition at a single point in time. Lifetime prevalence is the proportion of the population that will ever have a condition at any point in their lifetime.

The positivity rate is the proportion of tests that are positive (the marginal probability or base rate of positive tests; i.e., the selection ratio). The case fatality rate is the proportion of people with a condition who die as a result of that condition.

The Three S's: Signs, Symptoms, and Syndromes

The Three S's of assessment in clinical psychology are signs, symptoms, and syndromes. Signs are observable features (or manifestations) of a disorder (Lilienfeld et al., 2015). Therefore, signs can be perceived by a clinician. By contrast, symptoms are unobservable manifestations of a disorder that can only be perceived by the client (Lilienfeld et al., 2015). The term "sign" has a connotation of being somewhat more objective than symptom because a symptom is based on the client's perceptions, but at least in clinical psychology (and also in many medical fields), both signs and symptoms have considerable elements of subjectivity. A syndrome is a collection of signs and symptoms that co-occur and may reflect a particular disorder.

Errors of Pseudo-Prediction

There are many errors of pseudo-prediction that are relevant to clinical assessment. One error of pseudo-prediction is the confusion of inverse probabilities, as described in the source discussion. Another error of pseudo-prediction involves, when predicting a low base-rate phenomenon, only capitalizing on chance occurrences.

Another form of pseudo-prediction is constructing predictions that seem compelling but that have no basis, for instance using the date and place of birth to predict one's personality. Several techniques are used by people to make their predictions seem more compelling: make it true, make it positive, make it ambiguous, and make it mysterious or intimidating. Making it "true" means surrounding predictions with true statements, and using Barnum statements. Barnum statements are statements that are true for most people, such as: "You have a tendency to be critical of yourself." Including, in your prediction, outcomes that are a common occurrence for 80% of respondents (blanket statements) may make it more likely that people believe the prediction. Most of the statements are correct, so people tend to think the whole thing is correct. Making it "positive" means making the statement positive about the person. People tend to believe positive things about the self. However, if the statement is about a third person, people tend to believe it more if the statement is partially negative. Making it "ambiguous" means using vague language such as "it has not been unusual," "sometimes," "at times," "you tend to be," etc. Making it mysterious or intimidating could involve making the prediction from some organization or process with authority. An example of making statement mysterious is the PaTE report ("Psychodynamic and Therapeutic Evaluation") (Kriegman & Kriegman, 1965).

Unsupported fad treatments are another form of pseudo-prediction. There is unfortunately a large gap between science and practice. Many of the treatments known to work are not commonly implemented in practice; instead, many treatments implemented in practice have been shown not to be helpful (compared to placebo), or have been shown to be harmful (Lilienfeld, 2007), or have insufficient evidence supporting their use. Selecting a given treatment is a prediction by the therapist that the treatment will lead to the most positive outcomes for the client than the alternatives. There are likely a multitude of reasons for the disparity between practice and research. For instance, some practitioners feel that research is useless. Other practitioners are unaware that there are more effective treatments. Others may not have the time or money to learn new treatments on top of their already-busy schedules. Yet other therapists are greedy. In other cases, it is difficult to determine which treatments work due to a variety of effects.

For instance, some ineffective treatments may appear to "work" because of placebo effects, whereby merely receiving a treatment and expecting to improve yields positive benefits—not because of the specific treatment per se. In addition to client expectancy effects, there can also be therapist expectancy effects. There can also be demand effects, where participants form an interpretation of the study's purpose and subconsciously change their behavior to fit that interpretation. In addition, ineffective treatments can appear to work due to maturational effects, especially with children and long treatments. Ineffective treatments can appear to work due to differential dropout or attrition, where the people who are improving are the most likely to stay in treatment.

In addition, there are regression effects that can make an ineffective treatment appear to work. As a reminder, the true score formula in classical test theory is: X = T + e. There are random fluctuations in an observed score; part of the observed score is stable true score and part of it is random error. As an example, consider that we will use Norwegian acupuncture. We will select clients who are at the extremes in terms of high pain. They will ultimately converge or regress to their mean (i.e., their true score). So, on average, their pain will be lower the next time. In therapy, clients tend to come in at their worst, so they are bound to improve. The clients would have tended (on average) to get better, even without treatment. This is known as regression to the mean. For these reasons, it is important to include a control group to compare how clients would have done had they received no treatment, a placebo, or some other treatment. For example, in experiments that test the efficacy of cognitive behavioral therapy (CBT), it is important to include a control group, such as no treatment control, waitlist control, treatment as usual, or best available treatment.

Violating the extension rule (the conjunction fallacy) is another form of pseudo-prediction. The extension rule states that a special outcome cannot be more probable than a general one of which it is part. But in spite of this, we tend to assign high probabilities to prototypical combinations of characteristics, events, or sequences, and irrationally low probabilities to atypical single characteristics, events, or sequences. Here's an example from Dawes1986:, n.d.

"Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations." Which is more probable?

A) Linda is a bank teller.

B) Linda is a bank teller and is active in the feminist movement."

The probability that a woman is a bank teller and feminist is less likely than the probability that she is a bank teller. However, people tend to say that B is more probable. People tend to make judgments based on the representativeness heuristic—how similar to the prototypes the cases are.

Another form of pseudo-prediction is imposing structure (non-randomness) on a pattern where there are none. In addition, expecting alternation, known as the "belief in the law of small numbers", is a form of pseudo-prediction. Expecting alternation is believing that even small sequences should be representative of populations, even though that is false. For instance, it might involve expecting that one will get three heads if flipping a coin six times.

Spurious correlations are another form of pseudo-prediction. The expected value of R^2 is (K)/(n - 1), where K is the number of predictors. The more predictors in the model, the more variance is accounted for in the outcome. Thus, with many predictors and a small sample, there is pseudo-prediction, as described in the source discussion. However, as the sample size increases, spurious correlations decrease.

Multicollinearity is also a form of pseudo-prediction. As the correlation among the predictors increase, the chance of getting an arbitrary answer increases.

Another form of pseudo-prediction is the hot hand fallacy. The hot hand fallacy in basketball is believing that making a shot makes it more likely that you will make the next shot. Most research suggests that there is no such thing as a "hot hand" (Gilovich et al., 1985; Bar-Eli et al., 2006; Avugos et al., 2013). Some recent research, however, has suggested that there may be a small hot hand effect in some contexts (Bocskocsky et al., 2014; Miller & Sanjurjo, 2014).

Capitalizing on regression effects is pseudo-prediction. When underestimating regression effects, a person can capitalize on regression effects by predicting an extremity in one variable (e.g., amount of improvement over time) due to an extremity in another variable (e.g., pain severity), when in fact, regression to the mean is most likely.

Conclusion

Many of the origins of assessment came from attempts to address wartime needs. Since the 1980s, there has been a general decline in the frequency with which clinical assessments (and psychotherapy) are conducted by clinical psychologists, due to managed care and cost containment. Many errors of pseudo-prediction are relevant to clinical assessment.

Suggested Readings

Wood et al., 2002

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module evidence-based-assessment.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Evidence-Based Assessment

I was initially hesitant to include a chapter titled "evidence-based assessment" because this whole book is about doing assessment in the strongest, most scientifically supported (i.e., evidence-based) ways. However, I decided to include a chapter on evidence-based assessment to summarize many of the important considerations in ensuring our assessments are strong for their intended uses, especially in clinical psychology, and to discuss the value of some cutting-edge (and some not-so-cutting-edge) techniques to maximize the validity of inferences. A case study in evidence-based assessment is provided by Antony & Rowa, 2005.

Considerations

In general, we need to use assessment devices that are:

• clinically relevant (if used in a clinical domain): they inform decision-making processes

• culturally sensitive: they are relevant to individuals from various backgrounds, especially the backgrounds in the population of interest

• scientifically sound: strong psychometrics (reliability and validity)

Clinically Relevant

Effective treatment depends on accurate assessment. Thus, it is important to consider the treatment utility of assessment to determine whether the assessment has value. Several study methods were designed to examine the treatment utility of assessment, including manipulated assessment (compares assessment versus no assessment) and manipulated use (compares use versus no use of data from the assessment for treatment matching).

It is also important to consider the ease of dissemination of an assessment: is it time- and money-efficient, especially in terms of its cost–benefit ratio? Brief assessments, computerized testing, and adaptive testing can help assessments be clinically practical. It should be straightforward to score and interpret. And it should provide incremental validity, i.e., additional useful information above and beyond what is gained by other assessment devices.

We have already discussed how assessments can yield more effective outcomes indirectly through matching the right treatment approach to the client. It is also worth noting, however, that assessments can also lead directly to improved treatment outcomes. Conducting an assessment is an intervention—it can have carryover effects, and can result in self-reflection, self-awareness, and reactivity, i.e., change due to observing it.

Assessment is also an important component of measurement-based care. Measurement-based care is the approach of treatment monitoring and modifying treatment accordingly (Lutz et al., 2022). Assessment does not stop at the beginning of the treatment—it is continuous and ongoing. So, ongoing measurement is recommended throughout treatment so that the clinician can modify the treatment in response to measurements. Assessment of treatment progress needs to be sensitive to change, i.e., the assessment needs to show treatment sensitivity, and it needs to be actionable, i.e., it needs to have utility.

In clinical psychological assessment, it is important for assessments to:

• consider co-occurrence of multiple issues (comorbidity) and differential diagnosis

• accurately assess conditions that frequently co-occur or covary (e.g., depression and anxiety), and to

• differentiate between multiple possibilities for what may explain the client's difficulties.

In addition, it is valuable to:

• pose alternative or competing hypotheses and test them to rule out other potential explanations

• look for disconfirming evidence, not just evidence that confirms one's suspicions

• consider assessments from multiple informants (e.g., parents, teachers, peers) and multiple levels of analysis, including biological, psychological, and social-contextual factors

• consider functional impairment and not just diagnostic status, and to

• consider the context of the client's difficulties, including the onset, duration, and course of the problem, the client's treatment history, family history, familial and cultural context, and medical conditions.

Culturally Sensitive

It is important to mitigate cultural bias of instruments. It is also important for the measure to be useful across the population of interest. To accomplish this, one may have to modify an assessment approach to account for clinically significant moderating variables. We discuss culturally sensitive assessment more in the source discussion, and we discuss test bias in the source discussion.

Scientifically Sound

The development and selection of measures should be based on scientifically supported theories in psychopathology and basic psychological science. Measures should be standardized, with similar procedures across participants, clients, and examiners.

Reliability and validity of measures' scores are specific to a particular use of the test, and are specific for a given population and context. So, it is important to clearly specify the purpose of the assessment. As just a few examples, the purpose of the assessment could be screening for early identification of risk, for diagnosis, for treatment monitoring, for treatment evaluation, or for measuring a phenomenon of interest in research. In general, the purposes of an assessment can be summarized into the 3 Ps (Youngstrom et al., 2017): predict, prescribe, and process. The purpose of an assessment is to predict if the purpose is to relate an assessment to a criterion at a later point in time. An assessment is used to prescribe if it informs decision-making about the participant or client—for example, a decision about which treatment to give, identifying moderators of treatment effectiveness, or specifying potential confounds or alternative explanations that would warrant a different treatment. An assessment is used to understand process if it informs understanding of the participant or client—for example, identifying mediators of treatment effectiveness or tracking treatment progress.

Whichever the purpose of the assessment, it is important to test and evaluate the psychometrics of the assessment device for that particular purpose. Only use the assessment

device in a test battery if it advances that purpose. Youngstrom et al., 2017 describe the different core psychometric features that are especially relevant for each purpose. For instance, predictive and discriminative validity are crucial for prediction, whereas inter-rater reliability is crucial for prescription, and treatment sensitivity—a form of criterion-related validity—is crucial for understanding process.

Standard for Excellent Tests

Additionally, when making norm-referenced judgments, it is important that the assessment has appropriate norms and evidence of accuracy for any cut-scores (e.g., diagnostic thresholds). Per Youngstrom et al., 2017, below are the norms and psychometric standards for excellent tests:

• If the assessment has norms, the norms provide a mean and standard deviation for the total score (and any subscores) that were determined from multiple, large samples that are representative of the populations to which it is intended for the test to be administered.

• Internal consistency: Greater than.90 based on Cronbach's alpha or (better yet) omega.

• Inter-rater reliability: Cohen's kappa.85; intraclass correlation.90.

• Test–retest reliability: If the construct is stable, the measure shows stability of individual differences and repeatability.

However, note that not all constructs are expected to be stable. For stable constructs, the measure should show stability of individual differences (i.e., relative or rank-order stability): rs.70 over a year or longer. For stable constructs, the measure should also show repeatability, i.e., absolute stability in level: A Bland–Altman plot and corresponding regression shows no significant bias or trends, and the repeatability coefficient (also known as Smallest Real Difference, SRD, or limits of agreement, LOA) is small.

• Content validity: The test developers clearly defined the domain and ensured representation of the entire set of facets.

All elements of the assessment are evaluated by quantitative ratings by multiple groups of judges.

• Construct validity: The assessment has evidence in support of construct validity, including concurrent, predictive, convergent, and discriminant validity, that has been replicated by independent researchers.

• Incremental validity: The assessment shows incremental utility above other measures.

• Discriminative validity: for diagnostic assessment, the measure should have an area under the receiver operating characteristic curve (AUC) of.75 to.90 under clinically realistic conditions.

AUC is the single best index of discrimination accuracy, but it is also helpful to know the sensitivity, specificity, positive predictive value, and negative predictive value at particular cutoffs of interest to determine whether the assessment is aligned with its purpose.

• Prescriptive validity: An assessment that has strong prescriptive validity regarding which diagnosis to give would have good inter-rater reliability (kappa) for diagnosis in more than one sample.

An assessment that has strong prescriptive validity regarding which treatment to give would have a moderate effect size (or larger) for treatment moderation—i.e., there is evidence for treatment matching based on the characteristic.

• Validity generalization: The bulk of the evidence supports use of the assessment with more than one demographic group (if it is to be used with more than one group) and in multiple settings.

• Sensitivity to change and treatment sensitivity: Replications from independent researchers show evidence that the assessment detects (i.e., is sensitive to) change over time, change over the course of treatment, and differences as a function of different types of treatment.

• Clinical utility: After practical considerations, including costs, respondent burden, ease of administration and scoring, replication from independent researchers shows that the assessment data are likely to be clinically actionable and confer clinical benefits, such as better outcomes, lower attrition, and greater satisfaction in areas that are important to the stakeholders.

The whole assessment process should be empirically evaluated, including the selection, administration, scoring, and interpretation of an instrument, the integration of multiple sources of assessment data, and the decision-making process. When possible, it is valuable to evaluate the accuracy of the decision-making, ideally against some gold standard, for quality control improvement.

In general, it is better to include a wide range of assessment tools than a single strong measure. For instance, assessment tools could include self- and informant-report questionnaires, interviews, self-monitoring diaries, behavioral observation, performance-based techniques, and psychophysiological techniques. Then, each measure can be weighted according to its diagnostic likelihood ratio—i.e., its concurrent/predictive validity in the form of the positive likelihood ratio.

One challenge is how to handle a situation where different instruments yield different answers. For instance, if the results from parent report versus teacher report conflict with each other. To address potential discrepancies, one can weight the different instruments or informant sources according to their predictive accuracy. For instance, you might give more weight to youths' self-report for assessing anxiety but might give more weight to parent report for assessing ADHD. One way to handle informant discrepancies when there is not a clear primary informant who is best-positioned to rate the person most accurately is to count a symptom as present if it is endorsed by any of the informants (Hinshaw & Nigg, 1999). Alternatively, one can use a trifactor model to separate the effects of construct, item, and informant (Bauer et al., 2013; Shin et al., 2019), where construct reflects the "general" factor and "item" and "informant type" (e.g., parent, teacher, self) are two cross-cutting method factors. A trifactor model is similar to (and an extension of) a bifactor model; however, each rating loads onto three latent sources (in this case, construct, item, and informant type). Nevertheless, simpler models that include all informants often outperform (in terms of predictive validity) more complex models that integrate scores across informants, suggesting that each informant (e.g., parent, teacher, self) has an important perspective and provides valuable information (Aitken et al., 2024).

Bayesian Updating

As presented in the source discussion, the posttest (or posterior) odds are equal to the pretest odds multiplied by the likelihood ratio. Bayes' theorem is discussed in the source discussion. Using this formula, and converting odds to probabilities, we can use a Fagan probability nomogram to determine the posttest probability following a test result. The calculation of posttest probability is described in the source discussion. A probability nomogram is a way of visually applying Bayes' theorem to determine the posttest probability of having a condition based on the pretest (or prior) probability and likelihood ratio, as depicted in the source discussion. To use a probability nomogram, connect the dots from the starting probability (left line) with the likelihood ratio (middle line) to see the updated probability. The updated (posttest) probability is where the connecting line crosses the third, right line.

Probability Nomogram. (Figure retrieved from https://upload.wikimedia.org/wikipedia/commons/thumb/6/66/Fagan_nomogram.svg/945px-Fagan_nomogram.svg.png).

For instance, if the starting probability is.5% and the likelihood ratio is 10 (e.g., sensitivity =.90, specificity =.91: likelihood ratio = sensitivity1 - specificity = (.9)/(1-.91) = 10), the updated probability is less than 5%, as depicted in the source discussion. The formula and function for computing posttest probability are provided in the source discussion. The petersenlab package (Petersen, 2025) contains the petersenlab::posttestProbability function that calculates the posttest probability from the base rate and the likelihood ratio.

Probability Nomogram Example. (Figure adapted from https://upload.wikimedia.org/wikipedia/commons/thumb/6/66/Fagan_nomogram.svg/945px-Fagan_nomogram.svg.png)

A probability nomogram calculator can be found at the following link: http://araw.mede.uic.edu/cgi-bin/testcalc.pl (archived at https://perma.cc/X8TF-7YBX).

Dimensional Approaches to Psychopathology

The Diagnostic and Statistical Manual of Mental Disorders (DSM) takes primarily a categorical approach to psychopathology—it classifies mental disorders as discrete categories. That is, anxiety is classified as a distinct disorder from depression in the DSM, despite their strong covariation and co-occurrence. Moreover, disorders are generally classified in the DSM in a binary fashion—according to the DSM, you either "have" the disorder or "do not have" the disorder—there is no middle ground, and there is not a finer differentiation among those who are given the disorder.

Despite the categorical approach of the DSM, most disorders seem to be more accurately conceptualized as dimensional than categorical (Markon et al., 2011). As described in the source discussion, DSM disorders are fictive categories. They do not carve nature at its joints. Many disorders show high comorbidity with other disorders. Transdiagnostic dimensions can better account for the heterogeneity within and across disorders—and more parsimoniously—than DSM categories. For instance, depression and anxiety may share an underlying construct of negative emotionality. A dimensional approach considers a person's level on each dimension, even if the person is sub-threshold. Moreover a transdiagnostic approach may afford greater flexibility in treatment across disorders by focusing on dimensions of difficulties that may cut across traditional diagnostic boundaries. For

instance, the Unified Protocol is a form of psychotherapy designed to address a range of emotional disorders, not just depression or anxiety (Ellard et al., 2010).

Multi-Stage Approach to Assessment

Another way in which a dimensional approach to psychopathology can provide greater treatment flexibility than a categorical approach is in applying a multi-stage approach to assessment. Youngstrom & Van Meter, 2016 described a multi-stage approach to assessment in which the treatment provider calculates the posterior probability of disorder based on an instrument score, the extent of risk factors present, and the base rate of the disorder. Based on the posterior probability of disorder, the treatment provider places the client into one of three graded strata: the Green zone, Yellow zone, or Red zone (see the source discussion).

Multi-Stage Approach to Assessment.

The Green zone is the "Low Probability/Wait Zone". If the client is in the Green zone, the course of action would be to either do nothing or to engage the client in primary prevention techniques that are safe and inexpensive. The threshold between the Green zone and Yellow zone is the "wait–test" threshold. If the client exceeds the wait–test threshold, they move to the Yellow zone.

The Yellow zone is the "Moderate Probability/Assessment Zone". If the client is in the Yellow zone, secondary intervention would be indicated, including targeted intervention efforts, or nonspecific and low-risk treatment, such as skills-oriented psychotherapies. More assessment would be indicated until the posterior probability rises into the treatment zone (Red zone) or falls into the wait zone (Green zone). The threshold between the Yellow zone and Red zone is the "test–treat" threshold. If the client exceeds the test–treat threshold, they move to the Red zone.

The Red zone is the "High Probability/Acute Treatment Zone". If the client is in the Red zone, tertiary intervention would be indicated. The locations of the wait–test threshold and the test–treat threshold depend on the risks and benefits and the client's preferences.

To conduct a multi-stage approach to assessment, Youngstrom & Van Meter, 2016 recommend starting with an instrument with high sensitivity, such as tests with broad content coverage. Low scores on a measure with high sensitivity rule out diagnosis (i.e., the client is in the Green zone), but high scores are ambiguous (i.e., the client is in the Yellow zone) because the measure does not have high specificity. Tests in psychology do not have high accuracy, so cutoffs will commonly be set to prioritize either sensitivity or specificity, depending on the purposes of the assessment (and the relative costs of false negatives versus false positives). The mnemonic device that can be helpful for remembering the utility of sensitivity and specificity is "SNout" and "SPin". Low (i.e., negative) scores on a test with high sensitivity rule OUT a diagnosis, whereas high (i.e., positive) scores on a test with high specificity rule IN a diagnosis. If a person scores high on the test with high sensitivity and goes to the Yellow zone, follow up with tests that have higher specificity, including more focused tests, to reduce the number of false positives. Ideally, use multiple tests that have incremental validity and do not have high inter-correlation (r less than.3) so that they are not redundant (e.g., not all self-report) and so they meaningfully update the posterior probability. To avoid redundancy with rating scales, you can "take the best"—i.e., include

only the most valid rating scale from each informant to test each hypothesis to avoid high inter-correlations and multicollinearity.

To integrate the results of multiple assessments, use a probability nomogram or a diagnostic test calculator (http://araw.mede.uic.edu/cgi-bin/testcalc.pl; archived at https://perma.cc/X8TF-7YBX) to update the posterior probability after each assessment. The locations of the wait–test threshold and test–treat threshold are based on the risks and benefits of diagnosis and treatment, and can also be influenced by the client's preference for various approaches.

Reporting Guidelines for Publications

When publishing papers, it is important to report effect sizes, including the AUC, diagnostic likelihood ratios, and correlations between test scores and the criterion. Do not just rely on statistical significance. The effect size is important for determining the practical or clinical significance of the effects. Guidelines for reporting about and evaluating assessments in research reports are provided by Kazdin, 1995.

It is also important to report aspects of reliability and validity, both in the current sample and from prior studies. Describe the current sample in adequate detail to understand the population to whom the findings from the study may best generalize. Describe how many items were included in the assessment, the item content (or a description of the content, if proprietary), how the assessment was scored, how scores were aggregated (e.g., sum or mean), the range of possible values, the mean and standard deviation of scores, and what higher values represent.

Many Measures Are Available

Assessment in psychology is similar to the Wild West. There are many different measures available. Many measures in psychology do not have evidence of strong psychometrics (reliability/validity) and are not based on research in basic psychological science including cognitive psychology, social psychology, etc. Measures in clinical psychology often focus on complex syndromes and conditions rather than basic states or processes, unlike most tests in medicine.

Because there are so many measures, it can be challenging to know which ones to use. To learn about good tests or the strengths and weaknesses of assessment devices in psychology, there are resources that provide reviews of measures, including the Buros Mental Measurements Yearbook (Buros Center for Testing, 2021), which is published by the Buros Center for Testing, and the Handbook of Psychiatric Measures (Rush et al., 2009), which is published by the American Psychiatric Association. Another valuable resource for learning about available assessments is COSMIN (COnsensus-based Standards for the selection of health Measurement INstruments): https://www.cosmin.nl.

Conclusion

Evidence-based assessment involves using assessment devices that are clinically relevant, culturally sensitive, and scientifically sound. Standards are available for excellent tests. Bayesian updating is a key technique for evidence-based assessment. For assessing psychopathology, evidence-based techniques include dimensional and multi-stage assessment approaches. There are many measures available for assessment; the Buros Mental Measurements Yearbook (Buros Center for Testing, 2021) and the Handbook of Psychiatric Measures (Rush et al., 2009) provide reviews of measures.

Suggested Readings

Hunsley & Mash, 2007; Youngstrom et al., 2017

CHAPTER 02 / CAC.903

Ethics, Consent, Assent, and Test Security

Assessment has power. It can open doors to support, but it can also label, exclude, or expose a child. Ethics is not an add-on to technical skill. It is the frame that makes assessment trustworthy. In child and adolescent work, ethics is especially complex because the client is developing, caregivers have legal roles, and schools or courts may request information. The counselor must protect the young person's welfare while respecting lawful authority and confidentiality limits.

Consent, permission, and assent

In many settings, a parent or legal guardian provides permission for assessment. The child provides assent when developmentally able. Assent is not a legal formality. It is an ongoing process of explaining what will happen, why it may help, what will be shared, and what choice the child has. A seven-year-old may understand a simple explanation with pictures. An adolescent may understand a detailed discussion of privacy and report use.

A counselor might say, "We will do some activities to understand how school feels for you. I will talk with your parent about how to help. I will not share every private thing you say, but I will share if I think someone is unsafe." The language should match the child's development. The counselor should check understanding, not just obtain a signature.

Confidentiality and its limits

Confidentiality is central to trust, but it is not absolute. Limits include risk of harm, abuse or neglect reporting, court orders, and certain school or agency rules. These limits should be explained before assessment begins. Adolescents are especially sensitive to privacy. If a teen believes everything will be reported to parents, disclosure may stop. If a teen believes nothing will be shared, trust may break when safety requires action.

The counselor should distinguish private content from safety content and from decision-relevant content. A teen's music preferences may be private. A teen's plan to harm themselves is safety-relevant. A teen's sleep pattern may be decision-relevant and should be shared in a useful way. The counselor should share the minimum necessary information and, when possible, involve the adolescent in how it is shared.

Competence and scope

Counselors should use only methods they are trained to administer, score, and interpret. Some tests require specific qualifications. A counselor should not interpret a complex cognitive battery without training. Competence also includes cultural and linguistic competence. A test may be standardized, but the counselor may not be competent to interpret it for a child from a different language background without appropriate support.

Scope matters in interprofessional settings. A counselor may assess emotional and behavioral functioning but should not diagnose a medical condition. A physician may diagnose epilepsy but may need the counselor's behavioral data. Clear roles protect the client. The counselor should know when to refer, consult, or co-assess.

Test security and data protection

Standardized tests depend on security. If items become public, scores may lose meaning. Counselors should not post test items, answer keys, or protocols online. Reports should be stored securely and shared only with authorized persons. Digital assessment adds risks: data breaches, insecure platforms, and unclear ownership. The counselor should use approved platforms, limit access, and explain data handling.

Test security also includes responsible interpretation. A score should not be released without context that reduces misuse. A raw score without norms may be meaningless. A percentile without confidence intervals may be overinterpreted. The counselor should provide information in a form that helps the recipient make good decisions, not in a form that invites labeling.

Bias, fairness, and justice

Ethical assessment asks who benefits and who may be harmed. A tool validated mainly with one group may not be fair for another. A cutoff may produce disproportionate identification of a group for discipline or special education. The counselor should examine bias, not assume neutrality. Fairness does not mean identical treatment in every case. It means that interpretations and decisions are justified for the individual and context.

Justice also includes access. A family may not afford a private evaluation. A rural school may lack a specialist. The counselor should consider feasible alternatives and advocate for resources. An ethical assessment plan is not only psychometrically sound. It is also accessible and respectful.

Records, reports, and feedback

Assessment records should be accurate, timely, and limited to relevant information. Reports should distinguish observation from interpretation. They should include limitations and alternative explanations. Feedback should be understandable to the family. A report that only specialists can read may fail its ethical purpose. The counselor should explain results in plain language, invite questions, and check whether the family understands the plan.

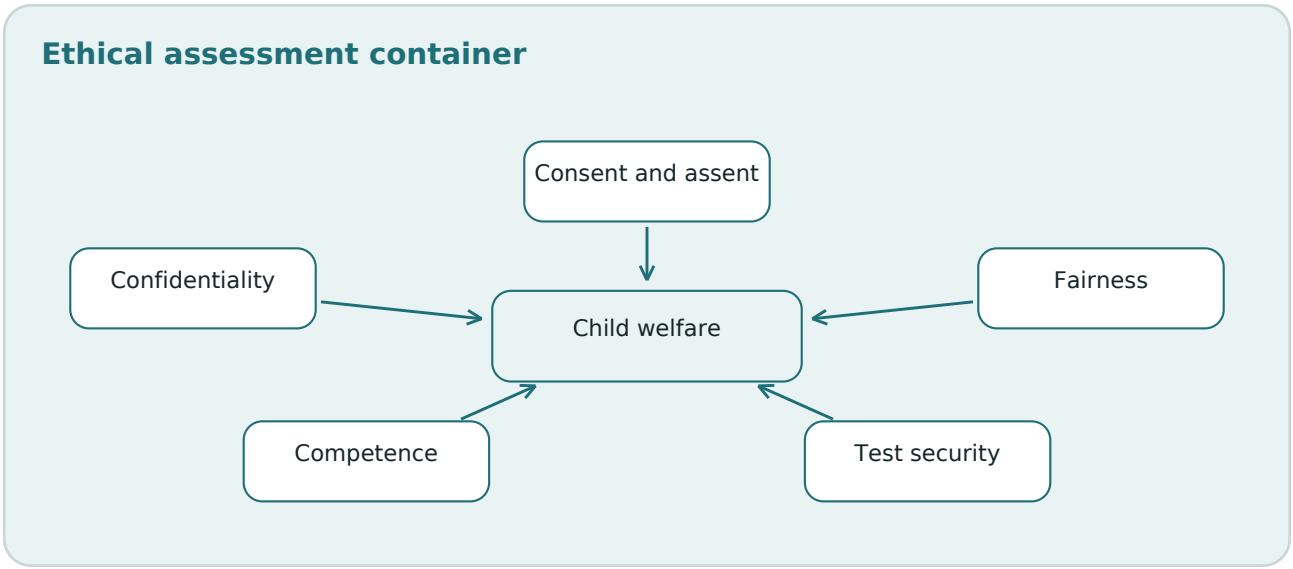
When results are uncertain, say so. A confident report based on weak evidence can harm a child. Ethical practice includes tolerance for uncertainty. It is better to recommend monitoring and follow-up than to force a premature conclusion.

Real-life application

A school administrator asks for a student's full assessment report to decide discipline. The counselor should not simply hand over the file. The counselor should clarify the lawful basis, share only information relevant to the decision, protect private therapy content, and involve the family as appropriate. If the report contains sensitive trauma information, the counselor should consider whether a summary of functional needs is safer and more useful.

Reflection

Think of a time when information about you was shared without your understanding. How did it affect trust? Children and adolescents experience the same. Ethical assessment is not only about following rules. It is about protecting a young person's developing agency.



Technical skill is held inside ethical responsibilities to the child and family.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module ethics.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Ethical Issues in Assessment

Ethics are relevant to all of our domains as psychologists, including (but not limited to) research, teaching, assessment, intervention, and supervision. Ethics provide a framework that guides us in moral behavior and knowing right from wrong (Curran & Hancock, 2020). Ethics are not always black and white; there is often considerable gray. Ethical behavior is probably better conceptualized along a spectrum than a binary. Nevertheless, there are many examples of highly problematic unethical behavior. Ethics focuses on the morality of the behavior, not whether the person is good or bad.

Belmont Report

The Belmont Report informs professional ethics. The principles from the Belmont Report include: respect for persons, beneficence, and justice.

Respect for Persons

A key component of respect for persons is informed consent. It also involves special considerations to protect the welfare of vulnerable people, including children, prisoners, pregnant women, people with intellectual disability, and people who are economically or educationally disadvantaged. It is important to get full informed consent from clients or participants before taking actions that will affect them.

Beneficence

The principle of beneficence is about maximizing benefit and minimizing harm, similar to the aphorism, "First, do no harm". Clients and participants have the prerogative to make choices about whether to receive a procedure, so they need to know the benefits and harm of each procedure. Just having the intent to help is not enough—you also need to be using a helpful technique and to have the knowledge of how to implement the technique.

Justice

Justice goes beyond the interaction with the individual client or participant; it deals with broader, societal concerns. For instance, justice involves ensuring that the people taking the risk will also benefit. For example, if a study to test the effectiveness of a medication enrolls primarily homeless people, the participant population may be taking the risk of trying the drug but not ultimately benefiting from the knowledge gained.

My Ethical Advice

My ethical advice is summarized in the following two pieces of advice:

1. Don't do stupid things

2. Don't do bad things

Don't Do Stupid Things

First, don't do stupid things. With respect to assessement, for instance, do not use weak assessment instruments. The assessment techniques you use should be reliable and valid for the tasks and population of interest. They should be based on scientific research, not by professional judgment. Reliability and validity have many facets and all facets must be used in evaluating assessment techniques. Reliability and validity are characteristics of uses of tests, not the tests themselves. So, identify what is important, for example test–retest reliability versus internal consistency. Reliability and validity may differ for populations that vary in characteristics (e.g., age, gender, education, ethnicity).

Don't Do Bad Things

Second, don't do bad things. For instance, do not administer services without first getting the person's consent. Informed consent for an assessment includes multiple components. Ensure that the person being assessed is informed about important aspects of the testing situation. Say what services will be administered (i.e., the nature of testing), why the testing will be administered (i.e., the purpose of testing), what the evidence is for the assessment devices, the costs and potential side effects of the assessment and the tests, the pros and cons of underdoing the assessment, and other options available. Describe the type of information that the client will receive. Describe any third-party involvement (e.g., insurance companies, schools, employers, court) in accessing the assessment information. Describe the limits of confidentiality, which typically would include when someone is in danger, as defined by unreported child abuse or neglect, elder abuse, or imminent suicidality or homicidality. If the assessment is court-ordered or the results of the assessment are subpoenaed by a court of law, the limits of confidentiality change. It is also important to describe the psychologist's role relative to the client. Be transparent about what services are experimental; do not pretend to know more than you know. Do not make false or misleading claims or promises.

Honor the agreement implied by the explanation provided in the informed consent. For an example of a document that I use with clients that describes the pros and cons of an intervention technique (parent management training) and other options available, see here: https://osf.io/qmnu6.

It is also important to maintain the integrity of assessment techniques. Do not teach people how to raise their scores. There has to be integrity to begin assessment.

Consult, Consult, Consult

If you are unsure how to handle an ethical dilemma, CONSULT, CONSULT, CONSULT! In notes and reports, provide documentation of the rationale for decisions made. Professional ethics boards want to know that you are acting in good faith of the client's best interests.

APA Ethics Code

The American Psychological Association (APA) publishes ethics guidelines in its document entitled, Ethical Principles of Psychologists and Code of Conduct (hereafter referred to as the "APA Ethics Code") (American Psychological Association, 2017). The latest version of the APA Ethics Code is available here: https://www.apa.org/ethics/code (archived at https://perma.cc/6F4Z-WQ57). The current version of the APA Ethics code was published in 2017 and is available here: https://www.apa.org/ethics/code/ethics-code-2017.pdf (archived at: https://perma.cc/PF6F-QZ5C). The APA also published "Guidelines for Psychological Assessment and Evaluation", available here: https://www.apa.org/about/policy/guidelines-psychological-assessment-evaluation.pdf (archived at https://perma.cc/FRB2-ND58).

Here, I emphasize a few of the guidelines from the APA Ethics Code with respect to assessment. However, you should read and follow the entire APA Ethics Code. That said, the APA Ethics Code is not up to date for clinical science. But there are many important components. Case illustrations of ethical guidelines from the APA Ethics Code are provided

by Campbell et al., 2010. Ethical issues in psychological assessment are also discussed by Bersoff et al., 2012 and Nagy, 2011.

Competence

Section 2 of the APA Ethics Code notes that you should only act withing the boundaries of your competence. It is also important to be aware of and report conflicts of interest.

Bases for Assessments

Section 9 of the APA Ethics Code describes ethics guidelines for assessment. Section 9.01 describes ethics guidelines regarding the bases for assessments. Use techniques that are sufficient to substantiate the findings, based on strong psychometrics. In a professional context, sit down face to face and evaluate the individual, as opposed to just relying on automated reports. If using computerized assessments, you should have the expertise to consider the appropriateness of the interpretations. Work to reduce or eliminate bias. If using record review, explain the sources of information used for conclusions and recommendations. Maintain a high standard when recording assessment information. Psychologists have responsibility for maintenance and retention of their records. Records should describe the nature, delivery, progress, and results of services, and related fees. Take reasonable steps to maintain confidentiality of records. When a client might want to know how records will be maintained, disclose record-keeping procedures in the informed consent. The duration that full records should be kept depends on whether they are adults or children. For adults, keep full records for 7 years after the last date of service. For children, keep full records for 7 years after the last date of service, or for 3 years after they become adults, whichever is later. The latest version of the APA record keeping guidelines is available here: https://www.apa.org/practice/guidelines/record-keeping (archived at https://perma.cc/S37L-5VNX). The current version of the APA record keeping guidelines was published in 2007 and is available here: https://www.apa.org/pubs/journals/features/record-keeping.pdf (archived at: https://perma.cc/7A8U-M4TQ).

Use of Assessments

Section 9.02 describes ethics guidelines regarding the use of assessments. It notes to use assessments "in light of research or evidence". But how would research and evidence not go together?! This phrase from the APA Ethics Code suggests that some forms of non-research-based evidence have equal footing with research in terms of deciding what assessments to use. I would argue that assessments should be selected based on the best available scientific evidence. And if scientific evidence does not yet exist to use an assessment device for a particular purpose with a particular population, the assessment device should be studied and evaluated before it is used to make important decisions.

The assessment should be reliable and valid for the population **and** purpose. When the reliability and validity has not been established, describe the strengths and weaknesses. Use language and competence level that is appropriate for the participant/client. Consider the potential social consequences of testing. High stakes decisions probably should not be made on the basis of the results of a single test.

Informed Consent in Assessments

Section 9.03 describes ethics guidelines regarding informed consent in assessments. Informed consent should be obtained unless:

• the assessment is government mandated (which changes things a bit),

• it is implied because it is considered voluntary (e.g., when applying for a job), or

• evaluating decisional capacity and the person is not capable of providing consent; in these cases, you should still inform them of what testing will include and the purpose of assessment.

Release of Test Data

Section 9.04 describes ethics guidelines regarding release of test data. "Test data" include raw and scaled scores, a client's responses to questions or stimuli, and the psychologist's notes about the participant. You are expected to release test data if the client provides a signed release, unless it would cause substantial harm or would be a misrepresentation of the data or test. You may also have to release test data if there is a law or court order, but you may be able to fight these.

Other Guidelines

Section 9.07 notes that you should not promote the use of assessment techniques by unqualified people. Section 9.08 notes that you should not use obsolete tests and outdated results.

Maintain Test Security (Integrity)

Section 9.11 describes ethics guidelines regarding maintaining test security (integrity). Maintaining test integrity involves protecting test instruments, protocols, questions or stimuli, and manuals from unauthorized access. If releasing test data, take care to release only the scores of copyrighted tests; do not release the copyrighted tests themselves. For example, if performing intellectual testing, try to release the subscale scores and summary scores (e.g., T scores) without releasing test stimuli. Also, do not coach people on how to perform better on questions, and do not let them take tests home.

Clinical Report Writing

Write every report as if it will be used in a court of law. You want it to be able to stand up in court. Always identify the source of each inference (e.g., "Per the client's report,..."). Do not write a self-reported statement as if it were a factual statement. Instead of writing, "The client cut herself", write "The client reported that she cut herself". Do not overstate findings. Information from reports can be used to make decisions that significantly impact people's lives, so it is important not to misrepresent information. Recognize factors, such as language, culture, and the situation, that could have impacted the results, and note them in the report. Recognize limitations in the recommendations.

Open Science

There has been a replication crisis in psychology and social science, in which the findings from many studies have failed to replicate when independent researchers attempt to replicate the studies' findings (Open Science Collaboration, 2015). Due to the replication crisis, there has been a movement toward open, transparent science where researchers pre-register their methods, and share their data and analysis scripts. Open practices reduce threats to replicability and increase the public accessibility of science. The movement to open practices in science is known as the open science movement.

The open science movement seeks to increase the replicability and accessibility of science through three primary methods. First, the open science movement seeks to improve transparency, i.e., what researchers are actually doing in their studies and analyses. Second, especially when researchers have confirmatory hypotheses, the open science movement seeks to restrict researcher degrees of freedom so that the researchers do what they said they would do rather than contingent analysis decisions to get a desired result. Similar to a garden of forking paths (see the source discussion), there are so many decision points in a study, and different decisions can lead to drastically different results. Third, the open science movement seeks to prevent ethically questionable research practices. Ethically questionable research practices may include (John et al., 2012):

• not reporting all dependent variables

• deciding to collect more data after looking at the data to see whether the results were significant

• not reporting all of a study's conditions

• stopping collecting data earlier than planned because the expected result is detected

• rounding down a p-value

• selectively reporting studies that worked

• deciding whether to exclude data after looking at the impact of doing so on the results

• reporting an unexpected finding as having been predicted from the start (known as HARK-ing: **h**ypothesizing **a**fter the **r**esults are **k**nown)

• claiming that results are unaffected by demographic variables when one is actually unsure

• falsifying data

• p-Hacking: multiple testing, selective reporting, and misreporting of the true effect size

Wicherts et al., 2016 provide a list of various researcher degrees of freedom, which can serve as a checklist against p-hacking.

To advance reproducibility and replicability in science, researchers are encouraged to provide greater transparency and sharing of their research. For instance, one way to provide greater transparency and to limit the researcher's degrees of freedom is to (pre-)register a study. There is a continuum of study registration that includes pre-registration, co-registration, and post-registration, depending on when the registration occurs relative to data collection and analysis (Benning et al., 2019). Study pre-registration involves publicly posting a study's design, hypotheses, methods, materials, and data analysis plan prior to

data collection. Co-registration occurs after data collection starts but before data analysis begins. Post-registration occurs after data analysis has begun.

Although pre-registration is most commonly used with confirmatory research in which you have a directional hypothesis, you can also pre-register exploratory research in which you have no directional hypothesis. Pre-registration helps to make it more explicit what tests are confirmatory (based on a priori directional hypotheses) and which are exploratory. It is important to interpret the results of confirmatory and exploratory tests differently, because you are more likely to get false positive results if conducting lots of exploratory tests. You can also pre-register decision trees that allow flexibility for multiple approaches depending on the data.

However, there may be instances when it makes sense to deviate from a preregistered plan, such as when encountering unanticipated events. Lakens, 2024 provides guidance on when to deviate from a preregistered plan and how to report such deviations.

Some journals see the value of pre-registered studies and publish registered reports. A registered report is a study that is evaluated based on the pre-registered methods, hypotheses, and data analysis plan, and if the pre-registered study is accepted, the eventual paper will be accepted (as long as the researchers followed the pre-registered plan) regardless of whether the findings supported the hypotheses. Registered reports of exploratory analyses are sometimes called exploratory reports.

There are many tools that aim to advance the open science movement. A key tool is the Open Science Framework (OSF; https://osf.io). The OSF is a free, open source platform that supports the transparency of scientific research. There are numerous ways that researchers can make use of the OSF to advance the aims of open science. Here is a template for a pre-registration of a study on the OSF: https://osf.io/2hrfy. Researchers can (pre-)register their study on the OSF by posting their hypotheses, study methods, and data analysis plan. In addition to the OSF, other website for pre-registration include AsPredicted (https://aspredicted.org), PROSPERO (for systematic reviews; https://www.crd.york.ac.uk/prospero), and ClinicalTrials.gov (https://clinicaltrials.gov). In addition, researchers can use the OSF to post manuals, protocols, lab notebooks, data, data processing syntax, statistical analysis code, detailed results, computational notebooks, and paper preprints (Tackett et al., 2019). A computational notebook (e.g., Jupyter notebook, R Markdown, Quarto) is a document in which the analysis code, output, and text are weaved together, so the reader can see how each statistical result was obtained for reproducibility. An example of an R Markdown computational notebook is on the book's page on the OSF: https://osf.io/nkyve (the associated .Rmd file that generated the html is here: https://osf.io/fxrzm). An example of a Quarto computational notebook is also on the book's page on the OSF: https://osf.io/8ybtk (the associated .qmd file that generated the html is here: https://osf.io/zbruf). In addition to the OSF, other website for sharing materials include Databrary (which is particularly useful for sharing videos; https://nyu.databrary.org), Zenodo (https://zenodo.org), the Dataverse Project (https://dataverse.org), and figshare (https://figshare.com).

After pre-registration, you can share a preprint of your paper on a preprint server like arXiv, PsyArXiv, bioRxiv, SocArXiv, EdArXiv, MetaArXiv, or medRxiv. For an example of a preprint, see here: https://psyarxiv.com/gtsvw. Posting a preprint disseminates the paper without waiting for peer review and the publishing process. It also provides free access to papers

that might otherwise be hidden behind journals' paywalls. Also, it helps establishes who was first to perform work on a topic.

There are now journals dedicated to publishing replication studies, including ReScienceX (http://rescience.org/x) and to publishing null findings, including the Journal of Articles in Support of the Null Hypothesis (https://www.jasnh.com).

Nevertheless, open science does not provide a one-size-fits-all solution to the vast diversity of scientific methods. For instance, there may be important ways needed to adapt open science and pre-registration practices to longitudinal research (Petersen et al., 2024).

Conclusion

According to the Belmont Report, professional ethics includes respect for persons, beneficence, and justice. It is also important to follow the APA Ethics Code (American Psychological Association, 2017) and AERA guidelines (American Educational Research Association et al., 2014). It is also important to produce a replicable science, which requires better science, including use of measures with strong psychometrics (reliability and validity), in addition to not engaging in questionable research practices. The open science movement, including pre-registrations, sharing data, preprints, and research materials may help improve the replicability of science.

Suggested Readings

American Psychological Association, 2017; American Educational Research Association et al., 2014; Bersoff et al., 2012; Nagy, 2011; Campbell et al., 2010

CHAPTER 03 / CAC.903

Cultural and Individual Diversity in Assessment

Every assessment happens in a cultural context. Language, family roles, religion, migration history, disability, gender, socioeconomic position, and experiences of discrimination shape how distress is expressed and how help is sought. A counselor who ignores culture may mistake difference for disorder. A counselor who stereotypes culture may miss the individual. The task is cultural humility: curiosity about the person's world, awareness of power, and willingness to correct assumptions.

Culture is not a checklist

Culture is often taught as a list of group traits. That approach is dangerous. A Nigerian adolescent in Lagos may value family obligation and personal ambition. A recent immigrant may feel connected to both heritage and peer culture. A deaf child may belong to a linguistic community with its own norms. The counselor should ask what matters to this child and family, not assume what matters to a category.

Cultural formulation questions include: How does the family understand the problem? What language do they prefer? Who makes decisions? What help have they tried? What role do faith or community play? What barriers exist? What strengths support the child? These questions turn culture from a label into a conversation.

Language and interpretation

Language affects assessment deeply. A child may understand everyday speech but struggle with abstract test language. A translated test may not be equivalent. An interpreter may help, but interpretation changes the interaction. The counselor should use qualified interpreters when needed, brief them beforehand, and speak to the child rather than the interpreter. The counselor should avoid using a child as an interpreter for a parent, because it reverses roles and may expose the child to adult content.

When a test is not available in the child's language, the counselor should consider nonverbal measures, dynamic assessment, curriculum-based measures, and observation. The report should state language limitations clearly. A score obtained through an interpreter should not be treated as fully standardized unless the tool supports that use.

Norms and reference groups

Norms compare a child's score to a reference group. If the reference group does not resemble the child, interpretation may be misleading. A child from a rural community may have different exposure to test content than a child from an urban sample. This does not mean the child lacks ability. It may mean the test measures familiarity as well as construct. The counselor should examine norm samples, exclusion criteria, and evidence for subgroup validity.

When norms are limited, the counselor should use multiple sources and avoid high-stakes decisions based on one score. The report should describe uncertainty. It may be more useful to describe functional performance than to assign a global label.

Discrimination and stress

Experiences of racism, bullying, poverty, and exclusion can produce symptoms that resemble anxiety, depression, or oppositionality. A counselor should not pathologize a reasonable response to an unsafe environment. Assessment should ask about discrimination and safety. It should also ask about community strengths, cultural pride, and coping. A formulation that includes oppression is not political bias. It is accurate context.

At the same time, culture should not be used to excuse harm. A cultural practice that endangers a child requires safeguarding. The counselor should balance respect with protection, consult when uncertain, and follow local law and ethical standards.

Disability and neurodiversity

Disability is part of diversity. Assessment should distinguish impairment from mismatch between the child and environment. A child with autism may struggle in a noisy classroom but thrive with structure. A child with ADHD may fail under long lectures but succeed with movement breaks. A neurodiversity-affirming assessment asks what supports allow the child to function, not only what deficits exist.

The counselor should avoid forcing eye contact, suppressing harmless stimming, or treating masking as health. Reports should describe strengths and needs. Language should respect the preferences of the child and community, whether person-first or identity-first.

Working with families

Families may have different explanatory models. A parent may understand distress as spiritual, disciplinary, or medical. The counselor should not dismiss these views. The counselor can explore meaning and collaborate with trusted supports while maintaining clinical responsibility. A family may prefer a religious leader, elder, or community healer. With consent, coordination can improve engagement. The counselor should still assess safety and evidence-based needs.

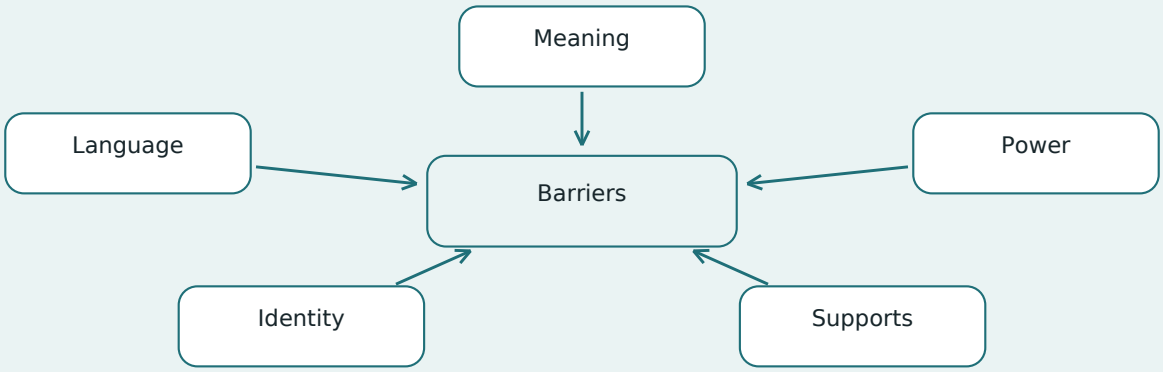
Real-life application

Fourteen-year-old Zainab is referred for defiance after refusing an after-school program. A culturally responsive assessment reveals that she cares for a younger sibling while her mother works nights. The refusal is not simple opposition. It is a negotiation of responsibility. The plan adjusts the schedule, honors her contribution, and protects her education. The assessment changed because the counselor asked about context.

Reflection

Write three assumptions you might bring to a family from a different background. For each, write a question that would test the assumption. Cultural humility is not knowing every culture. It is knowing that your first interpretation may be incomplete.

Cultural formulation questions



Culture is explored as a lived context, not a checklist or stereotype.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module diversity.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Cultural and Individual Diversity

Like ethics, considerations of cultural and individual diversity are relevant to all of our domains as psychologists, including (but not limited to) research, teaching, assessment, intervention, and supervision. As a result, we have discussed important considerations about diversity throughout the book, including the chapters on validity, evidence-based assessment, and test bias. At the same time, given the importance of cultural and individual diversity, we also provide this section, which is devoted more fully to these issues within the assessment context. Diverse samples provide stronger tests of theories and better generalizability of findings (external validity). On the assessment side, we need measures that provide us with the best information, and that are reliable and valid for all people we use them with. Diversity is a broad concept, and it spans many dimensions of differences among people.

Diversity

Diversity refers to all aspects of differences between people. Diversity is a characteristic of a group, not a characteristic of an individual. Thus, a person is not diverse. There are many aspects of human difference, including but not limited to race, ethnicity, creed, color, sex, gender, gender identity, sexual orientation, socioeconomic status, language, culture, national origin, religion/spirituality, age, (dis)ability, military/veteran status, personality, political perspective, and associational preferences.

Having a diverse group (e.g., of participants, clients, patients, etc.) is insufficient. It is also important to make sure that we take steps for all groups to feel included (inclusion) and to thrive (equity).

Equity

Equity refers to fair and just practices and policies that ensure everyone can thrive. Equity is different from equality. Equality involves treating everyone equally (i.e., treating everyone the same). However, due to structural inequities—historic and current (e.g., structural racism and patriarchy)—some people are disadvantaged and marginalized more so than others. Structural racism refers to "the normalization and legitimization of an array of dynamics—historical, cultural, institutional, and interpersonal—that routinely advantage non-Latine White persons while producing cumulative and chronic adverse outcomes for Black and other people with minoritized identities in housing, education, employment, health care, criminal justice, and psychology." (Byrd et al., 2021). Patriarchal societies reflect society's long history of wage gaps, gender discrimination, violence against women, etc. Treating people equally (equality) does not allow everyone to thrive given these different experiences and unequal access to opportunities. Thus, equity seeks to help everyone thrive, including those who are marginalized with fewer access to opportunities.

Inclusion

Inclusion refers to providing a community where all members are and feel respected, have a sense of belonging, and are able to participate and achieve their potential.

Aspects of Difference

There are many aspects of difference. However, identifying labels to capture these differences can be challenging. "Asian" is an overly broad term. There are more than 4 billion people from Asian countries, and considerable heterogeneity. "Hispanic" is based on language (i.e., a person from a Spanish speaking country), whereas "Latino/a/x/e" and "African American" are based on geography (i.e., a person from a Latin American country). Traditional classifications of race/ethnicity by the National Institutes of Health miss groups that may be important to distinguish, including Arab, Middle Eastern, or North African (AMENA) populations.

Labels by the 2020 U.S. Census attempt to be more precise than traditional classifications of race and ethnicity by the National Institutes of Health. For the classification of "Hispanic, Latino, or Spanish origin" in the 2020 U.S. Census, see the source discussion. Notice how the question distinguishes between identities of people from (or whose ancestors were from) various countries, including Mexico, Puerto Rico, Cuba, etc.

Question Asking About One's Hispanic Origin in the 2020 U.S. Census. (Figure reprinted from the American Community Survey (2020): https://censusreporter.org/topics/race-hispanic/ [archived at https://perma.cc/LRW6-4JAJ])

For the classification of "Race" in the 2020 U.S. Census, see the source discussion. Notice how the question distinguishes between identities of people from (or whose ancestors were from) various countries, including India, China, Philippines, etc.

Question Asking About One's Race in the 2020 U.S. Census. (Figure reprinted from the American Community Survey (2020): https://censusreporter.org/topics/race-hispanic/ [archived at https://perma.cc/LRW6-4JAJ])

Society treats people differently based on factors that they are born into and are outside of their control, including their race and their family's socioeconomic status. Socioeconomic status refers to the social standing or class of an individual, and it is often assessed as a combination of education, income, occupation, access to resources, privilege, power, and control (Suzuki & Hirabayashi, 1987). There is also growing attention to neighborhood- and community-level factors associated with deprivation.

There are also many other important aspects of difference to assess when thinking about individual differences in behavior. For instance, one important aspect of difference is family structure. Family structure includes lots of components, such as whether the parent is a single parent or whether there are two parents, whether the parents are working parents or whether the family has a stay-at-home parent, and whether there are other family members (e.g., grandparents) in the household.

Another important aspect of diversity includes one's sex, gender identity, and gender expression. Sex is the person's gender that was assigned at birth, and includes male, female, and intersex. Gender identity is how one self-identifies in terms of their gender. Gender identity is a spectrum in terms of the degree to which a person identifies as a female/woman/girl, a male/man/boy, or other genders, in terms of personality, likes/dislikes, jobs/hobbies, and roles and expectations. The person's gender identity may be non-binary or genderqueer if they do not identify as a man or woman (e.g., genderfluid). When the person's sex assigned at birth and their gender identity are aligned, the person is cis-gender. When the person's sex assigned at birth and their gender identity are not aligned, the person is transgender. Gender expression deals with how the person's gender is expressed to others, including their appearance.

Another dimension of diversity is whether someone is from a rural versus urban area. People from rural areas may have less access to resources. Another aspect of diversity is refugee or immigrant status. Refugee and immigrant groups may have less access to resources and may have language barriers, in addition to difficulty finding work, and fear of deportation.

It is also important to consider diversity in terms of sexual orientation and sexual behaviors. There is a distinction between one's sexual orientation (e.g., lesbian, gay, bisexual, heterosexual, pansexual, asexual)—that is, who one is sexually or romantically attracted to—versus whom one has sex with (i.e., sexual behavior). There are also differences in terms of the degree of people's sexual risk taking and sexual aggression.

Ability and disability are other important facets of diversity, including learning disabilities. Additionally, there are important brain-related differences that lead people to interact and

experience the world in different ways, as reflected in the term, neurodiversity. Neurodiversity can include differences related to autism and other conditions such as attention-deficit hyperactivity disorder.

Age is another important dimension of diversity. Consider the issue of heterotypic continuity. For example, externalizing problems look different at different ages and should be assessed in different ways with different measures at different ages (Chen & Jaffee, 2015; Miller et al., 2009; Moffitt, 1993; Patterson, 1993; Petersen et al., 2015; Petersen & LeBeau, 2022; Wakschlag et al., 2010).

Another important aspect of diversity is culture, which includes belief systems, value orientations, psychological processes, worldview, learned and transmitted beliefs, and practices (Suzuki & Hirabayashi, 1987). A person's culture has many facets, including but not limited to experiences related to geographic boundaries, language, religious belief, social class, gender, sexual orientation, and ability status (Suzuki & Hirabayashi, 1987). However, culture and one's related identities are dynamic and changing (Suzuki & Hirabayashi, 1987), which makes them challenging to assess. Culture tends to be ignored in psychological research relative to other facets of diversity, but culture is arguably the most important in terms of influencing behavior. Additional dimensions of difference include religion and spirituality.

Another aspect of diversity is intersectionality, the intersecting of multiple identities. Intersectionality deals with how the combination of various identities can combine to influence behavior and bias.

There are many other important aspects of difference, too. This was not an exhaustive list!

Assessing Cultural and Individual Diversity: Multicultural Assessment Frameworks

A review of frameworks for multicultural assessment is provided by Edwards et al., 2017.

The ADDRESSING Model

The ADDRESSING model (Hays, 2016) provides a framework to consider some of the commonly examined differences between people within a clinical context. ADDRESSING is an acronym that stands for:

• **A**ge and generational differences

• **D**evelopmental or other

Disability

• **R**eligion and spiritual orientation

• **E**thnic and racial identity

• **S**ocioeconomic status

• **S**exual orientation

• **I**ndigenous heritage

• **N**ational origin

• **G**ender

These are important aspects of differences between people that can be helpful to understand when working with a client and how they self-identify. For instance, the therapist could ask the client how they self-identify in each of these domains and how their identity is meaningful to them. In addition to the aspects of diversity identified in ADDRESSING, there are other aspects of diversity and difference that may be worth considering, including but not limited to personality, culture, and political beliefs.

DSM-5 Cultural Formulation Interview

The Diagnostic and Statistical Manual of Mental Disorders (DSM) provides a framework for clinicians to organize cultural information about a client using the Cultural Formulation Interview (Lewis-Fernández et al., 2014). The Cultural Formulation Interview is a semi-structured interview and is available here: https://www.psychiatry.org/File%20Library/Psychiatrists/Practice/DSM/APA_DSM5_Cultural-Formulation-Interview.pdf (archived at https://perma.cc/3EWR-LGEN)

Multicultural Assessment Procedure

The Multicultural Assessment Procedure was developed by Ridley et al., 1998. The framework includes four phases:

1. Gather clinical data, including cultural data, through history taking and multiple methods for the purpose of formulating a case conceptualization

2. Interpret clinical and cultural data to formulate a hypothesis, while keeping in mind to:

• differentiate the cultural data as being either idiosyncratic (i.e., unique to the individual client and would not necessarily be expected of other members of the client's culture) or culture-specific

• consider base rates

• differentiate dispositional from environmental stressors

• differentiate clinically significant data from data that are not clinically significant

1. Incorporate cultural data with other clinical information to test the hypotheses by ruling out medical explanations and using appropriate assessments and testing

2. Arrive at a sound assessment decision (case conceptualization)

The procedure also includes debiasing strategies to minimize the likelihood of clinical judgment errors. Ridley et al., 2001 expanded on this framework in consideration of ethical issues in multicultural assessment.

Multicultural Assessment–Intervention Process

The Multicultural Assessment–Intervention Process was developed by Dana, 1998. The process provides a flowchart for clinicians to conduct assessment by asking relevant questions about cultural orientation, type of assessment instrument (e.g., etic or emic), cultural formulation for diagnosis, and intervention approach (e.g., universal, culture-general, culture-specific, identity-specific, etc.). Etic assessments are culture-general, whereas emic assessments are culture-specific.

Methodological and Conceptual Challenges in Research

Methodological and conceptual challenges for conducting research with ethnic minorities are described by Okazaki & Sue, 1995. Guidelines for conducting research with diverse groups are described by Burlew et al., 2019.

Use of Terms: Race, Ethnicity, Culture

Use of terminology with respect to race, ethnicity, and culture is a key challenge in assessment of diverse samples. Many people refer to race, ethnicity, and culture interchangeably, which reflects conceptual confusion. There is no definition of race, ethnicity, and culture that is universally agreed on (Okazaki & Sue, 1995). As described by Okazaki & Sue, 1995, "the use of the term 'race' appears to imply biological factors, as races are typically defined by observable physiognomic features such as skin color, hair type and color, eye color, stature, facial features, and so forth" (p. 367). By contrast, ethnicity (ethnic status) has been defined as "an easily identifiable characteristic that implies a common cultural history with others possessing the same characteristic" (Eaton, 1980). Ethnic identifiers for evaluating whether people share a "common cultural history" include racial, national, tribal, religious, linguistic, or cultural origin or background. Thus, race and ethnicity, though conceptually distinct, are related as race can be an ethnic identifier.

Researchers have argued that race designations are arbitrary (Helms et al., 2005), that race is a social, cultural, and/or political construct rather than a biological construct (Robert J. Sternberg et al., 2005; Yudell et al., 2016), that race has no genetic basis (Yudell et al., 2016), that within-race differences are greater than between-race differences (Zuckerman, 1990), and that racial categories lack conceptual meaning (Helms et al., 2005). According to the American Anthropological Association (Executive Board of the American Anthropological Association, 1998), "physical variations in the human species have no meaning except the social ones that humans put on them". However, even though race does not appear to have a genetic or biological basis and therefore appears to be a social construct rather than a biological construct, racism as a social problem is real and is a major issue (Smedley & Smedley, 2005). People are born into different social reactions based on their race. Thus, race is important to consider from a perspective of racialized experiences and discrimination even if the racial categories do not represent underlying biological differences.

In terms of determining what to assess and analyze, it is important for the researcher to determine whether they are interested in evaluating race as a biological variable, ethnicity as a demographic variable, or an aspect of cultural experience as a psychological variable (Okazaki & Sue, 1995). Researchers often use race and/or ethnicity as proxy variables for psychological variables such as cultural values, self-concept, minority status, etc. However, race and ethnicity are distal to the variables that may be of most psychological interest. Some people identify their primary culture as being different from their self-identified ethnicity. It is thus important to assess the psychological processes of interest, rather than merely relying on racial or ethnic categories as proxies.

Culture refers to a psychological variable: i.e., the social context. Researchers often group people together based on race or ethnicity with the implicit assumption that they share some cultural experience. According to Okazaki and Sue (Okazaki & Sue, 1995), assumptions underlying the use of race or ethnicity in a study should be made explicit. Research should directly assess the underlying psychological variables associated with culture, such as different spiritual practices, that are hypothesized to produce the racial/ethnic/cultural

differences. That is, it is generally best not to rely solely on race or ethnicity, which are imprecise, to classify people unless the goal is to classify people based on racialized reactions and experiences. Thus, the researcher can classify people according to race/ethnicity, but can also examine the more proximal cultural variables that are thought to drive group-related differences.

Whether to Examine Individual Differences or Group Characteristics

One question the researcher should consider is whether they are interested in understanding individual differences or group characteristics. Both can be valuable and important questions to examine. However, researchers should not under-estimate within-group heterogeneity (Okazaki & Sue, 1995). The greater the within-group heterogeneity, the less accurate predictions tend to be. Moreover, inferences at the group-level cannot necessarily be applied to the individual level; the confounding of an individual with the individual's culture leads to stereotyping (Okazaki & Sue, 1995), which should be avoided.

Selecting Participants

One question is which groups to include in the research design and in what proportion. For group-related comparisons, the researcher would need a large enough sample size for every group examined to have adequate power to identify group-related differences. If the researcher is interested in studying a particular group, a question is whether to use a comparison group. A comparison group can provide a basis of comparison to better interpret the findings in the group of interest. However, inclusion of a comparison group can add time, cost, and as a result may lead to achieving a smaller sample in the group of interest. Moreover, the group comparison approach has been criticized for reinforcing racial stereotypes, reinforcing Whites as the standard group and non-White behavior as deviant, and overlooking within-group variation (Okazaki & Sue, 1995). Thus, decisions about which groups to include, in what proportion, and whether to include a comparison group should be guided by the questions and purpose of the study. Practical considerations can come into play, and having a comparison group should not be considered the necessary default for all studies. Moreover, White groups should not be considered the default.

Ensuring Fair Group Comparisons

Another issue is how to achieve fair group comparisons. One approach is to match groups by selecting participants a priori to be similar on relevant, secondary characteristics such as demographic characteristics (e.g., age, sex, socioeconomic status) or other abilities (e.g., intelligence). The goal of matching is for participants to be as similar as possible in the relevant characteristics apart from their group classification. However, there are challenges to matching groups on all relevant characteristics. Another approach is to control for these secondary characteristics post hoc in the statistical analysis. Which variables to include as control variables is an important question that depends on the goal. A key question is which variables to match participants on. However, there is not an agreed-upon list of matching variables. Generally, researchers recommend controlling for social and demographic characteristics such as educational attainment, income level, and language fluency, when group-related differences exist on those variables and the researcher believes that such differences may moderate the associations of interest (Okazaki & Sue, 1995). When comparing Black and White participants, it is important to control for group differences in socioeconomic status, especially in the U.S.

In addition to main effects of group status, it may also be important to consider interaction effects of group status, consistent with an intersectionality approach. However, power to detect interactions tends to be weaker than power to detect main effects, because of lower reliability of the product term, smaller effect sizes (archived at https://perma.cc/E3KA-SGB7), and smaller sample sizes of the intersecting groups.

Sampling

Researchers face important questions with how to identify, sample, and recruit ethnic minority samples. Groups, such as ethnic groups, may be classified at a broad level (e.g., Latine or Hispanic) or at a more specific level (e.g., Puerto Rican, Mexican American, etc.); however, these approaches do not need to be mutually exclusive. Researchers must also determine how to classify people of mixed racial or ethnic backgrounds. It is also important to note that it is an incorrect assumption that once people are identified as belonging to a particular ethnic-cultural group, that they share a common understanding of their own ethnicity or culture and identify with the ethnic-cultural group (Okazaki & Sue, 1995).

There are challenges of recruiting a sizeable sample size with specific ethnic-cultural groups (e.g., Japanese Americans). This is partly a challenge of small overall population size (e.g., American Indians). This is also partly a challenge of difficult-to-reach populations who are distrustful of science and researchers. Research has exploited under-privileged populations (e.g., the Tuskegee syphilis study in Black men), which has led to changes in the consent process to ensure that participation is voluntary. There may also be selection effects, such that the people who are willing to participate in research may differ in important ways from those less willing to participate in research, which is a challenge to generalizability. To address sample size challenges, researchers often combine data from multiple ethnic-cultural groups with some common origin. For instance, researchers might combine Chinese Americans, Japanese Americans, and Korean Americans into one group. Or, researchers might combine people from multiple tribal groups among American Indians. However, broadening the ethnic grouping increases the heterogeneity of the groups, so the researcher must decide which are the sources of variability that can and cannot be overlooked (Okazaki & Sue, 1995).

Much of the literature in general and on diversity, in particular, is from research on college students. There are limitations of conducting research on college students. College students are not representative of the broader population. Research using college students under-estimates the population heterogeneity in terms of demographic and psychosocial diversity.

One goal of research is to learn findings that apply to the broader population, with the goal of having a representative sample, where the mean of the sample is approximately the same as the mean of the population. However, having a population-representative sample is not as useful for comparing differences between groups. To do this, we would need large samples of each group—e.g., equally weighted samples. Most research is on participants from WEIRD countries (Henrich et al., 2010): Western, educated, industrialized, rich, and democratic (WEIRD) societies. This further limits the potential generalizability of findings to different cultures.

Provide a thorough description of the sample and sampling methodology used. For instance, it can be helpful to describe participants on additional dimensions to race and ethnicity,

including generational status, acculturation, self-identification, ethnic and cultural composition of the neighborhoods or communities, etc. (Okazaki & Sue, 1995). Acculturation deals with the change that individuals undergo due to contact with members of different cultures, and it includes strategies such as assimilation, separation, marginalization, and integration (Suzuki & Hirabayashi, 1987).

Establishing Equivalence of Measures Across Groups

Five key areas need to be empirically evaluated in using or adapting instruments across cultures or groups to reduce or eliminate cultural bias (Hunsley & Mash, 2007):

• Conceptual equivalence

• Linguistic (or translation) equivalence

• Psychological equivalence of the items

• Functional equivalence

• Metric and scalar equivalence (i.e., measurement invariance)

Conceptual equivalence refers to establishing that the construct has the same meaning across groups. Linguistic (or translation) equivalence refers to the measure having the same linguistic meaning in each language to which it has been translated. Best practices for establishing translation equivalence include having experts translate and back-translate the measure, comparing both the original and back-translated versions, and revising the translation accordingly (Okazaki & Sue, 1995). Psychological equivalence of the items deals with establishing that the psychological effect each item has for each group or in the different versions.

Functional equivalence deals with establishing similar predictive and concurrent criterion-related validity to prevent test bias (i.e., similar regression intercepts and slopes between measure and criterion). Functional equivalence can be difficult to establish when both the test and the external criteria of interest are both rooted in the same dominant universalist cultural epistemology or worldview; instead, it may be advantageous to validate tests against external criteria that are grounded in the nondominant group's culture (Suzuki & Hirabayashi, 1987).

Metric and scalar equivalence (i.e., measurement invariance) deal with establishing that the construct is measured on the same metric across groups, that is, a particular score reflects the same level on the construct for people in each group. Metric equivalence/invariance deals with establishing that items' discrimination parameters (or factor loadings) are the same across groups. Scalar equivalence/invariance deals with establishing that both items' discrimination parameters (or factor loadings) and difficulty parameters (or intercepts) are the same across groups.

Methods of Assessment

It is debatable whether some assessment methods may be more likely to result in cultural or ethnic bias compared to other assessment methods (Okazaki & Sue, 1995). Certain types of assessment techniques might be more likely to bias or obscure differences between cultural groups. This especially depends on who is doing the observing and their own beliefs. There may be selection biases regarding the types of people who are willing to participate in in-depth studies that involve observational or psychophysiological assessment. It may be worth considering supplementing quantitative measurement approaches with qualitative measurement approaches, such as interviews and life histories, which may be better able to capture cultural factors. In general, it is advantageous to assess people using multiple measures and methods to establish convergent validity of cultural constructs (Okazaki & Sue, 1995).

Interpretation of Data

Another challenge deals with how to interpret data from group comparison designs. Historically, group differences in scores were most commonly interpreted from a deficit-based narrative rather than a strengths-based narrative (Byrd et al., 2021).

APA Guidelines

There are many disparities in psychological services for under-served populations, including health disparities, diagnostic disparities, and disparities in the use and availability of psychological services (Rivera Mindt et al., 2010). To help address these disparities, the American Psychological Association (APA) Office of Ethnic Minority Affairs published guidelines for providers of psychological services to ethnic, linguistic, and culturally diverse populations (American Psychological Association Office of Ethnic Minority Affairs, 1993). Their guidelines included suggestions such as:

• Whenever possible, provide information in writing along with oral explanations.

• Whenever possible, provide the information and interact in the language that is understandable to the client.

If that is not feasible, make an appropriate referral. If this is not possible, provide a translator with cultural knowledge and an appropriate professional background. When no translator is available, then use a trained paraprofessional from the client's culture as a translator.

• If you do not possess knowledge and training about an ethnic group, seek consultation with, and/or make referrals to, appropriate experts as necessary.

• Consider the validity of a given instrument or procedure and interpret the resulting data, while keeping in mind the cultural and linguistic characteristics of the client.

• Be aware of the test's reference population and possible limitations of using the instrument with other populations.

• Seek to help clients determine whether a "problem" stems from racism or bias in others so that the client does not inappropriately personalize problems.

The APA has also published guidelines for working with particular groups, including:

• Groups based on race and ethnicity (archived at https://perma.cc/4LM9-X6JN)

• People with low income (archived at https://perma.cc/PZQ5-8YFG)

• People with disabilities (archived at https://perma.cc/5YZ8-FNJT; PDF: https://perma.cc/KYY4-CQFQ)

• People who are transgender or gender nonconforming (archived at https://perma.cc/E9VH-QG3W)

• Sexual minorities (archived at https://perma.cc/4BJD-QZCF)

• Girls and women (archived at https://perma.cc/AYA4-KB5W)

• Boys and men (archived at https://perma.cc/6HWK-W9R6)

• Older adults (archived at https://perma.cc/EQ95-Y78Y)

Seek Training and Consultation

Research has shown that relatively few practitioners receive extensive training in multicultural assessment and few use frameworks for multicultural assessment (Brickman et al., 2006; Edwards et al., 2017). Researchers have encouraged increasing multicultural education, training, awareness, and knowledge, increasing multicultural psychological research, and increasing the provision of culturally competent psychological services to ethnic minorities (Rivera Mindt et al., 2010). It can also be worthwhile to consult with informal support systems, including churches, ethnic clubs, family associations, and community leaders (Leong & Kalibatseva, 2016).

Develop Awareness of Cultural Differences

One goal should be to develop cultural competence with the groups, communities, and individuals with which one works or interacts. It can be helpful to be aware of cultural differences, including cultural differences in communication style and use of language (Leong & Kalibatseva, 2016).

Show Cultural Humility

Another goal should be to show cultural humility. Tervalon & Murray-Garcia, 1998 describe three key aspects of cultural humility: (1) a life-long commitment to self-evaluation and self-critique, (2) fixing power imbalances, and (3) developing mutually beneficial and nonpaternalistic partnerships with communities who advocate for others. For instance, it is important to be aware of one's biases and stereotypical beliefs. Cultural humility involves treating the client as the expert on their experience. Therefore, cultural humility involves collaboration between the clinician and client, because each brings important knowledge. The client brings their knowledge of their situation, and their preferences. The clinician brings their scientific knowledge about strengths and difficulties, and about behavior change.

Do Not Assume or Stereotype

It is important to consider cultural factors; at the same time, groups are not monolithic and every person is unique. Do not assume about someone based on their demographics, which is stereotyping. Just because a person comes from a particular group does not mean the person shares all beliefs, behaviors, etc. with the group (Okazaki & Sue, 1995). Thus, it is important to be aware of within-group differences, as well (Leong & Kalibatseva, 2016; Okazaki & Sue, 1995), which are often as large or larger than between-group differences (Zuckerman, 1990). Show genuine interest and curiosity to get to know a person as an individual.

Ensure Adequate Representation of Under-Represented Groups in Research

There are important issues of generalizability of research findings. Participant samples historically have been largely White and middle or upper middle class. This reduces the potential generalizability of findings, and limits effectiveness of treatments to under-represented populations, many of whom have experienced discrimination and social disadvantage. Many groups, including people who are Black or Latine, and gender and sexual minorities have not been adequately represented in research studies. As a result, the knowledge gained from unrepresentative studies may not generalize to other under-represented groups. Science plays an important role in advancing knowledge that can lead to the development of new treatments. However, groups who are under-represented in research are less likely to benefit from the new treatments because the knowledge upon which the treatments were developed and tested may not generalize to the under-represented groups. Moreover, the lack of inclusion of under-represented groups in studies leads to inappropriate normative data. Thus, it is crucial to have better representation of historically marginalized and minoritized groups in science, including people who are Black or Latine, and gender and sexual minorities.

People from Finland and Sweden are often studied to examine health-related questions. Because they have a socialized health care system, the government keeps medical records on all citizens, so researchers can examine millions of participants, and can study rare conditions and behaviors with a low base rate, like suicide (Lysell et al., 2018). However, there are questions of generalizability from the Scandinavian population and health care system to those of other countries, including the United States. One could hypothetically use propensity score matching for matching their sample to other populations, but one cannot really do that because they have too many empty cells for propensity score matching due to too few people of various under-represented racial/ethnic groups.

Do Not Use Offensive Content or Procedures

It is important to use stimuli that are valid for the populations being tested. It is important to avoid using content or procedures that are culturally or racially insensitive, or that are racist, sexist, heteronormative, and/or ableist (Byrd et al., 2021). When developing assessments, it may be important to evaluate stimuli in focus groups to ensure they are not offensive.

One of the most widely used standardized assessments in psychological assessment is the Boston Naming Test, a measure of visual confrontation naming that is used to assess cognitive impairment, aphasia, and dementia. The Boston Naming Test includes a noose as one of the items. Historically, lynchings with a noose were used in the United States as a

form of racial terror against the Black and African American community. Thus, the use of a noose as an item in the Boston Naming Test is racially insensitive and inappropriate for use because it is deeply offensive and has a harmful nature. It can have a strong negative impact on Black respondents in terms of distress, performance disparities, and health disparities (Byrd et al., 2021).

Culturally offensive stimuli can lead to stereotype threat and compromise the validity of the assessment. Stereotype threat occurs when people are or feel at risk of conforming themselves to stereotypes about their social group, thus leading them to show poorer performance in ways that are consistent with the stereotype. Moreover, the use of offensive procedures likely contributes to greater distrust of the scientific and medical fields among under-served communities, which acts as a barrier to health care and to health disparities. Thus, removing and replacing offensive stimuli may be necessary from a moral and ethical perspective to do no harm and to ensure valid assessment, even if it risks modifying the standardized procedure from which the norms were developed (Byrd et al., 2021). When deviating from standardized procedures, it is important to describe such deviations in resulting papers or reports.

Use Culturally Valid Measures

Most measures originated from Western societies (Suzuki & Hirabayashi, 1987) and have been validated only among non-Latine, English-speaking Whites (Manly, 2005). It is important to use measures that have been validated for the populations with whom they will be used. Many measures have been used for different populations than the populations with which the instruments were developed and validated, such as with immigrants, refugees, or when exporting measures to different countries or cultural contexts. Exporting measures for use with people from different cultures assumes universality (i.e., cross-cultural validity). More work should examine adaptation and validation of measures for other cultures, countries, languages, etc. (Leong & Kalibatseva, 2016). However, some measures may not be able to be applied validly across cultures. Using valid measures for the target population may in some cases mean using culturally specific (etic) instruments rather than universal (emic) instruments. If using measures whose validity and measurement equivalence has not been established, it is important to interpret results with caution and to provide disclaimers in any resulting papers or reports (Leong & Kalibatseva, 2016).

As described in the source discussion, it is also important for measures to have cultural validity, which refers to "the effectiveness of a measure or the accuracy of a clinical diagnosis to address the existence and importance of essential cultural factors." (Leong & Kalibatseva, 2016). Essential cultural factors may include values, beliefs, experiences, communication patterns, and approaches to knowledge (epistemologies).

Threats to cultural validity of assessments includes: pathoplasticity of psychological disorders, bias in clinical judgment, language capability of the client, biased measures, or inappropriate use or interpretation of measures (Leong & Kalibatseva, 2016).

Use Linguistically Appropriate Assessment

It is important to use assessment approaches that are linguistically appropriate for the examinee (Leong & Kalibatseva, 2016). First, the measure should be in a language that is understood by the examinee. Second, the measure should be consistent with the examinee's language ability. Sometimes interpreters or translators may be necessary. However, use of interpreters has been shown to lead to under-estimations of patients' emotion suffering and despair (Leong & Kalibatseva, 2016). In addition, clients and therapists can attach different meanings to the same words. It is thus important not to assume that you know what the client means by various terms and not to assume that the client knows what you mean by various terms.

Use Follow-Up As Necessary

As described by Suzuki & Hirabayashi, 1987, it may be necessary to use follow-up procedures to clarify people's responses to items or to test the limits by readministering items with additional supports (e.g., modifying instructions to involve more or fewer cues, adjusting the pace of information presented, adjusting memory demands, changing the response format) (Sattler & Hoge, 2006). When deviating from standardized procedures, it is important to describe such deviations in resulting papers or reports.

Examine and Correct for Test Bias

As described in the source discussion, there are two types of test bias: predictive test bias and test structure bias. It is important to evaluate the possibility of both when using assessments across groups.

It is important to establish measurement invariance across groups to ensure that scores have the same meaning for each group (Burlew et al., 2019). If measures are non-invariant, the means (i.e., level) and or associations between constructs could appear to (artifactually) differ across groups, when in reality they do not (Chen, 2008). Measurement invariance can be examined in terms of item intercepts, item factor loadings, and item residuals. Establishing measurement invariance of just intercepts or factor loadings (or vice versa) is not sufficient to establish measurement invariance because tests can show strong bias in one dimension (e.g., intercepts) even if the test does not show bias in other dimensions (e.g., factor loadings) (Wicherts & Dolan, 2010). Approaches for evaluating measurement invariance across groups are described by Han et al., 2019. To help minimize the potential for test bias, reseachers are encouraged to develop the scales simultaneously across the different groups or cultures (Fernández & Abe, 2018).

It is also important to determine whether the measure's scores show similar relations to meaningful external criteria (Burlew et al., 2019).

Fernández & Abe, 2018 provide strategies to address cross-cultural bias, such as excluding words or concepts that are specific to one language or culture. Use of North American neuropsychological tests in other countries has shown to lead to considerable false positives, and the extent of misdiagnosis differs by country (Daugherty et al., 2017).

It is important to examine factors that moderate the validity of assessment, e.g., groups for whom the assessment is biased. Ways to investigate and detect bias are described in the source discussion. The number of potential moderating variables is so large, that it is not realistic to develop an assessment base that encompasses all of these factors and their

interactions. Evidence should be used regarding which factors seem to matter for a given assessment purpose—that is, the factors that have a medium-to-large effect size. It is valuable to examine the effect size of moderators; if the effects are small, it might not be sufficient to scrap the instrument for the population. Gender and cultural differences have shown a number of statistically significant effects for a number of different assessment purposes, but many of the observed effects are quite small and likely trivial, and they do not present compelling reasons to change the assessment (Youngstrom & Van Meter, 2016). Nevertheless, if you do find evidence of bias, it is important to correct for it. Ways to correct for bias using score adjustment and other approaches are discussed in the source discussion.

Consider Pathoplasticity of Constructs

Constructs can manifest differently across cultures. Pathoplasticity of mental disorders refers to the variability in symptoms, course, outcome, and distribution of mental disorders among various cultural groups (Leong & Kalibatseva, 2016). Some syndromes are considered culture-bound—that is, they exist in some cultures but not in other cultures. Hwabyung is a Korean culture-bound syndrome in which suppressed negative emotions manifest themselves in bodily ways such as chest pains and difficulty breathing. In addition, features associated with a disorder, including the content, severity, or frequency of symptoms, may vary across cultures.

For instance, Asian Americans tend to report a greater degree of psychological disturbance than Whites, which could be due to several reasons (Leong & Kalibatseva, 2016). It is possible that Asian Americans (a) show higher rates of mental health problems than other groups, (b) under-utilize mental health services, and/or (c) are more likely than other groups to be misdiagnosed due to miscommunication, use of invalid instruments for Asian Americans, and/or lack of cultural knowledge on the part of therapists. In addition, how clients act in a diagnostic interview may depend on cultural factors, and it thus may be important to sample the clients more broadly than a diagnostic interview (Leong & Kalibatseva, 2016).

For a case example of pathoplasticity, there is a disorder among the traditional Navajo called "Moth Madness" (American Psychological Association Office of Ethnic Minority Affairs, 1993). Symptoms of Moth Madness include seizure-like behaviors. This disorder is believed by the Navajo to be the supernatural result of incestuous thoughts or behaviors. Both differential diagnosis and intervention should take into consideration the traditional values of Moth Madness.

Formulate a Culturally Informed Case Conceptualization

It can be helpful to discuss your diagnosis and case conceptualization with the client. For instance, you may learn that some behaviors are normative in their culture and may not be distressing or concerning to them. It can thus be helpful to consider client preferences in terms of what to address and how to address it, while adhering to evidence-based approaches.

Bias in Clinical Judgment

As reviewed by Garb, 1997, there is evidence of race bias, social class bias, and gender bias in clinical judgment. Race bias, social class bias, and gender bias in clinical judgment occurs when the accuracy of judgments varies as a function of client race, social class, or gender. That is, clinical judgment bias occurs when the accuracy of judgments varies as function of other variables (e.g., race, social class, gender). Bias is not when the judgments themselves vary as a function of other factors, but when the accuracy of those judgments varies. Examining how judgments differ across groups does not control for differences in clinical severity between the groups. If a clinician rates more people in one group as having a disorder than another group, this is not necessarily evidence of bias because the true prevalence of the condition could differ across groups. Thus, it is important to compare groups with similar levels of symptomatology or to control for level of symptomatology. If clinical judgments are more accurate for White clients than Black clients, we would say that race bias exists.

Accuracy of judgments could vary by the client's race, social class, or gender for several reasons (Garb, 1997). One way in which judgments could be biased is if the diagnostic criteria are biased—for instance, if the diagnoses are more valid for one group (e.g., White clients) than for another group (e.g., Black clients). Another reason that accuracy of judgments could vary as a function of client race, social class, or gender is if the judgments are made based on a biased instrument. Another way that accuracy of judgments could vary as a function of client race, social class, or gender is related to confirmation bias or biases introduced during the usage of a measure. This could occur, for instance if clinicians do not consider alternative hypotheses, and based on their hypotheses, they choose the data to collect that support their hypotheses rather than collecting data that could refute their hypotheses. Bias could also occur in the way that data are integrated from multiple measures or methods. Another form of bias could arise from how test results are used, in terms of inequitable social consequences (Suzuki & Hirabayashi, 1987).

Many studies do not find bias in clinical judgments as a function of race, social class, or gender, but some studies do find evidence of bias (Garb, 1997).

Race Bias

One pattern of race bias is that Black and Latine patients with a psychotic affective disorder are more likely than White patients to be misdiagnosed as having schizophrenia (Garb, 1997). That is, Black and Latine patients are less likely than White patients to be diagnosed with psychotic affective disorder (with the same symptoms) and are more likely to be diagnosed as having schizophrenia, even when the measures do not indicate that a diagnosis of schizophrenia is justified. Another pattern of race bias is that the risk of violence is over-estimated for Black patients but not for White patients (Garb, 1997). In addition, cases of child abuse are more likely to be reported when children are White than when they are Black (even when comparing similar behavior) (Garb, 1997). Race bias has also been observed with respect to antipsychotic medication. Antipsychotic medications are more often prescribed for Black patients than for other patients even for similar levels of psychosis (Garb, 1997). Moreover, affective symptoms in patients who are severely ill are more often under-treated for Black or Latine patients than for White patients (Garb, 1997). Another potential pattern of race bias is that, compared to non-Hispanic White children, racial/ethnic minorities appear to be more likely to be diagnosed with a disruptive behavior disorder rather than attention-deficit/hyperactivity disorder (ADHD) (Fadus et al., 2020); however, it is not yet known whether this holds when comparing individuals with the same symptom presentation.

Social Class Bias

One pattern of social class bias is that child abuse is more likely to be reported if a child is from a lower-class background rather than a middle- or upper-class background (even when comparing similar behavior) (Garb, 1997). In addition, middle-class children are more likely to be referred to special education programs compared to lower-class children (Garb, 1997). There is also social class bias in clinicians' decisions about psychotherapy. Referrals for psychotherapy are more often made for middle-class clients than for lower-class clients (Garb, 1997). Clinicians are more likely to recommend psychotherapy and expect a client to do well in psychotherapy when the client is from a middle- or upper-class background compared to when the client is from a lower-class background, leading to a tendency for clinicians to refer clients to different types of treatments based on the client's social class (Garb, 1997). Specifically, lower-class clients are more likely to be referred for supportive psychotherapy, whereas middle-class clients are more likely to be referred for insight-oriented psychotherapy.

Gender Bias

One pattern of gender bias is that women are more likely than men to be diagnosed with a histrionic personality disorder, whereas men are more likely than women to be diagnosed with antisocial personality disorder, even when female and male clients do not differ in symptomatology (Garb, 1997). Another pattern is that clinicians tend to over-estimate likelihood of violent behavior for male clients and under-estimate likelihood of violent behavior for female clients (Garb, 1997; McNiel & Binder, 1995). Although males tend to show more violence than females, studies have shown that false positives tend to be more common for male patients than female patients, whereas false negatives tend to be more common for female patients than male patients (McNiel & Binder, 1995).

In terms of autism, some conceptualizations of autism consider it to be a manifestation of "extreme male brain" (Baron-Cohen, 2002; Baron-Cohen, 2010; Greenberg et al., 2018).

Girls diagnosed with autism tend to be more severely affected than boys diagnosed with autism. People are less likely to notice autism in girls. This could possibly be due to girls with autism having fewer restricted interests and lower levels of repetitive behavior compared to boys. Research has tried to develop assessments to equate boys and girls, but girls and boys appear to be unequal in their underlying autism process—that is, boys and girls appear to show sex-specific autism phenotypes (Frazier et al., 2014).

Recommendations for Reducing Bias in Clinical Judgment

Below are recommendations for reducing bias in clinical judgment (Garb, 1997), including:

• Be aware of biases that have been reported in the clinical judgment literature.

• Give the clinician timely and specific feedback on their judgments.

• Use standardized and structured assessment approaches and attend strictly to diagnostic criteria when making diagnoses.

• Use actuarial approaches and statistical prediction rules.

Use Appropriate Normative Data

When group-referenced judgments will be used, it is important to consider what the appropriate norm is. The question of whether separate norms should be used for various racial/ethnic minority groups is of considerable controversy (Burlew et al., 2019). Pros and cons of using group-specific norms are discussed in the source discussion and reviewed by Manly, 2005. The question about which norms to use is complex, and psychologists should evaluate the cost and benefit of each norm, and use the norm with the greatest benefit and the least cost for the client (Manly & Echemendia, 2007). Burlew et al., 2019 recommend limiting the use of measures that require the use of norms to interpret, and to note in the limitations of a paper or report if a norm based on one group is used to evaluate a person from a different group.

Contextualize Group-Related Differences

When conducting research and identifying different scores between groups, it is important to be cautious not to assume that differences in the scores between groups are caused by the group membership. That is, correlation does not necessarily mean causation! For instance, if the researcher identifies higher rates of suicide in transgender than cisgender individuals (Toomey et al., 2018), it is important not to automatically assume the higher suicide rate is because of the person's transgender status, per se. Rather, there are likely many aspects of discrimination and victimization that transgender individuals experience that, unfortunately, may increase their suicide rate relative to cisgender individuals.

It is important to contextualize test score differences between groups within sociocultural and historical inequities rather than interpreting group-related differences in tests scores from a deficit-based narrative (Byrd et al., 2021). It is important to avoid interpretive bias and to pay attention not to over- or under-pathologize (Okazaki & Sue, 1995). Over-pathologizing can lead to stigmatization and institutionalization. Under-estimating psychopathology can deprive patients of treatments or services that they would benefit from. For instance, one should be aware of the danger of under-estimating psychopathology for clients from different cultures (from that of the therapist) through over-attributing bizarre behavior or thought patterns to the client's culture (Okazaki & Sue, 1995).

Dissemination and Implementation

Dissemination and implementation deals not just with finding differences between groups, but what to do about these differences. Aspirationally, we would successfully be able to custom-tailor treatments to individuals based on their individual characteristics. The goal is to identify the best treatment for each group, or ideally, each individual.

Use Inclusive and Bias-Free Language

It is important to use bias-free language, as discussed in the Publication manual of the American Psychological Association (American Psychological Association, 2020). Guidelines for bias-free language include the following:

1. Describe at the appropriate level of specificity

a. Focus on relevant characteristics b. Acknowledge relevant differences that exist c. Be appropriately specific

1. Be sensitive to labels

a. Acknowledge people's humanity b. Provide operational definitions and labels c. Avoid false hierarchies

When writing reports, manuscripts, or in conversation, it can be helpful to use inclusive and person-first language. The aim of people-first language is to reduce stigma because it treats everyone as a human first. People-first language puts the person before the disability and describes the difficulties a person has rather than who a person is. For instance, say "a child with intellectual disability" instead of "an intellectually disabled child". That said, some individuals may prefer identity-first labels (e.g., autistic person) rather than person-first labels for their own identities. Thus, it is important to use the labels preferred by the particular client. The APA guidelines for inclusive language are located here: https://www.apa.org/about/apa/equity-diversity-inclusion/language-guide.pdf (archived at https://perma.cc/82QH-ED96).

Conclusion

Diverse samples provide stronger tests of theories and better generalizability of findings (external validity). We need measures that are reliable and valid for all people we use them with. There are various multicultural frameworks to assessment. Guidelines for assessment of diverse populations are discussed. Recommendations include developing and using appropriate and valid measures for the target population(s), ensuring fair group comparisons, seeking training and consultation, developing awareness of cultural differences, showing cultural humility, avoiding assuming or stereotyping, ensuring adequate representation of under-represented minorities, considering pathoplasticity of constructs, formulating a culturally informed case conceptualization, using appropriate normative data, contextualizing group-related differences, and using inclusive language.

Suggested Readings

Burlew et al., 2019; Byrd et al., 2021; Okazaki & Sue, 1995

CHAPTER 04 / CAC.903

Constructs: What Are We Measuring?

Before choosing a test, the counselor must ask what construct is being measured. A construct is a concept, such as anxiety, attention, self-esteem, or family conflict. Constructs are not directly visible. We infer them from indicators: responses, behaviors, ratings, or task performance. If the construct is unclear, the score will be unclear. A common assessment error is treating a score as if it were the thing itself.

Reflective and formative models

In a reflective model, the construct causes the indicators. For example, an underlying anxiety may cause worry, restlessness, and avoidance. The indicators should correlate because they reflect the same construct. In a formative model, the indicators combine to form the construct. For example, socioeconomic stress may be formed by income, housing instability, and food insecurity. The indicators need not correlate strongly. They define the construct.

This distinction matters. A reflective scale should have internal consistency. A formative index may not. If a counselor demands high internal consistency for a formative index, the measure may be wrongly rejected. If a counselor sums unrelated items as if they reflected one trait, the score may be meaningless.

Symptoms, signs, and syndromes

A symptom is a subjective experience reported by the client, such as sadness. A sign is an observable behavior, such as crying. A syndrome is a pattern of symptoms and signs that tend to occur together. A diagnosis is a classification based on criteria. These levels are related but not identical. A child may report sadness without meeting a depressive disorder. A child may show irritability without reporting sadness. The counselor should assess multiple levels.

Impairment and context

A construct often includes impairment. Many symptoms are common. What makes a clinical concern is persistence, intensity, and interference with development. A child who fears dogs but lives where dogs are absent may have little impairment. A child who fears school and stops attending has major impairment. Assessment should measure functioning across home, school, peers, and self-care.

Context also changes meaning. High activity may be problematic in a quiet classroom but adaptive in sports. Sadness after a loss may be expected. The counselor should ask whether the behavior is developmentally and culturally situated, not only whether it is present.

Comorbidity and overlap

Constructs overlap. Anxiety and depression share distress. ADHD and trauma can both affect attention. Autism and anxiety can both affect social withdrawal. A score on one scale may reflect multiple constructs. The counselor should avoid assuming that a high score proves one disorder. Differential assessment asks what else could explain the pattern.

Dimensional approaches can help. Instead of only asking whether a disorder is present, the counselor can measure severity across domains. This supports monitoring and reduces false certainty. A child may have moderate anxiety and mild attention problems, both needing support.

Treatment targets

Assessment should identify targets that can change. A global label such as "behavior problem" is not a target. Specific targets include hitting, work refusal, panic attacks, sleep delay, or negative self-statements. Targets should be observable, measurable, and meaningful to the child and family. They should connect to intervention. If a target cannot guide action, it may not be worth measuring.

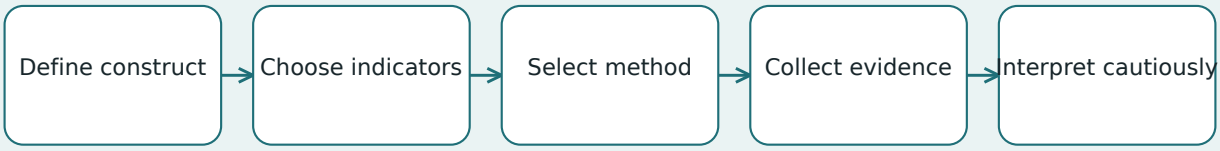
Real-life application

A teacher refers a child for "laziness." The counselor translates the construct into work initiation, task persistence, skill level, anxiety, and reinforcement. Observation shows the child starts tasks but stops when reading is required. Achievement testing shows weak decoding. The construct is not laziness. It is a reading difficulty with avoidance. The plan changes from punishment to reading intervention and support.

Reflection

Choose a referral label you have heard. Break it into observable indicators. Ask whether the indicators reflect one construct or form a composite. Then ask what intervention each indicator would suggest. Clear constructs make clear plans.

From construct to indicator



A score is an indicator of a construct, not the construct itself.

Attributed Foundational Reading

Attribution and critical-reading note

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Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module constructs.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Types of Constructs

A construct is a concept. A construct is generally considered a latent idea or phenomenon that is not directly observable. For example, depression could be considered a construct because we cannot directly measure someone's level of depression. Rather, we infer a person's level on the construct of depression by taking measurements of many indirect proxies that we think are indicators of the construct. An indicator is a measurement, and it could be thought of as a behavior or questionnaire item. Potential indicators of depression could include behaviors/items such as whether the person has low mood, low energy, sleep difficulties, loss of interest in formerly enjoyable activities, changes in weight or appetite, etc.

Because constructs are latent, unobservable phenomena, they are often estimated using latent variable models, as described in the source discussion.

There are two main types of constructs: reflective constructs and formative constructs.

Reflective Construct

With a reflective construct (or reflective model), the construct is the cause of the measures, and the construct is reflected by the indicators, see the source discussion (Bollen & Lennox, 1991).

Reflective and Formative Constructs in Structural Equation Modeling.

An example of a reflective construct is extraversion; see the source discussion. We conceptualize extraversion as a latent concept that, although not directly observable, influences the measurements of indicators of extraversion, such as whether the person enjoys talking to strangers, goes to many parties, is energetic, and is happy. In a reflective model, the answers that people give on the items are thought to reflect a certain disposition—in this case, extraversion. None of these indicators, in isolation, is considered a direct measure of extraversion. Nevertheless, we feel that we can get a better estimate of the concept of extraversion by capturing the variance that covaries across these indicators, i.e., the common (systematic) variance among the indicators. Thus, the latent factor is estimated to reflect the common variance among the indicators, as depicted in the source

discussion. Because a reflective construct influences the indicators, the indicators are called effect indicators. What we observe, theoretically, are the effects of the underlying construct, which influences scores on each individual measure.

Extraversion as a Reflective Construct.

Reflective Latent Variable as the Common Variance Among Indicators.

With a reflective construct, we would expect that the items would show strong internal consistency reliability. That is, we would expect that the indicators would all be correlated with each other, because all indicators are thought to be a reflection of the underlying construct. But in reality, there is often still residual correlation between items even after accounting for the latent factor.

In a measurement model of a reflective construct (see the source discussion), the lambda values are called factor loadings that represent regression coefficients from observable indicators to latent variables. A factor loading is the relative weight that the item is given in the estimation of the latent factor, similar to the correlation between the construct and the measure. The stronger an item's factor loading on the construct, the more relative weight a given item has in the estimation of the latent factor. The epsilon term reflects measurement error. Structural equation modeling is an application of the same way we think about measurement according to classical test theory, but in a latent variable framework. According to classical test theory, the observed score is thought to be some combination of the true score and error (observed score = true score + error; X = T + e). Or, in a multiple regression framework: extraversion = 1 · strangers + 2 · parties + 3 · happy + errors. That is, the true score influences the observed score on each measure. If a person's level on the construct (extraversion) changes, their score on each measure/indicator/item is expected to change (Edwards & Bagozzi, 2000).

Formative Construct

Formative constructs are the other fundamental way to understand the relation between a construct and a measure. With a formative construct (or formative model), the measures cause the construct; see the source discussion (Bollen & Lennox, 1991). That is, the construct is created by the measures. Because the indicators influence the formative construct, the indicators are called causal indicators.

Examples of formative constructs include constructs such as socioeconomic status (SES), stress, substance use, risk for cardiovascular disease, and exposure to discrimination. A formative construct is estimated as the total variance among the indicators, as depicted in the source discussion. For instance, we may define SES such that it is the linear combination of a person's educational attainment, occupational prestige, and income, as in the source discussion. With a formative construct, if a person changes in their level on the construct, we might not expect that their scores on all of the measures/indicators/items would change (Edwards & Bagozzi, 2000). If a person's SES changes, you would not necessarily expect the person's score on all of these measures/items to change. For instance, if a person's SES increases from T1 to T2, you would not necessarily expect that their educational attainment increased; it could be instead that their occupational prestige or income increased.

Socioeconomic Status as a Formative Construct.

Formative Latent Variable as the Total Variance Among Indicators.

In a structural equation model, the gamma values are regression coefficients from exogenous variables to endogenous variables. Endogenous variables are variables whose values are determined by the model (i.e., they are influenced by other variables), whereas exogenous variables are variables whose values are not determined by the model (i.e., they are not influenced by other variables). In a measurement model of a formative construct (see the source discussion), the gamma values reflect the relative weight that the item is given in the estimation of the latent factor, similar to the correlation between the construct and the measure. The stronger an item's factor loading on the construct, the more relative weight a given item has in the estimation of the latent factor.

With a formative construct, the disturbance term, zeta, represents the part of the construct that is not explained by the measures, i.e., the measurement error. However, the measures are thought (or at least assumed) to be error-free causes of the construct. The covariances between the indicators are permissive of item correlations, but unlike the reflective model, the formative model does not require that the items are correlated. In other words, for a formative model, it is not necessary that the items are correlated, even though the items could be correlated.

A formative construct is determined by the measures, so if you change the measures, you change the construct. A formative model requires that you include all facets (aspects) of the construct. This is not true in a reflective model because all items of a unidimensional reflective construct are interchangeable as long as the construct is unidimensional and the items are equally reliable because the construct influences the measure scores. With a reflective construct, all you need to do is find the items that reflect the underlying construct, and you could create parallel measures of the construct. This is impossible to do in a formative model.

Although formative conceptualizations of constructs may be useful in some cases, formative constructs can be controversial (Edwards, 2011; Hardin et al., 2011; Howell et al., 2007; Markus, 2018; Bollen & Lennox, 1991; Bollen & Bauldry, 2011; Bollen & Diamantopoulos, 2017; Diamantopoulos et al., 2008). It is important to think carefully about the construct of interest, as guided by theory, and how best to assess and estimate it.

Differences in Measurement Expectations

There are several differences in our measurement expectations based on the type of construct (formative or reflective) or indicator (causal or effect), as described by Bollen and Lennox (Bollen & Lennox, 1991).

One difference we would expect between effect versus causal indicators is the actual correlations between the indicators. Causal indicators may not be correlated, whereas effect indicators must be correlated. So, when we are assessing internal consistency reliability of a measure, and we assume that higher correlations between items suggests higher internal consistency reliability, we are assuming that indicators of the construct have an effects relation to the construct. A scale assessing causal indicators would likely have very low internal consistency reliability.

A second difference we would expect between effect versus causal indicators is whether to use a sample or a census of indicators. The suggestions that we sample all facets of a construct when determining indicators is necessary only for causal indicators, for which a failure to assess a certain formative indicator changes the overall meaning of the construct.

For effect indicators of a unidimensional construct, equally reliable measures are interchangeable.

A third difference we would expect between effect versus causal indicators is the optimal correlations between the indicators. For effect indicators, high correlations are desirable. For causal indicators, too high of a correlation between indicators is likely to introduce the problem of multicollinearity, so lower correlations are optimal. Multicollinearity is when two or more predictors are correlated such that their regression coefficients with the outcome variable (in this case, the factor loadings on the latent factor, i.e.,) are erratic.

Practical Issues

There are important practical issues to consider with both reflective and formative models.

As described in the source discussion in the chapter on structural equation modeling, it is challenging to use formative constructs in a latent variable framework. In a structural equation modeling (SEM) framework, formative constructs must be used in the context of a model that allows some constraints. But one can easily estimate formative constructs outside of a SEM context, for example, with an average or weighted average composite, similar to principal component analysis. However, unlike latent variable approaches, linear composite approaches (e.g., mean or sum scores) do not account for measurement error. That is, linear composite approaches do not include a disturbance/error term. Therefore, a linear composite is not the same as a latent variable. If no disturbance term is included in a structural equation model, the model is equivalent to a linear composite approach (e.g., sum score, weighted average). Linear composite approaches are likely more defensible for formative constructs than for reflective constructs.

How To Estimate

To estimate a reflective construct, a latent variable modeling approach is necessary, such as structural equation modeling, factor analysis, or item response theory.

To estimate a formative construct, one may estimate an average or sum score across items (which gives each item equal weight), estimate a weighted average, use principal component analysis, use confirmatory composite analysis, use partial least squares regression, or use structural equation modeling.

Thus, it is important to think about the nature of the construct before deciding how to estimate it. The nature of a construct is a theoretical question. Data alone cannot determine whether a construct is reflective or formative. Just because items are highly correlated does not mean that scientists think of the construct as reflective. By contrast, if items are not highly correlated, it suggests that a single unidimensional reflective construct may not adequately capture reality—the variance may be better captured by a formative or by multiple reflective constructs (i.e., a multidimensional model). In sum, although empiricism is relevant, theory is necessary to inform our understanding of the nature of constructs.

Latent Variable Modeling: IRT, SEM, and Factor Analysis

Item response theory (IRT), structural equation modeling (SEM), confirmatory composite analysis (CCA), and factor analysis are all approaches to latent variable modeling. Factor analysis and IRT most commonly estimate reflective latent variables rather than formative latent variables. CCA estimates formative latent variables. SEM can estimate both reflective and formative latent variables. IRT is essentially a form of confirmatory factor analysis with categorical, binary, or ordinal data. As we discuss in the source discussion, SEM is confirmatory factor analysis with regression paths that specify hypothesized causal relations between the latent variables (the structural component of the model). As we discuss in the source discussion, exploratory structural equation modeling (ESEM) is a form of SEM that allows for a combination of exploratory factor analysis and confirmatory factor analysis to estimate reflective latent variables and the relations between them.

Alternative Conceptualizations of Phenomena: A Network

Reflective and formative constructs are considered latent variables. However, there are alternative conceptualizations of phenomena and psychopathology. For instance, one alternative approach is to conceptualize psychopathology as a characteristic that emerges out of the local influences of symptoms on other symptoms. This would be consistent with a network analysis approach to conceptualization of psychopathology. In the network analysis conceptualization, "depression" as a latent construct does not exist; what we call "depression" merely represents the influences of various symptoms on one another. In this approach, if only one item is used to assess each symptom, then the symptoms are observed (manifest) variables that are assumed to be measured without error.

Note that the existence of latent constructs and local influences is not mutually exclusive. It could be for a given phenomenon, for instance, that there are underlying processes (representing a latent factor) that lead to a wide range of symptoms; and, simultaneously, that behaviors/symptoms/processes influence one another in a network.

Conclusion

It is important to think about the nature of a construct before deciding how to estimate it. For a formative construct, the indicators influence the construct, and a linear composite such as a weighted mean or sum score is justifiable. For a reflective construct, the construct influences the indicators, and it is more appropriate to estimate it with a reflective latent variable that reflects the common variance among the indicators using structural equation modeling, factor analysis, or item response theory.

Suggested Readings

Bollen & Lennox, 1991; Bollen & Bauldry, 2011; Bollen & Diamantopoulos, 2017

CHAPTER 05 / CAC.903

Scores, Scales, and Norms

Scores are translations. They turn behavior into numbers so that information can be compared, tracked, and communicated. But numbers can mislead if their meaning is not understood. A raw score may be the number of items endorsed. A percentile may show relative standing. A standard score may compare performance to a norm group. A confidence interval may show uncertainty. Each score type answers a different question.

Raw scores and derived scores

A raw score is often the simplest count. It may be the number correct, the number of symptoms, or the sum of ratings. Raw scores are useful for tracking within a person if the measure is stable. They are often poor for comparing across ages or forms. Derived scores adjust for age, grade, or norm group. Common derived scores include percentiles, standard scores, T scores, and stanines.

A percentile of 84 does not mean 84 percent correct. It means the child scored as high as or higher than about 84 percent of the norm group. A standard score of 100 with a standard deviation of 15 is average in many tests. A T score of 50 with a standard deviation of 10 is average in many rating scales. The counselor should know the metric before interpreting.

Norm-referenced and criterion-referenced interpretation

Norm-referenced interpretation compares a child to a group. Criterion-referenced interpretation compares performance to a standard or skill criterion. A child may be below average in reading compared to peers but still able to read simple sentences. A child may be above average in anxiety symptoms but not meet diagnostic criteria. Both interpretations can be useful. The question determines which is needed.

Confidence intervals and measurement error

No score is exact. Measurement error means the observed score is an estimate. A confidence interval gives a range likely to contain the true score. A child with a standard score of 98 and a confidence interval of 92 to 104 should not be treated as fundamentally different from a child with 102. Small differences may be noise. The counselor should interpret bands and patterns, not single points.

Cut scores and decisions

Cut scores divide scores into categories, such as at risk or not at risk. They are useful for decisions but create error near the boundary. A child just above a cut score may still need help. A child just below may not need intensive intervention. The counselor should consider consequences, base rates, and multiple data sources. High-stakes decisions should not rest on one cutoff.

Profile interpretation

Many tests produce profiles. A profile can show strengths and weaknesses, but scattered differences may occur by chance. The counselor should ask whether a difference is reliable and meaningful. A one-point difference between subtests is rarely important. A large, consistent pattern across methods may be meaningful. Profile interpretation should be guided by theory and referral questions, not by searching for any difference.

Communicating scores

Families often understand plain language better than numbers. The counselor can say, "Compared with children the same age, your child's worry score is in the high range. This does not mean your child is broken. It means worry is interfering enough that support is likely to help." The counselor should explain what the score does and does not mean. Visual aids can help, but they should not oversimplify uncertainty.

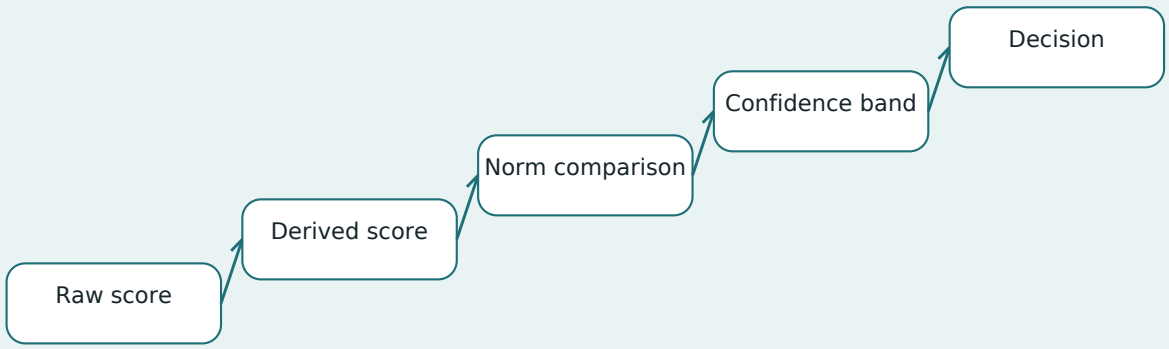
Real-life application

A parent hears that a child's IQ is 95 and concludes the child cannot succeed. The counselor explains that 95 is within the average range, that the confidence interval includes 100, and that achievement, motivation, instruction, and support also matter. The report shifts from a fixed label to a functional plan. The score becomes information, not destiny.

Reflection

Take a score you have seen. Identify its metric, norm group, and uncertainty. Write a plain-language explanation for a caregiver. Then write one decision the score supports and one decision it does not support. Scores serve decisions. They should not replace judgment.

Score interpretation ladder



Each step adds meaning and uncertainty. The final step is a responsible decision.

Attributed Foundational Reading

Attribution and critical-reading note

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Source module scores-scales.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Scores and Scales

Assessments yield information. The information is encoded in scores or in other types of data. It is important to consider the different types of data because the types of data restrict what options are available to analyze the data.

Load Libraries

First, we load the R packages used for this chapter:

Simulate Data

For this example, we simulate data below. For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Add Missing Data

Adding missing data to dataframes helps make examples more realistic to real-life data and helps you get in the habit of programming to account for missing data.

Data Types

There are four general data types: nominal, ordinal, interval, and ratio. Depending on the use of the variable, the data could fall into more than one category. The type of data influences what kinds of data analysis you can do. For instance, parametric statistical analysis (e.g., t test, analysis of variance [ANOVA], and linear regression) assumes that data are interval or ratio.

Nominal

Nominal data are distinct categories. They are categorical and unordered. Nominal data make no quantitative claims. Nominal data represent things that we can name (e.g., cat and dog). Nominal data can be represented with numbers. For example, zip codes are nominal. Numbers that represent a participant's sex, race, or ethnicity are also nominal. Categorical data are often dummy-coded into binary (i.e., dichotomous) variables, which represent nominal data. Higher numbers of nominal data do not reflect higher (or lower) levels of the construct because the numbers represent categories that do not have an order.

Ordinal

Ordinal data are ordered categories: they have a name and an order. They make no claim about the conceptual distance between the ranks, only that higher values represent higher (or lower) levels of the construct. For example, ranks following a race are ordinal—that is, the person with rank 1 finished before the person with rank 2, who finished before the person with rank 3 (1 > 2 > 3 > 4). Ordinal data make a limited claim because the conceptual distance between adjacent numbers is not the same. For instance, the person who finished the race first might have finished 10 minutes before the second-place finisher; whereas the third-place finisher might have finished 1 second after the second-place finisher.

That is, just because the numbers have the same mathematical distance does not mean that they represent the same conceptual distance on the construct. For example, if the respondent is asked how many drinks they had in the past day, and the options are 0 = 0 drinks; 1 = 1–2 drinks; 2 = 3 or more drinks, the scale is ordinal. Even though the numbers have the same mathematical distance (1, 2, 3), they do not represent the same conceptual distance. Most data in psychology are ordinal data even though they are often treated as if they were interval data.

Interval

Interval data are ordered and have meaningful distances (i.e., equal spacing between intervals). You can sum interval data (e.g., 2 is 2 away from 4), but you cannot multiply interval data (2 × 2 4). Examples of interval data are temperatures in Fahrenheit and Celsius—100 degrees Fahrenheit is not twice as hot as 50 degrees Fahrenheit. A person's number of years of education is interval, whereas educational attainment (e.g., high school degree, college degree, graduate degree) is only ordinal. Although much data in psychology involve numbers that have the same mathematical distance between intervals, the intervals likely do not represent the same conceptual distance. For example, the difference in severity of two people who have two symptoms and four symptoms of depression, respectively, may not be the same difference in depression severity as two people who have four symptoms and six symptoms, respectively.

Ratio

Ratio data are ordered, have meaningful distances, and have a true (absolute) zero that represents absence of the construct. With ratio data, multiplicative relationships are true. An example of ratio data is temperature in Kelvin—100 degrees Kelvin is twice as hot as 50 degrees Kelvin. There is a dream of having ratio scales in psychology, but we still do not have a true zero with psychological constructs—what does total absence of depression mean (apart from a dead person)?

Score Transformation

There are a number of score transformations, depending on the goal. Some score transformations (e.g., log transform) seek to make data more normally distributed to meet assumptions of particular analysis approaches. Score transformations alter the original (raw) data. If you change the data, it can change the results. Score transformations are not neutral. For instance, transformations change the structure of correlations between variables.

Raw Scores

Raw scores are the original data, or they may be aggregations (e.g., sums or means) of multiple items. Raw scores are the purest because they are closest to the original operation (e.g., behavior). A disadvantage of raw scores is that they are scale dependent, and therefore they may not be comparable across different measures with different scales. An example histogram of raw scores is in the source discussion.

Norm-Referenced Scores

Norm-referenced scores are scores that are referenced to some norm. A norm is a standard of comparison. For instance, you may be interested in how well a participant performed relative to other children of the same sex, age, grade, or ethnicity. However, interpretation of norm-referenced scores depends on the measure and on the normative sample. A person's norm-referenced score can vary widely depending on which norms are used. Which reference group should you use? Age? Sex? Age and sex? Grade? Ethnicity? The optimal reference group depends on the purpose of the assessment. The quality of the norms also depends on the representativeness of the reference group compared to the population of interest. Pros and cons of group-based norms are discussed in the source discussion. Guidelines for reporting on norm-referenced scores are provided by Timmerman et al., in press.

A standard normal distribution on various norm-referenced scales is depicted in the source discussion, as adapted from Bandalos, 2018.

Percentile Ranks

A percentile rank reflects what percent of people a person scored higher than, in a given group (i.e., norm). Percentile ranks are frequently used for tests of intellectual/cognitive ability, academic achievement, academic aptitude, and grant funding. They seem like interval data, but they are not intervals because the conceptual spacing between the numbers is not equal. The difference in ability for two people who scored at the 99th and 98th percentile, respectively, is not the same as the difference in ability for two people who scored at the 49th and 50th percentile, respectively. Percentile ranks are only judged against a baseline; there is no subtraction.

Percentile ranks have unusual effects. There are lots of people in the middle of a distribution, so a very small difference in raw scores gets expanded out in percentiles. For instance, a raw score of 20 may have a percentile rank of 50, but a raw score of 24 may have a percentile rank of 68. However, a larger raw score change at the ends of the distribution may have a smaller percentile change. For example, a raw score of 120 may have a percentile rank of 97, whereas a raw score of 140 may have a percentile rank of 99. Thus, percentile ranks stretch out differences for some people but constrict differences for others.

Here is an example of how to calculate percentile ranks using the dplyr package from the tidyverse (R-tidyverse, n.d.; tidyverse2019, n.d.):

A histogram of percentile rank scores is in the source discussion.

Deviation (Standardized) Scores

Deviation or standardized scores are the transformation of raw scores to a normal distribution using some norm. The norm could be a comparison group, or it could be the sample itself. With deviation scores, you have similar challenges as with percentile ranks, including which reference group to use, but there are additional challenges. Deviation scores are more informative when the scores are normally distributed compared to when the scores are skewed. If scores are skewed, it can lead to two z scores on the opposite side of the mean having different probabilities.

Many constructs we study in psychology are not normally distributed. For example, the frequency of hallucinations among people would show a positively skewed distribution with a truncation at zero, representing a floor effect—i.e., most people do not show hallucinations. For instance, consider a hypothetical distribution of hallucinations. It might follow a folded distribution, as depicted in the source discussion:, n.d.

Now consider the same distribution converted to a standardized (z) score, as depicted in the source discussion:, n.d.

Thus, you can compute a deviation score, but it may not be meaningful if the data and underlying construct are not normally distributed.

z scores The z score is the most common standardized score, and it can help put scores from different measures with different scales onto the same playing field. z scores have a mean of zero and a standard deviation of one. To get a z score that uses the sample as its own norm, subtract the mean from all scores and divide by the standard deviation. Every z score represents how far that person's score is from the (normed) average, represented in standard deviation units. With a standard normal curve, 68% of scores fall within one standard deviation of the mean. 95% of scores fall within two standard deviations of the mean. 99.7% of scores fall within three standard deviations of the mean.

The area under a normal curve within one standard deviation of the mean is calculated below using the pnorm function, which calculates the cumulative density function for a normal curve.

The area under a normal curve within one standard deviation of the mean is depicted in the source discussion.

The area under a normal curve within two standard deviations of the mean is calculated below:

The area under a normal curve within two standard deviations of the mean is depicted in the source discussion.

The area under a normal curve within three standard deviations of the mean is calculated below:

The area under a normal curve within three standard deviations of the mean is depicted in the source discussion.

Alternatively, if you want to determine the z score associated with a particular percentile in a normal distribution, you can use the qnorm function.

For instance, the z score associated with the 37th percentile is:

z scores are calculated using the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where x is the observed score, mu is the mean observed score, and sigma is the standard deviation of observed scores.

An example histogram of z scores is in the source discussion.

T scores T scores have a mean of 50 and a standard deviation of 10. T scores are frequently used with personality and symptom measures, where clinical cutoffs are often set at 70 (i.e., two standard deviations above the mean). For the Minnesota Multiphasic Personality Inventory (MMPI), you would examine peaks (elevations 70) and absences (30).

T scores are calculated using the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where z are z scores.

An example histogram of T scores is in the source discussion.

Standard scores Standard scores have a mean of 100 and a standard deviation of 15. Standard scores are frequently used for tests of intellectual ability, academic achievement, and cognitive ability. Intellectual disability is generally considered an I.Q. standard score less than 70 (two standard deviations below the mean), whereas giftedness is at 130 (two standard deviations above the mean).

Standard scores with a mean of 100 and standard deviation of 15 are calculated using the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where z are z scores.

An example histogram of standard scores is in the source discussion.

Scaled scores Scaled scores are raw scores that have been converted to a standardized metric. The particular metric of the scaled score depends on the measure. On tests of intellectual or cognitive ability, scaled scores commonly have a mean of 10 and a standard deviation of 3.

Using this metric, scaled scores can be calculated using the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where z are z scores.

An example histogram of scaled scores is in the source discussion.

Stanine scores Stanine scores (short for STANdard Nine) have a mean of 5 and a standard deviation of 2. The scores range from 1–9. Stanine scores are calculated using the bracketed proportions in the source discussion. The lowest 4% receive a stanine score of 1, the next 7% receive a stanine score of 2, etc.

A histogram of stanine scores is in the source discussion.

Conclusion

It is important to consider the types of data your data are because the types of data restrict what options are available to analyze the data. It is also important to consider whether the data were transformed because score transformations are not neutral—they can impact the results.

Suggested Readings

Childs2021:, n.d. https://dzchilds.github.io/stats-for-bio/data-transformations.html (archived at https://perma.cc/3YEB-QW2V)

CHAPTER 06 / CAC.903

Reliability and Measurement Error

Reliability asks whether a measurement is consistent enough to support an interpretation. A reliable scale gives similar results when the construct is stable and conditions are comparable. Reliability is not the same as truth. A scale can consistently measure the wrong thing. Still, without reliability, scores are too noisy to guide decisions. In child assessment, reliability is especially important because development, context, and informant differences already create variability.

Classical test theory

Classical test theory describes an observed score as a combination of a true score and error. The true score is a theoretical average over repeated measurements under identical conditions. Error includes random influences such as mood, fatigue, guessing, or unclear items. The standard error of measurement estimates how much observed scores may vary around the true score. A confidence interval uses this error to show a plausible range.

A child who scores 12 on a symptom scale may have a true score somewhere near 12, not exactly 12. If the standard error is large, a change from 12 to 10 may not be meaningful. The counselor should ask whether change exceeds measurement error before declaring improvement or deterioration.

Types of reliability

Test-retest reliability examines stability over time. It is appropriate when the construct should be stable. It is less appropriate for states that change quickly, such as panic or hunger. Inter-rater reliability examines agreement between observers. It matters for behavioral coding and clinical judgment. Intra-rater reliability examines consistency of the same observer over time. Parallel-forms reliability examines equivalence of different versions. Internal consistency examines whether items that reflect the same construct correlate.

No single reliability coefficient answers every question. A measure may have high internal consistency but poor test-retest stability. A behavioral code may have high inter-rater agreement but limited validity. The counselor should match the reliability evidence to the intended use.

How reliable is enough?

Higher-stakes decisions require stronger reliability. A brief screening tool may tolerate more error because it leads to further assessment. A diagnosis that affects placement or legal decisions requires stronger evidence. A rule of thumb is not enough. The counselor should consider the size of the error, the distance from cut scores, and the consequences of misclassification.

Reliability also depends on the sample. A measure may be reliable in a clinical sample but less reliable in a community sample where scores cluster near zero. Range restriction can lower correlations. The counselor should examine the population for which reliability was estimated.

Increasing reliability

Reliability can be improved by clear items, adequate length, standardized administration, trained raters, and multiple observations. Averaging across informants or occasions can reduce random error, but only if the sources are meaningfully related. Adding items does not help if they measure different constructs. Training raters helps only if the coding system is clear.

In child assessment, reliability also improves when the setting is appropriate. A hungry, tired, or frightened child may produce noisy data. The counselor should consider timing, breaks, rapport, and accessibility. Standardization should not become rigidity that harms validity.

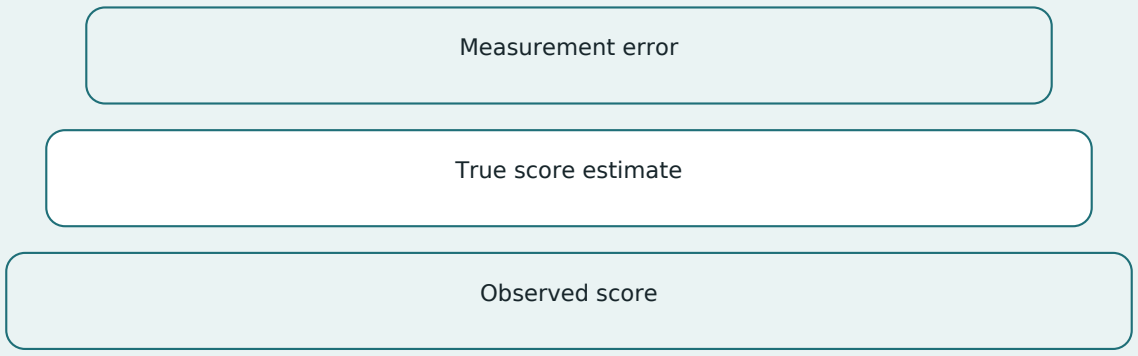
Real-life application

A counselor tracks a teen's depression score weekly. Scores move from 18 to 16 to 19 to 14. The counselor does not interpret each fluctuation as change. Instead, the counselor looks at the trend over several weeks, considers measurement error, and asks about life events. The teen reports better sleep and increased activity. The trend and context support cautious optimism. The counselor continues monitoring rather than declaring cure.

Reflection

Choose a measure you might use. Identify the type of reliability most relevant to your decision. Ask how much error is acceptable. Then write how you would explain uncertainty to a caregiver. Reliability is not a badge. It is a boundary around what a score can safely mean.

Observed score, true score, and error



Reliability describes how much error surrounds an observed score.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module reliability.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Overview

To understand the concept of reliability, it is helpful to understand classical test theory (CTT), which is also known as "true score theory." Classical test theory is one of various measurement theories, in addition to item response theory and generalizability theory. CTT has been the predominant measurement theory through the history of psychology. CTT is a theory of how test scores relate to a construct. A construct is the concept or characteristic that a measure is intended to assess.

Assume you take a measurement of the same object multiple times (i.e., repeated measure). For example, you assess the mass of the same rock multiple times. However, you obtain different estimates of the rock's mass each time. There are multiple possible explanations for this observation. One possible explanation could be that the rock is changing in its mass, which would be consistent with the idea proposed by the Greek philosopher Heraclitus that nothing is stable and the world is in flux. An alternative possibility, however, is that the rock is stable in its mass but the measurements are jittery—that is, they include error.

Based on the possibility that the rock is stable and that differences in scores across time reflect measurement error, CTT proposes the true score formula in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

X is the observed measurement or score, T is the classical (or psychometric) "true score," and e is the measurement error (i.e., error score).

This formula is depicted visually in the form of a path diagram in the source discussion.

Classical Test Theory Formula in a Path Diagram.

It is important to distinguish between the classical true score and the Platonic true score. The Platonic true score is the truth, and it does not depend on measurement. The Platonic true score is an abstract notion because it is not directly observable and is based on Platonic ideals and theories of the construct and what a person's "true" level is on the construct. In CTT, we attempt to approximate the Platonic true score with the classical true score, (T). If

we took infinite repeated observations (and the measurements had no carryover effect), the average score approaches the classical true score, T. That is, X = T as number of observations arrow infinity. CTT attempts to partition variance into different sources. Variance is an index of scores' variability, i.e., the degree to which scores differ. Variance is defined as the average squared deviation from the mean, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

According to CTT, any observed measurement includes both true score (construct) variance (sigma^2 T) and error variance (sigma^2 e). Given the true score formula (X = T + e), this means that their variance is as follows (see the source discussion):

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Nevertheless, the classical true score, T, is the expected value and is not necessarily the same thing as the Platonic true score (Borsboom, 2003; Klein & Cleary, 1969) because the expected value would need to be entirely valid/accurate (i.e., it would need to be the construct score) for it to be the Platonic true score. The expected score could also be influenced by systematic sources of error such as other constructs, which would not fall into the error portion of the CTT formula because, as described below, CTT assumes that error is random (not systematic). The distinctions between construct score, (classical) true score, and observed score, in addition to validity, reliability, systematic error, and random error are depicted in the source discussion.

Distinctions Between Construct Score, True Score, and Observed Score, in Addition to Reliability, Validity, Systematic Error, and Random Error. (Adapted from W. Joel Schneider.)

The true score formula is theoretically useful, but it is not practically useful because it is an under-identified equation and we do not know the values of T or e based on knowing the observed score (X). For instance, if we obtain an observed score of 10, our formula is 10 = T + e, and we do not know what the true score or error is. As a result, CTT makes several simplifying assumptions so we can estimate how stable (reliable) or noisy the measure is and what proportion of the observed score reflects true score versus measurement error.

1. E(e) = 0

The first assumption of CTT is that the expected value of the error (i.e., error scores) is zero. Basically, the error component of the observed scores is expected to be random with a mean of zero. The likelihood that the observed score is an overestimate of T is assumed to be the same as the likelihood that the observed score is an underestimate of T. In other words, the distribution of error scores above T is assumed to be the same as the distribution of error scores below T. In reality, this assumption is likely false in many situations. For instance, social desirability bias is a systematic error where people rate themselves as better than they actually are; thus, social desirability results in biasing scores in a particular direction across respondents, so such an error would not be entirely random. But using the assumption that the expected value of e is zero also informs that the expected value of the observed score (X) equals the expected value of the true score (T), as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

1. r T,e = 0

The second assumption of CTT is that the correlation between T and e is zero—that is, people's true scores are uncorrelated with the error around their measurement (i.e., people's error scores). However, this assumption is likely false in many situations. For instance, one can imagine that, on a paper-and-pencil intelligence test, scores may have greater error for respondents with lower true scores and may have less error for respondents with higher true scores.

1. r e 1, e 2 = 0

The third assumption of CTT is that the error is uncorrelated across time—that is, people's error scores at time 1 (e 1) are not associated with their error scores at time 2 (e 2). However, this assumption is also likely false in many situations. For example, if some people have a high social desirability bias at time 1, they are likely to also have a high social desirability bias at time 2. That is, the error around measurements of participants is likely to be related across time.

These three assumptions are implicit in the path analytic diagram in the source discussion, which depicts the CTT approach to understanding reliability of a measure across two time points.

Reliability of a Measure Across Two Time Points, as Depicted in a Path Diagram.

In path analytic (and structural equation modeling) language, rectangles represent variables we observe, and circles represent latent (i.e., unobserved) variables. The observed scores at time 1 (X 1) and time 2 (X 2) are entities we observe, so they are represented by rectangles. We do not directly observe the true scores (T 1, T 2) and error scores (e 1, e 2), so they are considered latent entities and are represented by circles. Single-headed arrows indicate regression paths, where conceptually, one variable is thought to influence another variable. As the model depicts, the observed scores are thought to be influenced both by true scores and by error scores. We also expect the true scores at time 1 (T 1) and time 2 (T 2) to be correlated, so we have a covariance path, as indicated by a double-headed arrow. A covariance is an unstandardized index of the strength of association between two variables. Because a covariance is unstandardized, its scale depends on the scale of the variables. The covariance between two variables is the average product of their deviations from their respective means, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The covariance of a variable with itself is equivalent to its variance, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

By contrast, a correlation is a standardized index of the strength of association between two variables. Because a correlation is standardized (fixed between [−1,1]), its scale does not

depend on the scales of the variables. In this figure, no other parameters (regressions, covariances, or means) are specified, so the following are implicit in the diagram:

• E(e) = 0

• r T,e = 0

• r e 1, e 2 = 0

The factor loadings reflect the magnitude that the latent factor influences the observed variable. In this case, the true scores influence the observed scores with a magnitude of sqrtr xx, which is known as the index of reliability. The index of reliability is the theoretical estimate of the correlation between the true scores and the observed scores. This is depicted in the source discussion.

Reliability of a Measure Across Two Time Points, as Depicted in a Path Diagram; Includes the Index of Reliability.

We can use path tracing rules to estimate the reliability of the measure, where the reliability of the measure, i.e., the coefficient of reliability (r xx), is estimated as the correlation between the observed score at time 1 (x 1) and the observed score at time 2 (x 2). According to path tracing rules (Pearl, 2013), the correlation between x 1 and x 2 is equal to the sum of the standardized coefficients of all the routes through which x 1 and x 2 are connected. The contribution of a given route to the correlation between x 1 and x 2 is equal to the product of all standardized coefficients on that route that link x 1 and x 2 that move in the following directions: (a) forward (e.g., T 1 to x 1) or (b) backward any number of times and then forward (e.g., x 1 to T 1 to T 2 to x 2). Path tracing does not allow moving forward and then backward—that is, it does not allow retracing (e.g., e to x 1 to T 1) in the same route. It also does not allow passing more than one curved arrow (covariance path) or through the same variable twice in the same route. Once you know the contribution of each route to the correlation, you can calculate the total correlation between the two variables as the sum of the contribution of each route. Therefore, using one route, we can calculate the association between x 1 and x 2 as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

When dealing with a stable construct, we would assume that the correlation between true scores across time is 1.0: r T 1,T 2 = 1.0, as depicted in the source discussion.

Reliability of a Measure of a Stable Construct Across Two Time Points, as Depicted in a Path Diagram.

Then, to calculate the association between x 1 and x 2 of a stable construct, we can use path tracing rules as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

That is, for a stable construct (i.e., whose true scores are perfectly correlated across time; r T 1,T 2 = 1.0), we estimate reliability as the correlation between the observed scores at time 1 (x 1) and the observed scores at time 2 (x 2). This is known as test–retest reliability. We therefore assume that the extent to which the correlation between x 1 and x 2 is less than

one reflects measurement error (an unstable measure), rather than people's changes in their true score on the construct (an unstable construct).

As described above, the reliability coefficient (r xx) is the association between a measure and itself over time or with another measure in the domain. By contrast, the reliability index (r xT) is the correlation between observed scores on a measure and the true scores (Nunnally & Bernstein, 1994). The reliability index is the square root of the reliability coefficient, as in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Four CTT Measurement Models

There are four primary measurement models in CTT (Graham, 2006):

1. parallel

2. tau-equivalent

3. essentially tau-equivalent

4. congeneric

Parallel

The parallel measurement model is the most stringent measurement model for use in estimating reliability. In CTT, a measure is considered parallel if the true scores and error scores are equal across items. That is, the items must be unidimensional and assess the same construct, on the same scale, with the same degree of precision, and with the same amount of error (Graham, 2006). Items are expected to have the same strength of association (i.e., factor loading or discrimination) with the construct.

Tau-Equivalent

The tau -equivalent measurement model is the same as the parallel measurement model, except error scores are allowed to differ across items. That is, a measure is tau-equivalent if the items are unidimensional and assess the same construct, on the same scale, with the same degree of precision, but with possibly different amounts of error (Graham, 2006). In other words, true scores are equal across items but each item is allowed to have unique error scores. Items are expected to have the same strength of association with the construct. Variance that is unique to a specific item is assumed to be error variance.

Essentially Tau-Equivalent

The essentially tau -equivalent model is the same as the tau-equivalent measurement model, except items are allowed to differ in their precision. That is, a measure is essentially tau-equivalent if the items are unidimensional and assess the same construct, on the same scale, but with possibly different degrees of precision, and with possibly different amounts of error (Graham, 2006). The essentially tau-equivalent model allows item true scores to differ by a constant that is unique to each pair of variables. The magnitude of the constant reflects the degree of imprecision and influences the mean of the item scores but not its variance or covariances with other items. Items are expected to have the same strength of association with the construct.

Congeneric

The congeneric measurement model is the least restrictive measurement model. The congeneric measurement model is the same as the essentially tau -equivalent model, except items are allowed to differ in their scale. That is, a measure is congeneric if the items are unidimensional and assess the same construct, but possibly on a different scale, with possibly different degrees of precision, and with possibly different amounts of error (Graham, 2006). Items are not expected to have the same strength of association with the construct.

Measurement Error

Measurement error is the difference between the measured (observed) value and the true value. All measurements come with uncertainty and measurement error. Even a measure of something as simple as whether someone is dead has error.

There are two main types of measurement error: systematic (nonrandom) error and unsystematic (random) error. In addition, measurement error can be within-person, between-person, or both.

Systematic (Nonrandom) Error

An example of systematic error is depicted in the source discussion. Systematic error is error that influences consistently for a person or across the sample. An error is systematic if the error always occurs, with the same value, when using the measure in the same way and in the same case. An example of a systematic error is a measure that consistently assesses constructs other than the construct the measure was designed to assess. For instance, if a test written in English to assess math skills is administered in a nonnative English-speaking country, some portion of the scores will reflect variance attributable to English reading skills rather than the construct of interest (math skills). Other examples of systematic error include response styles or subjective, idiosyncratic judgments by a rater—for instance, if the rater's judgments are systematically harsh or lenient. A systematic error affects the average score (i.e., resulting in bias), which makes the group-level estimates less accurate and makes the measurements for an individual less accurate.

As depicted in the source discussion, systematic error does not affect the variability of the scores but it does affect the mean of the scores, so the person-level mean and group-level mean are less accurate. In other words, a systematic error leads to a biased estimate of the average. However, multiple systematic errors may simultaneously coexist and can operate in the same direction (exacerbating the effects of bias) or in opposite directions (hiding the extent of bias).

Unsystematic (Random) Error

An example of unsystematic (random) error is depicted in the source discussion. Random error occurs due to chance. For instance, a random error could arise from a participant being fatigued on a particular testing day or from a participant getting lucky in guessing the correct answer. Random error does not have consistent effects for a person or across the sample, and it may vary from one observation to another. Random error does not (systematically) affect the average, i.e., the group-level estimate—random error affects only the variability around the average (noise). However, random error makes measurements for an individual less accurate. A large number of observations of the same construct cancels out random error but does not cancel out systematic error.

As depicted in the source discussion, random error does not affect the mean of the scores, but it does increase the variability of the scores. In other words, the group-level mean is still accurate, but individuals' scores are less precise.

Within-Person Error

Consider two data columns, one column for participants' scores at time 1 and another column for participants' scores at time 2. Adding within-person error would mean adding noise (e) within the given row (or rows) for the relevant participant(s). Adding between-person error would mean adding noise (e) across the rows within the column.

Within-person error occurs within a particular person. For instance, you could add within-person error to a data set by adding error to the given row (or rows) for the relevant participant(s).

Between-Person Error

Between-person error occurs across the sample. You could add between-person random error to a variable by adding error across the rows, within a column.

Types of Measurement Error

There are four nonmutually exclusive types of measurement error: within-person random error, within-person systematic error, between-person random error, and between-person systematic error. The four types of measurement error are depicted in the source discussion, as adapted from Willett, 2012.

Types of Measurement Error.

Within-Person Random Error

Adding within-person random error would involve adding random noise (e) to the given row (or rows) for the relevant participant(s). This could reflect momentary fluctuations in the assessment for a specific person. When adding within-person random error, the person's and group's measurements show no bias, i.e., there is no consistent increase or decrease in the scores from time 1 to time 2 (at least with a sample size large enough to cancel out the random error, according to the law of large numbers). A person's average approximates their true score if many repeated measurements are taken. A group's average approximates the sample mean's true score, especially when averaging the repeated measures across time. The influence of within-person random error is depicted in the source discussion.

Within-Person Random Error.

Within-Person Systematic Error

Adding within-person systematic error would involve adding systematic noise (e) (the same variance across columns) to the given row (or rows), reflecting the relevant participant(s). These are within-person effects that are consistent across time. For example, social desirability bias is high for some people and low for others. Another instance in which within-person systematic error could exist is when one or more people consistently misinterpret a particular question. Within-person systematic error increases person-level bias because the person's mean shows a greater difference from their true score. The influence of within-person systematic error is depicted in the source discussion.

Within-Person Systematic Error.

Between-Person Random Error

Adding between-person random error at time 2 would involve adding random noise (e) across the rows, within the column. Between-person random error would result in less accurate scores at the person level but would not result in bias at the group level. At a given timepoint, it results in over-estimates of the person's true score for some people and under-estimates for other people, i.e., there is no consistent pattern across the sample. Thus, the group average approximates the sample's mean true score (at least with a sample size large enough to cancel out the random error, according to the law of large numbers). In addition, the average of repeated measurements of the person's score would approximate the person's true score. However, the group's variance is inflated. The influence of between-person random error is depicted in the source discussion.

Between-Person Random Error.

Between-Person Systematic Error

Adding between-person systematic error at time 2 would involve adding systematic noise (e) across the rows, within the column. Between-person systematic error results from within-person error that tends to be negative or positive across participants. For instance, this could reflect an influence with a shared effect across subjects. For example, social desirability leads to a positive group-level bias for rating their socially desirable attributes. Another example would be when a research assistant enters values wrong at time 2 (e.g., adding 10 to all participants' scores). Between-person systematic error increases bias because it results in a greater group mean difference from the group's mean true score. The influence of between-person systematic error is depicted in the source discussion.

Between-Person Systematic Error.

Summary

In sum, all types of measurement error (whether systematic or random) lead to less accurate scores for an individual. But different kinds of error have different implications. Systematic and random error have different effects on accuracy at the group-level. Systematic error leads to less accurate estimates at the group-level, whereas random error does not.

CTT assumes that all error is random. According to CTT, as the number of measurements approaches infinity, the mean of the measurements gets closer to the true score, because the random errors cancel each other out. With more measurements, we reduce our uncertainty and increase our precision. According to CTT, if we take many measurements

and the average of the measurements is 10, we have some confidence that the true score (T) 10. In reality, however, error for a given measure likely includes both systematic and random error.

Overview of Reliability

The "Standards for Educational and Psychological Testing" set the standard for educational and psychological assessment and are jointly published by the American Educational Research Association, American Psychological Association, and National Council on Measurement in Education. According to the "Standards" (American Educational Research Association et al., 2014), reliability is the "consistency of the scores across instances of the testing procedure." In this book, we define reliability as how much repeatability, consistency, and precision the scores from a measure have. Reliability is not a property of the measure itself—the reliability of a measure's scores reflects an interaction of the properties of the measure with the population for whom it is designed and the sample and context in which it is administered. For these reasons, it is insufficient to cite evidence for the reliability of the measure in other work; it is also important to evaluate the reliability of the measure's scores in your particular sample.

Reliability (r xx) has been defined mathematically as the proportion of observed score variance (sigma^2 X) that is attributable to true score variance (sigma^2 T), as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

An alternative formulation is that reliability (r xx) is the lack of error variance or the degree to which observed scores are correlated with true scores or uncorrelated with error scores. In CTT, reliability can be conceptualized in four primary ways, as depicted in the source discussion (Furr, 2017).

Four Different Ways of Conceptualizing Reliability.

However, we cannot calculate reliability because we cannot measure the true score component of an observation. Therefore, we estimate reliability (the coefficient of reliability) based on the relation between two observations of the same measure (for test–retest reliability) or using other various estimates of reliability.

The coefficient of reliability can depend on several factors. Reliability is inversely related to the amount of measurement error. The coefficient of reliability, like correlation, depends on the degree of spread (variability) of the scores. If the scores at one or both time points show restricted range, the scores will show a weaker association and coefficient of reliability, as shown in the source discussion. An ideal way to estimate the reliability of a measure would be to take a person and repeatedly measure them many times to get an estimate of their true score and each measurement's deviation from their true score, and to do this for many people. However, this is rarely done in practice. Instead of taking one person and repeatedly measuring them many times, we typically estimate reliability by taking many people and doing repeated measures twice. This is a shortcut to estimate reliability, but even this shorter approach is not often done. In short, researchers rarely estimate the test–retest reliability of the measures they use.

Reliability can also be related to the number of items in the measure. In general, the greater the number of items, the more reliable the measure (assuming the items assess the same construct) because we are averaging out random error.

We never see the true scores or the error scores, so we cannot compute reliability—we can only estimate it from the observed scores. This estimate of reliability gives a probabilistic answer of the reliability of the measure, rather than an absolute answer.

How Reliable Is Reliable Enough?

As described by Nunnally & Bernstein, 1994, how reliable a measure should be depends on the proposed uses. If it is early in the research process, and the focus is on group-level inferences (e.g., associations or group differences), modest reliability (e.g.,.70) may be sufficient and save time and money. Then, the researcher can see what the associations would be when disattenuated for unreliability, as described in the source discussion of the chapter on validity. If the disattenuated associations are promising, it may be worth increasing the reliability of the measure. Associations are only weakly attenuated above a reliability of.80, so achieving a reliability coefficient of.80 may be an appropriate target for basic research.

However, when making decisions about individual people from their score on a measure, reliability and precision are more important (than when making group-level inferences) because small differences in scores can lead to different decisions. Nunnally & Bernstein, 1994 recommend that measures have at least a reliability of.90 and—when making important decisions about individual people—that measures preferably have a reliability of.95 or higher. Nevertheless, they also note that one should not switch to a less valid measure merely because it is more reliable.

Standard Error of Measurement (SEM)

The estimate of reliability gives a general idea of the degree of uncertainty you have of a person's true score given their observed score. From this, we can estimate the standard error of measurement, which estimates the extent to which an observed score deviates from a true score. The standard error of measurement indicates the typical distance of the observed score from the true score. The formula for the standard error of measurement is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where sigma x represents the standard deviation of scores. Thus, the standard error of measurement is directly related to the reliability of the measure. The higher the reliability, the lower the standard error of measurement. The standard error of measurement as a function of reliability of the measure and the standard deviation of scores is depicted in the source discussion.

The derivation of the SEM (from W. Joel Schneider) is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The SEM is equivalent to the standard deviation of measurement error (e) (Lek & Van De Schoot, 2018), as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Around 95% of observed scores would fall within ± 2 SEMs of the true score (or, more precisely, within ± [computed value omitted in this course adaptation] SEMs of the true score). Using the SEM, we can compute a confidence interval that represents a range of plausible values for the person's true score (with a specified confidence level, e.g., 95%), given measurement error. In other words, if we repeatedly observed this person and constructed intervals as the observed score ± 2 SEMs, about 95% of those intervals would contain the true score. Given an observed score of X = 15 and SEM = 2, the 95% confidence interval of the true score is [11, 19]. So if a person gets a score of 15 on the measure, it is likely that their true score falls within 11–19. The 95% confidence interval represents that, if we repeatedly observed this person under identical conditions and constructed a confidence interval each time, 95% of those intervals would contain the person's true score. However, the confidence interval does not indicate that there is a 95% probability that their true score falls within the interval. We provide an empirical example of estimating the SEM in the source discussion.

Based on the preceding discussion, consider the characteristics of measures that make them more useful from a reliability perspective. A useful measure would show wide variation across people (individual differences), so we can more accurately estimate its reliability. And we would expect a useful measure to show consistency, stability, precision, and reliability of scores.

Simulate Data

For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Add Missing Data

Adding missing data to dataframes helps make examples more realistic to real-life data and helps you get in the habit of programming to account for missing data.

Types of Reliability

Reliability is not one thing. There are several types of reliability. In this book, we focus on test–retest, inter-rater, intra-rater, parallel-forms, and internal consistency reliability. Moreover, one can estimate reliability at different levels (e.g., person-level versus group-level) (Geldhof et al., 2014; ten Hove et al., 2022).

Test–Retest Reliability

Test–retest reliability is defined as the consistency of scores across time. Typically, this is based on a two-week retest interval. The intent of a two-week interval between the original testing and the retest is to provide adequate time to pass to reduce any carryover effects from the original testing while not allowing too much time to pass such that the person's level on the construct (i.e., true scores) would change. A carryover effect is an effect of the experimental condition that affects the participant's behavior at a later time. Examples of carryover effects resulting from repeated measurement can include fatigue, boredom, learning (practice effects), etc. Another potential issue is that measurement error can be correlated across the two measurements.

Test–retest reliability controls for transient error and random response error. If the construct is not stable across time (i.e., people's true scores change), test–retest reliability is not relevant because the CTT approach to estimating reliability assumes that the true scores are perfectly correlated across time (see the source discussion).

The length of the optimal retest interval depends on the construct of interest. For a construct in which people's levels change rapidly, a shorter retest interval may be appropriate. But one should pay attention to ways to reduce potential carryover effects. By contrast, if the retest interval is too long, people's levels on the construct may change during that span. If people's levels on the construct change from test to retest, we can no longer assume that the true scores are perfectly correlated across time, which would violate CTT assumptions for estimating test–retest reliability of a measure. The longer the retest interval, the smaller the observed association between scores across time will tend to be. For weak associations obtained from a lengthy retest interval, it can be difficult to determine how much of this weak association reflects measurement unreliability versus people's change in their levels on the construct. Thus, when conducting studies to evaluate test–retest reliability, it is important to consider the length of the retest interval and ways to reduce carryover effects.

Coefficient of Stability (and Coefficient of Dependability)

The coefficient of stability is the most widely used index when reporting the test–retest reliability of a measure. It is estimated using a Pearson correlation of the scores at time 1 with the score at time 2. That is, the coefficient of stability assesses the stability of individual differences (i.e., rank-order stability). The Pearson correlation is called the coefficient of stability when the length of the retest interval (the delay between test and retest) is on the order of days or weeks. If the retest occurs almost at the same time as the original test (e.g., a 45-minute delay), the Pearson correlation is called the coefficient of dependability (Revelle & Condon, 2019).

We estimate the coefficient of stability below:

The r value of [computed value omitted in this course adaptation] indicates a strong positive association. the source discussion depicts a scatterplot of the time 1 scores on the x-axis and time 2 scores on the y-axis.

Considerations About the Correlation Coefficient The correlation coefficient ranges from −1.0 to +1.0. The correlation coefficient (r) tells you two things: (1) the direction of the association (positive or negative) and (2) the magnitude of the association. If the correlation coefficient is positive, the association is positive. If the correlation coefficient is negative, the association is negative. If the association is positive, as X increases, Y increases (or conversely, as X decreases, Y decreases). If the association is negative, as X increases, Y decreases (or conversely, as X decreases, Y increases). The smaller the absolute value of the correlation coefficient (i.e., the closer the r value is to zero), the weaker the association and the flatter the standardized slope of the best-fit line in a scatterplot. The larger the absolute value of the correlation coefficient (i.e., the closer the absolute value of the r value is to one), the stronger the association and the steeper the standardized slope of the best-fit line. However, the correlation coefficient does not reflect the unstandardized slope of the best-fit line. See the source discussion for a range of different correlation coefficients and what some example data may look like for each direction and strength of association.

Keep in mind that the Pearson correlation examines the strength of the linear association between two variables. If the association between two variables is nonlinear, the Pearson correlation provides the strength of the linear trend and may not provide a meaningful index of the strength of the association between the variables.

x1	y1	x2	y2	x3	y3	x4	y4	
10	8.04	10	9.14	10	7.46	8	6.58	
8	6.95	8	8.14	8	6.77	8	5.76	
13	7.58	13	8.74	13	12.74	8	7.71	
9	8.81	9	8.77	9	7.11	8	8.84	
11	8.33	11	9.26	11	7.81	8	8.47	
14	9.96	14	8.10	14	8.84	8	7.04	
6	7.24	6	6.13	6	6.08	8	5.25	
4	4.26	4	3.10	4	5.39	19	12.50	
12	10.84	12	9.13	12	8.15	8	5.56	
7	4.82	7	7.26	7	6.42	8	7.91	
5	5.68	5	4.74	5	5.73	8	6.89	: Anscombe's Quartet.

Property	Value	
Sample size	11	
Mean of X	9.0	
Mean of Y	~7.5	
Variance of X	11.0	
Variance of Y	~4.1	
Equation of regression line	Y = 3 + 0.5X	
Standard error of slope	0.118	
One-sample t-statistic	4.24	
Sum of squares of X	110.0	
Regression sum of squares	27.50	
Residual sum of squares of Y	13.75	
Correlation coefficient	.816	
Coefficient of determination	.67	: Descriptive Statistics of Anscombe's Quartet.

For instance, Anscombe's quartet includes four sets of data that have nearly identical basic descriptive statistics (see Tables (the source discussion) and (the source discussion)), including the same bivariate correlation, yet have very different distributions and whose association takes very different forms (see the source discussion).

For instance, Anscombe's quartet includes four sets of data that have nearly identical basic descriptive statistics (see Tables (the source discussion) and (the source discussion)),

including the same bivariate correlation, yet have very different distributions and whose association takes very different forms (see the source discussion).

As an index of correlation, the coefficient of stability (and dependability) has important weaknesses. It is a form of relative reliability rather than absolute reliability: It examines the consistency of scores across time relative to variation across people. Higher stability coefficients reflect greater stability of individual differences—not greater stability in people's absolute level on the measure. This is a major limitation. the source discussion depicts two example data sets that show strong relative reliability (a strong coefficient of stability) but poor absolute reliability based on inconsistency in people's absolute level across time.

Hypothetical Data Demonstrating Good Relative Reliability Despite Poor Absolute Reliability. The figure depicts two fictional data sets (black and red circles), which both exhibit a similar linear association. The line of best fit is the solid line in the graph and is the same for both data sets, but the black circles sit much closer to the line than the red circles, leading to a much higher coefficient of stability (r =.99 and.84, respectively). However, neither sets of circles are on the line of complete agreement represented by the dashed line in the graph. If the circles fall on the line of complete agreement, it indicates that the measure's scores show consistency in absolute scores across time. Thus, although the measures show strong relative reliability, they show poor absolute reliability. (Figure reprinted from Vaz et al., 2013, Figure 1, p. 3. Vaz, S., Falkmer, T., Passmore, A. E., Parsons, R., & Andreou, P. (2013). The case for using the repeatability coefficient when calculating test–retest reliability. PLoS ONE, 8(9), e73990. https://doi.org/10.1371/journal.pone.0073990)

Another limitation of the coefficient of stability (and dependability) is that it is sensitive to outliers. Additionally, if there are little or no individual differences in scores on a given measure, the coefficient of stability will be low relative to the true reliability because correlation coefficients tend to be attenuated in the presence of a restricted range, and it may not be a useful index of reliability depending on the purpose. For more information see the source discussion on the reliability paradox in the chapter on cognitive assessment. See the source discussion for an example of how a correlation coefficient tends to be attenuated when the range of one or both of the variables has restriction of range.

Here is the correlation with and without restriction of range on x (i.e., x is restricted to values between 55 and 65):

The observed correlation became much weaker due to restriction of range. Thus, when developing measures, it is important to ensure there is adequate variability of scores (i.e., individual differences) and that scores are not truncated due to ceiling or floor effects. Moreover, when interpreting associations with truncated variability, it is important to keep in mind that the true association is likely to be even stronger than what was observed if the measures did not have restricted range.

In sum, the correlation coefficient can be greatly impacted by the extent of variability in the data, differences in the shapes of the two distributions, nonlinearity, outliers, characteristics of the sample, and measurement error (Goodwin & Leech, 2006). These factors tend to have greater impact when the sample size is smaller (compared to when the sample size is larger). The Pearson correlation coefficient tends to be smaller when:

• there is low variability in one or both variables (i.e., restricted range)

• the two distributions show dissimilar shapes (in terms of skewness and kurtosis)

• the underlying association is nonlinear, such that Pearson correlation does not capture the association

• outliers weaken the apparent linear trend

• the association differs across subgroups of the sample (i.e., moderation)

• the data contain random measurement error

By contrast, the association between two variables can be artificially inflated by factors such as:

• outliers that happen to fall along a linear trend

• systematic measurement error (e.g., common method bias)

• third variable confounding

To address these limitations of the coefficient of stability, it is important to consider additional indices of test–retest reliability, such as the coefficient of repeatability and bias that are depicted in Bland-Altman plots, as described later.

Coefficient of Agreement

The intraclass correlation (ICC) can be used to evaluate the extent of absolute agreement of scores within a person across time. ICC ranges from 0–1, with higher scores indicating greater agreement of the scores across time within individuals (relative to the variability in scores across people). ICC reflects the proportion of variance in scores that is attributable to differences between people. That is, it reflects the extent to which scores are consistent across time within a person relative to the variability in scores across people. A higher ICC reflects greater between-person variability relative to within-person variability. ICC was estimated using the psych package (R-psych, n.d.).

There are various kinds of ICC coefficients. To read more about the various types of ICC coefficients, type the following command:

Below, absolute agreement of test–retest reliability was estimated using the irr package:

Coefficient of Repeatability

The coefficient of repeatability (CR), also known as the "smallest real difference," is a form of absolute reliability. That is, it reflects the consistency of scores across time within a person. When using a short time interval between measurements (e.g., two weeks), the coefficient of repeatability is often referred to as the smallest real difference (SRD) because it reflects the smallest difference between two measurements that can be considered a true change in the measure's score (rather than measurement error). For instance, if a person's change score falls within the smallest real difference, the difference between the two measurements could be due to measurement error rather than a true change in the measure's score.

The coefficient of repeatability is calculated as ~1.96 times the standard deviation of the mean differences between the two measurements (d 1 and d 2), as in the source discussion:,

n.d. https://www.medcalc.org/manual/bland-altman-plot.php (archived at https://perma.cc/MEH3-3TGN) (Bland & Altman, 1986; Bland & Altman, 1999). The CR is used for determining the lower and upper limits of agreement because it defines the range within which 95% of differences between the two measurement methods are expected to be. The precise value is [computed value omitted in this course adaptation] (i.e., qnorm(.975)) times the standard deviation of the within-subject differences between the two measurements. This precise value is based on the fact that 95% of the area under a standard normal distribution lies within ± [computed value omitted in this course adaptation] standard deviations of the mean. This is based on the quantile of the 2.5th ([computed value omitted in this course adaptation]) and 97.5th ([computed value omitted in this course adaptation]) percentile of the standard normal distribution (the 95% confidence interval spans the 92.5th percentile to the 97.5th percentile). However, some researchers round [computed value omitted in this course adaptation] to 1.96 or to 2 for simplicity.

The minimum possible coefficient of repeatability is zero, but there is no theoretical limit on the maximum possible coefficient of repeatability. The smaller the coefficient of repeatability (i.e., the closer to zero), the more consistent the scores are across time within a person. The coefficient of repeatability is tied to the units of the measure, so how small is considered "good" absolute reliability depends on the measure. In general, however, given the same units of measurement, a smaller coefficient of repeatability indicates stronger absolute test–retest reliability.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The petersenlab package (Petersen, 2025) contains the petersenlab::repeatability function that estimates the repeatability of a measure's scores and generates a Bland-Altman plot in the source discussion. The coefficient of repeatability is labeled cr in the output.

Bland-Altman Plot The Bland-Altman plot can be useful for visualizing the degree of (in)consistency of people's absolute scores on a measure across two time points (Bland & Altman, 1986; Bland & Altman, 1999).

Example of a Bland-Altman Plot. (Figure reprinted from Vaz et al., 2013, Figure 2, p. 4. Vaz, S., Falkmer, T., Passmore, A. E., Parsons, R., & Andreou, P. (2013). The case for using the repeatability coefficient when calculating test–retest reliability. PLoS ONE, 8(9), e73990. https://doi.org/10.1371/journal.pone.0073990)

In a Bland-Altman plot, the x-axis represents the mean of a person's score across the two time points, and the y-axis represents the within-subject differences across time points. The horizontal line at y = 0 is called the line of identity. Points would fall on the line of identity if people's scores showed perfect consistency in their absolute level across time points (i.e., not relative level as in a cross-time correlation). Another line reflects the bias estimate (i.e., mean difference from the measure across the two points). The bias plus or minus the coefficient of repeatability is the upper or lower limit of agreement (LOA), respectively, as in Equations the source discussion and the source discussion. The top dashed line reflects the upper LOA. The bottom dashed line reflects the lower LOA. The Bland-Altman plot was created using the blandr package (R-blandr, n.d.).

Bias Bias is the degree of systematic measurement error. Bias is also called "mean error" in the context of assessing accuracy against a criterion (see the source discussion). In

a Bland-Altman plot, (test–retest) bias is estimated as the average of the within-subject differences between the two measurements. Estimated (test–retest) bias values that are closer to zero reflect stronger absolute test–retest reliability at the group-level. However, (estimated) bias values can still be close to zero while individuals show large differences because large positive and negative differences could cancel each other out. As a result, it is also important to examine other indices of test–retest reliability, including the coefficient of stability and the coefficient of repeatability.

The bias of [computed value omitted in this course adaptation] indicates that the scores at time 2 are somewhat larger, on average, than the scores at time 1.

95% Limits of Agreement [Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

It is expected that the 95% limits of agreement (LOA) include 95% of differences between the two measurement methods (assuming the scores are normally distributed), based on how the coefficient of repeatability is calculated.

The limits of agreement indicate that 95% of scores at time 2 are expected to fall within [[computed value omitted in this course adaptation], [computed value omitted in this course adaptation]] of scores at time 1.

Inter-Rater Reliability

Inter-rater reliability is the consistency of scores across raters. For instance, two different raters may make a diagnostic decision for the same client, two different people may provide observational ratings of the same participant, or two different researchers may process psychophysiological data (e.g., electroencephalography, electrocardiography) from the same person. Consistency of scores can also be assessed within the same rater, which is known as intra-rater reliability and is described in the source discussion. For assessing inter-rater reliability, intraclass correlation (ICC) is recommended for continuous (interval/ratio) data, such as ratings of a participant's mood. By contrast, Cohen's kappa or Fleiss' kappa is recommended for categorical variables, such as making a diagnostic decision whether to give someone a diagnosis of depression or not. Cohen's weighted kappa or Kendall's coefficient of concordance (Kendall's W) is recommended for ordinal variables.

Continuous Data

Use intraclass correlation (ICC) for estimating inter-rater reliability of continuous (interval/ratio) data. ICC ranges from 0–1, with higher scores indicating greater agreement. Below, ICC was estimated using the psych package (R-psych, n.d.).

Below, ICC was estimated using the irrNA package:

Below, ICC was estimated using the irrICC package:

There are various kinds of ICC coefficients. To read more about the various types of ICC coefficients, see Shrout & Fleiss, 1979 and McGraw & Wong, 1996. You can also type the following commands to learn more:

Categorical Data

For estimating the inter-rater reliability of categorical data, use Cohen's kappa (Bakeman & Goodman, 2020), Fleiss's kappa, the S index (Falotico & Quatto, 2010), or Gwet's AC1 statistic (Gwet, 2008; Gwet, 2021; Gwet, 2021). Cohen's kappa accounts for chance agreement. For instance, if two coders were each to flip a coin 100 times and were to on agree on 50 of the flips (i.e., 50% agreement), that would not be particularly impressive. However, Cohen's kappa only allows two raters. For more than two raters, Fleiss's kappa, the S index, or Gwet's AC1 statistic is recommended.

Cohen's kappa (and weighted kappa) were estimated using the psych package (R-psych, n.d.). Fleiss's kappa and Gwet's AC1 statistic were estimated using the irrCAC package. These estimates of inter-rater reliability range from −1 to +1, with −1 indicating perfect disagreement, 0 indicating chance agreement, and 1 indicating perfect agreement. Larger, more positive scores indicate greater agreement.

Ordinal Data

For estimating the inter-rater reliability of ordinal data, use Cohen's weighted kappa (Cohen, 1968) or Kendall's W. Weighted kappa does not treat all errors as equal; when dealing with ordinal data, weighted kappa weights errors according to the distance of the error. For instance, if Coder 1 rated as person as 0 (no diagnosis) and Coder 2 rated a person as 2 (diagnosis), this would be a larger difference than if Coder 1 rated a person as 1 (subthreshold) and Coder 2 rated a person as 2 (diagnosis). However, Cohen's weighted kappa only allows two raters. For more than two raters, Kendall's W is recommended.

Cohen's kappa and weighted kappa were estimated using the psych package (R-psych, n.d.). Kendall's W was estimated using the irrNA package. The values range from −1 to +1, with −1 indicating perfect disagreement, 0 indicating chance agreement, and 1 indicating perfect agreement. Larger, more positive scores indicate greater agreement.

Intra-Rater Reliability

Principles of intra-rater reliability are similar to principles of inter-rater reliability (described in the source discussion), except the same rater provides ratings on two occasions, rather than using multiple raters.

Parallel- (or Alternate-) Forms Reliability

Parallel-forms reliability (also known as alternate-forms reliability) is the consistency of scores across two parallel forms. Parallel forms are two equivalent measures of the same construct that differ in content or format. The two measures can be administered in the same occasion, known as immediate parallel-forms reliability, or they can be administered separated by a delay, known as delayed parallel-forms reliability. Similar considerations of length of delay duration and ways to reduce carryover effects are relevant to parallel-forms reliability as they are to test–retest reliability.

Parallel-forms reliability is often estimated with the Pearson correlation, which is known as the coefficient of equivalence. The coefficient of equivalence is the consistency of scores across two parallel forms relative to variation across people. It is interpreted as the ratio of true score variance to observed score variance, consistent with the formal definition of reliability.

An example of parallel-forms reliability would be if we want to assess a participant twice with different measures of the same construct to avoid practice effects. This approach is often conducted in academic achievement testing, and it could involve retesting that is immediate or delayed. For instance, we could administer item set A at time 1 and item set B at time 2 for the same participant, to get rid of any possibility that the person's score differed across time because of improved performance that resulted merely from prior exposure to the same items. Many standardized academic achievement and aptitude tests have developed parallel forms, including the SAT, ACT, and GRE.

Parallel-forms reliability controls for specific error, i.e., error that is particular to a specific measure. However, parallel-forms reliability has the limitation that it assumes that the parallel forms are equivalent, such that it makes no difference which test you use. Two forms are considered equivalent when they assess the same construct, have the same mean and variance of scores, and have the same inter-correlations with external criteria. Technically, two instruments are only truly equivalent if they have the same true scores and variability of error scores, though this is difficult to establish empirically. Given these constraints, it can be difficult to assume that different forms are equivalent. Nevertheless, integrative data analysis and measurement harmonization may help make the assumption more tenable (Hussong et al., 2020; Hussong et al., 2013). In general, parallel forms can be difficult to create and they can more than double the time it takes to develop a single measure, often with limited benefit, depending on the intended use.

Internal Consistency Reliability

Internal consistency reliability is the consistency of scores across items, that is, their interrelatedness. Internal consistency is necessary, but insufficient, for establishing the homogeneity or unidimensionality of the measure, i.e., that the items on the measure assess the same construct.

There are many examples of internal consistency estimates, including Cronbach's coefficient alpha (alpha), Kuder-Richardson Formula 20 (which is a special case of Cronbach's alpha with dichotomous items), omega coefficients, average variance extracted, greatest lower bound, maximal reliability (coefficient H), split-half reliability, average inter-item correlation, and average item–total correlation. In general, internal consistency ranges from 0–1, with higher scores indicating greater consistency. There is no magic cutoff for determining adequate internal consistency; however, values below.70 suggest that the items may not assess the same construct.

Internal consistency can be affected by many factors, such as

• wording of the item

• a participant's comprehension of the item, which could be impacted if using items with dated language or double negatives

• interpretation

• attention of the participant

• compliance of the participant

Although establishing internal consistency can help ensure that items in the measure are interrelated, you do not want to select items just based on internal consistency (Revelle,

2024). If you did, you would end up with items that are too similar. For example, you could end up only with items that have similar item stems and are coded in the same direction. Therefore, selecting items solely based on internal consistency would likely not assess the breadth of the construct and would reflect strong method variance of the wording of the particular items. In other words, internal consistency can be too high. Internal consistency is helpful to a point, but you do not want items to be too homogeneous or redundant. For example, based on arguments by Cattell (Cattell, 1973), Cooper (Cooper, 2023) provides a conceptual example of how items could show zero correlation between them (and thus not be unidimensional or show internal consistency) but still be useful for assessing a construct like depression—because the scale should assess all of the various facets of the construct, which could lead an item that assesses a given facet (e.g., sad mood) to be only weakly associated with items that assess other facets (e.g., weight/appetite changes). Researchers have thus argued that the field has focused too much on internal consistency to the exclusion of other important psychometric considerations such as utility and predictive validity (Revelle, 2024).

Average Inter-Item Correlation

One estimate of internal consistency is the average (mean or median) inter-item correlation.

Here is the mean inter-item correlation for each item:

Here is the mean inter-item correlation across all items:

the source discussion is a histogram of the inter-item correlations.

Here is the median inter-item correlation for each item:

Here is the median inter-item correlation across all items:

Average Item–Total Correlation

Another estimate of internal consistency is the average item–total correlation.

When examining the item–total correlation for a given item, it is preferable to exclude the item of interest from the total score, so that the item is associated with the total score that is calculated from the remaining items. This reduces the extent to which the association reflects that the item is associated with itself.

Average Variance Extracted

The equivalent of the average item–total (squared) correlation when dealing with a reflective latent construct is the average variance extracted (AVE). AVE is estimated as the mean of the squared loadings of the indicators of a latent factor (i.e., the sum of the squared loadings divided by the number of indicators). When using unstandardized estimates, AVE is the item-variance weighted average of item reliabilities (Rönkkö & Cho, 2020). When using standardized estimates, AVE is the average indicator reliability (Rönkkö & Cho, 2020). AVE is equivalent to the communality of a latent factor. AVE was estimated using a confirmatory factor analysis model using the lavaan (R-lavaan, n.d.) and semTools (R-semTools, n.d.) packages.

Maximal Reliability

Maximal reliability is a model-based approach to estimating internal consistency. Maximal reliability is the highest possible internal consistency reliability estimate that can be achieved from a set of items by applying optimal weights for each item (i.e., an optimal linear combination) rather than equal weights for each item. Coefficient H is a common index of maximal reliability.

Maximal reliability was estimated using a confirmatory factor analysis model using the lavaan (R-lavaan, n.d.) and semTools (R-semTools, n.d.) packages. The confirmatory factor analysis model was fit above in the source discussion.

Split-Half Reliability

When estimating split-half reliability, we randomly take half of the items on the measure and relate scores on these items to the score on the other half of items. One challenge to split-half reliability is that, according to classical test theory, shorter tests tend to be less reliable than longer tests, so this would be an underestimate of the test's reliability. In general, the greater the number of items, the more reliable the test. To overcome this, the Spearman-Brown prediction (or prophecy) formula predicts the reliability of a test after changing the test length. Thus, the Spearman-Brown prediction formula accounts for the shorter test in predicting what the reliability of the full-length test would be when estimating split-half reliability.

The Spearman-Brown prediction formula is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where n = the number of tests combined (i.e., the ratio of the number of new items to the number of old items) and r xx' = the reliability of the original test.

So, according to the source discussion, doubling the length of a test whose reliability is.80 would be estimated to have a reliability of

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Another challenge to split-half reliability is that the reliability differs depending on which items go into which half.

One option is to calculate the split-half reliability of odd and even trials, while correcting for test length, as is performed with the item_split_half function of the performance package (Lüdecke et al., 2021):

Modern computational software allows for estimating the mean reliability estimate of all possible split-halves (Revelle & Condon, 2019), while correcting for test length, as seen below. Split-half reliability was estimated using the psych package (R-psych, n.d.).

Cronbach's Alpha (Coefficient Alpha)

Cronbach's alpha (alpha), also known as coefficient alpha, is approximately equal to the mean reliability estimate of all split-halves, assuming that item standard deviations are equal. Alpha considers the interrelatedness of a total set of items—giving us information about the extent to which each item in a set of items correlates with all other items. Alpha is the most commonly used estimate for internal consistency reliability, but it has many problems.

Cronbach's alpha assumes that the items are

• unidimensional: that there is one predominant dimension that explains the variance among the items

• essentially tau equivalent: that the items are equally related to the construct

Cronbach's alpha is affected by the

• number of items: The more items, the more inflated alpha is, even if correlations among the items are low.

• variance of the item scores: If the variances of item scores are low (e.g., dichotomous scores), alphas tend to be an underestimate of internal consistency reliability.

• violations of assumptions:

• Cronbach's alpha assumes items are essentially tau equivalent—that the items are equally related to the construct.

However, this is an unrealistic assumption.

• Cronbach's alpha depends on the dimensionality of the scale: Alpha is only valid when the scale is unidimensional.

If you use it with multidimensional scales, you know the extent to which items correlate with another but nothing about whether they are actually measuring the same constructs.

Cronbach's alpha is appropriate only in situations where a scale is unidimensional, where the items equally contribute to the scale, and you would like to determine whether its items are interrelated. However, Cronbach's alpha does not assess the dimensionality of the scale or any type of construct validity—it can only tell us how strongly the items hang together.

Cronbach's alpha (alpha) was estimated using the psych package (R-psych, n.d.).

Below is an estimate of standardized alpha (if the items are standardized before computing the scale score).

Below are item statistics:

However, because of the many weaknesses of Cronbach's alpha (Cortina, 1993; Dunn et al., 2014; Flora, 2020; Green & Yang, 2015; Hayes & Coutts, 2020; Kelley & Pornprasertmanit, 2016; McNeish, 2018; Peters, 2014; Raykov, 2001; Revelle & Condon, 2019; Sijtsma, 2008; Raykov & Marcoulides, 2019), current recommendations are to use coefficient omega instead, as described next.

Coefficient Omega

Current recommendations are to use McDonald's coefficient omega instead of Cronbach's alpha (alpha) for internal consistency reliability. Specifically, it is recommended to use omega total (t) for continuous items that are unidimensional, omega hierarchical (h) for continuous items that are multidimensional, and omega categorical (C) for categorical items (i.e., items with fewer than five ordinal categories), with confidence intervals calculated from a bias-corrected bootstrap (Flora, 2020; Kelley & Pornprasertmanit, 2016). In addition to estimating coefficient omega in the ways described below, omega can also be estimated in latent factor models, and is sometimes called composite reliability (see the source discussion for an example).

Omega Total (t)

Omega total (t) (or simply omega) is the proportion of the total variance that is accounted for by the common factor (Flora, 2020). Omega total, like Cronbach's alpha, assumes that the items are unidimensional. Omega total (t) was calculated using the MBESS package (R-MBESS, n.d.). This code quickly calculates a point estimate for omega total:

If you want to calculate a confidence interval for omega, you will need to specify an interval.type and specify how many bootstrap replications.

The following code calculates a confidence interval from a bias-corrected bootstrap. For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed. **Warning**: This code takes a while to run based on [computed value omitted in this course adaptation] bootstrap replications. You can reduce the number of replications to be faster. However, the number of bootstrap replications must be larger than the number of rows in the data.

Omega Hierarchical (h)

If the items are multidimensional and follow a bifactor structure, omega hierarchical (h) is preferred over omega total. Omega hierarchical is the proportion of the total variance that is accounted for by the general factor in a set of multidimensional items (Flora, 2020). Use omega hierarchical for estimating internal consistency of continuous items that are multidimensional and that follow a bifactor structure. If the multidimensional items follow a higher-order structure, use omega higher-order (Flora, 2020).

Omega hierarchical was calculated using the MBESS package (R-MBESS, n.d.).

The following code calculates a confidence interval from a bias-corrected bootstrap. For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed. **Warning**: This code takes a while to run based on [computed value omitted in this course adaptation] bootstrap replications. You can reduce the number of replications to be faster. However, the number of bootstrap replications must be larger than the number of rows in the data.

Omega Categorical (C)

Use omega categorical (C) for categorical items that are unidimensional. To handle categorical data, omega categorical is calculated using a polychoric correlation matrix instead of a Pearson correlation matrix. If the categorical items are multidimensional items and they follow a bifactor structure, use omega hierarchical categorical (Flora, 2020). If the

categorical items are multidimensional and they follow a higher-order structure, use omega higher-order categorical (Flora, 2020).

Omega categorical was calculated using the MBESS package (R-MBESS, n.d.).

The following code calculates a confidence interval from a bias-corrected bootstrap. For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed. **Warning**: This code takes a while to run based on [computed value omitted in this course adaptation] bootstrap replications. You can reduce the number of replications to be faster. However, the number of bootstrap replications must be larger than the number of rows in the data.

Comparison

Even though the types of reliability described in the source discussion are all called "reliability," they are very different from each other. Test–retest reliability and inter-rater reliability tend to be lower than immediate parallel-forms reliability and internal consistency reliability because they involve administering measures at different times or with different raters. However, delayed parallel-forms reliability tends to be lower than test–retest reliability because it involves assessing the construct at a different time point and with different items. Thus, if we put the different types of reliability in order from low to high, they tend to be (but not always) arranged like this (the source discussion):

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Reliability of a Linear Composite Score

A linear composite score is a linear combination of variables, each of which has a given weight in the linear composite. Unit weights (i.e., weights of 1) would reflect an unweighted composite.

We first specify the variables used to generate the correlation matrix, and the reliability, standard deviation, and weight of each variable. We specify items 2 and 3 to have twice the weight as item 1 in the composite, and item 4 to have 1.5 times the weight as item 1 in the composite.

Here is the reliability of an unweighted composite:

Here is the reliability of a weighted composite:

Reliability of a Difference Score

Difference scores tend to be lower in reliability than the individual indices that compose the difference score. This is especially the case when the two indices are correlated. If the correlation between the indices is equal to the reliability of the indices, the reliability of the difference score will be zero (Revelle & Condon, 2019). If the correlation between the two indices is large, it requires a very high reliability for the difference score to show a strong reliability (Trafimow, 2015). To the extent that the two indices have unequal variances (e.g., if one index has a standard deviation that is four times the standard deviation of the second index), the reliabilities do not need to be quite as high for the difference score to be reliable (Trafimow, 2015).

The reliability of a difference score can be estimated using the source discussion (Trafimow, 2015):

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The petersenlab package (Petersen, 2025) contains the petersenlab::reliabilityOfDifferenceScore function to estimate the reliability of a difference score:

For instance, in this case, the reliability of the difference score is.26, even though the reliability of both indices is.95.

In Figures (the source discussion) and (the source discussion), the reliability of a difference score is depicted as a function of (a) the reliability of the indices that compose the difference score and (b) the correlation between them, assuming that both indices have equal variances. As depicted, a variable would not be expected to be more strongly associated with other variables (i.e., the correlation between the indices) than it is with itself (i.e., its reliability).

Reliability of a Product Term

As with difference scores, product terms tend to have lower reliability than the individual indices that compose them (Aguinis et al., 2017; Busemeyer & Jones, 1983; Jaccard & Wan, 1995; McClelland & Judd, 1993). Product terms are widely used to test for statistical interaction or moderation effects. However, measurement error multiplies in product terms, often leading to lower reliability of the product term than the reliabilities of the individual indices that compose the product term. Similar challenges arise with ratio scores (i.e., scores that represent a ratio of two variables) because both variables included in the ratio calculation include error (Cronbach, 1941; Arndt et al., 1991).

The reliability of a product term from mean-centered variables can be estimated using the source discussion (Busemeyer & Jones, 1983; Aguinis et al., 2017):

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The reliability of a product term can also be used to find the reliability of ratio scores (e.g., (A)/(B)) because a ratio can be conceived of as a product term where one variable (e.g., A) is multiplied by the reciprocal of another variable (e.g., (1)/(B)).

If the two variables (x and y) are uncorrelated, the reliability of the product term simplifies to the product of their reliabilities, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The petersenlab package (Petersen, 2025) contains the petersenlab::reliabilityOfProductTerm function to estimate the reliability of a product term:

Power to detect interactions tends to be weaker than power to detect main effects because of lower reliability of the product term, smaller effect sizes (archived at https://perma.cc/E3KA-SGB7), and smaller sample sizes of the intersecting groups.

Options for increasing the reliability of interaction terms are presented by Hsiao et al., 2018, Ng & Chan, 2020, and Jaccard & Wan, 1995.

Applied Examples

Although reliability is an important consideration for measures, stronger reliability is not always better. Reliability is not one thing; it is not monolithic. There are multiple aspects of reliability, and which aspect(s) of reliability are most important depend on the construct of interest. That is, we want different kinds of reliability for measures of different constructs. In general, we want our measures to show strong inter-rater, intra-rater, immediate parallel-forms, and internal consistency reliability. However, we do not want internal consistency reliability to be too high so as to have redundant items that are correlated for reasons of shared method biases. Moreover, for some constructs (i.e., formative constructs such as stressful life events), we would not expect internal consistency reliability. And the extent to which our measures should show test–retest and delayed parallel-forms reliability depends on the stability of the construct and the time lag.

Consider the construct of reactive dysphoria. Assume that we assess people's reactive dysphoria with observations and several questionnaires today and one month from now. However, test–retest reliability is not meaningful because the phenomenon is not meant to be stable across time—it is supposed to be reactive to the situation, not a trait. Instead, we would expect that a strong measure of reactive dysphoria has high internal consistency reliability, a high coefficient of equivalence from immediate parallel-forms reliability, and high inter-/intra-rater reliability.

Consider the example of personality and intelligence. Personality is defined such that personality "traits" are thought to be stable across time and situation. Likewise, individual differences in intelligence are thought to be relatively stable in terms of individual differences across time (not in absolute level). Thus, strong measures of personality and intelligence should show high test–retest reliability, parallel-forms reliability, internal consistency reliability, and inter-/intra-rater reliability.

Just because one aspect of reliability is strong does not necessarily mean that other aspects of reliability will be strong. Measures of some constructs (e.g., depression) could be expected to show low test–retest reliability but high internal consistency reliability. You can also have measures that have low internal consistency reliability but that have high test–retest reliability (e.g., if our nonsensical measure averaged together z-scores of height and blood type).

Consider a questionnaire assessing people's extraversion. We would expect high stability of construct, and therefore a high coefficient of stability, but test and retest scores might not be equal. That is, the measures may show a somewhat lower repeatability coefficient because of random error—on a given day, the participant may be sick, stressed, or tired; they may have recently gotten into an argument; or they may be impacted by the season, the weather, or differences in the context (e.g., lab or online), just to name a few potential sources of random error.

Standard Error of Measurement

We presented and discussed the formula for estimating the standard error of measurement earlier in the source discussion. It is also below:

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where sigma x = standard deviation of scores on measure x and r xx = reliability of measure x

The reliability of the measure was estimated here based on the test–retest reliability of the measure at T1 with the measure at T2.

We can use the standard error of measurement to estimate the confidence interval around a person's score. A confidence interval indicates a range of plausible values for the person's true score (with a specified confidence level, e.g., 95%), given measurement error. The 95% confidence interval represents that, if we repeatedly observed this person under identical conditions and constructed a confidence interval each time (as observed score ± [computed value omitted in this course adaptation] \ SEMs), about 95% of those intervals would contain the person's true score.

The 95% CI is [[computed value omitted in this course adaptation], [computed value omitted in this course adaptation]].

Influences of Measurement Error on Test–Retest Reliability

This section describes the influence of measurement error on reliability, specifically test–retest reliability. There are different types of measurement error, and each can have different impacts on reliability and, ultimately, your findings. As discussed in the source discussion, a given error can be either systematic or random. In addition, error can be within-person, between-person, or both. However, error scores can include multiple errors simultaneously.

No Error

This section provides the coefficients of stability, repeatability, and bias that are used as a comparison when measurement error is added. For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Add within-person random error

In the example below, we added within-person random error. Specifically, we added error that is randomly distributed around a mean of 0 and a standard deviation of 3 to all participants' scores at all time points.

For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Findings

Adding within-person random error led to

- a weaker (lower) stability coefficient
- a weaker (higher) repeatability coefficient

Add within-person systematic error

In the example below, we added within-person systematic error. Specifically, to all participants, we added error that is consistent across time for a given participant: pulled from a distribution with a mean of 5 and a standard deviation of 3.

For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Findings

Adding within-person systematic error led to

• an artificially stronger (higher) stability coefficient

• no effect on the repeatability coefficient

• bias at the person level, but this bias would go undetected by the estimate of group-level bias in the Bland-Altman plot

Add between-person random error at T2

In the example below, we added between-person random error. Specifically, we added error that is randomly distributed around a mean of 0 and a standard deviation of 3 to all participants' scores at time 2 (see below).

For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Findings

Adding between-person random error led to

• a weaker (lower) stability coefficient

• a weaker (higher) repeatability coefficient

Add between-person systematic error at T2

In the example below, we added between-person systematic error. Specifically, we added constant variance of 10 to all participants' scores at T2.

For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Findings

Adding between-person systematic error led to

• no effect on the stability coefficient

• no effect on the repeatability coefficient

• bias at the person level and at the group level (detected by group-level estimate of bias in Bland-Altman plot)

Add constant variance of 10 to all participants at all time points

In the example below, we added constant variance to all participants' scores at all time points. Specifically, we added 10 to all participants' scores at all time points.

For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Findings

Adding constant variance to all participants' scores at all time points led to

• no effect on the stability coefficient

• no effect on the repeatability coefficient

• bias at the person level and at the group level, but this bias would go undetected by the estimate of group-level bias in the Bland-Altman plot

Summary

To summarize the effects of measurement error, random error (both within- and between-person) results in (a) a worse (lower) stability coefficient because a smaller proportion of the observed score variance is attributable to true score variance, and (b) a worse (higher) repeatability coefficient because the standard deviation of the difference between the two measurements is greater. Neither type of random error leads to bias at the person or group level.

In terms of systematic error, all types of systematic error lead to increased bias, i.e., reduced accuracy for the person-level average and/or group-level average. Though multiple biases can coexist and can work in opposite directions, which can obscure bias even though it exists. In terms of within-person systematic error, it (a) artificially increases the stability coefficient, even though the measure is not actually more reliable, (b) has no effect on the repeatability coefficient because it is not changing the difference between the two measurements for a person, and (c) results in bias at the person level because the person's mean systematically differs from their true score.

In terms of between-person systematic error at time 2 (an example of adding constant variance to all participants), it (a) had no effect on the stability coefficient because the rank order of individual differences stays the same, (b) had no effect on the repeatability coefficient because the standard deviation of the difference between the two measurements is not changed, (c) resulted in bias at the person level and at the group level that would be detected in a Bland-Altman plot because the group-level average deviates from the group's average true score.

In sum, just because a measure has high estimates of test–retest reliability does not mean that the measure is actually reliable because systematic error does not reduce the stability and repeatability coefficients. Indeed, within-person systematic error actually (and artificially) increases the coefficient of stability. The coefficient of stability index of test–retest reliability is affected by the measure and variance across people, and it can be thrown off by outliers or by range restriction because the Pearson correlation strongly depends on these, as discussed in the source discussion. And sample variation also has a strong influence, so it is important to cross-validate estimates of reliability in independent samples.

Effect of Measurement Error on Associations

As we describe in greater detail in the chapter on validity in the source discussion, (random) measurement error weakens (or attenuates) the association between variables. The greater the random measurement error, the weaker the association.

Attenuation of True Test–Retest Reliability Due to Measurement Error

In the context of examining the association between test and retest (i.e., test–retest reliability), the estimate of the stability of the construct (i.e., true test–retest association) is attenuated to the extent that the measure is unreliable, as described in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Find the observed test–retest association if the true test–retest association (i.e., the stability of the construct) is.7 and the reliability of the measure is.9:

The petersenlab package (Petersen, 2025) contains the petersenlab::attenuationCorrelation function that estimates the observed association given the true association and the reliability of the predictor and criterion. When dealing with test–retest reliability, the predictor and criterion are the same, so we use the same reliability estimate for each:

Disattenuation of Observed Test–Retest Reliability Due to Measurement Error

Then, to disattenuate the observed test–retest reliability due to random measurement error, we can rearrange the terms in the formula above to estimate what the true stability of the construct is, according to the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Find the true correlation of the construct across two time points given an observed correlation if the observed reliability of the measure is.9:

The petersenlab package (Petersen, 2025) contains the petersenlab::disattenuationCorrelation function that estimates the observed association given the true association and the reliability of the predictor and criterion. When dealing with test–retest reliability, the predictor and criterion are the same, so we use the same reliability estimate for each:

The next section discusses method biases, including what they are, why they are important, and how to account for them.

What Method Biases Are and Why They Are Important

Method bias is the influence of measurement on a person's score that is not due to the person's level on the construct. Method bias is a form of systematic, nonrandom measurement error because it involves errors/biases that are shared/common to/correlated for two measures.

Method bias is problematic because it can bias estimates of a measure's reliability and validity, and its association with other measures/constructs. There is a biasing effect of assessing two or more constructs with the same method on estimates of the association

between them—that is, some of the observed covariance may be due to the fact that they share the same method of measurement.

Although estimates vary widely from study to study, construct to construct, and measure to measure, estimates of method bias tend to be substantial. Meta-analyses suggest that around 20–40% of a given measure's variance reflects method variance (Podsakoff et al., 2012). According to Podsakoff et al., 2012, response styles inflate the magnitude of observed correlations by 54% if there is a positive correlation between the constructs and a positive correlation between the response styles. Response styles inflate the magnitude of observed correlations by even more (67%) if there is a negative correlation between the constructs and a negative correlation between the response styles. Moreover, the method biases can also weaken the association between measures. If the direction of correlation between the constructs (e.g., positive) is incongruent with the direction of the correlation between the response styles (e.g., negative), the association between the measures is attenuated.

Types of Method Biases

There are many different types of method biases, including the effects of

• same source: Whether ratings or information comes from the same source can lead to systematic method variance.

• response styles (response biases): Response styles are systematic ways of responding to items or questions that are not due to the construct of interest.

Different types of response styles can lead to systematic method variance across constructs. Examples of response styles include

• acquiescence: the tendency to agree with items (both positively and negatively worded), regardless of content

• disacquiescence: the tendency to disagree with items (both positively and negatively worded), regardless of content

• extreme: the tendency to endorse items on the extremes (positive or negative) of the response scale, regardless of content

• midpoint/central tendency: the tendency to endorse items on the middle scale category, regardless of content

• noncontingent: the tendency to respond to items carelessly, randomly, or nonpurposefully

• social desirability: the tendency to respond to items in a way that others will believe the respondent is better than they actually are.

Social desirability includes multiple components, including self-deception and impression management. Impression management is the conscious effort to control the way people view oneself or the way one presents oneself.

• proximity: Items (of unrelated constructs) that are in closer proximity to one another in a measure are more likely to be correlated than items that are further apart.

The associations of items assessing unrelated constructs are weakest when the pair of items are six or more items apart from each other. There is an exception for reversed items. For

reversed items (e.g., reverse scored), the further apart the reversed item is placed from another item, the higher the negative correlation.

• item wording: Whether items are positively or negatively worded can influence the association between items (i.e., like-worded items go with like).

• item context: Manipulating the context of a measure can have an effect on the perceived association.

You can observe a different association between constructs when you put the measure in a negative context versus in a positive context.

• item ambiguity: When respondents are less certain of how to accurately answer a question, it increases the likelihood that they will rely on response styles or succumb to the effect of context when responding, which biases scores.

• motivational factors: Researchers need to motivate participants to put in the cognitive effort to answer questions correctly, to decrease the likelihood that they will succumb to responding based on response style only.

Also, researchers should decrease pressures for "optimal responding" on respondents. That is, researchers should attempt to reduce social desirability effects.

• the ability of the respondent: Respondents with low cognitive abilities, less education, more fatigue (after lots of questionnaires), less experience with the topic being assessed, and more uncertainty about how to respond are more likely to succumb to biasing factors.

Potential Remedies

There are numerous potential remedies to help reduce the effects of method bias, including

• obtain measures of the predictor and criterion from multiple or different sources

• create temporal, physical, and/or psychological separation between the predictor and criterion

• eliminate common scale properties; vary the scale types and anchor labels

• remove ambiguity in items—keep questions simple, (behaviorally) specific, and concise

• reduce social desirability in item wording

• provide anonymity to respondents

• for estimating the prevalence of a sensitive behavior, use the randomized response model (Clark & Desharnais, 1998), as described below

• provide incentives (e.g., compensation) to respondents

• reverse score some items to balance positively and negatively worded (i.e., reverse-scored) items

• separate items on the questionnaire to eliminate proximity effects

• maximize respondent motivation to give an accurate response

• develop a good cover story and clear instructions, with reminders as necessary

• match the measure difficulty to the respondent's ability; minimize frustration

• use confirmatory factor analysis or structural equation modeling, as described in the source discussion

• use mixed models, as described below

Randomized Response Model

The randomized response model (Clark & Desharnais, 1998) is a technique to increase people's honesty in responding, especially around sensitive topics. The randomized response model sought to answer the question of how much information people withhold on surveys. It is no surprise that people lie on surveys, especially if the survey asks questions about sensitive topics, such as suicidality, criminal behavior (e.g., drug use), and child abuse or neglect. The randomized response model is based on the premise that respondents would give more honest answers if they could be certain that their survey answers would not reveal any sensitive information about themselves, and it is designed to reduce evasive-answer bias by guaranteeing privacy.

The randomized response model works by exploiting statistical probabilities. Consider a survey that asks respondents whether they have used various illegal drugs (e.g., "Have you used heroin?" or "Have you used cocaine?"). Each participant is given a coin. They are told to flip the coin before they answer each question. If the coin flip is "heads," they answer the question truthfully. If the coin flip is "tails," they answer the questions with a "yes" (i.e., the more sensitive answer) no matter what. That way, a yes could be due to the coin flip or could be the truth—so respondents feel as if they are divulging less.

Then, researchers can estimate the actual "no" rate by doubling the observed "no" rate for the sample. For instance, if 20% of people say they have never gambled and 80% say they have gambled, the actual "no" rate of gambling is 40%. That is, the actual gambling rate is 60%.

Findings using the randomized response model indicate that there are a lot of things that people are not sharing. The randomized response model is useful for estimating the true prevalence rate of sensitive behaviors. However, it is not great for individual differences research because you do not know whether any given individual truly engages in a behavior (because a "yes" response may have occurred merely because the coin flip was "heads"). Thus, the randomized response model is only helpful for knowing "true" overall rate of a given behavior in the sample.

Mixed Models

One approach to handle method biases is in mixed models, which are also called mixed-effects models, multilevel models, and hierarchical linear models (HLMs). Mixed models are similar to multiple regression in that they allow fitting a model that examines multiple predictor variables in relation to one outcome variable. One of the assumptions of multiple regression is that the data observations are independent from each other, as evidenced by the residuals being uncorrelated with each other. Mixed models are designed to handle nonindependent observations, that is, data that were collected from nonindependent units. Thus, mixed models can be fit to data that do not meet the nonindependence assumption of multiple regression.

Examples of data from nonindependent units include nested data, multiple membership data, and cross-classified data, as depicted in the source discussion.

Various Data Structures Involving Data from Nonindependent Units, Including Nested data, Multiple Membership Data, and Cross-Classified Data.

One example of nonindependent data is if your data are nested. When data are collected from multiple lower-level units (e.g., people) in an upper-level unit (e.g., married couple, family, classroom, school, neighborhood), and each lower-level unit belongs to one and only one higher-level unit, the data from the lower-level units are considered nested within the upper-level unit. For example, multiple participants in your sample may come from the same classroom. Data from multiple people can be nested within the same family, classroom, school, etc. Longitudinal data can also be considered nested data, in which time points are nested within the participant (i.e., the same participant provides an observation across multiple time points). And if you have multiple informants of a child's behavior (e.g., parent-, teacher-, friend-, and self-report), the ratings could also be nested within the participant (i.e., there are ratings from multiple informants of the same child).

Another form of nonindependent data is when data involve multiple membership. Data involve multiple membership when the lower-level units belong to more than one upper-level unit. As an example of multiple membership, children may have more than one teacher, and therefore, in a modeling sense, children "belong" to more than one teacher cluster.

Another form of nonindependent data is when data are cross-classified (also called crossed). Data are cross-classified when the lower-level units are classified by two or more upper-level units, but the upper-level units are not hierarchical or nested within one another. For instance, children may be nested within the crossing of schools and neighborhoods. That is, children are nested within schools, and children are also nested within neighborhoods; however, children attending the same school may not necessarily be from the same neighborhood. The data would still have a two-level hierarchy, but the data would be nested in specific school-neighborhood combinations. That is, children are nested within the cross-classification of schools and neighborhoods.

The general form of multiple regression is (see the source discussion):

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where b 0 is the intercept, e is the error term, and b 1 is the slope of the predictor, x, that states the relation between the predictor (x) and the outcome (y). However, the slope can exist at multiple levels when data are nested within groups or when time points are nested within people. For example, you can observe different associations of a predictor with an outcome at the between-individual level compared to the within-individual level. That is, the association between the predictor and outcome can differ when comparing across individuals versus when looking at the association between the predictor and outcome based on fluctuations within the individual. When the associations differ at multiple levels, this is known as Simpson's paradox. For an example of Simpson's paradox, see the source discussion (Kievit et al., 2013). Thus, it is important to consider that the association between variables may differ at different levels (person level, group level, population level, etc.).

Example of Simpson's paradox, where the association between the predictor and outcome differs at different levels. In this case, the association between dosage and recovery probability is positive at the population-level (upper-level unit), but the association is negative among men and women separately (lower-level unit). (Figure reprinted from Kievit

et al., 2013, Figure 1, p. 2. Kievit, R., Frankenhuis, W., Waldorp, L., & Borsboom, D. (2013). Simpson's paradox in psychological science: a practical guide. Frontiers in Psychology, 4(513). https://doi.org/10.3389/fpsyg.2013.00513)

To account for method biases in a mixed model, you could use scores from multiple measures and methods as the outcome variable. For instance, your mixed model could examine the association between income and optimism, where optimism is assessed by self-report, parent-report, and a performance-based measure. In the model, you could include, as a fixed effect, a factor that indicates the method of the measure, in addition to its interaction with income, to account for systematic ways in which the measurement method influences scores on the measures across the participants as a whole. You could also include a random effect of measurement method if you wanted to allow the effect of measurement method on the scores to differ from person to person.

[INCLUDE EXAMPLE]

Generalizability Theory (G-Theory)

Up to this point, we have discussed reliability from the perspective of CTT. However, as we discussed, CTT makes several assumptions that are unrealistic (e.g., that all error is random). There are other measurement theories that conceptualize reliability differently than the way that CTT conceptualizes reliability. One such measurement theory is generalizability theory (Brennan, 1992), also known as G-theory and domain sampling theory. G-theory is described in detail in the source discussion on generalizability theory and is also discussed in the chapters on validity (the source discussion, the source discussion), and structural equation modeling (the source discussion, the source discussion).

Item Response Theory

In addition to CTT and G-theory, item response theory (IRT) is another measurement theory. IRT also estimates reliability in a different way compared to CTT, as described in the source discussion. We discuss IRT in the source discussion.

The Problem of Low Reliability

Using measures that have low reliability can lead to several problems, including

• over- or under-estimation of the effects of interest.

When sample sizes are large, (random) measurement error tends to weaken effect sizes. However, when sample sizes are small to modest, random measurement error can actually lead to over-estimation of effect sizes as well as inaccurate estimates of the sign of the effect (positive or negative), due to the statistical significance filter (Loken & Gelman, 2017), as described in the source discussion. Within-person systematic error (e.g., common method bias) can also lead to over-estimation of effect sizes.

• reduced statistical power to detect the effects of interest

• failure to replicate findings

• invalid measurement for the proposed uses.

As we described in the chapter on validity in the source discussion, reliability constrains validity.

These concerns are especially important for individual differences research (e.g., correlational designs) or when making decisions about individual people based on their scores on measure. However, these concerns are also relevant for experimental designs.

Ways to Increase Reliability

Here are several potential ways to increase reliability of measurement:

1. Improve the clarity and homogeneity of items.

2. Use a structured response set and scoring system rather than unstructured ones.

3. Make sure participants know how to complete the task or items.

For example, make sure they have appropriate instructions that are easily understood.

1. Adhere closely to the standardized procedures for administering and scoring a measure.

2. Make scoring rules as explicit as possible.

3. Train raters to some criterion of inter-rater reliability.

Continue to monitor inter-rater reliability over time to reduce coder drift.

1. Use a more precise response scale.

The smaller the number of response options, the greater the attenuation of a correlation due to measurement error (Rigdon, 2010).

1. Have a sample with a full range of variation in the items and measure.

2. If possible, do not use a difference score.

This is discussed in greater detail in the source discussion and in the source discussion (the source discussion) on cognitive assessment.

1. Remove items that reduce reliability.

This may include items that have low variability because they are too easy or too difficult (i.e., items that show ceiling or floor effects), in addition to items that have a low item–total correlation.

1. Collect more information across people and time.

2. Aggregate: Add items, measures, observers, raters, and time points.

a. Having more items that assess the same construct in the measure tends to increase reliability unless the additional items cause fatigue because the instrument is too long. Aggregation of like things reduces measurement error, because averaging together more items helps cancel out random error. b. Having multiple measures that assess the same construct can be beneficial, especially measures from different methods that do not share the same systematic errors (e.g., observation in addition to parent report). c. Items that aggregate a large amount of information tend to have strong predictive validity. For example, when hot dog sales are high, there are more wins at Yankees games. Hot dog sales have no theoretical influence on the Yankees, but it predicts well. In general, for predictive

accuracy, it is often better to assess more things less well than to assess one thing extraordinarily well. Aggregation is a double-edged sword: It is your friend as a researcher, because of increased predictive utility, but it is your enemy as an interpreter because it is hard to interpret the theoretical meaning of predictive associations based on an aggregated variable because the measure is more general and nonspecific. If aggregation increases error, test–retest reliability will decrease. If error stays the same when aggregating, test–retest reliability will increase.

1. Use a latent variable approach to aggregate the multiple items, measures, observers, raters, or time points.

Latent variable approaches include structural equation modeling and item response theory. Estimating a latent variable allows identifying the variance that is common to all of the measures (i.e., a better approximate of true score variance), and it discards variance that is not shared by all measures (i.e., error variance). Thus, latent variables have the potential to get purer estimates of the constructs by reducing measurement error.

Conclusion

Reliability is how much repeatability, consistency, and precision a measure's scores have. Reliability is not one thing. There are multiple aspects of reliability, and the extent to which a given aspect of reliability is important depends on the construct of interest. In general, we want our measures to show strong inter-rater, intra-rater, immediate parallel-forms, and internal consistency reliability. However, the extent to which our measures should show test–retest and delayed parallel-forms reliability depends on the stability of the construct and the time lag. Moreover, although we want our measures to show strong internal consistency reliability, the internal consistency reliability of a measure can be too high, and for some constructs (i.e., formative constructs), internal consistency would not be expected. In addition, reliability is not a characteristic that resides in a test. The reliability of a measure's scores reflects an interaction of the properties of the test with the population for whom it is designed and the sample, situation, and context in which it is administered. Thus, when reporting reliability in papers, it is important to adequately describe the aspects of reliability that have been considered and the population, sample, and context in which the measure is assessed. However, it is not enough for measures to be consistent or reliable; our interpretation of the measures' scores should also be accurate, meaningful, and useful for the intended uses. In other words, our interpretation of the measures' scores should be valid for the intended uses. In the next chapter (the source discussion), we discuss validity of measurement.

Suggested Readings

For more information on inter-rater reliability, we suggest reading Gwet, 2021 and Gwet, 2021, for categorical and continuous ratings, respectively.

Questions

Note: Several of the following questions use data from the Children of the National Longitudinal Survey of Youth Survey (CNLSY). The CNLSY is a publicly available longitudinal data set provided by the Bureau of Labor Statistics (https://www.bls.gov/nls/nlsy79-children.htm#topical-guide; archived at https://perma.cc/EH38-HDRN). The CNLSY data file for these exercises is located on the book's page of the Open Science Framework (https://osf.io/3pwza). Children's behavior problems were rated in 1988 (time 1: T1) and then again in 1990 (time 2: T2) on the Behavior Problems Index (BPI).

1. What is the test–retest reliability of the Antisocial Behavior subscale of the BPI?

Provide the coefficient of stability and coefficient of repeatability, and interpret what they indicate about the Antisocial Behavior subscale of the BPI.

1. What was the estimate of bias of the Antisocial Behavior subscale of the BPI comparing the measurements at T1 and T2, and what does this indicate about the Antisocial Behavior subscale of the BPI and the construct?

2. If a person had a score of 6 on the Antisocial Behavior subscale of the BPI at T1, within what range would we expect their score to be at T2?

3. What interval is often considered the “standard” for evaluating test–retest reliability of measures in psychology?

What is a potential downside of having shorter intervals? What is a potential downside of having longer intervals? How would you expect the test–retest reliability and bias to differ had the BPI been re-assessed at that ("standard") interval?

1. For each of the following, what would the coefficient of stability, coefficient of repeatability, and bias be if the following measurement error were added?

Also summarize the effects of measurement error in each case: a. 3 points to all scores at only T2 b. 3 points to all scores at both T1 and T2 c. 3 points to the scores for the first 500 participants at only T1 d. 3 points to the scores for the first 500 participants at both T1 and T2

1. You and a colleague both conduct diagnostic interviews on the same five clients.

The data are in the source discussion below. What is your inter-rater reliability with your colleague?

1. You and a colleague both rate the same five clients in the extent to which they are likely to be suicidal, where 1 = very low likelihood and 5 = very high likelihood.

The data are in the source discussion below. What is your inter-rater reliability with your colleague?

1. What was the internal consistency of the items on the Antisocial Behavior subscale of the BPI at T1?

Provide Cronbach's alpha (alpha), omega total (t), omega hierarchical (h), and the average of all possible split-half reliabilities. What do the estimates of internal consistency indicate about the BPI subscale?

1. What is the standard error of measurement of the Antisocial Behavior subscale of the BPI?

If a child's observed score is 6, within what range would we expect their true score to be? How confident are we about a child's true score on the construct (antisocial behavior) if we know their observed score on the measure (Antisocial Behavior subscale of the BPI)?

1. For many ways of estimating test–retest reliability, we assume that the true association between the construct at T1 and the construct at T2 is 1.0 (that is, the association of the true scores is 1.0).

According to the attenuation formula (that attenuates the association due to measurement error), if the true association between the construct at T1 and the construct at T2 was 1.0 and the reliability of the measure was.8, the observed test–retest correlation would be r =.80. According to the attenuation formula, what would the observed test–retest correlation be if the true association was.9 and the reliability of the measure was.8?

1. According to the disattenuation formula (to correct for attenuation of the association due to measurement error), what would be the true test–retest association of the construct of antisocial behavior if the reliability of the Antisocial Behavior subscale was r =.6299494?

Answers

1. The coefficient of stability (r = [computed value omitted in this course adaptation]) indicates that the Antisocial subscale of the BPI shows moderate rank-order stability over a two-year interval at these ages in this sample.

The coefficient of repeatability ([computed value omitted in this course adaptation]) indicates that 95% of repeated assessments on the BPI over a two-year interval would be expected to fall within [computed value omitted in this course adaptation] points of the score at the first measurement occasion (after accounting for any bias). Given that the total score ranges from 0–14, a repeatability coefficient of [computed value omitted in this course adaptation] indicates that individual children showed considerable differences from T1 to T2.

1. The estimate of bias was [computed value omitted in this course adaptation], indicating that children's scores were slightly ([computed value omitted in this course adaptation] points) higher, on average, at T1 compared to T2.

This may indicate developmental change such that children tend to decrease in antisocial behavior as they get older.

1. Their score at T2 is expected to be within the range of [[computed value omitted in this course adaptation], [computed value omitted in this course adaptation]].

2. A two-week interval is typically considered standard for evaluating test–retest reliability.

Shorter intervals have the potential problem that there may be carryover effects from the original testing due to the participant's memory of their answers from the first testing (thus artificially increasing the estimate of test–retest reliability). Longer intervals have the potential problem that real developmental change could occur in between the measurement occasions, thus artificially decreasing the estimate of test–retest reliability. The BPI was assessed at a two-year interval. Had it been assessed at a two-week interval, we would expect test–retest reliability to be higher (higher coefficient of stability and lower coefficient of repeatability) and bias to be lower.

1. The estimates are as follows:

a. stability coefficient: r = [computed value omitted in this course adaptation]; repeatability coefficient = [computed value omitted in this course adaptation]; bias = [computed value omitted in this course adaptation] i. increased bias (greater absolute value); no detectable effect on stability or repeatability coefficient b. stability coefficient: r = [computed value omitted in this course adaptation]; repeatability coefficient = [computed value omitted in this course adaptation]; bias = [computed value omitted in this course adaptation] i. no detectable effect on stability coefficient, repeatability coefficient, or bias c. stability coefficient: r = [computed value omitted in this course adaptation]; repeatability coefficient = [computed value omitted in this course adaptation]; bias = [computed value omitted in this course adaptation] i. lower/worse stability coefficient; higher/worse repeatability coefficient; contributed bias (but in the opposite direction of the bias before the score adjustment) d. stability coefficient: r = [computed value omitted in this course adaptation]; repeatability coefficient = [computed value omitted in this course adaptation]; bias = [computed value omitted in this course adaptation] i. artificially higher/stronger stability coefficient; no detectable effect on repeatability coefficient or bias

1. Your inter-rater reliability was very poor (Cohen's kappa [] = [computed value omitted in this course adaptation]).

2. Your inter-rater reliability was good.

If you used only one judge's ratings in analysis, the scores would have an intra-class correlation of ICC(2,1) = [computed value omitted in this course adaptation]. If, however, you used the average of the judge's ratings in analysis, the scores would have an intra-class correlation of ICC(2,k) = [computed value omitted in this course adaptation].

1. The estimates of internal consistency (Cronbach's alpha [alpha] = [computed value omitted in this course adaptation], omega total [t] = [computed value omitted in this course adaptation], omega hierarchical [h] = [computed value omitted in this course adaptation], average split-half reliability = [computed value omitted in this course adaptation]) indicate that the ratings show moderate consistency across items on the Antisocial subscale (and that the items are somewhat similar to one another), but that the items on the Antisocial subscale are also somewhat distinct from one another.

2. The standard error of measurement is estimated based on the reliability of the measure.

Reliability can be evaluated in different ways, and there is no single index of reliability. If we were to take infinite observations of a child's score and average them, that average would approximate the child's true score. Then, the test–retest reliability of the measure would help tell us about how (im)precise our estimate was at any given occasion compared to the true score. For the Antisocial Behavior subscale of the BPI, the test–retest reliability was [computed value omitted in this course adaptation] as indexed by the coefficient of stability. If we used this as our estimate of the measure's reliability (though the stability coefficient would likely be higher if we used a retest interval of two weeks instead of two years), the standard error of measurement would be [computed value omitted in this course adaptation]. Using this as our estimate of the standard error of measurement, we would expect the child's true score to fall within the range of [[computed value omitted in this course adaptation], [computed value omitted in this course adaptation]]. That's a pretty wide

range of uncertainty for a scale that ranges from 0–14. So, we are not very confident of a child's true score on the construct from their observed score on this measure.

1. The observed test–retest correlation (coefficient of stability) would be r = [computed value omitted in this course adaptation].

2. The true test–retest association would be r = [computed value omitted in this course adaptation].

CHAPTER 07 / CAC.903

Validity: Evidence for Interpretation and Use

Validity asks whether evidence supports the interpretation and use of a score. It is not a property that a test possesses once and forever. A test may be valid for one purpose, population, and decision, but not for another. A depression scale may be valid for screening adolescents in a clinic but not for diagnosing bipolar disorder. A cognitive test may be valid for estimating reasoning in one language group but not another. Validity is an argument built from evidence.

Content validity

Content validity asks whether the measure adequately samples the construct. If a math test only measures computation, it may not represent problem-solving. If an anxiety scale only measures physical symptoms, it may miss worry. The counselor should examine whether items match the referral question and the child's experience. Content validity is especially important when adapting measures for culture or development.

Criterion validity

Criterion validity examines relations with an external standard. Concurrent validity looks at a criterion measured at the same time. Predictive validity looks at a future outcome. A screening tool has criterion validity if it identifies children who later receive a diagnosis. But the criterion itself may be imperfect. A diagnosis is not a perfect gold standard. The counselor should consider the quality of the criterion.

Construct validity

Construct validity asks whether the measure behaves as theory predicts. It should correlate with related measures and differ from unrelated ones. It should show expected developmental patterns. It should respond to intervention if the construct is changeable. Construct validity is built over many studies. A single correlation is not enough.

Consequential validity

Assessment has consequences. A tool may produce accurate scores but harmful decisions if used poorly. Consequential validity asks whether the use of the assessment benefits the client and avoids unfair harm. A risk tool that overidentifies a group for discipline may have poor consequential validity even if it predicts some outcomes. The counselor should consider the full decision pathway.

Incremental validity and clinical utility

Incremental validity asks whether a measure adds information beyond existing data. Clinical utility asks whether using the measure improves decisions or outcomes. A long battery may be impressive but useless if it does not change the plan. A brief functional interview may have high utility if it identifies a maintainable pattern. Doctoral assessment should justify each method by its contribution.

Validity threats in child assessment

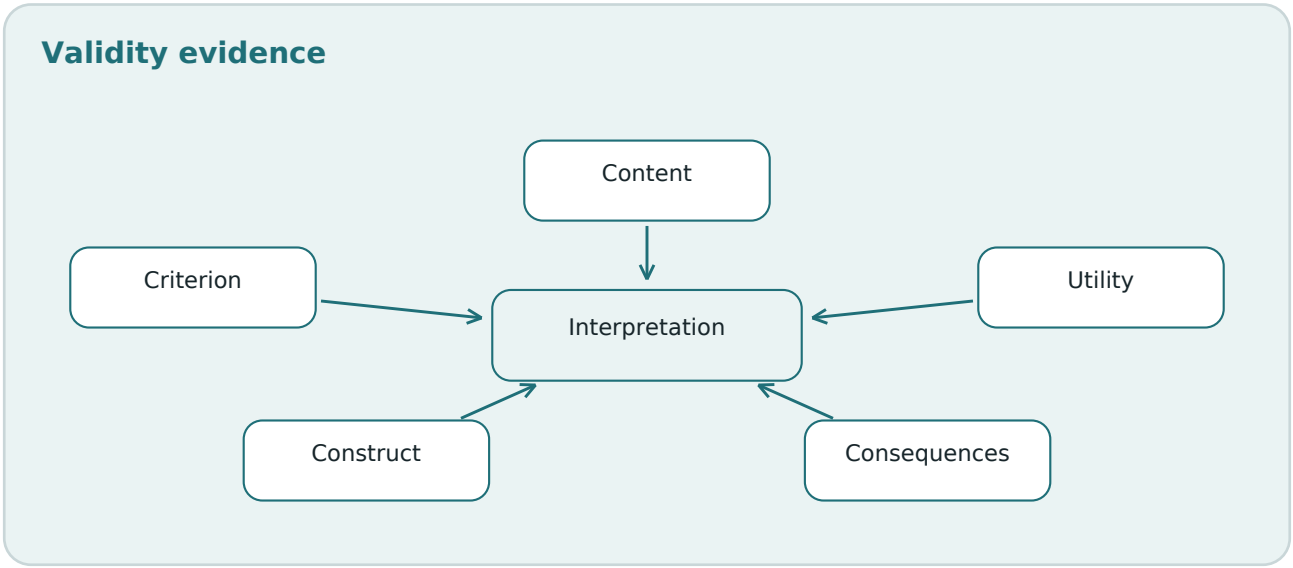
Threats include developmental mismatch, language barriers, informant bias, coaching, malingering, cultural misunderstanding, and comorbidity. A child may endorse items literally. A teen may minimize symptoms to avoid hospitalization. A parent may overreport to obtain services. The counselor should not assume invalidity from one inconsistency. Instead, the counselor should examine patterns and context.

Real-life application

A school uses a brief behavior screen to identify students for a social-skills group. The screen has reasonable sensitivity but many false positives. The counselor adds teacher observation and a brief interview before placement. This protects students from unnecessary labeling and improves group fit. The screen is used as a first step, not a verdict. Validity is protected by the pathway.

Reflection

Choose a test. Write the interpretation you want to make. List three pieces of validity evidence that would support it and two threats that would weaken it. Then ask whether the assessment will actually help the child. Validity is not only about scores. It is about responsible use.



Validity is an argument built from multiple sources of evidence for a specific use.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module validity.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Validity

"What we know depends on how we know it."

Overview of Validity

According to the Standards for Educational and Psychological Testing (American Educational Research Association et al., 2014), measurement validity is "the degree to which evidence and theory support the interpretations of test scores for proposed uses of tests." We summarized reliability with three words: repeatability, consistency, and precision. A summary of validity in three words is accuracy, utility, and meaningfulness. As with reliability, validity is not a property of the measure itself—the validity of a measure's scores reflects an interaction of the properties of the measure with the population for whom it is designed and the sample and context in which it is administered. Moreover, validity is tied to the interpretation of a measure's scores for the proposed uses, not (just) to the measure itself. The same set of scores can have different degrees of validity for different purposes. For instance, a measure's scores may have stronger validity for making a diagnostic decision than for making a prediction about future behavior. Thus, as the Standards indicate, it is incorrect to use the unqualified phrase "the validity of the measure" or "the measure is (in)valid," because these phrases do not specify which scores were used from the test, what the use is (e.g., predicting whether a person will succeed in a given job), and what interpretation was made of the test scores for this purpose.^[For shorthand and to avoid repetition, we sometimes refer to the validity of the measure (or research design), but in such instances, we are actually referring to the validity of the interpretation of a measure's scores for a given use (or of the inferences from a particular research design).] For these reasons, it is insufficient to cite evidence for the validity of the measure in other work; it is also important to evaluate the validity of the measure's scores for the intended use in your particular sample.

Below, we prepare the data to provide some validity-related examples throughout the rest of the chapter.

Simulate Data

For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Add Missing Data

Adding missing data to dataframes helps make examples more realistic to real-life data and helps you get in the habit of programming to account for missing data.

Types of Validity

Like reliability, validity is not one thing. There are many types of validity. In this book, we discuss the following types of validity:

- face validity
- content validity
- criterion-related validity
- concurrent validity
- predictive validity
- construct validity
- convergent validity
- discriminant (divergent) validity
- incremental validity
- treatment utility of assessment
- discriminative validity
- elaborative validity
- consequential validity
- representational validity
- factorial (structural) validity
- ecological validity
- process-focused validity
- diagnostic validity
- social validity
- cultural validity
- internal validity
- external validity
- (statistical) conclusion validity

We arrange these types of validity into two broader categories: measurement validity and research design validity.

Measurement Validity

Aspects of measurement validity involve the validity of a particular measure, or more specifically, the validity of interpretations of scores from that measure for the proposed uses. Aspects of measurement validity include:

• face validity

• content validity

• criterion-related validity

• concurrent validity

• predictive validity

• construct validity

• convergent validity

• discriminant (divergent) validity

• incremental validity

• treatment utility of assessment

• discriminative validity

• elaborative validity

• consequential validity

• representational validity

• factorial (structural) validity

• ecological validity

• process-focused validity

• diagnostic validity

• social validity

• cultural validity

Face Validity

The interpretation of a measure's scores has face validity (for a given construct and a given use) if a typical person—a nonexpert—who looks at the content of each item will believe that the item belongs in the scale for this construct and for this use. The measure, and each item, looks "on its face" like it assesses the target construct. There are several advantages of a measure having face validity. First, outside groups will be less likely to be critical of the measure because it is intuitive. Second, use of a face valid measure is rarely objected to on ethical and bias charges for selling the measure to the public or clinicians. Third, face validity can be helpful for dissemination because more people may be receptive to it.

However, face validity also has important disadvantages. First, judgments of face validity are not based on theory. Second, face validity is based on subjective judgment, which can be inaccurate. Third, these subjective judgments are made by laypeople whose judgments may be inaccurate because of biases and lack of awareness of scientific knowledge. Fourth, a face valid measure may be too simple because anybody can understand the questions and what the questions are intended to assess, so (presumably) respondents can easily fake responses to achieve their goals. Faking of responses may be more of a concern in situations when there is an incentive for the respondent to achieve a particular outcome (e.g., be deemed competent to stand trial, be judged competent to obtain a job or custody of child, be judged to have a disorder to receive accommodations or disability benefits).

It is disputed whether having face validity is good or bad. Whether face validity is important to a given measure depends on the construct that is intended to be assessed, the context in which the assessment will occur, who will be paying for and/or administering the assessment, whether the respondents have incentives to achieve particular scores, and the goals of the assessment (i.e., how the assessment will be used). There is also controversy about whether face validity is a true form of validity; many researchers have argued that it is not a true psychometric form of validity, because the appearance of validity is not validity (Royal, 2016).

Content Validity

Content validity involves a judgment about whether or not the content (items) of the measure theoretically matches the construct that is intended to be assessed—that is, whether the operationalization accurately reflects the construct. Content validity is developed based on items generated and selected by experts of the construct and based on the subjective determination that the measure adequately assesses and covers the construct of interest. Content validity differs from face validity in that, for face validity, a layperson determines whether or not the measure seems to assess the construct of interest. By contrast, for content validity, an expert determines whether or not the measure adheres to the construct of interest.

For a measure to have content validity, its items should span the breadth of the construct. For instance, the construct of depression has many facets, such as sleep disturbances, weight/appetite changes, low mood, suicidality, etc., as depicted in the source discussion.

Content Facets of the Construct of Depression.

For a measure to have content validity, there should be no gaps—facets of the construct that are not assessed by the measure—and there should be no intrusions—facets of different constructs that are assessed by the measure. Consider the construct of depression. If theory states the construct includes various facets such as sadness, loss of interest in activities, sleep disturbances, lack of energy, weight/appetite change, and suicidal thoughts, then a content-valid measure should assess all of these facets. If the measure does not assess sleep disturbances (a gap), the measure would lack content validity. If the measure assessed facets of other constructs, such as impulsivity (an intrusion), the measure would lack content validity.

With content validity, it is important to consider the population of interest. The same construct may look different in different populations and may require different content to assess it. For instance, it is important to consider the cultural relativity of constructs. The

content of a construct may depend on the culture, such as in the case of culture-bound syndromes. Culture-bound syndromes are syndromes that are limited to particular cultures. An example of a culture-bound syndrome among Korean women is hwa-byung, which is the feeling of an uncomfortable abdominal mass in response to emotional distress. Another important dimension to consider is development. Constructs can manifest differently at different points in development, known as heterotypic continuity, which is discussed in the source discussion of the source discussion on repeated assessments across time. When considering the different dimensions of your population, it can be helpful to remember the acronym ADDRESSING, which is described in the source discussion of the source discussion on cultural and individual diversity.

However, like face validity, content validity is based on subjective judgment, which can be inaccurate.

Criterion-Related Validity

Criterion-related validity examines whether a measure behaves the way it should given your theory of the construct. This is quantified by the correlation between a measure's scores and some (hopefully universally accepted) criterion we select. For instance, a criterion could be a diagnosis, a child's achievement in school, an employee's performance in a job, etc.

Below, we provide an example of criterion-related validity by examining the Pearson correlation between a predictor and a criterion.

In this case, the estimate of criterion-related validity is r = [computed value omitted in this course adaptation]. There are two types of criterion-related validity: concurrent validity and predictive validity.

Concurrent Validity Concurrent validity considers the concurrent association between the chosen measure and the criterion. That is, both the measure and the criterion are assessed at the same point in time. An example of concurrent validity would be examining self-report of court involvement in relation to current court records.

Sensitivity to Change (Responsiveness) An important aspect of validity that is related to concurrent validity is the degree to which a measure's scores show sensitivity to change (Fok & Henry, 2015). If a researcher conducts an intervention that improves people's mood, it is important to have measures that are sensitive enough to detect the improvements in people's mood.

Nonreactivity As noted in the source discussion, the reactivity of a measure is how a measure's scores change in response to being measured. In an intervention, reactivity can be valuable because having the person self-monitor their behavior can lead to behavior improvement, as described in the source discussion. However, when the goal is to measure change over time, reactivity can be a problem because it can lead to changes in the measure's scores that are not due to people's changes in the construct. Thus, when merely monitoring changes over time, it is important to use measures that show nonreactivity (Myers & Winters, 2002).

Predictive Validity Predictive validity considers the association between the chosen measure and the criterion at a later time point. An example of predictive validity would be examining the predictive association between children's scores on an academic achievement test in first grade and their eventual academic outcomes five years later. Revelle, 2024

argues that insufficient attention is paid to predictive validity due to excessive focus on reflective latent variables and internal consistency as a means to satisfy considerations of construct validity at the expense of the usefulness of the measure.

Empiricism and Theory Criterion-related validity arose out of a movement known as radical operationalism. Radical operationalism was a pushback against psychoanalysis. Psychoanalysis focused on grand theoretical accounts for how constructs relate. The goal of radical operationalism was to clarify concepts from a behavioristic perspective to allow predicting and changing behavior more successfully. An "operation" in radical operationalism refers to a fully described measurement.

Proponents of radical operationalism argued that all constructs in psychology that could not be operationally defined should be excluded from the field as "nonscientific." They asserted that operations should be well-defined enough to be able to replicate the findings. So, constructs had to be defined precisely according to this perspective, but how precisely? You could go on forever trying to more precisely describe a behavior in terms of its form, frequency, duration, intensity, situation, antecedents, consequences, biological substrates, etc. So, radical operationalists asserted that we should use theory of the construct to determine what is essential and what is not.

Radical operationalism was also related to radical behavioralism, which was espoused by B.F. Skinner. Skinner famously used a box (the "Skinner Box") to more directly control, describe, and assess behaviors. Skinner noted the major role that the environment played in influencing behavior. Skinner proposed a theory of implicit learning about a behavior or stimulus based on its consequences, known as operant conditioning. According to operant conditioning, something that increases the frequency of a given behavior is called a reinforcer (e.g., praise), and something that decreases the frequency of a behavior is called a punisher (e.g., loss of a privilege). Through this work, Skinner came to view everything an organism does (e.g., action, thought, feeling) as a behavior.

Related to these historical perspectives was a perspective known as dustbowl empiricism. Dustbowl empiricism focused on the empirical connections between things—how things were associated using data. It was a completely atheoretical perspective in which interpretation was entirely data driven. An example of dustbowl empiricism is the approach that was used to develop the first version of the Minnesota Multiphasic Personality Inventory (MMPI). The MMPI was developed using an approach known as empirical-criterion keying, where items were selected for the scale for no reason other than the items demonstrate an association with the criterion. That is, an item was selected if it showed a strong ability to discriminate (differentiate) between clinical and control groups. Using this method with hundreds of items (and thousands of inter-item correlations), the MMPI developed 10 clinical scales, which involved operational rules based on previously collected empirical evidence.

But what do you know with this abundance of correlations? You can use data reduction methods to reduce the many variables, based on their inter-correlations, down to a more parsimonious set of factors. But how do you name each factor, which is composed of many items? The developers originally numbered the MMPI clinical scales from 1 to 10. But numbered scales are not useful for other people, so the factors were eventually given labels (e.g., Paranoia). And if a client received an elevated score on a factor, many people would

label the clients as _____ [the name of the factor], such as "paranoid." The MMPI is discussed in further detail in the source discussion on objective personality testing.

The idea of dustbowl empiricism was to develop a strong empirical base that would provide a strong foundation to help build up to a broad understanding that was integrated, coherent, and systematic. However, this process was unclear when there was only a table of correlations. Radical operationalists were opposed to content validity because it allows intrusion of our flawed thinking. According to operationalists, there are no experts. According to this perspective, the content does not matter; we just need enough data to bootstrap ourselves to a better understanding of the constructs.

Although the atheoretical approach can perform reasonably well, it can be improved by making better use of theory. An empirical result (e.g., a correlation) might not necessarily have a lot of meaning associated with it. As the maxim goes, correlation does not imply causation.

Correlation Does Not Imply Causation Just because X is associated with Y does not mean that X causes Y. Consider that you find an association between variables X and Y, consistent with your hypothesis, as depicted in the source discussion.

Hypothesized Causal Effect Based on an Observed Association Between X and Y, Such That X Causes Y.

There are three primary reasons that an observed association between X and Y does not necessarily mean that X causes Y.

First, the association could reflect the opposite direction of effect, where Y actually causes X, as depicted in the source discussion.

Reverse (Opposite) Direction of Effect From the Hypothesized Effect, Where Y Causes X.

Second, the association could reflect the influence of a third variable. If a third variable is a common cause of each and accounts for their association, it is a confound. An observed association between X and Y could reflect a confound—i.e., a cause (Z) that influences both X and Y, which explains why X and Y are correlated even though they are not causally related. A third variable confound that is a common cause of both X and Y is depicted in the source discussion.

Confounded Association Between X and Y due to a Common Cause, Z.

Third, the association might be spurious. It might just reflect random variation (i.e., chance), and that when tested on an independent sample, what appeared as an association may not hold when testing whether the association generalizes.

However, even if the association between X and Y reflects a causal effect and that X causes Y, it does not necessarily mean that the effect is clinically actionable or useful. An association may reflect a static or unmodifiable predictor that is not practically useful as a treatment target.

Understanding the Causal System As Silver, 2012 notes, "The numbers have no way of speaking for themselves. We speak for them. We imbue them with meaning." (p. 9). If we understand the variables in the system and how they influence each other, we can predict things more accurately than predicting for the sake of predicting. For instance, we have made great strides in the last decades when it comes to more accurate weather

forecasts (see here; archived at https://perma.cc/PF8P-BT3D and here; archived at https://perma.cc/248N-9CW5), including extreme weather events like hurricanes. These great strides have more to do with a better causal understanding of the weather system and the ability to conduct simulations of the atmosphere than merely because of big data (Silver, 2012). By contrast, other events are still incredibly difficult to predict, including earthquakes, in large part because we do not have a strong understanding of the system (and because we do not have ways of precisely measuring those causes because they occur at a depth below which we are realistically able to drill) (Silver, 2012).

Model Over-Fitting Statistical models applied to big data (i.e., lots of variables and lots of samples) can over-fit the data, which means that the statistical model accounts for error variance (an overly specific prediction), which will not generalize to future samples. So, even though an over-fitting statistical model appears to be accurate, it is not actually that accurate—it will predict new data less accurately than how accurately it accounts for the data with which the model was built.

In the source discussion, the blue line represents the true distribution of the data, and the red line is an over-fitting model:

Criterion Contamination An important issue in predictive validity is the criterion problem—finding the right criterion. It is important to avoid criterion contamination, which is artificial commonality between the measure and the criterion. The criterion is not always a well-measured clear criterion to predict (like predicting death in the medical field). And you may not have access to a predictive criterion until a long time from now. So, what researchers often do is adopt intermediate assessments, which are not actually what they are interested in, but it is related to the criterion of interest, and it is in a window of time that allows for some meaningful prediction. For instance, intermediate graduate school markers of whether a graduate student will go on to have a successful career could include their grades in graduate school, whether they completed the dissertation, their performance in comprehensive/qualifying exams, etc. However, these intermediate assessments do not always indicate whether or not a student will go on to have a successful career (i.e., they are not always correlated with the real criterion of interest).

Using Theory as a Guide So, empiricism is often not enough. It is important to use theory to guide selection of an intermediate criterion that will relate to the real criterion of interest. In psychology, even our long-term criteria are not well defined relative to other sciences. In clinical psychology, for example, we are often predicting a diagnosis, which is not that much more valid compared to our measure/predictor.

At the same time, in psychology, our theories of the causal processes that influence outcomes are not yet very strong. Indeed, I have misgivings calling them theories because they do not meet the traditional scientific standard for a theory. A scientific theory is an explanation of the natural world that is testable and falsifiable, and that has withstood rigorous scientific testing and scrutiny. In psychology, our "theories" are more like conceptual frameworks (Stallworthy et al., in press). And these conceptual frameworks are often vague, do not make specific predictions of effects and noneffects, and do not hold up consistently when rigorously tested. As described by Meehl1978:, n.d.

I consider it unnecessary to persuade you that most so-called "theories" in the soft areas of psychology (clinical, counseling, social, personality, community, and school psychology) are

scientifically unimpressive and technologically worthless... Perhaps the easiest way to convince yourself is by scanning the literature of soft psychology over the last 30 years and noticing what happens to theories. Most of them suffer the fate that General MacArthur ascribed to old generals—They never die, they just slowly fade away. In the developed sciences, theories tend either to become widely accepted and built into the larger edifice of well-tested human knowledge or else they suffer destruction in the face of recalcitrant facts and are abandoned, perhaps regretfully as a "nice try." But in fields like personology and social psychology, this seems not to happen. There is a period of enthusiasm about a new theory, a period of attempted application to several fact domains, a period of disillusionment as the negative data come in, a growing bafflement about inconsistent and unreplicable empirical results, multiple resort to ad hoc excuses, and then finally people just sort of lose interest in the thing and pursue other endeavors. (pp. 806–807).

Even if we had strong theoretical understanding of the causal system that influences behavior, we would likely still have difficulty making accurate predictions because the field has largely relied on relatively crude instruments. According to one philosophical perspective known as LaPlace's demon, if we were able to know the exact conditions of everything in the universe, we would be able to know how the conditions would be in the future. This is an example of scientific determinism, where if you know the initial conditions, you also know the future. Other perspectives, such as quantum mechanics and chaos theory, would say that, even if we knew the initial conditions with 100% certainty, there would still be uncertainty in our understanding of the future. But assume, for a moment, that LaPlace's demon is true. The challenge in psychology is that we have a relatively poor understanding of the initial conditions of the universe. Thus, our predictions would necessarily be probabilistic, similar to weather forecasts. Despite having a strong understanding of how weather systems behave, we have imperfect understanding of the initial conditions (e.g., the position and movement of all molecules) (Silver, 2012).

Psychoanalysis Versus Empiricism We can consider the difference between psychoanalysts and empiricists in cultural references.

Here is an excerpt from Douglas Adams' The Restaurant at the End of the Universe (from the series, The Hitchhiker's Guide to the Galaxy):

To explain—since every piece of matter in the Universe is in some way affected by every other piece of matter in the Universe, it is in theory possible to extrapolate the whole of creation—every sun, every planet, their orbits, their composition and their economic and social history from, say, one small piece of fairy cake. --- Douglas Adams (1980, p. 76)

If we consider this idea that someone might be able to extrapolate a simulation of the universe from a single piece of cake, we find similarities to how psychoanalysts connect everything to everything else through grand theories. Psychoanalysts try to reconstruct the entire universe from a sparse bit of data with supposedly strong theoretical understanding (when, in reality, our theories are not so strong). Their "theories" make grand conceptual claims.

Let us contrast psychoanalysts with empiricists/radical operationalism. the source discussion presents a depiction of empiricism. Empiricists evaluate how an observed predictor relates to an observed outcome. The rectangles in the figure represent entities

that can be observed. For instance, an empiricist might examine the extent to which a person's blood pressure is related to the number of words they speak per minute.

Conceptual Depiction of Empiricism.

Contrast the empirical approach with psychoanalysis, as depicted in the source discussion.

Conceptual Depiction of Psychoanalysis.

Circles represent unobserved, latent entities. For instance a psychoanalyst may make a conceptual claim that one observed variable influences another observed variable through a complex chain of intervening processes that are unobservable.

There is a classic and hilarious Monty Python video (see the source discussion) that is an absurd example akin to radical operationalism taken to the extreme (BBC, 1973):

(BBC, 1973) provides a classic and hilarious Monty Python video that is an absurd example akin to radical operationalism taken to the extreme.

In the video, the researchers make lots of measurements and predictions. The researchers identify interspecies differences in intelligence where humans show better performance on the English-based intelligence test (who got an IQ score of 100) than the penguins, who got an IQ score of 2. But the penguins did not speak English and were unable to provide answers to the English-based intelligence test. So, the researchers also assessed a group of non-English speaking humans as an attempt to control for language ability. They found that the penguins' scores were equal to the scores of the non-English speaking humans. They argued that, based on their smaller brain and equal IQ when controlling for language ability, that penguins are smarter than humans. However, the researchers clearly made mistakes about confounding variables. And inclusion of a control group of non-English speaking humans does not solve the problem of validity or bias; it just obscures the problem. In summary, radical operationalism provides rich lower-level information, but lacks the broader picture. So, it seems, that we need both theory and empiricism. Theory and empiricism can—and should—inform each other.

Construct Validity

Construct validity is the extent to which a measure accurately assesses a target construct (Cronbach & Meehl, 1955). Construct validity is not quantified by a single index but rather consists of a "web of evidence" (the totality of evidence), which reflect the sum of inferences from multiple aspects of validity. That is, construct validity is the extent to which a measure assesses a target construct. Construct validity deals with the association between a measure and an unobservable criterion, i.e., a latent construct. By contrast, criterion-related validity deals with an observable criterion.

Construct validity encompasses all other types of measurement validity (e.g., content and criterion-related validity), in addition to

• scores on the measure show homogeneity—i.e., scores on the measure assess a single construct

• scores on the measure show theoretically expected developmental changes (i.e., criterion-related validity with respect to age or development)

• scores on the measure show theoretically expected group differences (i.e., criterion-related validity with respect to some group status)—this is also called known groups validity (when the groups are known to differ in their level on the construct)

• scores on the measure show theoretically expected intervention effects (i.e., criterion-related validity with respect to intervention)

• establishing the nomological network of a construct

Threats to construct validity are described by Shadish et al., 2002 (see Table 3.1).

Nomological Network A nomological network is the interlocking system of laws that constitute a theory. It describes how the concepts (constructs) of interest are causally linked, including their observable manifestations and the causal relations among and between them. An example of a nomological network is depicted in the source discussion.

Example of a Nomological Network. O.M. = Observable Manifestation.

With construct validity, we can judge the quality of a measure by how well or how sensibly it fits in a nomological network. Latent constructs and observed measures improve each other step by step. But there is no established way to evaluate the process. Approaches such as network analysis may be useful toward this aim.

Historically, construct validity became a way for some researchers to skip out on other types of validity. People found a correlation between a measure's scores and some group membership and argued, therefore, that the measure has construct validity because there was a theoretically expected correlation between some measure and _____ (insert whatever measure). People started finding a correlation of a measure with other measures, asserting that it provides evidence of construct validity, and saying "that's my nomological network." But a theoretically expected association is not enough! For example, consider a measure of how quickly someone can count backward by seven. Performance is impaired in those with schizophrenia, anxiety (due to greater distractibility), and depression (due to concentration difficulties and cognitive slowing). Therefore, it is a low-quality claim that counting backward is part of the phenomenology of these disorders because it lacks differential deficit or discriminant validity (Campbell & Fiske, 1959). This is related to the "glop problem," which asserts that every bad thing is associated with every other bad thing—there is high comorbidity. Therefore, researchers needed some way to distinguish method variance from construct variance. This led to the development of the multitrait-multimethod matrix.

Multitrait-Multimethod Matrix (MTMM) The multitrait-multimethod matrix (MTMM), as proposed by Campbell and Fiske (Campbell & Fiske, 1959), is a concrete way to evaluate the validity of a measure. The MTMM allows you to split the variance of measures' scores into variance that is due to the method (i.e., method variance or method bias) and variance that is due to the construct (trait) of interest (i.e., construct variance). To create an MTMM, you need at least two methods and at least two constructs. For example, an MTMM could include self-report and observation of depression and introversion. You would then examine the correlations across combinations of construct and method.

For an example of an MTMM, see the source discussion. Several aspects of psychometrics can be evaluated with an MTMM, including reliability, convergent validity, and discriminant validity. The reliability diagonal of an MTMM is the correlation of a variable with itself, i.e.,

the test–retest reliability or monotrait-monomethod correlations. The reliability coefficients should be the highest values in the matrix because each measure should be more correlated with itself than with anything else.

Multitrait-Multimethod Matrix. (Figure reprinted from Campbell and Fiske (Campbell & Fiske, 1959), Table 1, p. 82. Campbell, D. T., & Fiske, D. W. (1959). Convergent and discriminant validation by the multitrait-multimethod matrix. Psychological Bulletin, 56, 81–105. https://doi.org/10.1037/h0046016)

A multitrait-multimethod matrix can be organized by method and then by construct or vice versa. An MTMM organized by method then by construct is depicted in the source discussion.

Multitrait-Multimethod Matrix Organized by Method Then by Construct.

An MTMM organized by construct then by method is depicted in the source discussion.

Multitrait-Multimethod Matrix Organized by Construct Then by Method.

Convergent Validity Convergent validity is the extent to which a measure is associated with other measures of the same target construct. In an MTMM, convergent validity evaluates whether measures targeting the same construct, but using different methods, converge upon the same construct. These are observed in the validity diagonals, also known as the convergent correlations or the monotrait-heteromethod correlations. For strong convergent validity, we would expect the values in the validity diagonals to be significant and high-ish.

Discriminant (Divergent) Validity You can also evaluate the discriminant validity, also called divergent validity, of a measure in the context of an MTMM. Discriminant validity is the extent to which a measure is not associated (or less strongly associated) with measures of different constructs that are not theoretically expected to be related (compared to associations with measures of the same construct). In an MTMM, discriminant validity determines the extent to which a measure does not correlate with measures that share a method but assess different constructs. For strong discriminant validity, we would expect the discriminant correlations (heterotrait monomethod) to be low.

According to Campbell and Fiske (Campbell & Fiske, 1959), discriminant validity of measures can be established when three criteria are met:

1. The convergent (monotrait-heteromethod) correlations are stronger than heterotrait-heteromethod correlations.

This provides the weakest evidence for discriminant validity. That is, the convergent correlations are higher than the values in the same column or row in the same heteromethod block. This can be evaluated using the heterotrait-monotrait ratio (described below).

1. The convergent (monotrait-heteromethod) correlations are stronger than discriminant correlations (monomethod-heterotrait).

This provides stronger evidence of discriminant validity.

1. The patterns of intercorrelations between constructs are the same, regardless of which measurement method is used.

That is, the pattern of inter-trait associations is the same in all triangles. For example, if extraversion and anxiety are moderately correlated with each other but uncorrelated with achievement, we would expect this pattern of interrelations between constructs would hold, regardless of which method was used to assess the construct.

You can estimate a measure's degree of discriminant validity based on the heterotrait-monotrait ratio (Henseler et al., 2015; Roemer et al., 2021). HTMT is the average of the heterotrait-heteromethod correlations (i.e., the correlations of measures from different measurement methods that assess different constructs), relative to the average of the monotrait-heteromethod correlations (i.e., the correlations of measures from different measurement methods that assess the same construct). As described here (https://www.henseler.com/htmt.html) (archived at https://perma.cc/A9DU-WZWQ) based on evidence from Voorhees et al., 2016,

If the HTMT is clearly smaller than one, discriminant validity can be regarded as established. In many practical situations, a threshold of 0.85 reliably distinguishes between those pairs of latent variables that are discriminant valid and those that are not.

The authors updated the HTMT to use the geometric mean of the correlations rather than the arithmetic mean of the correlations to relax the assumption of tau-equivalence (Roemer et al., 2021). They called the updated index HTMT2. Unlike HTMT, HTMT2 can be used only if all inter-variable correlations are positive. HTMT2 was calculated below using the semTools package (R-semTools, n.d.). In this case, the HTMT values are less than.85, providing evidence of discriminant validity.

Other indexes of discriminant validity include the CI CFA(sys) (Rönkkö & Cho, 2020), ^2(sys) (Rönkkö & Cho, 2020), and the Fornell-Larcker Ratio (Claes Fornell & David F. Larcker, 1981). The FLR is based on the idea that a construct should be more strongly associated with its indicators than with other constructs, and is computed as the ratio of (a) the highest squared inter-correlation the construct has with other constructs to (b) the average variance extracted [AVE] by the construct. A criterion of FLR < 1 evaluates whether the AVE of a factor is greater than the highest squared inter-correlation between the factor and other factors in the model (Claes Fornell & David F. Larcker, 1981). Another index estimates the degree of correspondence of convergent and discriminant validity correlations with a priori hypotheses as an index of construct validity (Furr & Heuckeroth, 2019).

Using an MTMM, a researcher can learn a lot about the quality of a measure and build a nomological network. For instance, we can estimate the extent of method variance, or variance that is attributable to the measurement method rather than the construct of interest. Method variance is estimated by the difference between monomethod versus heteromethod blocks.

A poor example of an MTMM would be having two constructs such as height and depression and two methods such as Likert and true/false. This would be a poor MTMM for two reasons: (1) there are trivial differences in the methods, which would lead to obvious convergence, and (2) the differences between the constructs are not important—they are obviously discriminant. Using such an MTMM would find strong convergent associations because the methods are maximally similar and weak discriminant associations because the constructs are maximally different. It would be better to use maximally different measurement methods

(e.g., self-report and performance-based measures) and to use constructs that are important to distinguish (e.g., depression and anxiey).

The paper by Campbell and Fiske (Campbell & Fiske, 1959) that introduced the MTMM is one of the most widely cited papers in psychology of all time. Psychological Bulletin published a more recent paper by Fiske and Campbell (Fiske & Campbell, 1992), entitled "Citations Do Not Solve Problems." They noted how their original paper was the most widely cited paper in the history of Psychological Bulletin, but they argued that nothing came of their paper. The MTMM matrices published today show little-to-no improvement compared to the ones they published in 1959. In part, this may be because we need better measures (i.e., a higher ratio of construct to method variance).

Multitrait-Multimethod Correlation Matrix This example is courtesy of W. Joel Schneider. First, we simulate data for an MTMM model using a fixed model with population parameters using the simstandard package (R-simstandard, n.d.):

A multitrait-multimethod matrix correlation matrix is below.

Construct Validity Beyond Campbell and Fiske There are a number of other approaches that can be helpful for establishing construct validity in ways that go beyond the approaches proposed by Campbell and Fiske (Campbell & Fiske, 1959). One way is known as triangulation. Triangulation is conceptually depicted in the source discussion. Triangulation involves testing a hypothesis with multiple measures and/or methods to see if the findings are consistent—that is, whether the findings triangulate.

Using Triangulation to Arrive at a Closer Estimate of the Construct Using Multiple Measures and/or Methods.

A contemporary approach to multitrait-multimethod modeling uses confirmatory factor analysis, as described below.

MTMM in Confirmatory Factor Analysis (CFA) Using modern modeling approaches, there are even more advanced ways of examining an MTMM. For instance, you can use structural equation modeling (SEM) or confirmatory factor analysis (CFA) to derive a latent variable of a construct from multiple methods to be free from method-related error variance to generate purer assessments of the construct to see how it relates to other constructs. For an example of an MTMM in confirmatory factor analysis, see the source discussion and the source discussion in the source discussion on factor analysis. Guidelines for MTMM in CFA are provided by Marsh et al., 2026.

Multitrait-Multimethod Model in Confirmatory Factor Analysis With Three Constructs (Internalizing, Externalizing, and Thought-Disordered Problems) and Three Methods (Mother-, Father-, and Teacher-report).

The confirmatory factor analysis (CFA) model was fit in the lavaan package (R-lavaan, n.d.).

A path diagram of the model is depicted in the source discussion using the semPlot package (R-semPlot, n.d.).

Incremental Validity

Accuracy is not enough for a measure to be useful. Proposed by Sechrest, 1963, measures should also be judged by the extent to which they provide an increment in predictive efficiency over the information otherwise easily and cheaply available. Incremental validity deals with the incremental value or utility over a measure beyond other sources of information. It must be demonstrated that the addition of a measure will produce better predictions than are made on the basis of information ordinarily available. It is not enough to show that the measure is better than chance, and the measure should not just be capitalizing on shared method variance with the criterion or on increased reliability of the measure. That is, the measure should explain truly unique variance—variance that was not explained before. Incremental validity demonstrates added value, unless the other measure is cheaper or less time-consuming. Incremental validity is a specific kind of criterion-related validity: significantly increased R^2 in hierarchical regression. The incremental validity of a measure can be evaluated by examining whether the measure explains significant unique variance in the criterion when accounting for other information, such as easily accessible information, traditionally available measures, or the current gold-standard measure. The extent of incremental validity of a measure can be quantified with the change in the coefficient of determination (R^2) that compares (a) the model that includes the old predictor(s) to (b) the model that includes old predictors and the new predictor (measure).

R^2 was calculated using the rockchalk package (R-rockchalk, n.d.). The predictor shows significant incremental validity above the old predictor in predicting the criterion. Model 1 explains [computed value omitted in this course adaptation]% of the variance (R^2 = [computed value omitted in this course adaptation]), and Model 2 explains [computed value omitted in this course adaptation]% of the variance (R^2 = [computed value omitted in this course adaptation]). Thus, the predictor explains only [computed value omitted in this course adaptation]% additional variance in the criterion above the variance explained by the old predictor (R^2). Based on adjusted R^2 (R^2 adj), the predictor explains only [computed value omitted in this course adaptation]% additional variance in the criterion above the variance explained by the old predictor (R^2 adj).

Treatment Utility of Assessment

Hayes et al., 1987 argued that it is not enough for measures to be reliable and valid. The authors raised another important consideration for the validity of a measure: its treatment utility or usefulness. They asked the question: What goal do assessments accomplish in clinical psychology? In clinical psychology, the goal of assessment is to lead to better treatment outcomes. Therefore, the treatment utility of a measure is the extent to which a measure is shown to contribute to beneficial treatment outcomes. That is, if a clinician has the information/results from having administered this measure, do the clients have better outcomes? The treatment utility of assessment is a specific kind of criterion-related validity, with the idea that "all criteria are not created equal." And the criterion that is most important to optimize, from this perspective, when developing and selecting measures is a client's treatment outcomes.

Hayes et al., 1987 described different approaches to evaluating the extent to which a measure shows treatment utility. These are a priori group comparison approaches that examine whether a specific difference in the assessment approach relates to treatment

outcome. They distinguished between (a) manipulated assessment and (b) manipulated use. Manipulated assessment and manipulated use are depicted in the source discussion.

Research Designs That Evaluate the Treatment Utility of Assessment.

In manipulated assessment, a single group of subjects is randomly divided into two subgroups, and either the collection or availability of assessment data is varied systematically. Therapists then design or implement treatment in accord with the data available. As an example, the measure of interest may be administered in one condition, and the standard measure may be administered in the other condition. Then the treatment outcomes would be compared across the two conditions. An advantage of manipulated assessment is that this type of design is more realistic than manipulated use. Making the assessment data available but not assigning a certain treatment based on the assessment outcomes better simulates a realistic clinical environment. Also, because the control group has no access to the data, it might serve as a stronger control. A disadvantage of manipulated assessment is that it depends on whether and how clinicians use the measure, which depends on how positively the measure was received by the clinicians.

In manipulated use, the same assessment information is available for all subjects, but the researcher manipulates the way in which the assessment information is used. For example, one group gets a treatment matched to assessment outcomes, and the other group gets a cross-matched treatment that does not target the problem identified by the assessment. So, in one group, the assessment information is used to match the treatment to the client based on their assessment results, whereas the other group receives a standard treatment regardless of their assessment results. An advantage of this design is that you can be certain that the treatment decisions are explicitly informed by the assessment results because the researcher ensures this, whereas the decision of how the assessment information is used is up to the clinician in manipulated assessment. Disadvantages of manipulated use include that (a) it is not as naturally realistic in terms of how intervention occurs in the real world and (b) you cannot remove the assessment information from clinicians in Group 2 (the group for whom the assessment information is not used for treatment selection) once they have it.

Relatively few measures show evidence of treatment utility of assessment. For instance, a clinical trial using the manipulated assessment approach determined that the Development and Well-Being Assessment (DAWBA) did not show treatment utility as a diagnostic assessment for children and adolescents (Sayal et al., 2025). A review on the treatment utility of assessment is provided by Nelson-Gray, 2003.

Discriminative Validity

Discriminative validity is the degree to which a measure accurately identifies persons placed into groups on the basis of another measure. Discriminative validity is not to be confused with discriminant (divergent) validity and discriminant analysis. A measure shows discriminant validity if it does not correlate with things that it would not be theoretically expected to correlate with. A measure shows discriminative validity, by contrast, if the measure is able to accurately differentiate things (e.g., two groups such as men and women). Discriminant analysis is a model that combines predictors to differentiate between multiple groups.

Elaborative Validity

Elaborative validity involves the extent to which a measure increases our theoretical understanding of the target construct or of neighboring constructs. It deals with a measure's meaningfulness. Elaborative validity is a type of incremental theoretical validity. It is a combination of criterion-related validity and construct validity that examines how much a given measure increases our understanding of a construct's nomological network. However, I am unaware of strong examples of measures that show strong elaborative validity in psychology.

Consequential Validity

Consequential validity is a form of validity that differs from evidential validity, or a test's potential to provide accurate, useful information based on research. Consequential validity takes a more macro-view and deals with the community, sociological, and public policy perspective. Consequential validity evaluates the consequences of our measures beyond the circumstances of their development, based on their impact. Measures can have positive, negative, or mixed effects on society. An example of consequential validity would be asking what the impacts of aptitude testing are on society—how aptitude testing affects society in the long run. Some would argue that, even in cases where the aptitude tests have some degree of evidential validity (i.e., they accurately assess to some degree what they tend to assess), they are consequentially invalid—that is, they have had a net negative effect on society and, due to their low value to society, are invalid. The tests themselves may be fine in terms of their accuracy, but consequential validity says that their validity depends on what we do with the test, i.e., how the test is used.

Another example of consequential validity is when the validity of a measure changes over time due to changes in people's or society's response, as depicted in the source discussion. Judgments and predictions can change the way society reacts and the way people behave so that the predictions become either more or less accurate. A prediction that becomes more accurate as a result of people's response to a prediction is a self-fulfilling prediction or prophecy. For instance, if school staff make a prediction that a child will be held back a grade, the teacher may not provide adequate opportunities for the child to learn, which may lead to the child being more likely to be held back.

Invalidation of a Measure Due to Society's Response to the Use of the Measure.

A prediction that becomes less accurate as a result of people's response to a prediction is a self-canceling prediction. The most effective prediction about a flu outbreak might be one that leads people to safer behavior; therefore lowering flu rates, which does not correspond to the initial prediction (Silver, 2012). Society's response to a measure can invalidate the measure. For example, consider that an organization rates cities based on quality-of-life measures. If quality-of-life indices include the percent of solved cases by the city's police (i.e., the clearance rate), cities may try to improve their ratings of "quality-of-life" by increasing their clearance rate either by increasing the number of cases they mark as solved (such as by marking cases falsely as resolved) or by decreasing the number of cases (such as by investigating fewer cases). That is, cities may "game" the system based on the quality-of-life indices used by various organizations. In this case, the clearance rate becomes a less accurate measure of a city's quality of life.

Another example of a measure that becomes less valid due to society's response is the use of alumni donations as an indicator of the strength of a university that is used for generating university rankings. Such an indicator could lead schools to admit wealthier students who give the most donations, and students whose parents were alumni and provide lavish donations to the university. Yet another example could be standardized testing, where instructors may "teach to the test" such that better performance might not necessarily reflect better underlying competence.

Representational Validity

Representational validity examines the extent to which the items or content of a measure "flesh out" and mirror the true nature and mechanisms of the construct. There are many different types of validity, but many of them are overlapping and can be considered types of other forms of validity. For instance, representational validity is a type of elaborative validity and content validity.

Factorial (Structural) Validity

Factorial validity is examined in the source discussion on factor analysis. Factorial validity considers whether the factor structure of the measure(s) is consistent with the construct. According to factorial validity, if you claim that the measure is unidimensional with four items, factor analysis should support the unidimensionality. Thus, it involves testing empirically whether the measure has the same structure as would be expected based on theory. It is a type of construct validity.

Ecological Validity

Ecological validity examines the extent to which a measure provides scores that are indicative of the behavior of a person in the natural environment.

Process-Focused Validity

Process-focused validity attempts to get closer to the underlying mechanisms. Process-focused validity examines the degree to which respondents engage in a predictable set of psychological processes (which are specified a priori) when completing the measure (Bornstein, 2011; Furr, 2017). These psychological processes include effects of the instrument (e.g., observer versus third-party report versus projective test) as well as effects of the context (e.g., assessment setting, assessment instructions, affect state of the participant, etc.). To determine whether a test is valid in the process-focused technique, one experimentally manipulates variables that moderate the test score–criterion association—to better understand the underlying processes. The ideal outcome is a measure that both (1) has an adequate outcome (correlations where expected) as well as (2) adequate process validity—the psychological processes that one engages in are well hypothesized.

The idea of process-focused validity is that if a measure does not inform us about process or mechanisms, it is not worth doing. Process-focused validity is a type of elaborative validity and construct validity. For instance, consider the common finding that low socioeconomic status is associated with poorer outcomes. To make an impact, process-focused validity would argue that we need to know the mechanisms that underlie this association, and it involves how a measure helps us understand process.

Diagnostic Validity

Diagnostic validity is the extent to which the diagnostic category accurately captures the abnormal phenomenon of interest. It is a form of construct validity for diagnoses.

Social Validity

Social validity involves the extent to which the proposed procedures (assessment, intervention, etc.) will be well-liked and acceptable by those who receive and implement them.

Cultural Validity

Cultural validity refers to "the effectiveness of a measure or the accuracy of a clinical diagnosis to address the existence and importance of essential cultural factors" (Leong & Kalibatseva, 2016). Essential cultural factors may include values, beliefs, experiences, communication patterns, and approaches to knowledge (epistemologies).

Research Design (Experimental) Validity

Aspects of research design validity, also called experimental validity, involve the validity—i.e., the approximate truth of—an inference from a particular research design and approach (Shadish et al., 2002). Research design validity includes internal validity, external validity, and conclusion validity.

Internal Validity

Internal validity is the extent to which causal inference is justified from the research design. This encompasses multiple features including

• temporal precedence—does the (purported) cause occur before the (purported) effect?

• covariation of cause and effect—correlation is necessary (even if insufficient) for causality

• no plausible alternative explanations—such as third variables that could influence both variables and explain their covariation (i.e., confounds)

Even though a cause and its (direct) effect must be related (i.e., covary), it is not necessarily the case—for multiple reasons—that you would detect an association between a cause and an effect. First, the association may be nonlinear. If you are trying to detect an association using Pearson correlation, which assumes a linear association, the (nonlinear) association may go undetected. Second, from a cause to a downstream effect, multiple causal pathways with opposing signs may cancel out in the aggregate. A cause may have both beneficial and harmful effects on the outcome, and the mediating mechanisms can "cancel each other out". For instance, the cause may help via mechanism A, but may hurt via mechanism B. That is, consider if the cause, X, influences the effect, Y, via two mechanisms, A and B (i.e., mediation)—and that X has a positive influence on A (i.e., the association is positive), whereas X has a negative influence on B (i.e., the association is negative). In that case, the effects of the mediating mechanisms can cancel each other out, leaving no observable bivariate association between X and Y. Alternatively, a null association between X and Y can also arise if a confound influences both X and Y in an opposing direction to the effect of X on Y. For example, X may (positively) influence Y via mediator A, whereas B is a confound with a negative effect on Y. In that case, the positive effect through A and the negative effect from B can cancel, so no association may be observed unless the confound is controlled for; failing

to control for the confound is an example of omitted variable bias. So, a null association does not imply no causal effect. Wrongly inferring that a causal effect of one variable on another is not causal is an example of statistical conclusion invalidity arising from a failure to account for the underlying causal mechanisms of X on Y and/or omitted confounds.

There are number of potential threats to internal validity, which are important to consider when designing studies and interpreting findings. Threats to internal validity are other factors that may explain why the outcome could have occurred even if the hypothesized causal effect is noncausal. Examples of potential threats to internal validity include history, maturation, testing, instrumentation, regression, selection, attrition, and an interaction of threats (Slack & Draugalis, 2001; Shadish et al., 2002).

Research designs differ in the extent to which resulting causal inferences show internal validity versus external validity. An experiment is a research design in which one or more variables (independent variables) are manipulated to observe how the manipulation influences the dependent variable. In an experiment, the researcher has greater control over the variables and attempts to hold everything else constant (e.g., by standardization and random assignment). In correlational designs, however, the researcher has less control over the variables. They may be able to statistically account for potential confounds using covariates or for the reverse direction of effect using longitudinal designs. Nevertheless, we can have greater confidence about whether a variable influences another variable in an experiment. Thus, causal inferences from experiments tend to have higher internal validity than causal inferences from than correlational designs.

External Validity

External validity is the extent to which the findings can be generalized across person, place, and time—e.g., to the broader population and the real world. External validity is crucial to consider for studies that intend to make inferences to people outside of those who were assessed. For instance, norm-referenced assessments attempt to identify the distribution of scores for a given population from a sample within that population. The validity of the norms of a norm-referenced assessment are important to consider (Achenbach, 2001). The validity of norms and external validity, more broadly, can depend highly on how representative the sample is of the target population and how appropriate this population is to a given participant. Some measures are known to have poor norms, including the Exner Comprehensive System (Exner, 1974; Exner & Erdberg, 2005) for administering and scoring the Rorschach Inkblot Test, which has been known to over-pathologize (Wood et al., 2001; Wood et al., 2001). The Rorschach is discussed in greater detail in the source discussion. External validity can also depend on whether there is systematic missingness or attrition in the predictor or outcome variables of interest. For instance, if particular subgroups are more likely to drop out or fail to respond, the remaining sample may differ systematically from the intended population, biasing estimates and reducing generalizability. Threats to external validity are described by Shadish et al., 2002 (see Table 3.2).

Tradeoffs of Internal Validity and External Validity It is important to note that there is a tradeoff between internal and external validity—a single research design does not always yield inferences with both high internal and high external validity. Some research designs are better suited for making causal inferences, whereas other designs tend to be better suited for making inferences that generalize to the real world. The research design that is best suited to making causal inferences is an experiment, where the researcher

manipulates one variable (the independent variable) and holds all other variables constant to see how a change in the independent variable influences the outcome (dependent variable). Thus, experiments tend to yield inferences with higher internal validity than other research designs. However, by manipulating one variable and holding everything else constant, the research takes place in a very standardized fashion that can become like studying a process in a vacuum. So, even if a process is theoretically causal in a vacuum, it may act very differently in the real world when it interacts with other processes.

Inferences from observational designs have greater capacity for external validity than experimental designs because people can be observed in their natural environments to see how variables are related in the real world. However, the greater external validity comes at a cost of lower internal validity. Observational designs are not well-positioned to make causal inferences because they have multiple threats to internal validity, including issues of temporal precedence in cross-sectional observational designs, and there are numerous potential third variables (i.e., confounds) that could act as a common cause of both the predictor and outcome variables. Thus, just because two variables are associated does not necessarily mean that they are causally related.

As the internal validity of inferences from a study's design increases, the external validity of its inferences tend to decrease. The greater control we have over variables (and, therefore, have greater confidence about causal inferences), the lower the likelihood that the findings reflect what happens in the real world because it is studying things in a metaphorical vacuum. Because no single research design always yields inferences with both high internal and external validity, scientific inquiry needs a combination of many different research designs so we can be more confident in our inferences—experimental designs for making causal inferences and observational designs for making inferences that are more likely to reflect the real world.

Case studies, because they have smaller sample sizes, tend to have lower external validity than both experimental and observational studies. Case studies also tend to yield inferences with lower internal validity because they have less potential to control for threats to internal validity, such as potential confounds or temporal precedence. Nevertheless, case studies can still be useful for generating hypotheses that can then be tested empirically with a larger sample in experimental or observational studies.

(Statistical) Conclusion Validity

Conclusion validity, also called statistical conclusion validity, considers the extent to which conclusions are reasonable about the association among variables based on the data. That is, were the correct statistical analyses performed, and are the interpretations of the findings from those analyses correct? Threats to conclusion validity are described by Shadish et al., 2002 (see Table 2.2). Did you have adequate statistical power, adequate reliability of measures, accurate model estimation, and proper treatment of missing data? Did you not violate assumptions of the statistical tests used? Did any characteristics of the data distort the results (e.g., outliers, missingness, nonnormality, restriction of range, etc.)? Did you avoid fishing for significant results (i.e., p-hacking) and inflating the Type I error rate by performing multiple tests? Did you avoid making causal inferences from correlational data? Were effect sizes estimated accurately? To what extent are participants heterogeneous? If a treatment was administered, was it implemented with fidelity for all participants? To what extent was there extraneous variance in the research setting that may have contributed to error variance? All of these considerations are important for ensuring that the conclusions drawn from a study are valid.

Putting It All Together: An Organizational Framework

There are many types of measurement validity, but the central psychometric aspect of measurement validity is construct validity. That is, whether the measure accurately assesses the target construct is the most important consideration of measurement validity. As discussed earlier, construct validity includes the nomological network of the construct. Construct validity also subsumes other key types of measurement validity, including

• Convergent validity

• Discriminant (divergent) validity

• Criterion-related validity

• Concurrent validity

• Predictive validity

• Content validity

The organization of types of measurement validity that are subsumed by construct validity is depicted in the source discussion.

Organization of Types of Measurement Validity That Are Subsumed by Construct Validity.

Moreover, many different types of reliability and validity can be viewed through the lens of construct validity:

• Internal consistency, which can be estimated as the coefficient of internal consistency, where the criterion for criterion-related validity is other items on the same measure

• Test–retest reliability, which can be estimated as the coefficient of stability, where the criterion for criterion-related validity is the same measure at another time point

• Parallel-forms reliability or convergent validity, which can be estimated as the coefficient of equivalence, where the criterion for criterion-related validity is the parallel form

Validity Is a Process, Not an Outcome

Validity (and validation) is a continual process, not an outcome. Validation is never complete. When evaluating the validity of a measure, we must ask: Validity for what and to what degree? We would not just say that a measure is or is not valid. We would express the strength of evidence on a measure's validity across the different types of validity for a particular purpose, with a particular population, in a particular context (consistent with generalizability theory).

Reliability Versus Validity

Reliability and validity are not the same thing. Reliability deals with consistency, whereas validity deals with accuracy. A measure can be consistent but not accurate (see the source discussion). That is, a measure can be reliable but not valid. However, a measure cannot be accurate if it is inconsistent; that is, a measure cannot be valid and unreliable.

Traditional Depiction of Reliability (Consistency) Versus Validity (Accuracy).

The typical way of depicting the distinction between reliability and validity is in the source discussion, in which a measure can either have (a) low reliability and low validity, (b) high reliability and low validity, or (c) high reliability and high validity. However, it can be worth thinking about validity in terms of accuracy at the person level versus group level. When we distinguish between person- versus group-level accuracy, we can distinguish four general combinations of reliability and validity, as depicted in the source discussion:, n.d. (a) low reliability, low accuracy at the person level, and low accuracy at the group level, (b) low reliability, low accuracy at the person level, and high accuracy at the group level, (c) high reliability, low accuracy at the person level, and low accuracy at the group level, and (d) high reliability, high accuracy at the person level, and high accuracy at the group level. However, as discussed before, reliability and validity are not binary states of low versus high—they exist to varying degrees.

Depiction of Reliability Versus Validity, While Distinguishing Between Validity (Accuracy) at the Person Versus Group Level.

Even though reliability and validity are not the same thing, there is a relation between reliability and validity. Validity depends on reliability. Reliability is necessary but insufficient for validity. However, measurement error (unreliability) can be systematic or random. If measurement error is systematic, it reduces the validity of the measure. If measurement error is random, it reduces the precision of an individual's score on a measure, but the measure could still be a valid measure of the construct at the group level. However, random error would make it more difficult to make an accurate prediction for an individual person.

Reliability places the upper bound on validity because a measure can be no more valid than it is reliable. In other words, a measure should not correlate more highly with another variable than it correlates with itself. Therefore, the maximum validity coefficient is the square root of the product of the reliability of each measure, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

So, the maximum validity coefficient is based on the reliability of each measure. To the extent that one of the measures is unreliable, the validity coefficient will be attenuated relative to the true validity (i.e., the true strength of association of the constructs), as we describe below.

Effect of Measurement Error on Associations

the source discussion depicts the classical test theory approach to understanding the validity of a measure, i.e., its association with another measure, which is the validity coefficient (r xy).

The Criterion-Related Validity of a Measure, i.e., Its Association With Another Measure, as Depicted in a Path Diagram.

As described above, (random) measurement error weakens (or attenuates) the association between variables (Goodwin & Leech, 2006; Schmidt & Hunter, 1996). The greater the random measurement error, the weaker the association. Thus, the correlation between x and y depends on both the true correlation of x and y (r x ty t) and the reliabilities of x (r xx) and y (r yy). So, measurement error in x and y can reduce the observed correlation below the true correlation. This is known as the attenuation formula (the source discussion):

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The lower the reliability, the greater the attenuation of the validity coefficient relative to the true association between the constructs.

All of these r values (excluding the true correlation) are just estimates unless the sample size is infinite, so the observed association is an imperfect estimate. Hence, we need a correction for this attenuation (Schmidt & Hunter, 1996). This correction for the attenuation of an association due to measurement error (unreliability) is known as the disattenuation of a correlation, i.e., correction for the attenuation of an association due to measurement error to get a more accurate estimate of the true association between constructs. Rearranging the terms from the attenuation formula, the formula for disattenuation of a correlation (i.e., the disattenuation formula) is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

All of this is implied in the path diagram (see the source discussion). The attenuation and disattenuation formulas are based on classical test theory, and therefore assume that all measurement error is random, that errors are uncorrelated, etc. Nevertheless, the attenuation formula can be informative for understanding how imperiled your research is when your measures have low reliability (i.e., when there is instability in the measure). Unreliability is a strong threat to statistical power—i.e., being able to detect an effect if it exists. Many people do not account for reliability of the measures when calculating statistical power, but you should if you want your study to be adequately powered! For instance, one could determine what the expected observed association would be given the expected true association and the expected reliability of the measures, and then use that expected observed association to calculate statistical power. Researchers recommend accounting for measurement reliability, to better estimate the association between

constructs, either with the disattenuation formula (Schmidt & Hunter, 1996) or with structural equation modeling, as described in the next chapter.

Although unreliability tends to weaken bivariate associations, its effects are less predictable in the context of three or more variables, such as when including covariates in multiple regression (Shadish et al., 2002). Unreliability of a covariate has shown, in some cases, to lead to artificially stronger associations when between the predictor and outcome variables (Shadish et al., 2002).

Observed Association (After Adding Measurement Error)

Using simulated data, when the reliability of the predictor is [computed value omitted in this course adaptation], the reliability of the criterion is [computed value omitted in this course adaptation], and the true association between the predictor and criterion is r = [computed value omitted in this course adaptation], the observed association is attenuated to r = [computed value omitted in this course adaptation].

Attenuation of True Correlation Due to Measurement Error

The attenuation formula is presented in the source discussion. We extend it to a specific example in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where x = measure of construct \ A; y = measure of construct \ B.

Below is an example of how to find the observed correlation between the predictor and criterion if the true association between the constructs (i.e., correlation between true scores of constructs) is.70, the reliability of the predictor is.90, and the reliability of the criterion is.85:

The petersenlab package (Petersen, 2025) contains the petersenlab::attenuationCorrelation function that estimates the observed association given the true association and the reliability of the predictor and criterion:

The observed association (r = [computed value omitted in this course adaptation]) is attenuated relative to the true association (r =.70).

Disattenuation of Observed Correlation Due to Measurement Error

The disattenuation formula is presented in the source discussion. We extend it to a specific example in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where x = measure of construct \ A; y = measure of construct \ B

Find the true association between the construct assessed by the predictor and the construct assessed by the criterion given an observed association if the reliability of the predictor is.9, and the reliability of the criterion is.85:

The observed (attenuated) association is as follows:

The true (disattenuated) association is as follows:

The petersenlab package (Petersen, 2025) contains the petersenlab::disattenuationCorrelation function that estimates the observed association given the true association and the reliability of the predictor and criterion:

The disattenuation of an observed correlation due to measurement error can be demonstrated using structural equation modeling. For instance, consider the following observed association:

The observed association can be estimated in structural equation modeling in the lavaan package (R-lavaan, n.d.) using the following syntax:

A path diagram of the observed association is depicted in the source discussion.

Now consider when we account for the degree of unreliability of each measure. We can account for the (un)reliability of each measure by specifying the residual errors as 1 - reliability, as below:

A path diagram of the true association is depicted in Figures (the source discussion) and (the source discussion).

Disattenuation of an Observed Association Due to Measurement Error, as Depicted in a Structural Equation Model.

The observed association (beta = [computed value omitted in this course adaptation]) becomes beta = [computed value omitted in this course adaptation] when it is disattenuated for measurement error.

Generalizability Theory (G-Theory)

Generalizability theory (G-theory) is discussed in greater detail in the chapter on reliability in the source discussion and in the chapter on generalizability theory. As a brief reminder, G-theory is a measurement theory that, unlike classical test theory, does not treat all measurement differences across time, rater, or situation as "error" but rather as a phenomenon of interest. G-theory can simultaneously consider multiple aspects of reliability and validity in the same model, something that classical test theory cannot achieve.

Ways to Increase Validity

Here are potential ways to increase the validity of the interpretation of a measure's scores for a particular purpose:

• Make sure the measure's scores are reliable.

For potential ways to increase the reliability of measurement, see the source discussion. But do not switch to a less valid measure or to items that are less valid merely because they are more reliable.

• Use multiple measures and multiple methods to remedy the effects of method bias.

• Design the measure with a particular population and purpose in mind.

When describing the measure in papers or in public spheres, make it clear to others what the population and intended purposes are and what they are not.

• Make sure each item's scores are valid, based on theory and empiricism, for the particular population and purpose.

For instance, the items should show content validity—the items should assess facets of the target construct for the target population as defined by experts, without item intrusions from other constructs. The items' scores should show convergent validity—the items' scores should be related to other measures of the construct, within the population of interest. The items' scores should show discriminant validity—the items' scores should be more strongly related to measures that are intended to assess the same construct than they are to measures that are intended to assess other constructs.

• Combine the items with latent variable modeling, such as with item response theory (IRT), structural equation modeling (SEM), or factor analysis, to reduce the contribution of random error and to disattenuate associations for measurement error.

• Obtain samples that are as representative of the population as possible, paying attention to including people who are traditionally under-represented in research (if such groups are part of the target population).

• Make sure that people in the population can understand, interpret, and respond to each item in a meaningful and comparable way.

• Make sure the measure and its items are not biased against any subgroup within the population of interest.

Test bias is discussed in the source discussion.

• Be careful to administer the measure to the population of interest under the conditions in which it is designed.

If the measure must be administered to people from a different population or under different conditions from which it was designed, be careful to (a) note that the measure was not designed to be administered for these other populations or conditions, and (b) note that individuals' scores may not accurately reflect their level on the construct. If interpretations are made based on these scores, make them cautiously and say how the differences in population or condition may have influenced the scores.

• Continue to monitor the validity of the measure's scores for the given population and purpose.

The validity of measures' scores can change over time for a number of reasons. Cohort effects can lead items to become obsolete over time. If people or organizations change their behavior in response to a measure, this can invalidate a measure's scores for the intended purpose, as described in the source discussion when discussing consequential validity.

Conclusion

Validity is how much accuracy, utility, and meaningfulness the interpretation of a measure's scores have for a particular purpose. Like reliability, validity is not one thing. There are multiple aspects of validity. Validity is also not a characteristic that resides in a test. The validity of a measure's scores reflect an interaction of the properties of the test with the population for whom it is designed and the sample and context in which it is administered. Thus, when reporting validity in papers, it is important to adequately describe the aspects of validity that have been considered and the population, sample, and context in which the measure is assessed.

Suggested Readings

Cronbach & Meehl, 1955; Clark & Watson, 2019

Questions

Note: Several of the following questions use data from the Children of the National Longitudinal Survey of Youth (CNLSY). The CNLSY is a publicly available longitudinal data set provided by the Bureau of Labor Statistics (https://www.bls.gov/nls/nlsy79-children.htm#topical-guide; archived at https://perma.cc/EH38-HDRN). The CNLSY data file for these exercises is located on this book's page of the Open Science Framework (https://osf.io/3pwza). Children's behavior problems were rated in 1988 (time 1: T1) and then again in 1990 (time 2: T2) on the Behavior Problems Index (BPI).

1. What is the criterion-related validity of the Antisocial Behavior subscale of the BPI in relation to the Hyperactive subscale of the BPI?

2. Assume the true correlation between two constructs, antisocial behavior and hyperactivity, is r =.8.

And assume the reliability of the measure of antisocial behavior is r =.7, and the reliability of the measure of hyperactivity is r =.7090234. According to the attenuation formula (that attenuates the association due to measurement error), what would be the correlation between measures of antisocial behavior and hyperactivity that we would actually observe?

1. Assume the true correlation between two constructs, antisocial behavior and hyperactivity, is r =.8, the reliability of the Antisocial Behavior subscale of the BPI is r =.7, and the reliability of the measure of Hyperactive subscale of the BPI is r =.8.

According to the disattenuation formula (to correct for attenuation of the association due to measurement error), what would be the true association between the constructs of antisocial behavior and hyperactivity?

1. You are interested in whether a child's levels of anxiety/depression can explain unique variance in children's hyperactivity above and beyond their level of antisocial behavior.

Is the child's level of anxiety/depression (as rated on the Anxiety/Depression subscale of the BPI) significantly associated with the child's level of hyperactivity (as rated on the Hyperactive subscale of the BPI) above and beyond the variance accounted for by the child's level of antisocial behavior (as rated on the Antisocial Behavior subscale of the BPI). How much unique variance in hyperactivity is accounted for by their anxiety/depression?

1. In the source discussion, we simulated data for a multitrait-multimethod matrix. The simulated data includes data on participants' verbal, spatial, and quantitative abilities, each assessed in three subtests in each of three methods: oral, written, and manipulative.

a. Provide a multitrait-multimethod matrix of the data from the first subtest of each trait-by-method combination (VO1, SO1, QO1, VW1, SW1, QW1, VM1, SM1, QM1). Assume the reliability of the variables assessed orally is.7, the reliability of the variables assessed using the written method is.8, and the reliability of variables assessed using the manipulative method is.9. b. Interpret the multitrait-multimethod matrix you just created. c. What is the Heterotrait-Monotrait (HTMT) ratio for the measures of the verbal (VO1, VW1, VM1) and spatial (SO1, SW1, SM1) constructs? What does this indicate?

Answers

1. The criterion-related validity is r = [computed value omitted in this course adaptation].

2. The observed correlation would be r = [computed value omitted in this course adaptation].

3. The true association would be r = [computed value omitted in this course adaptation].

4. Yes, the child's level of anxiety/depression is significantly associated with their hyperactivity above and beyond their antisocial behavior (F[df = [computed value omitted in this course adaptation]] = [computed value omitted in this course adaptation], p less than.001).

The child's level of anxiety/depression accounted for [computed value omitted in this course adaptation]\% unique variance in hyperactivity above and beyond their antisocial behavior.
5. a. The multitrait-multimethod matrix is below:

Multitrait-Multimethod Matrix.

5. b. The convergent correlations (green cells) are statistically significant (ps less than.05) and moderate in magnitude (.45 less than rs less than.53), supporting the convergent validity of the measures. [The reliabilities of these measures are not provided, so we are not able to compare the magnitude of the convergent validities to the magnitude of reliability to see the extent to which the convergent validities may be attenuated due to measurement unreliability]. Evidence of discriminant validity is supported by three findings: First, the convergent correlations (green cells: rs =.46–.53) are stronger than the heteromethod-heterotrait correlations (pink cells: rs =.24–.34). Second, the convergent correlations (green cells: rs =.46–.53) are stronger than the discriminant correlations (orange cells: rs =.30–.43). Third, the patterns of intercorrelations between traits are the same, regardless of which measurement method is used. Verbal, spatial, and quantitative skills are intercorrelated for every measurement method used. c. The HTMT ratio for the measures of the verbal and spatial constructs is [computed value omitted in this course adaptation]. The HTMT ratio is the average of the heterotrait-heteromethod correlations, relative to the average of the monotrait-heteromethod correlations. Given that the HTMT ratio is considerably less than 1 (and less than the common cutoff of.85), it indicates that the monotrait-heteromethod correlations are considerably larger than the heterotrait-heteromethod correlations. Thus, the HTMT provides evidence that the measures of verbal and spatial constructs show discriminant validity.

CHAPTER 08 / CAC.903

Test Bias, Fairness, and Equity

Fairness is not the same as identical administration. A fair assessment gives each child a reasonable opportunity to show the construct being measured, and it uses scores in ways that are justified for that child. Bias occurs when a measure or decision systematically disadvantages a group for reasons unrelated to the construct. In child assessment, bias can affect identification, placement, discipline, and access to services.

Bias versus group differences

Group differences in average scores do not automatically prove bias. Differences may reflect real differences in exposure, opportunity, or construct levels. Bias is present when the score means something different across groups or when prediction differs. For example, if the same score predicts a different outcome for two groups, the test may have predictive bias. The counselor should examine evidence, not assume either bias or neutrality.

Predictive bias and measurement invariance

Predictive bias asks whether a score predicts a criterion equally across groups. Measurement invariance asks whether the construct is measured on the same scale across groups. Configural invariance means the same structure. Metric invariance means similar factor loadings. Scalar invariance means similar intercepts, allowing mean comparisons. Without scalar invariance, group mean comparisons may be misleading.

These concepts are technical, but the practical question is simple: can we compare scores fairly? If evidence is lacking, the counselor should avoid high-stakes comparisons and use multiple methods. The report should state limitations.

Adverse impact and fairness

Adverse impact occurs when a decision rule disproportionately affects a group. A cutoff may produce more referrals for one group. This does not automatically mean the test is biased, but it requires investigation. The counselor should ask whether the cutoff is justified, whether alternatives exist, and whether the decision pathway includes safeguards. Fairness includes the opportunity to learn, the quality of instruction, and the consequences of labeling.

Reducing bias

Bias can be reduced by using culturally appropriate measures, qualified interpreters, dynamic assessment, multiple informants, and local norms when justified. Training helps counselors notice stereotypes. Structured decision rules reduce arbitrary judgment. Feedback loops allow correction. The counselor should also advocate for systemic changes when assessment reveals inequity, such as unequal access to advanced coursework.

Score adjustment and its limits

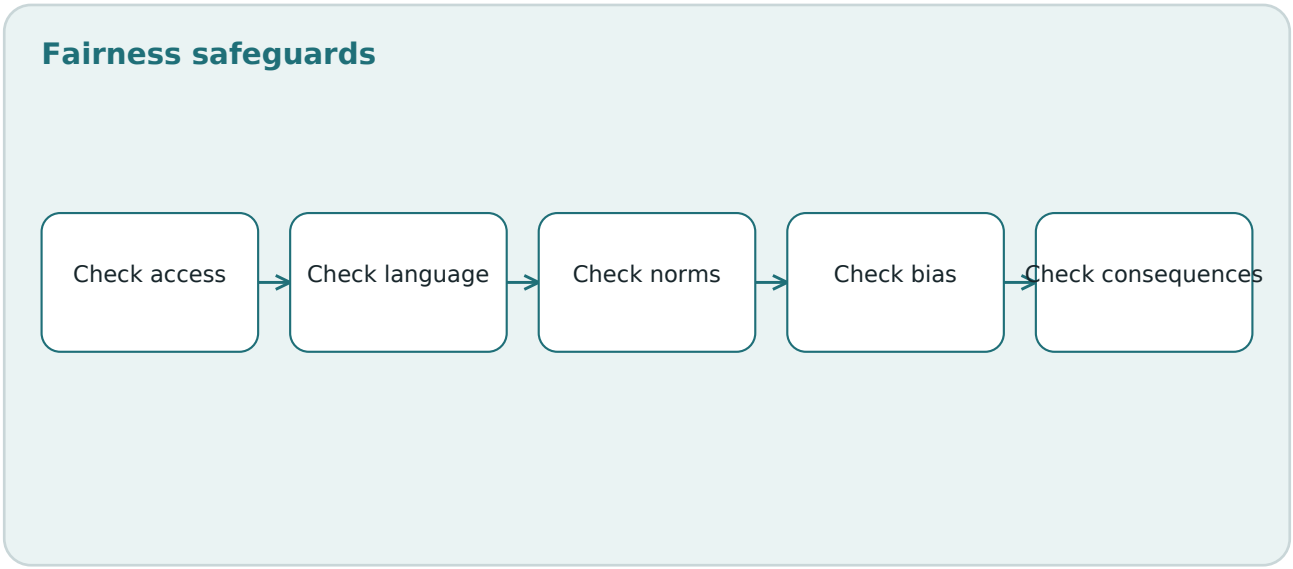
Some propose adjusting scores to correct bias. This is controversial. Adjusting scores may hide real differences in opportunity or may create new unfairness. The counselor should not adjust scores casually. Instead, the counselor should interpret scores in context, use multiple data sources, and address environmental barriers. If a test is invalid for a group, the ethical response may be to avoid using it for high-stakes decisions.

Real-life application

A district finds that English learners are overrepresented in emotional disturbance classifications. The assessment team reviews language access, trauma exposure, and instructional quality. They add bilingual interviews, classroom observation, and review of educational opportunity. Referrals become more accurate. The problem was not only the test. It was the pathway.

Reflection

Identify a decision in your setting that uses assessment scores. Ask who may be disadvantaged. List two safeguards you could add. Fairness is not a single statistical test. It is an ongoing responsibility.



Fairness requires safeguards across the whole assessment pathway.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module test-bias.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Overview of Bias

There are multiple definitions of the term "bias" depending on the context. In general, bias is a systematic error (Reynolds & Suzuki, 2012). Mean error is an example of systematic error, and is sometimes called bias. Cognitive biases are systematic errors in thinking, including confirmation bias and hindsight bias. Method biases are a form of systematic error that involve the influence of measurement on a person's score that is not due to the person's level on the construct. Method biases include response biases or response styles, including acquiescence and social desirability bias. Attentional bias refers to the tendency to process some types of stimuli more than others.

Sometimes bias is used to refer in particular to systematic error (in measurement, prediction, etc.) as a function of group membership, where test bias refers to the same score having different meaning for different groups. Under this meaning, a test is unbiased if a given test score has the same meaning regardless of group membership. For example, a test is biased if there is differential validity of test scores for groups (e.g., age, education, culture, race, sex). Test bias would exist, for instance, if a test is a less valid predictor for racial minorities or linguistic minorities. Test bias would also exist if scores on the Scholastic Aptitude Test (SAT) under-estimate women's grades in college, for instance.

There are some known instances of test bias, as described in the source discussion. Research has not produced much empirical evidence of test bias (Brown et al., 1999; Hall et al., 1999; Jensen, 1980; Kuncel & Hezlett, 2010; Reynolds & Suzuki, 2012; Reynolds et al., 2021; Sackett & Wilk, 1994; Sackett et al., 2008; Woo et al., 2023), though some item-level bias is not uncommon. Moreover, where test bias has been observed, it is often small, unclear, and does not always generalize (Cole, 1981). However, just because there is not much empirical evidence of test bias does not mean that test bias does not exist. Moreover, just because a test does not show bias does not mean that it should be used. Furthermore, just because a test does not show bias does not mean that there are not race-, social class-, and gender-related biases in clinical judgment during the assessment process.

It is also worth pointing out that group differences in scores do not necessarily indicate bias. Group differences in scores could reflect true group differences in the construct. For

instance, women have better verbal abilities, on average, compared to men. So, if women's scores on a verbal ability test are higher on average than men's scores, this would not be sufficient evidence for bias.

There are two broad categories of test bias:

1. predictive bias

2. test structure bias

Predictive bias refers to differences between groups in the relation between the test and criterion. As with all criterion-related validity tests, the findings depend on the strength and quality of the criterion. Test structure bias refers to differences in the internal test characteristics across groups.

Predictive Bias

Predictive bias exists when differences emerge between groups in terms of predictive validity to a criterion. It is assessed by using a regression line looking at the association between test score and job performance. For instance, consider a 2x2 confusion matrix used for the standard prediction problem. A confusion matrix for whom to select for a job is depicted in the source discussion.

2x2 Confusion Matrix for Job Selection. TP = true positive; TN = true negative; FP = false positive; FN = false negative.

We can also visualize the confusion matrix in terms of a scatterplot of the test scores (i.e., predicted job performance) and the "truth" scores (i.e., actual job performance), as depicted in the source discussion. The predictor (test score) is on the x-axis. The criterion (job performance) is on the y-axis. The quadrants reflect the cutoffs (i.e., thresholds) imposed from the 2x2 confusion matrix. The vertical line reflects the cutoff for selecting someone for a job. The horizontal line reflects the cutoff for good job performance (i.e., people who should have been selected for the job).

The data points in the top right quadrant are true positives: people who the test predicted would do a good job and who did a good job. The data points in the bottom left quadrant are true negatives: people who the test predicted would do a poor job and who would have done a poor job. The data points in the bottom right quadrant are false positives: people who the test predicted would do a good job and who did a poor job. The data points in the top left quadrant are false negatives: people who the test predicted would do a poor job and who would have done a good job.

the source discussion depicts a strong predictor. The best-fit regression line has a steep slope where there are lots of data points that are true positives and true negatives, with relatively few false positives and false negatives.

the source discussion depicts a poor predictor. The best-fit regression line has a shallow slope where there are just as many data points that are in the false cells (false positives and false negatives) as there are in the true cells (true positives and true negatives). In general, the steeper the slope, the better the predictor.

We can evaluate predictive bias using a best-fit regression line between the predictor and criterion for each group.

Types of Predictive Bias

There are three types of predictive bias:

1. Different slopes

2. Different intercepts

3. Different intercepts and slopes

The slope of the regression line is the steepness of the line. The intercept of the regression line is the y-value of the point where the line crosses the y-axis (i.e., when x = 0). If a measure shows predictive test bias, when looking at the regression line for each group, the groups' regression lines differ in either slopes and/or intercepts.

Different Slopes Predictive bias in terms of different slopes exists when there are differences in the slope of the regression line between minority and majority groups. The slope describes the direction and steepness of the regression line. The slope of a regression line is the amount of change in y for every unit change in x (i.e., rise over run). Differing slopes indicate differential predictive validity, in which the test is a more effective predictor of performance in one group over the other. Different slopes predictive bias is depicted in the source discussion. In the figure, the predictor performs well in the majority group. However, the slope is close to zero in the minority group, indicating that there is no association between the predictor and the criterion for the minority group.

Test Bias: Different Slopes. TP = true positive; TN = true negative; FP = false positive; FN = false negative.

Different slopes can especially occur if we develop our measure and criterion based on the normative majority group. Not much evidence has found empirical evidence of different slopes across groups. However, samples often do not have the power to detect differing slopes (Aguinis et al., 2010). Theoretically, to fix biases related to different slopes, you should find another measure that is more predictive for the minority group. If the predictor is a strong predictor in both groups but shows slight differences in the slope, within-group norming could be used.

Different Intercepts Predictive bias in terms of different intercepts exists when there are differences in the intercept of the regression line between minority and majority groups. The y-intercept describes the point on the y-axis that the line intersects with the y-axis (when x = 0). When the distributions have similar slopes, intercept differences suggest that the measure systematically under- or over-estimates group performance relative to the person's ability. The same test score leads to systematically different predictions for the majority and minority groups. In other words, minority group members get different tests scores than majority group members with the same ability. Different intercepts predictive bias is depicted in the source discussion.

Test Bias: Different Intercepts. TP = true positive; TN = true negative; FP = false positive; FN = false negative.

A higher intercept (relative to zero) indicates that the measure under-estimates a person's ability (at that test score)—i.e., the person's job performance is better than what the test score would suggest. A lower intercept (relative to zero) indicates that the measure over-estimates a person's ability (at that test score)—i.e., the person's job performance is

worse than what the test score would suggest. the source discussion indicates that the measure systematically under-estimates the job performance of the minority group.

Performance among members of a minority group could be under- or over-estimated. For example, historically, women's grades in math and engineering classes tended to be under-estimated by the Scholastic Aptitude Test (Clark & Grandy, 1984). However, where intercept differences have been observed, measures often show small over-estimation of school and job performance among minority groups (Reynolds & Suzuki, 2012). For example, women's physical strength and endurance is over-estimated based on physical ability tests (Sackett & Wilk, 1994). In addition, over-estimation of African Americans' and Hispanics' school and job performance has been observed based on cognitive ability tests (Cole, 1981; Reynolds & Suzuki, 2012; Sackett & Wilk, 1994; Sackett et al., 2008). At the same time, the Black–White difference in job performance is less than the Black–White difference in test performance.

The over-prediction of lower-scoring groups is likely mostly an artifact of measurement error (Gottfredson, 1994). The over-estimation of African Americans' and Hispanics' school and job performance may be due to measurement error in the tests. Moreover, test scores explain only a portion of the variation in job performance. Black individuals tend to be far less disadvantaged on the noncognitive determinants of job performance than on the cognitive ones. Nevertheless, the over-estimation that has been often observed is on average—the performance is not over-estimated for all individuals of the groups even if there is an average over-estimation effect. In addition, simulation findings indicate that lower intercepts (i.e., over-estimation) among minority groups compared to majority groups could be observed if there are different slopes but not different intercepts in the population, because different slopes are likely to go undetected due to low power (Aguinis et al., 2010). That is, if a test shows weaker validity for a minority group than the majority group, it could appear as different intercepts that favor the minority group when, in fact, it reflects shallower slopes of the minority group that go undetected.

Predictive biases in intercepts could especially occur if we develop tests that are based on the majority group, and the items assess constructs other than the construct of interest which are systematically biased in favor of the majority group or against the minority group. Arguments about reduced power to detect differences are less relevant for intercepts and means than for slopes.

To correct for a bias in intercepts, we could add bonus points to the scores for the minority group to correct for the amount of the systematic error, and to result in the same regression line. But if the minority group is over-predicted (as has often been the case where intercept differences have been observed), we would not want to use score adjustment to lower the minority group's scores.

Different Intercepts and Slopes Predictive bias in terms of different intercepts and slopes exists when there are differences in the intercept and slope of the regression line between minority and majority groups. In cases of different intercepts and slopes, there is both differential validity (because the regression lines have different slopes), as well as varying under- and over-estimation of groups' performance at particular scores. Different intercepts and slopes predictive bias is depicted in the source discussion.

Test Bias: Different Intercepts and Slopes. TP = true positive; TN = true negative; FP = false positive; FN = false negative.

In instances of different intercepts and slopes predictive bias, a measure can simultaneously over-estimate and under-estimate a person's ability at different test scores. For instance, a measure can under-estimate a person's ability at higher test scores and can over-estimate a person's ability at lower test scores.

Different intercepts and slopes across groups is possibly more realistic than just different intercepts or just different slopes. However, different intercepts and slopes predictive bias is more complicated to study, represent, and resolve. It is difficult to examine because of complexity, and it is not easy to fix. Currently, we have nothing to address different intercepts and slopes predictive bias. We would need to use a different measure or measures for each group.

Test Structure Bias

In addition to predictive bias, another type of test bias is test structure bias. Test structure bias involves differences in internal test characteristics across groups. Examining test structure bias is different from examining the total score, as is used when examining predictive bias. Test structure bias can be identified empirically or based on theory/judgment.

Empirically, test structure bias can be examined in multiple ways.

Empirical Approaches to Identification

In general, it is important to ensure that scores are on the same number line—and are thus able to be meaningfully compared—across groups. To empirically ensure that scores are on the same number line across groups, we must ensure two things:

• the scores have the same scale across groups—scale refers to how much the score changes per unit on the construct (number line)—that is, if scores have the same scale across groups, movement along the number line is equivalent.

• the scores have the same location across groups—location refers to the origin or zero point on the number line—that is, if scores have the same location across groups, the number lines have the same origin.

Two measures with different scales are depicted in the source discussion.

Two Measures (i.e., Rulers) With Different Scales.

Two measures with different locations are depicted in the source discussion.

Two Measures (i.e., Rulers) With Different Locations.

The concepts of location and scale for various model types are provided in the source discussion.

Model	Location	Scale	
Regression	intercept	slope	
Confirmatory Factor Analysis	intercept/threshold	factor loading	

Model	Location	Scale	
Item Response Theory	difficulty/severity	discrimination	: Concepts of Location and Scale in Various Model Types.

Item × Group tests (ANOVA) Item × Group tests in analysis of variance (ANOVA) examine whether the difference between groups on the overall score match comparisons among smaller items sets between groups. Item × Group tests are used to rule out that items are operating in different ways in different groups. If the items operate in different ways in different groups, they do not have the same meaning across groups. For example, if we are going to use a measure for multiple groups, we would expect its items to operate similarly across groups. So, if women show higher scores on a depression measure compared to men, would also expect them to show similar elevations on each item (e.g., sleep loss).

Item Response Theory Using item response theory, we can examine differential item functioning (DIF). Evidence of DIF, indicates that there are differences between group in terms of discrimination and/or difficulty/severity of items. Differences between groups in terms of the item characteristic curve (which combines the item's discrimination and severity) would be evidence against construct validity invariance between the groups and would provide evidence of bias. DIF examines stretching and compression of different groups. As an example, consider the item "bites others" in relation to externalizing problems. The item would be expected to show a weaker discrimination and higher severity in adults compared to children. DIF is discussed in the source discussion.

Confirmatory Factor Analysis Confirmatory factor analysis allows tests of measurement invariance (also called factorial invariance). Measurement invariance examines whether the factor structure of the underlying latent variables in the test is consistent across groups. It also examines whether the manifestation of the construct differs between groups. Measurement invariance is discussed in the source discussion.

Even if you find the same slope and intercepts across groups in a prediction model, the measure would still be assessing different constructs across groups if the measure has a different factor structure between the groups. A different factor structure across groups is depicted in the source discussion.

Different Factor Structure Across Groups.

An example of a different factor structure across groups is the differentiation of executive functions from two factors to three factors (inhibition, working memory, cognitive flexibility) across childhood (Lee et al., 2013).

There are different degrees of measurement invariance (Putnick & Bornstein, 2016):

• Configural invariance: same number of factors in each group, and which indicators load on which factors are the same in each group (i.e., the same pattern of significant loadings in each group).

• Metric ("weak factorial") invariance: items have the same factor loadings (discrimination) in each group.

• Scalar ("strong factorial") invariance: items have the same intercepts (difficulty/severity) in each group.

• Residual ("strict factorial") invariance: items have the same residual/unique variances in each group.

Structural Equation Modeling Structural equation modeling is a confirmatory factor analysis (CFA) model that incorporates prediction. Structural equation modeling allows examining differences in the underlying structure with differences in prediction in the same model.

Signal Detection Theory Signal detection theory is a dynamic measure of bias. It allows examining the overall bias in selection systems, including both accuracy and errors at various cutoffs (sensitivity, specificity, positive predictive value, and negative predictive value), as well as accuracy across all possible cutoffs (the area under the receiver operating characteristic curve). While there may be similar predictive validity between groups, the type of errors we are making across groups might differ. It is important to decide which types of error to emphasize depending on the fairness goals and examining sensitivity/specificity to adjust cutoffs.

Empirical Evidence of Test Structure Bias It is not uncommon to find items that show differences across groups in severity (intercepts) and/or discrimination (factor loadings). However, cross-group differences in item functioning tend to be small and not consistent across studies, suggesting that some of the differences may reflect Type I errors that result from sampling error and multiple testing. That said, some instances of cross-group differences in item parameters could reflect test structure bias that is real and important to address.

Facial Validity Bias Facial validity bias considers the extent to which an average person thinks that an item is biased—i.e., the item has differing validity between minority and majority groups. If so, the item should be reconsidered. Does an item disfavor certain groups? Is the language specific to a particular group? Is it offensive to some people? This type of judgment moves into the realm of whether or not an item should be used.

Content Validity Bias Content validity bias is determined by judgments of construct experts who look for items that do not do an adequate job assessing the construct between groups. A construct may include some content facets in one group, but may include different content facets in another group, as depicted in the source discussion.

Different Content Facets in a Given Construct for Two Groups.

Examples include information questions and vocabulary questions on the Wechsler Adult Intelligence Scale. If an item is linguistically complicated, grammatically complex or convoluted, or a double negative, it may be less valid or predictive for rural populations and those with less education.

Also, stereotype threat may contribute to content validity bias. Stereotype threat occurs when people are or feel at risk of conforming themselves to stereotypes about their social group, thus leading them to show poorer performance in ways that are consistent with the stereotype. Stereotype threat may partially explain why some women may perform more poorly on some math items than some men.

Another example of content validity bias is when the same measure is used to assess a construct across ages even though the construct shows heterotypic continuity. Heterotypic continuity occurs when a construct changes in its behavioral manifestation with development (Petersen et al., 2020). That is, the same construct may look different at different points in development. An example of a construct that shows heterotypic continuity is externalizing problems. In early childhood, externalizing problems often manifest in overt forms, including physical aggression (e.g., biting) and temper tantrums. By contrast, in adolescence and adulthood, externalizing problems more often manifest in covert ways, including relational aggression and substance use. Content validity and facial validity bias judgments are often related, but not always.

Examples of Bias

As described in the overview in the source discussion, there is not much empirical evidence of test bias (Brown et al., 1999; Hall et al., 1999; Jensen, 1980; Kuncel & Hezlett, 2010; Reynolds & Suzuki, 2012; Reynolds et al., 2021; Sackett & Wilk, 1994; Sackett et al., 2008; Woo et al., 2023). That said, some item-level bias is not uncommon. One instance of test bias is that, historically, women's grades in math and engineering classes tended to be under-estimated by the Scholastic Aptitude Test (Clark & Grandy, 1984). Fernández & Abe, 2018 review the evidence on other instances of test and item bias. For instance, test bias can occur if a subgroup is less familiar with the language, the stimulus material, or the response procedures, or if they have different response styles. In addition to test bias, there are known patterns of bias in clinical judgment, as described in the source discussion.

Test Fairness

There is interest in examining more than just the accuracy of measures. It is also important to examine the errors being made and differentiate the weight or value of different kinds of errors (and correct decisions). Consider an example of an unbiased test, as depicted in the source discussion, adapted from Gottfredson, 1994. Although the example is of a White group and a Black group, we could substitute any two groups into the example (e.g., males versus females).

Potential Unfairness in Testing. The ovals represent the distributions of individuals' performance both on a test and a job performance criterion. TP = true positive; TN = true negative; FP = false positive; FN = false negative. (Adapted from Gottfredson, 1994, Figure 1, p. 958. Gottfredson, L. S. (1994). The science and politics of race-norming. American Psychologist, 49(11), 955–963. https://doi.org/10.1037/0003-066X.49.11.955)

The example is of an unbiased test between White and Black job applicants. There are no differences between the two groups in terms of slope. If we drew a regression line, the line would go through the centroid of both ovals. Thus, the measure is equally predictive in both groups even though that the Black group failed the test at a higher rate than the White group. Moreover, there is no difference between the groups in terms of intercept. Thus, the performance of one group is not over-estimated relative to the performance of the other group. To demonstrate what a different intercept would look like, Group X shows a different intercept. In sum, there is no predictive validity bias between the two groups. But just because the test predicts just as well in both groups does not mean that the selection procedures are fair.

Although the test is unbiased, there are differences in the quality of prediction: there are more false negatives in the Black group compared to the White group. This gives the White group an advantage and the Black group additional disadvantages. If the measure showed the same quality of prediction, we would say the test is fair. The point of the example is that just because a test is unbiased does not mean that the test is fair.

There are two kinds of errors: false negatives and false positives. Each error type has very different implications. False negatives would be when the test predicts that an applicant would perform poorly and we do not give them the job even though they would have performed well. False negatives have a negative effect on the applicant. And, in this example, there are more false negatives in the Black group. By contrast, false positives would be when we predict that an applicant would do well, and we give them the job but they perform poorly. False positives are a benefit to the applicant but have a negative effect on the employer. In this example, there are more false positives in the White group, which is an undeserved benefit based on the selection ratio; therefore, the White group benefits.

In sum, equal accuracy of prediction (i.e., equal total number of errors) does not necessarily mean the test is fair; we must examine the types of errors. Merely ensuring accuracy does not ensure fairness!

Adverse Impact

Adverse impact is defined as rejecting members of one group at a higher rate than another group. Adverse impact is different from test validity. According to federal guidelines, adverse impact is present if the selection rate of one group is less than four-fifths (80%) the selection rate of the group with the highest selection rate.

There is much more evidence of adverse impact than test bias. Indeed, disparate impact of tests on personnel selection across groups is the norm rather than the exception, even when using valid tests that are unbiased, which in part reflect group-related differences in job-related skills (Gottfredson, 1994). Examples of adverse impact include:

• physical ability tests, which produce substantial adverse impact against women (despite over-estimation of women's performance),

• cognitive ability tests, which produce substantial impact against some ethnic minority groups, especially Black and Hispanic people (despite over-estimation of Black and Hispanic people's performance), even though cognitive ability tests tend to be among the strongest predictors of job performance (Sackett et al., 2008; Schmidt & Hunter, 1981), and

• personality tests, which produce higher estimates of dominance among men than women; it is unclear whether this has predictive bias.

Bias Versus Fairness

Whether a measure is accurate or shows test bias is a scientific question. By contrast, whether a test is fair and thus should be used for a given purpose is not just a scientific question; it is also an ethical question. It involves the consideration of the potential consequences of testing in terms of social values and consequential validity.

Operationalizing Fairness

There are many perspectives to what should be considered when evaluating test fairness (American Educational Research Association et al., 2014; Camilli, 2013; Dorans, 2017; Committee on the General Aptitude Test Battery et al., 1989; Fletcher et al., 2021; Gipps & Stobart, 2009; Helms, 2006; Jonson & Geisinger, 2022; Melikyan et al., 2019; Sackett et al., 2008; Thorndike, 1971; Zieky, 2006; Zieky, 2013). As described in Fletcher et al., 2021, there are three primary ways of operationalizing fairness:

1. Equal outcomes: the selection rate is the same across groups.

2. Equal opportunity: the sensitivity (true positive rate; 1 - false negative rate) is the same across groups.

3. Equal odds: the sensitivity is the same across groups and the specificity (true negative rate; 1 - false positive rate) is the same across groups.

For example, the job selection procedure shows equal outcomes if the proportion of men selected is equal to the proportion of women selected. The job selection procedure shows equal opportunity if, among those who show strong job performance, the proportion of classification errors (false negatives) is the same for men and women. Receiver operating characteristic (ROC) curves are depicted for two groups in the source discussion. A cutoff that represents equal opportunity is depicted with a horizontal line (i.e., the same sensitivity) in the source discussion. The job selection procedure shows equal odds if (a), among those who show strong job performance, the proportion of classification errors (false negatives) is the same for men and women, and (b), among those who show poor job performance, the proportion of classification errors (false positives) is the same for men and women. A cutoff that represents equal odds is depicted where the ROC curve for Group A intersects with the ROC curve from Group B in the source discussion. The equal odds approach to fairness is consistent with a National Academy of Sciences committee on fairness (Committee on the General Aptitude Test Battery et al., 1989; Gottfredson, 1994). Approaches to operationalizing fairness in the context of prediction models are described by Paulus & Kent, 2020.

Receiver Operating Characteristic (ROC) Curves for Two Groups. (Figure reprinted from Fletcher et al., 2021, Figure 2, p. 3. Fletcher, R. R., Nakeshimana, A., & Olubeko, O. (2021). Addressing fairness, bias, and appropriate use of artificial intelligence and machine learning in global health. Frontiers in Artificial Intelligence, 3(116). https://doi.org/10.3389/frai.2020.561802)

It is not possible to meet all three types of fairness simultaneously (i.e., equal selection rates, sensitivity, and specificity across groups) unless the base rates are the same across groups or the selection is perfectly accurate (Fletcher et al., 2021). In the medical context, equal odds is the most common approach to fairness. However, using the cutoff associated with equal odds typically reduces overall classification accuracy. And, changing the cutoff for specific groups can lead to negative consequences. In the case that equal odds results in a classification accuracy that is too low, it may be worth considering using separate assessment procedures/tests for each group. In general, it is best to follow one of these approaches to fairness. It is difficult to get right, so try to minimize negative impact. Many fairness supporters argue for simpler rules. In the 1991 Civil Rights Act, score adjustments

based on race, gender, and ethnicity (e.g., within-race norming or race-conscious score adjustments) were made illegal in personnel selection (Gottfredson, 1994).

Another perspective to fairness is that selection procedures should predict job performance and if they are correlated with any group membership (e.g., race, socioeconomic status, or gender), the test should not be used (Helms, 2006). That is, according to Helms, we should not use any test that assesses anything other than the construct of interest (job performance). Unfortunately, however, no measures like this exist. Every measure assesses multiple things, and factors such as poverty can have long-lasting impacts across many domains.

Another perspective to fairness is to make the selection procedures equal the number of successes within each group (Thorndike, 1971). According to this perspective, if you want to do selection, you should hire all people, then look at job performance. If among successful employees, 60% are White and 40% are Black, then set this selection rate for each group (i.e., hiring 80% White individuals and 20% Black individuals is not okay). According to this perspective, a selection system is only fair if the majority–minority differences on the selection device used are equal in magnitude to majority–minority differences in job performance. Selection criteria should be made based on prior distributions of success rates. However, you likely will not ever really know the true base rate in these situations. No one uses this approach because you would have a period where you have to accept everyone to find the percent that works. Also, this would only work in a narrow window of time because the selection pool changes over time.

There are lots of groups and subgroups. Ensuring fairness is very complex, and there is no way to accomplish the goal of being equally fair to all people. Therefore, do the best you can and try to minimize negative impact.

What to Do When Detecting Bias

When examining item bias (using differential item functioning/DIF or measurement non-invariance) with many items (or measures) across many groups, there can be many tests, which will make it likely that DIF/non-invariance will be detected, especially with a large sample. Some detected DIF may be artificial or trivial, but other DIF may be real and important to address. It is important to consider how you will proceed when detecting DIF/non-invariance. Considerations of effect size and theory can be important for evaluating the DIF/non-invariance and whether it is negligible or important to address.

When detecting bias, there are several steps to take. First, consider what the bias indicates. Does the bias present adverse impact for a minority group? For what reasons might the bias exist? Second, examine the effect size of the bias. If the effects are small, if the bias does not present adverse impact for a minority group, and if there is no compelling theoretical reason for the bias, the bias might not be sufficient to scrap the instrument for the population. Some detected bias may be artificial, but other bias may be real. Gender and cultural differences have shown a number of statistically significant effects for a number of different assessment purposes, but many of the observed effects are quite small and likely trivial, and they do not present compelling reasons to change the assessment (Youngstrom & Van Meter, 2016).

However, if you find bias, correct for it! There are a number of score adjustment and non-score adjustment approaches to correct for bias, as described in Sections (the source discussion) and (the source discussion). If the bias occurs at the item level (e.g., test

structure bias), it is generally recommended to remove or resolve items that show non-negligible bias. There are three primary options: (1) drop the item for both groups, (2) drop the item for one group but keep it for the other group, or (3) freely estimate the parameters for the item across groups. Addressing items that show larger bias can also reduce artificial bias in other items (Hagquist & Andrich, 2017). Thus, researchers are encouraged to handle item bias sequentially from high to low in magnitude. If the bias occurs at the test score level (e.g., predictive bias), score adjustments may be considered.

There are situations measurement invariance across time may not be expected (and may thus be problematic to require). For instance, if there is a treatment that occurs between two measurement occasions, the factor structure might change due to the treatment rather than due to the mere passage of time. Thus, it would be helpful to know whether the instrument shows measurement invariance across a similar timeframe when no treatment is occurring. Another situation where measurement invariance across time may not be expected is when the construct shows heterotypic continuity (Petersen et al., 2020). If the construct changes in its behavioral manifestation with development, then measurement invariance may not be expected across wide spans of time.

If you do not correct for bias, consider the impact of the test, procedure, and selection procedure when interpreting scores. Interpret scores with caution and provide necessary caveats in resulting papers or reports regarding the interpretations in question. In sum, it is important to examine the possibility of bias—it is important to consider how much "erroneous junk" you are introducing into your research.

Score Adjustment to Correct for Bias

Score adjustment involves adjusting scores for a particular group or groups.

Why Adjust Scores?

There may be several reasons to adjust scores for various groups in a given situation. First, there may be social goals to adjust scores. For example, we may want our selection device to yield personnel that better represent the nation or region, including diversity of genders, races, majors, social classes, etc. Score adjustments are typically discussed with respect to racial minority differences due to historical and systemic inequities. Our society aims to provide equal opportunity, including the opportunity to gain a fair share (i.e., proportional representation) of jobs. A diversity of perspectives in a job is a strength; a diversity of perspectives can lead to greater creativity and improved problem-solving. A second potential reason that we may want to apply score adjustment is to correct for bias. A third potential reason that we may want to apply score adjustment is to improve the fairness of a test.

Types of Score Adjustment

There are a number of potential techniques that have been used in attempts to correct for bias, i.e., to reduce negative impact of the test on an under-represented group. What is considered an under-represented group may depend on the context. For instance, men are under-represented compared to women as nurses, preschool teachers, and college students. However, men may not face the same systemic challenges compared to women, so even though men may show under-representation in some domains, it is arguable whether scores should be adjusted to increase their representation. Techniques for score adjustment include:

• Bonus points

• Within-group norming

• Separate cutoffs

• Top-down selection from different lists

• Banding

• Banding with bonus points

• Sliding band

• Separate tests

• Item elimination based on group differences

Bonus Points Providing bonus points involves adding a constant number of points to the scores of all individuals who are members of a particular group with the goal of eliminating or reducing group differences. Bonus points is used to correct for predictive bias differences in intercepts between groups. An example of bonus points is military veterans in placement for civil service jobs—points are added to the initial score for all veterans (e.g., add 5 points to test scores of all veterans). An example of using bonus points as a score adjustment is depicted in the source discussion.

Using Bonus Points as a Scoring Adjustment.

There are several pros of bonus points. If the distribution of each group is the same, this will effectively reduce group differences. Moreover, it is a simple way of impacting test selection and procedure without changing the test, which is therefore a great advantage. There are several cons of bonus points. If there are differences in group standard deviations, adding bonus points may not actually correct for bias. The use of bonus points also obscures what is actually being done to scores, so other methods like using separate cutoffs may be more explicit. In addition, the simplicity of bonus points is also a great disadvantage because it is easily understood and often not viewed as "fair" because some people are getting extra points that others do not.

Within-Group Norming A norm is the standard of performance that a person's performance can be compared to. Within-group norming treats the person's group in the sample as the norm. Within-group norming converts an individual's score to standardized scores (e.g., T scores) or percentiles within one's own group. Then, the people are selected based on the highest standard scores across groups. Withing-group norming is used to correct for predictive bias differences in slopes between groups. An example of using within-group norming as a score adjustment is depicted in the source discussion.

Using Within-Group Norming as a Scoring Adjustment.

There are several pros of within-group norming. First, it accounts for differences in group standard deviations and means, so it does not have the same problem as bonus points and is generally more effective at eliminating adverse impact compared to bonus points. Second, some general (non-group-specific) norms are clearly irrelevant for characterizing a person's functioning. Group-specific norms aim to describe a person's performance relative to people with a similar background, thus potentially reducing cultural bias. Third, group-specific norms may better reflect cultural, educational, socioeconomic, and other factors that may

influence a person's score (Burlew et al., 2019). Fourth, group-specific norms may increase specificity, and reduce over-pathologizing by preventing giving a diagnosis to people who might not show a condition (Manly & Echemendia, 2007).

There are several cons of within-group norming. First, group differences could be maintained if one decides to norm based on a reference sample or, when scores are skewed, a local sample, especially when using standardized scores. However, percentile scores will consistently eliminate adverse impact. Second, using group-specific norms may obscure background variables that explain underlying reasons for group-related differences in test performance (Manly, 2005; Manly & Echemendia, 2007). Third, group-specific norms do not address the problem if the measure shows test bias (Burlew et al., 2019). Fourth, group-specific norms may reduce sensitivity to detect conditions (Manly & Echemendia, 2007). For instance, they may prevent people from getting treatment who would benefit. It is worth noting that within-group norming on the basis of sex, gender, and ethnicity is illegal for the basis of personnel selection according to the 1991 Civil Rights Act.

As an example of within-group norming, the National Football League used to use race-norming for identification of concussions. The effect of race-norming, however, was that it lowered Black players' concussion risk scores, which prevented many Black players from being identified as having sustained a concussion and from receiving needed treatment. Race-norming compared the Black football players cognitive test scores to group-specific norms: the cognitive test scores of Black individuals in the general population (not to common norms). Using Black-specific norms assumed that Black football players showed lower cognitive ability than other groups, so a low cognitive ability score for a Black player was less likely to be flagged as concerning. Thus, the race-specific norms led to lower identified rates of concussions among Black football players compared to White football players. Due to the adverse impact, Black players sued the National Football League, and the league stopped the controversial practice of race-norming for identification of concussion (https://www.washingtonpost.com/sports/2021/06/03/nfl-concussion-settlement-race-norming/; archived at https://perma.cc/KN3L-5Z7R).

A common question is whether to use group-specific norms or common norms. Group-specific norms are a controversial practice, and the answer depends. If you are interested in a person's absolute functioning (e.g., for determining whether someone is concussed or whether they are suitable to drive), recommendations are to use common norms, not group-specific norms (Barrash et al., 2010; Silverberg & Millis, 2009). If, by contrast, you are interested in a person's relative functioning compared to a specific group, within-group norming could make sense if there is an appropriate reference group. The question about which norms to use are complex, and psychologists should evaluate the cost and benefit of each norm, and use the norm with the greatest benefit and the least cost for the client (Manly & Echemendia, 2007).

Separate Cutoffs Using separate cutoffs involves using a separate cutoff score per group and selecting the top number from each group. That is, using separate cutoffs involves using different criteria for each group. Using separate cutoffs functions the same as adding bonus points, but it has greater transparency—i.e., you are lowering the standard for one group compared to another group. An example of using separate cutoffs as a score adjustment is depicted in the source discussion.

Using Separate Cutoffs as a Scoring Adjustment. In this example, the cutoff for the majority group is 128; the cutoff for the minority group is 123.", fig.scap = "Using Separate Cutoffs as a Scoring Adjustment.

Top-Down Selection from Different Lists Top-down selection from different lists involves taking the best from two different lists according to a preset rule as to how many to select from each group. Top-down selection from different lists functions the same as within-group norming. An example of using top-down selection from different lists as a score adjustment is depicted in the source discussion.

Using Top-Down Selection From Different Lists as a Scoring Adjustment. In this example, the top three candidates are selected from each group.

Banding Banding uses a tier system that is based on the assumption that individuals within a specific score range are regarded as having equivalent scores. So that we do not over-estimate small score differences, scores within the same band are seen as equivalent—and the order of selection within the band can be modified depending on selection goals. The standard error of measurement (SEM) is used to estimate the precision (reliability) of the test scores, and it is used as the width of the band.

Consider an example: if a person received a score with confidence interval of 18–22, then scores between 18 to 22 are not necessarily different due to random fluctuation (measurement error). Therefore, scores in that range are considered the same, and we take a band of scores. However, banding by itself may not result in increased selection of lower scoring groups. The band provides a subsample of applicants so that we can use other criteria (other than the test) to select a candidate. Giving "minority preference" involves selecting members of minority group in a given band before selecting members of the majority group. An example of using banding as a score adjustment is depicted in the source discussion.

Using Banding as a Scoring Adjustment.

The problem with banding is that bands are set by the standard error of measurement: you can select the first group from the first band, but then whom do you select after the first band? There is no rationale where to "stop" the band because there are indistinguishable scores on the edges of each band to the next band. That is, 17 is indistinguishable from 18 (in terms of its confidence interval), 16 is indistinguishable from 17, and so on. Therefore, banding works okay for the top scores, but if you are going to hire a lot of candidates, it is a problem. A solution to this problem with banding is to use a sliding band, as described later.

Banding with Bonus Points Banding is often used with bonus points to reduce the negative impact for minority groups. An example of using banding with bonus points as a score adjustment is depicted in the source discussion.

Using Banding With Bonus Points as a Scoring Adjustment.

Sliding Band Using a sliding band is a solution to the problem of which bands to use when using banding. Using a sliding band can help increase the number of minorities selected. Using the top band, you select all members of a minority group in the top band, then select members of the majority group with the top score of the band, then slide the band down (based on SEM), and repeat. You work your way down with bands though groups

that are indistinguishable based on SEM, until getting a cell needed to select a relevant candidate.

For instance, if the top score is 22 and the SEM is 4 points, the first band would be: [18, 22]. Here is how you would proceed:

1. Select the minority group members who have a score between 18 to 22.

2. Select the majority group members who have a score of 22.

3. Slide the band down based on the SEM to the next highest score: [17, 21].

4. Select the minority group members who have a score between 17 to 21.

5. Select the majority group members who have a score of 21.

6. Slide the band down based on the SEM to the next highest score: [16, 20].

1....

1. And so on

An example of using a sliding band as a score adjustment is depicted in the source discussion.

Using a Sliding Band as a Scoring Adjustment.

In sum, using a sliding band, scores that are not significantly lower than the highest remaining score should not be treated as different. Using a sliding band has the same effects on decisions as bonus points that are the width of the band. For example, if the SEM is 3, it has the same decisions as bonus points of 3; therefore, any scores within 3 of the highest score are now considered equal.

A sliding band is popular because of its scientific and statistical rationale. Also, it is more confusing and, therefore, preferred by some because it may be less likely to be sued. However, a sliding band may not always eliminate adverse impact. A sliding band has never been overturned in court (or at least, not yet).

Separate Tests Using separate tests for each group is another option to reduce bias. For instance, you might use one test for the majority group and a different test for the minority group, making sure that each test is valid for the relevant group. Using separate tests is an extreme version of top-down selection and within-group norming. Using separate tests would be an option if a measure shows different slopes predictive bias.

One way of developing separate tests is to use empirical keying by group: different items for each group are selected based on each item's association with the criterion in each group. Empirical keying is an example of dustbowl empiricism (i.e., relying on empiricism rather than theory). However, theory can also inform the item selection.

Item Elimination based on Group Differences Items that show large group differences in scores can be eliminated from the test. If you remove enough items showing differences between groups, you can get similar scores between groups and can get equal group selection. A problem of item elimination based on group differences is that if you get rid of predictive items, then two goals, equal selection and predictive power, are not met. If you use this method, you often have to be willing for the measure to show decreases in predictive power.

Use of Score Adjustment

Score adjustment can be used in a number of different domains, including tests of aptitude and intelligence. Score adjustment also comes up in other areas. For example, the number of drinks it takes to be considered binge drinking differs between men (five) and women (four). Although the list of score adjustment options is long, they all really reduce to two ways:

1. Bonus points

2. Within-group norming

Bonus points and within-group norming are the techniques that are most often used in the real world. These techniques differ in their degree of obscurity—i.e., confusion that is caused not for scientific reasons, but for social, political, and dissemination and implementation reasons. Often procedures that are hard to understand are preferred because it is hard to argue against, critique, or game the system. Basically, you have two options for score adjustment. One option is to adjust scores by raising scores in one group or lowering the criterion in one group. The second primary option is to renorm or change the scores. In sum, you can change the scores, or you can change the decisions you make based on the scores.

Other Ways to Correct for Bias

Because score adjustment is controversial, it is also important to consider other potential ways to correct for bias that do not involve score adjustment. Strategies other than score adjustment to correct for bias are described by Sackett et al., 2001.

Use Multiple Predictors

In general, high-stakes decisions should not be made based on the results from one test. So, for instance, do not make hiring decisions based just on aptitude assessments. For example, college admissions decisions are not made just based on SAT scores, but also one's grades, personal statement, extracurricular activities, letters of recommendation, etc. Using multiple predictors works best when the predictors are not correlated with the assessment that has adverse impact, which is difficult to achieve.

There are larger majority–minority subgroup differences in verbal and cognitive ability tests than in noncognitive skills (e.g., motivation, personality, and interpersonal skills). So, it is important to include assessment of relevant noncognitive skills. Include as many relevant aspects of the construct as possible for content validity. For a job, consider as many factors as possible that are relevant for success, e.g., cognitive and noncognitive abilities.

Change the Criterion

Another option is to change the criterion so that the predictive validity of tests is less skewed. It may be that the selection instrument is not biased but the way in which we are thinking about selection procedures is biased. For example, for judging the quality of universities, there are many different criteria we could use. It could be valuable to examine the various criteria, and you might find what is driving adverse effects.

Remove Biased Items

Using item response theory or confirmatory factor analysis, you can identify items that function differently across groups (i.e., differential item functioning/DIF or measurement non-invariance). For instance, you can identify items that show different discrimination/factor loadings or difficulty/intercepts by group. You do not just want to remove items that show mean-level differences in scores (or different rates of endorsement) for one group than another, because there may be true group differences in their level on particular items. If an item is clearly invalid in one group but valid in another group, another option is to keep the item in one group, and to remove it in another group.

Be careful when removing items because removing items can lead to poorer content validity—i.e., items may no longer be a representative set of the content of the construct. Removing items also reduces a measure's reliability and ability to detect individual differences (Hagquist & Andrich, 2017; Hagquist, 2019). DIF effects tend to be small and inconsistent; removing items showing DIF may not have a big impact.

Resolve Biased Items

Another option, for items identified that show differential item functioning using IRT or measurement non-invariance using CFA, is to resolve instead of remove items. Resolving items involves allowing an item to have a different discrimination/factor loading and/or difficulty/intercept parameter for each group. Allowing item parameters to differ across groups has a very small effect on reliability and person separation, so it can be preferable to removing items (Hagquist & Andrich, 2017; Hagquist, 2019).

Use Alternative Modes of Testing

Another option is to use alternative modes of testing. For example, you could use audio or video to present test items, rather than requiring a person to read the items, or write answers. Typical testing and computerized exams are oriented toward the upper-middle class, which is therefore a procedure problem! McClelland's (McClelland, 1973) argument is that we need more real-life testing. Real-life testing could help address stereotype threats and the effects of learning disabilities. However, testing in different modalities could change the construct(s) being assessed.

Use Work Records

Using work records is based on McClelland's (McClelland, 1973) argument to use more realistic and authentic assessments of job-relevant abilities. Evidence on the value of work records for personnel selection is mixed. In some cases, use of work records can actually increase adverse impact on under-represented groups because the primary group typically already has an idea of how to get into the relevant job or is already in the relevant job; therefore, they have a leg up. It would be acceptable to use work records if you trained people first and then tested, but no one spends the time to do this.

Increase Time Limit

Another option is to allot people more testing time, as long as doing so does not change the construct. Time limits often lead to greater measurement error because scores conflate pace and quality of work. Increasing time limits requires convincing stakeholders that job performance is typically not "how fast you do things" but "how well you do them"—i.e., that time does not correlate with outcome of interest. The utility of increasing time limits depends on the domain. In some domains, efficiency is crucial (e.g., medicine, pilot). Increasing time limits is not that effective in reducing group differences, and it may actually increase group differences.

Use Motivation Sets

Using motivation sets involves finding ways to increase testing motivation for minority groups. It is probably an error to think that a test assesses just aptitude; therefore, we should also consider an individual's motivation to test. Thus, part of the score has to do with ability and some of the score has to do with motivation. You should try to maximize each examinee's motivation, so that the person's score on the measure better captures their true ability score. Motivation sets could include, for example, using more realistic test stimuli that are clearly applicable to the school or job requirements (i.e., that have face validity) to motivate all test takers.

Use Instructional Sets

Using instructional sets involves coaching and training. For instance, you could inform examinees about the test content, provide study materials, and recommend test-taking strategies. This could narrow the gap between groups because there is an implicit assumption that the primary group already has "light" training. Using instructional sets aims to reduce error variance due to test anxiety, unfamiliar test format, and poor test-taking skills.

Giving minority groups better access to test preparation is based on the assumption that group differences emerge because of different access to test preparation materials. This could theoretically help to systematically reduce test score differences across groups. Standardized tests like the SAT/GRE/LSAT/GMAT/MCAT, etc. embrace coaching/training. For instance, the organization ETS gives training materials for free. After training, scores on standardized tests show some but minimal improvement. In general, training yields some improvement on quantitative subscales but minimal change on verbal subscales. However, the improvements tend to apply across groups, and they do not seem to lessen group differences in scores.

Load Data

cnlsy is a subset of a data set from the Children of the National Longitudinal Survey of Youth Survey (CNLSY). The CNLSY is a publicly available longitudinal data set provided by the Bureau of Labor Statistics (https://perma.cc/EH38-HDRN). The CNLSY data file for these examples is located on the book's page of the Open Science Framework (https://osf.io/3pwza).

Simulate Data

For reproducibility, I set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Add Missing Data

Adding missing data to dataframes helps make examples more realistic to real-life data and helps you get in the habit of programming to account for missing data.

HolzingerSwineford1939 is a data set from the lavaan package (R-lavaan, n.d.) that contains mental ability test scores (x1–x9) for seventh- and eighth-grade children.

Unbiased test where males and females have equal means on predictor and criterion

the source discussion depicts an example of an unbiased test where males and females have equal means on the predictor and criterion. The test is unbiased because there are no significant differences in the regression lines (of predictor predicting criterion) between males and females.

Unbiased test where females have higher means than males on predictor and criterion

the source discussion depicts an example of an unbiased test where females have higher means than males on the predictor and criterion. The test is unbiased because there are no differences in the regression lines (of predictor predicting criterion) between males and females.

Unbiased test where males have higher means than females on predictor and criterion

the source discussion depicts an example of an unbiased test where males have higher means than females on the predictor and criterion. The test is unbiased because there are no differences in the regression lines (of predictor predicting criterion) between males and females.

Example of unbiased prediction (no differences in intercepts or slopes)

the source discussion depicts an example of an unbiased test where males and females have equal means on the predictor and criterion. The test is unbiased because there are no differences in the regression lines (of predictor predicting criterion1) between males and females.

Example of intercept bias

the source discussion depicts an example of a biased test due to intercept bias. There are differences in the intercepts of the regression lines (of predictor predicting criterion2) between males and females: males have a higher intercept than females. That is, the same score on the predictor results in higher predictions for males than females.

Example of slope bias

the source discussion depicts an example of a biased test due to slope bias. There are differences in the slopes of the regression lines (of predictor predicting criterion3) between males and females: males have a higher slope than females. That is, scores have stronger predictive validity for males than females.

Example of intercept and slope bias

the source discussion depicts an example of a biased test due to intercept and slope bias. There are differences in the intercepts and slopes of the regression lines (of predictor predicting criterion4) between males and females: males have a higher slope than females. That is, scores have stronger predictive validity for males than females. Females have a higher intercept than males. That is, at lower scores, the same score on the predictor results in higher predictions for females than males; at higher scores on the predictor, the same score results in higher predictions for males than females.

Example of different measurement reliability/error across groups

In the example depicted in the source discussion, there are differences in the measurement reliability/error on the criterion between males and females: males' scores have a lower reliability (higher measurement error) on the criterion than females' scores. That is, we are more confident about a female's level on the criterion given a particular score on the predictor than we are about a male's level on the criterion given a particular score on the predictor.

Differential Item Functioning (DIF)

Differential item functioning (DIF) indicates that one or more items functions differently across groups. That is, one or more of the item parameters for an item differs across groups. For instance, the severity or discrimination of an item could be higher in one group compared to another group. If an item functions differently across groups, it can lead to biased scores for particular groups. Thus, when observing group differences in level on the measure or group differences in the measure's association with other measures, it is unclear whether the observed group differences reflect true group differences or differences in the functioning of the measure across groups.

For instance, consider an item such as "disobedience to authority", which is thought to reflect externalizing behavior in childhood. However, in adulthood, disobedience to authority could reflect prosocial functions, such as protesting against societally unjust actions, and may show weaker construct validity with respect to externalizing problems, as operationalized by a weaker discrimination coefficient in adulthood than in childhood. Tests of differential item functioning are equivalent to tests of measurement invariance.

Approaches to addressing DIF are described in the source discussion. If we identify that an item shows non-negligible DIF, we have three primary options (described above): (1) drop the item for both groups, (2) drop the item for one group but keep it for the other group, or (3) freely estimate the parameters (discrimination and difficulty) for the item across groups.

Tests of DIF were conducted using the mirt package (R-mirt, n.d.).

Model Summary

Items that appear to differ:

• items 5 and 6 are more discriminating (a parameter) among females than males

• item 4 has higher difficulty/severity (b 1 and b 2 parameters) among females than males

Constrained Model

Constrain item parameters to be equal across groups (to use as baseline model for identifying DIF).

Compare model fit of constrained model to unconstrained model

The constrained model and the unconstrained model are considered "nested" models. The constrained model is nested within the unconstrained model because the unconstrained model includes all of the terms of the constrained model along with additional terms. Model fit of nested models can be compared with a chi-square difference test.

The constrained model fits significantly worse than the unconstrained model, which suggests that item parameters differ between males and females.

Identify DIF by iteratively removing constraints from fully constrained model

One way to identify DIF is to iteratively remove constraints from a model in which all parameters are constrained to be the same across groups. Removing constraints allows individual items to have different item parameters across groups, to identify which items yield a significant improvement in model fit when allowing their discrimination and/or severity to differ across groups. Items 1, 3, 4, 5, 6, and 7 showed DIF in discrimination and/or severity, based on a significant chi-square difference test.

Items that differ in discrimination

Items 1, 3, 4, and 5 showed DIF in discrimination, based on a significant chi-square difference test.

Items that differ in severity

Items 1, 3, 4, 5, and 7 showed DIF in difficulty/severity, based on a significant chi-square difference test.

Items that differ in discrimination and/or severity

Another way to identify DIF is to iteratively add constraints to a model in which all parameters are allowed to differ across groups. Adding constraints forces individual items to have the same item parameters across groups, to identify which items yield a significant worsening in model fit when constraining their discrimination and/or severity to be the same across groups. Items 1, 2, 3, 4, 5, and 6 showed DIF in discrimination and/or severity, based on a significant chi-square difference test. The DIF in discrimination and/or severity is depicted in the source discussion.

Items that differ in discrimination

Items 6 and 7 showed DIF in discrimination, based on a significant chi-square difference test. The DIF in discrimination is depicted in the source discussion.

Items that differ in severity

Items 1, 2, 3, 4, 5, and 6 showed DIF in difficulty/severity, based on a significant chi-square difference test. The DIF in difficulty/severity is depicted in the source discussion

Compute effect size of DIF

Effect size measures of DIF were computed based on expected scores (Meade, 2010) using the mirt package (R-mirt, n.d.). Some researchers recommend using a simulation-based procedure to examine the impact of DIF on screening accuracy (Gonzalez & Pelham, 2021).

Test-level DIF

In addition to consideration of differential test functioning, we can also consider whether the test as a whole (i.e., the collection of items) differs in its functioning by group—called differential test functioning. Differential test functioning is depicted in the source discussion. Estimates of differential test functioning are in the source discussion.

In general, the measure showed greater difficulty for females than for males. That is, at a given construct level, males were more likely than females to endorse the items. Otherwise said, it takes a higher construct level for females to obtain the same score on the measure as males.

Item-level DIF

Differential item functioning is depicted in the source discussion. Estimates of differential item functioning are in the source discussion.

The measure-level differences in functioning appear to be largely driven by greater difficulty of items 4 and 5 for females than males. That is, at a given construct level, males were more likely than females to endorse items 4 and 5, in particular. Otherwise said, it takes a higher construct level for females than males to endorse items 4 and 5.

Item plots

Plots of item response category characteristic curves by sex are in Figures (the source discussion)–(the source discussion) below.

Plots of item information functions by sex are in Figures (the source discussion)–(the source discussion) below. Items 1, 5, and 6 showed greater information for females than for males. Items 2, 4, and 7 showed greater information for females than for males.

Plots of expected item scores by sex are are in Figures (the source discussion)–(the source discussion) below.

Addressing DIF

Based on the analyses above, item 5 shows the largest magnitude of DIF, followed by item 4. Specifically, item 5 has a stronger discrimination parameter for women than men. So, we should handle item 5 first. If we deem the DIF for item 5 to be non-negligible, we have three primary options: (1) drop item 5 for both men and women, (2) drop item 5 for men but keep it for women, or (3) freely estimate the parameters (discrimination and difficulty) for item 5 across groups.

Drop item for both groups

The first option is to drop item 5 for both men and women groups. You might do this if you want to use a measure that has only those items that function equivalently across groups. However, this can lead to lower reliability and weaker ability to detect individual differences.

Drop item for one group but not another group

A second option is to drop item 5 for men but to keep it for women. You might do this if the item is invalid for men, but still valid for women. The coefficients show an item parameter for item 5 for men, but it was constrained to be equal to the parameter for women, and data were removed from the estimation of item 5 for men.

Freely estimate item to have different parameters across groups

Alternatively, we can resolve DIF by allowing the item to have different parameters (discrimination and difficulty) across both groups. You might do this if the item is valid for both men and women, and it has importantly different item parameters nonetheless.

Compare model fit

We can compare the model fit against the fully constrained model. In this case, all three of these approaches for handling DIF resulted in improvement in model fit based on AIC and, for the two nested models, the chi-square difference test.

Next steps

Whichever of these approaches we select, we would then identify and handle the remaining non-negligent DIF sequentially by magnitude. For instance, if we chose to resolve DIF in item 5 by allowing the item to have different parameters across both groups, we would then iteratively drop constraints to see which items continue to show non-negligible DIF.

Estimates of DIF after resolving item 5 are in the source discussion. Of the remaining DIF, item 4 appears to have the largest DIF. If we deem item 4 to show non-negligible DIF, we would address it using one of the three approaches above and then see which items continue to show the largest DIF, address it if necessary, then identify the remaining DIF, etc.

Measurement/Factorial Invariance

Before making comparisons across groups in terms of associations between constructs or their level on the construct, it is important to establish measurement invariance (also called factorial invariance) across the groups (Millsap, 2011). Tests of measurement invariance are equivalent to tests of differential item functioning. Measurement invariance can help provide greater confidence that the measure functions equivalently across groups, that is, the items have the same strength of association with the latent factor, and the latent factor is on the same metric (i.e., the items have the same level when accounting for the latent factor). For instance, If you observe differences between groups in level of depression without establishing measurement invariance across the groups, you do not know whether the differences observed reflect true group-related differences in depression or differences in the functioning of the measure across the groups.

Measurement invariance can be tested using many methods, including confirmatory factor analysis and IRT. CFA approaches to testing measurement invariance include multi-group CFA (MGCFA), multiple-indicator, multiple-causes (MIMIC) models, moderated nonlinear factor analysis (MNLFA), score-based tests (Wang et al., 2014), and the alignment method (Lai, 2021). The IRT approach to testing measurement (non-)invariance is to examine whether items show differential item functioning.

The tests of measurement invariance in confirmatory factor analysis (CFA) models were fit in the lavaan package (R-lavaan, n.d.). The examples were adapted from lavaan documentation: https://lavaan.ugent.be/tutorial/groups.html (archived at https://perma.cc/2FBK-RSAH). Procedures for testing measurement invariance are outlined in Putnick & Bornstein, 2016. In addition to nested chi-square difference tests (^2 diff), I also demonstrate permutation procedures for testing measurement invariance, as described by Jorgensen et al., 2018. Also, you are encouraged to read about MIMIC models (Cheng et al., 2016; Wang et al., 2009), score-based tests (Wang et al., 2014), and MNLFA that allows for testing measurement invariance across continuous moderators (Bauer et al., 2020; Curran et al., 2014; Gottfredson et al., 2019). MNLFA is implemented in the mnlfa package (R-mnlfa, n.d.). You can also generate syntax for conducting MNLFA in Mplus software (Muthén & Muthén, 2019) using the aMNLFA package (R-aMNLFA, n.d.). The alignment method allows many groups to be compared (Han et al., 2019).

Model fit of nested models can be compared with a chi-square difference test, as a way of testing measurement invariance. In this approach to testing measurement invariance, first model fit is evaluated in a configural invariance model, in which the same number of factors is specified in each group, and which indicators load on which factors are the same in each group. Then, successive constraints are made across groups, including constraining factor loadings, intercepts, and residuals across groups. The metric ("weak factorial") invariance model is similar to the configural invariance model, but it constrains the factor loadings to be the same across groups. The scalar ("strong factorial") invariance model keeps the constraints of the metric invariance model, but it also constrains the intercepts to be the same across groups. The residual ("strict factorial") invariance model keeps the constraints of the scalar invariance model, but it also constrains the residuals to be the same across groups. The fit of each constrained model is compared to the previous model without such constraints. That is, the fit of the metric invariance model is compared to the fit of the configural invariance model, the fit of the scalar invariance model is compared to the fit of the metric invariance model, and the fit of the residual invariance model is compared to the fit of the scalar invariance model.

The chi-square difference test is sensitive to sample size, and trivial differences in fit can be detected with large samples (Cheung & Rensvold, 2002). As a result, researchers also recommend examining change in additional criteria, including CFI, RMSEA, and SRMR (Chen, 2007). Cheung & Rensvold, 2002 recommend a cutoff of CFI ≥ -.01 for identifying measurement non-invariance. Chen, 2007 recommends the following cutoffs for identifying measurement non-invariance:

• with a small sample size (total N ≤ 300):

• testing invariance of factor loadings:

• CFI ≥ -.005

• supplemented by RMSEA ≥.010 or SRMR ≥.025

• testing invariance of intercepts or residuals:

• CFI ≥ -.005

• supplemented by RMSEA ≥.010 or SRMR ≥.005

• with an adequate sample size (total N greater than 300):

• testing invariance of factor loadings:

• CFI ≥ -.010

• supplemented by RMSEA ≥.015 or SRMR ≥.030

• testing invariance of intercepts or residuals:

• CFI ≥ -.005

• supplemented by RMSEA ≥.015 or SRMR ≥.010

Little et al., 2007 suggested that researchers establish at least partial invariance of factor loadings (metric invariance) to compare covariances across groups, and at least partial invariance of intercepts (scalar invariance) to compare mean levels across groups. Partial invariance refers to invariance with some but not all indicators. So, to examine associations with other variables, invariance of at least some factor loadings would be preferable. To examine differences in level or growth, invariance of at least some factor loadings and intercepts would be preferable. Residual invariance is considered overly restrictive, and it is not generally expected that one establish residual invariance (Little, 2013).

Although measurement invariance is important to test, it is also worth noting that tests of measurement invariance and DIF rest on various fundamentally untestable assumptions related to scale setting (Raykov et al., 2020). For instance, to give the latent factor units, oftentimes researchers set one item's factor loading to be equal across time (i.e., the marker variable). Results of factorial invariance tests can depend highly on which item is used as the anchor item for setting the scale of the latent factor (Belzak & Bauer, 2020). And subsequent tests of measurement invariance then rest on the fundamentally untestable assumption that changes in the level of the latent factor precipitates the same amount of change in the item across groups. If this scaling is not proper, however, it could lead to improper conclusions. Researchers are, therefore, not conclusively able to establish measurement invariance.

Several possible approaches may help address this. First, it can be helpful to place greater focus on degree of measurement invariance and confidence (rather than presence versus absence of measurement invariance). To the extent that non-invariance is trivial in effect size, it provides the researcher with greater confidence that they can use the measure to assess a construct in a comparable way over time.

Second, there may be important robustness checks. I describe each in greater detail below. One important sensitivity analysis could be to see if measurement invariance holds when using different marker variables. This would provide greater evidence that their apparent measurement invariance was not specific to one marker variable (i.e., one particular set of assumptions). A second robustness check would be to use effects coding, in which the average of items' factor loadings is equal to 1, so the metric of the latent variable is on the metric of all of the items rather than just one item (Little et al., 2006). A third robustness check would be to use regularization, which is an alternative method to select the anchor items and to identify differential item functioning based on a machine learning technique that applies penalization to remove parameters that have little impact on model fit (Bauer et al., 2020; Belzak & Bauer, 2020).

Another issue with measurement invariance is that the null hypothesis that adding constraints across groups will not worsen fit is likely always false, because there are often at least slight differences across groups in factor loadings, intercepts, or residuals that reflect sampling variability. However, we are not interested in trivial differences across groups that reflect sampling variability. Instead, we are interested in the extent to which there are substantive, meaningful differences across the groups of large enough magnitude to be practically meaningful. Thus, it can also be helpful to consider whether the measures show approximate measurement invariance (Van De Schoot et al., 2015). For details on how to test approximate measurement invariance, see Van De Schoot et al., 2013.

When detecting measurement non-invariance in a given parameter (e.g., factor loadings, intercepts, or residuals), one can identify the specific items that show measurement non-invariance in one of two primary ways: (1) starting with a model that allows the given parameter to differ across groups, a researcher can iteratively add constraints to identify the item(s) for which measurement invariance fails, or (2) starting with a model that constrains the given parameter to be the same across groups, a researcher can iteratively remove constraints to identify the item(s) for which measurement invariance becomes established (and by process of elimination, the items for which measurement invariance does not become established).

Approaches to addressing measurement non-invariance are described in the source discussion. If we identify that an item shows non-negligible non-invariance, we have three primary options (described above): (1) drop the item for both groups, (2) drop the item for one group but keep it for the other group, or (3) freely estimate the parameters (factor loadings and/or intercepts) for the item across groups.

Using the traditional chi-square difference test, tests of measurement invariance compare the model fit to a model with perfect fit. As the sample size grows larger, smaller differences in model fit will be detected as significant, thus rejecting measurement invariance even if the model fits well. So, it can also be useful to compare the model fit to a null hypothesis that the fit is poor, instead of the null hypothesis that the model is a perfect fit. Chi-square equivalence tests use the poor model fit as the null hypothesis. Thus, a significant chi-square equivalence test suggests that the equality constraints are plausible, providing support for the alternative hypothesis that the model fit is acceptable (Counsell et al., 2020).

Effect Size of Measurement Non-Invariance

A number of studies describe how to test the effect size of measurement invariance (Liu et al., 2017; Gunn et al., 2020; Newsom, 2015; Nye et al., 2019) or differential item functioning (Meade, 2010; Gonzalez & Pelham, 2021). Indices of effect size of measurement noninvariance include d MACS (Nye et al., 2019), w (Newsom, 2015), and M C (Newsom, 2015).

w is interpreted such that.1,.3, and.5 indicate a small, medium, and large effect size, respectively. w is computed as in the source discussion (Newsom, 2015):

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

McDonald's centrality index (M C) is computed as in the source discussion (Newsom, 2015):

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Then, M C is computed as in the source discussion (Newsom, 2015):

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Null Model

Fix residual variances and intercepts of manifest variables to be equal across groups.

Configural Invariance

Specify the same number of factors in each group, and which indicators load on which factors are the same in each group.

Metric ("Weak Factorial") Invariance

Specify invariance of factor loadings across groups.

Scalar ("Strong Factorial") Invariance

Specify invariance of factor loadings and intercepts across groups.

Residual ("Strict Factorial") Invariance

Specify invariance of factor loadings, intercepts, and residuals across groups.

Configural Invariance

Specify the same number of factors in each group, and which indicators load on which factors are the same in each group.

Model Fit

You can specify the null model as the baseline model using: baseline.model = nullModelFit

Effect Size of Non-Invariancee

The effect size of measurement non-invariance, as described by Nye et al., 2019, was calculated using the dmacs package (R-dmacs, n.d.).

The w and M C indices of the effect size of measurement non-invariance, as described by Newsom, 2015, were calculated using the petersenlab::measurementNoninvarianceEffectSize function of the petersenlab package (Petersen, 2025).

Equivalence Test

The petersenlab package (Petersen, 2025) contains the petersenlab::equiv_chi function from Counsell et al., 2020 that performs an equivalence test: https://osf.io/cqu8v. An equivalence test evaluates the null hypothesis that a model is equivalent to another model. In this case, the equivalence test evaluates the null hypothesis that a model is equivalent to another model in terms of (poor) fit. Here, we operationalize a mediocre-fitting model as a model whose RMSEA is.08 or greater. So, the equivalence test evaluates whether our invariance model fits significantly better than a mediocre-fitting model. Thus, a statistically significant p-value indicates that our invariance model fits significantly better than the mediocre-fitting model (i.e., that invariance is established), whereas a non-significant p-value indicates that our invariance model does not fit significantly better than the mediocre-fitting model (i.e., that invariance is failed).

The chi-square equivalence test is non-significant, suggesting that the model fit of the configural invariance model is not acceptable (i.e., it is not better than the mediocre-fitting model). In other words, configural invariance failed.

Permutation Test

Permutation procedures for testing measurement invariance are described in Jorgensen et al., 2018. The permutation test evaluates the null hypothesis that the model fit is not worse than the best-possible fitting model. A significant p-value indicates that the model fits significanlty worse than the best-possible fitting model. A non-significant p-value indicates that the model does not fit significantly worse than the best-possible fitting model.

For reproducibility, I set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed. You can specify the null model as the baseline model using: baseline.model = nullModelFit. **Warning**: this code takes a while to run based on [computed value omitted in this course adaptation] iterations. You can reduce the number of iterations to be faster.

The p-values are non-significant, indicating that the model does not fit significantly worse than the best-possible fitting model. In other words, configural invariance held.

Internal Consistency Reliability

Internal consistency reliability of items composing the latent factors, as quantified by omega, average variance extracted (AVE), and maximal reliability, was estimated using the semTools package (R-semTools, n.d.).

Path Diagram

A path diagram of the model generated using the semPlot package (R-semPlot, n.d.) is in the figures below ("1" = group 1; "2" = group 2).

Metric ("Weak Factorial") Invariance Model

Specify invariance of factor loadings across groups.

Model Fit

You can specify the null model as the baseline model using: baseline.model = nullModelFit

Nested Model (^2) Difference Test The configural invariance model and the metric ("weak factorial") invariance model are considered "nested" models. The metric invariance model is nested within the configural invariance model because the configural invariance model includes all of the terms of the metric invariance model along with additional terms. Model fit of nested models can be compared with a chi-square difference test, also known as a likelihood ratio test or deviance test. A significant chi-square difference test would indicate that the simplified model with additional constraints (the metric invariance model) is significantly worse fitting than the more complex model that has fewer constraints (the configural invariance model).

The metric invariance model did not fit significantly worse than the configural invariance model, so metric invariance held. This provides evidence of measurement invariance of factor loadings across groups. Measurement invariance of factor loadings across groups provides support for examining whether the groups show different associations of the factor with other constructs (Little et al., 2007).

The petersenlab package (Petersen, 2025) contains the petersenlab::satorraBentlerScaledChiSquareDifferenceTestStatistic function that performs a Satorra-Bentler scaled chi-square difference test: Below is a Satorra-Bentler scaled chi-square difference test, where c0 and c1 are the scaling correction factor for the nested model and comparison model, respectively; d0 and d1 are the degrees of freedom of the nested model and comparison model, respectively; and T0 and T1 are the chi-square values of the nested model and comparison model, respectively.

Score-Based Test Score-based tests of measurement invariance are implemented using the strucchange package and are described by Wang et al., 2014.

A score-based test and expected parameter change (EPC) estimates (Oberski, 2014; Oberski et al., 2015) are provided by the lavaan package (R-lavaan, n.d.).

Effect Size of Non-Invariance The effect size of measurement non-invariance, as described by Nye et al., 2019, was calculated using the dmacs package (R-dmacs, n.d.).

The w and M C indices of the effect size of measurement non-invariance, as described by Newsom, 2015, were calculated using the petersenlab::measurementNoninvarianceEffectSize function of the petersenlab package (Petersen, 2025).

Equivalence Test The petersenlab package (Petersen, 2025) contains the petersenlab::equiv_chi function from Counsell et al., 2020 that performs an equivalence test: https://osf.io/cqu8v. The chi-square equivalence test is non-significant, suggesting that the model fit is not acceptable.

Moreover, the equivalence test of the chi-square difference test is non-significant, suggesting that the degree of worsening of model fit is not acceptable. In other words, metric invariance failed.

Permutation Test Permutation procedures for testing measurement invariance are described in (Jorgensen et al., 2018).

For reproducibility, I set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed. You can specify the null

model as the baseline model using: baseline.model = nullModelFit. **Warning**: this code takes a while to run based on [computed value omitted in this course adaptation] iterations. You can reduce the number of iterations to be faster.

The p-values are non-significant, indicating that the model does not fit significantly worse than the configural invariance model. In other words, metric invariance held.

Internal Consistency Reliability

Internal consistency reliability of items composing the latent factors, as quantified by omega, average variance extracted (AVE), and maximal reliability, was estimated using the semTools package (R-semTools, n.d.).

Path Diagram

A path diagram of the model generated using the semPlot package (R-semPlot, n.d.) is below.

Scalar ("Strong Factorial") Invariance Model

Specify invariance of factor loadings and intercepts across groups.

Model Fit

You can specify the null model as the baseline model using: baseline.model = nullModelFit

Nested Model (^2) Difference Test The metric invariance model and the scalar ("strong factorial") invariance model are considered "nested" models. The scalar invariance model is nested within the metric invariance model because the metric invariance model includes all of the terms of the metric invariance model along with additional terms. Model fit of nested models can be compared with a chi-square difference test, also known as a likelihood ratio test or deviance test. A significant chi-square difference test would indicate that the simplified model with additional constraints (the scalar invariance model) is significantly worse fitting than the more complex model that has fewer constraints (the metric invariance model).

The scalar invariance model fit significantly worse than the configural invariance model, so scalar invariance did not hold. This provides evidence of measurement non-invariance of indicator intercepts across groups. Measurement non-invariance of indicator intercepts poses challenges to being able to meaningfully compare levels on the latent factor across groups (Little et al., 2007).

The petersenlab package (Petersen, 2025) contains the petersenlab::satorraBentlerScaledChiSquareDifferenceTestStatistic function that performs a Satorra-Bentler scaled chi-square difference test:

Score-Based Test Score-based tests of measurement invariance are implemented using the strucchange package and are described by Wang et al., 2014.

A score-based test and expected parameter change (EPC) estimates (Oberski, 2014; Oberski et al., 2015) are provided by the lavaan package (R-lavaan, n.d.).

Effect Size of Non-Invariance The effect size of measurement non-invariance, as described by Nye et al., 2019, was calculated using the dmacs package (R-dmacs, n.d.).

The w and M C indices of the effect size of measurement non-invariance, as described by Newsom, 2015, were calculated using the petersenlab::measurementNoninvarianceEffectSize function of the petersenlab package (Petersen, 2025).

Equivalence Test The petersenlab package (Petersen, 2025) contains the petersenlab::equiv_chi function from Counsell et al., 2020 that performs an equivalence test: https://osf.io/cqu8v. The chi-square equivalence test is non-significant, suggesting that the model fit is not acceptable.

Moreover, the equivalence test of the chi-square difference test is non-significant, suggesting that the degree of worsening of model fit is not acceptable. In other words, scalar invariance failed.

Permutation Test Permutation procedures for testing measurement invariance are described in (Jorgensen et al., 2018). For reproducibility, I set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed. You can specify the null model as the baseline model using: baseline.model = nullModelFit

Warning: this code takes a while to run based on [computed value omitted in this course adaptation] iterations. You can reduce the number of iterations to be faster.

The p-values are significant, indicating that the model fit significantly worse than the metric invariance model. In other words, scalar invariance failed.

Internal Consistency Reliability

Internal consistency reliability of items composing the latent factors, as quantified by omega, average variance extracted (AVE), and maximal reliability, was estimated using the semTools package (R-semTools, n.d.).

Path Diagram

A path diagram of the model generated using the semPlot package (R-semPlot, n.d.) is below.

Residual ("Strict Factorial") Invariance Model

Specify invariance of factor loadings, intercepts, and residuals across groups.

Model Fit

You can specify the null model as the baseline model using: baseline.model = nullModelFit

Nested Model (^2) Difference Test The scalar invariance model and the residual invariance model are considered "nested" models. The residual invariance model is nested within the scalar invariance model because the scalar invariance model includes all of the terms of the residual invariance model along with additional terms. Model fit of nested models can be compared with a chi-square difference test, also known as a likelihood ratio test or deviance test. A significant chi-square difference test would indicate that the simplified model with additional constraints (the residual invariance model) is significantly worse fitting than the more complex model that has fewer constraints (the scalar invariance model).

In this instance, you do not need to examine whether residual invariance held because the preceding level of measurement invariance (scalar invariance) did not hold.

The petersenlab package (Petersen, 2025) contains the petersenlab::satorraBentlerScaledChiSquareDifferenceTestStatistic function that performs a Satorra-Bentler scaled chi-square difference test:

Score-Based Test Score-based tests of measurement invariance are implemented using the strucchange package and are described by Wang et al., 2014.

A score-based test and expected parameter change (EPC) estimates (Oberski, 2014; Oberski et al., 2015) are provided by the lavaan package (R-lavaan, n.d.).

Effect Size of Non-Invariance The effect size of measurement non-invariance, as described by Nye et al., 2019, was calculated using the dmacs package (R-dmacs, n.d.).

The w and M C indices of the effect size of measurement non-invariance, as described by Newsom, 2015, were calculated using the petersenlab::measurementNoninvarianceEffectSize function of the petersenlab package (Petersen, 2025).

Equivalence Test The petersenlab package (Petersen, 2025) contains the petersenlab::equiv_chi function from Counsell et al., 2020 that performs an equivalence test: https://osf.io/cqu8v. The chi-square equivalence test is non-significant, suggesting that the model fit is not acceptable. The equivalence test of the chi-square difference test is non-significant, suggesting that the degree of worsening of model fit is not acceptable.

Moreover, the equivalence test of the chi-square difference test is non-significant, suggesting that the degree of worsening of model fit is not acceptable. In other words, residual invariance failed.

Permutation Test Permutation procedures for testing measurement invariance are described in (Jorgensen et al., 2018). For reproducibility, I set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed. You can specify the null model as the baseline model using: baseline.model = nullModelFit.

Warning: this code takes a while to run based on [computed value omitted in this course adaptation] iterations. You can reduce the number of iterations to be faster.

The p-values are non-significant, indicating that the model did not fit significantly worse than the scalar invariance model.

Internal Consistency Reliability

Internal consistency reliability of items composing the latent factors, as quantified by omega, average variance extracted (AVE), and maximal reliability, was estimated using the semTools package (R-semTools, n.d.).

Path Diagram

A path diagram of the model generated using the semPlot package (R-semPlot, n.d.) is below.

Addressing Measurement Non-Invariance

In the example above, we detected measurement non-invariance of intercepts, but not factor loadings. Thus, we would want to identify which item(s) show non-invariant intercepts across groups. When detecting measurement non-invariance in a given parameter (e.g., factor loadings, intercepts, or residuals), one can identify the specific items that show measurement non-invariance in one of two primary ways: (1) starting with a model that allows the given parameter to differ across groups, a researcher can iteratively add constraints to identify the item(s) for which measurement invariance fails, or (2) starting with a model that constrains the given parameter to be the same across groups, a researcher can iteratively remove constraints to identify the item(s) for which measurement invariance becomes established (and by process of elimination, the items for which measurement invariance does not become established).

Iteratively add constraints to identify measurement non-invariance

In the example above, we detected measurement non-invariance of intercepts. One approach to identify measurement non-invariance is to iteratively add constraints to a model that allows the given parameter (intercepts) to differ across groups. So, we will use the metric invariance model as the baseline model (in which item intercepts are allowed to differ across groups), and iteratively add constraints to identify which item(s) show non-invariant intercepts across groups.

Variable 1 Variable 1 does not have non-invariant intercepts.

Variable 2 Variable 2 has non-invariant intercepts.

Variable 2 appears to have larger intercepts in group 2 than in group 1:

Thus, in subsequent models, we would allow variable 2 to have different intercepts across groups.

Variable 3 Variable 3 has non-invariant intercepts.

Variable 4 Variable 4 has non-invariant intercepts.

Variable 5 Variable 5 has non-invariant intercepts.

Variable 6 Variable 6 has non-invariant intercepts.

Variable 7 Variable 7 has non-invariant intercepts.

Variable 8 Variable 8 has non-invariant intercepts.

Variable 9 Variable 9 has non-invariant intercepts.

Summary When iteratively adding constraints, item 1 showed measurement invariance of intercepts, but all other items showed measurement non-invariance of intercepts. This could be due to the selection of starting with item 1 as the first constraint. We could try applying constraints in a different order to see the extent to which the order influences which items are identified as showing measurement non-invariance.

Iteratively drop constraints to identify measurement non-invariance

Another approach to identify measurement non-invariance is to iteratively drop constraints to a model that constrains the given parameter (intercepts) to be the same across groups. So, we will use the scalar invariance model as the baseline model (in which item intercepts are constrained to be the same across groups), and we will iteratively drop constraints to identify the items for which measurement invariance becomes established (relative to the metric invariance model which allows item intercepts to differ across groups), and therefore, identifies which item(s) show non-invariant intercepts across groups.

Variable 1 Measurement invariance does not yet become established when freeing intercepts of variable 1 across groups. This suggests that, whether or not variable 1 shows non-invariant intercepts, there are other items that show non-invariant intercepts.

Variable 2 Measurement invariance does not yet become established when freeing intercepts of variable 2 across groups. This suggests that, whether or not variables 1 and 2 show non-invariant intercepts, there are other items that show non-invariant intercepts.

Variable 3 Measurement invariance does not yet become established when freeing intercepts of variable 3 across groups. This suggests that, whether or not variables 1, 2, and 3 show non-invariant intercepts, there are other items that show non-invariant intercepts.

Variable 4 Measurement invariance does not yet become established when freeing intercepts of variable 4 across groups. This suggests that, whether or not variables 1, 2, 3, and 4 show non-invariant intercepts, there are other items that show non-invariant intercepts.

Variable 5 Measurement invariance does not yet become established when freeing intercepts of variable 5 across groups. This suggests that, whether or not variables 1, 2, 3, 4, and 5 show non-invariant intercepts, there are other items that show non-invariant intercepts.

Variable 6 Measurement invariance does not yet become established when freeing intercepts of variable 6 across groups. This suggests that, whether or not variables 1, 2, 3, 4, 5, and 6 show non-invariant intercepts, there are other items that show non-invariant intercepts.

Variable 7 Measurement invariance becomes established when freeing intercepts of variable 7 across groups. This suggests that item 7 shows non-invariant intercepts across groups, and that measurement invariance holds when constraining items 8 and 9 to be invariant across groups. There may also be other items, especially among the preceding items (variables 1–6), that show non-invariant intercepts across groups.

Variable 8 Measurement invariance continues to hold when freeing intercepts of variable 8 across groups. This suggests that variable 8 does not show non-invariant intercepts across groups, at least when the other intercepts have been freed.

Variable 9 Measurement invariance continues to hold when freeing intercepts of variable 9 across groups (even after constraining intercepts of variable 8 across groups). This suggests that variable 9 does not show non-invariant intercepts across groups, at least when the other intercepts have been freed.

Summary When iteratively dropping constraints, measurement invariance became established when freeing the intercepts for items 1–7 across groups. This suggests that items 7, 8, and 9 show measurement invariance of intercepts across groups. This could be due to the selection of starting with item 1 as the first constraint to be freed. We could try freeing constraints in a different order to see the extent to which the order influences which items are identified as showing measurement non-invariance.

Next Steps

After identifying which item(s) show measurement non-invariance, we would evaluate the measurement non-invariance according to theory and effect sizes. We might start with the item with the largest measurement non-invariance. If we deem the measurement non-invariance for this item is non-negligible, we have three primary options: (1) drop the item for both groups, (2) drop the item for one group but keep it for the other group, or (3) freely estimate parameters for the item across groups. We would do this iteratively for the remaining items, by magnitude, that show non-negligible measurement non-invariance. Thus, our model might show partial scalar invariance—some items might have intercepts that are constrained to be the same across groups, whereas other items might have intercepts that are allowed to differ across groups. Then, we would test subsequent measurement invariance models (e.g., residual invariance) while keeping the partially freed constaints from the partial scalar invariance model. Once we establish the best fitting model that makes as many constraints that are theoretically and empirically justified for the purposes of the study, we would use that model in subsequent tests. As a reminder, full or partial metric invariance is helpful for comparing associations across groups, whereas full or partial scalar invariance is helpful for comparing mean levels across groups.

Conclusion

Bias is a systematic error. Test bias refers to a systematic error (in measurement, prediction, etc.) as a function of group membership. The two broad categories of test bias include predictive bias and test structure bias. Predictive bias refers to differences between groups in the relation between the test and criterion—i.e., different intercepts and/or slopes of the regression line between the test and the criterion. Test structure bias refers to differences in the internal test characteristics across groups, and can be evaluated with approaches such as measurement invariance and differential item function. When detecting test bias, it is important to address it, and there are a number of ways of correcting for bias, including score adjustment and other approaches that do not involve score adjustment. However, even if a test does not show bias, it does not mean that the test is fair. Fairness is not a scientific question but rather a moral, societal, and ethical question; there are many different ways of operationalizing fairness. Fairness is a complex question, so do the best you can and try to minimize any negative impact of the assessment procedures.

Suggested Readings

Putnick & Bornstein, 2016; Cole, 1981; Fernández & Abe, 2018; Reynolds & Suzuki, 2012; Sackett & Wilk, 1994; Jonson & Geisinger, 2022

Questions

Note: Several of the following questions use data from the Children of the National Longitudinal Survey of Youth Survey (CNLSY). The CNLSY is a publicly available longitudinal data set provided by the Bureau of Labor Statistics (https://www.bls.gov/nls/nlsy79-children.htm#topical-guide; archived at https://perma.cc/EH38-HDRN). The CNLSY data file for these exercises is located on the book's page of the Open Science Framework (https://osf.io/3pwza). Children's behavior problems were rated in 1988 (time 1: T1) and then again in 1990 (time 2: T2) on the Behavior Problems Index (BPI). Below are the items corresponding to the Antisocial subscale of the BPI:

1) cheats or tells lies 2) bullies or is cruel/mean to others 3) does not seem to feel sorry after misbehaving 4) breaks things deliberately 5) is disobedient at school 6) has trouble getting along with teachers 7) has sudden changes in mood or feeling

1. a. Determine whether there is predictive bias, as a function of sex, in predicting antisocial behavior at T2 using antisocial behavior at T2. Describe any predictive bias observed. b. How could you address this predictive bias?

1. Assume that children are selected to receive preventative services if their score on the Antisocial scale at T1 is > 6.

Assume that children at T2 have an antisocial disorder if their score at T2 is > 6. Does the Antisocial scale show fairness in terms of equal outcomes between boys and girls?

1. A lawyer argues in court that a test that is used to select students for college admissions should be able to be used because it has been demonstrated to show no evidence of bias against particular groups.

However, you know that just because a measure may be unbiased does not mean that using the measure's scores for that purpose is fair. How could such a measure be unbiased yet unfair?

1. Fit a multi-group item response theory model to the Antisocial subscale at time 2, with the participant's sex as a grouping factor.

a. Does a model with item parameters constrained across sexes fit significantly worse than a model with item parameters that are allowed to vary across sexes? b. Starting with an unconstrained model that allows item parameters to vary across sexes, sequentially add constraints to item parameters across groups. Which items differed in discrimination across groups? Which items differed in severity across groups? Describe any differences. c. How could any differential item functioning be addressed?

1. Examine the longitudinal measurement invariance of the Antisocial scale across T1 and T2.

Use full information maximum likelihood (FIML) to account for missing data. Use robust standard errors to account for non-normally distributed data. Which type(s) of longitudinal measurement invariance failed? Which items and parameters appeared to differ across ages?

Answers

1. a. There is predictive bias in terms of different slopes across sexes. The slope is steeper for boys than girls, indicating that the measure is not as predictively accurate for girls as it is for boys. b. You could address this predictive bias by using a different predictive measure for girls than boys, changing the criterion, or by using within-group norming.

1. No, the Antisocial scale does not show equal outcomes in terms of selection rate.

The selection ratio is much lower in girls ([computed value omitted in this course adaptation]) compared to boys ([computed value omitted in this course adaptation]).

1. The measure could be unbiased if it shows the same intercepts and slopes across groups in predicting college performance.

However, a measure can still be unfair even it is unbiased. If there are different types of errors in different groups, the measure may not be fair. For instance, if there are more false negative errors in Black applicants compared to White applicants and more false positive errors in White applicants than Black applicants, the measure would be unfair against Black applicants. Even if the percent accuracy of prediction is the same for Black and White applicants, the errors have different implications for each group—White applicants who would have failed in college are more likely to be accepted whereas strong Black applicants who would have succeeded are less likely to be accepted. To address the unfairness of the test, it will be important to use one of the various operationalizations of fairness: equal outcomes, equal opportunity, or equal odds. 4. a. The model with item parameters constrained across sexes fit significantly worse than a model with item parameters that were allowed to vary across sexes, ^2[[computed value omitted in this course adaptation]] = [computed value omitted in this course adaptation], p less than.001. This suggests that the item parameters show differential item functioning across boys and girls. b. No items showed significant differences in discrimination across sexes, suggesting that items did not differ in their relation to the latent factor across boys and girls. Items 1, 2, 3, 4, 5, and 6 differed in severity across sexes. Items showed greater severity for girls than boys. It took a higher level of antisocial behavior for a girl to be endorsed as engaging in one of the following behaviors: cheats or tells lies, bullies or is cruel/mean to others, does not seem to feel sorry after misbehaving, breaks things deliberately, is disobedient at school, and has trouble getting along with teachers. c. The differences in item severity across boys and girls could be addressed by resolving DIF. That is, the items that show severity DIF (items 1–6) would be allowed to differ in their severity parameter across boys and girls, but these items would be constrained to have the same discrimination across groups; non-DIF items would have severity and discrimination parameters constrained to be the same across groups.

1. The configural invariance model fit well according to RMSEA ([computed value omitted in this course adaptation]) and SRMR ([computed value omitted in this course adaptation]), but fit less well according to CFI ([computed value omitted in this course adaptation]).

The metric invariance model did not fit significantly worse than the configural invariance model (^2[[computed value omitted in this course adaptation]] = [computed value omitted in this course adaptation], p = [computed value omitted in this course adaptation]). However, the scalar invariance model fit significantly worse than the metric invariance model (^2[[computed value omitted in this course adaptation]] = [computed value omitted in this course adaptation], p = [computed value omitted in this course adaptation]), and the

residual invariance model fit significantly worse than the scalar invariance model (^2[[computed value omitted in this course adaptation]] = [computed value omitted in this course adaptation], p less than.001). Therefore, scalar and residual invariance failed. The following items showed larger intercepts at T1 compared to T2: items 1, 2, 3, 4, 5, and 7. Only item 6 showed larger intercepts at T2 than T1. The following items showed a larger residual at T1 compared to T2: items 2–7. Only item 1 showed a larger residual at T2 compared to T1.

CHAPTER 09 / CAC.903

The Clinical Interview and Diagnostic Systems

The clinical interview is the most flexible assessment method. It allows the counselor to hear the child's story, observe behavior, test hypotheses, and build rapport. It is also vulnerable to bias, suggestion, and uneven coverage. A good interview is both relational and structured enough to answer the referral question. With children and adolescents, the interview must adapt to development, language, and comfort.

Interview goals

The interview should clarify the concern, history, context, strengths, and goals. It should assess safety, functioning, and possible maintaining factors. It should also establish a working relationship. A child who feels interrogated may shut down. A child who feels understood may disclose more. The counselor should balance warmth with purpose.

Developmental adaptation

Young children may not answer abstract questions. They may show distress through play, drawings, or behavior. The counselor can use simple language, concrete choices, and visual aids. Adolescents may need privacy and respect for autonomy. The counselor should explain confidentiality limits clearly and avoid moralizing. A teen may test whether the counselor is an agent of parents. Transparency helps.

Structured and unstructured interviews

Unstructured interviews are flexible but may miss important domains. Structured interviews improve coverage and reliability but may feel rigid. Semi-structured interviews combine core questions with flexibility. The choice depends on purpose. A diagnostic research assessment may need structure. A clinical formulation may need flexibility. The counselor should know what each approach gains and loses.

Diagnostic systems

The DSM and ICD provide common language for classification. They help communication, research, and service access. They also have limits. Categories may not capture developmental change, culture, or comorbidity. A diagnosis should not replace formulation. The counselor should ask what the diagnosis explains and what it leaves out. A child may meet criteria for ADHD and anxiety, but the plan must address sleep, instruction, and family stress.

Differential assessment

Many symptoms overlap. Inattention may reflect ADHD, anxiety, trauma, sleep loss, or learning difficulty. Irritability may reflect depression, trauma, autism-related overload, or family conflict. The interview should test alternatives. It should ask about onset, course, settings, triggers, and impairment. It should also ask about strengths and protective factors.

Safety assessment

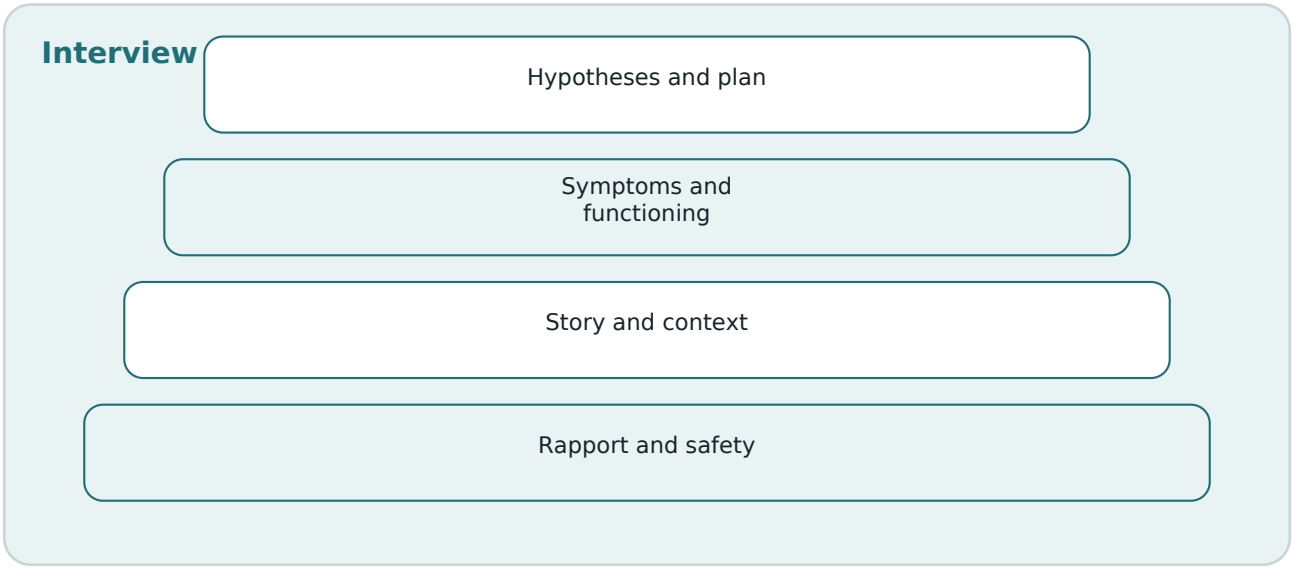
Safety assessment is part of many interviews. The counselor should ask about self-harm, suicidality, abuse, neglect, and violence in a calm, direct manner. Questions should be developmentally appropriate. A positive response requires further assessment and action according to law and ethics. The counselor should not avoid safety questions out of fear of suggestion. Direct questions do not create risk; they allow help.

Real-life application

A twelve-year-old says school is "fine" but avoids eye contact and gives short answers. The counselor shifts to a drawing of a school day. The child draws a large figure labeled "noise" near the classroom. The interview reveals sensory overload and bullying. The formulation changes from oppositional avoidance to anxiety and environmental stress. The plan includes sensory supports and anti-bullying action.

Reflection

Write five questions you would ask a child about school refusal. Include one strength question, one safety question, and one context question. Then write how you would adapt the questions for a six-year-old and a sixteen-year-old. The interview is a conversation with a purpose.



The interview is relational and purposeful at the same time.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module interview-dsm.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Overview of Clinical Interviews

Clinical interviews are a form of conversation with a client, in which the conversation has explicit clinical goals including information gathering on the client. Sharp et al., 2013 provide an overview of the clinical interview. Interviews are the most widely used assessment technique in clinical psychology, yet there is very little done to explore their validity or ways to improve their validity. There is very little data on the validity of interviews. It is likely that there has been little research on the validity of interviews because the field has been riding the coattails of biological psychiatry, in which the tradition has been that the clinical interview is just assumed to be the "gold standard". However, this is based on theory, not data.

The problem is that we do not have a "gold standard" (Lilienfeld et al., 2015). We do not have any robust measure to compare interviews against to verify their accuracy—i.e., we do not have true knowledge about the pathophysiology as a strong indicator of disease status. Diagnoses are not directly observable concepts. Comparisons from one interviewer to another interviewer establishes inter-rater reliability (i.e., diagnostic agreement), but it does not establish diagnostic validity—i.e., the correspondence to truth about illness status.

To estimate diagnostic accuracy in the absence of a gold standard, one can use latent class models (Faraone & Tsuang, 1994), where you identify latent (unmeasured) class membership among participants using multiple observed variables. For example, you can identify a latent class membership based on multiple symptoms, raters or observers, and/or at multiple time points. High scores across symptoms, raters, and time points are more likely to reflect true illness. This is consistent with making diagnostic decisions using the LEAD standard: LEAD is an acronym for: **L**ongitudinal, **E**xpert, **A**ll **D**ata. So, the best approach may be to use LEAD data to evaluate diagnostic validity.

The latent class would be the illness status—i.e., whether the client "has" the disorder or not. Illness status is not directly observable, so we use observables to infer latent classes. Each person is assigned to a latent class with different probabilities to account for uncertainty. For example, Person A may be assigned a 75% probability of "having" depression, whereas the probability may be 13% for Person B. Then, we could treat the

latent class estimate of illness status as the latent gold standard. Diagnoses based on LEAD data are not 100% valid; thus, they are a LEAD standard, not a gold standard. However, LEAD approaches are expensive and time-consuming, and they are not widely used.

Two Traditions: Unstructured and Structured Interviews

There are two primary traditions to the clinical interview: the unstructured interview and the structured interview.

Unstructured Interview

The unstructured interview is the oldest tradition of the clinical interview. It was the most widely used interview method until the late 1970s. The unstructured clinical interview came from a psychoanalytic theoretical orientation. In an unstructured interview, the clinician keeps the framework "in the head" of the information they want to extract from the client, including presenting problems, past treatments obtained, comorbid symptoms, etc.

An unstructured interview is an open-ended, free flowing conversation between the clinician and client. According to psychoanalysis, the nondirective interview was believed to serve as a catalyst for the client's expression of their unconscious, for example by means of transference and free associations. Transference is when the client treats the therapist as if the therapist was an important figure in the client's past—for example, if the client treats the therapist as if the therapist were the client's parent. Notes were frowned upon because they were viewed as disrespectful to the client and a disruption of the flow of session.

A problem of the unstructured interview is that taking a different "conversation path" (i.e., asking different questions) or having a different clinician led to different information extracted from the client and to low inter-rater and test–retest reliability. The unstructured interview is susceptible to multiple forms of bias, such as halo effects when dealing with a polite client. Another important form of (cognitive) bias is confirmation bias, in which clinicians look for evidence that confirms their hypothesis about the client, which leads to the tendency to stop the interview once the first diagnosis is identified, and to fewer diagnoses. This is known as diagnostic overshadowing. For instance, intellectual disability can overshadow other mental health conditions (Dell'Armo & Tassé, 2025). Diagnostic overshadowing is analogous to challenges that radiologists have in identifiying a second tumor after already identifying one tumor, known as "satisfaction of search" (Fleck et al., 2010). The unstructured interview is also susceptible to race, gender, class bias, etc.

In general, unstructured interviews show low reliability. The low reliability of unstructured interviews is a result of several factors. One factor is information variance: different information is extracted from the client in each interview. Another factor that contributes to the low reliability of unstructured interviews is criterion variance, where different criteria are used to determine the presence or absence of a condition. Criterion variance was inadequate once criterion-based diagnosis came along. A third factor that contributes to the low reliability of unstructured interviews is varying levels of skill and knowledge of interviewers.

A potential advantage of the unstructured interview is that it can be good for non-specific aspects of interview. For instance, an unstructured interview can potentially be helpful for building rapport because the client feels the clinician's first goal is to understand them and their situation in order to provide help. Classic examples of an unstructured interview are

provided in "The Psychiatric Interview" (1970) by Harry Stack Sullivan, who is a psychodynamically oriented clinician. Sullivan, 1970's book provides a guide to understanding the unstructured interview. He described processes such as "reciprocal emotion," which he described as how the interviewer and interviewee influence each other and responses of the other through one's emotional tone.

Structured Interview

An important movement in the development of interviews was behavioral interviews that began in the 1960s. Behavioral interviews came from a cognitive-behavioral theoretical orientation. They started out as unstructured. However, behavioral interviews had a different set of goals from traditional unstructured interviews. Behavioral interviews sought to quantify aspects of behavior—for example, how often the symptoms occur. Behavioral interviews applied the A-B-C (antecedent-behavior-consequence) model. The A-B-C model tries to identify patterns of sequences that include the problem behavior. Specifically, the A-B-C model identifies things that came before the behavior (antecedents), and things that came after the behavior (consequences). This helps to identify precipitating and maintaining factors for the problem behavior, as well as strengths to bolster in treatment. It provides helpful ways to generate information and helps with case conceptualization. The behavior interviews eventually led to more structured versions, called structured interviews.

An example of a structured clinical interview is the Structured Clinical Interview for DSM (SCID). Structured interviews were developed to solve the problem of reliability. In a structured interview, a pre-determined domain of questions is asked in a pre-defined way. A fully structured interview provides a script to use to ask each question (to standardize wording), and in a particular order, with a decision tree and branching logic to choose subsequent questions based on the responses given.

For a structured diagnostic interview, the diagnostic criteria for a disorder are codified, and the exact language is provided to ask about each symptom. Then, the responses are scored and integrated in a systematic way, and the diagnosis is determined in a systematic way.

Structured interviews show higher reliability compared to unstructured interviews due to: less information variance because the clinician extracts similar information from a given client, and less criterion variance because the clinician makes diagnostic decisions using the same criteria. Interviews differ in their continuum or degree of structure, from semi-structured to structured. In general, convergent validity improves as the degree of structure for an interview increases (Widiger, 2002).

When very structured, even a layperson could perform the interview. Taken to the extreme, a human interviewer becomes unnecessary; the entire interview can be recorded and done by a computer. In these cases, inter-rater reliability approaches perfection. However, this removes the role of clinical judgment, which can lead to errors if the client misinterprets the computer-administered question or describes a behavior that is not actually clinically concerning. Therefore, structured interviews reduce the amount of training necessary for a clinician to administer, but they lose the potential positive qualities of unstructured interview: e.g., the capacity for better rapport; clients may not feel understood, cared about, or that they are working with an expert. Therefore, semi-structured interviews were developed.

Semi-Structured Interview

Semi-structured interviews provide some leeway in how the clinician asks questions and the pace of the questions, so the clinician can ask questions in their own words and ask follow-up questions or restate questions using the client's own words. Semi-structured interviews are still structured in that they are still asking about the same range of problems as a structured interview, to avoid confirmation bias, and they provide a structured way to score diagnostic criteria and assign diagnoses, to ensure consistency across therapists. A well-designed semi-structured interview balances improvements in validity—due to greater clinical judgment and probing—with the tradeoff in lower inter-rater reliability due to idiosyncratic decisions. A semi-structured interview also allows clarifying the meaning of questions and response choices.

Conducting semi-structured interviews requires more training and there is frequent discussion, in the literature, of their problems, and it takes a long time to become really skilled. This can become an issue with highly trained individuals because they sometimes "screw around" with the interview (i.e., do things differently because they believe their approach is superior) and bypass the structure, leading to an unstructured interview.

In addition to degree of structure, interviews also differ in their degree of specialization. Some interviews are general/broad, including the SCID. Other interviews are specific to particular domains, such as the Semi-Structured Assessment for the Genetics of Alcoholism (SSAGA), which assesses alcohol use disorder. As you get further away from the construct, you lose valued information and the interview becomes less and less helpful. Specialized interviews provide the maximum amount of information in the area of interest, and they take less time, which is why they are popular in research.

A review of structured and semi-structured diagnostic interviews is provided by Summerfeldt et al., 2010.

Structured Versus Unstructured Interviews

Structured interviews show higher reliability than unstructured interviews, in terms of both inter-rater and test–retest reliability. Inter-rater reliability of a clinical interview is typically estimated based on the degree of agreement between an interviewer and an observer of the interview. That is, most often inter-rater reliability of interviews is estimated based on two raters of the same interview content. However, this does not necessarily capture different styles of different interviewers. Therefore, the typical estimates of inter-rater reliability of interviews do not capture all parts of the reliability of the measure.

Test–retest reliability of a clinical interview, is usually evaluated by conducting interviews of the same client separated (typically) by 2 weeks, often by a different interviewer. However, different ratings might be made at each time point if the diagnosis (e.g., depression) fluctuates over time or especially if the disorder is influenced by variation of the context. Therefore, changes in symptoms can contribute to differences in reliability over time—or, alternatively, differences could be due to differences in interviewers. Because of conflating differences in raters and time, test–retest reliability tends to be lower than inter-rater reliability in evaluating the reliability of interviews.

Fully structured diagnostic interviews typically deliver inter-rater reliability Cohen's kappa estimates of.80 or higher. By contrast, estimates of inter-rater agreement between clinicians conducting unstructured interviews usually are near zero, which indicates that their

agreement is no better than chance. Cohen's kappa ranges from -1 (perfect disagreement) to +1 (perfect agreement). A kappa greater than.75 would be considered good inter-rater reliability.

Despite the increased reliability of using structured and semi-structured interviews compared to unstructured interviews, many clinicians are reluctant to use structured and semi-structured interviews. Clinicians often feel that structured interviews impinge upon their professional autonomy and believe that structured interviews will damage rapport, even though semi-structured interviews have been shown to increase rapport. Clients prefer structured approaches, feeling that the clinician has a more comprehensive understanding of the client's needs. Clinicians prefer to use their expertise, insight, and judgment. However, their judgment is what results in lower reliability and validity. Unstructured interviews are much more widely used in practice despite the advantages of structured interviews in terms of reliability and validity, because clinicians tend to prefer them.

Clinicians using unstructured approaches tend to diagnose fewer conditions than structured interviews detect. This likely reflects confirmation bias and diagnostic overshadowing.

Other Findings Regarding Interviews

There are other notable patterns regarding clinical interviews. Inter-rater reliability of Diagnostic and Statistical Manual of Mental Disorders (DSM) diagnoses is better for some conditions than others, but in general it is not very good. This provides further evidence that DSM-defined diagnostic categories are not "real" in a phenomenological sense. Moreover, there is considerable evidence that these diagnostic categories are better conceptualized as dimensional than categorical (Markon et al., 2011). Therefore, a better understanding of the structure of psychopathology may lead to a more valid diagnostic system.

Low inter-rater reliability of interviews could be a result of either low sensitivity or low specificity. Even if one is high but the other is low, you will get low inter-rater reliability.

It is worth noting that the clinical interview is not merely a tool for assessment. A clinical interview can be used as a form of intervention (i.e., therapeutic interview). For instance, motivational interviewing is a therapeutic interview that is often used as a treatment for substance use.

Best Practice for Diagnostic Assessment

There are several best practice approaches for conducting diagnostic assessment. As discussed earlier, when possible, it is best to follow the LEAD standard: **l**ongitudinal, **e**xpert, **a**ll **d**ata. Have the assessment conducted by an expert or multiple experts in the domain(s) of interest. Have multiple people who are experts on the client (e.g., parent, teacher, friend, sibling, self) report on the person's behavior. Use rating scales to generate the contending hypotheses, and then pick structured or semi-structured interview modules to use for follow-up to examine and rule out hypotheses. Incorporate multiple forms of information, e.g., interviews, rating scales, observations, objective/direct/performance-based measures, and school/work/legal records. If possible, include an observational measure and an objective/direct/performance-based measure of diagnostic-relevant behavior. Examples of observational measures include parent–child play in the clinic paired with a cleanup command to see how parents give commands, how children respond to the parent's commands, and how parents respond to their child's noncompliance. Examples of more objective measures could include biological assessments, such as polysomnography, or performance-based assessments, such as intelligence tests and academic achievement tests.

There are many important skills when conducting a clinical interview, including:

- Taking a nonjudgmental stance
- Active listening
- Empathy
- Authenticity
- Effective use of silence
- Paraphrasing
- Summarizing
- Providing a sense of hope

The DSM and ICD

The Diagnostic and Statistical Manual of Mental Disorders (DSM) provides the list of mental disorders and the diagnostic criteria for mental health treatment providers. It is published by the American Psychiatric Association and is used in the United States. The equivalent list of mental disorders and diagnostic criteria for mental health providers outside of the United States is the International Classification of Diseases (ICD). However, the ICD also includes diseases that are not considered mental disorders. Collectively, the DSM and ICD represent the diagnostic system for mental disorders. A goal of DSM and ICD is to define different "mental disorders" to make sure people get the services they need.

Strengths

There are several strengths of the DSM and ICD. First, they can facilitate communication about disorders with other mental health providers; they provide a common language to describe complex behavioral presentations. Second, ideally, the DSM and ICD also guide treatment selection, especially if people are homogeneous within a disorder in terms of their symptoms, course, etiology, and treatment response. Third, diagnosis is used to justify payment for services from third-party payers (e.g., insurance companies, government). Fourth, diagnosis can be normalizing and empowering to some clients. It may help people learn that other people are experiencing similar challenges, and it may give them hope that there is something that can be done to address it. Fifth, the DSM/ICD promotes research in psychopathology, in terms of epidemiology (the disorder distribution in the population), etiology (causes of the disorder), course (how the disorder plays out over time), and treatment (including development and evaluation of interventions).

Concerns

However, there are also key concerns with the DSM and ICD. One potential concern with the DSM and ICD is stigmatization. The goal of the DSM/ICD is not to label people. Labeling people can be an unfortunate consequence, however, and it can be stigmatizing because mental disorders often carry a stigma. A second concern is that providing a label to a client may lead the client to see their difficulties as stable and unchangeable; they may mistakenly interpret the label as something that they "have" (rather than as description of what they "do"), a misconception which is clarified later in this section. For instance, Ahuvia et al., 2024 found that college students who self-labeled themselves as having depression had less perceived control over depression and experienced more catastrophizing, even after controlling for depression symptom severity. Another concern of the DSM and ICD that they may be pathologizing normality. Mental disorders are not infrequent. Estimates of lifetime prevalence of mental disorders are around 75% (Schaefer et al., 2017). That is, three out of four people will experience a mental disorder at some point in their life. Thus, abnormality as defined by the DSM and ICD are normal. Another concern with the DSM and ICD is that they medicalize and pathologize problems in living as mental illnesses, and they obscure the role of environmental factors such as poverty (Gambrill, 2014). Moreover, diagnoses specified in the DSM and ICD ignore causes and etiology—they do not provide explanations for behavior (Eiko I. Fried, 2022). It would thus be incorrect (and tautological) to say that someone engages in a particular behavior because they have ADHD (or insert whatever diagnosis; archived at https://perma.cc/5972-PUQD).

Ideally, a diagnostic system should have validity and utility. For instance, the diagnoses in a diagnostic system should have construct validity. That is, the diagnostic categories should reflect truth, reality, or the existence of the construct (i.e., diagnostic validity). Diagnostic validity is the extent to which the diagnostic category accurately captures the abnormal phenomenon of interest. Typical indicators of diagnostic validity are homogeneity (similarity) across people receiving a given diagnosis in terms of etiology, course, treatment response, etc. Utility means that the diagnostic system should help clinical decision-making, in terms of ease of use (e.g., saving time), facilitating communication with others, and in treatment planning. However, diagnostic systems (DSM and ICD) have serious concerns with both validity and utility. The diagnostic system has serious concerns with validity, in terms of whether the diagnostic categories defined by the DSM/ICD are actually real. There is great heterogeneity (variability) between people with the same diagnosis, in terms of

symptom profiles, etiologies, course, and treatment outcomes (Eiko I. Fried, 2022). For instance, there are over 600,000 ways in terms of differing symptom profiles that a person can meet criteria for post-traumatic stress disorder (Isaac R. Galatzer-Levy & Richard A. Bryant, 2013). As another example, two people diagnosed with conduct disorder can share zero symptoms. Subtypes can attempt to address this problem, but there is little support for the validity of these subtypes. Many disorder subtypes and features are not based on empirical data—instead, they are based on expert judgment, consensus, and politics. Moreover, which diagnoses a person experiences at a given time do not portend the types of diagnoses they will experience in the future—people show considerable diagnosis switching from one disorder to another (and even across diagnostic families—e.g., from an externalizing disorder to an internalizing or thought disorder) across development (Caspi et al., 2020).

There is also strong comorbidity (co-occurence) across disorders. A person who meets criteria for one disorder is likely to also meet criteria for other disorders (Caspi et al., 2020). Many disorders often co-occur within an individual, including major depressive disorder and generalized anxiety disorder, which suggests that they may share an underlying liability and do not reflect distinct categories. This complicates treatment and challenges the validity of the diagnostic system.

Another challenge is regarding the coverage of the diagnostic system, including the number of categories and which are represented. How many disorders/categories should there be? Do we have too many categories? There are hundreds of possible diagnoses, including 541 disorders in the DSM-5 (Blashfield et al., 2014). Do we have too few diagnoses? Clinicians often use "Not Otherwise Specified" or "Unspecified" for diagnoses, which suggests that the current diagnostic categories do not cover the variability of clients' presentations. How thinly should we "split" the categories? Some people are "splitters" and tend to split categories into as many categories as possible. Other people are "lumpers" and tend to lump categories together so there are fewer categories.

The diagnostic system also has serious concerns with utility, in terms of lengthy criterion sets that are time-consuming and difficult to assess in practice (Mullins-Sweatt & Widiger, 2009). One possible way to address the time-consuming nature of using lengthy criterion sets is to use prototypal matching instead of criterion sets. Prototypal matching involves having the clinician rate the similarity of the client to a narrative description of a prototypical person with the disorder. However, prototypal matching could result in lower inter-rater reliability (compared to using criterion sets) because of idiosyncratic judgments. There is a tradeoff between assessment time and validity. The clinician's time can be saved by using computerized interviews or computerized tests. Another way to save the clinician's time is to use self-report inventories as a screening device followed by an interview that is specific to the most relevant conditions identified by the screen.

The diagnostic system includes arbitrary thresholds that do not facilitate social and clinical decision-making. Current diagnostic categories do not provide much utility in terms of determining the appropriate treatment.

No single biological substrate has been identified for any psychological disorder that has high enough sensitivity and specificity to be useful for diagnosis or for predicting response to treatment (Jeggan Tiego et al., 2023). This is because the DSM/ICD-based psychological disorders are fictive categories. The DSM/ICD-defined categories do not exist in nature. This

does not dismiss the importance of what people are experiencing, which is very real. A person's experience of depression is real, but the construct of major depressive disorder—as defined by the DSM/ICD—is not real. Mental disorders are not a concrete thing. Psychological disorders are social constructs. Mental disorders are not something that people "have"; rather, they are something that people "do" or "experience". The mental disorders in the DSM/ICD are merely a description of behaviors. They are fuzzy concepts with vague boundaries. They are subjective, often without a clear "yes" or "no" as to whether someone meets criteria for a particular disorder. Mental disorders are defined in relation to cultural, social, and familial norms and values. That is, they are socially defined not biologically. The boundaries between normality and abnormality vary across cultures. Thus, the diagnostic categories in the DSM/ICD may not involve "biological pathology"—they do not carve nature at its joints.

Researchers have examined biology-related differences in people with versus without mental disorders, and they have found some differences, but these biology-related differences are not able to classify perfectly because the categories are defined socially in terms of behaviors, and not in terms of biology.

Another concern with the DSM and ICD is their conceptualization of psychopathology as binary. The DSM and ICD treat most disorders as categorical phenomena that are binary in nature: either you have a disorder or you do not. There is no gray in this conceptualization of psychopathology. However, research has shown that the difference between "normal" and "abnormal" behavior frequently is one of degree rather than kind (Markon et al., 2011). Thus, a dimensional approach to classification may provide more valid portrayal of many clinical phenomena than the categorical approach used by the DSM and ICD.

There is also potential for bias in the diagnostic system or its application, especially against people with particular disorders. For instance, some therapists will not treat people with particular disorders, especially certain types of personality disorders (e.g., borderline personality disorder, antisocial personality disorder), because of perceived challenges. The diagnostic system also might overestimate abnormality in women (e.g., premenstrual dysphoric disorder) or racial/ethnic minorities.

In general, reliability of diagnoses is low. The inter-rater reliability of diagnoses tends to be low (Lobbestael et al., 2011). Moreover, test–retest reliability is also relatively low. A client might cycle back and forth between diagnosis and no diagnosis based on one or a few symptoms. For instance, major depressive disorder is given a diagnosis with 5 or more symptoms, but no diagnosis is given with 4 or fewer symptoms. The fluctuation of symptoms and disorder status across time suggests that a binary approach to diagnosis is inaccurate.

It will thus be important to improve our diagnostic system. For instance, it will be important for the diagnostic system to account for the dimensional nature of constructs so that the diagnostic system better carves nature at its joints (better validity) and is flexible to allow different optimal thresholds for different social and clinical decisions (better utility). One argument against dimensional diagnostic systems is that treatment decisions are often binary (e.g., whether or not to provide treatment). However, decisions in practice can be dimensional too, and can reflect different levels of treatment, including, for example, the dosage of medication, the frequency and intensity of therapy, and the degree of hospitalization.

Alternative Structures of Psychopathology

Emerging evidence suggests that alternative models of psychopathology may provide a better fit to its structure. Models that have captured the greatest attention include hierarchical (higher-order) models of psychopathology, including the p-factor (Caspi et al., 2014; Smith et al., 2020) and the hierarchical taxonomy of psychopathology (Kotov et al., 2017; Kotov et al., 2021). The p-factor is the general factor of psychopathology, akin to the general factor of intelligence (g), and it attempts to account for the fact that many forms of psychopathology show strong covariation and co-occurrence. The p-factor (Caspi et al., 2014; Smith et al., 2020) is a hierarchical model of psychopathology in which the p-factor subsumes three lower-order dimensions of psychopathology: internalizing problems (e.g., depression, anxiety, obsessive-compulsive disorder), externalizing problems (e.g., conduct disorder, ADHD), and thought-disordered problems (e.g., autism, schizophrenia).

In the HiTOP model (Kotov et al., 2017; Kotov et al., 2021), a p-factor influences super spectra, including emotional dysfunction, psychosis, and externalizing problems. Emotional dysfunction subsumes somatoform and internalizing problems (e.g., sexual problems, eating pathology, fear, distress, and mania). Psychosis subsumes thought disorder and detachment. Externalizing problems are subdivided into disinhibited externalizing and antagonistic externalizing. Both disinhibited and antagonistic externalizing subsume antisocial behavior, whereas disinhibited externalizing subsumes heavy substance use.

A third alternative conceptualization of psychopathology is the National Institute of Mental Health (NIMH) Research Domain Criteria (RDoC). RDoC is described in the source discussion.

Another emerging technique is network approaches that model the covariation among symptoms (McNally, 2021).

Conclusion

Interviews can be administered (a) in a free-flowing way as unstructured interviews, (b) in a way that asks questions in a pre-defined way with a pre-defined order and pre-defined scoring criteria as structured interviews, or (c) as semi-structured interviews that blend structured interviews with freedom to ask follow-up questions. In general, reliability and validity improves as the degree of structure for an interview increases.

The Diagnostic and Statistical Manual of Mental Disorders (DSM) and the International Classification of Diseases (ICD) provide the list of mental disorders and the diagnostic criteria for mental health treatment providers. The DSM and ICD have potential strengths, including (ideally) facilitating communication, guiding treatment selection, providing justification for payment for services, providing a normalizing and empowering effect for some clients, and promoting research in psychopathology. However, there are key concerns of the DSM and ICD, including concerns with stigmatization, pathologizing normality, poor coverage, binary classification, obscuring environmental factors, no biological criteria, potential bias, and low reliability, validity, and utility. There are alternatives to the DSM and ICD for conceptualizing psychopathology. Alternative structures of psychopathology include hierarchical structures such as the p-factor and the hierarchical taxonomy of psychopathology (HiTOP).

Suggested Readings

Sommers-Flanagan & Sommers-Flanagan, 2016

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

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According to the American Psychiatric Association, a **psychological disorder**, or mental disorder, is “a syndrome characterized by clinically significant disturbance in an individual's cognition, emotion regulation, or behavior that reflects a dysfunction in the psychological, biological, or developmental processes underlying mental functioning. Mental disorders are usually associated with significant distress in social, occupational, or other important activities” (2013). **Psychopathology** is the study of psychological disorders, including their symptoms, **etiology** (i.e., their causes), and treatment. The term psychopathology can also refer to the manifestation of a psychological disorder. Although consensus can be difficult, it is extremely important for mental health professionals to agree on what kinds of thoughts, feelings, and behaviors are truly abnormal in the sense that they genuinely indicate the presence of psychopathology. Certain patterns of behavior and inner experience can easily be labeled as abnormal and clearly signify some kind of psychological disturbance. The person who washes their hands 40 times per day and the person who claims to hear the voices of demons exhibit behaviors and inner experiences that most would regard as abnormal: beliefs and behaviors that suggest the existence of a psychological disorder. But, consider the nervousness a young man feels when talking to an attractive person or the loneliness and longing for home a first-year student experiences during her first semester of college—these feelings may not be regularly present, but they fall in the range of normal. So, what kinds of thoughts, feelings, and behaviors represent a true psychological disorder? Psychologists work to distinguish psychological disorders from inner experiences and behaviors that are merely situational, idiosyncratic, or unconventional.

Definition of a Psychological Disorder

Perhaps the simplest approach to conceptualizing psychological disorders is to label behaviors, thoughts, and inner experiences that are atypical, distressful, dysfunctional, and sometimes even dangerous, as signs of a disorder. For example, if you ask a classmate for a date and you are rejected, you probably would feel a little dejected. Such feelings would be normal. If you felt extremely depressed—so much so that you lost interest in activities, had difficulty eating or sleeping, felt utterly worthless, and contemplated suicide—your feelings would be **atypical**, would deviate from the norm, and could signify the presence of a psychological disorder. Just because something is atypical, however, does not necessarily mean it is disordered.

For example, only about 4% of people in the United States have red hair, so red hair is considered an atypical characteristic, but it is not considered disordered, it's just unusual. And it is less unusual in Scotland, where approximately 13% of the population has red hair ("DNA Project Aims," 2012). As you will learn, some disorders, although not exactly typical, are far from atypical, and the rates in which they appear in the population are surprisingly high.

Source figure description (original figure omitted): Red hair is considered unusual, but not abnormal. (a) Isla Fischer, (b) Prince Harry, and (c) Marcia Cross are three natural redheads. (credit a: modification of work by Richard Goldschmidt; credit b: modification of work by Glyn Lowe; credit c: modification of work by Kirk Weaver)

If we can agree that merely being atypical is an insufficient criterion for a having a psychological disorder, is it reasonable to consider behavior or inner experiences that differ from widely expected cultural values or expectations as disordered? Using this criterion, a person who walks around a subway platform wearing a heavy winter coat in July while screaming obscenities at strangers may be considered as exhibiting symptoms of a psychological disorder. Their actions and clothes violate socially accepted rules governing appropriate dress and behavior; these characteristics are atypical.

Cultural Expectations

Violating cultural expectations is not, in and of itself, a satisfactory means of identifying the presence of a psychological disorder. Since behavior varies from one **culture** to another, what may be expected and considered appropriate in one culture may not be viewed as such in other cultures. For example, returning a stranger's smile is expected in the United States because a pervasive social norm dictates that we reciprocate friendly gestures. A person who refuses to acknowledge such gestures might be considered socially awkward—perhaps even disordered—for violating this expectation. However, such expectations are not universally shared. Cultural expectations in Japan involve showing reserve, restraint, and a concern for maintaining privacy around strangers. Japanese people are generally unresponsive to smiles from strangers (Patterson et al., 2007). Eye contact provides another example. In the United States and Europe, eye contact with others typically signifies honesty and attention. However, most Latin-American, Asian, and African cultures interpret direct eye contact as rude, confrontational, and aggressive (Pazain, 2010). Thus, someone who makes eye contact with you could be considered appropriate and respectful or brazen and offensive, depending on your culture.

Source figure description (original figure omitted): Eye contact is one of many social gestures that vary from culture to culture. (credit: Joi Ito)

Hallucinations (seeing or hearing things that are not physically present) in Western societies is a violation of cultural expectations, and a person who reports such inner experiences is readily labeled as psychologically disordered. In other cultures, visions that, for example, pertain to future events may be regarded as normal experiences that are positively valued (Bourguignon, 1970). Finally, it is important to recognize that cultural norms change over time: what might be considered typical in a society at one time may no longer be viewed this way later, similar to how fashion trends from one era may elicit quizzical looks decades later—imagine how a headband, legwarmers, and the big hair of the 1980s would go over on your campus today.

Note: In the 1950s and 1960s, the concept of mental illness was widely criticized. One of the major criticisms focused on the notion that mental illness was a “myth that justifies psychiatric intervention in socially disapproved behavior” (Wakefield, 1992). Thomas Szasz (1960), a noted psychiatrist, was perhaps the biggest proponent of this view. Szasz argued that the notion of mental illness was invented by society (and the mental health establishment) to stigmatize and subjugate people whose behavior violates accepted social and legal norms. Indeed, Szasz suggested that what appear to be symptoms of mental illness are more appropriately characterized as “problems in living” (Szasz, 1960). In his 1961 book, The Myth of Mental Illness: Foundations of a Theory of Personal Conduct, Szasz expressed his disdain for the concept of mental illness and for the field of psychiatry in general (Oliver, 2006). The basis for Szasz’s attack was his contention that detectable abnormalities in bodily structures and functions (e.g., infections and organ damage or dysfunction) represent the defining features of genuine illness or disease, and because symptoms of purported mental illness are not accompanied by such detectable abnormalities, so-called psychological disorders are not disorders at all. Szasz (1961/2010) proclaimed that “disease or illness can only affect the body; hence, there can be no mental illness” (p. 267). Today, we recognize the extreme level of psychological suffering experienced by people with psychological disorders: the painful thoughts and feelings they experience, the disordered behavior they demonstrate, and the levels of distress and impairment they exhibit. This makes it very difficult to deny the reality of mental illness. However controversial Szasz’s views and those of his supporters might have been, they have influenced the mental health community and society in several ways. First, lay people, politicians, and professionals now often refer to mental illness as mental health “problems,” implicitly acknowledging the “problems in living” perspective Szasz described (Buchanan-Barker & Barker, 2009). Also influential was Szasz’s view of sexual orientation. Szasz was perhaps the first psychiatrist to openly challenge the idea that same-sex attraction represented a form of mental illness or disease (Szasz, 1965), which helped pave the way for the social and civil rights that gay and lesbian people now have (Barker, 2010). His work also inspired legal changes that protect the rights of people in psychiatric institutions and allow such individuals a greater degree of influence and responsibility over their lives (Buchanan-Barker & Barker, 2009).

Harmful Dysfunction

If none of the criterion discussed so far is adequate by itself to define the presence of a psychological disorder, how can a disorder be conceptualized? Many efforts have been made to identify the specific dimensions of psychological disorders, yet none is entirely satisfactory. No universal definition of psychological disorder exists that can apply to all situations in which a disorder is thought to be present (Zachar & Kendler, 2007). However, one of the more influential conceptualizations was proposed by Wakefield (1992), who defined psychological disorder as a **harmful dysfunction**. Wakefield argued that natural internal mechanisms—that is, psychological processes honed by evolution, such as cognition, perception, and learning—have important functions, such as enabling us to experience the world the way others do and to engage in rational thought, problem solving, and communication. For example, learning allows us to associate a fear with a potential danger in such a way that the intensity of fear is roughly equal to the degree of actual danger. Dysfunction occurs when an internal mechanism breaks down and can no longer perform its normal function. But, the presence of a dysfunction by itself does not determine a disorder. The dysfunction must be harmful in that it leads to negative consequences for the individual or for others, as judged by the standards of the individual's culture. The harm may include significant internal anguish (e.g., high levels of anxiety or depression) or problems in day-to-day living (e.g., in one's social or work life).

To illustrate, Janet has an extreme fear of spiders. Janet's fear might be considered a dysfunction in that it signals that the internal mechanism of learning is not working correctly (i.e., a faulty process prevents Janet from appropriately associating the magnitude of fear with the actual threat posed by spiders). Janet's fear of spiders has a significant negative influence on daily life: she avoids all situations in which she suspects spiders to be present (e.g., the basement or a friend's home), and she quit her job last month because she saw a spider in the restroom at work and is now unemployed. According to the harmful dysfunction model, Janet's condition would signify a disorder because (a) there is a dysfunction in an internal mechanism, and (b) the dysfunction has resulted in harmful consequences. Similar to how the symptoms of physical illness reflect dysfunctions in biological processes, the symptoms of psychological disorders presumably reflect dysfunctions in mental processes. The internal mechanism component of this model is especially appealing because it implies that disorders may occur through a breakdown of biological functions that govern various psychological processes, thus supporting contemporary neurobiological models of psychological disorders (Fabrega, 2007).

The American Psychiatric Association (APA) Definition

Many of the features of the harmful dysfunction model are incorporated in a formal definition of psychological disorder developed by the **American Psychiatric Association** (APA). According to the APA (2013), a psychological disorder is a condition that is said to consist of the following:

• **There are significant disturbances in thoughts, feelings, and behaviors**. A person must experience inner states (e.g., thoughts and/or feelings) and exhibit behaviors that are clearly disturbed—that is, unusual, but in a negative, self-defeating way. Often, such disturbances are troubling to those around the individual who experiences them. For example, an individual who is uncontrollably preoccupied by thoughts of germs spends hours each day bathing, has inner experiences, and displays behaviors that most would

consider atypical and negative (disturbed) and that would likely be troubling to family members.

• **The disturbances reflect some kind of biological, psychological, or developmental dysfunction**. Disturbed patterns of inner experiences and behaviors should reflect some flaw (dysfunction) in the internal biological, psychological, and developmental mechanisms that lead to normal, healthy psychological functioning. For example, the hallucinations observed in schizophrenia could be a sign of brain abnormalities.

• **The disturbances lead to significant distress or disability in one's life**. A person's inner experiences and behaviors are considered to reflect a psychological disorder if they cause the person considerable distress, or greatly impair their ability to function as a normal individual (often referred to as functional impairment, or occupational and social impairment). As an illustration, a person's fear of social situations might be so distressing that it causes the person to avoid all social situations (e.g., preventing that person from being able to attend class or apply for a job).

• **The disturbances do not reflect expected or culturally approved responses to certain events**. Disturbances in thoughts, feelings, and behaviors must be socially unacceptable responses to certain events that often happen in life. For example, it is perfectly natural (and expected) that a person would experience great sadness and might wish to be left alone following the death of a close family member. Because such reactions are in some ways culturally expected, the individual would not be assumed to signify a mental disorder.

Some believe that there is no essential criterion or set of criteria that can definitively distinguish all cases of disorder from nondisorder (Lilienfeld & Marino, 1999). In truth, no single approach to defining a psychological disorder is adequate by itself, nor is there universal agreement on where the boundary is between disordered and not disordered. From time to time we all experience anxiety, unwanted thoughts, and moments of sadness; our behavior at other times may not make much sense to ourselves or to others. These inner experiences and behaviors can vary in their intensity, but are only considered disordered when they are highly disturbing to us and/or others, suggest a dysfunction in normal mental functioning, and are associated with significant distress or disability in social or occupational activities.

Summary

Psychological disorders are conditions characterized by abnormal thoughts, feelings, and behaviors. Although challenging, it is essential for psychologists and mental health professionals to agree on what kinds of inner experiences and behaviors constitute the presence of a psychological disorder. Inner experiences and behaviors that are atypical or violate social norms could signify the presence of a disorder; however, each of these criteria alone is inadequate. Harmful dysfunction describes the view that psychological disorders result from the inability of an internal mechanism to perform its natural function. Many of the features of harmful dysfunction conceptualization have been incorporated in the APA's formal definition of psychological disorders. According to this definition, the presence of a psychological disorder is signaled by significant disturbances in thoughts, feelings, and behaviors; these disturbances must reflect some kind of dysfunction (biological, psychological, or developmental), must cause significant impairment in one's life, and must not reflect culturally expected reactions to certain life events.

Review Questions

In the harmful dysfunction definition of psychological disorders, dysfunction involves _____.

1. the inability of a psychological mechanism to perform its function

1. the breakdown of social order in one's community

1. communication problems in one's immediate family

1. all the above

Patterns of inner experience and behavior are thought to reflect the presence of a psychological disorder if they _____.

1. are highly atypical

1. lead to significant distress and impairment in one's life

1. embarrass one's friends and/or family

1. violate the norms of one's culture

Critical Thinking Question

Discuss why thoughts, feelings, or behaviors that are merely atypical or unusual would not necessarily signify the presence of a psychological disorder. Provide an example.

Personal Application Question

Identify a behavior that is considered unusual or abnormal in your own culture; however, it would be considered normal and expected in another culture.

Attribution and critical-reading note

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A first step in the study of psychological disorders is carefully and systematically discerning significant signs and symptoms. How do mental health professionals ascertain whether or not a person’s inner states and behaviors truly represent a psychological disorder? Arriving at a proper **diagnosis**—that is, appropriately identifying and labeling a set of defined symptoms—is absolutely crucial. This process enables professionals to use a common language with others in the field and aids in communication about the disorder with the patient, colleagues and the public. A proper diagnosis is an essential element to guide proper and successful treatment. For these reasons, classification systems that organize psychological disorders systematically are necessary.

The Diagnostic and Statistical Manual of Mental Disorders (DSM)

Although a number of classification systems have been developed over time, the one that is used by most mental health professionals in the United States is the **Diagnostic and Statistical Manual of Mental Disorders (DSM-5)**, published by the American Psychiatric Association (2013). (Note that the American Psychiatric Association differs from the American Psychological Association; both are abbreviated APA.) The first edition of the DSM, published in 1952, classified psychological disorders according to a format developed by the U.S. Army during World War II (Clegg, 2012). In the years since, the DSM has undergone numerous revisions and editions. The most recent edition, published in 2013, is the DSM-5 (APA, 2013). The DSM-5 includes many categories of disorders (e.g., anxiety disorders, depressive disorders, and dissociative disorders). Each disorder is described in detail, including an overview of the disorder (diagnostic features), specific symptoms required for diagnosis (diagnostic criteria), prevalence information (what percent of the population is thought to be afflicted with the disorder), and risk factors associated with the disorder. shows lifetime prevalence rates—the percentage of people in a population who develop a disorder in their lifetime—of various psychological disorders among U.S. adults. These data were based on a national sample of 9,282 U.S. residents (National Comorbidity Survey, 2007).

Source figure description (original figure omitted): The graph shows the breakdown of psychological disorders, comparing the percentage prevalence among adult males and adult females in the United States. Because the data is from 2007, the categories shown here are from the DSM-IV, which has been supplanted by the DSM-5. Most categories remain the same; however, alcohol abuse now falls under a broader Alcohol Use Disorder category.

The DSM-5 also provides information about **comorbidity**; the co-occurrence of two disorders. For example, the DSM-5 mentions that 41% of people with obsessive-compulsive disorder (OCD) also meet the diagnostic criteria for major depressive disorder. Problematic substance use is highly comorbid with other mental illnesses; 6 out of 10 people who have a substance use disorder also suffer from another form of mental illness (National Institute on Drug Abuse [NIDA], 2007).

Source figure description (original figure omitted): Obsessive-compulsive disorder and major depressive disorder frequently occur in the same person.

Note: As you've learned in the text, comorbidity refers to situations in which an individual suffers from more than one disorder, and often the symptoms of each can interact in negative ways. Co-occurrence and comorbidity of psychological disorders are quite common, and some of the most pervasive comorbidities involve substance use disorders that co-occur with psychological disorders. Indeed, some estimates suggest that around a quarter of people who suffer from the most severe cases of mental illness exhibit substance use disorder as well. Conversely, around 10 percent of individuals seeking treatment for substance use disorder have serious mental illnesses. Observations such as these have important implications for treatment options that are available. When people with a mental illness are also habitual drug users, their symptoms can be exacerbated and resistant to treatment. Furthermore, it is not always clear whether the symptoms are due to drug use, the mental illness, or a combination of the two. Therefore, it is recommended that behavior is observed in situations in which the individual has ceased using drugs and is no longer experiencing withdrawal from the drug in order to make the most accurate diagnosis (NIDA, 2018). Obviously, substance use disorders are not the only possible comorbidities. In fact, some of the most common psychological disorders tend to co-occur. For instance, more than half of individuals who have a primary diagnosis of depressive disorder are estimated to exhibit some sort of anxiety disorder. The reverse is also true for those diagnosed with a primary diagnosis of an anxiety disorder. Further, anxiety disorders and major depression have a high rate of comorbidity with several other psychological disorders (Al-Asadi, Klein, & Meyer, 2015).

The DSM has changed considerably in the half-century since it was originally published. The first two editions of the DSM, for example, listed homosexuality as a disorder; however, in 1973, the APA voted to remove it from the manual (Silverstein, 2009). While the DSM-III did not list homosexuality as a disorder, it introduced a new diagnosis, ego-dystonic homosexuality, which emphasized same-sex arousal that the patient viewed as interfering with desired heterosexual relationships and causing distress for the individual. This new diagnosis was considered by many as a compromise to appease those who viewed homosexuality as a mental illness. Other professionals questioned how appropriate it was to have a separate diagnosis that described the content of an individual's distress. In 1986, the diagnosis was removed from the DSM-III-R (Herek, 2012). Additionally, beginning with the DSM-III in 1980, mental disorders have been described in much greater detail, and the number of diagnosable conditions has grown steadily, as has the size of the manual itself. DSM-I included 106 diagnoses and was 130 total pages, whereas DSM-III included more than 2 times as many diagnoses (265) and was nearly seven times its size (886 total pages) (Mayes & Horowitz, 2005). Although DSM-5 is longer than DSM-IV, the volume includes only 237 disorders, a decrease from the 297 disorders that were listed in DSM-IV. The latest edition, DSM-5, includes revisions in the organization and naming of categories and in the diagnostic criteria for various disorders (Regier, Kuhl, & Kupfer, 2012), while emphasizing careful consideration of the importance of gender and cultural difference in the expression of various symptoms (Fisher, 2010).

Some believe that establishing new diagnoses might overpathologize the human condition by turning common human problems into mental illnesses (The Associated Press, 2013). Indeed, the finding that nearly half of all Americans will meet the criteria for a DSM disorder

at some point in their life (Kessler et al., 2005) likely fuels much of this skepticism. The DSM-5 is also criticized on the grounds that its diagnostic criteria have been loosened, thereby threatening to “turn our current diagnostic inflation into diagnostic hyperinflation” (Frances, 2012, para. 22). For example, DSM-IV specified that the symptoms of major depressive disorder must not be attributable to normal bereavement (loss of a loved one). The DSM-5, however, has removed this bereavement exclusion, essentially meaning that grief and sadness after a loved one’s death can constitute major depressive disorder.

The International Classification of Diseases

A second classification system, the **International Classification of Diseases (ICD)**, is also widely recognized. Published by the World Health Organization (WHO), the ICD was developed in Europe shortly after World War II and, like the DSM, has been revised several times. The categories of psychological disorders in both the DSM and ICD are similar, as are the criteria for specific disorders; however, some differences exist. Although the ICD is used for clinical purposes, this tool is also used to examine the general health of populations and to monitor the prevalence of diseases and other health problems internationally (WHO, 2013). The ICD is in its 10th edition (ICD-10); however, efforts are now underway to develop a new edition (ICD-11) that, in conjunction with the changes in DSM-5, will help harmonize the two classification systems as much as possible (APA, 2013).

A study that compared the use of the two classification systems found that worldwide the ICD is more frequently used for clinical diagnosis, whereas the DSM is more valued for research (Mezzich, 2002). Most research findings concerning the etiology and treatment of psychological disorders are based on criteria set forth in the DSM (Oltmanns & Castonguay, 2013). The DSM also includes more explicit disorder criteria, along with an extensive and helpful explanatory text (Regier et al., 2012). The DSM is the classification system of choice among U.S. mental health professionals, and this chapter is based on the DSM paradigm.

The Compassionate View of Psychological Disorders

As these disorders are outlined, please bear two things in mind. First, remember that psychological disorders represent extremes of inner experience and behavior. If, while reading about these disorders, you feel that these descriptions begin to personally characterize you, do not worry—this moment of enlightenment probably means nothing more than you have had a range of experiences and emotions. Each of us experiences episodes of sadness, anxiety, and preoccupation with certain thoughts—times when we do not quite feel ourselves. These episodes should not be considered problematic unless the accompanying thoughts and behaviors become extreme and have a disruptive effect on one’s life. Second, understand that people with psychological disorders are far more than just embodiments of their disorders. We do not use terms such as schizophrenics, depressives, or phobics because they are labels that objectify people who have these conditions, thus promoting biased and disparaging assumptions about them. It is important to remember that a psychological disorder is not what a person is; it is something that a person has—through no fault of their own. As is the case with cancer or diabetes, those with psychological disorders suffer debilitating, often painful conditions that are not of their own choosing. These individuals deserve to be viewed and treated with compassion, understanding, and dignity.

Summary

The diagnosis and classification of psychological disorders is essential in studying and treating psychopathology. The classification system used by most U.S. professionals is the DSM-5. The first edition of the DSM was published in 1952, and has undergone numerous revisions. The 5th and most recent edition, the DSM-5, was published in 2013. The diagnostic manual includes a total of 237 specific diagnosable disorders, each described in detail, including its symptoms, prevalence, risk factors, and comorbidity. Over time, the number of diagnosable conditions listed in the DSM has grown steadily, prompting criticism from some. Nevertheless, the diagnostic criteria in the DSM are more explicit than that of any other system, which makes the DSM system highly desirable for both clinical diagnosis and research.

Review Questions

The letters in the abbreviation DSM-5 stand for _____.

1. Diseases and Statistics Manual of Medicine

1. Diagnosable Standards Manual of Mental Disorders

1. Diseases and Symptoms Manual of Mental Disorders

1. Diagnostic and Statistical Manual of Mental Disorders

A study based on over 9,000 U. S. residents found that the most prevalent disorder was _____.

1. major depressive disorder

1. social anxiety disorder

1. obsessive-compulsive disorder

1. specific phobia

Critical Thinking Questions

Describe the DSM-5. What is it, what kind of information does it contain, and why is it important to the study and treatment of psychological disorders?

The International Classification of Diseases (ICD) and the DSM differ in various ways. What are some of the differences in these two classification systems?

Attribution and critical-reading note

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Scientists, mental health professionals, and cultural healers may adopt different perspectives in attempting to understand or explain the underlying mechanisms that contribute to the development of a psychological disorder. The specific perspective used in explaining a psychological disorder is extremely important. Each perspective explains psychological disorders, their causes or etiology, and effective treatments from a different viewpoint. Different perspectives provide alternate ways for how to think about the nature of psychopathology.

Supernatural Perspectives of Psychological Disorders

For centuries, psychological disorders were viewed from a **supernatural** perspective: attributed to a force beyond scientific understanding. Those afflicted were thought to be practitioners of black magic or possessed by spirits (Maher & Maher, 1985). For example, convents throughout Europe in the 16th and 17th centuries reported hundreds of nuns falling into a state of frenzy in which the afflicted foamed at the mouth, screamed and convulsed, sexually propositioned priests, and confessed to having carnal relations with devils or Christ. Although, today, these cases would suggest serious mental illness; at the time, these events were routinely explained as possession by devilish forces (Waller, 2009a). Similarly, grievous fits by young girls are believed to have precipitated the witch panic in New England late in the 17th century (Demos, 1983). Such beliefs in supernatural causes of mental illness are still held in some societies today; for example, beliefs that supernatural forces cause mental illness are common in some cultures in modern-day Nigeria (Aghukwa, 2012).

Source figure description (original figure omitted): In The Extraction of the Stone of Madness, a 15th century painting by Hieronymus Bosch, a practitioner is using a tool to extract an object (the supposed “stone of madness”) from the head of an afflicted person.

Note: Between the 11th and 17th centuries, a curious epidemic swept across Western Europe. Groups of people would suddenly begin to dance with wild abandon. This compulsion to dance—referred to as **dancing mania**—sometimes gripped thousands of people at a time. Historical accounts indicate that those afflicted would sometimes dance with bruised and bloody feet for days or weeks, screaming of terrible visions and begging priests and monks to save their souls (Waller, 2009b). What caused dancing mania is not known, but several explanations have been proposed, including spider venom and ergot poisoning (“Dancing Mania,” 2011). Source figure description (original figure omitted): Although the cause of dancing mania, depicted in this painting, was unclear, the behavior was attributed to supernatural forces. Historian John Waller (2009a, 2009b) has provided a comprehensive and convincing explanation of dancing mania that suggests the phenomenon was attributable to a combination of three factors: psychological distress, social contagion, and belief in supernatural forces. Waller argued that various disasters of the time (such as famine, plagues, and floods) produced high levels of psychological distress that could increase the likelihood of succumbing to an involuntary trance state. Waller indicated that anthropological studies and accounts of possession rituals show that people are more likely to enter a trance state if they expect it to happen, and that entranced individuals behave in a ritualistic manner, their thoughts and behavior shaped by the spiritual beliefs of their culture. Thus, during periods of extreme physical and mental distress, all it took were a few people—believing themselves to have been afflicted with a dancing curse—to slip into a

spontaneous trance and then act out the part of one who is cursed by dancing for days on end.

Biological Perspectives of Psychological Disorders

The biological perspective views psychological disorders as linked to biological phenomena, such as genetic factors, chemical imbalances, and brain abnormalities; it has gained considerable attention and acceptance in recent decades (Wyatt & Midkiff, 2006). Evidence from many sources indicates that most psychological disorders have a genetic component; in fact, there is little dispute that some disorders are largely due to genetic factors. The graph in shows **heritability** estimates for schizophrenia.

Source figure description (original figure omitted): A person’s risk of developing schizophrenia increases if a relative has schizophrenia. The closer the genetic relationship, the higher the risk.

Findings such as these have led many of today’s researchers to search for specific genes and genetic mutations that contribute to mental disorders. Also, sophisticated neural imaging technology in recent decades has revealed how abnormalities in brain structure and function might be directly involved in many disorders, and advances in our understanding of neurotransmitters and hormones have yielded insights into their possible connections. The biological perspective is currently thriving in the study of psychological disorders.

The Diathesis-Stress Model of Psychological Disorders

Despite advances in understanding the biological basis of psychological disorders, the psychosocial perspective is still very important. This perspective emphasizes the importance of learning, stress, faulty and self-defeating thinking patterns, and environmental factors. Perhaps the best way to think about psychological disorders, then, is to view them as originating from a combination of biological and psychological processes. Many develop not from a single cause, but from a delicate fusion between partly biological and partly psychosocial factors.

The **diathesis-stress model** (Zuckerman, 1999) integrates biological and psychosocial factors to predict the likelihood of a disorder. This diathesis-stress model suggests that people with an underlying predisposition for a disorder (i.e., a diathesis) are more likely than others to develop a disorder when faced with adverse environmental or psychological events (i.e., stress), such as childhood maltreatment, negative life events, trauma, and so on. A diathesis is not always a biological vulnerability to an illness; some diatheses may be psychological (e.g., a tendency to think about life events in a pessimistic, self-defeating way).

The key assumption of the diathesis-stress model is that both factors, diathesis and stress, are necessary in the development of a disorder. Different models explore the relationship between the two factors: the level of stress needed to produce the disorder is inversely proportional to the level of diathesis.

Summary

Psychopathology is very complex, involving a plethora of etiological theories and perspectives. For centuries, psychological disorders were viewed primarily from a supernatural perspective and thought to arise from divine forces or possession from spirits. Some cultures continue to hold this supernatural belief. Today, many who study psychopathology view mental illness from a biological perspective, whereby psychological disorders are thought to result largely from faulty biological processes. Indeed, scientific advances over the last several decades have provided a better understanding of the genetic, neurological, hormonal, and biochemical bases of psychopathology. The psychological perspective, in contrast, emphasizes the importance of psychological factors (e.g., stress and thoughts) and environmental factors in the development of psychological disorders. A contemporary, promising approach is to view disorders as originating from an integration of biological and psychosocial factors. The diathesis-stress model suggests that people with an underlying diathesis, or vulnerability, for a psychological disorder are more likely than those without the diathesis to develop the disorder when faced with stressful events.

Review Questions

The diathesis-stress model presumes that psychopathology results from _____.

1. vulnerability and adverse experiences

1. biochemical factors

1. chemical imbalances and structural abnormalities in the brain

1. adverse childhood experiences

Dr. Anastasia believes that major depressive disorder is caused by an over-secretion of cortisol. His view on the cause of major depressive disorder reflects a _____ perspective.

1. psychological

1. supernatural

1. biological

1. diathesis-stress

Critical Thinking Question

Why is the perspective one uses in explaining a psychological disorder important?

Personal Application Question

Even today, some believe that certain occurrences have supernatural causes. Think of an event, recent or historical, for which others have provided supernatural explanation.

CHAPTER 10 / CAC.903

Behavioral Observation and Functional Assessment

Behavioral observation records what a child does in context. It can confirm or challenge reports from interviews and rating scales. It is especially useful when the referral question concerns behavior frequency, triggers, consequences, or skill use. Observation is not neutral. The observer chooses what to record, when to record, and how to interpret. A good observation system makes these choices explicit.

Naturalistic and structured observation

Naturalistic observation occurs in everyday settings such as classroom, home, or playground. It has ecological validity but limited control. Structured observation occurs in a planned situation, such as a play task or problem-solving interaction. It allows comparison but may not reflect daily life. The counselor should choose based on the question. A behavior that occurs only during math instruction may need classroom observation, not a clinic play task.

Behavioral coding

Coding turns behavior into data. Event coding records each occurrence of a discrete behavior. Interval coding records whether a behavior occurs during set intervals. Duration coding records how long a behavior lasts. Latency coding records time between a cue and behavior. The coding system should define behaviors clearly. "Aggression" is too vague. "Hitting with open hand" is clearer.

Functional behavioral assessment

Functional assessment asks what maintains a behavior. The ABC model records antecedents, behavior, and consequences. A child may scream to escape difficult work, gain attention, access a tangible item, or regulate sensory input. The same behavior can serve different functions. Intervention should match function. If screaming is escape-maintained, sending the child out may reinforce it. If it is attention-maintained, planned attention for appropriate communication may help.

Reactivity and observer effects

People may change behavior when observed. A child may behave better because the counselor is present. A teacher may change instruction. Reactivity can be reduced by repeated observation, unobtrusive positioning, and habituation. The counselor should note possible reactivity in the report. A single observation is a sample, not the whole truth.

Self-monitoring

Older children and adolescents can monitor their own behavior, thoughts, or feelings. A teen may record panic episodes, study time, or mood. Self-monitoring increases awareness and provides data. It also changes behavior, which can be therapeutic. The counselor should keep the method simple and review it collaboratively. A complex log may become another failure experience.

Rating scales

Behavior rating scales summarize informant perceptions across time and settings. They are efficient and normed, but they are indirect. A parent rating may reflect parent stress. A teacher rating may reflect classroom demands. The counselor should use rating scales as one source, not as direct observation. Discrepancies should be explored, not averaged away.

Real-life application

A child hits peers during recess. Observation shows that hitting occurs when peers exclude him from a game, and it ends when an adult intervenes. The hypothesis is that hitting produces adult attention and escape from social confusion. The plan teaches a request for help, rehearses entry skills, and changes adult response. The behavior decreases because the function is addressed.

Reflection

Choose a behavior. Write an ABC hypothesis. Design an observation plan with setting, time, behavior definition, and coding method. Then write how you would check whether the hypothesis is wrong. Observation is disciplined curiosity.

ABC functional assessment



Functional assessment links behavior to context and designs a matching support.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module behavioral-assessment.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Overview of Behavioral Assessment

In other chapters, we discuss using psychophysiological assessment, cognitive assessment, and other assessment techniques. Psychology is often concerned about understanding behavior. To assess behavior, one approach would be to observe people. Observing people allows seeing the dynamic of the behavior unfold in relation to other processes. Observations might also allow getting around examinees' response biases such as social desirability.

Behavioral observation provides a temporal stream of behavior to see which behaviors precede other behaviors. An overview of observational assessment is provided by Girard & Cohn, 2016. When conducting observations, it is important to consider the context and the environment.

Approaches

Examples of behavioral assessment include naturalistic behavioral observation, analogue (structured) observation, self-monitoring/diary, behavior ratings (e.g., informant report), and self-report. Self-report is essentially retrospective self-monitoring. I published a practice brief for the assessment of externalizing behavior in school-aged children that integrates these different methods of behavioral assessment (Petersen, 2024).

Naturalistic behavioral observation involves observing a person in their natural context or habitat. For instance, it might involve a person in their home, at school, at work, or during their everyday interpersonal interactions. The observer is like a "fly on the wall" in that they want to exert as little effect as possible on the person's behavior. The observation occurs within normal situational contexts and under naturally occurring reinforcements and consequences. Naturalistic observations were the origins of behavioral assessment.

Analogue observation, also called structured observation, is the observation of a person in a contrived environment (Haynes, 2001). The goal of analogue observation is for the contrived situation to be analogous to the situations that the client is likely to encounter in their natural environment. For instance, the examiner might set up similar conditions in a lab or therapy room. Analogue observations can differ in the degree to which the observation is

structured, but they are typically more structured than naturalistic observation. Analogue observations might involve role-plays, family or marital interaction tasks, and compliance (do/don't) tasks. As an example, the observer might observe role-played scenarios in which the person engages in one or more simulated social interactions.

Contexts for Observing

There are multiple contexts in which we can observe people. Here are just a few examples:

• Classrooms

• Homes

• In the community; for example, you could take a client out into the world to see how they interact with others

• Group homes

• Psychiatric hospitals

Token economies for people with schizophrenia were based on behavioral assessment in psychiatric hospitals. A token economy is a contingency management system based on operant conditioning. Operant conditioning occurs when the frequency of a behavior is influenced by the outcomes of that behavior. In a token economy, a secondary reinforcer, such as a token or symbol, is given to the client for matching the target behavior, which can be exchanged for backup reinforcers. For instance, a token may be given to the client for making their bed. The tokens could then be turned in for backup reinforcers. Back-up reinforcers include material reinforcers such as food, money, and cigarettes; services, such as having your room cleaned; and privileges such as getting to make a phone call. Token economies have been shown to work and to influence people's behavior, especially the negative symptoms of schizophrenia. However, token economies are not widely used because of political issues.

Costs of Behavioral Observation

Behavioral observations are costly due to people and time. The cost limits the amount of observation time that is possible, which raises questions about how typical and representative the participant's behavior was during the observation period relative to their everyday life. The limited amount of observation time raises questions about reliability and external validity, that is the generalizability of the person's behavior in the situation to their typical functioning.

A gold standard for naturalistic observations would be to observe someone in their natural context across more than one day, time, situation, and context with more than one rater to get a more robust sampling of the person's behavior. But it is costly to do this, and is therefore uncommon in practice.

Dependent Variable

After (or during) the observation, we need to turn the observation into information that can be useful. To do that, we create a dependent variable using behavioral coding. Behavioral coding involves observing behavior and quantifying it, that is turning behavior into numbers that can be analyzed to make inferences. During observation, you can measure the frequency of the behavior (how many times a behavior occurs), which is a useful measure of individual differences. But you may want to know the stream of behavior, so you know when the behavior occurred. So, you might make a coding sample at regular intervals (e.g., every 6 seconds). Coding systems can differ in their granularity with which they treat time. In event-based coding, the coder codes whenever a discrete behavior occurs—i.e., the behavior has discrete start and stop points. For example, an event-based coding system may determine when a person is talking. Event-based coding allows assessing the number, duration, and sequence of behaviors with high temporal precision.

In interval-based coding, the coder codes during specified spans of time that are called intervals. Intervals can be of any length. Shorter intervals provide more temporal precision but are more burdensome to code. Longer intervals provide less temporal precision but are less burdensome to code. An example of interval-based coding would be coding the extent of positive affect that a person showed during each 10-second interval.

Coding can focus on the small units of behavior or what the behavior means. Sign-based coding involves focusing on small units of behavior, such as movements. Message-based coding involves focusing on the meaning of the behavior using theory. For example, message-based coding might attempt to code examinees' emotional states. Sign-based coding requires a lesser degree of inference, has less subjectivity, and has higher inter-rater reliability than message-based coding. On the other hand, message-based coding is potentially more theoretically interesting than sign-based coding.

Functional Behavioral Assessment/Analysis

One goal of behavioral assessment is to determine the function of the problem behavior. Every behavior has a function, reason, or purpose (or multiple ones). If you can determine the function(s) of the behavior, you may be able to help the client learn more adaptive ways of accomplishing their goals. To determine the function(s) of behavior, functional behavioral assessments (FBAs) are commonly conducted, especially in schools. FBA involves identifying the patterns of sequences that involve the problem behavior, noting the following:

- The time
- The context
- What led up to the problem behavior (antecedents)
- What problem behavior occurred
- What happened after the problem behavior (consequences)
- How the situation resolved

An FBA is based on the antecedent-behavior-consequence (A-B-C) model. The idea of the A-B-C model is that every behavior is a function of its antecedents and consequences. The goal of the A-B-C model is to identify common patterns of antecedents and consequences.

Examples of antecedents include insufficient sleep, a change in routine, and an ineffective command. Examples of consequences include giving attention, and getting to play without having to pick up the toys. By identifying the patterns of antecedents and consequences, we can generate hypotheses for the function of the behavior. For example, a possible hypothesis for a function of the given problem behavior might be to get attention, to get out of doing what they do not want to do (escape or avoidance), to get access to a preferable activity or object, or to engage in sensory stimulation (e.g., rocking).

You can design treatment to change the antecedents and consequences to show them more adaptive ways of behaving. One approach to case conceptualization or case formulation is the "6 Ps": presenting problems, protective factors, predisposing factors, precipitating factors, perpetuating factors, and prognosis. Treatment would address the presenting problems by addressing the perpetuating (maintaining) factors, and helping the client identify the triggering (precipitating) factors and how to handle those, while building on their strengths (protective factors) and considering their vulnerabilities (predisposing factors), with the intensity of treatment matched to the prognosis.

Mental Status Exam

A mental status exam is a structured assessment of a client's cognition, affect, etc. A mental status exam is often used in psychiatric hospitals to quickly "size up" a patient. Mental status exams are also useful in outpatient settings. For training clinicians, it can be useful to get in the habit of performing mental status exams.

Reliability

It is important to consider the reliability of behavioral observations. From a generalizability perspective, we aim to generalize behavioral observations across raters, contexts, and time points. Thus, inter-rater reliability and test–retest reliability are especially relevant. Averaging ratings across multiple raters can yield scores that are more reliable than the scores from the individual raters, assuming both raters have inter-rater reliability.

Validity

Validity is also an important consideration of behavioral observations. In general, little attention has been paid to validity of observational measures, particularly those used in clinical practice. This is a content validity problem. When coding, the examiner makes choices about what to code; they are not "writing down the real thing".

Reactivity

Another concern of observational measures is response biases, such as faking good, faking bad, and impression control. For instance, people may act differently because they are being observed, which is a form of reactivity. Reactivity is the change in a behavior for the mere reason that the behavior is being observed or measured. People could behave better or worse than they typically do—observation tends to push people to the extremes.

Coding

Regarding coders and observers, it is ideal during training to provide them immediate corrective feedback. It is ideal for them to be blind to hypotheses, so they are not biased in favor of a particular outcome. It can be helpful to periodically compare coders' ratings to criterion ratings that are accepted as the gold-standard ratings to prevent coder drift. Coder drift occurs when there are changes in coders' ratings over time. You can evaluate intra-observer reliability by having them rate the same (video-recorded) behavior at two different time points. You can have two or more raters code at least a subset of the participants, to establish inter-rater reliability.

Forms of Measurement

There are several forms of measurement that behavioral assessments can take. One form of measurement from observation is notes, which involves qualitative data. For rating scales, the form of measurement is often Likert scales. Likert scales assess the observer's impression of the examinee's behavior, which is a simpler way of getting the same information as many approaches to video coding. Rating scales can be completed in the context of, for example, a home observation, parent report, teacher report, or school observation.

It can be helpful to examine correlations between these different methods and contexts. Having multiple forms of measurement is great for research but is difficult to do in clinical practice because it comes at a tremendous expense. There is also difficulty in observing what you want to observe. For example, aggression is a relatively low base rate phenomenon, so the client may not engage in aggressive behavior during the observation window. There are also questions about the incremental validity of naturalistic observational assessments because simpler measures (including analogue assessments) could get at the same thing.

Analogue (Structured) Observational Assessments

Analogue or structured assessments use tasks that evoke individual differences in the behavior of interest. For instance, in our lab we have forbidden toys on a shelf, and we see how the children and parents handle that situation when they are in the room together, then when the parent is occupied on the phone, and then when the parent leaves the room and the child is alone in the room. In the clinic, we have a parent tell their child to clean up the toys so we can observe how the parent gives the command and how the child responds. To study marital conflict, researchers may have the couple provide a list of topics they disagree about, and have them have a discussion about one of those topics to evoke an argument in the lab. To study stress and anxiety, or to perform exposure in the clinic, the clinician or researcher may have the client/participant give a public speech in a simulated session.

The goal of analogue assessments is that the tasks should have content validity. That is, the tasks should be relevant to the target construct, and they should be representative of the tasks the person is likely to face in their natural environment by sampling all facets that affect the behavior problem in the natural environment.

Analogue assessment can lead to a greater frequency of the target behavior through the use of contrived situations or stimuli, compared to naturalistic observation. For instance, noncompliance might occur more often when doing a compliance task in the lab than when

observing a child in their natural environment. Therefore, the frequency or rate of behavior may not generalize to the natural context. But the functional relations of the behavior in relation to other behaviors, such as antecedents and consequences, may generalize. Thus, they can help identify potential intervention targets.

Analogue assessment is most useful for assessing behaviors that are difficult or rare to observe in the natural environment but that are frequent in the analogue environment. For example, analogue assessment is particularly useful for assessing socially sensitive behavior problems such as verbal aggression, panic, and oppositional behavior.

Analogue assessment can be sensitive to change, and therefore can be a clinically useful measure of treatment gains or failure. A client who cannot demonstrate a skill in the clinic may not be able to demonstrate a skill in the real world. But just because they show a skill in a session does not necessarily mean they will be able to use it in the real world.

In terms of reliability and validity, analogue/structured assessments tend to show greater reliability than naturalistic observations. Analogue assessments tend to show greater internal validity than naturalistic observations. That is, analogue assessments are more likely to reflect causal factors because they have more control of when and how the stimuli are presented. However, analogue assessments show weaker ecological validity than naturalistic observations and therefore tend to show weaker external validity. That is, analogue assessments yield lower generalizability than naturalistic observations because analogue assessments may not be as reflective of their functioning in the real world. Nevertheless, analogue assessments tend to show greater clinical utility, in terms of incremental validity and cost-effectiveness than naturalistic observations, when they are done well.

Self-Monitoring

Self-monitoring involves observing and recording one's own behaviors, thoughts, emotions, bodily sensations, events, etc. Given the expense of conducting behavioral observations, you can train people to be their own observers, which has better portability. Phone apps can now be used to track many different processes, such as mood, sleep, and nutrition. These are examples of electronic diaries. An overview of research on self-monitoring and electronic diaries is provided by Korotitsch & Nelson-Gray, 1999 and Piasecki et al., 2007.

Self-monitoring provides a great deal of information at a low cost. There are several benefits of self-monitoring. Self-monitoring can help characterize the problem and the presumed causal or maintaining factors. Self-monitoring can also contribute to greater objective self-awareness. In addition, self-monitoring can be used to track treatment progress. Furthermore, self-monitoring can be helpful for assessing covert behaviors that are hidden to outside observers, such as obsessions, paranoid thoughts, and sexual problems.

Self-monitoring is a component of most empirically supported treatments. In addition, the extent to which clients complete self-monitoring is predictive of treatment effectiveness. As an example of self-monitoring in therapy, in parent management training, we have parents track (tally) the frequency of their child's compliance, noncompliance, and aggression each day during the week between sessions as homework. Self-monitoring provides a more objective way to track treatment progress than retrospective reports or satisfaction ratings because clients track the specific behavior the moment it occurs. We modify treatment in response to the information gained from tracking. We also become more aware of the

situations and contexts in which the behavior occurs, so it is predictable and the client can better anticipate, prepare for, and control the problem.

Based on principles of measurement-based care, assessment does not just occur at the outset of treatment; it should continue throughout treatment. And self-monitoring helps parents become more aware of the actual frequency of the problem behavior. For instance, they might have over- (or under-)estimated how often a behavior occurs. Thus, self-monitoring can lead to a change in people's perceptions of the behavior. Also, some children will misbehave less at home or school due to reactivity because they know they are being observed, so this can be a way of kick-starting some behavioral improvement.

There are often changes in a behavior in response to self-monitoring—typically, the client experiences behavioral improvement. Self-monitoring might increase awareness of how often a problem behavior is happening and their perceived predictability and control of the behavior. Self-monitoring is now considered an empirically supported treatment for a number of conditions, including academic/conduct problems, addictive behaviors, anxiety and depression, disordered eating, physical health problems, psychosis, and insomnia. That is, assessment is a form of intervention!

However, there are challenges to self-monitoring. One challenge is how to evaluate the reliability of self-monitoring. Another challenge is how to handle if the client fills out the whole tracking sheet retrospectively, all at once, right before session—this is called "backfilling". In addition, there is a possibility of inaccuracy and distortion with self-monitoring due to the respondent not having been trained on how to observe the phenomenon of interest. To counter this, in parent management training, we explicitly define what counts as an instance of compliance, noncompliance, and aggression. Another challenge with self-monitoring is that it involves respondent burden because it requires compliance by the client. To handle noncompliance with self-monitoring, it can be helpful to "get procedural" and to discuss with the client when they will track, where they will put the tracking sheet, etc. It can also be helpful to perform advance troubleshooting: help the client anticipate obstacles and how they will address them. For instance, if the client feels that they might forget to do the tracking, have them set a timer. Self-monitoring has lower reliability and validity than behavioral observation, but it is still useful and widely used because it is cheap and easy to obtain lots of information that can lead to behavioral improvement.

Behavior Rating Scales

Behavior rating scales include self- or informant ratings on a checklist of behaviors assessing particular construct(s). An example assessment system that uses behavior rating scales is the Achenbach System of Empirically Based Assessment (ASEBA) that includes the Child Behavior Checklist (CBCL), Teacher's Report Form (TRF), Adult Self-Report, etc. The ASEBA is an empirically based assessment of psychopathology, rather than a DSM-focused assessment.

Behavior rating scales are widely used in research and clinical settings. They are relatively cheap and easy to administer and score. Item response theory techniques reveal issues with different levels of Likert scale anchors of rating scales. For example, people assign different meanings to the levels "never", "sometimes", and "a lot". It is often better to have the rater

describe or provide the numeric amount (Petersen et al., 2026). For example: "How many times during the last week did your child physically hit someone else?"

Behavior rating scales are typically retrospective where the rater has to reflect back and estimate the frequency of behaviors and people's responses to the environment. Retrospective ratings can have challenges with validity due to inaccurate memory recall.

Assessment of Therapeutic Process

Other behavioral assessments include assessments of therapeutic process. Example assessments of therapeutic process include having a video recording of a session and then coding it in a naturalistic setting. For a therapist, the naturalistic setting is the therapeutic context.

An example of assessment of the therapeutic process could include assessing the clinician's adherence and competence. Assessing adherence would involve coding whether the therapist is following the treatment protocol and manual, that is, the degree to which the clinician shows treatment fidelity. Assessing the clinician's competence would involve coding how skilled the therapist is in providing the treatment. Coding clinician competence in most treatments, including cognitive behavioral therapy, would not be a moment-to-moment coding; instead, the coder would examine the sequence of interactions between client and therapist.

Conclusion

Examples of behavioral assessment include naturalistic behavioral observation, analogue (structured) observation, self-monitoring/diary, behavior ratings (e.g., informant report), and self-report. A gold standard for naturalistic observations is to observe people in their natural context across more than one day, time, situation, and context with more than one rater to get a more robust sampling of the person's behavior. Compared to naturalistic observation, analogue assessments tend to show greater reliability and internal validity but lower external validity. A useful approach to identify the function of a given behavior is to identify patterns of sequences according to the antecedent-behavior-consequence (A-B-C) model as part of a functional behavior assessment. Another useful approach is self-monitoring, which provides a lot of information at low cost and is therefore a component of most empirically supported treatments.

Suggested Readings

Girard & Cohn, 2016

CHAPTER 11 / CAC.903

Assessment of Cognition and Thought Processes

Cognitive assessment examines how a young person thinks, remembers, attends, plans, and solves problems. It is not only about IQ. A child may have strong reasoning but weak working memory. Another may have adequate memory but slow processing under anxiety. Cognitive assessment should describe a pattern of functioning that can guide support, not produce a single label that closes conversation.

Domains of cognition

Common domains include attention, working memory, processing speed, language, visual-spatial reasoning, executive function, and memory. Executive function includes planning, inhibition, cognitive flexibility, and goal maintenance. These domains interact. A child who cannot hold instructions in mind may appear noncompliant. A child who processes slowly may appear unmotivated. The counselor should ask what cognitive demand the setting places on the child.

Methods

Cognitive assessment may use standardized tasks, curriculum-based measures, think-aloud protocols, observation, and self-report. Standardized tasks allow comparison to norms. Think-aloud methods reveal strategy use. Observation shows how the child approaches difficulty. Self-report can reveal perceived effort or confusion, though children may have limited insight. Multiple methods reduce the risk of mistaking one task for the whole mind.

Anxiety, motivation, and performance

Test performance is not pure ability. Anxiety can consume working memory. Low motivation can reduce effort. Fatigue, hunger, and sleep loss matter. A child from a unfamiliar testing culture may need more rapport and explanation. The counselor should create conditions that allow a fair sample of functioning. Standardization should be preserved where required, but accommodations should be documented and interpreted carefully.

Cognitive errors and content

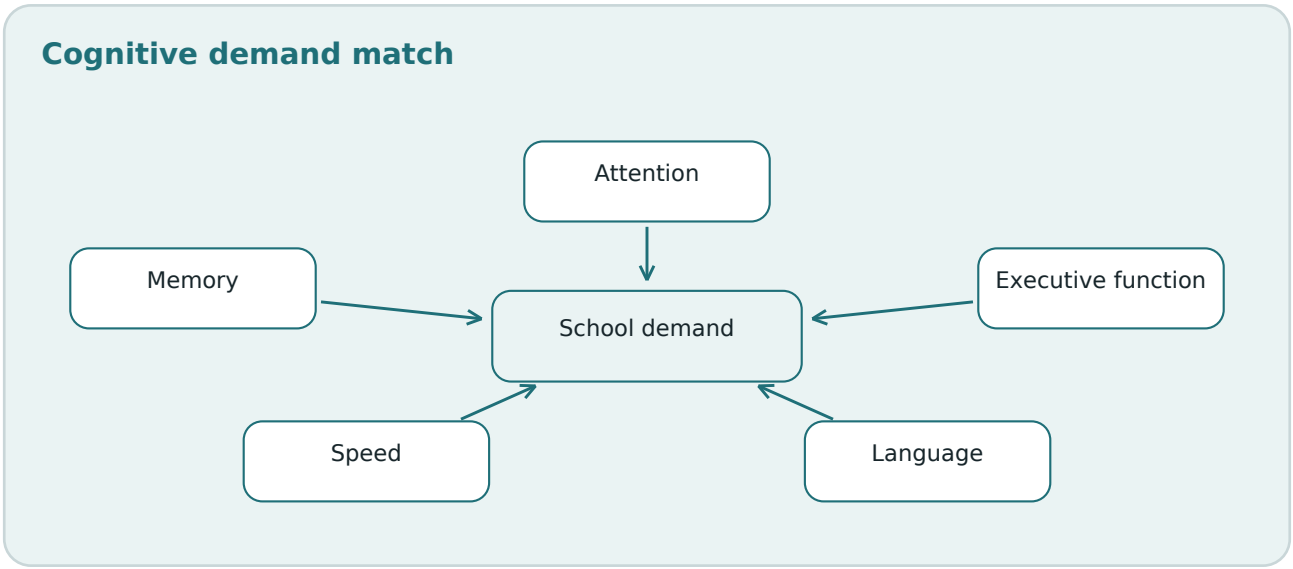
Cognitive assessment can also examine thought content. A child with depression may show negative automatic thoughts. A child with anxiety may show threat bias. A child with trauma may show intrusive images. Cognitive-behavioral assessment uses thought records, downward arrow, and behavioral experiments. With younger children, thoughts may be externalized as a "worry voice" or drawn as pictures.

Real-life application

A nine-year-old is referred for "not listening." Cognitive assessment shows average reasoning but weak working memory. Classroom observation shows the child fails after multi-step instructions. The plan includes short instructions, visual checks, and repetition. The label changes from defiance to a support need. The child's behavior improves because the demand is adjusted.

Reflection

Choose a classroom behavior. List two cognitive demands that might explain it. Design a brief assessment plan that includes a task, an observation, and a caregiver or teacher question. Cognitive assessment should lead to support, not blame.



Behavior may reflect a mismatch between cognitive demands and supports.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module cognitive-assessment.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Overview of Cognition

Cognitive-behavioral therapy (CBT) is the most effective treatment we have for many mental health conditions, especially depression and anxiety. The cognitive therapy component of CBT assumes that thoughts mediate the influences of the environment on emotions and behavioral responses. In general, cognitions and cognitive processes play a key role in many psychological theories. Therefore, cognitions and distorted cognitions are assumed to be a key construct to assess. However, there are many challenges to cognitive assessment. Some consider cognitions to be fictitious entities or epiphenomenal to people's experience—i.e., that is, they are not a causal process, but rather they are a by-product of neural and biological processes.

Aspects of Cognition Assessed

Multiple aspects of cognition can be assessed, including cognitive products, processes, and structures or organization. Cognitive products include a person's conscious thoughts or imaginal images. Cognitive processes include how a person transforms the environmental input and how they take meaning from it. Cognitive structures and organization include the hypothesized structures that a person has to guide information processing.

Consider an example of a person with an anxiety disorder. Cognitive products for the person may include having the thought, "I'm going to make a fool of myself, and everyone will think I'm stupid". The person's cognitive processes may involve the over-estimation of personal risk. The person's cognitive structures may include "danger" schemas that, when activated, lead to attention bias toward threat stimuli. Though, this finding is under question because of challenges to the reliability of the dot-probe task, which has been frequently used to examine attentional bias.

Approaches to Assessing Cognition

The general approaches to assessing cognition include endorsement methods and production methods. Endorsement methods for assessing cognition contain a predetermined set of thoughts that participants identify or rate, such as a checklist. Using production methods, participants generate or recall their thoughts, i.e., the method involves free response. The psychometric status is currently stronger for endorsement methods; however, increasing attention is being given to production methods. Approaches to assessing cognition are reviewed by Dunkley et al., 2019.

Self-Reports of Cognition

Self-reports of cognition can either be in written or interview format and comprise most of the cognitive assessment techniques that are regularly used in clinical practice. Few of the other approaches to assessing cognition are used routinely in clinical practice. In a self-report of cognition, the participant must reflect on and report about their own cognitive style. Thus, self-report of cognition involves introspection.

Self-reports of cognition are still used in CBT, to learn about people's cognitive styles, their attributions to life events or a specific negative event, and their expectancies. Expectancies include, for instance, what the person anticipates happening.

Pros

Pros of self-reports of cognition include:

• They are cheap, easy to score and administer, and are able to be useful in clinical practice.

• They are sensitive to treatment effects.

Cognitive therapy advocates would argue that changes in cognitions mediate treatment changes (i.e., they are thought to be a causal mechanism that explains how therapy works).

• They are standardized, normed, and have been psychometrically validated.

Cons

Despite the fact that these techniques are so popular, there are still some major drawbacks to these approaches. Cons of self-reports of cognition include:

• They require a participant to have insight into their cognitions that they can report on—such insight is not always typical.

Nisbett & Wilson, 1977 published a classic study titled, "Telling more than we can know: Verbal reports on mental processes". The study was based on the idea that people believe they were thinking particular things; however, they reported thoughts on things that would not have been possible for them to know. Findings such as these challenge the validity of self-reported cognition.

• They require a person to reflect upon their cognitions, retrospectively recalling what they were thinking in various situations.

• At least in the written format, which is often used as an endorsement method, participants are limited to responding to the experimenter-generated choice options—which can lead people into having certain thoughts and make people endorse thoughts that they did not actually have.

Think-Aloud Approaches (Thought Listing)

In think-aloud approaches, the participant is asked to verbalize their cognitions in the form of a continuous monologue while performing some task, and responses are recorded for later evaluation. In this way, thoughts are assessed concurrently with their occurrence. In some approaches, such as Articulated Thoughts in Simulated Situations (ATSS), thoughts are assessed in simulated situations that are geared towards eliciting certain emotions—the "situations" can be presented in video, audiotape, or virtual reality format. In think-aloud approach, the participant lists their thoughts, so the approach is also called "thought listing".

The purpose of think-aloud approaches is to avoid retrospection, and to create a situation in the lab that resembles a real-world situation to see what thoughts or what stream of thoughts are generated. Think-aloud approaches are reviewed by Davison et al., 1997.

Examples

Examples of think-aloud assessments of cognition include video replay and private speech.

Video Replay Video replay is also called videotape thought reconstruction. In videotape thought reconstruction, the examiner video records the interaction, then replays the video for the examinee, and asks the examinee what they were thinking during that moment. It is more proximal than retrospective report. The act of re-watching a video of oneself can be strange, and there is commonly reactivity to that.

Private Speech Another example of think-aloud cognition is children's private speech. Young children sometimes talk out loud to themselves to guide their behavior, and the speech is not serving a communicative function. This is known as private or self-directed speech. Private speech is most common among 2- to 7-year-olds, and it provides insight into their cognition, often while performing a challenging task.

Pros

Pros of think-aloud approaches include:

• Participants are asked to report on cognitions at the same time or just after the thoughts are occurring, so they are less sensitive to retrospective recall bias.

• Responses are not limited to experimenter-selected choices—all cognitions can be assessed.

The constraints are made in terms of how the thoughts are coded and analyzed by the experimenter, not in terms of the unstructured and open-ended thoughts that the participant can provide.

• Specific situations of interest to the research team can be explored.

For example, the examiner can examine the respondent in situations that are thought to evoke particular cognitive processes, including situations that might be impractical due to low frequency or that are unethical to assess in their natural context. For example, an examiner could examine a person's self-reported thoughts in response to social criticism for someone with social anxiety.

Cons

Cons of think-aloud approaches include:

• Performance is susceptible to observer effects.

Observer effects are changes in the person's response because they are being watched or recorded. That is, response biases may be present because the person's responses are not anonymous.

• Social desirability bias: people may censor negative or disturbing thoughts and may describe their thoughts in a way that portrays themselves in a more positive light.

• It can be difficult to generate the thoughts.

There are individual differences in people's awareness of their cognitions.

• Think-aloud approaches can interrupt the thought flow—we think faster than we can speak, so verbalizing thoughts can alter the thoughts.

That is, talking about thoughts interferes with the task itself.

• It can be difficult to get a whole narrative of thoughts from think-aloud approaches.

• It can be difficult to code the qualitative thoughts and reduce them down to meaningful data.

And coders may attribute different meaning to a thought than was intended by the respondent.

• Think-aloud approaches tend to be lower in ecological validity because they typically occur in the lab rather than in a naturalistic context.

Random Thought Sampling

The purpose of thought sampling is to quantify characteristics or aspects of thinking in an ecologically valid context. By ecologically valid, we mean that the situation closely represents the specific situations under which the person usually exists. In a random thought sampling procedure, participants are given a beeper or some other device that beeps randomly. When the person hears the beep, they must record what they are doing and what they are thinking—either quantitatively or qualitatively. The person hears a beep, and jots down the thought they were having when the beep occurred, so it is retrospective back a second.

In the experience sampling method (ESM), the examiner can also ask participants to report the context in which the experience occurred. This can be helpful for those with schizophrenia who are hallucinating to help link the hallucinations to particular situations or contexts. Thought sampling and ESM are described by Hurlburt, 1997.

Pros

Pros of random thought sampling include:

• The delay of recall is instantaneous—hence participants are not forced to recall what they were thinking.

• The situations are ecologically valid in that they are occurring in situations that the person actually experiences, rather than a lab.

• Random thought sampling gets at what people are actually thinking about rather than what they think they are thinking about.

• Random thought sampling is better able to detect fluctuations in thoughts than retrospective approaches, and it is better able to detect co-variation with other processes, such as mood.

• Random thought sampling involves less interference with the interaction.

Presumably, the "flow" of a situation is not interrupted…although it very well might be.

• A client's reactivity could show therapeutic benefit.

Cons

Cons of random thought sampling include:

• It assesses narrow slices of time.

• Because it assesses narrow slices of time, it is hard to capture the whole sequence of events.

• It is odd to be interrupted by a beeper.

• Random thought sampling could elicit reactivity such that thinking about one's thoughts could change thoughts or create thoughts.

• Random thought sampling only accesses the contents of the consciousness, like the approaches discussed earlier.

That is, the respondent has to be aware of the thoughts in order to write them down. There are likely many cognitive processes that nobody is aware of.

• The written-down thoughts have to be coded.

• Because random thought sampling typically occurs in an ecologically valid context, the experimenter has less experimental control of the situation, context, and stimuli.

Consequently, the experimenter is less likely to discover relations between rare but theoretically important processes.

Cognitive Science Approaches

Cognitive science approaches to assessing cognition include a range of non-mutually exclusive approaches, including performance-based measures, cognitive modeling, and cognitive neuroscience approaches. These cognitive science approaches are different from the approaches described earlier in that they do not rely on respondents reporting their thoughts. Cognitive science approaches try to assess a person's cognitions based on behavior or neural functioning. For instance, using performance-based measures and cognitive modeling, the approaches are based on behavioral performance and then use a model to try to estimate what is happening cognitively—the participants just have to engage and process information.

Performance-Based Measures

Performance-based measures are used to assess many domains and constructs, including attention, executive functions, inhibitory control, memory, decision-making, categorization, etc. Examples of performance-based measures include the Stroop test, Iowa Gambling Task, and implicit cognition tasks such as the Implicit Association Test. In many performance-based tasks, respondents' accuracy and reaction times are evaluated when there are competing demands.

Cognitive Modeling

Cognitive modeling attempts to decompose behavioral performance into different indices that reflect different sub-processes, such as working memory and processing speed. Behavior is complex and is influenced by many different processes, such as cognitive, motivational, and response processes. If you observe that a person shows behavioral deficits, it can be helpful to know what specific process is responsible for the behavioral deficits.

It can be difficult to use multiple behavioral tasks to decompose basic processes because any task taps multiple cognitive processes, including likely overlapping processes, and each is affected by measurement error. Cognitive modeling seeks to identify the "hidden processes" that underlie behavioral performance in complex tasks. Cognitive modeling generally requires a lot of data—for example, many trials—but it can be used with small sample sizes.

Busemeyer & Stout, 2002 provide an example of cognitive modeling of the Iowa Gambling Task to identify the sub-processes that account for behavioral performance deficits in Huntington's Disease. The Iowa Gambling Task (IGT) attempts to simulate real-life decision-making based on learning from decisions that have different probabilities of rewards and punishments. In the Iowa Gambling Task, participants are presented with four virtual decks on a computer screen. They pick a card each turn, and have to select from one of the four decks. Some decks reward the player more often and some decks penalize the

player more often. The player has to figure out which decks to pick from based on statistical probabilities.

Busemeyer & Stout, 2002 compared and tested competing models for the task based on different theoretical possibilities. With cognitive models, the aim is to achieve a combination of accuracy and parsimony (simplicity). Busemeyer & Stout, 2002 selected a model with three parameters: a cognitive process, a motivational process, and a response mechanism. A cognitive process was examined with an updating rate parameter, which describes a person's memory for past sequences produced by each deck. A motivational process was examined with the attention weight parameter, which describes the amount of attention a person allocated to gains versus losses. A response mechanism, such as recklessness and/or impulsivity, was assessed with threshold parameter, which describes the sensitivity of the choice mechanisms to the expectancies.

Busemeyer & Stout, 2002 found that people with Huntington's Disease showed deficits in the updating rate and threshold parameters but not in the attention weight parameter. In particular, the people with Huntington's Disease were more reactive to recent information and forgot old information more rapidly, and they became less sensitive with training and produced more random behavior. But people with Huntington's Disease did not show deficits in how much attention they allocated to losses.

Other examples of clinically relevant cognitive model include cognitive modeling approaches to better understand cognitive processes in sexual aggression and eating disorders (Treat et al., 2007). Emerging approaches for cognitive modeling include use of artificial intelligence (Frank & Goodman, 2026).

Challenges

Despite the advantages of not relying on participants' recall of their thoughts, there are challenges to using cognitive science approaches to assess cognition.

Reliability of Difference Scores One challenge related to the use of performance-based assessment of cognition is that many such tasks involve difference scores. A difference score involves the subtraction of one score from another score. In performance-based assessments of cognition, many tasks subtract scores, especially accuracy or reactive time, in one condition from scores in another condition. For example, the most-often used dependent variable in the Flanker task, Stroop task, stop-signal task, and dot-probe task is a difference score. For instance, in the Flanker task, the reaction time to incongruent (interference) stimuli is subtracted from the reaction time to congruent stimuli.

The problem is that difference scores tend to be lower in reliability than other scores because differences depend on the reliability of both indices in the subtraction, as described in the source discussion. Difference scores tend to be lower in reliability than each of the indices that compose it, especially when the two indexes are correlated. Therefore, scores on these tasks tend to be less reliable than scores on other tasks that do not involve difference scores. To be reliable, difference scores require high reliability of the individual indices compared to the correlation between them. Otherwise said, the more two things are the same thing, the more likely that subtracting one from the other leaves measurement error rather than construct variance.

As an alternative to difference scores, researchers can examine residualized change scores. In addition, researchers have developed accuracy-based and adaptive assessments for attention control that do not use difference scores or speeded responding and thus show stronger reliability (Draheim et al., 2024).

Because of the reliance of difference scores in the dot-probe task, Rodebaugh et al., 2016 challenged whether attention bias toward threat exists or whether it is stable across time for those with anxiety.

Reliability Paradox Another challenge with performance-based assessments of cognition is known as the reliability paradox (Hedge et al., 2018). Many basic cognitive paradigms, such as the Flanker task, were designed for detecting normative cognitive effects. However, a downside of that is that not all of them are good for assessing individual differences. That is, their scores are not reliable for a particular person. Low reliability precludes making strong inferences and decisions about individuals and calls into question the validity of inferences regarding individual differences, or people's change, in scores on those tasks.

The reliability paradox is that despite some robust cognitive tasks showing well-established experimental effects, many of these tasks do not produce reliable individual differences, including the Flanker task, Stroop task, and stop-signal task, etc. Experimental effects become well-established, and therefore those tasks become widely used, when between-subject variability is low. However, low between-subject variability leads to low reliability of individual differences. The stability coefficient, i.e., test–retest reliability, an index of reliability of individual differences relies on a Pearson correlation. As described in the source discussion, correlation requires variability: restricted range leads to weaker associations. When a measure has low between-subject variability, it prevents us from detecting consistent rank ordering of people across time. That is, the very reason such tasks produce robust and easily replicable experimental effects due to low between-person variability makes their use as correlational tools problematic. Moreover, the poor test–retest reliability of many performance-based assessments is exacerbated by the use of difference scores. Because many tasks using reaction times use difference scores, their reliability tends to be poor (Draheim et al., 2019). Moreover, reaction times are subject to speed-accuracy tradeoffs, which are ignored when examining only reaction time (Draheim et al., 2019). Researchers are thus advised against using reaction time-based measures (Draheim et al., 2019).

The implications of the reliability paradox are that many well-established approaches in experimental, cognitive, and neuropsychology may not translate well to the study of individual differences, even though the measures can be useful for making group-level inferences. It is important to know the reliability of your measures, especially when dealing with issues of trying to understand where a particular person stands on the construct relative to other people.

Unreliability and measurement error are threats to science and knowledge. Unreliability leads to false inferences and failures to replicate findings. As described in the source discussion, and formalized with the attenuation formula (the source discussion), associations with other variables are weakened to the extent that measurement error exists. If you know the degree of unreliability, you can account for it using the disattenuation formula (the

source discussion), to get more accurate estimates of the true associations with other variables.

It is important to test the reliability of your measures and report the reliability in papers. Continually work to improve the reliability of your measures. There are attempts to improve the reliability of measures of basic cognitive processes like those assessing attention control (Burgoyne et al., 2023). Ways to improve the reliability of measures are described in the source discussion. Another way to improve reliability of scores is to aggregate multiple measures using a multimethod approach in structural equation modeling.

Cognitive Neuroscience Approaches

Another branch of cognitive science approaches includes cognitive neuroscience approaches. For example, cognitive neuroscience approaches connect cognitive processes to measures of brain functioning, such as electroencephalography (EEG) or functional magnetic resonance imaging (fMRI). An example of linking cognitive processes to measures of brain functioning might be examining people's digit span in relation to fMRI measures. Cognitive neuroscience techniques are discussed in the source discussion. Cognitive neuroscience is a high priority of the National Institute of Mental Health with their emphasis on the Research Domain Criteria (RDoC). RDoC seeks to assess the underlying substrates of illness across multiple levels of analysis and the neurodevelopmental trajectories rather than just behavioral symptoms, which are heterogeneous.

Conclusion

Multiple aspects of cognition can be assessed, including cognitive products, processes, and structures or organization. The general approaches to assessing cognition include endorsement methods and production methods. Cognitive assessments include self-report, think-aloud approaches, random thought sampling, performance-based measures, cognitive modeling, and cognitive neuroscience approaches. One challenge related to the use of performance-based assessment of cognition is that many such tasks involve difference scores, which tend to be lower in reliability than each of the indices that compose it. Another challenge with performance-based assessments of cognition is known as the reliability paradox that despite some robust cognitive tasks showing well-established experimental effects, many of these tasks do not produce reliable individual differences due to low between-subject variability and the use of difference scores.

Suggested Readings

Dunkley et al., 2019; Treat et al., 2007

CHAPTER 12 / CAC.903

Intellectual and Adaptive Functioning Assessment

Intellectual assessment estimates general cognitive ability and specific cognitive strengths. Adaptive functioning assessment examines how a child manages everyday life: communication, self-care, social skills, safety, and responsibility. Both are important in developmental assessment. A child may have cognitive ability that is not reflected in daily independence because of environment, anxiety, or lack of opportunity. Another may have adaptive skills that exceed test performance because of strong support.

Intelligence as a construct

Intelligence is a contested construct. Some models emphasize a general factor. Others emphasize broad abilities such as fluid reasoning, crystallized knowledge, visual processing, and processing speed. The counselor should know the model behind a test. A full-scale score may hide meaningful variation. A profile may show strengths that can be used in intervention.

History and caution

Intelligence testing has a harmful history in some contexts, including eugenics and discriminatory classification. Modern ethical practice requires humility, cultural caution, and focus on support. A score should not be used to limit a child's future. It should be used to understand current functioning and needed accommodations. The counselor should explain limits clearly.

Adaptive behavior

Adaptive behavior is measured through caregiver or teacher report because it concerns daily functioning. It includes conceptual, social, and practical skills. A child with intellectual disability is identified by limitations in both intellectual and adaptive functioning, with onset during development. A low IQ alone is not enough. A child may score low because of language, culture, or testing conditions. Adaptive data help prevent misclassification.

Dynamic assessment

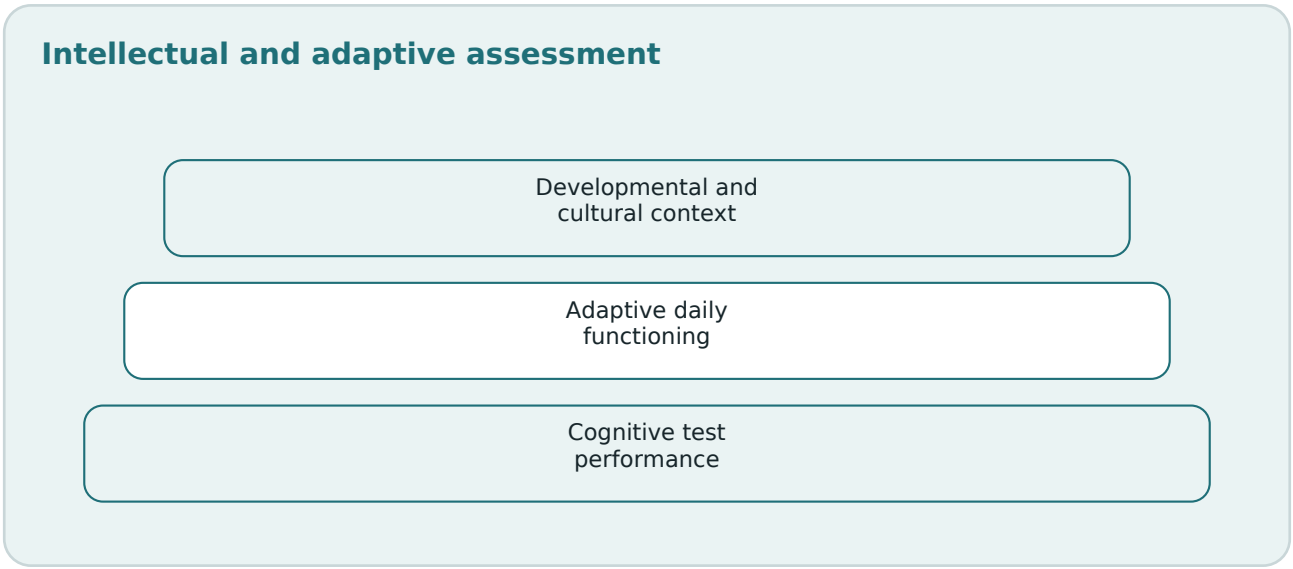
Dynamic assessment examines how a child learns with support. The counselor may teach a strategy and observe change. This approach can reveal learning potential and reduce cultural bias. It is especially useful when standardized norms are questionable. Dynamic assessment does not replace standardized testing when required, but it adds functional information.

Real-life application

A child scores low on a verbal cognitive test after recent migration. Adaptive reports show strong practical skills in the home language. The counselor uses nonverbal reasoning tasks, dynamic assessment, and interpreter-supported interview. The report avoids a global deficit label and recommends language support and continued monitoring. The assessment protects the child from premature placement.

Reflection

Write why a full-scale score alone is insufficient for a high-stakes decision. List three additional sources of information. Intellectual assessment should describe functioning, not define worth.



Intellectual assessment should be interpreted alongside daily functioning and context.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module intellectual-assessment.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Defining Intelligence

Definitions for intelligence are many. Some define intelligence as problem-solving abilities and the capacity for learning, where general intelligence is called "g" and reflects the ability to use this capacity. We can also consider emotional aspects of intelligence. For instance, social cognition reflects adaptive functioning. We can consider intelligence to be what reflects adaptive functioning. We can also consider whatever we value to be intelligence. For example, you might hear about a player's "basketball IQ". Some conceptualizations of intelligence emphasize it as a brain thing that involves speed of processing, processing capacity, and connectedness. From this perspective, intelligence includes components such as working memory, processing speed, visual and auditory perception, etc. We could also consider the products of these skills and components in interaction with the world. This is the goal of behavioral measures that are thought to reflect the underlying neural mechanism of general intelligence ("g").

History of Intelligence Research

A history of intelligence research is described in Sechrest et al., 1998. The assessment of intelligence is considered one of the key successes in psychology. Intelligence is one of the most widely assessed constructs in research and practice. Information from intelligence tests has important uses. Many people have contributed in important ways to the assessment of intelligence. Here, we outline just a few key figures as a way to introduce different models of intelligence. Please note that our discussion of the history of intelligence and key figures is no endorsement of their views. For instance, historically, some of the researchers on intelligence were proponents of eugenics.

Sir Francis Galton (1822–1911)

Sir Francis Galton was known for his work on lots of topics, including correlation, regression, twin studies, heredity, individual differences, the fax machine, and the telegraph. His work discovered the importance of reaction time, and conceptualized it as reflecting an underlying construct. He examined individual differences in many domains: vision, hearing, reaction time, other sensorimotor functions, and physical characteristics. Many of these individual differences would fall into the construct of "intelligence" now.

James Cattell (1860–1944)

James Cattell was Wihelm Wundt's first American student. Cattell measured sensory acuity and psychophysical characteristics. Cattell's work followed Galton's, and he examined whether reaction time might shed light on consciousness. Cattell used factor analysis to formally approach Galton's questions about why scores correlated—why people who tended to be good at one thing also tended to be good at lots of other things. He was credited with initiating the individual differences movement and with the first published reference to a mental test in the psychological literature.

Alfred Binet (1857–1911)

Alfred Binet was a French psychologist, and his work was more applied than the work by Galton and Cattell. He tried to apply principles of psychology to ameliorate mental problems. In France, he was forward-thinking at the time: he sought to identify children with an intellectual disability so that they could receive training and live a productive life. He and his colleague, Théodore Simon, asked who is trainable and when. And with their work, so begins clinical assessment.

In 1905, Binet and Simon developed the Binet–Simon Intelligence Scale for children. It was developed from a government commission to study the education of children with an intellectual disability (called mentally retarded or "arriéré" at the time). Items on the scales were administered from easiest to hardest, until the participant could not do three-quarters of the items. This approach is similar to modern-day item response theory approaches to adaptive testing. Items were selected for inclusion in the scales if they showed criterion-related validity in relation to two criteria:

• age—discriminative validity (i.e., age differentiation); items were identified for which older children performed better than younger children, and

• performance in school; items were selected that differentiated children who showed success versus failure in school.

This approach of using ceiling criteria to establish stop points in testing is still widely used today. The Binet–Simon Intelligence Scale included subtests that are still widely used today involving:

• Naming objects in pictures (similar to the Peabody Picture Vocabulary Test): phone, squash, etc.

• Digit span and reverse digit span: having a respondent repeat numbers of a progressively longer and longer length, then having respondents repeat numbers in the reverse number; examines how much information a person can store in working memory and their underlying capacity.

• A vocabulary test: asking children to define words to show they know the meaning of words; a concrete measure reflective of ability to learn words.

• Similarities: examines abstraction and understanding.

• Block design: having the respondent manipulate blocks to make different designs; examines visuospatial reasoning.

• Comprehension questions: how much the person has learned information about life and culture because of their capacity.

Binet and Simon assessed performance of typically developing children and children with intellectual disability. They quantified children's performance on the tests with what they called "mental age" (as opposed to chronological age). The child's mental age was estimated by correcting for the expected (median) score at that age. For instance, if a child has a mental age of a 7-year-old, that means that the child's performance is the same as the average (50th percentile) 7-year-old.

The Binet–Simon Intelligence Scale was validated with the opinions of teachers of children's academic functioning. Teachers' reports used to show stronger criterion-related validity compared to intelligence tests. Now, intelligence tests are the gold standard; they show stronger criterion-related validity compared to teacher reports.

Lewis Terman (1877–1956)

Lewis Terman at Stanford University revised the Binet–Simon Intelligence Scale, and the revised test became known as the Stanford–Binet Intelligence Scales. After the release of the Stanford–Binet, academics became excited about intelligence testing.

William Stern (1871–1938)

A key challenge that intelligence researchers had was how to quantify children's performance on the tests. Binet had introduced the concept of the mental age—the typical age for participants with that score. Mental age had a generalizable notion that was informative. It provided meaning compared to saying that someone received a score of 8 on the measure. Mental age conceptualizes intelligence as a measure of current functioning to inform how to improve their functioning to efficiently use resources.

However, performance does not increase across all domains with age through adulthood. That is, the same level of performance could have multiple mental ages. Therefore, William Stern changed the scoring to a trait measure by developing the Intelligence Quotient (IQ) that Terman used in the Stanford–Binet. IQ was calculated using the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

This helped keep IQ to be a more stable as a characteristic of the person. IQ was more trait-like and stable than mental age, and it does not change as much with age because it accounts for age.

Today, this index is known as the "Ratio IQ". It involves taking a ratio of two numbers, but it is not a ratio measure because it does not have a meaningful zero, and the gaps between the scores might not be equidistant.

Stern was a German psychologist who moved to America and promoted Binet's ideas. Compared to England where you became what your parents were, in America, your future was thought to be determined by your underlying ability. This was based on the idea that "everyone who is able could become successful". Therefore, intelligence took hold with the idea that individual abilities determine success. Intelligence took hold, then World War I occurred.

Robert Yerkes (1876–1956)

World War I involved using lots of technology. In 1917, under the lead of Robert Yerkes and others, the United States Army developed the Army Alpha and Beta Test, a precursor to the eventual Armed Services Vocational Aptitude Battery (ASVAB). The U.S. Army sought to improve the army by classifying people in rank and tasks according to their capabilities—it is important to have the best army possible or you will lose the war.

The content was similar to the Binet scales, but it was changed from a clinical test to a multiple-choice, paper-and-pencil test that a clerk could give to large groups. The Alpha Test included words and letters for literate recruits. The Beta Test was a picture-based

intelligence test without words for illiterate recruits or recent immigrants. For instance, the Beta Test involved having recruits code digits with symbols, which is still part of the Wechsler Adult Intelligence Scales today.

The Army Alpha and Beta Tests are among the first examples of affirmative action because anybody could get picked based on their test performance compared to only doing what one's parents did. However, the Army Alpha and Beta Tests turned out to be biased against rural, non-whites because they were less experienced with standardized setting. The Army Alpha and Beta Tests were the precursors to much of the modern psychometric movement in psychology, and to the development of more advanced intelligence tests.

Charles Spearman (1863–1945)

Charles Spearman found that performance on many tasks was highly correlated. Using factor analysis, he proposed that an underlying "force" ("g") drove performance on all tests. However, he found that "g" could not explain all variance in tasks. He proposed a two factor-theory. According to Spearman two-factor theory, the variation between people on all cognitive tasks is a function of two factors: general intelligence ("g") and test-specific variance ("s"), as depicted in the source discussion.

Depiction of Spearman's Two-Factor Theory of Intelligence.

Louis Leon Thurstone (1887–1955)

Louis Leon Thurstone had restricted range in his samples and observed lower correlations among the skills than Charles Spearman observed. Thurstone hypothesized that the tests tapped specific abilities or faculties, not just a general factor. Thus, he argued that intelligence was not one thing—it was many things. He argued that the intelligences consisted of independent, lower-order processes, such as memory, verbal skills, and spatial skills, with a clearer psychological meaning than "g" or test-specific skills, and that these lower-order processes might better explain the pattern of correlations among cognitive tests than a general factor. Examples of independent abilities are depicted in the source discussion.

Depiction of Thurstone's Theory of Intelligence (With Only Two Factors Depicted for Simplicity).

Raymond Cattell (1905–1998)

Raymond Cattell proposed splitting "g" into two factors: fluid intelligence (G f) and crystallized intelligence (G c), as depicted in the source discussion. Fluid intelligence (G f) was considered to be the capacity to respond adaptively in novel situations. It was thought to be based on neural integrity (versus noise), neural connectedness, processing speed, memory size, spatial processing, abstract reasoning, and one's capacity for novel learning. Crystallized intelligence (G c) was considered to be preserved with age and was based on life experience (e.g., quality of schools, number of words spoken to child, etc.), history, learned skills, cultural knowledge, verbal skills, and vocabulary. Crystallized intelligence could be further sub-divided into historical G c and current G c. Historical G c was considered what a person had learned prior to receiving specialized training or practice. Current G c was considered the domains of expert knowledge and skills. Most assessment tests assess historical G c rather than current G c. Tests include both G f and G c, but we do not know how much.

Depiction of Cattell's G f-G c Theory of Intelligence. G f = fluid intelligence; G c = crystallized intelligence.

Fluid intelligence was hypothesized by Cattell to be one's inborn and physiologically based cognitive processing capacity which develops rapidly in childhood and is then invested in developing crystallized skills. Fluid intelligence was thought to be necessary for acquiring knowledge that becomes crystallized intelligence. Fluid intelligence increases with age from childhood to adulthood, peaks in the early 20s, and decreases in middle and older age. Fluid intelligence is more likely to be impaired (compared to crystallized intelligence) following brain trauma.

Crystallized intelligence is theoretically expected to continue expanding with age, but historical G c does seem to peak around late 30s. Age-related changes in crystallized and fluid intelligence are depicted in the source discussion. Given extensive age-related changes in intelligence (including fluid and crystallized intelligence), performance needs to be compared to same-aged peers.

Cross-sectional aging data showing behavioral performance on measures of speed of processing (ie, Digit Symbol, Letter Comparison, Pattern Comparison), working memory (ie, Letter rotation, Line span, Computation Span, Reading Span), Long-Term Memory (ie, Benton, Rey, Cued Recall, Free Recall), and world knowledge (ie, Shipley Vocabulary, Antonym Vocabulary, Synonym Vocabulary). Almost all measures of cognitive function (fluid intelligence) show decline with age, except world knowledge (crystallized intelligence), which may even show some improvement. (Figure reprinted from Park & Bischof, 2013, Figure 1, p. 111. Park, D. C., & Bischof, G. N. (2013). The aging mind: neuroplasticity in response to cognitive training. Dialogues in Clinical Neuroscience, 15(1), 109–119. https://doi.org/10.31887/DCNS.2013.15.1/dpark Copyright (c) Journal World Federation of Societies of Biological Psychiatry, reprinted by permission of Taylor & Francis Ltd, https://www.tandfonline.com on behalf of (c) Journal World Federation of Societies of Biological Psychiatry. No license is provided to any third party permission to reproduce this copyrighted material.)

John Horn (1926–2006)

John Horn extended Cattell's G f-G c model to include 10 broad abilities, including:

- G f: fluid reasoning (fluid intelligence)
- G c: comprehension-knowledge (crystallized intelligence)
- G a: auditory processing
- G lr: long-term storage and retrieval
- G q: quantitative knowledge
- G rw: reading and writing ability
- G s: processing speed
- G sm: short-term memory
- G t: reaction time/decision speed
- G v: visual processing

John Carroll (1916–2003)

John Carroll proposed a three-stratum hierarchical model that included narrow abilities (stratum 1), broad abilities (stratum 2), and general abilities (stratum 3), as depicted in the source discussion.

Depiction of the Cattell-Horn-Carroll Hierarhical Theory of Intelligence.

This model of intelligence became known as the Cattell-Horn-Carroll (CHC) model, which is now the most prominent contemporary theory of intelligence.

David Wechsler (1896–1981)

David Wechsler developed the Wechsler Adult Intelligence Scale (WAIS) and Wechsler Intelligence Scale for Children (WISC), which are among the most widely used intelligence tests. These tests sub-divide intelligence into verbal IQ and performance IQ, along with more narrowly defined abilities.

Alternative Conceptualizations of Intelligence

As noted earlier, the CHC model (the source discussion) is the most prominent contemporary theory of intelligence. However, there are alternative conceptualizations of intelligence. One alternative conceptualization of intelligence is the bifactor model, which is depicted in the source discussion. In a bifactor model, "g" is modeled in a non-hierarchical fashion where it is extracted first, and the primary factors are extracted separately from the remaining common variance among the tests. In this model, the resulting factors are independent of variance due to "g". Variation in performance on subtests can be due to g and/or to primary factors, which are independent. Evidence has provided support for the bifactor model of intelligence when fit to several intelligence measures, though the structure often depends on the measure used (Dombrowski et al., 2021).

Depiction of Bifactor Model of Intelligence.

Purposes of Intelligence Tests

Intelligence tests have multiple purposes, though these uses are limited. Scores on intelligence tests have been shown to predict future academic achievement. They can also be helpful for characterizing functional impairment. For instance, they can be helpful for academic planning—i.e., whether a person is gifted or in need of remediation. Intelligence tests can also be helpful for identifying change in functioning over time, for example that may be related to dementia or brain injury.

Scores on intelligence tests can also influence approaches to clinical assessment. Many assessment strategies assume average intelligence for question comprehension. A clinician might use assessments with more basic instructions or nonverbal assessments for those with poorer intelligence. Moreover, intelligence scores can influence the approaches to treatment, including the vocabulary used by the clinician, the extent of use of written material, and the clinician's expectations that the client engages in abstract reasoning. For example, a clinician might not focus on the cognitive side of therapy in cognitive-behavior therapy; instead, they might focus more on the behavioral side of cognitive-behavior therapy: exposure therapy for anxiety and behavioral activation for depression.

Intelligence Versus Achievement Versus Aptitude

In theory, intelligence is what you "have" (in terms of ability), whereas achievement is "what you do with it". Aptitude is what you will be able to do with maximum training and opportunity.

Tests that are intended to assess intelligence include the Wechsler Adult Intelligence Scale (WAIS), Wechsler Intelligence Scale for Children (WISC), Wechsler Abbreviated Scale of Intelligence (WASI), Wechsler Preschool and Primary Scale of Intelligence (WPPSI), Woodcock-Johnson Tests of Cognitive Abilities, Stanford–Binet Intelligence Scale, Differential Ability Scales (DAS), and Kaufman Brief Intelligence Test (KBIT).

Tests that are intended to assess achievement include the Wechsler Individual Achievement Test (WIAT), Wide Range Achievement Test (WRAT), and Woodcock-Johnson Tests of Achievement.

Tests that are intended to assess aptitude include the Graduate Record Examination (GRE), Scholastic Aptitude Test (SAT), American College Testing (ACT), Graduate Management Admission Test (GMAT), and Law School Admission Test (LSAT).

Achievement presumably depends on both intelligence and aptitude, as well as motivation and effort. In reality, these constructs are difficult to differentiate at a measurement level. Assessment instruments that are intended to assess one of these likely assess elements of all of these.

Theory Influences Intepretation of Scores

Despite the considerable research on intelligence, it is still difficult to define intelligence. What is intelligence? According to the 1994 academic workgroup, "Mainstream Science on Intelligence", intelligence "involves the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly, and learn from experience" (Arvey et al., 1994; Gottfredson, 1997). Some of the most widely used assessments of intelligence lack a clear theoretical foundation. They are more based on psychometrics than theory. Thus, it has been argued that "intelligence is what intelligence tests measure" (Boring, 1923).

Theory plays an important role in interpretation of scores from tests. With the Iowa Tests of Basic Skills (ITBS) and other tests, they have divergent goals from traditional intelligence tests: evaluating schools as opposed to evaluating children. For example, some stakeholders may be more interested in how much learning occurs in particular schools rather than merely a person's level of ability. Thus, the content that comprises "intelligence" differs depending on the purpose of assessment. You would also prioritize different predictive criteria depending on the goal, with items and subtests selected accordingly. For example, predictive criteria could focus on school performance versus job performance versus life skills.

Time-Related Influences

Individual differences in intelligence scores tend to be relatively stable as people age, despite changes in absolute performance. Nevertheless, there are time-related influences on intelligence scores. One time-related influence on intelligence scores is called the Flynn effect. The Flynn effect is the phenomenon that there are age cohort differences in IQ scores such that IQ scores rise around 3 points every decade; younger generations perform better on intelligence tests than older generations, when they were at the same age. So your children's generation is likely to be more intelligent than your generation, at least as assessed with the currently available instruments.

Concerns with Intelligence Tests

Despite the utility of intelligence tests, there are numerous concerns with intelligence tests. For instance, there are concerns about the content validity of intelligence tests. That is, there are concerns that intelligence tests do not assess all important aspects of the construct of intelligence. Intelligence tests typically do not assess creativity, emotional intelligence, empathy, etc.

Another concern is that intelligence tests encourage participants to put forth maximum mental effort. However, the typical mental effort needed to solve problems may be more important for many assessment questions than what someone can answer with maximum mental effort. Fluid intelligence is most likely to be affected by effort and attention.

Another concern is that scores can be influenced by factors other than intelligence. For instance, IQ scores may be affected, at least indirectly, by socioeconomic status and educational background. IQ scores are moderately associated with socioeconomic status and level of education. IQ scores might also be influenced by cultural background (due to the potential for intelligence tests to have cultural biases), motivation, intimidation or anxiety, and the physical testing conditions.

There are also concerns about the predictive utility of intelligence tests. Scores from intelligence tests have been shown to predict some aspects of functioning, but only in limited ways. For instance, IQ scores have been shown to predict grades in K–12 school, number of school years completed, academic performance through freshman year of college, and whether someone obtains a job. However, there are limits to what IQ scores can predict and the (weak) magnitude in which they predict these phenomena. Moreover, IQ scores do not appear to predict how successfully one performs in their job, the quality of their interpersonal relationships, or their happiness.

Aptitude Testing

The primary origin of considerations of the predictive utility of aptitude testing was World War I. Robert Ladd Thorndike (1910–1990) developed a test to assess aptitude for becoming a pilot. The tests examined individual differences in keeping track of all things to predict who will be a good pilot—including skills involving stress, vision, attention, reaction time, working memory, navigation, being upside down, etc. Intelligence and aptitude testing took off from there.

McClelland (McClelland, 1973; McClelland, 1994) examined the data on aptitude predicting outcomes. He examined the question of whether grades and intelligence tests predict life

and occupation success. He says no—or at least, intelligence does not influence life and occupation success, but rather intelligence opens up access to education, which serves as an important credential for access to good jobs. That is, McClelland argued that there is criterion contamination in the evaluation of criterion-related validity of intelligence tests.

He argued that intelligence and aptitude tests are culturally and socioeconomically biased. That is, the associations between intelligence test scores and job scores that have been observed in prior research may be an artifact given that socioeconomic status (SES) influences both. Others argue that IQ scores do predict later life and occupation success, and that they have shown validity across groups and SES levels—i.e., that the association is not an artifact of SES. Moreover, others have argued that the observed associations are weaker than true associations because of restriction of range. For example, students with lowest SAT/GRE scores are typically not admitted, so one cannot examine their outcomes in association with the full range of SAT/GRE scores because we do not have access to outcomes for the people who are not admitted.

McClelland argues that, instead of performing aptitude tests, we should look at life characteristics and job-related competencies. To understand McClelland's position, it can be helpful to distinguish between signs and samples. Signs are instances where human behavior is viewed as a sign of some underlying characteristic. For instance, the Rorschach Inkblot Test is all sign and no sample. Psychodynamic theorists view responses as signs of underlying personality dynamics. By contrast, samples focus on operations (behaviors). Samples come from a behavioral perspective in which behaviors are just what they are; no underlying causes are inferred—i.e., there is less inference for samples than for signs. Examples of a sample for an occupation as a pilot could involve evaluating how well the person flies an airplane. A sample from a driving test would be to literally sample driving behavior. These are examples of criterion sampling. Criterion sampling involves sampling of the behavior in the area of interest to use as the criterion. For example, criterion sampling could involve assessing a sample of abilities relevant for a position (i.e., competency testing) based on a job analysis of the duties and responsibilities of a particular job and the skills and abilities necessary to perform the job well. McClelland's goal is to use the criterion sample to reduce bias for minorities. However, using a criterion sample actually increased bias.

McClelland was interested in shifting the field from using signs to using samples. McClelland argued that use of intelligence tests for job selection involves too much inference about an underlying ability—i.e., the test is used as a sign rather than as a sample. McClelland would argue that, instead, employers should do testing for competencies such as communication skills, goal setting, ego development, and patience. However, he does not say how to assess these skills. He would argue that it is important to sample multiple aspects of the construct of interest; a wider array of talents—not just academic performance. For a relevant criterion sample of competencies for a particular job or position, you need strong theory. However, criterion sampling is very difficult.

Consider tests for selection of firefighters. Evaluative tests for selecting firefighters include fitness tests, among other tests. Fitness tests may include activities like running with heavy objects, climbing ladders, and willingness to go into burning buildings—skills which translate to the activities that many firefighters perform on the job. However, these fitness tests have been shown to lead to adverse impact toward women—i.e., they result in lower selection rates of women than men. The fitness tests are also biased in favor of people who

had practiced, and who had previous training. Criterion sampling without pre-training makes things worse. You could lead bootcamps of pre-training to level the playing field for job applicants and then conduct testing. However, pre-training takes time and money, and many organizations may not be willing to do that.

Range restriction leads to artifactually lower validity estimates of intelligence/aptitude tests with school and job performance. After accounting for range restriction, predictive validity coefficients are around.5 with school and job performance. Thus, many intelligence/aptitude tests tend to show some predictive validity for academic/job performance. However, it is also important to consider whether test bias exists. We discuss test bias in the next chapter.

Scales

IQ has had various scales.

Ratio IQ

Ratio IQ is mental age divided by chronological age, known as the intelligence quotient (see the source discussion). Ratio IQ changes with age, and it is not widely used today. Deviation IQ is more widely used today.

Deviation IQ

Deviation IQ (also called Standard Scores) is a standardized norm-referenced score with a mean of 100 and a standard deviation of 15. Deviation IQ was developed by David Wechsler. Scores are norm-referenced to same-aged peers. It has no reference to mental age. Using Deviation IQ scores, intellectual disability is defined as a score of 70 or below (along with functional impairment) because the score represents two standard deviations below the mean. However, this cutoff is based on an arbitrary statistical decision and tradition. Giftedness is defined as a score of 130 or above, which is two standard deviations above the mean.

Nominal, Ordinal, Interval, or Ratio?

Intelligence scores are ordinal or interval, not ratio. You cannot talk about someone as being twice as intelligent as someone else because we do not know what zero intelligence is or what the maximum intelligence is.

Conclusion

Intelligence has been defined in many ways, and intelligence assessment has a long history. The assessment of intelligence is considered one of the key successes in psychology. The most prominent contemporary theory of intelligence is the CHC model, which is a three-stratum hierarchical model that includes narrow abilities (stratum 1), broad abilities (stratum 2), and general abilities (stratum 3). Assessments of intelligence have utility for various purposes, including predicting future academic achievement, characterizing functional impairment, and identifying change in functioning over time. However, there are many things that intelligence does not predict well, such as how successfully one will perform in their job, the quality of their interpersonal relationships, or their happiness. Moreover, there are potential weaknesses of intelligence tests, including content validity because they do not assess creativity or emotional aspects of intelligence such as empathy.

Suggested Readings

Ackerman, 2013

Attribution and critical-reading note

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Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

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A four-and-a-half-year-old boy sits at the kitchen table with his father, who is reading a new story aloud to him. He turns the page to continue reading, but before he can begin, the boy says, “Wait, Daddy!” He points to the words on the new page and reads aloud, “Go, Pig! Go!” The father stops and looks at his son. “Can you read that?” he asks. “Yes, Daddy!” And he points to the words and reads again, “Go, Pig! Go!”

This father was not actively teaching his son to read, even though the child constantly asked questions about letters, words, and symbols that they saw everywhere: in the car, in the store, on the television. The dad wondered about what else his son might understand and decided to try an experiment. Grabbing a sheet of blank paper, he wrote several simple words in a list: mom, dad, dog, bird, bed, truck, car, tree. He put the list down in front of the boy and asked him to read the words. “Dad, dog, bird, bed, truck, car, tree,” he read, slowing down to carefully pronounce birdand truck.Then, “Did I do it, Daddy?” “You sure did! That is very good.” The father gave his little boy a warm hug and continued reading the story about the pig, all the while wondering if his son’s abilities were an indication of exceptional intelligence or simply a normal pattern of linguistic development. Like the father in this example, psychologists have wondered what constitutes intelligence and how it can be measured.

Classifying Intelligence

What exactly is intelligence? The way that researchers have defined the concept of intelligence has been modified many times since the birth of psychology. British psychologist Charles Spearman believed intelligence consisted of one general factor, called g, which could be measured and compared among individuals. Spearman focused on the commonalities among various intellectual abilities and de-emphasized what made each unique. Long before modern psychology developed, however, ancient philosophers, such as Aristotle, held a similar view (Cianciolo & Sternberg, 2004).

Others psychologists believe that instead of a single factor, intelligence is a collection of distinct abilities. In the 1940s, Raymond Cattell proposed a theory of intelligence that divided general intelligence into two components: crystallized intelligence and fluid intelligence (Cattell, 1963). **Crystallized intelligence** is characterized as acquired knowledge and the ability to retrieve it. When you learn, remember, and recall information,

you are using crystallized intelligence. You use crystallized intelligence all the time in your coursework by demonstrating that you have mastered the information covered in the course. **Fluid intelligence** encompasses the ability to see complex relationships and solve problems. Navigating your way home after being detoured onto an unfamiliar route because of road construction would draw upon your fluid intelligence. Fluid intelligence helps you tackle complex, abstract challenges in your daily life, whereas crystallized intelligence helps you overcome concrete, straightforward problems (Cattell, 1963).

Other theorists and psychologists believe that intelligence should be defined in more practical terms. For example, what types of behaviors help you get ahead in life? Which skills promote success? Think about this for a moment. Being able to recite all 45 presidents of the United States in order is an excellent party trick, but will knowing this make you a better person?

Robert Sternberg developed another theory of intelligence, which he titled the **triarchic theory of intelligence** because it sees intelligence as comprised of three parts (Sternberg, 1988): practical, creative, and analytical intelligence.

Source figure description (original figure omitted): Sternberg’s theory identifies three types of intelligence: practical, creative, and analytical.

Practical intelligence, as proposed by Sternberg, is sometimes compared to “street smarts.” Being practical means you find solutions that work in your everyday life by applying knowledge based on your experiences. This type of intelligence appears to be separate from traditional understanding of IQ; individuals who score high in practical intelligence may or may not have comparable scores in creative and analytical intelligence (Sternberg, 1988).

Analytical intelligence is closely aligned with academic problem solving and computations. Sternberg says that analytical intelligence is demonstrated by an ability to analyze, evaluate, judge, compare, and contrast. When reading a classic novel for literature class, for example, it is usually necessary to compare the motives of the main characters of the book or analyze the historical context of the story. In a science course such as anatomy, you must study the processes by which the body uses various minerals in different human systems. In developing an understanding of this topic, you are using analytical intelligence. When solving a challenging math problem, you would apply analytical intelligence to analyze different aspects of the problem and then solve it section by section.

Creative intelligence is marked by inventing or imagining a solution to a problem or situation. Creativity in this realm can include finding a novel solution to an unexpected problem or producing a beautiful work of art or a well-developed short story. Imagine for a moment that you are camping in the woods with some friends and realize that you’ve forgotten your camp coffee pot. The person in your group who figures out a way to successfully brew coffee for everyone would be credited as having higher creative intelligence.

Multiple Intelligences Theory was developed by Howard Gardner, a Harvard psychologist and former student of Erik Erikson. In Gardner’s theory, each person possesses at least eight intelligences. The eight intelligences are linguistic intelligence, logical-mathematical intelligence, musical intelligence, bodily kinesthetic intelligence, spatial intelligence, interpersonal intelligence, intrapersonal intelligence, and naturalistic intelligence. Among cognitive psychologists, Gardner’s theory has been heavily criticized for lacking empirical

evidence. However, educators continue to study and use Gardner's theory, with some colleges even discussing how they integrate Gardner's theory into their classrooms. Gottfredson describes one possible reason for the continued use of Gardner's theory: "... that there are multiple independent intelligences, suggesting that everyone can be smart in some way. This is, understandably, a very attractive idea in democratic societies" (2004).

Gardner's inter- and intrapersonal intelligences are often combined into a single type: emotional intelligence. **Emotional intelligence** encompasses the ability to understand the emotions of yourself and others, show empathy, understand social relationships and cues, and regulate your own emotions and respond in culturally appropriate ways (Parker, Saklofske, & Stough, 2009). People with high emotional intelligence typically have well-developed social skills. Some researchers, including Daniel Goleman, the author of Emotional Intelligence: Why It Can Matter More than IQ, argue that emotional intelligence is a better predictor of success than traditional intelligence (Goleman, 1995). However, emotional intelligence has been widely debated, with researchers pointing out inconsistencies in how it is defined and described, as well as questioning results of studies on a subject that is difficult to measure and study empirically (Locke, 2005; Mayer, Salovey, & Caruso, 2004)

The most comprehensive theory of intelligence to date is the Cattell-Horn-Carroll (CHC) theory of cognitive abilities (Schneider & McGrew, 2018). In this theory, abilities are related and arranged in a hierarchy with general abilities at the top, broad abilities in the middle, and narrow (specific) abilities at the bottom. The narrow abilities are the only ones that can be directly measured; however, they are integrated within the other abilities. At the general level is general intelligence. Next, the broad level consists of general abilities such as fluid reasoning, short-term memory, and processing speed. Finally, as the hierarchy continues, the narrow level includes specific forms of cognitive abilities. For example, short-term memory would further break down into memory span and working memory capacity.

Intelligence can also have different meanings and values in different cultures. If you live on a small island, where most people get their food by fishing from boats, it would be important to know how to fish and how to repair a boat. If you were an exceptional angler, your peers would probably consider you intelligent. If you were also skilled at repairing boats, your intelligence might be known across the whole island. Think about your own family's culture. What values are important for Latinx families? Italian families? In Irish families, hospitality and telling an entertaining story are marks of the culture. If you are a skilled storyteller, other members of Irish culture are likely to consider you intelligent.

Some cultures place a high value on working together as a collective. In these cultures, the importance of the group supersedes the importance of individual achievement. When you visit such a culture, how well you relate to the values of that culture exemplifies your **cultural intelligence**, sometimes referred to as cultural competence.

Note: Watch this video that compares different theories of intelligence to learn more.

Creativity

Creativity is the ability to generate, create, or discover new ideas, solutions, and possibilities. Very creative people often have intense knowledge about something, work on it for years, look at novel solutions, seek out the advice and help of other experts, and take risks. Although creativity is often associated with the arts, it is actually a vital form of intelligence that drives people in many disciplines to discover something new. Creativity can be found in every area of life, from the way you decorate your residence to a new way of understanding how a cell works.

Creativity is often connected to a person's ability to engage in **divergent thinking**. Divergent thinking can be described as thinking "outside the box;" it allows an individual to arrive at unique, multiple solutions to a given problem. In contrast, **convergent thinking** describes the ability to provide a correct or well-established answer or solution to a problem (Cropley, 2006; Gilford, 1967)

Note: Dr. Tom Steitz, former Sterling Professor of Biochemistry and Biophysics at Yale University, spent his career looking at the structure and specific aspects of RNA molecules and how their interactions could help produce antibiotics and ward off diseases. As a result of his lifetime of work, he won the Nobel Prize in Chemistry in 2009. He wrote, "Looking back over the development and progress of my career in science, I am reminded how vitally important good mentorship is in the early stages of one's career development and constant face-to-face conversations, debate and discussions with colleagues at all stages of research. Outstanding discoveries, insights and developments do not happen in a vacuum" (Steitz, 2010, para. 39). Based on Steitz's comment, it becomes clear that someone's creativity, although an individual strength, benefits from interactions with others. Think of a time when your creativity was sparked by a conversation with a friend or classmate. How did that person influence you and what problem did you solve using creativity?

Summary

Intelligence is a complex characteristic of cognition. Many theories have been developed to explain what intelligence is and how it works. Sternberg generated his triarchic theory of intelligence, whereas Gardner posits that intelligence is comprised of many factors. Still others focus on the importance of emotional intelligence. Finally, creativity seems to be a facet of intelligence, but it is extremely difficult to measure objectively.

Review Questions

Fluid intelligence is characterized by _____.

1. being able to recall information

1. being able to create new products

1. being able to understand and communicate with different cultures

1. being able to see complex relationships and solve problems

Which of the following is not one of Gardner's Multiple Intelligences?

1. creative

1. spatial

1. linguistic

1. musical

Which theorist put forth the triarchic theory of intelligence?

1. Goleman

1. Gardner

1. Sternberg

1. Steitz

When you are examining data to look for trends, which type of intelligence are you using most?

1. practical

1. analytical

1. emotional

1. creative

Critical Thinking Questions

Describe a situation in which you would need to use practical intelligence.

Describe a situation in which cultural intelligence would help you communicate better.

Personal Application Question

What influence do you think emotional intelligence plays in your personal life?

Attribution and critical-reading note

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Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

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While you’re likely familiar with the term “IQ” and associate it with the idea of intelligence, what does IQ really mean? IQ stands for **intelligence quotient** and describes a score earned on a test designed to measure intelligence. You’ve already learned that there are many ways psychologists describe intelligence (or more aptly, intelligences). Similarly, IQ tests—the tools designed to measure intelligence—have been the subject of debate throughout their development and use.

When might an IQ test be used? What do we learn from the results, and how might people use this information? While there are certainly many benefits to intelligence testing, it is important to also note the limitations and controversies surrounding these tests. For

example, IQ tests have sometimes been used as arguments in support of insidious purposes, such as the eugenics movement (Severson, 2011). The infamous Supreme Court Case, Buck v. Bell, legalized the forced sterilization of some people deemed “feeble-minded” through this type of testing, resulting in about 65,000 sterilizations (Buck v. Bell, 274 U.S. 200; Ko, 2016). Today, only professionals trained in psychology can administer IQ tests, and the purchase of most tests requires an advanced degree in psychology. Other professionals in the field, such as social workers and psychiatrists, cannot administer IQ tests. In this section, we will explore what intelligence tests measure, how they are scored, and how they were developed.

Measuring Intelligence

It seems that the human understanding of intelligence is somewhat limited when we focus on traditional or academic-type intelligence. How then, can intelligence be measured? And when we measure intelligence, how do we ensure that we capture what we’re really trying to measure (in other words, that IQ tests function as valid measures of intelligence)? In the following paragraphs, we will explore how intelligence tests were developed and the history of their use.

The IQ test has been synonymous with intelligence for over a century. In the late 1800s, Sir Francis Galton developed the first broad test of intelligence (Flanagan & Kaufman, 2004). Although he was not a psychologist, his contributions to the concepts of intelligence testing are still felt today (Gordon, 1995). Reliable intelligence testing (you may recall from earlier chapters that reliability refers to a test’s ability to produce consistent results) began in earnest during the early 1900s with a researcher named Alfred Binet. Binet was asked by the French government to develop an intelligence test to use on children to determine which ones might have difficulty in school; it included many verbally based tasks. American researchers soon realized the value of such testing. Louis Terman, a Stanford professor, modified Binet’s work by standardizing the administration of the test and tested thousands of different-aged children to establish an average score for each age. As a result, the test was normed and standardized, which means that the test was administered consistently to a large enough representative sample of the population that the range of scores resulted in a bell curve (bell curves will be discussed later). **Standardization** means that the manner of administration, scoring, and interpretation of results is consistent. **Norming** involves giving a test to a large population so data can be collected comparing groups, such as age groups. The resulting data provide norms, or referential scores, by which to interpret future scores. Norms are not expectations of what a given group shouldknow but a demonstration of what that group doesknow. Norming and standardizing the test ensures that new scores are reliable. This new version of the test was called the Stanford-Binet Intelligence Scale (Terman, 1916). Remarkably, an updated version of this test is still widely used today.

Source figure description (original figure omitted): French psychologist Alfred Binet helped to develop intelligence testing. (b) This page is from a 1908 version of the Binet-Simon Intelligence Scale. Children being tested were asked which face, of each pair, was prettier.

In 1939, David Wechsler, a psychologist who spent part of his career working with World War I veterans, developed a new IQ test in the United States. Wechsler combined several subtests from other intelligence tests used between 1880 and World War I. These subtests tapped into a variety of verbal and nonverbal skills, because Wechsler believed that intelligence encompassed “the global capacity of a person to act purposefully, to think

rationally, and to deal effectively with his environment" (Wechsler, 1958, p. 7). He named the test the Wechsler-Bellevue Intelligence Scale (Wechsler, 1981). This combination of subtests became one of the most extensively used intelligence tests in the history of psychology. Although its name was later changed to the Wechsler Adult Intelligence Scale (WAIS) and has been revised several times, the aims of the test remain virtually unchanged since its inception (Boake, 2002). Today, there are three intelligence tests credited to Wechsler, the Wechsler Adult Intelligence Scale-fourth edition (WAIS-IV), the Wechsler Intelligence Scale for Children (WISC-V), and the Wechsler Preschool and Primary Scale of Intelligence—IV (WPPSI-IV) (Wechsler, 2012). These tests are used widely in schools and communities throughout the United States, and they are periodically normed and standardized as a means of recalibration. As a part of the recalibration process, the WISC-V was given to thousands of children across the country, and children taking the test today are compared with their same-age peers.

The WISC-V is composed of 14 subtests, which comprise five indices, which then render an IQ score. The five indices are Verbal Comprehension, Visual Spatial, Fluid Reasoning, Working Memory, and Processing Speed. When the test is complete, individuals receive a score for each of the five indices and a Full Scale IQ score. The method of scoring reflects the understanding that intelligence is comprised of multiple abilities in several cognitive realms and focuses on the mental processes that the child used to arrive at their answers to each test item.

Interestingly, the periodic recalibrations have led to an interesting observation known as the Flynn effect. Named after James Flynn, who was among the first to describe this trend, the **Flynn effect** refers to the observation that each generation has a significantly higher IQ than the last. Flynn himself argues, however, that increased IQ scores do not necessarily mean that younger generations are more intelligent per se (Flynn, Shaughnessy, & Fulgham, 2012).

Ultimately, we are still left with the question of how valid intelligence tests are. Certainly, the most modern versions of these tests tap into more than verbal competencies, yet the specific skills that should be assessed in IQ testing, the degree to which any test can truly measure an individual's intelligence, and the use of the results of IQ tests are still issues of debate (Gresham & Witt, 1997; Flynn, Shaughnessy, & Fulgham, 2012; Richardson, 2002; Schlinger, 2003).

Note: The case of Atkins v. Virginia was a landmark case in the United States Supreme Court. On August 16, 1996, two men, Daryl Atkins and William Jones, robbed, kidnapped, and then shot and killed Eric Nesbitt, a local airman from the U.S. Air Force. A clinical psychologist evaluated Atkins and testified at the trial that Atkins had an IQ of 59. The mean IQ score is 100. The psychologist concluded that Atkins had an intellectual disability. The jury found Atkins guilty, and he was sentenced to death. Atkins and his attorneys appealed to the Supreme Court. In June 2002, the Supreme Court reversed a previous decision and ruled that executions of people with intellectual disabilities are 'cruel and unusual punishments' prohibited by the Eighth Amendment. Using the accepted language of the time, the court wrote in their decision: Clinical definitions of mental retardation require not only subaverage intellectual functioning, but also significant limitations in adaptive skills. Mentally retarded persons frequently know the difference between right and wrong and are competent to stand trial. Because of their impairments, however, by definition they have

diminished capacities to understand and process information, to communicate, to abstract from mistakes and learn from experience, to engage in logical reasoning, to control impulses, and to understand others' reactions. Their deficiencies do not warrant an exemption from criminal sanctions, but diminish their personal culpability (Atkins v. Virginia, 2002, par. 5). The court also decided that there was a state legislature consensus against the execution of people with intellectual disabilities and that this consensus should stand for all of the states. The Supreme Court ruling left it up to the states to determine their own definitions of intellectual disability. The definitions vary among states as to who can be executed. In the Atkins case, a jury decided that because he had many contacts with his lawyers and thus was provided with intellectual stimulation, his IQ had reportedly increased to a level where the state could execute him. He was given an execution date and then received a stay of execution after it was revealed that lawyers for co-defendant, William Jones, coached Jones to "produce a testimony against Mr. Atkins that did match the evidence" (Liptak, 2008). After the revelation of this misconduct, Atkins was re-sentenced to life imprisonment. Atkins v. Virginia (2002) highlights several issues regarding society's beliefs around intelligence. In the Atkins case, the Supreme Court decided that intellectual disability does affect decision making and therefore should affect the nature of the punishment such criminals receive. Where, however, should the lines of intellectual disability be drawn? In May 2014, the Supreme Court ruled in a related case (Hall v. Florida) that IQ scores cannot be used as a final determination of a prisoner's eligibility for the death penalty (Roberts, 2014).

The Bell Curve

The results of intelligence tests follow the bell curve, a graph in the general shape of a bell. When the bell curve is used in psychological testing, the graph demonstrates a normal distribution of a trait, in this case, intelligence, in the human population. Many human traits naturally follow the bell curve. For example, if you lined up all your female schoolmates according to height, it is likely that a large cluster of them would be the average height for an American woman: 5'4"–5'6". This cluster would fall in the center of the bell curve, representing the average height for American women. There would be fewer women who stand closer to 4'11". The same would be true for women of above-average height: those who stand closer to 5'11". The trick to finding a bell curve in nature is to use a large sample size. Without a large sample size, it is less likely that the bell curve will represent the wider population. A **representative sample** is a subset of the population that accurately represents the general population. If, for example, you measured the height of the women in your classroom only, you might not actually have a representative sample. Perhaps the women's basketball team wanted to take this course together, and they are all in your class. Because basketball players tend to be taller than average, the women in your class may not be a good representative sample of the population of American women. But if your sample included all the women at your school, it is likely that their heights would form a natural bell curve.

Source figure description (original figure omitted): Are you of below-average, average, or above-average height?

The same principles apply to intelligence tests scores. Individuals earn a score called an intelligence quotient (IQ). Over the years, different types of IQ tests have evolved, but the way scores are interpreted remains the same. On most IQ tests, the average (or mean) IQ

score is 100. **Standard deviations** describe how data are dispersed in a population and give context to large data sets. The bell curve uses the standard deviation to show how all scores are dispersed from the average score. In modern IQ testing, one standard deviation is 15 points. So a score of 85 would be described as “one standard deviation below the mean.” How would you describe a score of 115 and a score of 70? Any IQ score that falls within one standard deviation above and below the mean (between 85 and 115) is considered average, and 68% of the population has IQ scores in this range. An IQ score of 130 or above is considered a superior level.

Source figure description (original figure omitted): The majority of people have an IQ score between 85 and 115.

Only 2.2% of the population has an IQ score below 70 (American Psychiatric Association [APA], 2013). If a person earns a score approximately two standard deviations below the mean on an intelligence test, (about 70 on a test with a mean of 100), has major deficits in adaptive functioning, and these cognitive and adaptive deficits were present before the age of 18, they could be diagnosed as having an intellectual disability (ID). Formerly known as mental retardation, the accepted term now is intellectual disability, and it has four subtypes: mild, moderate, severe, and profound. The Diagnostic and Statistical Manual of Psychological Disorders lists criteria for each subgroup (APA, 2013).

Characteristics of Cognitive Disorders

Intellectual Disability Subtype	Percentage of Population with Intellectual Disabilities	Description
Mild	85%	3rd- to 6th-grade skill level in reading, writing, and math; may be employed and live independently
Moderate	10%	Basic reading and writing skills; functional self-care skills; requires some oversight
Severe	5%	Functional self-care skills; requires oversight of daily environment and activities
Profound	<1%	May be able to communicate verbally or nonverbally; requires intensive oversight

On the other end of the intelligence spectrum are those individuals whose IQs fall into the highest ranges. Consistent with the bell curve, about 2% of the population falls into this category. People are considered to have a higher aptitude for learning (and may be classified as "gifted" within educational systems) if they have an IQ score of 130 or higher, or superior intelligence in a particular area. Long ago, popular belief suggested that people of high intelligence were maladjusted. This idea was disproven through a groundbreaking study of these children. In 1921, Lewis Terman began a longitudinal study of over 1500 children with IQs over 135 (Terman, 1925). His findings showed that these children became well-educated, successful adults who were, in fact, well-adjusted (Terman & Oden, 1947). Additionally, Terman’s study showed that the subjects were above average in physical build

and attractiveness, dispelling an earlier popular notion that highly intelligent people were “weaklings.” Some people with very high IQs elect to join Mensa, an organization dedicated to identifying, researching, and fostering intelligence. Members must have an IQ score in the top 2% of the population, and they may be required to pass other exams in their application to join the group.

Note: In the past, individuals with IQ scores below 70 and significant adaptive and social functioning delays were diagnosed with mental retardation. When this diagnosis was first named, it was replacing more negative and insensitive terms, and the title held no social stigma; several prominent research and support organizations even used the word in their names and mission statements. However, members of those populations as well as their families and supporting professionals found that the term was not only inaccurate, but demeaning and insulting. As such, the DSM-5 now labels this diagnosis as “intellectual disability.” Many states once had a Department of Mental Retardation to serve those diagnosed with such cognitive delays, but most have changed their name to Department of Developmental Disabilities or something similar in language. Due to the passage of "Rosa's Law" in 2010 and to the growing support for changing the terminology, most U.S. federal agencies formally adopted the words "intellectual disability." While the change was widely supported, you can view in the Federal Register several counterpoints from parents of people with intellectual disabilities, who felt that the new term was imprecise and less applicable to their children. Earlier in the chapter, we discussed how language affects how we think. Do you think changing the title of this department has any impact on how people regard those with developmental disabilities? Does a different name give people more dignity, and if so, how? Do you think the terminology is likely to change again? Why or why not?

Why Measure Intelligence?

The value of IQ testing is most evident in educational or clinical settings. Children who seem to be experiencing learning difficulties or severe behavioral problems can be tested to ascertain whether the child’s difficulties can be partly attributed to an IQ score that is significantly different from the mean for their age group. Without IQ testing—or another measure of intelligence—children and adults needing extra support might not be identified effectively. In addition, IQ testing is used in courts to determine whether a defendant has special or extenuating circumstances that preclude them from participating in some way in a trial. People also use IQ testing results to seek disability benefits from the Social Security Administration.

The following case study demonstrates the usefulness and benefits of IQ testing. Candace, a 14-year-old girl experiencing problems at school in Connecticut, was referred for a court-ordered psychological evaluation. She was in regular education classes in ninth grade and was failing every subject. Candace had never been a stellar student but had always been passed to the next grade. Frequently, she would curse at any of her teachers who called on her in class. She also got into fights with other students and occasionally shoplifted. When she arrived for the evaluation, Candace immediately said that she hated everything about school, including the teachers, the rest of the staff, the building, and the homework. Her parents stated that they felt their daughter was picked on, because she was of a different ethnicity than the teachers and most of the other students. When asked why she cursed at her teachers, Candace replied, “They only call on me when I don’t know the answer. I don’t

want to say, ‘I don’t know’ all of the time and look like an idiot in front of my friends. The teachers embarrass me.” She was given a battery of tests, including an IQ test. Her score on the IQ test was 68. What does Candace’s score say about her ability to excel or even succeed in regular education classes without assistance? Why were her difficulties never noticed or addressed?

Summary

In this section, we learned about the history of intelligence testing and some of the challenges regarding intelligence testing. Intelligence tests began in earnest with Binet; Wechsler later developed intelligence tests that are still in use today: the WAIS-IV and WISC-V. The Bell curve shows the range of scores that encompass average intelligence as well as standard deviations.

Review Questions

In order for a test to be normed and standardized it must be tested on _____.

1. a group of same-age peers

1. a representative sample

1. children with mental disabilities

1. children of average intelligence

The mean score for a person with an average IQ is _____.

1. 70

1. 130

1. 85

1. 100

Who developed the IQ test most widely used today?

1. Sir Francis Galton

1. Alfred Binet

1. Louis Terman

1. David Wechsler

The DSM-5 now uses _____ as a diagnostic label for what was once referred to as mental retardation.

1. autism and developmental disabilities

1. lowered intelligence

1. intellectual disability

1. cognitive disruption

Critical Thinking Questions

Why do you think different theorists have defined intelligence in different ways?

Compare and contrast the benefits of the Stanford-Binet IQ test and Wechsler's IQ tests.

Personal Application Question

In thinking about the case of Candace described earlier, do you think that Candace benefitted or suffered as a result of consistently being passed on to the next grade?

Attribution and critical-reading note

Adapted from Psychology 2e by Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Rice University (2020), CC BY 4.0. Selection, organization, and PDF layout modified for this course; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

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A young girl, born of teenage parents, lives with her grandmother in rural Mississippi. They have a very low income, but they do their best to get by with what they have. She learns to read when she is just 3 years old. As she grows older, she longs to live with her mother, who now resides in Wisconsin. She moves there at the age of 6 years. At 9 years of age, she is raped. During the next several years, several different male relatives repeatedly molest her. Her life unravels. She turns to drugs and sex to fill the deep, lonely void inside her. Her mother then sends her to Nashville to live with her father, who imposes strict behavioral expectations upon her, and over time, her wild life settles once again. She begins to experience success in school, and at 19 years old, becomes the youngest and first African-American female news anchor ("Dates and Events," n.d.). The woman—Oprah Winfrey—goes on to become a media giant known for both her intelligence and her empathy.

High Intelligence: Nature or Nurture?

Where does high intelligence come from? Some researchers believe that intelligence is a trait inherited from a person's parents. Scientists who research this topic typically use twin studies to determine the **heritability** of intelligence. The Minnesota Study of Twins Reared Apart is one of the most well-known twin studies. In this investigation, researchers found that identical twins raised together and identical twins raised apart exhibit a higher correlation between their IQ scores than siblings or fraternal twins raised together (Bouchard, Lykken, McGue, Segal, & Tellegen, 1990). The findings from this study reveal a genetic component to intelligence. At the same time, other psychologists believe that intelligence is shaped by a child's developmental environment. If parents were to provide their children with intellectual stimuli from before they are born, it is likely that they would absorb the benefits of that stimulation, and it would be reflected in intelligence levels.

Source figure description (original figure omitted): The **correlations** of IQs of unrelated versus related persons reared apart or together suggest a genetic component to intelligence.

The reality is that aspects of each idea are probably correct. In fact, one study suggests that although genetics seem to be in control of the level of intelligence, the environmental influences provide both stability and change to trigger manifestation of cognitive abilities

(Bartels, Rietveld, Van Baal, & Boomsma, 2002). Certainly, there are behaviors that support the development of intelligence, but the genetic component of high intelligence should not be ignored. As with all heritable traits, however, it is not always possible to isolate how and when high intelligence is passed on to the next generation.

Range of Reaction is the theory that each person responds to the environment in a unique way based on their genetic makeup. According to this idea, your genetic potential is a fixed quantity, but whether you reach your full intellectual potential is dependent upon the environmental stimulation you experience, especially in childhood. Think about this scenario: A couple adopts a child who has average genetic intellectual potential. They raise her in an extremely stimulating environment. What will happen to the couple's new daughter? It is likely that the stimulating environment will improve her intellectual outcomes over the course of her life. But what happens if this experiment is reversed? If a child with an extremely strong genetic background is placed in an environment that does not stimulate him: What happens? Interestingly, according to a longitudinal study of highly gifted individuals, it was found that "the two extremes of optimal and pathological experience are both represented disproportionately in the backgrounds of creative individuals"; however, those who experienced supportive family environments were more likely to report being happy (Csikszentmihalyi & Csikszentmihalyi, 1993, p. 187).

Another challenge to determining origins of high intelligence is the confounding nature of our human social structures. It is troubling to note that some ethnic groups perform better on IQ tests than others—and it is likely that the results do not have much to do with the quality of each ethnic group's intellect. The same is true for socioeconomic status. Children who live in poverty experience more pervasive, daily stress than children who do not worry about the basic needs of safety, shelter, and food. These worries can negatively affect how the brain functions and develops, causing a dip in IQ scores. Mark Kishiyama and his colleagues determined that children living in poverty demonstrated reduced prefrontal brain functioning comparable to children with damage to the lateral prefrontal cortex (Kishiyama, Boyce, Jimenez, Perry, & Knight, 2009).

The debate around the foundations and influences on intelligence exploded in 1969, when an educational psychologist named Arthur Jensen published the article "How Much Can We Boost I.Q. and Achievement" in the Harvard Educational Review. Jensen had administered IQ tests to diverse groups of students, and his results led him to the conclusion that IQ is determined by genetics. He also posited that intelligence was made up of two types of abilities: Level I and Level II. In his theory, Level I is responsible for rote memorization, whereas Level II is responsible for conceptual and analytical abilities. According to his findings, Level I remained consistent among the human race. Level II, however, exhibited differences among ethnic groups (Modgil & Routledge, 1987). Jensen's most controversial conclusion was that Level II intelligence is prevalent among Asian people, then White people, then Black people. Robert Williams was among those who called out racial bias in Jensen's results (Williams, 1970).

Obviously, Jensen's interpretation of his own data caused an intense response in a nation that continued to grapple with the effects of racism (Fox, 2012). While even some who took issue with Jensen's findings indicated that they did not detect overt racism in his work, Jensen himself made a number of racist statements during an interview with a White nationalist publication, American Renaissance. He indicated his belief that many Black

people were not "educable up to a level for which there’s any economic demand," while also having a higher birth rate than other groups, which would lead to a shift in population and a deterioration of the nation (Taylor, 1992). However, Jensen’s ideas were not solitary or unique; rather, they represented one of many examples of psychologists asserting racial differences in IQ and cognitive ability. In fact, Rushton and Jensen (2005) reviewed three decades worth of research on the relationship between race and cognitive ability. Jensen’s belief in the inherited nature of intelligence and the validity of the IQ test to be the truest measure of intelligence are at the core of his conclusions.

In a related story, parents of African American students filed a case against the State of California in 1979, because they believed that the testing method used to identify students with learning disabilities was culturally unfair as the tests were normed and standardized using White children (Larry P. v. Riles). The testing method used by the state disproportionately identified African American children as “mentally retarded," which resulted in many students being incorrectly classified. According to a summary of the case, Larry P. v. Riles:

In violation of Title VI of the Civil Rights Act of 1964, the Rehabilitation Act of 1973, and the Education for All Handicapped Children Act of 1975, defendants have utilized standardized intelligence tests that are racially and culturally biased, have a discriminatory impact against Black children, and have not been validated for the purpose of essentially permanent placements of Black children into educationally dead-end, isolated, and stigmatizing classes for the so-called educable mentally retarded. Further, these federal laws have been violated by defendants' general use of placement mechanisms that, taken together, have not been validated and result in a large over-representation of Black children in the special E.M.R. classes. (Larry P. v. Riles, par. 6)

Once again, the limitations of intelligence testing were revealed.

What are Learning Disabilities?

Learning disabilities are cognitive disorders that affect different areas of cognition, particularly language or reading. It should be pointed out that learning disabilities are not the same thing as intellectual disabilities. Learning disabilities are considered specific neurological impairments rather than global intellectual or developmental disabilities. A person with a language disability has difficulty understanding or using spoken language, whereas someone with a reading disability, such as dyslexia, has difficulty processing what they are reading.

Often, learning disabilities are not recognized until a child reaches school age. One confounding aspect of learning disabilities is that they most often affect children with average to above-average intelligence. In other words, the disability is specific to a particular area and not a measure of overall intellectual ability. At the same time, learning disabilities tend to exhibit comorbidity with other disorders, like attention-deficit hyperactivity disorder (ADHD). Anywhere between 30–70% of individuals with diagnosed cases of ADHD also have some sort of learning disability (Riccio, Gonzales, & Hynd, 1994). Let’s take a look at three examples of common learning disabilities: dysgraphia, dyslexia, and dyscalculia.

Dysgraphia

Children with **dysgraphia** have a learning disability that results in a struggle to write legibly. The physical task of writing with a pen and paper is extremely challenging for the person. These children often have extreme difficulty putting their thoughts down on paper (Smits-Engelsman & Van Galen, 1997). This difficulty is inconsistent with a person's IQ. That is, based on the child's IQ and/or abilities in other areas, a child with dysgraphia should be able to write, but can't. Children with dysgraphia may also have problems with spatial abilities.

Students with dysgraphia need academic accommodations to help them succeed in school. These accommodations can provide students with alternative assessment opportunities to demonstrate what they know (Barton, 2003). For example, a student with dysgraphia might be permitted to take an oral exam rather than a traditional paper-and-pencil test. Treatment is usually provided by an occupational therapist, although there is some question as to how effective such treatment is (Zwicker, 2005).

Dyslexia

Dyslexia is the most common learning disability in children. An individual with **dyslexia** exhibits an inability to correctly process letters. The neurological mechanism for sound processing does not work properly in someone with dyslexia. As a result, dyslexic children may not understand sound-letter correspondence. A child with dyslexia may mix up letters within words and sentences—letter reversals, such as those shown in, are a hallmark of this learning disability—or skip whole words while reading. A dyslexic child may have difficulty spelling words correctly while writing. Because of the disordered way that the brain processes letters and sound, learning to read is a frustrating experience. Some dyslexic individuals cope by memorizing the shapes of most words, but they never actually learn to read (Berninger, 2008).

Source figure description (original figure omitted): These written words show variations of the word "teapot" as written by individuals with dyslexia.

Dyscalculia

Dyscalculia is difficulty in learning or comprehending arithmetic. This learning disability is often first evident when children exhibit difficulty discerning how many objects are in a small group without counting them. Other symptoms may include struggling to memorize math facts, organize numbers, or fully differentiate between numerals, math symbols, and written numbers (such as "3" and "three").

Summary

Genetics and environment affect intelligence and the challenges of certain learning disabilities. The intelligence levels of all individuals seem to benefit from rich stimulation in their early environments. Highly intelligent individuals, however, may have a built-in resiliency that allows them to overcome difficult obstacles in their upbringing. Learning disabilities can cause major challenges for children who are learning to read and write. Unlike developmental disabilities, learning disabilities are strictly neurological in nature and are not related to intelligence levels. Students with dyslexia, for example, may have extreme difficulty learning to read, but their intelligence levels are typically average or above average.

Review Questions

Where does high intelligence come from?

1. genetics

1. environment

1. both A and B

1. neither A nor B

Arthur Jensen believed that _____.

1. genetics was solely responsible for intelligence

1. environment was solely responsible for intelligence

1. intelligence level was determined by race

1. IQ tests do not take socioeconomic status into account

What is a learning disability?

1. a developmental disorder

1. a neurological disorder

1. an emotional disorder

1. an intellectual disorder

Which of the following statements is true?

1. Poverty always affects whether individuals are able to reach their full intellectual potential.

1. An individual’s intelligence is determined solely by the intelligence levels of his siblings.

1. The environment in which an individual is raised is the strongest predictor of her future intelligence

1. There are many factors working together to influence an individual’s intelligence level.

Critical Thinking Questions

What evidence exists for a genetic component to an individual’s IQ?

Describe the relationship between learning disabilities and intellectual disabilities to intelligence.

Personal Application Question

Do you believe your level of intelligence was improved because of the stimuli in your childhood environment? Why or why not?

CHAPTER 13 / CAC.903

Objective Personality and Symptom Measures

Objective measures use structured items and scoring rules to assess personality, symptoms, or behavior. They include broadband inventories, narrow symptom scales, and behavior rating systems. They are efficient and can provide norms. They are also limited by self-report bias, informant bias, item ambiguity, and cultural meaning. The counselor should use them as evidence, not as truth machines.

Self-report and informant report

Children and adolescents can report inner experience, but insight varies with age. Young children may interpret items literally. Adolescents may minimize or exaggerate for reasons that make sense to them. Caregivers and teachers observe behavior but may not see inner distress. A multi-informant approach is often needed. Discrepancies should be explored: Does the child hide distress? Is the parent overwhelmed? Does the behavior occur only at school?

Response styles

Response styles include acquiescence, social desirability, random responding, and defensiveness. Validity scales can help detect unusual patterns, but they are not perfect. A high defensiveness score may reflect privacy concerns, not pathology. A high symptom score may reflect a cry for help, not malingering. The counselor should interpret response styles in context.

Narrow versus broadband measures

Broadband measures screen many domains. They are useful for case finding but may produce false positives. Narrow measures focus on a construct such as anxiety or ADHD. They are useful for monitoring and depth. A multi-stage approach may begin with a broadband screen and follow with narrow measures and interview. This balances efficiency and accuracy.

Personality assessment caution

Personality is developing in childhood and adolescence. Labels should be used cautiously. Traits such as negative emotionality or effortful control can describe patterns without pathologizing. Personality disorder diagnoses in youth require careful consideration of development, duration, and context. The counselor should avoid fixed labels that follow a child unnecessarily.

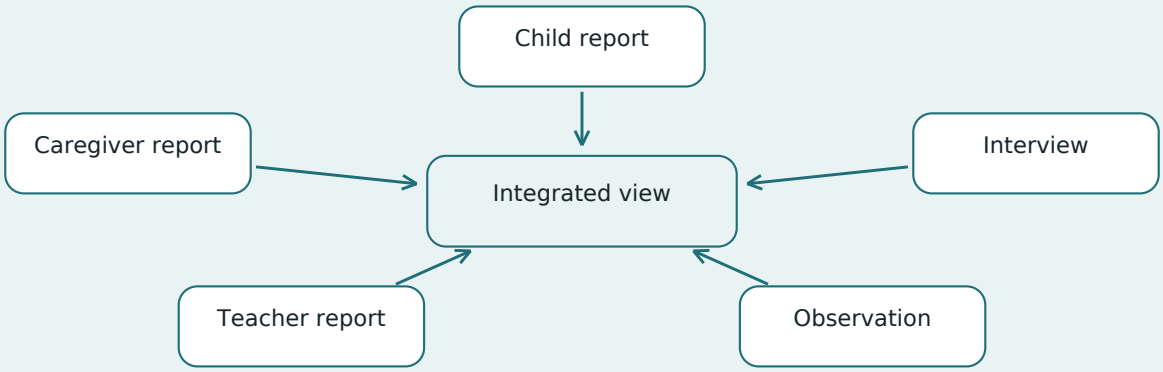
Real-life application

A teen's self-report shows high depression, while the parent report shows low depression. The interview reveals that the teen hides sadness at home and experiences hopelessness privately. The counselor does not dismiss the teen because the parent disagrees. The formulation includes masked distress and family communication. Safety assessment follows. The discrepancy becomes clinically meaningful.

Reflection

Choose a symptom scale. Identify who should complete it and why. Write two possible reasons for informant disagreement. Objective measures are useful because they are structured. They are limited because they are still human reports.

Multi-informant symptom assessment



Different informants see different contexts. Discrepancies are data, not noise.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

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Overview of Personality Tests

There are two main types of personality tests: objective personality tests and projective personality tests. Of course, no measure is truly "objective" but some measures are more or less so. In a so-called objective personality test (or structured personality test), a stimulus is presented to a respondent, who makes a closed-ended (constrained) response, such as True/False or Likert ratings. Examples of objective personality or symptomatology tests include the Minnesota Multiphasic Personality Inventory (MMPI) and the Beck Depression Inventory (BDI). In a projective personality test, an ambiguous stimulus is presented to a respondent, who is asked to make an open-ended response. Examples of projective personality tests include the Rorschach Inkblot Test and the Thematic Apperception Test (TAT). Projective tests have largely fallen by the wayside now, but it is still helpful to think about their potential advantages. Projective tests are described in further detail in the source discussion.

Projective Personality Tests

In a projective personality test, the client's response is not limited. Projective personality tests were designed from a psychodynamic perspective, and they are supposed to have limitless variability and therefore a freer access to the client's internal world. However, Card V of the Rorschach Inkblot Test looks like a moth or a bat, and around 90% of respondents likely give that response (Wiggins, 1973), so projective tests are not completely limitless. Projective personality tests are designed to have ambiguous content. This is, in part, to make them hard to figure out what is being assessed. That is, they tend to have low face validity. However, even so-called objective personality tests can have items that are ambiguous and that function similarly to a projective test. For instance, one of the items on the original MMPI asks respondents whether they like mechanics magazines. But in modern times, many people have never looked at a mechanics magazine. So, it becomes like a Rorschach question because the client starts to think about other factors.

Scoring of projective tests does not rely on the client's insight, so projective tests might get past social desirability of clients' responses and potential defensiveness. Therefore, projective tests are potentially difficult to fake. Faking good means to present oneself as better (or in a more positive light) than one actually is, whereas faking bad means to present oneself as worse (or in a more negative light) than one actually is. Faking or feigning responses often happens for incentives due to external reasons. For instance, a client may want to fake good on a test if it allows them to get a job or to win custody of a child. By contrast, a client may want to fake bad to receive disability insurance or to be pronounced not guilty by reason of insanity.

As of 1995, the TAT and Rorschach were the #5 and #6 most widely used assessments, respectively, by clinical psychologists (Watkins et al., 1995). However, they and other projective techniques have lost considerable ground since then.

Objective Personality Tests

In contrast to projective personality tests, objective personality tests, the client's responses are substantially constrained to the possible answers. Moreover, objective personality tests tend to be cheap and fast to administer, and they can be scored by computers now. In addition, objective tests have more reliable scoring than projective tests. It takes a very long time to score the Rorschach Inkblot Test, and it still has very low reliability. With an objective test, by contrast, scoring reliability approaches perfection. The earliest examples of objective personality tests were the MMPI and the Strong Vocational Interest Blank. The MMPI assesses personality, whereas the Strong Vocational Interest Blank assessed preferences for different jobs or professions. An overview of objective personality tests is provided by Wiggins, 1973.

Example of an Objective Personality Test: MMPI

The MMPI is an example of a so-called "objective" personality test. But the MMPI and other objective personality tests are not truly objective. Consider an example of a True/False item from the MMPI: "I hardly ever notice my heart pounding and I am seldom short of breath". The item is intended to assess somatic symptom disorder, with a response of "false" being more indicative of disorder. But there are other factors that could influence a person's response to the question besides whether the person has somatic symptom disorder. the source discussion depicts how a person's response to the question could be influenced by several factors. A comorbidity of somatic symptom disorder could make it more likely that a person answers with a response of "false", in line with somatic symptom disorder. In addition, a person's insight ability may influence whether they answer with a response of "false". Also, people who exercise more may experience a pounding heart and shortness of breath more frequently (i.e., while exercising), and therefore answer with a response of "false". In sum, the item and the test as a whole is clearly not "objective" because constructs in the questions are not clearly defined. What does a "pounding heart" mean? How much is "hardly ever"? People may define "seldom" differently.

Various Factors That Could Influence a Respondent's Answer to the True/False Question: "I hardly ever notice my heart pounding, and I am seldom short of breath".

There are multiple forms of measurement error for a given item. For instance, there are situational sources of measurement error. Consider the item "I hardly ever notice my heart pounding, and I am seldom short of breath". For example, how many stairs the participant climbed to get to the lab could influence their response to the question. Other situational factors could include whether they are dehydrated, whether they have a cold or are sick, and what they did over the past week. Another form of measurement error could result from the purpose of test taking. It is important to consider the purpose of the assessment. For example, if the assessment is for disability payments or to get a physical job, such as a firefighter, it may change a person's answers. Response biases, such as social desirability, could also influence a person's answers. In addition, memory limitations of the person could influence their responses.

There have been multiple versions of the MMPI. Graham et al., 2022 provide an overview of how the MMPI was developed. The original MMPI was developed using the external approach to scale construction. The second version of the MMPI (MMPI-2) placed greater emphasis on content validity of the items, on removing items with outdated or offensive

language, and on updating the norms to be more representative. Later, restructured versions of the MMPI-2 were created, which became known as the MMPI-2-Restructured Forms (MMPI-2-RF). Evidence on the MMPI-2-RF is reviewed by Sellbom, 2019. The latest version of the MMPI is the MMPI-3.

Problems with Objective True/False Measures

There are a number of problems with objective True/False measures. However, the problems are not necessarily unique to objective personality measures or to True/False measures.

One problem is related to the response biases of acquiescence and disacquiescence. Acquiescence occurs when the person agrees in response to items regardless of item content. Acquiescence occurs oftentimes when the participant is just going along because they think the experimenter may want them to have certain characteristics, therefore they often just say "TRUE" a lot. That is, they may want to please the investigators. Disacquiescence, by contrast, is when the person disagrees in response to the items regardless of item content. Disacquiescence is also called opposition bias. Disacquiescence may occur if the person does not think they have the disorder or problem of the characteristics being explored.

True/False measures, just like other questionnaire formats, are influenced by multiple sources of variation. In an objective personality test, there is a high demand on respondents. For instance, on the MMPI, respondents must be aware of recall over an unspecified period of time, which can cause confusion. The high demand on respondents can lead to inaccuracy of individuals' reading of the items. Additional sources of information could influence the respondents' answers to the item, "I hardly ever notice my heart pounding, and I am seldom short of breath". Some respondents may be intoxicated while answering questions, which could influence physiological experiences. Some respondents may have comorbordities that influence their responses. Psychological and medical comorbidities could cause difficulty in interpreting the questions. For example, the respondent may wonder whether the investigator is inquiring about heart beating that reflects anxiety versus heart beating that reflects high blood pressure.

In addition, there are effort differences that influence people's responding. There has been a recent resurgence in how to assess respondents' effort. There are often limits to predicting phenomena due to individuals not putting effort into answering questions. Other potential sources of variation in an item could include psychopathology, stress, physical fitness, age, and gender. For instance, older people are more likely to notice shortness of breath because their lung capacity is less than what it used to be. Moreover, women are more likely to say "FALSE" to this item than men, but the item has less social desirability bias than some other items, so women could be putting in more effort and giving different responses than men.

In sum, lots of factors beyond the construct of interest can influence a response on an item. It is valuable to scan through your measures and consider what goes into a person's answers. It is important to keep the content in mind of all things contributing to scores, and you should consider these potential factors when interpreting results!

Another problem with some questions is compound questions, also called double-barreled questions. Double-barreled questions following the structure: T/F: X + Y. An example of a compound question is, "True or False: Cars should be faster and safer". It is unclear which components of a compound question people are responding to. In conclusion, every objective

test is partly projective—that is, the stimuli are interpreted in different ways by different people.

Approaches to Developing Personality Measures

There are three primary approaches to developing personality measures and other scales:

1. An external approach to scale construction, also called an empirical approach or criterion-keyed approach

2. A deductive approach to scale construction, also called a rational, theoretical, or intuitive approach

3. An inductive approach to scale construction, also called an internal or item-metric approach

However, these approaches are not mutually exclusive and can be combined.

External Approach to Scale Construction

The external approach to developing a personality measure is also called the empirical approach or the criterion-keyed approach. The external approach relies on an external criterion. In general, a criterion-keyed approach examines scores on items in relation to the criterion, and selects items that are associated with the criterion, regardless of the item content. The MMPI is an example of a scale that was developed using the external approach. For the MMPI, the criterion was a group (i.e., a criterion group approach), or more accurately, multiple groups: patients with different disorders and controls. The original MMPI was not developed based on theory (e.g., theoretical understanding of the construct of depression); instead, it was developed based on items' empirical associations with a criterion group. Because an external approach relies on an external criterion, if people lose interest in the scale criterion, the scale loses interest.

Consider the development of the MMPI as an example of a measure that was developed using the external approach to scale construction. The developers grouped people together based on their criterion status, for example one group of people with schizophrenia and a control group that does not have mental disorders. How did they develop the test? The idea of the external approach is to let nature decide what goes into the test. So, the MMPI developers sampled thousands of questions very broadly from pre-existing questionnaires of personality, symptoms, etc. Then, they had the criterion groups (e.g., schizophrenia) and control groups answer the questions. They examined the item responses to determine which items discriminate between groups (i.e., which items are associated with criterion group status). And if an item(s) is good at discriminating between the criterion and control groups, the items are selected for the measure. It was purely empirical business—i.e., dustbowl empiricism. In sum, using an external approach, item selection depends on the discriminatory power of each item to inform about an external criterion of interest. Such an approach does not require having any theoretical assumptions about item functioning.

Pros

There are several pros of using the external approach to develop measures:

• You do not need to know anything—it requires no theory or theoretical knowledge about disorders and their etiology

• There is likely to be some generalizability of the utility of the measure in the future because there is some carryover of criterion-related validity

• The measure has some practical utility given its relation to criteria of interest

• "Subtle items" with poor face validity can be selected solely based on their discriminatory power.

Subtle items are items that neither you nor the respondent predicted would show differences between groups. Subtle items provide an advantage that the data are moving beyond our ignorance. It is also an advantage that clients cannot fake subtle items as well as they can more obvious items.

Cons

There are also cons of using the external approach to develop measures:

• Measures developed using the external approach have lower content validity and/or face validity because they include subtle items.

For example, the original version of the MMPI had low content validity and face validity. Using the external approach, it is possible to construct a scale that makes no sense due to lack of consideration of face validity.

• The success of the external approach depends highly on the quality of the criterion and control groups—if a criterion falls out of favor as an indicator of the construct (or interest fades), then the utility of the test decreases because it is no longer applicable.

• Items will not always generalize because the generalizability depends on the representativeness of the sample and the quality of the groups—if you select poor groups, it is not a representative sample, which results in a biased measure.

The validity of the scale depends on the representativeness of your groups in regard to the criterion of interest. It is possible that your findings might not be generalizable if your sample is not—this is especially problematic when you rely only on data and not on theory.

• An example of the importance of the representativeness of the sample comes from the original MMPI.

The norms of the original MMPI were based on largely White participants from Scandinavian, German, and Irish descent in the Midwestern U.S., with an average of an 8th grade education. The norms became known as "Minnesota farmers". Therefore, "Minnesota normals" (i.e., the control group) were pretty "dull" normals. In addition, to be included in the norms, the respondents for the MMPI had to be waiting in the hospital (not everyone does that for family in the hospital) and had to be kind enough to take a 4–5 hour measure for psychologists (not all would do this). For all of these reasons, there were issues with poor generalizability of the original MMPI norms in the broader population. A scale is most likely to be valid when it is used with similar populations and in similar conditions.

• Shrinkage often occurs when using a measure developed using an external approach.

As described in the source discussion, shrinkage is when variables with stronger predictive power in the original data set tend to show somewhat smaller predictive power (smaller validity coefficients) when applied to new groups. When variables are selected empirically, they tend to show less predictive power (smaller validity coefficients) when applied to new

groups during cross-validation. Shrinkage reflects a model over-fitting, because it is somewhat capitalizing on chance in selecting items. Many subtle items may be instances of Type I error (false positives). Shrinkage is especially likely when the original sample is small and/or unrepresentative and the number of variables considered for inclusion is large.

• Externally developed measures also have a problem of communicability (Burisch, 1984).

For something to have meaning to others, it should have connection to constructs, which may not be true for many measures developed using the external approach.

• It can take a long time to develop measures using the external approach, and they tend to be longer to administer because of having more items.

For instance, there are 567 items in the MMPI-2.

MMPI Examples

Paul Meehl wrote his dissertation to develop the K scale of the MMPI to attempt to detect faking good. In the context of the MMPI, faking good would involve under-reporting of symptoms. He developed the K scale based on all positive qualities that around half of people typically endorse. So, it is not obvious that the items reflect faking good, and if someone is trying to respond in a socially desirable way, they endorse more of these positive qualities. There is also a "faking bad" version of the scale, too. The "faking bad" (F) scale was developed as an attempt to detect malingering (over-reporting of symptoms).

However, the MMPI could pathologize normality in some cases. Some psychotic patients may have been identified as "faking bad" because they actually have had a lot of odd experiences. And very healthy people may have gotten higher score on "faking good", but they just may be very positive and well-adjusted. As examples from the original MMPI from 1940, male teenagers tended to have elevated scores on the psychopathy and mania scales. Graduate students, including female graduate students, tended to have higher "masculine" scores on the Masculinity/Femininity (Mf) scale. This shows that non-clinical samples can still have "clinical" scores. In the 1930s–1940s, there was an emphasis on developing empirically based measures, where the researchers rely on data, not theory as had previously been emphasized.

Deductive Approach to Scale Construction

The deductive approach to developing a personality measure is also called a rational, theoretical, or intuitive approach. Using a deductive approach, the choice and definition of constructs precede and govern the formulation of items. That is, when using a deductive approach, we deduce the items from a pre-existing theory or conceptual framework. Item pools are generated using theoretical considerations, and item selection depends on possessing a rich theoretical knowledge about the construct and selecting which items assess the construct the best. In a deductive approach, the measure developer deduces the content; they do not rely on criterion data to select the content. Deducing the content involves thinking and talking about the construct, and having experts write items that they think would do well in eliciting information about the construct. The measure developer deduces from the construct which items to use. Therefore, the deductive approach completely depends on our ability to understand a given construct and translate this understanding to the generation of item content that will be understood by the examinees in such a way that it elicits accurate ratings for the construct of interest. Most assessments are developed using the deductive approach.

Pros

There are several pros of using the deductive approach to develop measures:

• Using the deductive approach is fast, easy, and short.

Generating such scales requires few people, and is often fast and accurate. It does not typically take as much time to develop a measure using the deductive approach. Moreover, it allows the possibility of short scales that are quick to administer. By contrast to short scales developed by the deductive approach, the MMPI (developed using the external approach) is very long.

• Measures developed (well) using the deductive approach are always content valid because of the reliance on theory—items tend to be prototypical of the construct.

• Measures developed using the deductive approach are usually face valid, more likely than external approach.

Face validity is often an advantage, but not always. Face validity can help with disseminability because others may be more likely to adopt it if it appears to assess what it claims to assess. However, face validity is not desirable when trying to prevent faking good or faking bad.

• Measures developed using the deductive approach tend to have better communicability—i.e., how comprehensible the information communicated is to the examiner based on the responses.

Cons

There are also cons of using the deductive approach to develop measures:

• If your theory or understanding is wrong, your scale will be wrong!

• Additionally, if the construct itself is vague, and there is overlap between constructs, scales may be difficult to differentiate between them, making it difficult to establish discriminant validity.

Many theories and constructs overlap. Therefore, measures often overlap—even measures with very different names! For example, consider the Rosenberg Self-Esteem Scale and the Spielberger State–Trait Anxiety Inventory:

Items on the Rosenberg Self-Esteem Scale include:

• "At times I think I am no good at all"

• "I certainly feel useless at times"

• "On the whole, I am satisfied with myself"

Items on the Spielberger State–Trait Anxiety Inventory (STAI):

• "I lack self-confidence"

• "I feel inadequate"

• "I feel satisfied with myself"

Inductive Approach to Scale Construction

The inductive approach to developing a personality measure is also called an internal or item-metric approach. The inductive approach is an empirical, data-driven approach for scale construction, in which scales are derived from the pre-existing internal associations between items. That is, when using an inductive approach, we induce the construct(s) from the pattern of associations (and non-associations) among the items. A large pool of items is selected, and scales are generated from the item pools based on the structure of the internal association between items. The inductive approach assumes that universal laws exist for personality structure, that is, that there is a natural structure. The hope is that personality has simple structure: that each item loads onto (i.e., reflects) one and only one factor. It is the hope of the inductive approach that there is simple structure because it makes the natural structure easier to detect using available methods.

The inductive approach is empirical because the answers come from within the data. But the inductive approach differs from the external approach. In the inductive approach, the data come from the internal structure of the measure's items. The empirical approach is also based on empirical data. Science involves prediction and both the inductive and empirical approach use prediction. The empirical approach examines how items predict some external criterion. By contrast, the inductive approach examines how items predict or relate to each other. The goal of the inductive approach is to "describe nature at its joints".

According to the inductive approach, once you understand constructs, you can understand how they are connected to each other. The external approach does not really care about the items themselves. In the inductive approach, you need to use theoretical knowledge to interpret findings. The deductive approach uses theory up front to make the scale. The inductive approach concerns itself with the items and what inferences can be made. The method helps group the large set of items into subscales based on clusters of items that covary most strongly with each other using factor analysis, and it drops items with a low item–total correlation, factor loading, or discrimination parameter.

Factor analysis is used for the inductive approach to developing measures. It is used to evaluate the internal structure of a measure, and ideally, the structure of a construct. Factor analysis is considered to be a "pure" data-driven method for structuring data, but as noted in

the source discussion, the "truth" that we get depends heavily on the decisions we make regarding the parameters of our factor analysis. In sum, factor analysis is not purely inductive because the result is influenced by many decisions by the investigator. Even though the inductive approach (factor analysis) is empirical, theory and interpretability should also inform decisions.

Pros

There are several pros of using the inductive approach to develop measures:

• The inductive approach yields estimates of associations between items and can arrive at estimates of a simple, homogeneous construct.

• You do not need to know much to use the inductive approach (relative to the deductive approach): just use the items you have and use a data reduction approach.

• The inductive approach can derive short, homogeneous scales.

Then, you can see how constructs relate to other constructs.

• The inductive approach is a "purer" method of scale construction because it relies on the natural structure of the data, and no theoretical knowledge or validation to a criterion is required on the front end of scale development.

• The data are allowed to "speak for themselves".

Cons

There are also cons of using the inductive approach to develop measures:

• The inductivists (i.e., users of the inductive approach) hope that a simple structure exists within a set of items and that this structure can meaningfully differentiate between constructs.

If there is no simple structure, interpretations of scales that emerge can be difficult.

• In addition, this approach is not "pure" because the structure we get depends on the analysis decisions we make.

The answers you get depend on the decisions you make, and there really is no basis on which to make decisions. This is called indeterminacy. There are a number of decisions in factor analysis, including decisions such as the number of factors and the nature of factors—i.e., how to interpret them.

• Factor analysis is not straightforward, and depends on decisions made along the way.

Therefore, some argue because so much is in the hands of the investigator that factor analysis is really a semi-empirical approach.

• SPSS likely contributes to the problem because it makes so many decisions for you, and many have no idea what they are doing!

SPSS is for ease of use, but it is limiting. In SPSS, you can use principal component analysis (PCA) for item extraction. Investigators often determine how many components/factors to keep based on the criterion of keeping components with eigenvalues greater than 1, often use orthogonal rotation of data, etc. Factor analysis and PCA are described in the source discussion.

Hybrid Approach

The preceding discussion described the three primary approaches to developing objective personality measures. However, the approaches can be mixed. For instance, one could write a large set of items based on theory (i.e., the deductive approach) and then pick items to keep based on their criterion-related validity (i.e., the external approach), and group them into scales based on their internal structure (i.e., the inductive approach). There is not strong evidence for the superiority of any of the approaches compared to the others.

Measure Development and Item Selection

Despite not having strong evidence for the superiority of any of the approaches to scale construction, best practices to measure development include:

• Start with theory to define the construct and create item pools, using a deductive approach.

• Be inclusive at this stage.

Create more items than you will actually use—even if items are only tangentially related—so you have a broad pool of items.

• Include items of other constructs to establish the boundaries of the construct, i.e., discriminant validity.

• Then, test these item pools, and consider their empirical relations to revise and/or drop items.

Ideally, you would test the items in large and heterogeneous samples that are representative of the population.

• Examine the items in relation to external criteria, using an empirical approach.

• Examine items in relation to each other, using an inductive approach.

This likely involves factor analysis and/or item response theory.

• We want items, collectively, to span the full range of difficulty/severity of the construct in the target range of interest.

It is important for the items to have accuracy (i.e., strong discrimination and information) in the target range of interest on the construct (e.g., low, medium, and/or high). The target range of interest depends on the purpose of the assessment, as described in the source discussion. For example, items used for diagnosis should focus on higher levels of the construct, whereas items used for screening should identify those with elevated risk but might not need to discriminate at higher levels. For assessing individual differences, you would want items that discriminate across the full range, including at the lower end.

• Items should show some internal consistency, as evidenced by an inter-item or item–total correlation, but items should not be too highly inter-correlated.

If items are too highly correlated, then they are redundant and do not provide unique information. Inter-item correlations should only be moderate, e.g., should range from approximately.15 to.50. But items should be highly correlated with the latent factor

representing the target construct. That is, the items should have a high discrimination or a strong factor loading.

• Then, interpret the results and label the factors based on theory.

• Evaluate multiple aspects of reliability and validity of the scale.

Other ideas in scale development are discussed by Burisch, 1984, Clark & Watson, 1995, Clark & Watson, 2019, and Loevinger, 1957.

The Response Scale

Evidence suggests that there may not benefits of having more than six response options for likert-scale items that assess personality (Simms et al., 2019).

Emerging Techniques

One emerging technique for developing personalized models of personality is the group iterative multiple model estimation (GIMME) model (Wright et al., 2019), as described in the source discussion. In addition, artificial intelligence approaches can be leveraged for scale development and validation (Austin et al., in press).

The Flawed Nature of Self-Assessments

It is a common finding that people tend to over-estimate their skill and abilities—most people tend to describe themselves as "above average", which is statistically impossible. People are over-confident, and they over-estimate the likelihood of achieving desirable outcomes and underestimate how long it will take to complete future projects. Self-report is only weakly associated with people's actual behavior.

There are there prominent ways in which self-assessment has been shown to be flawed:

• response bias

• ambiguity of items

• lack of insight by people to rate themselves

Response Bias

One way that self-assessment has been shown to be flawed is in terms of response bias. Bias involves a systematic measurement error. For instance, it is not uncommon for participants to fake good or fake bad on assessments due to situational reasons or due to how questions are worded. When answering questions, many people desire to seem better than they are. This phenomenon is called social desirability bias, which is a form of method bias in which people systematically adjust their responses to reflect more socially desirable attributes. The degree of a person's faking has been shown to be related to social desirability bias (Bensch et al., 2019). If you want to want to know the true prevalence of a given behavior, you can deal with social desirability bias using a randomized response model, as described in the source discussion. There are indicators of response bias that are worth considering (Burchett & Ben-Porath, 2019).

Ambiguity of Items

Another way that self-assessment has been shown to be flawed is in the ambiguity of items. People tend to have a difficult time providing an accurate characterization of their skills on tasks that are poorly defined or ambiguous. For example, what does it mean to be a "warm" parent? Beyond this, even the language on self-assessments can be highly ambiguous. For example, how often is "rarely"? Such questions can systematically skew answers to self-assessments.

Lack of Insight by People to Rate Themselves

A third way that self-assessment has been shown to be flawed is due to respondents' lack of insight into the required information to make good self-assessments. People often make poor judgments because they lack the required skills and information necessary to have insight into their actual performance, or they neglect it when it is available. The lack of insight into poor judgments can also be caused by errors of omission—in which they do not know that they have made a mistake because they do not know what the best alternative would have been. Also, people infrequently get feedback from others on the constructs we are attempting to assess—as a result, their self-views are not informed by objective feedback.

Ways to Improve Self-Assessment

Ways to improve self-assessment are described by Dunning2004:, n.d.

• Use clear items that are behaviorally specific

• Use an expanded response format rather than a Likert format (Zhang & Savalei, 2024)

• Provide frequent, timely, objective, and individualized feedback

• Use self-testing, following some delay of time after studying the material

• Review one's past performance

• Use peer assessment

• Target the motivational basis of the over-confidence

• Benchmark—compare one's performance against others' performance

• Introduce "desirable difficulties" to instruction, such as spreading training over several sessions and varying the circumstances of the training.

These challenges can harm the speed that students learn but leaves them better able to retain what they learned and to transfer it to new situations in the future.

• To account for over-confidence, add in safety factors and buffer time.

People tend to under-estimate how long it will take to complete a project, known as the planning fallacy (Kahneman, 2011). For instance, add 30%–50% extra time to all time-completion estimates for projects.

Satisficing (Versus Optimizing)

There have been new developments in gaining insight into the cognitive processes by which respondents generate answers to survey questions (Krosnick, 1999).

Optimizing

Optimizing involves a respondent responding optimally to a question—i.e., responding in an unbiased and thorough manner. There are four cognitive steps or stages that respondents must complete to answer a question optimally:

1. They interpret the question

2. They search their memory for relevant information

3. They integrate all relevant information into a single judgment

4. They use that judgment to select a response

The complexity of the cognitive processes one must engage in when giving an optimal answer requires a lot of cognitive effort. Giving such effort can happen for a variety of reasons: desire of self-expression, being altruistic, desire for gratification, etc. The extent to which such motivations inspire a person to engage in the cognitive requirements to respond to questions in an unbiased and thorough manner is referred to as optimization. Optimizing deals with how hard the person worked and how much they care about giving their best response.

Satisficing

If, for some reason, a person is not motivated to expend the cognitive effort required to make an optimal response, and instead settle for a satisfactory response, they are said to be satisficing. That is, they are expending less effort that prevents obtaining the optimal answer. The word satisfice is a portmanteau of satisfy and suffice—the respondent picks a "good enough" answer that will satisfy the requirements of the task and suffice for their purposes, rather than expending the effort needed to identify the optimal answer. There are two types of satisficing: weak satisficing and strong satisficing. In weak satisficing, respondents execute all four cognitive steps, they are just less diligent about doing so, and they settle on selecting a satisfactory answer rather than the optimal answer. Weak satisficing may lead to selecting earlier response options without careful evaluation of later response options, which is susceptible to confirmation bias. In strong satisficing, a respondent skips the retrieval and judgment steps, interprets the question superficially and selects an answer based on what they think will be a reasonable (or "satisfactory") response. Their answer is not reflective of a person's actual feelings about the construct of interest. Answers could also be selected arbitrarily.

There are several conditions when satisficing is most likely to occur:

• The greater the task difficulty

• The lower the respondent's ability

• The lower the respondent's motivation

• When "No opinion" responses are an option

In sum, to reduce the likelihood of satisficing, match the task to the participant's ability, make sure they have motivation to respond correctly, and do not provide a "no opinion" response option.

Observational Assessments

Given the challenges with self- and informant-report of personality, it can also be worth considering observational assessments. For example, one observational approach to assessing personality involves a "thin-slice" approach, in which observers briefly assess people's personality via observations across a range of situations or contexts (Tackett et al., 2019). Observational ratings of personality can then be combined with self- and informant-rated personality in a multitrait-multimethod matrix (Tackett et al., 2019).

Structure of Personality

The most well-supported structure of personality is the five-factor model of personality. The highest-order dimensions of the five-factor model are defined by the Big Five. The Big Five are known by the acronym OCEAN: **O**penness to experience (versus closed-mindedness), **C**ontientiousness (versus disorganization), **E**xtraversion (versus intraversion), **A**greeableness (versus disagreeableness), and **N**euroticism (versus emotional stability).

Personality Across the Lifespan

As described by Costa Jr. et al., 2019, individual differences (i.e., rank order) in personality are relatively stable from middle childhood to old age. On average, neuroticism tends to decline, and agreeableness and contientiousness tend to increase with age (Costa Jr. et al., 2019). It is unclear how extraversion and openness to new experiences change across the life span.

Conclusion

In an objective personality test, a stimulus is presented to a respondent, who makes a closed-ended response, such as True/False or Likert ratings. An example of an objective personality test is the Minnesota Multiphasic Personality Inventory (MMPI). There are three primary approaches to developing personality measures and other scales. One is the external approach, in which items are selected based on their association with an external criterion. A second approach is the deductive approach, in which items are deduced based on theory. A third approach is the inductive approach, in which items are selected based on the internal association between items that are intended to assess the same construct. Guidelines for measurement development are provided. Nevertheless, relying on self-report and self-assessment is prone to key weaknesses including response bias, ambiguity of items, lack of insight, and satisficing. There are ways to improve self-assessment, but it can also be helpful to supplement self-assessments with informants' ratings and with observational assessments.

Suggested Readings

Burisch, 1984; Dunning et al., 2004; Krosnick, 1999

CHAPTER 14 / CAC.903

Projective, Expressive, and Play-Based Methods

Projective and expressive methods invite a child to use images, stories, play, or art to communicate experience. They can be valuable when words are limited or when direct questions feel threatening. They are also vulnerable to overinterpretation. The counselor should distinguish between using a method as a relational and hypothesis-generating tool and using it as a definitive diagnostic test.

The projective hypothesis

The projective hypothesis suggests that ambiguous stimuli allow inner needs, fears, and conflicts to be expressed. Some projective tests have scoring systems and research support for specific uses. Others are used clinically with limited evidence. The counselor should know the evidence for each method. A drawing should not be treated as proof of abuse or a specific disorder.

Play-based assessment

Play assessment observes themes, affect, organization, flexibility, and interaction. A child may enact separation, danger, repair, or control. The counselor tracks patterns over time rather than interpreting one symbol. Play can reveal relational expectations and coping strategies. It can also assess change during therapy. The counselor should use a consistent frame and document observations carefully.

Expressive arts

Drawing, sandtray, music, and movement can help children express experience. The counselor should ask the child about the creation rather than impose meaning. "Tell me about this picture" is safer than "This dark color means depression." The process may be more informative than the product. The counselor should consider motor skills, culture, and access to materials.

Ethical limits

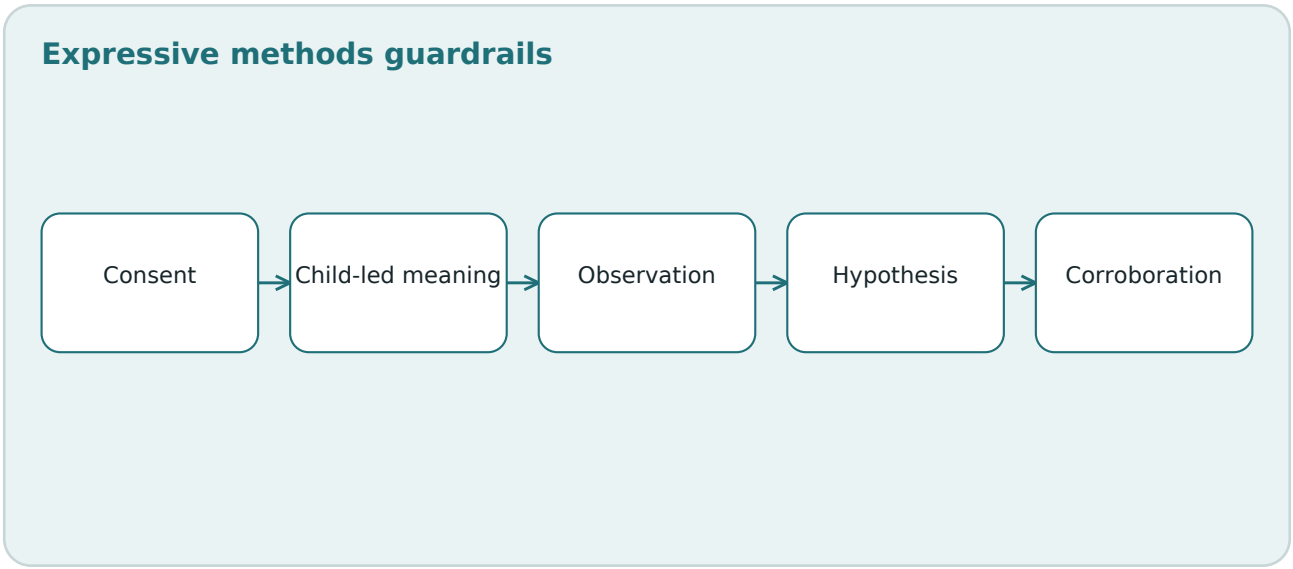
Expressive methods should not be used to make forensic conclusions without proper training and corroboration. Suggestive questions can contaminate reports. The counselor should avoid leading interpretations. In child protection contexts, standardized forensic protocols should be followed. The counselor's role may be to support, not to investigate.

Real-life application

A six-year-old repeatedly draws a locked house after a hospitalization. The counselor does not announce trauma. The counselor asks about the house and follows the child's play. Over sessions, the child introduces a key and a helper. The play suggests growing mastery. The counselor shares themes with the caregiver in general terms and supports medical follow-up. The method guides relationship and hypothesis, not verdict.

Reflection

Write three rules for using expressive methods responsibly. Include consent, interpretation limits, and documentation. Expressive work can open a door. It should not be used to force a conclusion.



Expressive methods generate hypotheses and support relationship; they do not replace evidence.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module projective-personality-testing.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Overview of Projective Personality Testing

Surveys consistently show projective measures to be among the most commonly used assessment devices in clinical psychology, school psychology, and neuropsychology (Youngstrom & Van Meter, 2016). There has been some decline in their usage, in part due to managed care and in part due to criticisms, but they are still widely used.

Frank, 1939 presents a problem to solve—how best to assess personality—and his proposed solution. He sought to develop a predictive model of people's personality and behavior based on how people process information. The approach of predictive modeling has worked in other areas of psychology. According to their conceptual framework, people exist in multiple spheres:

1. in the common public world of nature (i.e., as humans)

2. as members of their social and cultural groups

3. as individuals in their private worlds of highly idiosyncratic meanings, significances, and feelings (i.e., personality)

When psychologists try to assess people's private worlds (personalities), they are not seeking to assess the cultural and social norms—instead they seek to assess the peculiarities of the individual. Psychodynamic theorists viewed standardized tests as a measure of how much a person conforms to the expectations of a cultural group, not as a measure of the person as an individual. Psychodynamic theorists viewed personality as a dynamic process of how a person organizes their experience according to the unique individual's private world. They viewed projective measures as an "x-ray of the soul" that reveals its components and organization, including the state, condition, and maturation of the organism.

Projective measures are an indirect method used to assess the internal organization and composition of an individual organism. According to the conceptual framework, projective measures allowed examiners to get a glimpse into this dynamic process and their private world by observing the examinee's response to various "fields." Fields are ambiguous stimuli consisting of objects, materials, and experiences with relatively little structure and cultural patterning. Psychodynamic theorists hypothesized that an individual's personality projects onto the field their way of seeing life, their meanings, significances, patterns, and feelings.

Projection, according to Freud, 1911, is a defense mechanism by which people unconsciously attribute their negative personality traits and impulses to others. According to a psychodynamic framework, psychopathology is hypothesized to be caused by unconscious motives, conflicts, and memories. In this framework, defense mechanisms keep us unaware of the painful contents of the unconscious. However, attributing negative characteristics to others does not appear to be effective in reducing anxiety or keeping one unaware of one's own negative characteristics.

Projective measures were thought to be a way to access a person's unconscious. When presented with a field (an ambiguous stimulus), the person has to organize the field, interpret the material, and react affectively to it. Therefore, according to the framework, projective measures elicit a projection of the individual person's private world, that is how they organize their life space. Psychoanalysts seek to assess "what one cannot or will not say" because the person does not know themselves well enough, and is unaware of what they are revealing about themselves through their projections. According to this perspective, the person rarely has an understanding of themselves or awareness of what their activities signify.

Categorization of Responses

The psychoanalysis separates projections (i.e., responses to the ambiguous stimuli) into categories, such as:

• Constitution: forming a holistic entity or "Gestalt"

• Interpretive: describing what a stimulus-situation means to them

• Cathartic: discharging affect or feeling upon the stimulus-situation for emotional release

• Constructive: building with the materials in a way that reflects something meaningful about their life

Projective Hypothesis

Psychodynamic theorists are less concerned with psychometrics (reliability and validity) from statistical correlations across many people. Instead, they seek to assess one person in many ways to see how they structure their life space. They would argue that the projective measures are valid if the examinee provides similar response configurations across stimuli and situations.

Projective measures are based on the projective hypothesis. According to the projective hypothesis, whatever a person does when exposed to an ambiguous stimulus will reveal important aspects of the person's personality. An ancillary hypothesis to the projective hypothesis is that indirect responses are more valid than direct responses. Indirect responses are responses to ambiguous stimuli, whereas direct responses are responses to interviews or questionnaires.

Projective Measures

Projective measures are measures in which an ambiguous stimulus is presented to the respondent, who is asked to make an open-ended response. There is theoretical strength behind projective measures in that they have good goals, but they are based on poor implementation. The goal of projective measures is to discover things that do not depend on self-knowledge and the person's ability or willingness to share it. Projective measures aim to bypass defense mechanisms so that they assess characteristics the person has and which the person is not consciously aware of. For example, projective measures aim to assess characteristics the person does not recognize or wants to hide.

Examples of projective measures include the Rorschach Inkblot Test, the Thematic Apperception Test (TAT), Draw-A-Person Test, and drawing a family (Kinetic Family Drawing).

Most projective techniques do **not** have:

• standardized stimuli and testing instructions,

• systematic algorithms for scoring responses to stimuli, or

• well-calibrated norms for comparing responses with those of other people

There are various types of projective measures, including measures that use:

• association techniques, often with an ambiguous form—e.g., inkblot, clouds, or word association tests

• construction techniques, often with artistic media—e.g., human figure drawing or story creation methods like the Thematic Apperception Test (TAT).

Thematic perception involves writing or telling stories about a series of pictures.

• completion techniques—e.g., sentence completion tests

• arrangement or selection techniques

• expression techniques, often with movement or play—e.g., projective doll play, puppetry, or hand-writing analysis.

Play is often used by psychoanalysts as a projective measure with children because it is thought that children have fewer defenses to hide behind (compared to adults) and are less aware of how much they are revealing through their play. For instance, psychoanalysts believe that children who have been abused will reveal that in their play configurations with dolls.

The Rorschach Inkblot Test is a classic example of a projective measure. It was designed by Swiss psychiatrist Hermann Rorschach in 1921. It was initially designed as an instrument for studying perception, but it became to be conceived of as a test of personality pathology. It was conceived by psychoanalysts as an "x-ray of the soul." The Rorschach Inkblot Test has low face validity, and is therefore potentially difficult to fake. For instance, a person may fake good (present as if they are better than they are) in the context of a custody evaluation or job interview, whereas a person may fake bad (present as if they are worse than they are) in the context of an evaluation for disability claims or of the person's sanity to stand trial.

In 1968, a projectively oriented, classic textbook was published, titled The Interpretation of Psychological Tests. The textbook provides case examples of a patient. Examiners were trying to decide if the patient was psychotic, and they interpret her crystallized intelligence from the Wechsler Adult Instelligence Scale based on Rorschach scores ("insufficiency of ego strength"). The examiners attributed her lack of consistency across items not to the inconsistency in the test (and its stimuli), but to her own inconsistencies.

Why Not Use Projective Measures?

Projective measures can be considered examples of performance-based assessments. It can be helpful to include performance-based assessments as part of a broader assessment battery to avoid exclusive reliance on self-report. From this perspective, one might think that it could be beneficial to use projective measures. However, there are many reasons not to use projective measures. First, projective measures can be very time consuming: both for the client and the psychologist, in terms of administration, scoring, and interpretation. Second, unlike other observational and performance-based measures, projective measures provide very little information. You could just ask clients questions on a questionnaire or in an interview. Consider the example of assessing a client for depression. Instead of relying on the client to report relatively color-less items in the Rorschach, why not ask them (and others) how their mood is and observe their energy and affect? Thus, projective measures have an incremental validity problem. The scientific status of projective measures is described in further detail later in the chapter in the source discussion.

Examples of Projective Measures

In this chapter, two projective measures are described in detail. The first is the Rorschach Inkblot Test and the second is the Thematic Apperception Test (TAT).

Rorschach Inkblot Test

The Rorschach Inkblot Test was designed by Hermann Rorschach in 1921 and has ten inkblots, five of which are black and white, and five of which contain color. The measure is composed of three phases, depending on whether one is using the Comprehensive System (CS) (Exner, 1974; Exner & Erdberg, 2005) or the Rorschach Performance Assessment System (R-PAS) (Meyer et al., 2011). The phases include: (1) the Free Association (CS) or Response (R-PAS) Phase, (2) the Inquiry (CS) or Clarification (R-PAS) Phase, and (3) the Follow-Up Phase.

In the "Free Association" or "Response" phase, the examinee is handed one inkblot card at a time and is asked what they see. The stimuli were designed with an aim to provide the least amount of intrinsic information; the test developers did not want to impose intrinsic information from concrete stimuli. The inkblots start completely black and white, then they move to grayscale, which was considered to be "easier emotionally" for clients, and then color inkblots are shown, which was considered more challenging emotionally. The clinician intends not to guide the examinee to particular responses, but they also want to hear more than one response. The idea is that eventually, with additional responses, the examinee will get to be less censored.

After finishing the response phase, the clinician moves to the "Inquiry" or "Clarification" phase, in which the examiner reminds the examinee of the examinee's responses and asks where on the card the examinee saw their responses and what about the inkblot made it look like that. The clinician inquires what part of the card includes the image the examinee had described. The examinee then identifies the location in the card, i.e., whether it is the whole card or particular parts of the card, or excluded parts (negative space) of the card. It involves asking the examinee what exactly they saw, and what (determinants) made them see that. It also involves examining whether they use integration to complete the whole picture. Or if they are more defensive, it is hypothesized that the examinee will pick out a little part of the picture when describing what they see. Then, they are asked to flip the card around. The inquiry phase can last a long time, sometimes 1 hour or more.

In the open-ended "Follow-Up" phase, the examiner tries to learn what the examinee was thinking and the process that may have led to the response obtained (Choca & Rossini, 2018). The examiner may use a technique called "testing the limits". This may involve additional questions or observations. For instance, questions might ask why a respondent may not see what many others see. Did the person see two insects or two people? Were they males or females? Observations might include noting that the examinee provided many responses.

Variables

There are a number of variables that are often scored in a Rorschach administration. Here are some of the ones that can be scored:

• Latency to respond

• Let the examinee stew for a while because the first response is the least interesting because it is the most obvious.

• How are they holding the card?

How do they turn the card? How many times do they turn the card?

• What content do they see and describe in the inkblot?

• Do they see sexual content?

Violent content?

• Location: did the examinee see the whole blot as a picture or just one particular area of the blot, or excluded parts (negative space) of the card?

• Determinants: the characteristics of what the image looked like to the respondent

• Form: form or shape features

• Movement: hypothesized to reflect intelligence

• Color: lack of color is hypothesized to reflect depression

• Shading: e.g., texture

• Form dimension: e.g., three-dimensionality

• Pairs and reflections: identical objects

Scoring

The clinician then can spend several hours scoring, and there are many different potential indices. Whether the content described by the examinee is considered good or bad is based on textbook judgments. This could represent an example of an illusory correlation. An illusory correlation is the tendency to perceive associations between signs and psychopathology even when they do not exist.

John Exner (Exner, 1974) developed standardized rules for administration and scoring, which improved the reliability of scores. The Exner system, also known as the "Comprehensive System" (CS) (Exner, 1974; Exner & Erdberg, 2005) is the most widely used scoring system for the Rorschach. But there are still major problems with the system. A more recent scoring system has been developed, known as the Rorschach Performance Assessment System (R-PAS) (Meyer et al., 2011). R-PAS has limited research on its use by those outside of its developers, but it likely has similar problems with validity and utility.

Thematic Apperception Test (TAT)

The Thematic Apperception Test (TAT) is described by Lindzey, 1952. The TAT was developed by Christiana Morgan and Henry Murray in 1935 (Morgan & Murray, 1935). The TAT is more about storytelling, and having the examinee tell a story. So, unlike the Rorschach, storytelling is allowed in the TAT. It was developed by scientists, and there has been a lot of research on the TAT.

The TAT aims to get a better understanding of underlying motivations (e.g., achievement motivation), and it is thought by psychoanalysts to assess implicit motives, unlike self-attributed motives assessed by self-report instruments. Most stimuli in the TAT involve humans, but the situations are ambiguous.

Examinees' responses on the TAT are strongly predictive of various motivations, but they are no more predictive than just asking people their motivations. Motivations that are intended to be assessed by the TAT include: achievement motivation, need for power, need for

affiliation, and object relations. Object relations are people's mental representations of other people.

The administrator picks cards to administer that "pull" at issues of interest related to the respondent's presenting difficulties. As a result, the TAT is non-standardized. The examiner asks the examinee what is going on in the picture, what the characters are thinking and feeling, what led up to it, and how it ended.

The examiner might examine the respondent's response latency, and writes down all words of the response. They would examine who the examinee is relating to and identifying with. The examiner seeks to identify patterns and commonalities across stories in response to different TAT cards.

Towards the end of the story, there is very little information from the card, so the respondent has to fill in the story. Thus, psychoanalysts believe that responses at the end of the story are the most "pure", and come from the inside (as opposed to external factors).

The examiner pays attention for themes that are consistent across cards. However, fantasy behavior is not the same as actual behavior. For instance, telling violent stories is not the same as being violent. The horror author Stephen King is an apt example of how describing violence does not mean that the person is violent.

Most Widely Used Assessments for Children

Projective assessments are widely used. As of 2002, the top 30 most widely used assessments for children (Cashel, 2002) include:

• Wechsler Intelligence Scale for Children

• Child Behavior Checklist

• And then many projective techniques, including:

• Sentence completion

• Draw-A-Person test (with psychodynamic oriented scoring)

• House–Tree–Person Technique

• Kinetic Family Drawing

• Rorschach Inkblot Test

• Thematic Apperception Test (TAT)

• Children's Apperception Test (Children's TAT)

In sum, seven of the most widely used assessments for children are projective tests.

Evaluating the Scientific Status of Projective Measures

Given that projective assessments are so widely used, it would be expected that they have a strong scientific status. Lilienfeld et al., 2000 evaluated the scientific status of projective techniques. To evaluate the scientific status of the Rorschach Inkblot Test (or any other projective measure), we would assess across several domains of evidence, as described below. Evaluation of the Rorschach Inkblot Test is provided by Wood and colleagues (Wood et al., 1996; Wood et al., 1996; Wood et al., 2001).

Utility of Norms

One consideration to evaluate is the utility of the norms. This includes assessing if the normative information on which scoring of the test is based sets the cut-off accurately for those with and without maladaptive behavior, across cultural and minority groups. Of note, the Rorschach does not correctly distinguish between those with and without maladaptive behavior, and it does not lead to unbiased estimates for ethnic minorities. In addition, we cannot evaluate the norms because they are not published.

In terms of the Exner norms, they showed somewhat higher reliability than the original Rorschach norms, but they did not show strong evidence for validity. The Exner norms are not well-calibrated. The Exner norms result in over-diagnosis and over-pathologizing of normality. And they are not representative and generalizable: there are ethnic and cultural biases.

Reliability of Scores

Another important consideration of projective measures is the reliability of scores. This would include evaluating both inter-rater reliability and test–retest reliability. For the Rorschach, the inter-rater reliability depends on the subscale of diagnosis, and has to be much lower than that of objective techniques, especially in field settings. Field reliability refers to reliability in actual practice rather than ideal reliability in optimal conditions of a controlled lab study. The reliability and standardization of administration of the Rorschach is also questionable. The test–retest reliability is also low—but high enough to be passable for research settings—but it should be used with caution in clinical diagnostic settings.

Influence of Measurement Error on Scores

Another important consideration is the influence of measurement error on scores: one aspect of the Rorschach in particular that can lead to measurement error is the effect of response frequency on test validity. If people provide more answers, which is correlated with intelligence, these people are more likely to express a deviant answer indicative of maladjustment.

Validity

A crucial consideration of the scientific status is their validity. Validity is not a yes/no thing. It is important to ask what the validity is for what purpose? Rorschach responses can be moderately valid indicators of creativity. Projective drawings have moderate validity for drawing ability. But are projective tests valid for their typical purposes in clinical psychology? For instance, do they have diagnostic accuracy? Their accuracy for their typical purposes in clinical psychology is discussed below.

Criterion-Related Validity

Criterion-related validity would consider whether the diagnosis given on the basis of the Rorschach aligns with an external criterion of an actual diagnosis. For the Rorschach, the answer is: not really. Scores have some relation to psychosis and thought disorders, but they are not strongly related to other aspects of psychopathology. Research designs comparing diagnostic versus control groups often over-estimate predictive validity of the Rorschach compared to what it would be in practice because clinicians are typically interested in low base rate phenomena for which predictive accuracy will be lower.

Convergent Validity

Convergent validity would consider whether Rorschach findings correspond to the findings of more reliable objective tests such as the Minnesota Multiphasic Personality Inventory? The answer is also no.

Incremental Validity

In terms of incremental validity, a frequent argument by proponents of projective tests is that, "I don't use the Rorschach/TAT in isolation but in combination with other measures". But does the inclusion of the Rorschach lead to improvements in the validity of diagnoses of clinicians? No. In many cases, it has decremental validity—it decreases predictive or diagnostic accuracy (Howard N. Garb et al., 2005).

Treatment Utility

Treatment utility considers whether it is worth the vast amount of time spent administering, scoring, and interpreting the measure (2–3 hours)? The answer is also no. You can assess psychopathology more accurately with more reliable and less time-consuming measures, including interviews, questionnaires, and demographic data. Projective measures present actual costs in addition to opportunity costs. Actual costs of projective measures include that they are expensive and time-consuming for training, and are time-consuming to administer, score, and interpret. Opportunity costs of projective measures include that they take time away from other, more useful procedures.

Summary

Hence, we would not say the scientific status of the Rorschach is very positive. The evidence does not provide the evidence necessary to call projective measures "tests". No known projective measures have enough power to justify their lengthy use. Considering the important role that assessments can play in clinical decision-making that influences people's lives, it seems unethical to use projective techniques, as they are currently designed and used.

Conclusion

Projective measures are measures in which an ambiguous stimulus is presented to the respondent, who is asked to make an open-ended response. Projective measures were developed based on psychodynamic theory and the projective hypothesis that whatever a person does when exposed to an ambiguous stimulus will reveal important aspects of the person's personality. The goal of projective measures is to discover things that do not depend on self-knowledge and the person's ability or willingness to share it. Projective measures aim to bypass defense mechanisms so that they assess characteristics the person has and which the person is not consciously aware of. Examples of projective measures include the Rorschach Inkblot Test, the Thematic Apperception Test (TAT), Draw-A-Person Test, and drawing a family (Kinetic Family Drawing). However, there are many problems with projective measures: most projective techniques do **not** have standardized stimuli and testing instructions, systematic algorithms for scoring responses to stimuli, or well-calibrated norms for comparing responses with those of other people. Their scientific status is not strong in terms of utility of norms, reliability, influence of measurement error, or validity. In addition, they can be very time-consuming, and they provide very little information. Thus, there are many reasons not to use projective measures.

Suggested Readings

Lilienfeld et al., 2000

CHAPTER 15 / CAC.903

Clinical Judgment and Algorithmic Prediction

Clinical judgment is the counselor's reasoning about assessment data. Algorithmic prediction uses a formal rule or statistical model to combine data. Research often finds that simple algorithms outperform unstructured clinical judgment for many prediction tasks. This does not mean counselors are useless. It means that human judgment is vulnerable to bias and that structure can help.

Why judgment fails

Common errors include anchoring on first information, confirming a favorite hypothesis, ignoring base rates, overweighing vivid details, and being influenced by liking or disliking a client. A counselor may remember dramatic cases and overestimate risk. A counselor may see a well-dressed parent and underestimate neglect. These errors are human. The solution is not shame but structure.

Actuarial and structured professional judgment

Actuarial methods apply a fixed rule. Structured professional judgment uses a guide to ensure relevant factors are considered while allowing professional reasoning. In risk assessment, structured tools can improve consistency. The counselor should know the tool's population, outcome, and limits. A tool validated for adult violence should not be used casually with a young child.

Combining data

The counselor should combine data transparently. A report can state which sources support a conclusion and which conflict. A decision rule can be written before assessment: if certain patterns appear, recommend further evaluation. This reduces post-hoc storytelling. The counselor should also seek feedback on predictions to calibrate judgment.

Artificial intelligence and decision support

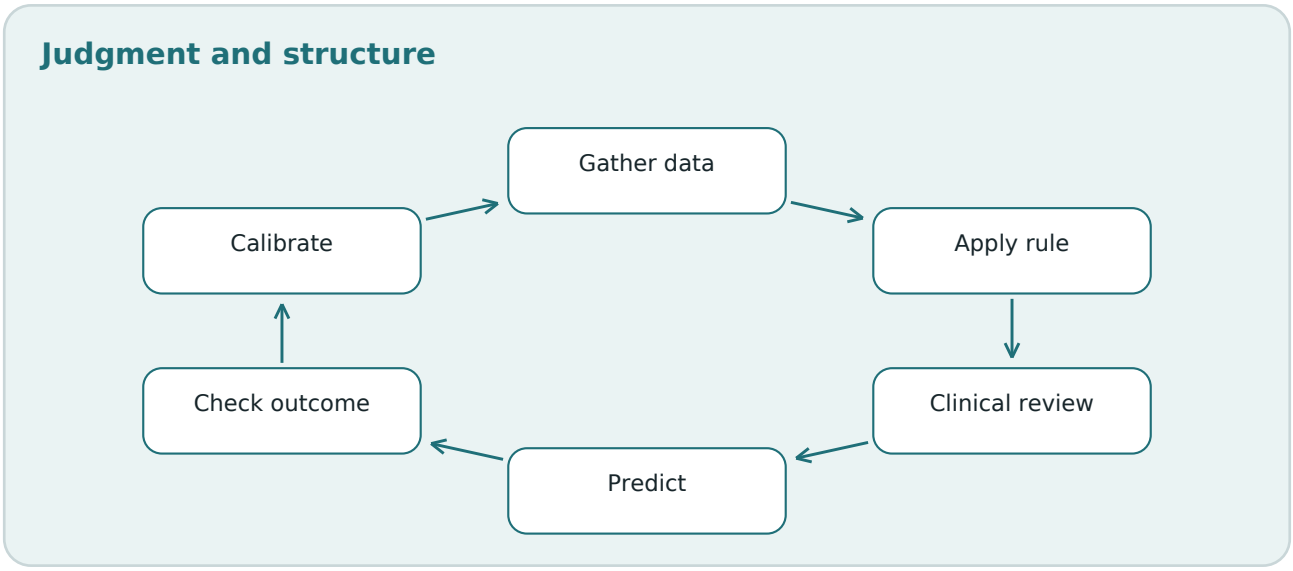
Digital tools and AI may assist scoring or pattern detection. They introduce risks: opaque algorithms, biased training data, privacy breaches, and overreliance. The counselor remains responsible. AI should not replace clinical interview, cultural formulation, or ethical judgment. Any tool should be validated for the population and purpose.

Real-life application

A counselor predicts that a teen will drop out based on a vivid interview. A structured review shows attendance, credits, support, and mental health. The algorithmic risk score is moderate, not high. The counselor adjusts the plan to include monitoring and support rather than a fatalistic label. Structure protects the teen from overprediction.

Reflection

Write a prediction you might make about a client. List the data you would use. Then write a simple decision rule. Ask how you would check whether the prediction was accurate. Clinical judgment improves when it can be corrected.



Structured prediction and feedback improve clinical judgment over time.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module clinical-actuarial-judgment.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Approaches to Prediction

There are two primary approaches to prediction: human (i.e., clinical) judgment and the actuarial (i.e., statistical) method.

Human/Clinical Judgment

Using the clinical judgment method of prediction, all gathered information is collected and formulated into a diagnosis or prediction in the clinician's mind. The clinician selects, measures, and combines risk factors and produces risk estimate solely according to clinical experience and judgment.

Actuarial/Statistical Method

In the actuarial or statistical method of prediction (i.e., statistical prediction rules), information is gathered and combined systematically in an evidence-based statistical prediction formula, and established cutoffs are used. The method is based on equations and data, so both are needed.

An example of a statistical method of prediction is the Violence Risk Appraisal Guide (Rice et al., 2013). The Violence Risk Appraisal Guide is used in an attempt to predict violence and is used for parole decisions. For instance, the equation might be something like the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Then, based on their score and the established cutoffs, a person is given a "low risk", "medium risk", or "high risk" designation.

Combining Human Judgment and Statistical Algorithms

There are numerous ways in which humans and statistical algorithms could be involved. On one extreme, humans make all judgments (consistent with the clinical judgment approach). On the other extreme, although humans may be involved in data collection, a statistical formula makes all decisions based on the input data, consistent with an actuarial approach. However, the clinical judgment and actuarial approaches can be combined in a hybrid way (Dana & Thomas, 2006). For example, to save time and money, a clinician might use an actuarial approach in all cases, but might only use a clinical approach when the actuarial approach gives a "positive" test. Or, the clinician might use both human judgment and an actuarial approach independently to see whether they agree. That is, the clinician may make a prediction based on their judgment and might also generate a prediction from an actuarial approach.

The challenge is what to do when the human and the algorithm disagree. Hypothetically, humans reviewing and adjusting the results from the statistical algorithm could lead to more accurate predictions. However, human input also could lead to the possibility or exacerbation of biased predictions. In general, with very few exceptions, actuarial approaches are as accurate or more accurate than clinical judgment (Dawes et al., 1989; óttir et al., 2006; Baird & Wagner, 2000; Grove & Meehl, 1996; Grove et al., 2000). Moreover, the superiority of actuarial approaches to clinical judgment tends to hold even when the clinician is given more information than the actuarial approach (Dawes et al., 1989). In addition, actuarial predictions outperform human judgment even when the human is given the result of the actuarial prediction (Kahneman, 2011). Allowing clinicians to override actuarial predictions consistently leads to lower predictive accuracy (Garb & Wood, 2019).

In general, validity tends to increase with greater structure (e.g., structured or semi-structured interviews as opposed to free-flowing, unstructured interviews), in terms of administration, responses, scoring, interpretation, etc. Unstructured interviews are susceptible to confirmatory bias, such as exploring only the diagnoses that confirm the clinician's hypotheses rather than attempting to explore diagnoses that disconfirm the clinician's hypotheses, which leads to fewer diagnoses. Unstructured interviews are also susceptible to bias relating to race, gender, class, etc. (Garb, 1997; Garb, 2005).

There is sometimes a misconception that formulas cannot account for qualitative information. However, that is not true. Qualitative information can be scored or coded to be quantified so that it can be included in statistical formulas. That said, the quality of predictions rests on the quality and relevance of the assessment information for the particular prediction/judgment decision. If the assessment data are lousy, it is unlikely that a

statistical algorithm (or a human for that matter) will make an accurate prediction: "Garbage in, garbage out". A statistical formula cannot rescue inaccurate assessment data.

Errors in Clinical Judgment

Clinical judgment is naturally subject to errors. Below, I describe a few errors to which clinical judgment seems particularly prone.

When operating freely, clinicians tend to over-estimate exceptions to the established rules (i.e., the broken leg syndrome). Meehl, 1957 acknowledged that there may be some situations where it is glaringly obvious that the statistical formula would be incorrect because it fails to account for an important factor. He called these special cases "broken leg" cases, in which the human should deviate from the formula (i.e., broken leg countervailing). The example goes like this:

"If a sociologist were predicting whether Professor X would go to the movies on a certain night, he might have an equation involving age, academic specialty, and introversion score. The equation might yield a probability of.90 that Professor X goes to the movie tonight. But if the family doctor announced that Professor X had just broken his leg, no sensible sociologist would stick with the equation. Why didn't the factor of 'broken leg' appear in the formula? Because broken legs are very rare, and in the sociologist's entire sample of 500 criterion cases plus 250 cross-validating cases, he did not come upon a single instance of it. He uses the broken leg datum confidently, because 'broken leg' is a subclass of a larger class we may crudely denote as 'relatively immobilizing illness or injury,' and movie-attending is a subclass of a larger class of 'actions requiring moderate mobility.' " (Meehl, 1957)

However, people too often think that cases where they disagree with the statistical algorithm are broken leg cases. People too often think their case is an exception to the rule. As a result, they too often change the result of the statistical algorithm and are more likely to be wrong than right in doing so. Because actuarial methods are based on actual population levels (i.e., base rates), unique exceptions are not over-estimated.

Actuarial predictions are perfectly reliable—they will always return the same conclusion given an identical set of data. The human judge is likely to both disagree with others and with themselves given the same set of symptoms.

A clinician's decision is likely to be influenced by past experiences, and given the sample of humanity that the clinician is exposed to, the clinician is likely (based on prior experience) to over-estimate the likelihood of occurrence of infrequent phenomena. Actuarial methods are based on objective algorithms, and past personal experience and personal biases do not factor into any decisions. Clinicians give weight to less relevant information, and often give too much weight to singular variables (e.g., Graduate Record Examination scores). Actuarial formulas do a better job of focusing on relevant variables. Computers are good at factoring in base rates, inverse conditional probabilities, etc. Humans ignore base rates (base rate neglect), and tend to show confusion of the inverse.

Computers are better at accurately weighing risk factors and calculating unbiased risk estimates. In an actuarial formula, the relevant risk factors are weighted according to their predictive power. This stands in contrast to the Diagnostic and Statistical Manual of Mental Disorders (DSM), in which each symptom is equally weighted.

Humans are typically given no feedback on their judgments. To improve accuracy of judgments, it is important for feedback to be clear, consistent, and timely. It is especially unlikely for feedback in clinical psychology to be timely, because we will have to wait a long time to see the outcomes of predictions. The feedback that clinicians receive regarding their clients' long-term outcomes tends to be sparse, delayed or nonexistent, and too ambiguous to support learning from experience (Kahneman, 2011). Moreover, it is important to distinguish short-term anticipation in a therapy session (e.g., what the client will say next) from long-terms predictions predictions. Although clinicians may have strong intuitions dealing with short-term anticipation in therapy sessions, their long-term predictions are not accurate (Kahneman, 2011).

Clinicians are susceptible to representative schema biases (Dawes, 1986). Clinicians are exposed to a skewed sample of humanity, and they make judgments based on a prototype from their (biased) experiences. This is known as the representativeness heuristic. Different clinicians may have different prototypes, leading to lower inter-rater reliability.

Intuition is a form of recognition-based judgment (i.e., recognizing cues that provide access to information in memory). Development of strong intuition depends on the quality and speed of feedback, in addition to having adequate opportunities to practice (Kahneman, 2011). The quality and speed of the feedback tend to benefit anesthesiologists who often quickly learn the results of their actions. By contrast, radiologists tend not to receive quality feedback about the accuracy of their diagnoses, including their false-positive and false-negative decisions (Kahneman, 2011).

In general, many so-called experts are "pseudo-experts" who do not know the boundaries of their competence—that is, they do not know what they do not know; they have the illusion of validity of their predictions and are overconfident about their predictions (Kahneman, 2011). Yet, many people arrogantly proclaim to have predictive powers, including in low-validity environments such as clinical psychology. True experts know their limits in terms of knowledge and ability to predict.

Intuitions tend to be skilled when a) the environment is regular and predictable, and b) there is opportunity to learn the regularities, cues, and contingencies through extensive practice (Kahneman, 2011). Example domains that meet these conditions supporting intuition include activities such as chess, bridge, and poker, and occupations such as medical providers, athletes, and firefighters. By contrast, clinical psychology and other domains such as stock-picking and other long-terms forecasts are low-validity environments that are irregular and unpredictable. In environments that do not have stable regularities, intuition cannot be trusted (Kahneman, 2011).

Advantages of Computers

Here are some advantages of computers over humans, including "experts":

• Computers can process lots of information simultaneously.

So can humans. But computers can to an even greater degree.

• Computers are faster at making calculations.

• Given the same input, a formula will give the exact same result everytime.

Humans' judgment tends to be inconsistent both across raters and within rater across time, when trying to make judgments or predictions from complex information (Kahneman, 2011). As noted in the source discussion, reliability sets the upper bound for validity, so unreliable judgments cannot be valid.

• Computations by computers are error-free (as long as the computations are programmed correctly).

• Computers' judgments will not be biased by fatigue or emotional responses.

• Computers' judgments will tend not to be biased in the way that humans' cognitive biases are, such as with anchoring bias, representativeness bias, confirmation bias, or recency bias.

Computers are less likely to be over-confident in their judgments.

• Computers can more accurately weight the set of predictors based on large data sets.

Humans tend to give too much weight to singular predictors. Experts may attempt to be clever and to consider complex combinations of predictors, but doing so often reduces validity (Kahneman, 2011). Simple combinations of predictions often outperform more complex combinations (Kahneman, 2011).

Advantages of Humans

Computers are bad at some things too. Here are some advantages of humans over computers (as of now):

• Humans can be better at identifying patterns in data (but also can mistakenly identify patterns where there are none).

• Humans can be flexible and take a different approach if a given approach is not working.

• Humans are better at tasks requiring creativity and imagination, such as developing theories that explain phenomena.

• Humans have the ability to reason, which is especially important when dealing with complex, abstract, or open-ended problems, or problems that have not been faced before (or for which we have insufficient data).

• Humans are better able to learn.

• Humans are better at holistic, gestalt processing, including facial and linguistic processing.

There may be situations in which a clinical judgment would do better than an actuarial judgment. One situation where clinical judgment would be important is when no actuarial method exists for the judgment or prediction. For instance, when no actuarial method exists for the diagnosis or disorder (e.g., suicide), it is up to the clinical judge. However, we could collect data on the outcomes or on clinicians' judgments to develop an actuarial method that will be more reliable than the clinicians' judgments. That is, an actuarial method developed based on clinicians' judgments will be more accurate than clinicians' judgments. In other words, we do not necessarily need clients' outcome data to develop an actuarial method. We could use the client's data as predictors of the clinicians' judgments to develop a structured approach to prediction that weighs factors similarly to clinicians, but with more reliable predictions.

Another situation in which clinical judgment could outperform a statistical algorithm is in true "broken leg" cases, e.g., important and rare events (edge cases) that are not yet accounted for by the algorithm.

Another situation in which clinical judgment could be preferable is if advanced, complex theories exist. Computers have a difficult time adhering to complex theories, so clinicians may be better suited. However, we do not have any of these complex theories in psychology that are accurate. We would need strong theory informed by data regarding causal influences, and accurate measures to assess them. If theories alone were true and could explain everything, the psychoanalytic tradition would give us all the answers, as depicted in the source discussion. However, no theories in psychology are that good. Nevertheless, predictive accuracy can be improved when considering theory (Silver, 2012; Garb & Wood, 2019).

Conceptual Depiction of the Psychoanalytic Tradition.

If the diagnosis/prediction requires complex configural relations that a computer will have a difficult time replicating, a clinician's judgment may be preferred. Although the likelihood that the clinician can accurately work through these complex relations is theoretically possible, it is highly unlikely. Holistic pattern recognition (such as language and faces) tends to be better by humans than computers. But computers are getting better with holistic pattern recognition through machine learning. An example of a simple decision (as opposed to a complex decision) would be whether to provide treatment A (e.g., cognitive behavior therapy) or treatment B (e.g., medication) for someone who shows melancholy.

In sum, the clinician seeks to integrate all information to make a decision, but is biased.

Comparison of Evidence

Hundreds of studies have examined clinical versus actuarial prediction methods across many disciplines. Findings consistently show that actuarial methods are as accurate or more accurate than clinical prediction methods (Dawes et al., 1989; óttir et al., 2006; Baird & Wagner, 2000; Grove & Meehl, 1996; Grove et al., 2000). "There is no controversy in social science that shows such a large body of qualitatively diverse studies coming out so uniformly…as this one" (Meehl, 1986). In general, actuarial methods tend to be about 10% more accurate than clinical judgment (Grove et al., 2000). Thus, actuarial methods have incremental validity over clinical judgment. As noted earlier, the superiority of actuarial approaches to clinical judgment tends to hold even when the clinician is given more information than the actuarial approach (Dawes et al., 1989). In addition, actuarial predictions outperform human judgment even when the human is given the result of the actuarial prediction (Kahneman, 2011). Allowing clinicians to override actuarial predictions consistently leads to lower predictive accuracy (Garb & Wood, 2019).

Actuarial methods are particularly valuable for criterion-referenced assessment tasks, in which the aim is to predict specific events or outcomes (Garb & Wood, 2019). For instance, actuarial methods have shown promise in predicting violence, criminal recidivism, psychosis onset, course of mental disorders, treatment selection, treatment failure, suicide attempts, and suicide (Garb & Wood, 2019). Actuarial methods are especially important to use in low-validity environments (like clinical psychology) in which there is considerable uncertainty and unpredictability (Kahneman, 2011). By contrast, for norm-referenced assessment tasks, different approaches may be preferred over actuarial methods and

statistical prediction rules. For instance, for norm-referenced assessment tasks such as describing personality and psychopathology (Garb & Wood, 2019), where the criterion is a latent construct (derived from the internal pattern of associations among the items) rather than a specific event or outcome, psychometric methods of scale construction such as factor analysis may be preferred (Garb & Wood, 2019)—i.e., the inductive approach to scale construction.

Another benefit of actuarial methods is that they are explicit; they can be transparent and lead to informed scientific criticism to improve them. By contrast, clinical judgment methods are not typically transparent; clinical judgment relies on mental processes that are often difficult to specify.

Accuracy of Different Statistical Models

It is important to evaluate the accuracy of different types of statistical models to provide guidance on which types of models may be most accurate for a given prediction problem. Simpler statistical formulas are more likely to generalize to new samples than complex statistical models due to model over-fitting. As described in the source discussion, over-fitting occurs when the statistical model accounts for error variance (an overly specific prediction), which will not generalize to future samples.

Youngstrom et al., 2018 discussed the benefits of the probability nomogram. The probability nomogram combines the risk ratio with the base rate (according to Bayes' theorem) to generate a prediction. The risk ratio (diagnostic likelihood ratio) is an index of the predictive validity of an instrument: it is the ratio of the probability that a test result is correct to the probability that the test result is incorrect. There are two types of diagnostic likelihood ratios: the positive likelihood ratio and the negative likelihood ratio. As described in the source discussion, the positive likelihood ratio is the probability that a person with the disease tested positive for the disease (true positive rate) divided by the probability that a person without the disease tested positive for the disease (false positive rate). As described in the source discussion, the negative likelihood ratio is the probability that a person with the disease tested negative for the disease (false negative rate) divided by the probability that a person without the disease tested negative for the disease (true negative rate). Using a probability nomogram, you start with the base rate, then plot the diagnostic likelihood ratio corresponding to a second source of information (e.g., positive test), then connect the base rate (pretest probability) to the posttest probability through the likelihood ratio. These approaches are described in greater detail in the source discussion.

Youngstrom et al., 2018 ordered actuarial approaches to diagnostic classification from less to more complex:

- Predicting from the base rate
- Take the best
- The probability nomogram
- Naïve Bayesian algorithms
- Logistic regression with one predictor
- Logistic regression with multiple predictors

• Least absolute shrinkage and selection option (LASSO)

As described in the source discussion, predicting from the base rate is selecting the most likely outcome in every case. "Take the best" refers to focusing on the single variable with the largest validity coefficient, and making a decision based on it (based on some threshold). The probability nomogram combines the base rate (prior probability) with the information offered by an assessment finding to revise the probability. Naïve Bayesian algorithms use the probability nomogram with multiple assessment findings. Basically, you continue to calculate a new posttest probability based on each assessment finding, revising the pretest probability for the next assessment finding based on the posttest probability of the last assessment finding. However, this approach assumes that all predictors are uncorrelated, which is probably not true in practice.

Logistic regression is useful for classification problems because it deals with a dichotomous dependent variable, but one could extrapolate it to a continuous dependent variable with multiple regression. The effect size from logistic regression is equivalent to receiver operating characteristic (ROC) curve analysis. Logistic regression with multiple predictors combines multiple predictors into one regression model. Including multiple predictors in the same model optimizes the weights of multiple predictors, but it is important for predictors not to be collinear.

LASSO is a form of regression that can handle multiple predictors (like multiple regression). However, it can handle more predictors than multiple regression. For instance, LASSO can accommodate situations where there are more predictors than there are participants. Moreover, unlike multiple regression, it can handle collinear predictors. LASSO uses internal cross-validation to avoid over-fitting.

Youngstrom et al., 2018 found that model complexity improves accuracy but only to a point; some of the simpler models (Naïve Bayesian and Logistic regression) did just as well and in some cases better than LASSO models in classifying bipolar disorder. Although LASSO models showed the highest discrimination accuracy in the internal sample, they showed the largest shrinkage in the external sample. Moreover, the LASSO models showed poor calibration in the internal sample; they over-diagnosed bipolar disorder. By contrast, simpler models showed better calibration in the internal sample. The probability nomogram with one or multiple assessment findings showed better calibration than LASSO models because they accounted for the original base rate, so they did not over-diagnose bipolar disorder.

In summary, the best models are those that are relatively simple (parsimonious), that can account for one or several of the most important predictors and their optimal weightings, and that account for the base rate of the phenomenon. Multiple regression and/or prior literature can be used to identify the weights of various predictors. Even unit-weighted formulas (formulas whose predictors are equally weighted with a weight of one) can sometimes generalize better to other samples than complex weightings (Garb & Wood, 2019; Wainer, 1976). Differential weightings sometimes capture random variance and over-fit the model, thus leading to predictive accuracy shrinkage in cross-validation samples (Garb & Wood, 2019), as described below. The choice of predictive variables often matters more than their weighting.

In general, there is often shrinkage of estimates from training data set to a test data set. Shrinkage is when variables with stronger predictive power in the original data set tend to

show somewhat smaller predictive power (smaller validity coefficients) when applied to new groups. Shrinkage reflects a model over-fitting (i.e., fitting to error by capitalizing on chance). Shrinkage is especially likely when the original sample is small and/or unrepresentative and the number of variables considered for inclusion is large. Cross-validation with large, representative samples can help evaluate the amount of shrinkage of estimates, particularly for more complex models such as machine learning models (Ursenbach et al., 2019). Ideally, cross-validation would be conducted with a separate sample (external cross-validation) to see the generalizability of estimates. However, you can also do internal cross-validation. For example, you can perform k-fold cross-validation, where you:

• split the data set into k groups

• for each unique group:

• take the group as a hold-out data set (also called a test data set)

• take the remaining groups as a training data set

• fit a model on the training data set and evaluate it on the test data set

• after all k-folds have been used as the test data set, and all models have been fit, you average the estimates across the models, which presumably yields more robust, generalizable estimates

An emerging technique that holds promise for increasing predictive accuracy of actuarial methods is machine learning (Garb & Wood, 2019). However, one challenge of some machine learning techniques is that they are like a "black box" and are not transparent, which raises ethical issues (Garb & Wood, 2019). Machine learning may be most valuable when the data available are complex and there are many predictive variables (Garb & Wood, 2019).

Load Data

aSAH is a data set from the pROC package (R-pROC, n.d.) that contains test scores (s100b) and clinical outcomes (outcome) for patients.

Simulate Data

For reproducibility, I set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Fitting the Statistical Models

You can find an array of statistical models for classification and prediction available in the caret package (R-caret, n.d.) at the following link: https://topepo.github.io/caret/available-models.html (archived at https://perma.cc/E45U-YE6S)

Dichotomous Outcome

Below is an example predicting a dichotomous outcome with one predictor, using the stats::glm function:

Ordinal Outcome

Below are examples predicting an ordinal outcome with one predictor, using the MASS::polr function of the MASS package (Venables & Ripley, 2002), the rms::lrm function of the rms package (R-rms, n.d.), and the caret::train function of the caret package (R-caret, n.d.):

Continuous Outcome

Below is an example predicting a continuous outcome with one predictor, using the stats::lm function:

Dichotomous Outcome

Below is an example predicting a dichotomous outcome with multiple predictors, using the stats::glm function:

Ordinal Outcome

Below are examples predicting an ordinal outcome with multiple predictors, using the MASS::polr function of the MASS package (Venables & Ripley, 2002), the rms::lrm function of the rms package (R-rms, n.d.), and the caret::train function of the caret package (R-caret, n.d.):

Continuous Outcome

Below is an example predicting a continuous outcome with multiple predictors, using the stats::lm function:

Least Absolute Shrinkage and Selection Option (LASSO)

The least absolute shrinkage and selection option (LASSO) models were fit in the caret package (R-caret, n.d.).

Dichotomous Outcome

Below is an example predicting a dichotomous outcome with LASSO, using the caret::train function of the caret package (R-caret, n.d.):

Continuous Outcome

Below is an example predicting a continuous outcome with LASSO, using the caret::train function of the caret package (R-caret, n.d.):

Ridge Regression

The ridge regression models were fit in the caret package (R-caret, n.d.).

Dichotomous Outcome

Below is an example predicting a dichotomous outcome with ridge regression, using the caret::train function of the caret package (R-caret, n.d.):

Continuous Outcome

Below is an example predicting a continuous outcome with ridge regression, using the caret::train function of the caret package (R-caret, n.d.):

Elastic Net

The elastic net models were fit in the caret package (R-caret, n.d.).

Below is an example predicting a continuous outcome with elastic net regression, using the caret::train function of the caret package (R-caret, n.d.):

Random Forest Machine Learning

The random forest models were fit in the caret package (R-caret, n.d.).

Dichotomous Outcome

Below is an example predicting a dichotomous outcome with random forest machine learning, using the caret::train function of the caret package (R-caret, n.d.):

Ordinal Outcome

Below is an example predicting an ordinal outcome with random forest machine learning, using the caret::train function of the caret package (R-caret, n.d.):

Continuous Outcome

Below is an example predicting a continuous outcome with random forest machine learning, using the caret::train function of the caret package (R-caret, n.d.):

k-Fold Cross-Validation

The k-fold cross-validation models were fit in the caret package (R-caret, n.d.). We specify the method of cross-validation using the the caret::trainControl function of the caret package (R-caret, n.d.).

Dichotomous Outcome

Below is an example predicting a dichotomous outcome with k-fold cross-validation, using the caret::train function of the caret package (R-caret, n.d.):

Continuous Outcome

Below is an example predicting a continuous outcome with k-fold cross-validation, using the caret::train function of the caret package (R-caret, n.d.):

Leave-One-Out (LOO) Cross-Validation

The leave-one-out (LOO) cross-validation models were fit in the caret package (R-caret, n.d.). We specify the method of cross-validation using the the caret::trainControl function of the caret package (R-caret, n.d.).

Dichotomous Outcome

Below is an example predicting a dichotomous outcome with LOO cross-validation, using the caret::train function of the caret package (R-caret, n.d.):

Continuous Outcome

Below is an example predicting a continuous outcome with LOO cross-validation, using the caret::train function of the caret package (R-caret, n.d.):

Examples of Using Actuarial Methods

There are many examples of actuarial methods used in research and practice. As just a few examples, actuarial methods have been used to predict violence risk (Rice et al., 2013), to predict treatment outcome (DeRubeis et al., 2014; Cohen & DeRubeis, 2018), and to screen for attention-deficit/hyperactivity disorder (Lindhiem et al., 2015). Machine learning is becoming more widely used as another form of approaches to actuarial methods (Dwyer et al., 2018). For an empirical example, we have examined actuarial methods to predict arrests, illegal drug use, and drunk driving (Petersen et al., 2015).

Use of Artificial Intelligence and Large Language Models

Emerging work uses large language models (LLMs) for real-time assessment (Aafjes-van Doorn et al., in press). There are two broad classes of approaches using LLMs for assessment. One class of approaches use LLMs to infer symptom severity or diagnostic likelihood from static inputs (Aafjes-van Doorn et al., in press). A second class of approaches use LLMs to engage clients in a real-time assessment interaction akin to an interview, allowing the LLM to clarify ambiguity, adapt questions, and generate individualized output (Aafjes-van Doorn et al., in press). LLMs tend to show stronger performance when they use structured (or semi-structured) interviewing logic, domain-specific adaptation, and clinically anchored reference standards (Aafjes-van Doorn et al., in press).

Why Clinical Judgment is More Widely Used Than Statistical Formulas

Despite actuarial methods being generally more accurate than clinical judgment, clinical judgment is much more widely used. There are several reasons why actuarial methods have not caught on; one reason is professional traditions. Experts in any field do not like to think that a computer could outperform them. Some practitioners argue that judgment/prediction is an "art form" and that using a statistical formula is treating people like a number. However, using an approach (i.e., clinical judgment) that systematically leads to less accurate decisions and predictions is an ethical problem.

Some clinicians do not think that group averages (e.g., in terms of which treatment is most effective) apply to an individual client. This invokes the distinction between nomothetic (group-level) inferences and idiographic (individual-level) inferences. However, the scientific evidence and probability theory strongly favor the nomothetic approach to clinical prediction—it is better to generalize from group-level evidence than throwing out all the evidence and taking the approach of "anything goes." Clinicians frequently believe the broken leg fallacy, i.e., thinking that your client is an exception to the algorithmic prediction. In most cases, deviating from the statistical formula will result in less accurate predictions. People tend to over-estimate the probability of low base rate conditions and events.

Another reason why actuarial methods have not caught on is the belief that receiving a treatment is the only thing that matters. But it is an empirical question which treatment is most effective for whom. What if we could do better? For example, we could potentially use a formula to identify the most effective treatment for a client. Some treatments are no better than placebo; other treatments are actually harmful (Lilienfeld, 2007; Williams et al., 2021).

Another reason why clinical methods are more widely used than actuarial methods is the over-confidence in clinicians' predictions—clinicians think they are more accurate than they

actually are. We see this when examining their calibration; their predictions tend to be mis-calibrated. For example, things they report with 80% confidence occur less than 80% of the time. Clinicians will sometimes be correct by chance, and they tend to mis-attribute that to their assessment; clinicians tend to remember the successes and forget the failures. Note, however, that it is not just clinicians who are over-confident; humans in general tend to be over-confident in their predictions.

Another argument against using actuarial methods is that "no methods exist". In some cases, that is true—actuarial methods do not yet exist for some prediction problems. However, one can always create an algorithm of the clinicians' judgments, even if one does not have access to the clients' outcome information. A model of clinicians' responses tends to be more accurate than clinicians' judgments themselves because the model gives the same outcome with the same input data—i.e., it is perfectly reliable.

Another argument from some clinicians is that, "My job is to understand, not to predict". But what kind of understanding does not involve predictions? Accurate predictions help in understanding. Knowing how people would perform in different conditions is the same thing as good understanding.

Another potential reason clinicians may not use actuarial judgment is time. This can be a legitimate challenge to using actuarial models in some contexts. Actuarial methods can take time to apply—they require assessing the relevant criteria, putting the assessment scores into an actuarial formula, and interpreting the output score. In some clinical situations, such as risk assessments in terms of whether someone is imminently suicidal or homicidal and how to proceed, clinicians must make immediate in-session decisions (Jacinto et al., 2018). Thus, it can be challenging or impractical in some situations to conduct all of the relevant assessments needed to apply actuarial approaches. Moreover, in some workplace settings, such as health care settings, clinicians may be pushed beyond their limits in terms of productivity expectations. In these situations, one possible approach is to exercise one's best possible judgment in the moment based on the available evidence (Jacinto et al., 2018), and to check it against an actuarial model after the fact. This is like comparing your decisions in a chess match, after the game ends, to how a computer would have moved in each of the situations. It is valuable to receive feedback on the accuracy of one's judgments and, when time permits, to evaluate one's intuitive decision-making against analytical decision-making. However, there are structured approaches to assessing suicide risk that are quick to administer (Posner et al., 2011).

Clinical psychology is not the only domain in which there has been pushback against the use of computers and statistical/actuarial formulas. For a long time, many people in sports (e.g., coaches, general managers, scounts) were hesitant to use analytics, for many similar (flawed) reasons. However, after considerable pushback, analytics is increasingly being used in many sports—likely because it works (Petersen, 2026). Indeed, many changes in sports have occurred due to the use of analytics. As described in PetersenFantasyFootballText:, n.d.

[In baseball], in addition to teams using Sabermetrics (e.g., on-base percentage) to evaluate player talent, teams also began to frequently use statistical analysis to inform decision making during games, which led to key changes in the style of play. For instance, defensive shifts—where defensive players moved to locations on the field where particular hitters were most likely to hit the ball—became more common, attempts to steal bases became less

common, there were fewer bunts, batters took more pitches (i.e., watched more pitches without swinging), there were more frequent pitching changes (for particular pitcher–batter matchups; such as to have a right-handed pitcher face a left-handed batter or vice versa), and a greater focus on velocity and spin rate among pitchers. Some of these analytics-driven changes in play style eventually led Major League Baseball (MLB) to make rule changes in an attempt to make the game more exciting to watch, including banning defensive shifts, reducing the number of pitching changes allowed, and making the bases larger and easier to steal... in basketball, there has been a greater focus on three points and buckets close to the rim (e.g., dunks and layups), with way fewer midrange shots (Partnow, 2021). Moreover, star players rest more games ("load management")... Examples of changes in football due to the use of analytics include more often going for it on fourth down, a greater emphasis on the passing game, drafting Running Backs later in the draft (and, more generally, valuing Running Backs less), and trading down in the draft to obtain more low picks (rather than having fewer high picks) because top picks are frequently overvalued (Massey & Thaler, 2013).

Thus, despite the pushback in clinical psychology, there holds promise for implementing statistical/actuarial formulas.

Conclusion

In general, it is better to develop and use structured, actuarial approaches than informal approaches that rely on human or clinical judgment. Actuarial approaches to prediction tend to be as accurate or more accurate than clinical judgment. Nevertheless, clinical judgment tends to be much more widely used than actuarial approaches, which is a major ethical problem.

Suggested Readings

Dawes et al., 1989; Garb & Wood, 2019; Grove et al., 2000

CHAPTER 16 / CAC.903

Prediction, Base Rates, and Decision Errors

Assessment often aims to predict: Will a child benefit from this intervention? Is a teen at risk of self-harm? Will a student succeed in a program? Prediction is difficult because behavior is multiply determined and base rates matter. A tool that seems accurate can perform poorly when the outcome is rare. Counselors need to understand sensitivity, specificity, predictive values, and decision errors.

Sensitivity and specificity

Sensitivity is the proportion of true cases correctly identified. Specificity is the proportion of non-cases correctly identified. A highly sensitive screen catches many cases but may produce false positives. A highly specific tool rules in cases but may miss some. The choice depends on consequences. In suicide risk, missing a true case may be costly, so sensitivity matters. But false positives can lead to unnecessary hospitalization or stigma. Balance is required.

Positive and negative predictive value

Predictive values depend on base rate. Positive predictive value is the probability that a positive screen is a true case. Negative predictive value is the probability that a negative screen is a true non-case. When a condition is rare, even a good test may have low positive predictive value. A counselor should not interpret a positive screen as certainty. Follow-up assessment is needed.

Confusion matrix

A simple table helps: true positives, false positives, true negatives, false negatives. Each cell has a human meaning. A false negative may mean a child does not receive help. A false positive may mean a child is labeled unnecessarily. The counselor should consider which error is more harmful in the context and design a pathway that reduces harm. A two-stage assessment can use a sensitive screen followed by a specific evaluation.

Regression to the mean

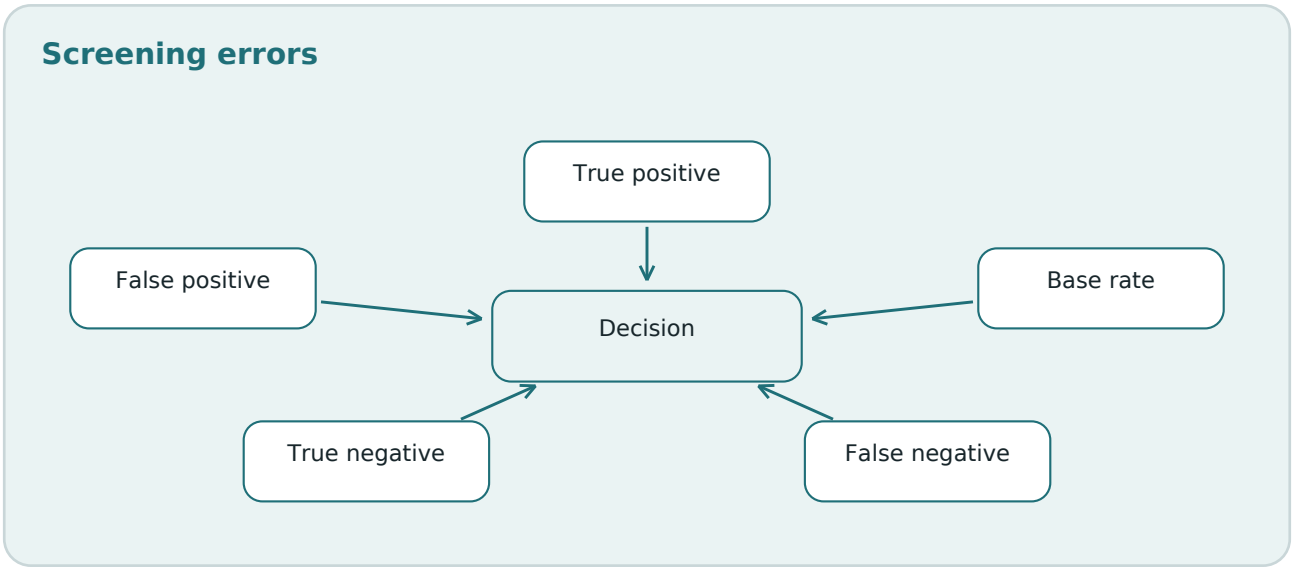
Extreme scores tend to move toward the average on retesting. A child selected for very high symptoms may improve partly due to regression, not treatment. The counselor should use reliable change indices, multiple baselines, or comparison data when evaluating outcomes. Otherwise, assessment may credit or blame interventions incorrectly.

Real-life application

A school screens all students for depression. Many positive screens appear. The counselor does not diagnose all. Instead, the counselor conducts brief follow-up interviews, assesses safety, and prioritizes students with impairment. The pathway respects base rates and avoids overloading services. Prediction becomes a process, not a single score.

Reflection

Choose a screening decision. Write the possible errors and their consequences. Decide whether sensitivity or specificity should be prioritized. Then design a follow-up step. Good prediction includes a plan for being wrong.



Every screening decision has possible errors. Base rates shape predictive values.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module prediction.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Prediction

"It is very difficult to predict—especially the future." --- Neils Bohr

Overview of Prediction

In psychology, we are often interested in predicting behavior. Behavior is complex. Behavior is multiply determined—it is probabilistically influenced by many processes, including processes internal to the person in addition to external processes. In addition, the same behavior can occur for different reasons—that is, behavior can be considered overdetermined. Moreover, people's behavior occurs in the context of a dynamic system with nonlinear, probabilistic, and cascading influences that change across time. The ever-changing system makes behavior challenging to predict. And, similar to chaos theory (or the butterfly effect), one small change in the system can lead to large differences later on. Another example of a chaotic system is the atmosphere and resulting weather (archived at https://perma.cc/248N-9CW5).

When something is overdetermined, it has multiple causes, any of which could be sufficient to result in the outcome. In other words, the same behavior can occur for different reasons, consistent with the principle of developmental psychopathology known as equifinality—i.e., multiple causes can result in the same outcome. For instance, aggression could be caused by either excessively low anxiety for some children or excessively high anxiety for others; or for other children, it could be caused by deficient social information processing or another cognitive process. Multifinality refers to the idea that the same cause can result in different outcomes. For instance, substance use might lead to aggressive behavior for some people, whereas it might lead to depressive behavior for others. In contrast to overdetermination, underdetermination refers to the idea that the identified causes are insufficient for explaining the resulting outcome. All of these challenges make behavior difficult to predict.

Predictions can come in different types. Some predictions involve categorical data, whereas other predictions involve continuous data. When dealing with categorical data, we can evaluate predictions using a 2x2 table known as a confusion matrix (see the source discussion), or with logistic regression models. When dealing with continuous data, we can evaluate predictions using multiple regression or similar variants such as structural equation modeling and mixed models.

Let's consider a prediction example, assuming the following probabilities:

• The probability of contracting HIV is.3%

• The probability of a positive test for HIV is 1%

• The probability of a positive test if you have HIV is 95%

What is the probability of HIV if you have a positive test?

As we will see, the probability is: (95\% ×.3\%)/(1\%) = 28.5\%. So based on the above probabilities, if you have a positive test, the probability that you have HIV is 28.5%. Most people tend to vastly over-estimate the likelihood that the person has HIV in this example. Why? Because they do not pay enough attention to the base rate (in this example, the base rate of HIV is.3%).

Types of Probabilities

It is important to distinguish between different types of probabilities: marginal probabilities, joint probabilities, and conditional probabilities.

Base Rate (Marginal Probability) A base rate is the probability of an event. Base rates are marginal probabilities. A marginal probability is the probability of an event irrespective of the outcome of another variable. For instance, we can consider the following marginal probabilities:

P(C i) is the probability (i.e., base rate) of a classification, C, independent of other things. A base rate is often used as the "prior probability" in a Bayesian model. In our example above, P(C i) is the base rate (i.e., prevalence) of HIV in the population: P(HIV) =.3\%. P(R i) is the probability (base rate) of a response, R, independent of other things. In the example above, P(R i) is the base rate of a positive test for HIV: P(positive test) = 1\%. The base rate of a positive test is known as the positivity rate or selection ratio.

Joint Probability A joint probability is the probability of two (or more) events occurring simultaneously. For instance, the probability of events A and B both occurring together is P(A, B).

If the events depend on each other, rearranging the terms for the calculation of conditional probability, their joint probability can be calculated using the conditional probability and marginal probability, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

If the events are independent, the formula simplifies down, and their joint probability can be calculated by multiplying the marginal probability of each event, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Conditional Probability A conditional probability is the probability of one event occurring given the occurrence of another event. Conditional probabilities are written as: P(A | B). This is read as the probability that event A occurs given that event B occurred. For instance, we can consider the following conditional probabilities:

P(C | R) is the probability of a classification, C, given a response, R. In other words, P(C | R) is the probability of having HIV given a positive test: P(HIV | positive test). P(R | C) is the probability of a response, R, given a classification, C. In the example above, P(R | C) is the probability of having a positive test given that a person has HIV: P(positive test | HIV) = 95\%.

A conditional probability can be calculated using the joint probability and marginal probability (base rate), as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Confusion of the Inverse

A conditional probability is not the same thing as its reverse (or inverse) conditional probability. Unless the base rate of the two events (C and R) are the same, P(C | R) ≠ P(R | C). However, people frequently make the mistake of thinking that two inverse conditional probabilities are the same. This mistake is known as the "confusion of the inverse", or the "inverse fallacy", or the "conditional probability fallacy". The confusion of inverse probabilities is the logical error of representative thinking that leads people to assume that the probability of C given R is the same as the probability of R given C, even though this is not true. As a few examples to demonstrate the logical fallacy, if 93% of breast cancers occur in high-risk women, this does not mean that 93% of high-risk women will eventually get breast cancer. As another example, if 77% of car accidents take place within 15 miles of a driver's home, this does not mean that you will get in an accident 77% of times you drive within 15 miles of your home.

Which car is the most frequently stolen? It is often the Honda Accord or Honda Civic—probably because they are among the most popular/commonly available cars. The probability that the car is a Honda Accord given that a car was stolen (p(Honda Accord | Stolen)) is what the media reports and what the police care about. However, that is not what buyers and car insurance companies should care about. Instead, they care about the probability that the car will be stolen given that it is a Honda Accord (p(Stolen | Honda Accord)).

Bayes' Theorem

An alternative way of calculating a conditional probability is using the inverse conditional probability (instead of the joint probability). This is known as Bayes' theorem. Bayes' theorem can help us calculate a conditional probability of some classification, C, given some response, R, if we know the inverse conditional probability and the base rate (marginal probability) of each. Bayes' theorem is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Or, equivalently (rearranging the terms):

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Or, equivalently (rearranging the terms):

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

More generally, Bayes' theorem has been described as:

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where H is the hypothesis, and E is the evidence—the new information that was not used in computing the prior probability.

In Bayesian terms, the posterior probability is the conditional probability of one event occurring given another event—it is the updated probability after the evidence is

considered. In this case, the posterior probability is the probability of the classification occurring (C) given the response (R). The likelihood is the inverse conditional probability—the probability of the response (R) occurring given the classification (C). The prior probability is the marginal probability of the event (i.e., the classification) occurring, before we take into account any new information. The model evidence is the marginal probability of the other event occurring—i.e., the marginal probability of seeing the evidence.

In the HIV example above, we can calculate the conditional probability of HIV given a positive test using three terms: the conditional probability of a positive test given HIV (i.e., the sensitivity of the test), the base rate of HIV, and the base rate of a positive test for HIV. The conditional probability of HIV given a positive test is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The petersenlab package (Petersen, 2025) contains the petersenlab::pAgivenB function that estimates the probability of one event, A, given another event, B.

Thus, assuming the probabilities in the example above, the conditional probability of having HIV if a person has a positive test is 28.5%. Given a positive test, chances are higher than not that the person does not have HIV.

Bayes' theorem can be depicted visually (Ballesteros-Pérez et al., 2018). If we have 100,000 people in our population, we would be able to fill out a 2-by-2 confusion matrix, as depicted in the source discussion.

Confusion Matrix: 2x2 Prediction Matrix. TP = true positives; TN = true negatives; FP = false positives; FN = false negatives; BR = base rate; SR = selection ratio.

We know that.3% of the population contracts HIV, so 300 people in the population of 100,000 would contract HIV. Therefore, we put 300 in the marginal sum of those with HIV (.003 × 100,000 = 300), i.e., the base rate of HIV. That means 99,700 people do not contract HIV (100,000 - 300 = 99,700). We know that 1% of the population tests positive for HIV, so we put 1,000 in the marginal sum of those who test positive.01 × 100,000 = 1,000, i.e., the marginal probability of a positive test (the selection ratio). That means 99,000 people test negative for HIV (100,000 - 1,000 = 99,000). We also know that 95% of those who have HIV test positive for HIV. Three hundred people have HIV, so 95% of them (i.e., 285 people;.95 × 300 = 285) tested positive for HIV (true positives). Because we know that 300 people have HIV and that 285 of those with HIV tested positive, that means that 15 people with HIV tested negative (300 - 15 = 285; false negatives). We know that 1,000 people tested positive for HIV, and 285 with HIV tested positive, so that means that 715 people without HIV tested positive (1,000 - 285 = 715; false positives). We know that 99,000 people tested negative for HIV, and 15 with HIV tested negative, so that means that 98,985 people without HIV tested negative (99,000 - 15 = 98,985; true negatives). So, to answer the question of what is the probability of having HIV if you have a positive test, we divide the number of people with HIV who had a positive test (285) by the total number of people who had a positive test (1000), which leads to a probability of 28.5%.

This can be depicted visually in Figures (the source discussion) and (the source discussion).^[Please note that the areas in the figure are not drawn to scale; otherwise, some regions would be too small to include text.]

Bayes' Theorem (and Confusion Matrix) Depicted Visually, Where the Marginal Probability is the Base Rate (BR). The four boxes represent the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Note: Boxes are not drawn to scale; otherwise, some regions would be too small to include text.

Bayes' Theorem (and Confusion Matrix) Depicted Visually, where the Marginal Probability is the Selection Ratio (SR). The four boxes represent the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Note: Boxes are not drawn to scale; otherwise, some regions would be too small to include text.

Now let's see what happens if the person tests positive a second time. We would revise our "prior probability" for HIV from the general prevalence in the population (0.3%) to be the "posterior probability" of HIV given a first positive test (28.5%). This is known as Bayesian updating. We would also update the "evidence" to be the marginal probability of getting a second positive test.

If we do not know a marginal probability (i.e., base rate) of an event (e.g., getting a second positive test), we can calculate a marginal probability with the law of total probability using conditional probabilities and the marginal probability of another event (e.g., having HIV). According to the law of total probability, the probability of getting a positive test is the probability that a person with HIV gets a positive test (i.e., sensitivity) times the base rate of HIV plus the probability that a person without HIV gets a positive test (i.e., false positive rate) times the base rate of not having HIV, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

In this case, we know the marginal probability (P(R i)), and we can use that to solve for the unknown conditional probability that reflects the false positive rate (P(R | not C)), as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

We can then estimate the marginal probability of the event, substititing in P(R | not C), using the law of total probability. The petersenlab package (Petersen, 2025) contains the petersenlab::pA function that estimates the marginal probability of one event, A.

The petersenlab package (Petersen, 2025) contains the petersenlab::pBgivenNotA function that estimates the probability of one event, B, given that another event, A, did not occur.

With this conditional probability (P(R | not C)), the updated marginal probability of having HIV (P(C i)), and the updated marginal probability of not having HIV (P(not C i)), we can now calculate an updated estimate of the marginal probability of getting a second positive test. The probability of getting a second positive test is the probability that a person with HIV gets a second positive test (i.e., sensitivity) times the updated probability of HIV plus the probability that a person without HIV gets a second positive test (i.e., false positive rate) times the updated probability of not having HIV, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The petersenlab package (Petersen, 2025) contains the petersenlab::pB function that estimates the marginal probability of one event, B.

We then substitute the updated marginal probability of HIV (P(C i)) and the updated marginal probability of getting a second positive test (P(R i)) into Bayes' theorem to get the probability that the person has HIV if they have a second positive test (assuming the errors of each test are independent, i.e., uncorrelated), as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The petersenlab package (Petersen, 2025) contains the petersenlab::pAgivenB function that estimates the probability of one event, A, given another event, B.

Thus, a second positive test greatly increases the posterior probability that the person has HIV from 28.5% to over 98%.

As seen in the rearranged formula in the source discussion, the ratio of the conditional probabilities is equal to the ratio of the base rates. Thus, it is important to consider base rates. People have a strong tendency to ignore (or give insufficient weight to) base rates when making predictions. The failure to consider the base rate when making predictions when given specific information about a case is a cognitive bias known as the base-rate fallacy or as base rate neglect. For example, people tend to say that the probability of a rare event is more likely than it actually is given specific information.

As seen in the rearranged formula in the source discussion, the inverse conditional probabilities (P(C | R) and P(R | C)) are not equal unless the base rates of C and R are the same. If the base rates are not equal, we are making at least some prediction errors. If P(C i) greater than P(R i), our predictions must include some false negatives. If P(R i) greater than P(C i), our predictions must include some false positives.

Using the law of total probability, we can substitute the calculation of the marginal probability (P(R i)) into Bayes' theorem to get an alternative formulation of Bayes' theorem, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Instead of using marginal probability (base rate) of R, as in the original formulation of Bayes' theorem, it uses the conditional probability, P(R|not C). Thus, it uses three terms: two conditional probabilities—P(R|C) and P(R|not C)—and one marginal probability, P(C i). This alternative formulation of Bayes' theorem can be used to calculate positive predictive value, based on sensitivity, specificity, and the base rate, as presented in the source discussion.

Let us see how the alternative formulation of Bayes' theorem applies to the HIV example above. We can calculate the probability of HIV given a positive test using three terms: the conditional probability that a person with HIV gets a positive test (i.e., sensitivity), the conditional probability that a person without HIV gets a positive test (i.e., false positive rate), and the base rate of HIV. Using the P(R|not C) calculated in the source discussion, the conditional probability of HIV given a single positive test is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The petersenlab package (Petersen, 2025) contains the petersenlab::pAgivenB function that estimates the probability of one event, A, given another event, B.

To calculate the conditional probability of HIV given a second positive test, we update our priors because the person has now tested positive for HIV. We update the prior probability of HIV (P(C i)) based on the posterior probability of HIV after a positive test (P(C | R)) that we calculated above. We can calculate the conditional probability of HIV given a second positive test using three terms: the conditional probability that a person with HIV gets a positive test (i.e., sensitivity; which stays the same), the conditional probability that a person without HIV gets a positive test (i.e., false positive rate; which stays the same), and the updated marginal probability of HIV. The conditional probability of HIV given a second positive test is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The petersenlab package (Petersen, 2025) contains the petersenlab::pAgivenB function that estimates the probability of one event, A, given another event, B.

If we want to compare the relative probability of two outcomes, we can use the odds form of Bayes' theorem, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

In sum, the marginal probability, including the prior probability or base rate, should be weighed heavily in predictions unless there are sufficient data to indicate otherwise, i.e., to update the posterior probability based on new evidence. Bayes' theorem provides a powerful tool to anchor predictions to the base rate unless sufficient evidence changes the posterior probability (by updating the evidence and prior probability).

Decision Outcomes

To consider how we can evaluate the accuracy of predictions, consider an example adapted from Meehl & Rosen, 1955. The military conducts a test of its prospective members to screen out applicants who would likely fail basic training. To evaluate the accuracy of our predictions using the test, we can examine a confusion matrix. A confusion matrix is a matrix that presents the predicted outcome on one dimension and the actual outcome (truth) on the other dimension. If the predictions and outcomes are dichotomous, the confusion matrix is a 2x2 matrix with two rows and two columns that represent four possible predicted-actual combinations (decision outcomes): true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

When discussing the four decision outcomes, "true" means an accurate judgment, whereas "false" means an inaccurate judgment. "Positive" means that the judgment was that the person has the characteristic of interest, whereas "negative" means that the judgment was that the person does not have the characteristic of interest. A true positive is a correct judgment (or prediction) where the judgment was that the person has (or will have) the characteristic of interest, and, in truth, they actually have (or will have) the characteristic. A true negative is a correct judgment (or prediction) where the judgment was that the person

does not have (or will not have) the characteristic of interest, and, in truth, they actually do not have (or will not have) the characteristic. A false positive is an incorrect judgment (or prediction) where the judgment was that the person has (or will have) the characteristic of interest, and, in truth, they actually do not have (or will not have) the characteristic. A false negative is an incorrect judgment (or prediction) where the judgment was that the person does not have (or will not have) the characteristic of interest, and, in truth, they actually do have (or will have) the characteristic.

An example of a confusion matrix is in the source discussion.

Confusion Matrix: 2x2 Prediction Matrix. TP = true positives; TN = true negatives; FP = false positives; FN = false negatives; BR = base rate; SR = selection ratio.

With the information in the confusion matrix, we can calculate the marginal sums and the proportion of people in each cell (in parentheses), as depicted in the source discussion.

Confusion Matrix: 2x2 Prediction Matrix With Marginal Sums. TP = true positives; TN = true negatives; FP = false positives; FN = false negatives.

That is, we can sum across the rows and columns to identify how many people actually showed poor adjustment (n = 100) versus good adjustment (n = 1,900), and how many people were selected to reject (n = 508) versus retain (n = 1,492). If we sum the column of predicted marginal sums (508 + 1,492) or the row of actual marginal sums (100 + 1,900), we get the total number of people (N = 2,000).

Based on the marginal sums, we can compute the marginal probabilities, as depicted in the source discussion.

Confusion Matrix: 2x2 Prediction Matrix With Marginal Sums And Marginal Probabilities. TP = true positives; TN = true negatives; FP = false positives; FN = false negatives; BR = base rate; SR = selection ratio.

The marginal probability of the person having the characteristic of interest (i.e., showing poor adjustment) is called the base rate (BR). That is, the base rate is the proportion of people who have the characteristic. It is calculated by dividing the number of people with poor adjustment (n = 100) by the total number of people (N = 2,000): BR = (FN + TP)/(N). Here, the base rate reflects the prevalence of poor adjustment. In this case, the base rate is.05, so there is a 5% chance that an applicant will be poorly adjusted. The marginal probability of good adjustment is equal to 1 minus the base rate of poor adjustment.

The marginal probability of predicting that a person has the characteristic (i.e., rejecting a person) is called the selection ratio (SR). The selection ratio is the proportion of people who will be selected (in this case, rejected rather than retained); i.e., the proportion of people who are identified as having the characteristic. The selection ratio is calculated by dividing the number of people selected to reject (n = 508) by the total number of people (N = 2,000): SR = (TP + FP)/(N). In this case, the selection ratio is.25, so 25% of people are rejected. The marginal probability of not selecting someone to reject (i.e., the marginal probability of retaining) is equal to 1 minus the selection ratio.

The selection ratio might be something that the test dictates according to its cutoff score. Or, the selection ratio might be imposed by external factors that place limits on how many people you can assign a positive test value. For instance, when deciding whether to treat a

client, the selection ratio may depend on how many therapists are available and how many cases can be treated.

Percent Accuracy

Based on the confusion matrix, we can calculate the prediction accuracy based on the percent accuracy of the predictions. The percent accuracy is the number of correct predictions divided by the total number of predictions, and multiplied by 100. In the context of a confusion matrix, this is calculated as: 100\% × TP + TNN. In this case, our percent accuracy was 78%—that is, 78% of our predictions were accurate, and 22% of our predictions were inaccurate.

Percent Accuracy by Chance

78% sounds pretty accurate. And it is much higher than 50%, so we are doing a pretty good job, right? Well, it is important to compare our accuracy to what accuracy we would expect to get by chance alone, if predictions were made by a random process rather than using a test's scores. Our selection ratio was 25.4%. How accurate would we be if we randomly selected 25.4% of people to reject? To determine what accuracy we could get by chance alone given the selection ratio and the base rate, we can calculate the chance probability of true positives and the chance probability of true negatives. The probability of a given cell in the confusion matrix is a joint probability—the probability of two events occurring simultaneously. To calculate a joint probability, we multiply the probability of each event.

So, to get the chance expectancies of true positives, we would multiply the respective marginal probabilities, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

To get the chance expectancies of true negatives, we would multiply the respective marginal probabilities, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

To get the percent accuracy by chance, we sum the chance expectancies for the correct predictions (TP and TN):.0127 +.7087 =.7214. Thus, the percent accuracy you can get by chance alone is 72%. This is because most of our predictions are to retain people, and the base rate of poor adjustment is quite low (.05). Our measure with 78% accuracy provides only a 6% increment in correct predictions. Thus, you cannot judge how good your judgment or prediction is until you know how you would do by random chance.

The chance expectancies for each cell of the confusion matrix are in the source discussion.

Chance Expectancies in 2x2 Prediction Matrix. TP = true positives; TN = true negatives; FP = false positives; FN = false negatives; BR = base rate; SR = selection ratio.

Predicting from the Base Rate

Now, let us consider how well you would do if you were to predict from the base rate. Predicting from the base rate is also called "betting from the base rate", and it involves setting the selection ratio by taking advantage of the base rate so that you go with the most likely outcome in every prediction. Because the base rate is quite low (.05), we could predict from the base rate by selecting no one to reject (i.e., setting the selection ratio at zero). Our percent accuracy by chance if we predict from the base rate would be calculated by multiplying the marginal probabilities, as we did above, but with a new selection ratio, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

We sum the chance expectancies for the correct predictions (TP and TN): 0 +.95 =.95. Thus, our percent accuracy by predicting from the base rate is 95%. This is damning to our measure because it is a much higher accuracy than the accuracy of our measure. That is, we can be much more accurate than our measure simply by predicting from the base rate and selecting no one to reject.

Going with the most likely outcome in every prediction (predicting from the base rate) can be highly accurate (in terms of percent accuracy) as noted by Meehl & Rosen, 1955, especially when the base rate is very low or very high. This should serve as an important reminder that we need to compare the accuracy of our measures to the accuracy by (1) random chance and (2) predicting from the base rate. There are several important implications of the impact of base rates on prediction accuracy. One implication is that using the same test in different settings with different base rates will markedly change the accuracy of the test. Oftentimes, using a test will actually decrease the predictive accuracy when the base rate deviates greatly from.50. But percent accuracy is not everything. Percent accuracy treats different kinds of errors as if they are equally important. However, the value we place on different kinds of errors may be different, as described next.

Different Kinds of Errors Have Different Costs

Some errors have a high cost, and some errors have a low cost. Among the four decision outcomes, there are two types of errors: false positives and false negatives. The extent to which false positives and false negatives are costly depends on the prediction problem. So, even though you can often be most accurate by going with the base rate, it may be advantageous to use a screening instrument despite lower overall accuracy because of the huge difference in costs of false positives versus false negatives in some cases.

Consider the example of a screening instrument for HIV. False positives would be cases where we said that someone is at high risk of HIV when they are not, whereas false negatives are cases where we said that someone is not at high risk when they actually are. The costs of false positives include a shortage of blood, some follow-up testing, and potentially some anxiety, but that is about it. The costs of false negatives may be people getting HIV. In this case, the costs of false negatives greatly outweigh the costs of false positives, so we use a screening instrument to try to identify the cases at high risk for HIV because of the important consequences of failing to do so, even though using the screening instrument will lower our overall accuracy level.

Another example is when the Central Intelligence Agency (CIA) used a screen for protective typists during wartime to try to detect spies. False positives would be cases where the CIA believes that a person is a spy when they are not, and the CIA does not hire them. False negatives would be cases where the CIA believes that a person is not a spy when they actually are, and the CIA hires them. In this case, a false positive would be fine, but a false negative would be really bad.

How you weigh the costs of different errors depends considerably on the domain and context. Possible costs of false positives to society include: unnecessary and costly treatment with side effects and sending an innocent person to jail (despite our presumption of innocence in the United States criminal justice system that a person is innocent until proven guilty). Possible costs of false negatives to society include: setting a guilty person free, failing to detect a bomb or tumor, and preventing someone from getting treatment who needs it.

The differential costs of different errors also depend on how much flexibility you have in the selection ratio in being able to set a stringent versus loose selection ratio. Consider if there is a high cost of getting rid of people during the selection process. For example, if you must hire 100 people and only 100 people apply for the position, you cannot lose people, so you need to hire even high-risk people. However, if you do not need to hire many people, then you can hire more conservatively.

Any time the selection ratio differs from the base rate, you will make errors. For example, if you reject 25% of applicants, and the base rate of poor adjustment is 5%, then you are making errors of over-rejecting (false positives). By contrast, if you reject 1% of applicants and the base rate of poor adjustment is 5%, then you are making errors of under-rejecting or over-accepting (false negatives).

A low base rate makes it harder to make predictions, and tends to lead to less accurate predictions. For instance, it is very challenging to predict low base rate behaviors, including suicide (Kessler et al., 2020). The difficulty in predicting events with a low base rate is apparent with the true score formula from classical test theory: $X = T + e$. As described in the source discussion, reliability is the ratio of true score variance to observed score variance. As true score variance increases, reliability increases. If the base rate is.05, the maximum variance of the true scores is.05. The lower true score variance makes the measure less reliable and hard to make accurate predictions.

Sensitivity, Specificity, PPV, and NPV

As described earlier, percent accuracy is not the only important aspect of accuracy. Percent accuracy can be misleading because it is highly influenced by base rates. You can have a high percent accuracy by predicting from the base rate and saying that no one has the condition (if the base rate is low) or that everyone has the condition (if the base rate is high). Thus, it is also important to consider other aspects of accuracy, including sensitivity (SN), specificity (SP), positive predictive value (PPV), and negative predictive value (NPV). We want our predictions to be sensitive to be able to detect the characteristic but also to be specific so that we classify only people actually with the characteristic as having the characteristic.

Let us return to the confusion matrix in the source discussion. If we know the frequency of each of the four predicted-actual combinations of the confusion matrix (TP, TN, FP, FN), we can calculate sensitivity, specificity, PPV, and NPV.

Confusion Matrix: 2x2 Prediction Matrix. TP = true positives; TN = true negatives; FP = false positives; FN = false negatives.

Sensitivity is the proportion of those with the characteristic (TP + FN) that we identified with our measure (TP): TPTP + FN = (86)/(86 + 14) =.86. Specificity is the proportion of those who do not have the characteristic (TN + FP) that we correctly classify as not having the characteristic (TN): TNTN + FP = (1,478)/(1,478 + 422) =.78. PPV is the proportion of those who we classify as having the characteristic (TP + FP) who actually have the characteristic (TP): TPTP + FP = (86)/(86 + 422) =.17. NPV is the proportion of those we classify as not having the characteristic (TN + FN) who actually do not have the characteristic (TN): TNTN + FN = (1,478)/(1,478 + 14) =.99.

Sensitivity, specificity, PPV, and NPV are proportions, and their values therefore range from 0 to 1, where higher values reflect greater accuracy. With sensitivity, specificity, PPV, and NPV, we have a good snapshot of how accurate the measure is at a given cutoff. In our case, our measure is good at finding whom to reject (high sensitivity), but it is rejecting too many people who do not need to be rejected (lower PPV due to many FPs). Most people whom we classify as having the characteristic do not actually have the characteristic. However, the fact that we are over-rejecting could be okay depending on our goals, for instance, if we do not care about over-dropping (i.e., the PPV being low).

Some Accuracy Estimates Depend on the Cutoff Sensitivity, specificity, PPV, and NPV differ based on the cutoff (i.e., threshold) for classification. Consider the following example. Aliens visit Earth, and they develop a test to determine whether a berry is edible or inedible.

the source discussion depicts the distributions of scores by berry type. Note how there are clearly two distinct distributions. However, the distributions overlap to some degree. Thus, any cutoff will have at least some inaccurate classifications. The extent of overlap of the distributions reflects the amount of measurement error of the measure with respect to the characteristic of interest.

the source discussion depicts the distributions of scores by berry type with a cutoff. The red line indicates the cutoff—the level above which berries are classified by the test as inedible. There are errors on each side of the cutoff. Below the cutoff, there are some false negatives (blue): inedible berries that are inaccurately classified as edible. Above the cutoff, there are some false positives (green): edible berries that are inaccurately classified as inedible. Costs of false negatives could include sickness or death from eating the inedible berries. Costs of false positives could include taking longer to find food, finding insufficient food, and starvation.

Based on our assessment goals, we might use a different selection ratio by changing the cutoff. the source discussion depicts the distributions of scores by berry type when we raise the cutoff. There are now more false negatives (blue) and fewer false positives (green). If we raise the cutoff (to be more conservative), the number of false negatives increases and the number of false positives decreases. Consequently, as the cutoff increases, sensitivity and NPV decrease (because we have more false negatives), whereas specificity and PPV increase

(because we have fewer false positives). A higher cutoff could be optimal if the costs of false positives are considered greater than the costs of false negatives. For instance, if the aliens cannot risk eating the inedible berries because the berries are fatal, and there are sufficient edible berries that can be found to feed the alien colony.

the source discussion depicts the distributions of scores by berry type when we lower the cutoff. There are now fewer false negatives (blue) and more false positives (green). If we lower the cutoff (to be more liberal), the number of false negatives decreases and the number of false positives increases. Consequently, as the cutoff decreases, sensitivity and NPV increase (because we have fewer false negatives), whereas specificity and PPV decrease (because we have more false positives). A lower cutoff could be optimal if the costs of false negatives are considered greater than the costs of false positives. For instance, if the aliens cannot risk missing edible berries because they are in short supply relative to the size of the alien colony, and eating the inedible berries would, at worst, lead to minor, temporary discomfort.

In sum, sensitivity and specificity differ based on the cutoff for classification. if we raise the cutoff, sensitivity and PPV increase (due to fewer false positives), whereas and sensitivity and NPV decrease (due to more false negatives). If we lower the cutoff, sensitivity and NPV increase (due to fewer false negatives), whereas specificity and PPV decrease (due to more false positives). Thus, the optimal cutoff depends on how costly each type of error is: false negatives and false positives. If false negatives are more costly than false positives, we would set a low cutoff. If false positives are more costly than false negatives, we would set a high cutoff.

Signal Detection Theory

Signal detection theory (SDT) is a probability-based theory for the detection of a given stimulus (signal) from a stimulus set that includes non-target stimuli (noise). SDT arose through the development of radar (**RA**dio **D**etection **A**nd **R**anging) and sonar (**SO**und **N**avigation **A**nd **R**anging) in World War II based on research on sensory-perception research. The military wanted to determine which objects on radar/sonar were enemy aircraft/submarines, and which were noise (e.g., different object in the environment or even just the weather itself). SDT allowed determining how many errors operators made (how accurate they were) and decomposing errors into different kinds of errors. SDT distinguishes between sensitivity and bias. In SDT, sensitivity (or discriminability) is how well an assessment distinguishes between a target stimulus and non-target stimuli (i.e., how well the assessment detects the target stimulus amid non-target stimuli). Bias is the extent to which the probability of a selection decision from the assessment is higher or lower than the true rate of the target stimulus.

Some radar/sonar operators were not as sensitive to the differences between signal and noise, due to factors such as age, ability to distinguish gradations of a signal, etc. People who showed low sensitivity (i.e., who were not as successful at distinguishing between signal and noise) were screened out because the military perceived sensitivity as a skill that was not easily taught. By contrast, other operators could distinguish signal from noise, but their threshold was too low or high—they could take in information, but their decisions tended to be wrong due to systematic bias or poor calibration. That is, they systematically over-rejected or under-rejected stimuli. Over-rejecting leads to many false negatives (i.e., saying that a stimulus is safe when it is not). Under-rejecting leads to many false positives

(i.e., saying that a stimulus is harmful when it is not). A person who showed good sensitivity but systematic bias was considered more teach-able than a person who showed low sensitivity. Thus, radar and sonar operators were selected based on their sensitivity to distinguish signal from noise, and then were trained to improve the calibration so they reduce their systematic bias and do not systematically over- or under-reject.

Although SDT was originally developed for use in World War II, it now plays an important role in many areas of science and medicine. A medical application of SDT is tumor detection in radiology. SDT also plays an important role in psychology, especially cognitive psychology. For instance, research on social perception of sexual interest has shown that men tend to show lack of sensitivity to differences in women's affect—i.e., they have relative difficulties discriminating between friendliness and sexual interest (Farris et al., 2008). Men also tend to show systematic bias (poor calibration) such that they tend to over-estimate women's sexual interest in them—i.e., men tend to have too low of a threshold for determining that a women is showing sexual interest in them (Farris et al., 2006).

SDT metrics of sensitivity include d' ("d-prime"), A (or A'), and the area under the receiver operating characteristic (ROC) curve. SDT metrics of bias include beta (beta), c, and b.

Receiver Operating Characteristic (ROC) Curve The x-axis of the ROC curve is the false alarm rate or false positive rate (1 - specificity). The y-axis is the hit rate or true positive rate (sensitivity). We can trace the ROC curve as the combination between sensitivity and specificity at every possible cutoff. At a cutoff of zero (top right of ROC curve), we calculate sensitivity (1.0) and specificity (0) and plot it. At a cutoff of zero, the assessment tells us to make an action for every stimulus (i.e., it is the most liberal). We then gradually increase the cutoff, and plot sensitivity and specificity at each cutoff. As the cutoff increases, sensitivity decreases and specificity increases. We end at the highest possible cutoff, where the sensitivity is 0 and the specificity is 1.0 (i.e., we never make an action; i.e., it is the most conservative). Each point on the ROC curve corresponds to a pair of hit and false alarm rates (sensitivity and specificity) resulting from a specific cutoff value. Then, we can draw lines or a curve to connect the points.

the source discussion depicts an empirical ROC plot where lines are drawn to connect the hit and false alarm rates.

the source discussion depicts an ROC curve where a smoothed and fitted curve is drawn to connect the hit and false alarm rates.

Area Under the ROC Curve ROC methods can be used to compare and compute the discriminative power of measurement devices free from the influence of selection ratios, base rates, and costs and benefits. An ROC analysis yields a quantitative index of how well an index predicts a signal of interest or can discriminate between different signals. ROC analysis can help tell us how often our assessment would be correct. If we randomly pick two observations, and we were right once and wrong once, we were 50% accurate. But this would be a useless measure because it reflects chance responding.

The geometrical area under the ROC curve reflects the discriminative accuracy of the measure. The index is called the **a**rea **u**nder the **c**urve (AUC) of an ROC curve. AUC quantifies the discriminative power of an assessment. AUC is the probability that a randomly selected target and a randomly selected non-target is ranked correctly by the assessment method. AUC values range from 0.0 to 1.0, where chance accuracy is 0.5 as indicated by

diagonal line in the ROC curve. That is, a measure can be useful to the extent that its ROC curve is above the diagonal line (i.e., its discriminative accuracy is above chance).

As an example, given an AUC of.75, this says that the overall score of an individual who has the characteristic in question will be higher 75% of the time than the overall score of an individual who does not have the characteristic. In lay terms, AUC provides the probability that we will classify correctly based on our instrument if we were to randomly pick one good and one bad outcome. AUC is a stronger index of accuracy than percent accuracy, because you can have high percent accuracy just by going with the base rate. AUC tells us how much better than chance a measure is at discriminating outcomes. AUC is useful as a measure of general discriminative accuracy, and it tells us how accurate a measure is at all possible cutoffs. Knowing the accuracy of a measure at all possible cutoffs can be helpful for selecting the optimal cutoff, given the goals of the assessment. In reality, however, we may not be interested in all cutoffs because not all errors are equal in their costs.

If we lower the base rate, we would need a larger sample to get enough people to classify into each group. SDT/ROC methods are traditionally about dichotomous decisions (yes/no), not graded judgments. SDT/ROC methods can get messy with ordinal data that are more graded because you would have an AUC curve for each ordinal grouping.

Load Data

aSAH is a data set from the pROC package (R-pROC, n.d.) that contains test scores (s100b) and clinical outcomes (outcome) for patients.

Simulate Data

For reproducibility, I set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Add Missing Data

Adding missing data to dataframes helps make examples more realistic to real-life data and helps you get in the habit of programming to account for missing data.

Receiver Operating Characteristic (ROC) Curve

The receiver operating characteristic (ROC) curve shows the combination of hit rate (sensitivity) and false alarm rate (1 - specificity) at every possible cutoff. It depicts that, as the cutoff increases (i.e., becomes more conservative), sensitivity decreases and specificity increases. It also depicts that, as the cutoff decreases (i.e., becomes more liberal), sensitivity increases and specificity decreases.

Receiver operating characteristic (ROC) curves were generated using the pROC package (R-pROC, n.d.). The examples depict ROC curves that demonstrate that the measure is moderately accurate—the measure is more accurate than chance but there remains considerable room for improvement in predictive accuracy.

Empirical ROC Curve

The syntax used to generate an empirical ROC plot is below, and the plot is in the source discussion.

An empirical ROC plot with cutoffs overlaid is in the source discussion.

Youden's J Statistic

The threshold at the Youden's J statistic is the threshold where the test has the maximum combination (i.e., sum) of sensitivity and specificity: max(sensitivity + specificity - 1)

For this test, the Youden's J Statistic is at a threshold of [computed value omitted in this course adaptation], where sensitivity is [computed value omitted in this course adaptation] and specificity is [computed value omitted in this course adaptation].

The point closest to the top-left part of the ROC curve with perfect sensitivity and specificity

The point closest to the top-left part of the ROC plot with perfect sensitivity and specificity: min[(1 - sensitivity)^2 + (1 - specificity)^2]

For this test, the combination of sensitivity and specificity is closest to the top left of the ROC plot at a threshold of [computed value omitted in this course adaptation], where sensitivity is [computed value omitted in this course adaptation] and specificity is [computed value omitted in this course adaptation].

Prediction Accuracy Across Cutoffs

There are two primary dimensions of accuracy: (1) discrimination (e.g., sensitivity, specificity, area under the ROC curve) and (2) calibration. Some general indexes of accuracy combine discrimination and calibration, as described in the source discussion. This section (the source discussion) describes indexes of accuracy that span all possible cutoffs. That is, each index of accuracy described in this section provides a single numerical index of accuracy that aggregates the accuracy across all possible cutoffs. Aspects of accuracy at a particular cutoff are described in the source discussion.

The petersenlab package (Petersen, 2025) contains the petersenlab::accuracyOverall function that estimates the prediction accuracy across cutoffs.

General Prediction Accuracy

There are many metrics of general prediction accuracy. When thinking about which metric(s) may be best for a given problem, it is important to consider the purpose of the assessment. The estimates of general prediction accuracy are separated below into scale-dependent and scale-independent accuracy estimates.

Scale-Dependent Accuracy Estimates

The estimates of prediction accuracy described in this section are scale-dependent. These accuracy estimates depend on the unit of measurement and therefore cannot be compared across measures with different scales or across data sets.

Mean Error Here, "error" (e) is the difference between the predicted and observed value for a given individual (i). Mean error (ME; also known as bias, see the source discussion) is the mean difference between the predicted and observed values across individuals (i), that is, the mean of the errors across individuals (e i). Values closer to zero reflect greater accuracy. If mean error is above zero, it indicates that predicted values are, on average, greater than observed values (i.e., over-estimating errors). If mean error is below zero, it indicates that predicted values are, on average, less than observed values (i.e., under-estimating errors). If both over-estimating and under-estimating errors are present,

however, they can cancel each other out. As a result, even with a mean error of zero, there can still be considerable error present. Thus, although mean error can be helpful for examining whether predictions systematically under- or over-estimate the actual scores, other forms of accuracy are necessary to examine the extent of error. The formula for mean error is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

In this case, the mean error is negative, so the predictions systematically under-estimate the actual scores.

Mean Absolute Error (MAE) Mean absolute error (MAE) is the mean of the absolute value of differences between the predicted and observed values across individuals, that is, the mean of the absolute value of errors. Smaller MAE values (closer to zero) reflect greater accuracy. MAE is preferred over root mean squared error (RMSE) when you want to give equal weight to all errors and when the outliers have considerable impact. The formula for MAE is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Mean Squared Error (MSE) Mean squared error (MSE) is the mean of the square of the differences between the predicted and observed values across individuals, that is, the mean of the squared value of errors. Smaller MSE values (closer to zero) reflect greater accuracy. MSE penalizes larger errors more heavily than smaller errors (unlike MAE). However, MSE is sensitive to outliers and can be impacted if the errors are skewed. The formula for MSE is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Root Mean Squared Error (RMSE) Root mean squared error (RMSE) is the square root of the mean of the square of the differences between the predicted and observed values across individuals, that is, the root mean squared value of errors. Smaller RMSE values (closer to zero) reflect greater accuracy. RMSE penalizes larger errors more heavily than smaller errors (unlike MAE). However, RMSE is sensitive to outliers and can be impacted if the errors are skewed. The formula for RMSE is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Scale-Independent Accuracy Estimates

The estimates of prediction accuracy described in this section are intended to be scale-independent (unit-free) so the accuracy estimates can be compared across measures with different scales or across data sets (Hyndman & Athanasopoulos, 2018).

Mean Percentage Error (MPE) Mean percentage error (MPE) values closer to zero reflect greater accuracy. The formula for percentage error is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

We then take the mean of the percentage errors to get MPE. The formula for MPE is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Note: MPE is undefined when one or more of the observed values equals zero, due to division by zero. I provide the option in the function to drop undefined values so you can still generate an estimate of accuracy despite undefined values, but use this option at your own risk.

Mean Absolute Percentage Error (MAPE) Smaller mean absolute percentage error (MAPE) values (closer to zero) reflect greater accuracy. The formula for MAPE is in the source discussion:, n.d. MAPE is asymmetric because it overweights underestimates and underweights overestimates. MAPE can be preferable to symmetric mean absolute percentage error (sMAPE) if there are no observed values of zero and if you want to emphasize the importance of underestimates (relative to overestimates).

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Note: MAPE is undefined when one or more of the observed values equals zero, due to division by zero. I provide the option in the function to drop undefined values so you can still generate an estimate of accuracy despite undefined values, but use this option at your own risk.

Symmetric Mean Absolute Percentage Error (sMAPE) Unlike MAPE, symmetric mean absolute percentage error (sMAPE) is symmetric because it equally weights underestimates and overestimates. Smaller sMAPE values (closer to zero) reflect greater accuracy. The formula for sMAPE is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Note: sMAPE is undefined when one or more of the individuals has a prediction–observed combination such that the sum of the absolute value of the predicted value and the absolute value of the observed value equals zero (|predicted i| + |observed i|), due to division by zero. I provide the option in the function to drop undefined values so you can still generate an estimate of accuracy despite undefined values, but use this option at your own risk.

Mean Absolute Scaled Error (MASE) Mean absolute scaled error (MASE) is described by (Hyndman & Athanasopoulos, 2018). Values closer to zero reflect greater accuracy.

The adapted formula for MASE with non-time series data is described here (https://stats.stackexchange.com/a/108963/20338) (archived at https://perma.cc/G469-8NAJ). Scaled errors are calculated using the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Then, we calculate the mean of the absolute value of the scaled errors to get MASE, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Note: MASE is undefined when the scaling factor is zero, due to division by zero. With non-time series data, the scaling factor is the average of the absolute value of individuals' observed scores minus the average observed score ((1)/(n) sum i = 1^n |observed i - observed|).

Root Mean Squared Log Error (RMSLE) The squared log of the accuracy ratio is described by Tofallis, 2015. The accuracy ratio is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

However, the accuracy ratio is undefined with observed or predicted values of zero, so it is common to modify it by adding 1 to the predictor and denominator, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Squaring the log values keeps the values positive, such that smaller values (values closer to zero) reflect greater accuracy. Then we take the mean of the squared log values, which keeps the values positive, and calculate the square root of the mean squared log values to put them back on the (pre-squared) log metric. This is known as the root mean squared log error (RMSLE). Division inside the log is equal to subtraction outside the log. So, the formula can be reformulated with the subtraction of two logs, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

RMSLE can be preferable when the scores have a wide range of values and are skewed. RMSLE can help to reduce the impact of outliers. RMSLE gives more weight to smaller errors in the prediction of small observed values, while also penalizing larger errors in the prediction of larger observed values. It overweights underestimates and underweights overestimates.

There are other variations of prediction accuracy metrics that use the log of the accuracy ratio. One variation makes it similar to median symmetric percentage error (Morley et al., 2018).

Note: Root mean squared log error is undefined when one or more predicted values or actual values equals -1. When predicted or actual values are -1, this leads to log(0), which is undefined. I provide the option in the function to drop undefined values so you can still generate an estimate of accuracy despite undefined values, but use this option at your own risk.

Coefficient of Determination (R^2) The coefficient of determination (R^2) is a general index of accuracy that combines both discrimination and calibration. It reflects the proportion of variance in the outcome (dependent) variable that is explained by the model predictions: R^2 = variance explained in Ytotal variance in Y. Larger values indicate greater accuracy. The formula for the coefficient of determination is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where y is the outcome variable, y i is the observed value of the outcome variable for the ith observation, y i is the model predicted value for the ith observation, and y is the mean of the observed values of the outcome variable. The total sum of squares is an index of the total variation in the outcome variable.

R^2 is commonly estimated in multiple regression, in which multiple predictors are allowed to predict one outcome. Multiple regression can be conceptualized with overlapping circles in what is called a Ballantine graph (archived at https://perma.cc/C7CU-KFVG). R^2 in multiple regression is depicted conceptually with a Ballantine graph in the source discussion.

Conceptual Depiction of Proportion of Variance Explained (R^2) in an Outcome Variable (Y) by Multiple Predictors (X1 and X2) in Multiple Regression. The size of each circle represents the variable's variance. The proportion of variance in Y that is explained by the predictors is depicted by the areas in orange. The dark orange space (G) is where multiple predictors explain overlapping variance in the outcome. Overlapping variance that is explained in the outcome (G) will not be recovered in the regression coefficients when both predictors are included in the regression model.

R^2:

The predictor (testScore) explains [computed value omitted in this course adaptation]% of the variance (R^2 = [computed value omitted in this course adaptation]) in the outcome (disorder status).

The coefficient of determination typically ranges from 0 to 1, but it can be negative. A negative coefficient of determination indicates that the predicted values are less accurate than using the mean of the actual values as the predicted values (i.e., predicting the mean of the observed values for every case).

An issue with R^2 is that it can be artificially inflated by adding more predictors to the model, even if those predictors do not improve the model fit. R^2 never decreases when adding predictors to a model, even if the predictors do not provide any predictive power. Thus, to account for the number of predictors in the model, we can use adjusted R^2.

Adjusted R^2 (R^2 adj) Adjusted R^2 is similar to the coefficient of determination, but it accounts for the number of predictors included in the regression model to penalize overfitting. Specifically, adjusted R^2 penalizes the number of predictors in the model. Adjusted R^2 reflects the proportion of variance in the outcome (dependent) variable that is explained by the model predictions over and above what would be expected to be accounted for by chance, given the number of predictors in the model. Thus, adjusted R^2 will increase only if the new predictor improves the model (i.e., explains additional variance in the outcome variable) more than would be expected by chance. Larger values indicate greater accuracy. The formula for adjusted R^2 is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where p is the number of predictors in the model, and n is the sample size.

Adjusted R^2 is described further in the source discussion.

Predictive R^2 Predictive R^2 is described here: https://tomhopper.me/2014/05/16/can-we-do-better-than-r-squared/ (archived at https://perma.cc/BK8J-HFUK) and https://mfatihtuzen.netlify.app/posts/2025-04-30_rsquared/ (archived at https://perma.cc/EPP8-VG23). Predictive R^2 penalizes overfitting, unlike traditional R^2. Larger values indicate greater accuracy. The formula for predictive R^2 is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where PRESS is the predictive residual sum of squares, and SS total is the total sum of squares.

It is preferable to examine predictive R^2 using new data that the model was not trained on—i.e., cross-validation, either in an independent sample (i.e., external cross-validation) or in a hold-out sample (i.e., internal cross-validation on test data).

For example, here is an example of estimating predictive R^2 with 10-fold cross-validation using the caret package (R-caret, n.d.).

Here is an example of estimating predictive R^2 with a training and testing data set using the caret package (R-caret, n.d.).

Discrimination

When dealing with a categorical outcome, discrimination is the ability to separate events from non-events. When dealing with a continuous outcome, discrimination is the strength of the association between the predictor and the outcome. Aspects of discrimination at a particular cutoff (e.g., sensitivity, specificity) are described in the source discussion.

Area under the ROC curve (AUC)

The area under the ROC curve (AUC) is a general index of discrimination accuracy for a categorical outcome. It is also called the concordance (c) statistic. Larger values reflect greater discrimination accuracy. AUC was estimated using the pROC package (R-pROC, n.d.).

Coefficient of Predictive Ability (CPA)

The coefficient of predictive ability (CPA) is a generalization of AUC to handle non-binary outcomes including ordinal and continuous outcomes (Gneiting & Walz, 2021). Larger values reflect greater accuracy. It was calculated using the uroc package (Gneiting & Walz, 2021).

Spearman's Rho (rho) Rank Correlation

When the coefficient of predictive ability is applied to an ordinal or continuous outcome, it is linearly related to Spearman's rho (rho) rank correlation (Gneiting & Walz, 2021). Larger values reflect greater accuracy. Spearman's rho was estimated in the rms package (R-rms, n.d.).

Somers' D xy Rank Correlation

Somers D xy is an index of discrimination for an ordinal outcome (Harrell, 2015). Larger values reflect greater accuracy. Somers' D xy was estimated in the rms package (R-rms, n.d.).

Kendall's Tau-a (A) Rank Correlation

Kendall's tau-a (A) is an index of the discrimination for an ordinal outcome (Harrell, 2015). Larger values reflect greater accuracy. Kendall's tau-a was estimated in the rms package (R-rms, n.d.).

c Index

The c index, also called the concordance probability, is a generalization of the area under the ROC curve that applies to a continuous outcome (Harrell, 2015). The c index is the proportion of all pairs of predicted-actual values whose actual value can be ordered such that the pair with the higher predicted value is the one who had the higher actual value. Larger values reflect greater accuracy. The Brown–Hollander–Korwar non-parametric test of association was estimated in the rms package (R-rms, n.d.).

Effect Size (beta) of Regression

The effect size of a predictor, i.e., the standardized regression coefficient is called a beta (beta) coefficient, is a general index of discrimination accuracy for a continuous outcome. Larger values reflect greater accuracy. We can obtain standardized regression coefficients by standardizing the predictors and outcome using the scale function in R.

beta:

Calibration

When dealing with a categorical outcome, calibration is the degree to which a probabilistic estimate of an event reflects the true underlying probability of the event. When dealing with a continuous outcome, calibration is the degree to which the predicted values are close in value to the outcome values. The importance of examining calibration (in addition to discrimination) is described by Lindhiem et al., 2020.

Calibration came to be considered a central aspect of weather forecast accuracy. For instance, on the days that the meteorologist says there is a 60% chance of rain, it should rain about 60% of the time. Through improvements in scientific understanding of weather systems, rain forecasts have become more accurate. Meteorologists produce "ensemble forecasts", where they run many different models (archived at https://perma.cc/248N-9CW5). If all of the models predict a similar outcome (e.g., no rain), the metereologists have greater confidence in the forecast. However, if the models predict different outcomes (e.g., some models predict rain and others predict no rain), the meteorologists have less confidence in the forecast, which is why the meteorologists might predict a "10% chance of rain".

With these advancements, rain forecasts from the National Weather Service are well calibrated (the source discussion; Charba & Klein, 1980). However, forecasts of rain may be exaggerated by local TV meteorologists to boost ratings (Silver, 2012). Interestingly, rain forecasts from The Weather Channel are somewhat miscalibrated under certain conditions. For instance, on days when they forecast a 20% chance of rain, the actual chance of rain is

around 5% (the source discussion; Bickel & Kim, 2008). However, this miscalibration is deliberate. People tend to be more angry when the meteorologist says it will not rain when it actually does (false negative) compared to when the meteorologist says it will rain when it actually does not (false positive). As Silver, 2012 notes, "If it rains when it isn't supposed to, [people] curse the weatherman for ruining their picnic, whereas an unexpectedly sunny day is taken as a serendipitous bonus. It isn't good science, but as Dr. Rose at The Weather Channel acknowledged to [Mr. Silver]: 'If the forecast was objective, if it has zero bias in precipitation, we'd probably be in trouble.' " (p. 135).

Calibration Plot Of Same-Day Probability Of Precipitation (PoP) Forecasts From The Weather Channel. The plot depicts that the rain forecasts are generally well calibrated, but shows some miscalibration. For instance, the forecasted probability of rain is lower than the actual probability of rain when the forecasted probability of rain is 20%. (Figure reprinted from Bickel & Kim, 2008, Figure 2, p. 4872. Bickel, J. E., & Kim, S. D. (2008). Verification of The Weather Channel probability of precipitation forecasts. Monthly Weather Review, 136(12), 4867–4881. doi: https://doi.org/10.1175/2008MWR2547.1 Copyright (c) American Meteorological Society. Used with permission.)

Calibration Plot Of Local Probability Of Precipitation (PoP) Forecasts for 87 Stations From the United States National Weather Service. Numbers next to the plotted points are the sample sizes. (Figure reprinted from Charba & Klein, 1980, Figure 6, p. 1550. Charba, J. P., & Klein, W. H. (1980). Skill in precipitation forecasting in the National Weather Service. Bulletin of the American Meteorological Society, 61(12), 1546–1555. https://doi.org/10.1175/1520-0477(1980)061. Copyright (c) American Meteorological Society. Used with permission.)

Calibration is not just important for weather forecasts. It is also important for psychological assessment. Calibration can be examined in several ways, including Brier Scores (see the source discussion), the Hosler-Lemeshow test (see the source discussion), Spiegelhalter's z (see the source discussion), and the mean difference between predicted and observed values at different binned thresholds as depicted graphically with a calibration plot (see the source discussion).

Calibration Plot

Calibration plots can be helpful for identifying miscalibration. A calibration plot depicts the predicted probability of an event on the x-axis, and the actual (observed) probability of the event on the y-axis. The predictions are binned into a certain number of groups (commonly 10). The diagonal line reflects predictions that are perfectly calibrated. To the extent that predictions deviate from the diagonal line, the predictions are miscalibrated. There are four general patterns of miscalibration: overextremity, underextremity, overprediction, and underprediction (see the source discussion). Overextremity exists when the predicted probabilites are too close to the extremes (zero or one). Underextremity exists when the predicted probabilities are too far away from the extremes. Overprediction exists when the predicted probabilities are consistently greater than the observed probabilities. Underprediction exists when the predicted probabilities are consistently less than the observed probabilities. For a more thorough description of these types of miscalibration, see Lindhiem et al., 2020.

This calibration plot was generated using the PredictABEL package (R-PredictABEL, n.d.), and is depicted in the source discussion.

This calibration plot (and ROC curve) was generated using the rms package (R-rms, n.d.), and is depicted in the source discussion. The calibration plot is consistent with underprediction. That is, the predicted probabilities were consistently lower than the actual probabilities. To re-calibrate the predicted probabilities, it would be necessary to increase them so they are consistent with observed probabilties.

This calibration plot was adapted from Darren Dahly's example, and is depicted in the source discussion:, n.d. https://darrendahly.github.io/post/homr/ (archived at https://perma.cc/6J3J-69G7)

Brier Scores

Brier scores were calculated using the rms package (R-rms, n.d.). Smaller values reflect greater calibration accuracy.

Hosler-Lemeshow Test

The Hosler-Lemeshow goodness of fit (GOF) test evaluates the null hypothesis that there is no difference between the predicted versus observed frequencies of an event. The test requires specifying how many groups/bins (g) to use to calculate quantiles. Smaller ^2 values (and larger p-values) reflect greater calibration accuracy. A statistically significant ^2 ($p < .05$) indicates a significant degree of miscalibration. The Hosler-Lemeshow GOF test was calculated using the ResourceSelection package (R-ResourceSelection, n.d.). The table of predicted versus observed frequencies was generated using the PredictABEL package (R-PredictABEL, n.d.).

For each number of bins (g) specified below, the predictions show statistically significant miscalibration.

g = 2

g = 4

g = 6

g = 8

g = 10

Spiegelhalter's z

Spiegelhalter's z was calculated using the rms package (R-rms, n.d.). Smaller z values (and larger associated p-values) reflect greater calibration accuracy. A statistically significant Spiegelhalter's z ($p < .05$) indicates a significant degree of miscalibration.

In this case, the predictions show statistically significant miscalibration.

Calibration for predicting a continuous outcome

When predicting a continuous outcome, calibration of the predicted values in relation to the outcome values can be examined in multiple ways including:

• in a calibration plot, the extent to which the intercept is near zero and the slope is near one

• in a calibration plot, the extent to which the 95% confidence interval of the observed value, across all values of the predicted values, includes the diagonal reference line with an intercept of zero and a slope of one.

• mean error

• mean absolute error

• mean squared error

• root mean squared error

With a plot of the predictions on the x-axis, and the outcomes on the y-axis (i.e., a calibration plot), calibration can be examined graphically as the extent to which the best-fit regression line has an intercept (alpha) close to zero and a slope (beta) close to one (Steyerberg & Vergouwe, 2014; Stevens & Poppe, 2020). The intercept is also called "calibration-in-the-large", whereas "calibration-in-the-small" refers to the extent to which the predicted values match the observed values at a specific predicted value (e.g., when the weather forecaster says that there is a 10% chance of rain, does it actually rain 10% of the time?). For predictions to be well calibrated, the intercept should be close to zero and the slope should be close to one. If the slope is close to one but the intercept is not close to zero (or the intercept is close to zero but the slope is not close to one), the predictions would not be considered well calibrated. The 95% confidence interval of the observed value, across all values of the predicted values, should include the diagonal reference line whose intercept is zero and whose slope is one.

For instance, based on the intercept and slope of the calibration plot in the source discussion, the predictions are not well calibrated, despite having a slope near one, because the 95% confidence interval of the intercept does not include zero. The best-fit line is the yellow line. The intercept from the best-fit line is positive, as shown in the regression equation. This is a case of underprediction, where the predicted values are consistently less than the observed values. The confidence interval of the observed value (i.e., the purple band) is the interval within which we have 95% confidence that the true observed value would lie for a given predicted value, based on the model. The 95% prediction interval of the observed value (i.e., the dashed red lines) is the interval within which we would expect that 95% of future observations would lie for a given predicted value. The black diagonal line indicates the reference line with an intercept of zero and a slope of one. The predictions would be significantly miscalibrated at a given level of the predicted values if the 95% confidence interval of the observed value does not include the reference line at that level of the predicted value. In this case, the 95% confidence interval of the observed value does not include the reference line (i.e., the actual observed value) at lower levels of the predicted values, so the predictions are miscalibrated lower levels of the predicted values.

Gold-standard recommendations include examining the predicted values in relation to the observed values using locally estimated scatterplot smoothing (LOESS) (Austin & Steyerberg, 2014), such as in the source discussion. We can examine whether the LOESS-based 95% confidence interval of the observed value at every level of the predicted values includes the diagonal reference line (i.e., the actual observed value). In this case, the 95% confidence interval of the observed value does not include the reference line at lower levels of the predicted values, so the predictions are miscalibrated at lower levels of the predicted values.

Set a Cutoff

Here, I set a cutoff at the Youden's J Statistic to calculate the accuracy statistics at that cutoff:

Accuracy at a Given Cutoff

The petersenlab package (Petersen, 2025) contains the petersenlab::accuracyAtCutoff function that estimates the prediction accuracy at a given cutoff.

There are also test calculators available online:

- http://araw.mede.uic.edu/cgi-bin/testcalc.pl
- https://dlrs.shinyapps.io/shinyDLRs/

Confusion Matrix aka 2x2 Accuracy Table aka Cross-Tabulation aka Contingency Table

A confusion matrix (aka 2x2 accuracy table, cross-tabulation table, or contigency table) is a matrix for categorical data that presents the predicted outcome on one dimension and the actual outcome (truth) on the other dimension. If the predictions and outcomes are dichotomous, the confusion matrix is a 2x2 matrix with two rows and two columns that represent four possible predicted-actual combinations (decision outcomes). In such a case, the confusion matrix provides a tabular count of each type of accurate cases (true positives and true negatives) versus the number of each type of error (false positives and false negatives), as shown in the source discussion. An example of a confusion matrix is in the source discussion.

Confusion Matrix.

True Positives (TP)

True positives (TPs) are instances in which a positive classification (e.g., a disorder present) is correct—that is, the test says that a classification is present, and the classification is present. True positives are also called valid positives (VPs) or hits. Higher values (relative to the same sample size) reflect greater accuracy. The formula for true positives is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

True Negatives (TN)

True negatives (TNs) are instances in which a negative classification (e.g., absence of a disorder) is correct—that is, the test says that a classification is not present, and the classification is actually not present. True negatives are also called valid negatives (VNs) or correct rejections. Higher values (relative to the same sample size) reflect greater accuracy. The formula for true negatives is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

False Positives (FP)

False positives (FPs) are instances in which a positive classification (e.g., a disorder present) is incorrect—that is, the test says that a classification is present, and the classification is not present. False positives are also called false alarms (FAs). Lower values (relative to the same sample size) reflect greater accuracy. The formula for false positives is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

False Negatives (FN)

False negatives (FNs) are instances in which a negative classification (e.g., absence of a disorder) is incorrect—that is, the test says that a classification is not present, and the classification is present. False negatives are also called misses. Lower values (relative to the same sample size) reflect greater accuracy. The formula for false negatives is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Selection Ratio (SR)

The selection ratio (SR) is the marginal probability of selection, independent of other things: P(R i). In clinical psychology, the selection ratio is the proportion of people who test positive for the disorder, as in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Base Rate (BR)

The base rate (BR) of a classification is its marginal probability, independent of other things: P(C i). In clinical psychology, the base rate of a disorder is its prevalence in the population, as in the source discussion. Without additional information, the base rate is used as the initial pretest probability.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Pretest Odds

The pretest odds of a classification can be estimated using the pretest probability (i.e., base rate). To convert a probability to odds, divide the probability by one minus that probability, as in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Percent Accuracy

Percent Accuracy is also called overall accuracy. Higher values reflect greater accuracy. The formula for percent accuracy is in the source discussion. Percent accuracy has several problems. First, it treats all errors (FP and FN) as equally important. However, in practice, it is rarely the case that false positives and false negatives are equally important. Second, percent accuracy can be misleading because it is highly influenced by base rates. You can have a high percent accuracy by predicting from the base rate and saying that no one has the characteristic (if the base rate is low) or that everyone has the characteristic (if the base rate is high). Thus, it is also important to consider other aspects of accuracy.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Percent Accuracy by Chance

The formula for calculating percent accuracy by chance is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Percent Accuracy Predicting from the Base Rate

Predicting from the base rate is going with the most likely outcome in every prediction. It is also called "betting from the base rate". If the base rate is less than.50, it would involve predicting that the condition is absent for every case. If the base rate is.50 or above, it would involve predicting that the condition is present for every case. Predicting from the base rate is a special case of percent accuracy by chance when the selection ratio is set to either one (if the base rate ≥.5) or zero (if the base rate <.5).

Relative Improvement Over Chance (RIOC)

Relative improvement over chance (RIOC) is a prediction's improvement over chance as a proportion of the maximum possible improvement over chance, as described by Farrington & Loeber, 1989. Higher values reflect greater accuracy. The formula for calculating RIOC is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Relative Improvement Over Predicting from the Base Rate

Relative improvement over predicting from the base rate is a prediction's improvement over predicting from the base rate as a proportion of the maximum possible improvement over predicting from the base rate. Higher values reflect greater accuracy. The formula for calculating relative improvement over predicting from the base rate is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Sensitivity (SN)

Sensitivity (SN) is also called true positive rate (TPR), hit rate (HR), or recall. Sensitivity is the conditional probability of a positive test given that the person has the condition: P(R|C). Higher values reflect greater accuracy. The formula for calculating sensitivity is in the source discussion. As described in the source discussion and as depicted in the source discussion, as the cutoff increases (becomes more conservative), sensitivity decreases. As the cutoff decreases, sensitivity increases.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Below I compute sensitivity and specificity at every possible cutoff.

Specificity (SP)

Specificity (SP) is also called true negative rate (TNR) or selectivity. Specificity is the conditional probability of a negative test given that the person does not have the condition: P(not R|not C). Higher values reflect greater accuracy. The formula for calculating specificity is in the source discussion. As described in the source discussion and as depicted in the source discussion, as the cutoff increases (becomes more conservative), specificity increases. As the cutoff decreases, specificity decreases.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

False Negative Rate (FNR)

The false negative rate (FNR) is also called the miss rate. The false negative rate is the conditional probability of a negative test given that the person has the condition: P(not R|C). Lower values reflect greater accuracy. The formula for calculating false negative rate is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

False Positive Rate (FPR)

The false positive rate (FPR) is also called the false alarm rate (FAR) or fall-out. The false positive rate is the conditional probability of a positive test given that the person does not have the condition: P(R|not C). Lower values reflect greater accuracy. The formula for calculating false positive rate is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Positive Predictive Value (PPV)

The positive predictive value (PPV) is also called the positive predictive power (PPP) or precision. Many people confuse sensitivity (P(R|C)) with its inverse conditional probability, PPV (P(C|R)). PPV is the conditional probability of having the condition given a positive test: P(C|R). Higher values reflect greater accuracy. The formula for calculating positive predictive value is in the source discussion.

PPV can be low even when sensitivity is high because it depends not only on sensitivity, but also on specificity and the base rate. Because PPV depends on the base rate, PPV is not an intrinsic property of a measure. The same measure will have a different PPV in different contexts with different base rates (Treat & Viken, 2023). As described in the source discussion and as depicted in the source discussion, as the base rate increases, PPV increases. As the base rate decreases, PPV decreases. PPV also differs as a function of the cutoff. As described in the source discussion and as depicted in the source discussion, as the cutoff increases (becomes more conservative), PPV increases. As the cutoff decreases (becomes more liberal), PPV decreases.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Below I compute PPV and NPV at every possible base rate given the sensitivity and specificity at the current cutoff.

Below I compute PPV and NPV at every possible cutoff.

Negative Predictive Value (NPV)

The negative predictive value (NPV) is also called the negative predictive power (NPP). Many people confuse specificity (P(not R|not C)) with its inverse conditional probability, NPV (P(not C| not R)). NPV is the conditional probability of not having the condition given a negative test: P(not C| not R). Higher values reflect greater accuracy. The formula for calculating negative predictive value is in the source discussion.

NPV can be low even when specificity is high because it depends not only on specificity, but also on sensitivity and the base rate. Because NPV depends on the base rate, NPV is not an intrinsic property of a measure. The same measure will have a different NPV in different contexts with different base rates (Treat & Viken, 2023). As described in the source discussion and as depicted in the source discussion, as the base rate increases, NPV decreases. As the base rate decreases, NPV increases. NPV also differs as a function of the cutoff. As described in the source discussion and as depicted in the source discussion, as the cutoff increases (becomes more conservative), NPV decreases. As the cutoff decreases (becomes more liberal), NPV decreases.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

False Discovery Rate (FDR)

Many people confuse the false positive rate (P(R|not C)) with its inverse conditional probability, the false discovery rate (P(not C| R)). The false discovery rate (FDR) is the conditional probability of not having the condition given a positive test: P(not C| R). Lower values reflect greater accuracy. The formula for calculating false discovery rate is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

False Omission Rate (FOR)

Many people confuse the false negative rate (P(not R|C)) with its inverse conditional probability, the false omission rate (P(C|not R)). The false omission rate (FOR) is the conditional probability of having the condition given a negative test: P(C|not R). Lower values reflect greater accuracy. The formula for calculating false omission rate is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Youden's J Statistic

Youden's J statistic is also called Youden's Index or informedness. Youden's J statistic is the sum of sensitivity and specificity (and subtracting one). Higher values reflect greater accuracy. The formula for calculating Youden's J statistic is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Balanced Accuracy

Balanced accuracy is the average of sensitivity and specificity. Higher values reflect greater accuracy. The formula for calculating balanced accuracy is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

F-Score

The F-score combines precision (positive predictive value) and recall (sensitivity), where beta indicates how many times more important sensitivity is than the positive predictive value. If sensitivity and the positive predictive value are equally important, beta = 1, and the F-score is called the F 1 score. Higher values reflect greater accuracy. The formula for calculating the F-score is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

The F 1 score is the harmonic mean of sensitivity and positive predictive value. The formula for calculating the F 1 score is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Matthews Correlation Coefficient (MCC)

The Matthews correlation coefficient (MCC) is also called the phi coefficient. It is a correlation coefficient between predicted and observed values from a binary classification. Higher values reflect greater accuracy. The formula for calculating the MCC is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Diagnostic Odds Ratio

The diagnostic odds ratio is the odds of a positive test among people with the condition relative to the odds of a positive test among people without the condition. Higher values reflect greater accuracy. The formula for calculating the diagnostic odds ratio is in the source discussion. If the predictor is bad, the diagnostic odds ratio could be less than one, and values can go up from there. If the diagnostic odds ratio is greater than 2, we take the odds ratio seriously because we are twice as likely to predict accurately than inaccurately. However, the diagnostic odds ratio ignores/hides base rates. When interpreting the diagnostic odds ratio, it is important to keep in mind the clinical significance, because otherwise it is not very meaningful. Consider a risk factor that has a diagnostic odds ratio of 3 for tuberculosis, i.e., it puts you at 3 times as likely to develop tuberculosis. The prevalence of tuberculosis is relatively low. Assuming the prevalence of tuberculosis is less than 1/10th of 1%, your risk of developing tuberculosis is still very low even if the risk factor (with a diagnostic odds ratio of 3) is present.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Diagnostic Likelihood Ratio

A likelihood ratio is the ratio of two probabilities. It can be used to compare the likelihood of two possibilities. The diagnostic likelihood ratio is an index of the predictive validity of an instrument: it is the ratio of the probability that a test result is correct to the probability that the test result is incorrect. The diagnostic likelihood ratio is also called the risk ratio. There are two types of diagnostic likelihood ratios: the positive likelihood ratio and the negative likelihood ratio.

Positive Likelihood Ratio (LR+)

The positive likelihood ratio (LR+) compares the true positive rate to the false positive rate. Positive likelihood ratio values range from 1 to infinity. Higher values reflect greater accuracy, because it indicates the degree to which a true positive is more likely than a false positive. The formula for calculating the positive likelihood ratio is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Negative Likelihood Ratio (LR−)

The negative likelihood ratio (LR−) compares the false negative rate to the true negative rate. Negative likelihood ratio values range from 0 to 1. Smaller values reflect greater accuracy, because it indicates that a false negative is less likely than a true negative. The formula for calculating the negative likelihood ratio is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Posttest Odds

As presented in the source discussion, the posttest (or posterior) odds are equal to the pretest odds multiplied by the likelihood ratio. The posttest odds and posttest probability can be useful to calculate when the pretest probability is different from the pretest probability (or prevalence) of the classification. For instance, you might use a different pretest probability if a test result is already known and you want to know the updated posttest probability after conducting a second test. The formula for calculating posttest odds is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

For calculating the posttest odds of a true positive compared to a false positive, we use the positive likelihood ratio, described later. We would use the negative likelihood ratio if we wanted to calculate the posttest odds of a false negative compared to a true negative.

Posttest Probability

The posttest probability is the probability of having the disorder given a test result. When the base rate is used as the pretest probability, the posttest probability given a positive test is equal to positive predictive value. To convert odds to a probability, divide the odds by one plus the odds, as is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Consider the following example: Assume the base rate of the condition is.03%. We have two tests. Test A has a sensitivity of.95 and a specificity of.80. Test B has a sensitivity of.70 and a specificity of.90. What is the probability of having the condition if a person has a positive test on Test A? Assuming the errors of the two tests are independent, what is the probability of having the condition if the person has a positive test on Test B after having a positive test on Test A?

The probability of having the condition if a person has a positive test on Test A is [computed value omitted in this course adaptation]%. The probability of having the condition if the person has a positive test on Test B after having a positive test on Test A is [computed value omitted in this course adaptation]%.

Probability Nomogram

The petersenlab package (Petersen, 2025) contains the petersenlab::nomogrammer function that creates a probability nomogram plot, adapted from https://github.com/achekroud/nomogrammer. In the source discussion, the probability nomogram is generated using the number of true positives, true negatives, false positives, and false negatives at a given cutoff.

The blue line indicates the posterior probability of the condition given a positive test. The pink line indicates the posterior probability of the condition given a negative test. One can also generate the probability nomogram from the base rate and the sensitivity and specificity of the test at a given cutoff:

One can also generate the probability nomogram from the base rate, positive likelihood ratio, and negative likelihood ratio at a given cutoff:

d' Sensitivity from Signal Detection Theory

d' (d prime) is an index of sensitivity from signal detection theory, as described by Stanislaw & Todorov, 1999. Higher values reflect greater accuracy. The formula for calculating d' is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

A (Non-Parametric) Sensitivity from Signal Detection Theory

A is a non-parametric index of sensitivity from signal detection theory, as described by Zhang & Mueller, 2005. Higher values reflect greater accuracy. The formula for calculating A is in the source discussion.

https://sites.google.com/a/mtu.edu/whynotaprime/ (archived at https://perma.cc/W2M2-39TJ)

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where H is the hit rate and F is the false alarm rate.

beta Bias from Signal Detection Theory

beta is an index of bias from signal detection theory, as described by Stanislaw & Todorov, 1999. Smaller values reflect greater accuracy. The formula for calculating beta is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where H is the hit rate, F is the false alarm rate, and (phi) is a mathematical function that converts a z score to a probability by determining the portion of the normal distribution that lies to the left of the z score.

c Bias from Signal Detection Theory

c is an index of bias from signal detection theory, as described by Stanislaw & Todorov, 1999. Smaller values reflect greater accuracy. The formula for calculating c is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where H is the hit rate, F is the false alarm rate, and (phi) is a mathematical function that converts a z score to a probability by determining the portion of the normal distribution that lies to the left of the z score.

b (Non-Parametric) Bias from Signal Detection Theory

b is a non-parametric index of bias from signal detection theory, as described by Zhang & Mueller, 2005. Smaller values reflect greater accuracy. The formula for calculating b is in the source discussion.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Mean Difference between Predicted Versus Observed Values (Miscalibration)

The mean difference between predicted values versus observed values at a given cutoff is an index of miscalibration of predictions at that cutoff. It is called "calibration-in-the-small" (as opposed to calibration-in-the-large, which spans all cutoffs). Values closer to zero reflect greater accuracy. Values above zero indicate that the predicted values are, on average, greater than the observed values. Values below zero indicate that the observed values are, on average, greater than the predicted values.

Optimal Cutoff Specification

There are two ways to improve diagnostic performance (Swets et al., 2000). One way is to increase the diagnostic accuracy of the assessment. The second way is to increase the utility of the diagnostic decisions that are made, based on where we set the cutoff. The optimal cutoff depends on the differential costs of false positives versus false negatives, as applied in decision theory. When differential costs of false positives versus false negatives cannot be specified, an alternative approach to specifying the optimal cutoff is to use information theory.

Decision Theory

According to the decision theory approach to picking the optimal cutoff, the optimal cutoff depends on the value/importance placed on each of the four decision outcomes ([true positives, true negatives, false positives, false negatives); Treat & Viken, 2023]. Utility is the relative value placed on a specific decision-making outcome (i.e., user-perceived benefit or cost): utilities typically range between zero and one, where a value of zero represents the least desired outcome, and a value of one indicates the most desired outcome. According to the decision theory approach, the optimal cutoff is the cutoff with the highest overall utility.

Overall utility of a specific cutoff value

The overall utility of a specific cutoff value is a utilities-weighted sum of the probabilities of the four decision-making outcomes (hits, misses, correct rejections, false alarms). That is, overall utility is the sum of the product of the probability of a particular outcome (TP, TN, FP, FN; e.g., BR × TP rate) and the utility of that outcome (e.g., how much we value TPs relative to other outcomes). Higher values reflect greater utility, so you would pick the cutoff with the highest overall utility. The formula for calculating overall utility is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where BR = base rate, HR = hit rate (true positive rate), FAR = false alarm rate (false positive rate), U H = utility of hits (true positives), U M = utility of misses (false negatives), U FA = utility of false alarms (false positives), U CR = utility of correct rejections (true negatives).

Utility ratio

The utility ratio is the user-perceived relative importance of decisions about negative versus positive cases. If the utility ratio value is one, it indicates that identifying negative cases and positive cases is equally important. Values above one indicate greater relative importance of identifying negative cases than positive cases. Values below one indicate greater relative importance of identifying positive cases than negative cases. Values of one indicate that you are maximizing percent accuracy. The formula for calculating the utility ratio is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where U H = utility of hits (true positives), U M = utility of misses (false negatives), U FA = utility of false alarms (false positives), U CR = utility of correct rejections (true negatives).

Decision theory has key advantages, because it identifies the cutoff that would help you best achieve the goals/purpose of the assessment. However, it can be challenging to specify the relative costs of errors. If you cannot decide values for outcomes (relative importance between FP and FN), you can use information theory to identify the optimal cutoff.

Information Theory

When the user does not differentially weigh the value/importance of the four decision-making outcomes (hits, misses, correct rejections, false alarms), the information theory approach can be useful for specifying the optimal cutoff. According to the information theory approach, the optimal cutoff is the cutoff that provides the greatest information gain (Treat & Viken, 2023).

Information Gain

Information gain (I gain) is the reduction of uncertainty about the true classification of a case that results from administering an assessment or prediction measure (Treat & Viken, 2023). Greater values reflect greater reduction of uncertainty, so the optimal cutoff can be specified as the cutoff with the highest information gain.

Formula from Treat & Viken, 2023 The formula from Treat & Viken, 2023 for calculating information gain is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where BR = base rate, HR = hit rate (true positive rate), FAR = false alarm rate (false positive rate), G = selection ratio = BR (HR) + (1 - BR) (FAR), as reported in Somoza et al., 1989 (see below).

Alternative formula from Metz et al., 1973 The alternative formula from Metz et al., 1973 for calculating information gain is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where p(S|s) = sensitivity (hit rate or true positive rate); i.e., the conditional probability of deciding a signal is present (S) when the signal is in fact present(s), p(S|n) = false positive rate (false alarm rate); i.e., the conditional probability of deciding a signal is present (S)

when the signal is in fact absent (n), p(s) = base rate, i.e., the probability that the signal is in fact present (s).

Alternative formula from Somoza et al., 1989 The alternative formula from Somoza et al., 1989 for calculating information gain is in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where TP = true positive rate (hit rate), Pr = prevalence (base rate), FP = false positive rate (false alarm rate), G = Pr (TP) + (1 - Pr) (FP) = selection ratio

Examples Case A from Exhibit 38.2 (Treat & Viken, 2023):

Case B from Exhibit 38.2 (Treat & Viken, 2023):

Case C from Exhibit 38.2 (Treat & Viken, 2023):

Case B from Exhibit 38.3 (Treat & Viken, 2023):

Effect of Base Rate Information gain depends on the base rate (Treat & Viken, 2023), as depicted in the source discussion. The maximum reduction of uncertainty (i.e., greatest information) occurs when the base rate is 0.5. A base rate tells us little a priori about the condition if the probability of the condition is 50/50, so the measure can provide more benefit. If the base rate is 0.3 or 0.7, we can do better than going with the base rate. If the base rate is 0.9 or 0.1, it is difficult for our measure to do better than going with the base rate. If the base rate is 0.05 of 0.95 (or more extreme), it is likely that our measure will do almost nothing in terms of information gain.

All Accuracy Statistics

The petersenlab package (Petersen, 2025) contains the petersenlab::accuracyAtEachCutoff R function that estimates the prediction accuracy at every possible cutoff.

Threshold

Threshold at maximum combination of sensitivity and specificity:

max(sensitivity + specificity)

Threshold

Threshold where the ROC plot is closest to the Top Left:

Cutoff that optimizes each of the following criteria:

The petersenlab package (Petersen, 2025) contains the petersenlab::optimalCutoff R function that identifies the cutoff that optimizes each of various accuracy estimates.

Regression for Prediction of Continuous Outcomes

When predicting a continuous outcome, regression is particularly relevant (or multiple regression, when dealing with multiple predictors). Regression takes the general form in the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where y is the outcome, b 0 is the intercept, b 1 is the slope of the association between the predictor (x 1) and outcome, and e is the error term.

If the predictors reflect what causes the outcome, then the error term reflects what cannot be accounted for by the predictors. It has been argued that human freedom resides in the error term (Kenny, 1979).

Pseudo-Prediction

Consider the following example where you have one predictor and one outcome, as shown in the source discussion.

Using the data, the best fitting regression model is: y = [computed value omitted in this course adaptation] + [computed value omitted in this course adaptation] · x 1. In this example, the R^2 is [computed value omitted in this course adaptation]. The equation is not a perfect prediction, but with a single predictor, it captures the majority of the variance in the outcome.

Now consider the following example where you add a second predictor to the data above, as shown in the source discussion.

With the second predictor, the best fitting regression model is: y = [computed value omitted in this course adaptation] + [computed value omitted in this course adaptation] · x 1 + [computed value omitted in this course adaptation] · x 2. In this example, the R^2 is [computed value omitted in this course adaptation]. The equation with the second predictor provides a perfect prediction of the outcome.

Providing perfect prediction with the right set of predictors is the dream of multiple regression. So, in psychology, we often add predictors to incrementally improve prediction. Knowing how much variance would be accounted for by random chance follows the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

where E(R^2) is the expected value of R^2 (the proportion of variance explained), K is the number of predictors, and n is the sample size. The formula demonstrates that the more predictors in the regression model, the more variance will be accounted for by chance. With many predictors and a small sample, you can account for a large share of the variance merely by chance—this would be an example of pseudo-prediction.

As an example, consider that we have 13 predictors to predict behavior problems for 43 children. Assume that, with 13 predictors, we explain 38% of the variance (R^2 =.38; r =.62). Explaining more than 20–30% of the variance can be a big deal in psychology. We explained a lot of the variance in the outcome, but it is important to consider how much variance could have been explained by random chance: E(R^2) = (K)/(n-1) = (13)/(43 - 1) =.31. We expect to explain 31% of the variance, by chance, in the outcome. So, 82% of the variance explained was likely spurious. As the sample size increases, the spuriousness decreases. Adjusted R^2 accounts for the number of predictors in the model, based on how much would be expected to be accounted for by chance. But adjusted R^2 also has its problems.

Multicollinearity

Multicollinearity occurs when two or more predictors in a regression model are highly correlated. The problem is that it makes it challenging to estimate the regression coefficients accurately.

Multicollinearity in multiple regression is depicted conceptually in the source discussion.

Conceptual Depiction of Multicollinearity in Multiple Regression.

Consider the following example where you have two predictors and one outcome, as shown in the source discussion.

The second measure is not very good—it is exactly twice the value of the first measure. This means that there are different prediction equation possibilities that are equally good—see the source discussion:, n.d.

[Formal equation omitted in this course adaptation; see the source for the mathematical expression.]

Then, what are the regression coefficients? We do not know, and we could come up with arbitrary estimates with an enormous standard error around each estimate. Any predictors that have a correlation above ~ r =.30 with each other could have an impact on the confidence interval of the regression coefficient. As the correlations among the predictors increase, the chance of getting an arbitrary answer increases, sometimes called "bouncing betas." So, it is important to examine a correlation matrix of the predictors before putting them in the same regression model. You can also examine indices such as variance inflation factor (VIF).

Generalized VIF (GVIF) values are estimated below using the `car` package (R-car, n.d.).

To address multicollinearity, you can drop a redundant predictor or you can also use principal component analysis or factor analysis of the predictors to reduce the predictors down to a smaller number of meaningful predictors. For a meaningful answer in a regression framework that is precise and confident, you need a low level of intercorrelation among predictors, unless you have a very large sample size.

However, multicollinearity does not bias parameter estimates (i.e., multicollinearity does not lead to mean error). Instead, multicollinearity increases the uncertainty (i.e., standard errors) around the parameter estimates (archived at https://perma.cc/DJ7L-TCUK), which makes it more challenging to detect an effect as statistically significant. Some forms of multicollinearity are ignorable (archived at https://perma.cc/2JV5-2QEZ), including when the multicollinearity is among (a) the control variables rather than the variables of interest, (b) the powers (e.g., quadratic term) or products (e.g., interaction term) of other variables, or (c) dummy-coded categories. Ultimately, it is important to examine the question of interest, even if that means inclusion of predictors that are inter-correlated in a regression model (archived at https://perma.cc/DJ7L-TCUK). However, it would not make sense to include two predictors that are perfectly correlated, because they are redundant.

Ways to Improve Prediction Accuracy

On the whole, experts' predictions are inaccurate. Experts' predictions from many different domains tend to be inaccurate, including political scientists (Tetlock, 2017), physicians (Koehler et al., 2002), clinical psychologists (Oskamp, 1965), stock market traders and corporate financial officers (Skala, 2008), seismologists' predictions of earthquakes (Hough, 2016), economists' predictions about the economy (Makridakis et al., 2009), lawyers (Koehler et al., 2002), and business managers (Russo & Schoemaker, 1992). The most common pattern of experts' predictions is that they show overextremity, that is, their predictions have probability judgments that tend to be too extreme, as described in the source discussion. Overextremity of experts' predictions likely reflects over-confidence. The degree of confidence of a person's predictions is often not a good indicator of the accuracy of their predictions (Silver, 2012). Cognitive biases including the anchoring bias (Tversky & Kahneman, 1974), the confirmation bias (Hoch, 1985; Koriat et al., 1980), and base rate neglect (Eddy, 1982; Koehler et al., 2002) could contribute to over-confidence of predictions. Poorly calibrated predictions are especially likely when the base rate is very low (e.g., suicide), as is often the case in clinical psychology, or when the base rate is very high (Koehler et al., 2002).

Nevertheless, there are some domains that have shown greater predictive accuracy, from which we may learn what practices may lead to greater accuracy. For instance, experts have shown stronger predictive accuracy in weather forecasting (Murphy & Winkler, 1984), horse race betting (Johnson & Bruce, 2001), and playing the card game of bridge (Keren, 1987), but see Koehler et al., 2002 for exceptions.

Here are some potential ways to improve the accuracy (and honesty) of predictions and judgments:

• Provide appropriate anchoring of your predictions to the base rate of the phenomenon you are predicting.

To the extent that the base rate of the event you are predicting is low, more extreme evidence should be necessary to consistently and accurately predict that the event will occur. Applying Bayes' theorem and Bayesian approaches can help you appropriately weigh base rate and evidence.

• Include multiple predictors, ideally from different measures and measurement methods.

Include the predictors with the strongest validity based on theory of the causal process and based on criterion-related validity.

• When possible, aggregate multiple perspectives of predictions, especially predictions made independently (from different people/methods/etc.).

The "wisdom of the crowd" is often more accurate than individuals' predictions, including predictions by so-called "experts" (Silver, 2012).

• Instead of aggregating scores from multi-item scales, another option is to "unamalgamate" the predictors, that is, to use each item in the scale as a separate predictor, to maximize predictive power (Trafimow et al., 2025).

• A goal of prediction is to capture as much signal as possible and as little noise (error) as possible (Silver, 2012).

Parsimony (i.e., not having too many predictors) can help reduce the amount of error variance captured by the prediction model. However, to accurately model complex systems like human behavior, the brain, etc., complex models may be necessary. Nevertheless, strong theory of the causal processes and dynamics may be necessary to develop accurate complex models.

• Although incorporating theory can be helpful, provide more weight to empiricism than to theory, until our theories and measures are stronger.

Ideally, we would use theory to design a model that mirrors the causal system, with accurate measures of each process in the system, so we could make accurate predictions. However, as described in the source discussion, our psychological theories of the causal processes that influence outcomes are not yet very strong. Until we have stronger theories that specify the causal process for a given outcome, and until we have accurate measures of those causal processes, actuarial approaches are likely to be most accurate, as discussed in the source discussion. At the same time, keep in mind that measures in psychology, and their resulting data, are often noisy. As a result, theoretically (conceptually) informed empirical approaches may lead to more accuracy than empiricism alone.

• Use an empirically validated and cross-validated statistical algorithm to combine information from the predictors in a formalized way.

Give each predictor appropriate weight in the statistical algorithm, according to its strength of association with the outcome. Use measures with strong reliability and validity for assessing these processes to be used in the algorithm. Cross-validation will help reduce the likelihood that your model is fitting to noise and will maximize the likelihood that the model predicts accurately when applied to new data (i.e., the model's predictions accurately generalize), as described in the source discussion.

• When presenting your predictions, acknowledge what you do not know.

• Express your predictions in terms of probabilistic estimates and present the uncertainty in your predictions with confidence intervals (Silver, 2012).

• Qualify your predictions by identifying and noting counter-examples that would not be well fit by your prediction model, such as extreme cases, edge cases, and "broken leg" (Meehl, 1957) cases.

• Provide clear, consistent, and timely feedback on the outcomes of the predictions to the people making the predictions (Bolger & Önkal-Atay, 2004).

• Be self-critical about your predictions.

Update your judgments based on their accuracy, rather than trying to confirm your beliefs (Atanasov et al., 2020).

• In addition to considering the accuracy of the prediction, consider the quality of the prediction process, especially when random chance is involved to a degree (such as in poker) (Silver, 2012).

• Work to identify and mitigate potential blindspots; be aware of cognitive biases, such as confirmation bias and base rate neglect.

• Evaluate for the possibility of bias in the predictions or in the tests from which the predictions are derived.

Correct for any test bias.

Conclusion

Human behavior is challenging to predict. People commonly make cognitive pseudo-prediction errors, such as the confusion of inverse probabilities. People also tend to ignore base rates when making predictions. When the base rate of a behavior is very low or very high, you can be highly accurate in predicting the behavior by predicting from the base rate. Thus, you cannot judge how accurate your prediction is until you know how accurate your predictions would be by random chance. Moreover, maximizing percent accuracy may not be the ultimate goal because different errors have different costs. Though there are many types of accuracy, there are two broad types: discrimination and calibration—and they are orthogonal. Discrimination accuracy is frequently evaluated with the area under the receiver operating characteristic curve, or with sensitivity and specificity, or standardized regression coefficients. Calibration accuracy is frequently evaluated graphically and with various indices. Sensitivity and specificity depend on the cutoff. Therefore, the optimal cutoff depends on the purposes of the assessment and how much one weights the various costs of the different types of errors: false negatives and false positives. It is important to evaluate both discrimination and calibration when evaluating prediction accuracy.

Suggested Readings

Steyerberg et al., 2010; Meehl & Rosen, 1955; Treat & Viken, 2023; Wiggins, 1973

Questions

Note: Several of the following questions use data from the survivorship of the Titanic accident. The data are publicly available (https://hbiostat.org/data/; archived at https://perma.cc/B4AV-YH4V). The Titanic data file for these exercises is located on the book's page of the Open Science Framework (https://osf.io/3pwza) and in the GitHub repo (https://github.com/isaactpetersen/Principles-Psychological-Assessment/tree/main/Data). Every record in the data set represents a passenger—including the passenger's age, sex, passenger class, number of siblings/spouses aboard (sibsp), number of parents/children aboard (parch) and, whether the passenger survived the accident. I used these variables to create a prediction model (based on a logistic regression model using Leave-10-out cross-validation) for whether the passenger survived the accident. The model's prediction for the passenger's likelihood of survival are in the variable called "prediction".

1. What are the two main types of prediction accuracy?

Define each. How can you quantify each? What is an example of an index that combines both main types of prediction accuracy?

1. Provide the following indexes of overall prediction accuracy for the prediction model in predicting whether a passenger survived the Titanic:

a. Mean error (bias) b. Mean absolute error (MAE) c. Mean squared error (MSE) d. Root mean squared error (RMSE) e. Mean percentage error (MPE) f. Mean absolute percentage error (MAPE) g. Symmetric mean absolute percentage error (sMAPE) h. Mean absolute

scaled error (MASE) i. Root mean squared log error (RMSLE) j. Coefficient of determination (R^2) k. Adjusted R^2 (R^2 adj) l. Predictive R^2

1. Based on the mean error for the prediction model you found in 2a, what does this indicate?

2. Create two receiver operating characteristic (ROC) curves for the prediction model in predicting whether a passenger survived the Titanic: one empirical ROC curve and one smooth ROC curve.

3. What is the area under the ROC curve (AUC) for the prediction model in predicting whether a passenger survived the Titanic?

What does this indicate?

1. Create a calibration plot.

Are the predictions well-calibrated? Provide empirical evidence and support your inferences with interpretation of the calibration plot. If the predictions are miscalibrated, describe the type of miscalibration present.

1. You predict that a passenger survived the Titanic if their predicted probability of survival is.50 or greater.

Create a confusion matrix (2x2 matrix of prediction accuracy) for Titanic survival using this threshold. Make sure to include the marginal cells. Label each cell. Enter the number and proportion in each cell.

1. Using the 2x2 prediction matrix, identify or calculate the following:

a. Selection ratio b. Base rate c. Percent accuracy d. Percent accuracy by chance e. Percent accuracy predicting from the base rate f. Relative improvement over chance (ROIC) g. Relative improvement over predicting from the base rate h. Sensitivity (true positive rate [TPR]) i. Specificity (true negative rate [TNR]) j. False negative rate (FNR) k. False positive rate (FPR) l. Positive predictive value (PPV) m. Negative predictive value (NPV) n. False discovery rate (FDR) o. False omission rate (FOR) p. Diagnostic odds ratio q. Youden's J statistic r. Balanced accuracy s. F 1 score t. Matthews correlation coefficient (MCC) u. Positive likelihood ratio v. Negative likelihood ratio w. d' sensitivity x. A (non-parametric) sensitivity y. beta bias z. c bias aa. b (non-parametric) bias ab. Miscalibration (mean difference between predicted and observed values; based on 10 groups) ac. Information gain

1. In terms of percent accuracy, how much more accurate are the predictions compared to predicting from the base rate?

What would happen to sensitivity and specificity if you raise the selection ratio? What would happen if you lower the selection ratio?

1. For your assessment goals, it is 4 times more important to identify survivors than to identify non-survivors.

Consistent with these assessment goals, you specify the following utility for each of the four outcomes: hits: 1; misses: 0; correct rejections: 0.25; false alarms: 0. What is the utility ratio? What is the overall utility (U overall) of the current cutoff? What cutoff has the highest overall utility?

1. Find the optimal cutoff threshold that optimizes each of the following criteria:

a. Youden's J statistic b. Closest to the top left of the ROC curve c. Percent accuracy d. Percent accuracy by chance e. Relative improvement over chance (ROIC) f. Relative improvement over predicting from the base rate g. Sensitivity (true positive rate [TPR]) h. Specificity (true negative rate [TNR]) i. Positive predictive value (PPV) j. Negative predictive value (NPV) k. Diagnostic odds ratio l. Balanced accuracy m. F 1 score n. Matthews correlation coefficient (MCC) o. Positive likelihood ratio p. Negative likelihood ratio q. d' sensitivity r. A (non-parametric) sensitivity s. beta bias t. c bias u. b (non-parametric) bias v. Miscalibration (mean difference between predicted and observed values; based on 10 groups) w. Overall utility x. Information gain

1. A company develops a test that seeks to determine whether someone has been previously infected with a novel coronavirus (COVID-75) based on the presence of antibodies in their blood.

You take the test and your test result is negative (i.e., the test says that you have not been infected). Your friend takes the test and their test result is positive for coronavirus (i.e., the test says that your friend has been infected). Assume the prevalence of the coronavirus is 5%, the sensitivity of the test is.90, and the specificity of the test is.95. a. What is the probability that you actually had the coronavirus? b. What is the probability that your friend actually had the coronavirus? c. Your friend thinks that, given their positive test, that they have the antibodies (and thus may have immunity). What is the problem with your friend's logic? d. What logical fallacy is your friend demonstrating? e. Why is it challenging to interpret a positive test in this situation?

1. You just took a screening test for a genetic disease.

Your test result is positive (i.e., the tests says that you have the disease). Assume the probability of having the disease is 0.5%, the probability of a positive test is 2%, and the probability of a positive test if you have the disease is 99%. What is the probability that you have the genetic disease?

1. You just took a screening test (Test A) for a virus.

Your test result is positive. Assume the base rate of the virus is.05%. Test A has a sensitivity of.95 and a specificity of.90. a. What is your probability of having the virus after testing positive on Test A? b. After getting the results back from Test A, the physician wants greater confidence regarding whether you have the virus given its low base rate, so the physician orders a second test, Test B. You test positive on Test B. Test B has a sensitivity of.80 and a specificity of.95. Assuming the errors of the Tests A and B are independent, what is your updated probability of having the virus?

Answers

1. The two main types of prediction accuracy are discrimination and calibration.

Discrimination refers to the ability of a prediction model to separate/differentiate data into classes (e.g., survived versus died). Calibration for a prediction model refers to how well the predicted probability of an event (e.g., survival) matches the true probability of an event. You can quantify discrimination with AUC; you can quantify various aspects of discrimination with sensitivity, specificity, positive predictive value, and negative predictive value. You can quantify degree of (poor) calibration with Spiegelhalter's z, though other

metrics also exist (e.g., Brier scores and the Hosmer-Lemeshow goodness-of-fit statistic). R^2 is an overall index of accuracy that combines both discrimination and calibration. 2. a. Mean error (bias): [computed value omitted in this course adaptation] b. Mean absolute error (MAE): [computed value omitted in this course adaptation] c. Mean squared error (MSE): [computed value omitted in this course adaptation] d. Root mean squared error (RMSE): [computed value omitted in this course adaptation] e. Mean percentage error (MPE): undefined, but when dropping undefined values: [computed value omitted in this course adaptation]\% f. Mean absolute percentage error (MAPE): undefined, but when dropping undefined values: [computed value omitted in this course adaptation]\% g. Symmetric mean absolute percentage error (sMAPE): [computed value omitted in this course adaptation]\% h. Mean absolute scaled error (MASE): [computed value omitted in this course adaptation] i. Root mean squared log error (RMSLE): [computed value omitted in this course adaptation] j. Coefficient of determination (R^2): [computed value omitted in this course adaptation] k. Adjusted R^2 (R^2 adj): [computed value omitted in this course adaptation] l. Predictive R^2: [computed value omitted in this course adaptation] 3. The small mean error/bias ([computed value omitted in this course adaptation]) indicates that predictions did not consistently under- or over-estimate the actual values to a considerable degree.

1. Empirical ROC curve:

Smooth ROC curve:

1. The AUC is [computed value omitted in this course adaptation].

AUC indicates the probability that a randomly selected case has a higher test result than a randomly selected control. Thus, the probability is [computed value omitted in this course adaptation]\% that a randomly selected passenger who survived had a higher predicted probability of survival (based on the prediction model) than a randomly selected passenger who died. The AUC of [computed value omitted in this course adaptation] indicates that the prediction model was moderately accurate in terms of discrimination.

1. Calibration plot:

1. In general, the predictions are well-calibrated.

There is not significant miscalibration according to Spiegehalter's z ([computed value omitted in this course adaptation], p = [computed value omitted in this course adaptation]). This is verified graphically in the calibration plot, in which the predicted probabilities fall mostly near the actual/observed probabilities. However, based on the calibration plot, there does appear to be some miscalibration. When the predicted probability of survival was ~60%, the actual probability of survival was lower (~40%). This pattern of miscalibration is known as overprediction, as depicted in Figure 1 of Lindhiem et al., 2020.

1. The 2x2 prediction matrix is below:

Exercise 6: 2x2 Prediction Matrix. TP = true positives; TN = true negatives; FP = false positives; FN = false negatives; BR = base rate; SR = selection ratio.

8. a. Selection ratio: [computed value omitted in this course adaptation] b. Base rate: [computed value omitted in this course adaptation] c. Percent accuracy: [computed value omitted in this course adaptation]\% d. Percent accuracy by chance: [computed value

omitted in this course adaptation]\% e. Percent accuracy predicting from the base rate: [computed value omitted in this course adaptation]\% f. Relative improvement over chance (ROIC): [computed value omitted in this course adaptation] g. Relative improvement over predicting from the base rate: [computed value omitted in this course adaptation] h. Sensitivity (true positive rate [TPR]): [computed value omitted in this course adaptation] i. Specificity (true negative rate [TNR]): [computed value omitted in this course adaptation] j. False negative rate (FNR): [computed value omitted in this course adaptation] k. False positive rate (FPR): [computed value omitted in this course adaptation] l. Positive predictive value (PPV): [computed value omitted in this course adaptation] m. Negative predictive value (NPV): [computed value omitted in this course adaptation] n. False discovery rate (FDR): [computed value omitted in this course adaptation] o. False omission rate (FOR): [computed value omitted in this course adaptation] p. Diagnostic odds ratio: [computed value omitted in this course adaptation] q. Youden's J statistic: [computed value omitted in this course adaptation] r. Balanced accuracy: [computed value omitted in this course adaptation] s. F 1 score: [computed value omitted in this course adaptation] t. Matthews correlation coefficient (MCC): [computed value omitted in this course adaptation] u. Positive likelihood ratio: [computed value omitted in this course adaptation] v. Negative likelihood ratio: [computed value omitted in this course adaptation] w. d' sensitivity: [computed value omitted in this course adaptation] x. A (non-parametric) sensitivity: [computed value omitted in this course adaptation] y. beta bias: [computed value omitted in this course adaptation] z. c bias: [computed value omitted in this course adaptation] aa. b (non-parametric) bias: [computed value omitted in this course adaptation] ab. Miscalibration (mean difference between predicted and observed values): [computed value omitted in this course adaptation] ac. Information gain: [computed value omitted in this course adaptation]

1. Predicting from the base rate would have a percent accuracy of [computed value omitted in this course adaptation]\%.

So, the predictions increase the percent accuracy by [computed value omitted in this course adaptation]\%. If you raise the selection ratio (i.e., predict more people survived) sensitivity will increase whereas specificity will decrease. If you lower the selection ratio, specificity will increase and sensitivity will decrease.

1. The utility ratio is [computed value omitted in this course adaptation].

The overall utility (U overall) of the cutoff is [computed value omitted in this course adaptation]. The cutoff with the highest overall utility is [computed value omitted in this course adaptation].

1. The cutoff that optimizes each of the following criteria:

a. Youden's J statistic: [computed value omitted in this course adaptation] b. Closest to the top left of the ROC curve: [computed value omitted in this course adaptation] c. Percent accuracy: [computed value omitted in this course adaptation] d. Percent accuracy by chance: 1 (i.e., predicting from the base rate—that nobody will survive) e. Relative improvement over chance (ROIC): [computed value omitted in this course adaptation] f. Relative improvement over predicting from the base rate: [computed value omitted in this course adaptation] g. Sensitivity (true positive rate [TPR]): 0 (i.e., predicting that everyone will survive will minimize false negatives) h. Specificity: 1 (i.e., predicting that nobody will survive will minimize false positives) i. Positive predictive value (PPP): [computed value omitted in this

course adaptation] j. Negative predictive value (NPV): [computed value omitted in this course adaptation] k. Diagnostic odds ratio: [computed value omitted in this course adaptation] l. Balanced accuracy: [computed value omitted in this course adaptation] m. F 1 score: [computed value omitted in this course adaptation] n. Matthews correlation coefficient (MCC): [computed value omitted in this course adaptation] o. Positive likelihood ratio: [computed value omitted in this course adaptation] p. Negative likelihood ratio: [computed value omitted in this course adaptation] q. Miscalibration (mean difference between predicted and observed values): [computed value omitted in this course adaptation]–[computed value omitted in this course adaptation] r. d' sensitivity: [computed value omitted in this course adaptation] s. A (non-parametric) sensitivity [computed value omitted in this course adaptation] t. beta bias [computed value omitted in this course adaptation] u. c bias [computed value omitted in this course adaptation] v. b (non-parametric) bias [computed value omitted in this course adaptation] w. Overall utility: [computed value omitted in this course adaptation] x. Information gain: [computed value omitted in this course adaptation]

12. a. Based on negative predictive value (i.e., the probability of no disease given a negative test), the probability that you actually had the coronavirus is less than 1 in 100 ([computed value omitted in this course adaptation]). b. Based on positive predictive value (i.e., the probability of disease given a positive test), the probability that your friend actually had the coronavirus is less than 50% ([computed value omitted in this course adaptation]). c. The problem with your friend's logic is that your friend is assuming they have the antibodies when chances are more likely that they do not. d. Your friend is likely demonstrating the fallacy of base rate neglect. e. A positive test on a screening device is hard to interpret in this situation because of the low base rate. "Even with a very accurate test, the fewer people in a population who have a condition, the more likely it is that an individual's positive result is wrong." For more info, see here: https://www.scientificamerican.com/article/coronavirus-antibody-tests-have-a-mathematical-pitfall/ (archived at https://perma.cc/GL9F-EVPH)

1. According to Bayes' theorem, the probability that you have the disease is [computed value omitted in this course adaptation]\%.

For more info, see here: https://www.scientificamerican.com/article/what-is-bayess-theorem-an/ (archived at https://perma.cc/GM6B-5MEP) 14. a. According to Bayes' theorem, the probability that you have the virus after testing positive on Test A is [computed value omitted in this course adaptation]\%. b. According to Bayes' theorem, the updated probability that you have the virus after testing positive on both Tests A and B is [computed value omitted in this course adaptation]\%.

CHAPTER 17 / CAC.903

Repeated Assessment and Progress Monitoring

Assessment does not end at intake. Children develop, contexts change, and interventions should be evaluated. Repeated assessment tracks progress, detects deterioration, and guides adjustment. It should be brief enough to be feasible and meaningful enough to inform decisions. A progress measure that no one uses is a burden, not a service.

Purposes of monitoring

Monitoring can evaluate symptom change, functioning, alliance, goal attainment, and side effects. It can also detect when a plan is not working. A counselor should choose targets that matter to the child and family. A teen may care about panic attacks and school attendance more than a global score. A parent may care about sleep and aggression. Targets should be specific.

Frequency and timing

Weekly monitoring may suit active treatment. Monthly monitoring may suit stable maintenance. Too frequent assessment can create fatigue. Too infrequent assessment can miss deterioration. The counselor should consider the speed of change and the risk level. High-risk cases may need more frequent contact. The method should fit the setting.

Reliable change

A change score should exceed measurement error to be considered reliable. Some measures provide reliable change indices. The counselor should also consider clinical significance: has the child moved into a functional range? A small improvement may matter if it restores school attendance. A large score change may not matter if impairment remains.

Feedback-informed treatment

Sharing monitoring data with the client can strengthen alliance. A counselor might say, "Your worry score went down, but school avoidance went up. What do you think is happening?" The client becomes a collaborator. Feedback can reveal ruptures or new stressors. The counselor should not use data punitively.

Attrition and missing data

Families may stop completing measures. Missing data may signal disengagement, burden, or improvement. The counselor should ask rather than assume. A simplified measure or different format may help. Digital reminders should respect privacy. The counselor should document why data are missing and how decisions are made without them.

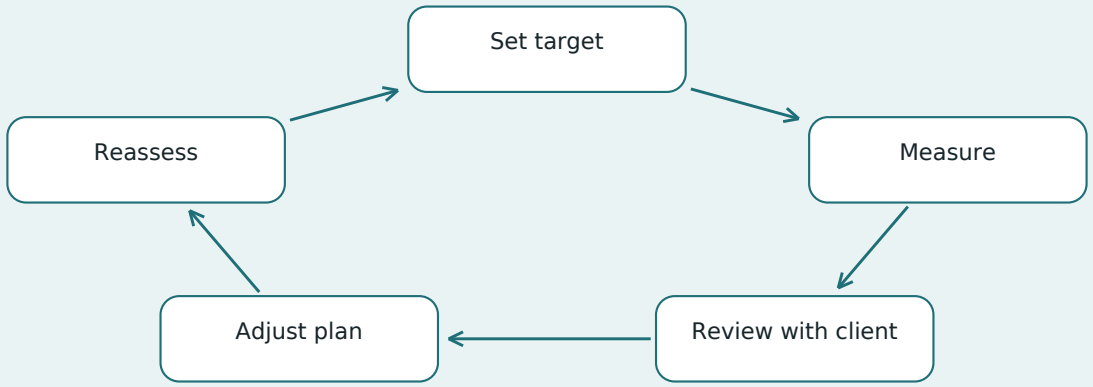
Real-life application

A teen with depression completes a brief weekly measure. Scores improve for three weeks, then rise after a family conflict. The counselor reviews the data with the teen and adjusts the plan to include family communication. The monitoring data prevent a premature conclusion that treatment failed. They guide a timely change.

Reflection

Design a monitoring plan for a case. Include target, measure, frequency, feedback method, and decision rule. Progress monitoring is not paperwork. It is a conversation with evidence.

Progress monitoring loop



Monitoring turns assessment into an ongoing conversation with evidence.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module repeated-assessments.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Overview of Repeated Measurement

Repeated measurement enables examining within-person change in constructs. Relating within-person change in a construct to within-person change in another construct provides a stronger test of causality compared to simple bivariate correlations. Identifying that within-person change in construct X predicts later within-person change in construct Y does not necessarily indicate that X causes Y but it provides stronger evidence consistent with causality than a simple bivariate association between X and Y. As described in the source discussion, there are three primary reasons a correlation between X and Y does not necessarily mean that X causes Y: First, the association could reflect the opposite direction of effect, where Y actually causes X. Second, the association could reflect the influence of a third variable that influences both X and Y (i.e., a confound). Third, the association might be spurious.

If you find that within-person changes in sleep predict later within-person changes in mood, that is a stronger test of causality because (a) it demonstrates temporal precedence; it attenuates the possibility that the association is due to the opposite direction of effect; and (b) it accounts for all time-invariant confounds that differ between people (e.g., birth order) because the association is examined within the individual. That is, the association examines whether the person's mood is better when they get more sleep compared to when that same person gets less sleep. The association could still be due to time-varying confounds, but you can control for specific time-varying confounds if you measure them.

Repeated measurement also enables examination of the developmental timing of effects and sensitive periods. Furthermore, repeated measures designs are important for tests of mediation, which are tests of hypothesized causal mechanisms.

Repeated measurement involves more complex statistical analysis than cross-sectional measurement because multiple observations come from the same person, which would violate assumptions of traditional analyses that the observations (i.e., the residuals) are independent and uncorrelated. Mixed-effects modeling (i.e., mixed modeling, multilevel modeling, hierarchical linear modeling) and other analyses, such as structural equation modeling, can handle the longitudinal dependency or nesting of repeated measures data.

Repeated measurement can encompass many different timescales, such as milliseconds, days, or years.

Comparison of Approaches

Research designs can be compared in terms of their internal validity—the extent to which we can be confident about causal inferences. A cross-sectional association is depicted in the source discussion:, n.d.

Cross-Sectional Association.

Among observational/correlational research designs, cross-sectional designs tend to have the weakest internal validity. That is, if we observe a cross-sectional association between X and Y, we have little confidence that X causes Y. Consider a lagged association, which is a slightly better approach:

Lagged Association.

A lagged association has somewhat better internal validity than a cross-sectional association because we have greater evidence of temporal precedence—that the influence of the predictor precedes the outcome because the predictor was assessed before the outcome and it shows a predictive association. However, part of the association between the predictor with later levels of the outcome could be due to prior levels of the outcome that are stable across time. Thus, consider an even stronger alternative—a lagged association that controls for prior levels of the outcome:

Lagged Association, Controlling for Prior Levels of the Outcome.

A lagged association controlling for prior levels of the outcome has better internal validity than a lagged association that does not control for prior levels of the outcome. When we control for prior levels of the outcome in the prediction, we are essentially predicting (relative) change. Predicting change provides stronger evidence consistent with causality because it uses the individual as their own control and controls for many time-invariant confounds. However, consider an even stronger alternative—lagged associations that control for prior levels of the outcome and that simultaneously test each direction of effect:

Lagged Association, Controlling for Prior Levels of the Outcome, Simultaneously Testing Both Directions Of Effect.

Lagged associations that control for prior levels of the outcome and that simultaneously test each direction of effect provide the strongest internal validity among correlational/observational designs, especially when examining whether within-person changes in the predictor predict later within-person changes in the outcome (and vice versa). Such a design can help better clarify which among the variables is the chicken and the egg—which variable is more likely to be the cause and which is more likely to be the effect. Or, if there are bidirectional, transactional effects, what the magnitude of each direction of effect are. The cross-lagged panel model and dual latent change score model are examples of models that include lagged associations that control for prior levels of the outcome and that simultaneously test each direction of effect. However, consider an even stronger model—lagged associations that control for prior levels of the outcome and for random intercepts and that simultaneously test each direction of effect:

Lagged Association, Controlling for Prior Levels of the Outcome and Random Intercepts, Simultaneously Testing Both Directions Of Effect.

Some researchers have called into question the cross-lagged panel model (Berry & Willoughby, 2017; Hamaker et al., 2015). A stronger alternative may be a cross-lagged model that controls for stable between-person differences (i.e., random intercepts) (Hamaker et al., 2015). This is known as a random intercepts cross-lagged panel model.

Examples of Repeated Measurement

Below are some examples of repeated measurement.

Alcohol Timeline Followback

A clinically relevant example of repeated measurement is the Alcohol Timeline Followback, published by Sobell & Sobell, 2008. The Alcohol Timeline Followback is a method for retrospectively assessing the quantity of alcohol consumption on a daily basis. It provides retrospective estimates of people's drinking using a calendar of the number of drinks consumed each day. The time frame can vary from 30 days to 360 days. The participant is given detailed instructions to enhance the accuracy of recall and reporting of their behavior.

Moreover, a specific definition is provided of what counts as a drink. The measure is intended to provide a detailed record of patterns of alcohol use to guide treatment and assess treatment outcome. The measure provides many summary statistics and norms are available.

The measure has several clinical uses. First, it can provide personalized feedback to the client regarding their drinking behavior compared to the population as part of enhancing their motivation to change, in a motivational interviewing framework. Second, the measure can help identify high- and low-risk periods for a client to help them prevent relapse.

Because the measure provides detailed information with norms, it can be more useful than global estimates of drinking derived from less structured clinical interviews. The detailed instructions help respondents remember more accurately, so the clinician or researcher can examine how people's substance abuse influences other processes like parenting.

Ambulatory Assessment

We discussed ambulatory assessment in the source discussion. Ambulatory assessment includes an array of methods used to study people in their natural environment. There is a proliferation of smartphone apps for ambulatory assessment. And the ambulatory assessments often, because of their temporal resolution, provide repeated measurements.

For instance, Carpenter et al., 2016 depicted the examination, during a drinking episode, of self-reported drinking in relation to physiological indices including skin temperature, heart rate, breathing rate, physical activity, and in relation to self-reported sadness and mood dysregulation. The self-reported indices were assessed on various intervals (e.g., every 15 minutes) and upon various events (e.g., when they take a drink). However, the physiological indices were assessed continuously from moment to moment. These different timescales pose challenges for how to examine them in relation to each other.

Matthews et al., 2016 provided an example of how they developed an ambulatory assessment using a smartphone app to assess the dynamic process of bipolar disorder. Regularity of social rhythms is disrupted in bipolar disorder, so the authors developed an app to help assess social rhythms. The authors used a participatory design in which they involved the patients in the development of the app to ensure it worked well for them. The app tracked the occurrence and timing of daily events, such as getting out of bed, starting your day, having dinner, and going to bed. It also passively detected daily events ("social rhythms") with many sensors on the smartphone using information from the light, accelerometer, and microphone of the smartphone. And the app used a machine learning algorithm to estimate level of physical activity and social interaction.

Test Revisions

An important part of repeated measurements deals with test revisions. Tests are revised over time. Some test revisions are made to be consistent with improvements in the understanding of the construct. Other test revisions are made to make the norms more up-to-date. For instance, the Weschler Adult Intelligence Scale has multiple versions (WAIS-3, WAIS-IV, etc.). There are important considerations when dealing with test revisions.

When you assess people at different times, whether the same people at multiple time points (such as in a longitudinal study) or different people at different times, challenges may result in comparing scores across time. There can be benefits of keeping the same measure across ages or across time for comparability of scores. However, it is not necessarily good to use the same measure across ages or at different times merely for the sake of mathematical comparability of scores. If a measure was revised to improve its validity or if a measure is not valid at particular ages, it may make sense to use different measures across ages or across time. You would not want to use a measure at a given age if it is not valid at that age. Likewise, it does not make sense to use an invalid measure at a later time point if a more valid version becomes available, even if it was used at an earlier time point. That is, even if a measure's scores are mathematically comparable across ages or time, they may not be conceptually comparable across ages or time. If different measures are used across ages or across time, there may be ways to link scores from the different measures to be on the same scale, as described in the source discussion.

One interesting phenomenon of time-related changes in scores is the Flynn effect, in which the population scores on intelligence tests rise around 3 points every decade. That is, intelligence, as measured by tests of cognitive abilities, increases from generation to generation. Test revisions can hide effects like the Flynn effect if we just keep using score transformations instead of raw scores.

To observe a person's (or group's) growth, it is recommended to use raw scores rather than standardized scores that are age-normed, such as T scores, z scores, standard scores, and percentile ranks (Moeller, 2015). Age-normed scores prevent observing growth at the person level or at the group level (e.g, changes in means or variances over time). T scores (M = 50, SD = 10), z scores (M = 0, SD = 1), and standard scores (M = 100, SD = 15) have a fixed mean and standard deviation. Percentile ranks have a fixed range (0–100).

Change and Stability

A key goal of developmental science is to assess change and stability of people and constructs. Stability is often referred to in a general way. For instance, you may read that intelligence and personality show "stability" over time. However, this is not a useful way of describing the stability of people and/or constructs because stability is not one thing. There are multiple types of stability, and it is important to refer to the types of stability considered.

Types of stability include, for instance, stability of level, rank order (individual differences), and structure. Stability of level refers to whether people show the same level on the construct over time. For example, stability of level considers whether people stay the same in their level of neuroticism across childhood to adulthood. Stability of people's level is evaluated with the coefficient of repeatability.

Mean-level stability refers to whether the mean level stays the same across time. For example, mean-level stability considers whether the population average stays the same in their level of neuroticism across childhood to adulthood. Mean-level stability is evaluated by examining mean-level age-related differences; for instance, with a t-test or ANOVA, or by examining whether the construct is associated with age.

Rank-order stability refers to whether individual differences stay the same across time. For instance, rank-order stability considers whether people who have higher neuroticism than their peers as children also tend to have higher neuroticism than their peers as adults. Rank-order stabiliy is examined by examining the association of people's level at T1 with people's level at T2 (e.g., correlation coefficient or regression coefficent).

Stability of structure considers whether the structure of the construct stays the same across ages. For instance, stability of structure considers whether personality is characterized by the same five factors across ages (i.e., openness, conscientiousness, extraversion, agreeableness, and neuroticism). Stability of structure is examined using longitudinal factorial invariance.

People and constructs can show stability in some types and not others. For instance, although intelligence and personality show relative stability in structure and rank-order, they do not show absolute stability in level. Even though individual differences in intelligence and personality show relative stability, people tend to change across time in their absolute level of intelligence and various personality factors across ages. For instance, neuroticism decreases in later adulthood. Examining age-normed scores would preclude identifying mean-level change because age-normed scores have a fixed mean. Moreover, examining a strong correlation of intelligence and personality scores across time merely indicates that individual differences show relative stability; it does not indicate whether individuals show absolute stability in level.

Assessing Change

There are many important considerations for assessing change, that is changes in a person's level on a construct across time. There is a key challenge in assessing change between two time points. If a person's score differs between two time points, how do you know that the differences across time reflect a person's true change in their level on the construct rather than measurement error, such as regression to the mean? Regression to the mean occurs if an extreme observation on a measure at time 1 (T1) more closely approximates the person's mean at later time points, including time 2 (T2). For instance, clients with high levels of symptomatology tend to get better with the mere passage of time, which has been known as the self-righting reflex.

Inferring Change

The inference of change is strengthened to the extent that:

a. the magnitude of the difference between the scores at the two time points is large (i.e., a large effect size) b. the measurement error (unreliability) at both time points is small c. the measure has the same meaning and is on a comparable scale at each time point (i.e., the measure's scores show measurement invariance across time) d. evidence suggests that the differences across time are not likely to be due to potential confounds of change such as practice effects, cohort effects, and time-of-measurement effects.

Measurement error can be reduced by combining multiple measures of a construct into a latent variable, such as in structural equation modeling (SEM) or item response theory (IRT).

To detect change, it is important to use measures that are sensitive to change. For instance, the measures should not show range restriction owing to ceiling effects or floor effects.

At the individual level, it is common to infer that a person's level changed from T1 from T2 if the magnitude of their change is greater than the smallest real difference. However, just because a person (or a group) changed in their level on the construct does not necessarily mean that the change is meaningful. Thus, it is also important to consider the effect size of the change and whether the change is greater than the minimal clinically important difference (MCID).

Potential Confounds of Change

Potential confounds of change include practice effects, cohort effects, and time-of-measurement effects.

Practice Effects

It is important to consider the potential for practice or retest effects (Hertzog & Nesselroade, 2003). Practice/retest effects occur to the extent that a person improves on a task assessed at multiple time points for reasons due to the repeated assessment rather than true changes in the construct. That is, a person is likely to improve on a task if they complete the same task at multiple time points, even if they are not changing in their level on the construct. Practice and retest effects are especially likely to be larger the closer in time the retest is to the prior testing. Improvement in scores across time might not reflect people's true change in the construct; thus, you might not be able to fairly compare people's scores who have completed the task different numbers of times.

In sum, the ages and frequency of assessment should depend on theory.

Cohort Effects

It is also important to consider the potential for cohort effects (Hertzog & Nesselroade, 2003). Cohorts often refer to birth cohorts—people who were born at the same time (e.g., the same year). Cohorts tend to have some similar experiences at similar ages, including major events (e.g., 9/11 terrorist attacks, assassination of John F. Kennedy, bombing of Pearl Harbor) and economic or socio-political climates (e.g., the Great Depression, coronavirus pandemic, Cold War, World War II). These collective differences in the different experiences between different cohorts may lead to apparent age-related differences between cohorts that are not actually due to age differences per se, but rather to differences in the experiences between the cohorts.

Time-of-Measurement Effects

It is also important to consider time-of-measurement effects. For instance, people's general anxiety may be higher, on average, during the coronavirus pandemic. Thus, some apparent age-related differences might be due to the time of measurement (i.e., a pandemic) rather than age.

When and How Frequently to Assess

Key questions arise in terms of when (at what ages or times) and how often to assess the construct. If the measurement intervals are too long, you might miss important change in between. If the measurement intervals are too frequent, differences might reflect only measurement error. The Nyquist theorem provides guidance in terms of how frequently to assess a construct. According to the Nyquist theorem, in order to accurately reproduce a signal, you should sample it at twice the highest frequency you wish to detect. For instance, to detect a 15 Hz wave, you should sample it at least at 30 Hz (Hertz means times per second). Thus, the frequency of assessment should be as frequent as, if not more frequent than, the highest frequency of change you seek to observe.

Difference Scores

As described in the source discussion, difference scores (also called change scores) can have limitations because they can be less reliable than each of the individual measures that compose the difference score. The reliability of individual differences in change scores depends on the extent to which: (a) the measures used to calculate the change scores are reliable, and (b) the variability of true individual differences in change is large (Rogosa & Willett, 1983)—that is, there are true individual differences in change.

Superior Approaches to Difference Scores

There are several options that are often better than traditional difference scores or change scores. Instead of difference scores, use residualized change (e.g., autoregressive cross-lagged models), latent change score models, or growth curve models.

Residualized Change

One alternative to using difference scores is to examine residualized change. Residualized change means regressing the score at timepoint 2 (i.e., the outcome variable) on the score at timepoint 1 (i.e., the predictor variable) and saving the residual. Examining residualized change is equivalent to examining difference scores when the unstandardized regression coefficient of the score at timepoint 1 in predicting the score at timepoint 2 is 1.0—i.e., for every one-unit increase in the score at timepoint 1, there is a one-unit increase in the score at timepoint 2. An example of residualized change is the autoregressive cross-lagged panel model.

Autoregressive Cross-Lagged Models Autoregressive cross-lagged models (aka cross-lagged panel models) can be fit in path analysis or SEM. In a cross-lagged panel model, the predictor at timepoint 1 is allowed to predict the outcome at timepoint 2 while controlling for prior levels of the outcome at timepoint 1. Cross-lagged panel models examine relative change in a variable, that is, change in individual differences in the variable, but not changes in level on the variable. Such a model allows examining whether one variable predicts relative change in another variable. The traditional cross-lagged panel model tests whether individuals with low levels of X relative to others experience subsequent rank-order increases in Y (Orth et al., 2021; Evans & Shaughnessy, 2024). To disaggregate within-person from between-person effects, it is recommended to include the estimation of random intercepts, a so-called random-intercepts cross-lagged panel model (Hamaker et al., 2015). The random-intercepts cross-lagged panel model tests whether, when individuals experience lower-than-their-own usual levels of X, they experience subsequent increases in Y relative to their usual level (Orth et al., 2021; Evans &

Shaughnessy, 2024). As an example of a cross-lagged panel model, I conducted a collaborative study that examined whether language ability predicts later behavior problems controlling for prior levels of behavior problems—that is, whether language ability predicts a person's change in behavior problems relative to the change among other people in the sample (Petersen et al., 2013; Petersen & LeBeau, 2021). Cross-lagged panel models predict relative (rank-order) change, not absolute change in level. For tests of absolute change, models such as change score models or growth curve models are necessary.

Latent Change Score Models Latent change score models can be fit in SEM. Unlike autoregressive cross-lagged models, latent change score models examine absolute change—that is, people's changes in level. With latent change score models, you can examine change in latent variables that reflect the common variance among multiple measures. Latent variables are theoretically error-free (at least of random error), so latent change scores can be perfectly reliable, unlike traditional difference scores, whose reliability is strongly influenced by the measurement unreliability of the specific measures that compose them. Latent change score models are described by Kievit et al., 2018.

Growth Curve Models Growth curve models can be fit in mixed-effects modeling or SEM. Growth curve models require three or more time points to be able to account for measurement error in terms of the shape of a linear trajectory. Growth curve models examine absolute change, i.e., changes in level. Growth curve models include an intercept (starting point) and slope (amount of change), but you can model other nonlinear forms of change as well, such as quadratic, logistic, etc. with more time points. Growth curve models correct for measurement error in the shape of the person's trajectory. In addition, growth curve models can examine risk and protective factors as predictors of the intercepts and slopes. For instance, in collaboration, I examined risk and protective factors that predict the development of externalizing problems (Petersen et al., 2015).

Nomothetic Versus Idiographic

The approaches described above (autoregressive cross-lagged models, latent change score models, and growth curve models) tend to be nomothetic in nature. That is, they are interested in examining group-level phenomena and assume that all people come from the same population and can be described by the same parameters of change. That is, they assume a degree of homogeneity, even though they allow each person to differ in their intercept and slope. This nomothetic (population) approach differs from an idiographic (individual) approach. In clinical work, we are often interested in making inferences or predictions at the level of the individual person, consistent with an idiographic approach. But fully ignoring the population estimates is unwise because they tend to be more accurate than individual-level judgments. Further, individual-level predictions may not generalize to the population. Nomothetic modeling approaches can be adapted to examine change for different subgroups if there are theoretically different populations. Approaches that examine different subgroups include group-based trajectory models, latent class growth models, and growth mixture models (Fontaine & Petersen, 2017; van der Nest et al., 2020).

For instance, Moffit (Moffitt, 1993; Moffitt, 2006; Moffitt, 2006) created the developmental taxonomy of antisocial behavior with the support of these approaches. Moffitt identified one subgroup that showed very low levels of antisocial behavior across the lifespan, which she labeled the "low" group. She identified one subgroup that showed antisocial behavior only in childhood, which she labeled the "childhood-limited" group. She hypothesized that the

antisocial behavior among the childhood-limited group was influenced primarily by genetic factors and the early environment. She identified another subgroup that showed antisocial behavior only in adolescence, which she labeled the "adolescent-limited"group. She hypothesized that the antisocial behavior among the adolescent-limited group was largely socially mediated. She identified another subgroup that showed antisocial behavior across the lifespan, which she labeled the "life-course persistent" group. She hypothesized that the "life-course persistent" group reflected neuropsychological deficits.

Idiographic approaches for personalized modeling of psychopathology are described by Wright & Woods, 2020. An example of an approach that combines an idiographic (person-centered) approach with a nomothetic (population) approach to understand change is the group iterative multiple model estimation (GIMME) model, as described by Beltz et al., 2016 and Wright et al., 2019. The GIMME model is a data-driven approach to identifying person-specific time-series models. The approach first identifies the associations among constructs that replicate across most participants. The group-level associations are then used as a starting point for identifying person-specific associations. Thus, the GIMME model provides an approach for developing personalized time-series models. The choices on which model to use depend on the research question and should be guided by theory.

Structural Equation Modeling

Structural equation modeling (SEM) allows multiple dependent variables. This provides a key advantage in longitudinal designs, such that two constructs can each be represented by a dependent variable that is predicted by the other. If examining changes in two constructs in the context of SEM, you could examine for instance whether one precedes the other in prediction, to better understand which might cause which and to help tease apart the chicken and the egg. For instance, in collaboration, I found that low language ability predicts later behavior problems stronger than behavior problems predict later language ability, suggesting that, if they are causally related, that language ability is more likely to have a causal effect on behavior problems than the reverse (Petersen et al., 2013).

Attrition and Systematic Missingness

Rates of attrition in longitudinal studies can be substantial and missingness is often not completely at random. There is often selective and systematic attrition. For example, attrition often differs as a function of socioeconomic status and other important factors. It is important to use approaches that examine and account for systematic missingness. It can be helpful to use full-information maximum likelihood (FIML) in SEM to make use of all available data. Multiple imputation is another approach that can be helpful for dealing with missingness. With multiple imputation, you can use predictors that might account for missingness, such as socioeconomic status and demographic factors. You can help account for practice effects with planned missingness designs in which participants are randomly assigned to receive or not receive a given measurement occasion.

Measurement Invariance

Just like measurement or factorial invariance can be tested and established across genders, races, ethnicities, cultures, etc. to ensure measurement equivalence, it can also be tested across ages or times. The goal is to establish longitudinal measurement invariance, i.e., measurement invariance across ages, so that you know that you are assessing the same construct over time in a comparable way so that you know that differences across time reflect true change. To examine longitudinal measurement invariance, you can use SEM or IRT. In SEM, you could examine whether measures have the same intercepts, factor loadings, and residuals across ages. In IRT, you could examine whether items show differential item functioning, versus whether they show the same difficulty and discrimination across ages.

But there are times when we would expect a construct would, theoretically, fail to show longitudinal factorial invariance, as in the case of heterotypic continuity, described in the source discussion.

Types of Research Designs

Schaie, 1965, Baltes, 1968, and Schaie & Baltes, 1975 proposed various types of research designs, in terms of the various combinations of age, period (i.e., time of measurement), and cohort. Age refers to a person's chronological age at the time of measurement. Period refers to the time of measurement (e.g., April 12, 2014 at 7:53 AM). Cohort refers to a person's time (e.g., year) of birth (e.g., 2003). If you know two of these (e.g., age and period), you can determine the third (e.g., cohort). That is, age = period - cohort.

Here are the types of research designs based on combinations of age, period, and cohort:

- cross-sectional
- cross-sectional sequences
- time-lag
- longitudinal
- longitudinal sequences
- time-sequential
- cross-sequential
- cohort-sequential

These research designs are depicted by age and cohort in the source discussion and by time of measurement and cohort in the source discussion. Although the depictions of cohort-sequential and cross-sequential designs differ from the depictions by Schaie, 2005 (i.e., they are reversed), they are consistent with contemporary definitions of these designs (Little, 2013; Masche & van Dulmen, 2004; Whitbourne, 2019).

Research Designs by Age and Cohort. Values in the cells are the times (years) of measurement. Dashed line indicates different participants were assessed at each time of measurement.

Research Designs by Time Of Measurement and Cohort. Values in the cells are ages of the participants. Dashed line indicates different participants were assessed at each time of measurement.

Cross-Sectional Design

A cross-sectional design involves multiple participants, often spanning different ages, assessed at one time of measurement. A cross-sectional design is a single factor design, where the researcher is interested in comparing effects of one factor: age. However, in a cross-sectional design, cohort (i.e., birth year) is confounded with age differences. Thus, observed age-related differences could reflect cohort differences rather than true age differences. Additionally, sampling differences at each age could yield non-comparable age groups.

Cross-sectional designs are thus limited in their ability to describe developmental processes including change and stability over time. Any age-related differences are confounded with cohort differences and are strongly influenced by between age-group sampling variability. Cross-sectional studies are useful for preliminary data on validity of measures for the age groups, and for examining whether the age differences are in the expected direction. Cross-sectional studies are less costly and time-consuming than longitudinal studies. Therefore, they are widely used and provide a useful starting point.

However, findings from cross-sectional studies can differ from findings examining the same people over time, which violates the convergence assumption. In cross-sectional studies, the convergence assumption is the assumption that cross-sectional age differences and longitudinal age changes converge onto a common trajectory. However, it has been shown, for example, that cross-sectional studies over-estimate age-related cognitive declines compared to following the same people over time in a longitudinal study (Ackerman, 2013).

Cross-Sectional Sequences Design

A cross-sectional sequences design involves successive cross-sectional studies of different participants at different times of measurement. In a cross-sectional sequences design, cohort and time-of-measurement are confounded with age differences. Thus, observed age-related differences could reflect cohort differences or time-of-measurement differences rather than true age differences. This poses challenges for using cross-sectional sequences to estimate age-related changes due to time-related effects such as the Flynn effect.

Time-Lag Design

A time-lag design involves participants from different cohorts assessed at the same age. The time-lag design is a single factor design: the researcher specifies one factor: cohort. However, time of measurement is confounded with cohort. Thus, any cohort differences could reflect different times of measurement.

Longitudinal Design

A longitudinal design involves following the same participants over time. The single-cohort longitudinal design is a single factor design: the researcher specifies one factor: age. However, in a single-cohort longitudinal design, cohort and time of measurement are confounded with age. Thus, any observed changes with age could be due to time-of-measurement effects or cohort effects (instead of people's true change).

Longitudinal Sequences Design

A longitudinal sequences design involves following multiple cohorts across time. There are three types of longitudinal sequences designs: time-sequential, cross-sequential, and cohort-sequential designs. The three types of longitudinal sequences are depicted in the source discussion as a function of which two factors are specified by the researcher.

Types of Longitudinal Sequences as a Function of Which Two Factors are Specified by the Researcher.

Time-Sequential Design

A time-sequential design is depicted in the source discussion. In a time-sequential design, multiple age groups are assessed at multiple times (Whitbourne, 2019). It is a repeated cross-sectional design, with some participants followed longitudinally. Additional age groups are added to a time-lag design, with some of the participants assessed at more than time point. In other words, the age range is kept the same and is repeatedly assessed (with only some participants being repeatedly assessed). The two factors a researcher specifies in a time-sequential design are age and time of measurement. A time-sequential design can be helpful for identifying age differences while controlling for time-of-measurement differences, or for identifying time-of-measurement differences while controlling for age differences. However, cohort effects are confounded with the interaction of age and time of measurement (Whitbourne, 2019). Thus, observed differences as a function of age or time of measurement could reflect cohort effects.

Time-Sequential Research Design.

Cross-Sequential Design

A cross-sequential design is depicted in the source discussion. In a cross-sequential design, multiple cohorts are assessed at multiple times (Whitbourne, 2019). It is a cross of a cross-sectional and longitudinal design. It starts as a cross-sectional study with participants from multiple cohorts, and then all participants are followed longitudinally (typically across the same duration). It is also called an accelerated longitudinal design. The two factors a researcher specifies in a cross-sequential design are time of measurement and cohort. A cross-sequential design can be helpful for identifying cohort differences while controlling for time-of-measurement differences, or for identifying time-of-measurement differences while controlling for cohort differences. However, age differences are confounded with the interaction between cohort and time of measurement. Thus, observed differences as a function of cohort or time of measurement could reflect age effects.

Cross-Sequential Research Design.

Cohort-Sequential Design

A cohort-sequential design is depicted in the source discussion. In a cohort-sequential design, multiple cohorts are assessed at multiple ages (Whitbourne, 2019). It starts multiple cohorts at the same age and then follows them longitudinally (typically across the same duration). It is like starting a longitudinal study at the same age over and over again. The two factors a researcher specifies in a cohort-sequential design are age and cohort. A cohort-sequential design can be helpful for identifying age differences while controlling for cohort differences, or for identifying cohort differences while controlling for age differences. However, time-of-measurement effects are confounded with the interaction of age and cohort (Whitbourne, 2019). Thus, observed differences as a function of age or cohort could reflect time-of-measurement effects.

Cohort-Sequential Research Design.

Using Sequential Designs To Make Developmental Inferences

Which research design you select should depend on which change function you are most interested in: age, cohort, or time of measurement. However, age is not the only metric for development; time after a transition (e.g., school entry, quitting smoking) or an exposure (e.g., puberty onset) can also be an important time metric. There are many other potential metrics of time for studying development.

To have greater confidence that age-related differences reflect true change (development) rather than effects of cohort or time of measurement, we would conduct all three sequential designs. However, that is not often practical. Confidence that observed age-related differences reflect developmental change increases when there is converging evidence across multiple research designs. To the extent that the age-related effects in the time-sequential and cohort-sequential designs are stronger than the cohort- and time-of-measurement effects in the cross-sequential design, we have confidence that the observed age-related differences reflect development (Whitbourne, 2019). In addition, there may be ways in a given study of accounting for period or cohort effects. For instance, in an accelerated longitudinal study (cross-sequential design), it would be valuable to evaluate the extent to which the differing cohorts are comparable—i.e., that they show similar scores at the same ages. Moreover, there is an entire set of analytic approaches, known as age-period-cohort analysis (Yang & Land, 2013), whose goal is to disentangle the effects of age, period, and cohort. Rohrer, 2025 provides additional suggestions for considering age, period, and cohort effects.

Stability Versus Continuity

Continuity addresses the trajectory of development within individuals. By contrast, stability refers to the maintenance of between-individual or group characteristics (Petersen, 2024; Schulenberg & Zarrett, 2006; Schulenberg et al., 2014). When describing stability or continuity, it is important to specify which aspects of stability (e.g., of mean, variance, or rank order) and continuity (e.g., of causes, courses, forms, functions, or environments) one is referring to. For example, the source discussion depicts the distinction between mean-level stability and rank-order stability; the source discussion depicts the distinction between continuity in course and rank-order stability.

Mean-Level Stability Versus Rank-Order Stability. C1, C2, and C3 are three individual children assessed at two timepoints: T1 and T2. The gray arrow depicts the mean (average) trajectory. (Figure reprinted from Petersen, 2024, Figure 1, Petersen, I. T. (2024). Reexamining developmental continuity and discontinuity in the 21st century: Better aligning behaviors, functions, and mechanisms. Developmental Psychology, 60(11), 1992–2007. https://doi.org/10.1037/dev0001657 Copyright (c) American Psychological Association. Used with permission.)

Continuity in Course Versus Rank-Order Stability. C1, C2, and C3 are three individual children assessed at two timepoints: T1 and T2. Continuity in course is sometimes called "stability in level." (Figure reprinted from Petersen, 2024, Figure 2, Petersen, I. T. (2024). Reexamining developmental continuity and discontinuity in the 21st century: Better aligning behaviors, functions, and mechanisms. Developmental Psychology, 60(11), 1992–2007. https://doi.org/10.1037/dev0001657 Copyright (c) American Psychological Association. Used with permission.)

The various types of continuity, including homotypic continuity, heterotypic continuity, and phenotypic continuity, in addition to discontinuity are depicted in Figures (the source discussion) and (the source discussion).

The Latin square depicts Kagan's -Kagan, 1969 typology of continuity as a function of behavior form and function. However, it is also important to consider continuity versus discontinuity of the mechanisms underlying a given behavior–function pair. (Figure reprinted from Petersen, 2024, Figure 3, Petersen, I. T. (2024). Reexamining developmental continuity and discontinuity in the 21st century: Better aligning behaviors, functions, and mechanisms. Developmental Psychology, 60(11), 1992–2007. https://doi.org/10.1037/dev0001657 Copyright (c) American Psychological Association. Used with permission.)

Three Types of Continuity in Addition to Discontinuity in the Form of a 2 (Behavioral Manifestation, Underlying Processes) x 2 (Same Versus Different Across Time) Latin Square. 'T1,' 'T2,' and 'T3' reflect time points 1, 2, and 3, respectively. The illustrations above the lines are buildings, representing the surface structure (i.e., behavioral manifestation). The illustrations below the lines depict the underlying processes supporting the buildings at each time point. The squares on the buildings are windows. The black windows represent content facets that are active across all time points (i.e., age-common content). The windows that contain X's represent content facets that are active at some but not all time points (i.e., age-unique content). The white windows represent content facets that are inactive, and therefore are not part of the construct at that time point. The increasing size of the buildings at later time points reflects growth with development. The top row of the Latin square involves the same underlying processes across time, whereas the bottom row involves different underlying proccesses across time. The left column of the Latin square involves the same behavioral manifestation across time, whereas the right column involves a different behavioral manifestation across time. Homotypic continuity (top left) describes the same behaviorial manifestation with the same underlying process (i.e., construct) across development. Heterotypic continuity (top right) describes the same underlying process with a different behaviorial manifestation across development. Phenotypic continuity (or functional discontinuity) (bottom left) describes the same behavior with different underlying processes across development. Discontinuity (bottom right) describes different behavioral

manifestations with different underlying processes across development. Thus, in both homotypic continuity and heterotypic continuity, the active content facets reflect the same construct or underlying process across time, whereas in phenotypic continuity and discontinuity, the active content facets do not reflect the same construct across time. (Figure reprinted from Petersen et al., 2020, Figure 1, p. 2. Petersen, I. T., Choe, D. E., & LeBeau, B. (2020). Studying a moving target in development: The challenge and opportunity of heterotypic continuity. Developmental Review, 58, 100935. https://doi.org/10.1016/j.dr.2020.100935 Copyright (c) Elsevier. Used with permission.)

Heterotypic Continuity

Heterotypic continuity is the persistence of an underlying construct with behavioral manifestations that change across development (Caspi & Shiner, 2006; Cicchetti & Rogosch, 2002; Petersen et al., 2020; Petersen, 2024). Patterson, 1993 described antisocial behavior like a chimera. A chimera is a mythical creature with a body of a goat that, with development, grows the head of a lion and then a tail of a snake. Patterson, 1993 conceptualized antisocial behavior across development as having an underlying essence of oppositionality that is maintained while adding more mature features with age, including the ability to inflict serious damage with aggression and violence. Other analogies of heterotypic continuity include the change from water to ice or steam, and the metamorphosis of a caterpillar to a butterfly—the same construct shows different manifestations across development.

How to Identify Heterotypic Continuity

Heterotypic continuity can be identified or quantified as changes in the factor structure (i.e., content) of a construct across time. The content of a measure include the facets assessed by a given measure, such as individual behaviors, questionnaire items, or sub-dimensions of a broader construct. The content of the measure can be compared to content of a construct to evaluate the content validity of the measure.

Heterotypic continuity is present to the extent that the content of a construct show cross-time changes in: people's rank-order stability, the content's level on the construct, and how strongly the content reflect the construct. Cross-time changes in people's rank-order stability indicate changes in the degree of stability of individual differences in the construct across time, and are examined with correlation of the content across time.

The content's level on the construct is similar to the inverse of frequency, prevalence, or base rate of the content. The content's level on the construct is examined with item difficulty (severity) in IRT or with intercepts in SEM. For example, the item "sets fires" has a higher item severity than the item "argues" because setting fires is less frequent than arguing.

How strongly the content reflect the construct, conceptually is the degree of a content's construct validity, and is examined with item discrimination in IRT or with factor loadings in SEM. For example, the item "hits others" has a higher item discrimination than "eats ice cream" for the construct of externalizing problems because hitting others is more strongly related to the construct (compared to eating ice cream).

Externalizing problems are an example of a construct that changes in behavioral manifestation with development (Chen & Jaffee, 2015; Miller et al., 2009; Moffitt, 1993; Patterson, 1993; Petersen et al., 2015; Petersen & LeBeau, 2022; Wakschlag et al., 2010).

The content of externalizing problems show changes in the magnitude of rank-order stability, changes in the content's level on the construct, and changes in how strongly the construct reflect the construct. For example, inattention shows increases in the magnitude of rank-order stability with age (Arnett et al., 2012). Drinking alcohol becomes more frequent in adolescence compared to childhood, so its item severity decreases. Threatening other people is more strongly associated with externalizing problems in adolescence than in early childhood (Lubke et al., 2018). Moreover, externalizing content, including disobedience and biting, likely change in meaning with developoment. In childhood, disobedience could be an indicator of externalizing problems. In adulthood, disobedience could serve prosocial functions, including protesting against societally unjust actions, and thus shows weaker construct validity for externalizing problems. In general, in early childhood, externalizing problems often manifest in overt forms, such as physical aggression and temper tantrums. Later in development, externalizing problems tends to manifest in covert forms, such as indirect aggression, relational aggression, and substance use (Miller et al., 2009).

In addition to externalizing problems, other constructs also likely show heterotypic continuity across development. For instance, internalizing problems may show heterotypic continuity (Weems, 2008; Weiss & Garber, 2003) such that anxiety precedes depression (Garber & Weersing, 2010). In addition, somatic complaints such as headaches, stomachaches, and heart pounding are more strongly associated with and more common in those with internalizing problems in childhood than adulthood (Petersen et al., 2018). Other constructs that show heterotypic continuity include inhibitory control (Petersen et al., 2021; Petersen et al., 2016), substance use (Schulenberg & Maslowsky, 2009), sleep states (Blumberg, 2013), and temperament (Putnam et al., 2008). In addition, constructs that are typically assessed with different measures across ages also likely show heterotypic continuity, including language ability (Petscher et al., 2018), nonverbal ability (McArdle & Grimm, 2011), working memory (McArdle et al., 2009), and academic skills (Tonnquist-Uhlen et al., 1995). In sum, many constructs likely show heterotypic continuity.

However, it is important rule out potential alternatives to heterotypic continuity, as depicted in the source discussion.

Heterotypic Continuity Versus Alternative Developmental Processes. Arrows reflect causal influence. Dotted lines reflect persistence (i.e., continuity) of the same behavior or construct across time. "T1" and "T2" reflect time points 1 and 2, respectively. (Figure reprinted from Petersen, 2024, Figure 6, Petersen, I. T. (2024). Reexamining developmental continuity and discontinuity in the 21st century: Better aligning behaviors, functions, and mechanisms. Developmental Psychology, 60(11), 1992–2007. https://doi.org/10.1037/dev0001657 Copyright (c) American Psychological Association. Used with permission.)

The Problem

Heterotypic continuity poses challenges for longitudinal assessment. According to the maxim, "what we know depends on how we know it". If a construct changes in manifestation across time, and the measures do not align with the changing manifestation, the scores will not validly assess individual's growth. Failing to account for heterotypic continuity leads to misidentified trajectories of people's growth (Chen & Jaffee, 2015; Petersen et al., 2026) and to trajectories that are less able to detect people's change (Petersen et al., 2018; Petersen et al., 2021). Thus, it is important that our measurement and statistical schemes account for heterotypic continuity when a construct changes in manifestation across time. These measurement and statistical schemes are discussed below.

Ways to Assess a Construct Across Development

There are three ways of assessing a construct across development:

1. All possible content

2. Only the common content

3. Only the construct-valid content

Assume content A is construct-valid at T1, content B is content-valid at T1 and T2, and content C is content-valid at T2. All possible content would assess: ABC at T1, ABC at T2. Only the common content would assess: B at T1, B at T2. Only the construct-valid content would assess: AB at T1, BC at T2. A depiction of assessing only the construct-valid content is in the source discussion.

Using Only the Construct-Valid Content at Each Age. Content set A corresponds to content that is construct-valid at only T1. Content set B corresponds to content that is construct-valid at both T1 and T2. Content set C corresponds to content that is construct-valid at only T2. The common content (content set B) is highlighted in gray. The three approaches to assessing a construct over time are as follows: (1) using all possible content across all ages: ABC at T1 and T2, (2) using only common content across all ages: B at T1 and T2, or (3) using only construct-valid content at each age: AB at T1 and BC at T2. (Figure reprinted from Petersen et al., 2021, Figure 1, p. e3. Petersen, I. T., LeBeau, B., & Choe, D.E. (2021). Creating a developmental scale to account for heterotypic continuity in development: A simulation study. Child Development, 92(1), e1–e19. https://doi.org/10.1111/cdev.13433 Copyright (c) John Wiley and Sons. Used with permission.)

All Possible Content

There are advantages and disadvantages of each. Advantages of using all possible content include that it is comprehensive and allows examining change in each content facet. However, it also has key disadvantages. First, it is inefficient and assesses lots of items. Second, it could assess developmentally inappropriate content. And in the case of heterotypic continuity, it assesses content that lack construct validity, i.e., there are item intrusions.

Only the Common Content

Advantages of using only the common content include that it is efficient and may exclude developmentally inappropriate content. Moreover, it permits examining consistent content across ages. However, it also has key disadvantages. First, it results in fewer items and a loss of information. In particular, it may involve a systematic loss of content that reflects very low or very high levels on the construct. Third, and due to the loss of information, it may be less sensitive to developmental change. And in the case of heterotypic continuity, it assesses content that lack content validity because all important facets of the construct are not assessed—i.e., there are item gaps.

Only the Construct-Valid Content

There are disadvantages to using only the construct-valid content. First, it is more time-intensive than assessing only the common content. Second, it uses a different measure across time, which can pose challenges to ensuring scores across time are comparable. Nevertheless, the approach has important advantages. First, it is more efficient than using all possible content. Second, it retains content validity and construct validity. And in the case of heterotypic continuity, it is recommended. Developmental theory requires that we account for changes in the construct's expression!

Many measures are used outside the age range for which they were designed and validated to use. Many measures for children and adolescents are little more than downward extensions of those developed for use with adults (Hunsley & Mash, 2007). Use of the same measure across ages does not ensure measurement equivalence. The problem is that even if the same measure is used across ages, the measure may lack construct validity at some ages. Thus heterotypic continuity may require different measures at different ages to retain construct validity. But then there is the challenge of how to assess change when using different measures across time.

Accounting for Heterotypic Continuity

To account for heterotypic continuity, it is recommended to use different measures across time. Specifically, it is recommended to use (only) the construct-valid content at each age when the content is valid for assessing the construct. To ensure scores are comparable across time, despite the differing measures, it is important to take steps to ensure statistical and theoretical equivalence of scores across time.

Ensuring Statistical Equivalence

Historically, there have been various attempts to handle different measures across time to ensure statistical equivalence of scores—i.e., that measures are on the same mathematical metric.

Age-norming Age-norming includes standardized scores (e.g., T scores, z scores, standard scores) and percentile ranks. However, as discussed in the source discussion, standardized scores prevent observing individuals' or groups' absolute change over time. Moreover, despite putting scores on the same mathematical metric, they do not ensure scores are on the same conceptual metric. Thus, age-norming is not recommended for putting different measures onto the same score to account for heterotypic continuity and examine individuals' growth.

Average or Proportion Scores Average and proportion scores account for the different numbers of items in each measure. However, average and proportion scores assume that measures do not differ in difficulty or discrimination. Thus, average and proportion scores are not recommended to account for heterotypic continuity and examine individuals' growth.

Developmental Scaling A third approach to ensure statistical equivalence of measures is developmental scaling, which is also called vertical scaling (see the source discussion). In developmental scaling, measures that differ in difficulty and discrimination are placed on the same scale using the common content. The common content set the scale, but all content (including both age-common content and age-differing content) estimate each person's score on that scale. There are several approaches to developmental scaling, including Thurstone scaling, factor analysis, and IRT.

Illustrative Example of a Vertical Scaling Design That Uses Common Content to Link the Different Measures at Adjacent Ages to be on the Same Scale. For example, content set B is administered at both kindergarten and 1st grade, and is the common content used to link scores at 1st grade to the same scale as kindergarten scores. Content set A is the unique content at kindergarten; content set C is the unique content at 1st grade (but it is common content with content set C at 2nd grade). The unique content represents age-specific manifestations of the construct. The procedure of linking measures that differ in difficulty to be on the same scale is called vertical scaling (as opposed to horizontal scaling) because the measures, especially in educational settings, tend to increase in difficulty with age (relative to a given level of ability; as depicted above with the upward-trend). Thus, vertical scaling is particularly useful for linking different measures across ages to be on the same scale. Horizontal scaling, by contrast, links different measures that have the same difficulty (commonly at the same age). To better align with the breadth of constructs in developmental psychology, we use the term 'developmental scaling' instead of vertical scaling to refer to putting measures across development on the same scale. (Figure reprinted from Petersen et al., 2020, Figure 3, p. 9. Petersen, I. T., Choe, D. E., & LeBeau, B. (2020). Studying a moving target in development: The challenge and opportunity of heterotypic continuity. Developmental Review, 58, 100935. https://doi.org/10.1016/j.dr.2020.100935 Copyright (c) Elsevier. Used with permission.)

Thurstone scaling aligns the percentile scores of two measures based on z scores on the common content. It is an observed score approach. An example of Thurstone scaling is provided by Petersen et al., 2018.

Factor analysis allows estimation of a latent variable using different content over time. An example of a factor analysis approach to developmental scaling is provided by Wang et al., 2013. IRT finds scaling parameters that put people's scores from the different measures on the same metric, and it links the measures' scales based on the difficulty and discrimination of the common content. Examples of an IRT approach to developmental scaling are provided by Petersen et al., 2018, Petersen et al., 2021, Petersen & LeBeau, 2022, Petersen & LeBeau, 2021, and Petersen et al., 2026.

The IRT approach to developmental scaling is depicted in the source discussion.

Effect of Linking the Latent Externalizing Problems Scores, theta, Across Ages, Using Mother-Reported Externalizing Problems at Ages 4 and 5 as an Example. The left panel

illustrates the test characteristic curves representing the model-implied proportion out of total possible scores across the latent externalizing problems score at age 4 and 5, before the linking process. The right panel illustrates the test characteristic curves after the linking process. The shading between the age 4 and age 5 test characteristic curves represents differences between the two test characteristic curves in terms of discrimination and/or severity, where larger differences reflect scores that are less comparable. Linking minimizes differences between the discrimination and severity of the common items. Discrimination is depicted by the steepness of the slope at the inflection point of the test characteristic curve. Severity is represented by the value on the x-axis at the inflection point of the test characteristic curve. The left panel indicates that the externalizing problem items showed higher severity at age 5 than at age 4. The right panel shows considerably smaller differences between the two test characteristic curves, which provides empirical evidence that the linking successfully placed the latent externalizing problem scores across age on a more comparable scale (i.e., more similar discrimination and severity of the common items). (Figure reprinted from Petersen & LeBeau, 2021, Figure 1, p. 7. Petersen, I. T., & LeBeau, B. (2021). Language ability in the development of externalizing behavior problems in childhood. Journal of Educational Psychology, 113(1), 68–85. https://doi.org/10.1037/edu0000461 Copyright (c) American Psychological Association. Used with permission.)

Ensuring Theoretical Equivalence

A number of pieces of evidence provide support for the theoretical equivalence of measures. The goal is for the measures to have construct validity invariance—the measure should assess the same construct across time. The content that are assessed at each age should be based on theory. The assessed content should reflect the same construct at relevant ages—i.e., the content should show construct validity at each age they are assessed. Second, the content should adequately sample all aspects of construct—i.e., the content should show content validity. Second, assuming the construct is thought to be relatively stable, the measures should show test–retest reliability, at least in the short term. In addition, the measures should show convergent and discriminant validity. Furthermore, the measures assessed should show a similar factor structure across time, based on factor analysis. In addition, the measures should have high internal consistency reliability.

Are There Measures that Can Span the Lifespan?

There are measures that attempt to span the lifespan, including the NIH Toolbox (Weintraub et al., 2013) and the Minnesota Executive Function Scale (Carlson & Zelazo, 2014). However, I would argue that a single (non-adaptive) measure would not be able to validly assess the same construct across development if the construct shows heterotypic continuity. The content must change with changes in the construct's expression in order to remain valid for the construct given its changing behavioral manifestation.

Adaptive testing may help to achieve this goal. However, just because something is on the same mathematical metric does not mean it is on the same conceptual metric. The same challenge can be relevant for many different types of data, including questionnaire items, reaction time, event-related potentials (ERPs), electrocardiography (ECG/EKG), and functional magnetic resonance imaging (fMRI).

Why not Just Study Development in a Piecewise Way?

If heterotypic continuity poses such challenges, why not just study development in a piecewise way? Although there can be benefits to examining development in a piecewise way, relying on a fixed set of content restricts our ability to see growth over a longer time, and to understand key developmental transitions. The pathways to outcomes can be just as important as the outcomes themselves. Consistent with the principle of multifinality, the same genetic and environmental influences can lead to different outcomes for different people. Conversely, consistent with the principle of equifinality, different genetic and environmental influences can lead to the same outcome for different people.

Longitudinal studies allow understanding antecedent-consequent effects, better estimation of change than difference scores derived from two time points, and separating age-related from cohort effects.

Why not Just Study Individual Behaviors?

If higher-order constructs show heterotypic continuity, why not just study individual behaviors? Individual behaviors can also show changes in meaning. Moreover, there is utility in examining constructs rather than individual behaviors. Theoretical constructs have better psychometric properties, including better reliability than individual behaviors due to the principle of aggregation (Rushton et al., 1983). Moreover, constructs are theoretically and empirically derived. You can always reduce to a lower-level unit, and this reductionism can result in a loss of utility.

Summary

In summary, it is important to pay attention to whether constructs change in expression across development. Use different measures across time to align with changes in the manifestation of the construct. Use developmental scaling approaches to link different measures on the same scale. Use theoretical considerations to ensure the measures reflect the same construct across development. In doing so, we will be better able to study development across the lifespan.

Conclusion

Repeated measurement is a design feature that can provide stronger tests of causality compared to concurrent associations. By repeatedly assessing multiple constructs, we can examine lagged associations that control for prior levels of the outcome and that simultaneously test each direction of effect. There are many important considerations for assessing change. The inference of change is strengthened to the extent that (a) the magnitude of the difference between the scores at the two time points is large, (b) the measurement error at both time points is small, (c) the measure has the same meaning and is on a comparable scale at each time point, (d) the differences across time are not likely to be due to potential confounds of change such as practice effects, cohort effects, and time-of-measurement effects. Difference scores, though widely used, can have limitations because they can be less reliable than each of the individual measures that compose the difference score. Instead of using traditional difference scores to represent change, consider other approaches such as autoregressive cross-lagged models, latent change score models, or growth curve models.

There are a variety of types of research designs based on combinations of age, period, and cohort: cross-sectional, cross-sectional sequences, time-lag, longitudinal, and longitudinal sequences, such as time-sequential, cross-sequential, and cohort-sequential designs. Which research design you select should depend on which change function you are most interested in: age, cohort, or time of measurement. However, any single research design has either age, cohort, or time of measurement confounded, so the strongest inferences come from inferences drawn from more than one research design.

When considering people's change, it is important to consider changes in the behavioral manifestation of constructs, known as heterotypic continuity. Heterotypic continuity can be identified as changes in the factor structure (i.e., content) of a construct across time. Use different measures across time to align with changes in the manifestation of the construct. Use developmental scaling to link different measures on the same scale. Use theoretical considerations to ensure the measures reflect the same construct across development.

Suggested Readings

Petersen et al., 2020; Hertzog & Nesselroade, 2003

CHAPTER 18 / CAC.903

Generalizability, Technology, and Ambulatory Assessment

Assessment is moving beyond the office. Phones, wearables, and online platforms allow data collection in daily life. These methods can capture context and change, but they also raise questions about reliability, validity, privacy, and equity. Generalizability theory helps counselors think about how scores depend on occasions, raters, settings, and items.

Generalizability theory

Generalizability theory asks how much of a score reflects the person versus the measurement conditions. A child's mood rating may vary by time of day, informant, and item. A generalizability study estimates sources of variation. A decision study asks how many observations are needed for a dependable score. The practical lesson is that one observation may not represent a pattern. Multiple samples improve confidence.

Computerized and adaptive testing

Computerized assessment can standardize administration, score quickly, and adapt item difficulty. Adaptive testing selects items based on previous responses, reducing length. These methods can improve efficiency. They also require validation. A child may respond differently to a screen than to a person. Accessibility matters: visual, motor, and language needs must be considered.

Ambulatory assessment

Ambulatory assessment collects data in daily life, such as mood prompts, sleep logs, or activity tracking. It can reveal patterns that retrospective interviews miss. A teen may report panic only in social settings. A child's sleep may vary on school nights. The counselor should keep prompts brief and relevant. Data should be reviewed collaboratively, not used to surveil.

Privacy and equity

Digital data are sensitive. Location, contacts, and mood patterns can expose a young person. The counselor should use secure platforms, obtain consent, and limit collection. Families may lack devices or data plans. A digital-only approach may exclude some clients. Alternatives should be available. The counselor should not assume that technology is neutral.

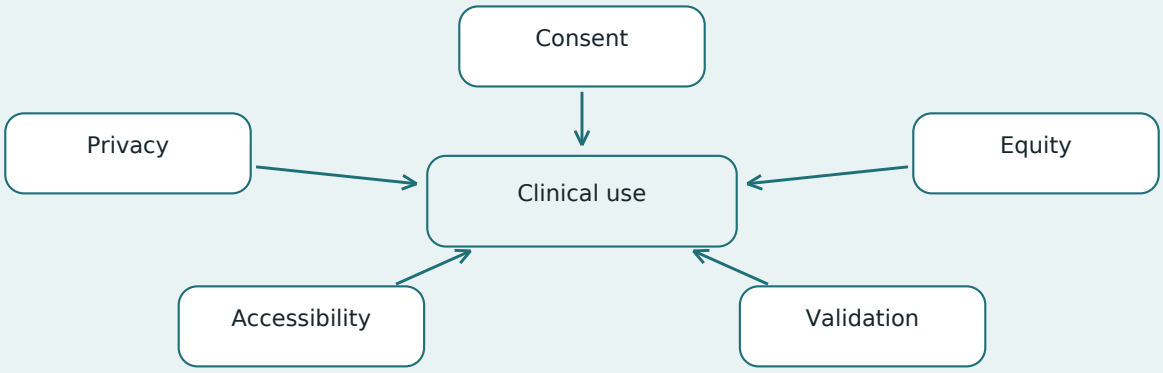
Real-life application

A teen with panic uses a phone log to record situations, thoughts, and body sensations. The data show panic peaks during unstructured lunch periods. The counselor designs exposure and coping practice for that context. The log is reviewed weekly and then faded. Technology serves the formulation.

Reflection

Choose a digital assessment idea. List two benefits and two risks. Write a consent statement for a teen. Technology should expand understanding, not replace relationship.

Digital assessment safeguards



Technology should expand understanding while protecting the young person.

Attributed Foundational Reading

Attribution and critical-reading note

Adapted from Principles of Psychological Assessment: With Applied Examples in R by Isaac T. Petersen, University of Iowa Libraries (2026, version 1.0.9), CC BY 4.0. Selection, organization, and PDF layout modified for this course; R code and original figures omitted; new lectures and course diagrams are separate additions.

Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module generalizability-theory.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Generalizability Theory

Up to this point, we have discussed reliability from the perspective of classical test theory (CTT). However, as we discussed, CTT makes several assumptions that are unrealistic. For example, CTT assumes that all error is random. In CTT, there is an assumption that there exists a true score that is an accurate measure of the trait under a specific set of conditions. There are other measurement theories that conceptualize reliability differently than the way that CTT conceptualizes reliability. One such measurement theory is generalizability theory (Brennan, 1992), also known as G-theory and domain sampling theory. G-theory is also discussed in the chapters on reliability (the source discussion, the source discussion), validity (the source discussion, the source discussion) and structural equation modeling (the source discussion, the source discussion).

Overview of Generalizability Theory (G-Theory)

G-theory is an alternative measurement theory to CTT that does not treat all measurement differences across time, rater, or situation as "error" but rather as a phenomenon of interest (Wiggins, 1973). G-theory is a measurement theory that is used to examine the extent to which scores are consistent across a specific set of conditions. In G-theory, the true score is conceived of as a person's universe score—the mean of all observations for a person over all conditions in the universe—this allows us to estimate and recognize the magnitude of multiple influences on test performance. These multiple influences on test performance are called facets.

The Universe of Generalizability

Instead of conceiving of all variability in a person's scores as error, G-theory argues that we should describe the details of the particular test situation (universe) that lead to a specific test score. The universe is described in terms of its facets:

• settings

• observers (e.g., amount of training they had)

• instruments (e.g., number of items in test)

• occasions (time points)

• attributes (i.e., what we are assessing; the purpose of test administration)

Measures with strong reliability show a high ratio of variance as a function of the person relative to the variance as a function of other facets or factors. To the extent that variance in scores is attributable to different settings, observers, instruments, occasions, attributes, or other facets, the reliability of the measure is weakened.

Universe Score

A person's universe score is the average of a person's scores across all conditions in the universe. According to G-theory, given the exact same conditions of all the facets in the universe, the exact same test score should be obtained. This is the universe score, which is analogous to the true score in CTT. G-theory assumes that the true score does not vary across levels of a facet (e.g., across occasions, or across raters, or across items).

G-Theory Perspective on Reliability

G-theory asserts that the reliability of a test does not reside within the test itself; a test's reliability depends on the circumstances under which it is developed, administered, and interpreted. A person's test scores vary from testing to testing because of (many) variables in the testing situation. By assessing a person in multiple facets of the universe, this allows us to estimate and recognize the magnitude of multiple sources of measurement error. Such measurement error includes:

• day-to-day variation in performance (stability of the construct, test–retest reliability)

• variance in the item sampling (coefficient of internal consistency)

• variance due to both day-to-day and item sampling (coefficient of equivalence from parallel-forms reliability, or convergent validity, as discussed in the source discussion)

In G-theory, all sources of measurement error (facets) are considered simultaneously—something CTT cannot achieve (Shavelson et al., 1989). This occurs through specifying many different variance facets in the estimation of the true score, rather than just one source of error variance as in CTT. This specification allows us to take into consideration variance due to occasion effects, item effects, and occasion × item effects (i.e., main effects of the facets in addition to their interaction), as in the source discussion and the source discussion.

Venn Diagram of Variance Components from a Generalizability Theory Model with Person, Item, and Occasion Facets. e refers to the residual. Adapted from https://wmmurrah.github.io/AdvancedMeasurementTheoryNotebook/generalizability.html (archived at https://perma.cc/2EXS-R9MN).

A score's usefulness largely depends on the extent to which it allows us to generalize accurately to behavior in a wider set of situations—i.e., a universe of generalization. The G-Theory equivalent of the CTT reliability coefficient of a measure is the generalizability coefficient or dependability coefficient.

G-Theory Perspective on Validity

G-theory can simultaneously consider multiple aspects of reliability and validity in the same model. For instance, internal consistency reliability, test–retest reliability, inter-rater reliability, parallel-forms reliability, and convergent validity (Campbell & Fiske, 1959) can all be incorporated into a G-theory model.

For example, a G-theory model could assess each participant across the following facets:

• time: e.g., T1 and T2 (test–retest reliability)

• items: e.g., questions within the same instrument (internal consistency reliability) and questions across different instruments (parallel-forms reliability)

• rater: e.g., self-report and other-report (inter-rater reliability)

• method: e.g., questionnaire and observation (convergent validity)

Using such a G-theory model, we can determine the extent to which scores on a measure generalize to other conditions, measures, etc. Measures with strong convergent validity show a high ratio of variance as a function of the person relative to the variance as a function of measurement method. To the extent that variance in scores is attributable to different measurement methods, convergent validity is weakened.

An example data structure that could leverage G-theory to partition the variance in scores as a function of different facets (person, time, item, rater, method) and their interactions is in the source discussion.

Person	Time	Item	Rater	Method	Score	
1	1	"hits others"	1	questionnaire	10	
1	1	"hits others"	1	observation	15	
1	1	"hits others"	2	questionnaire	8	
1	1	"hits others"	2	observation	13	

Person	Time	Item	Rater	Method	Score	
1	1	"argues"	1	questionnaire	4	
1	1	"argues"	1	observation	2	
1	1	"argues"	2	questionnaire	5	
1	1	"argues"	2	observation	7	
1	2	"hits others"	1	questionnaire	8	
1	2	"hits others"	1	observation	10	
1	2	"hits others"	2	questionnaire	6	
1	2	"hits others"	2	observation	7	
1	2	"argues"	1	questionnaire	2	
1	2	"argues"	1	observation	2	
1	2	"argues"	2	questionnaire	4	
1	2	"argues"	2	observation	6	
2	1	"hits others"	1	questionnaire	5	
...	...	...	...	...	...	: Example Data Structure for Generalizability Theory With the Following Facets: Person, Time, Item, Rater, Method.

In sum, G-theory can be a useful way of estimating the degree of reliability and validity of a measure's scores in the same model.

Generalizability Study

In G-theory, the goal is to conduct a generalizability study (G study) and a decision study (D study). A generalizability study examines the extent of variance in the scores that is attributable to various facets. The researcher must specify and define the universe (set of conditions) to which they would like to generalize their observations and in which they would like to study the reliability of the measure. For instance, it might involve randomly sampling from within that universe in terms of people, items, observers, conditions, timepoints, measurement methods, etc.

Decision Study

After conducting a generalizability study, one can then use the estimates of the extent of variance in scores that are attributable to various facets (estimated from the generalizability study) to conduct a decision study (D study). A decision study examines how generalizable scores from a particular test are if the test is administered in different situations. In G-theory, reliability is estimated with the generalizability coefficient and the dependability coefficient.

Analysis Approach

Traditionally, a generalizability theory approach would test the generalizability study and decision study using a factorial analysis of variance (Brennan, 1992), as exemplified in the source discussion, (as opposed to simple ANOVA in CTT). However, ANOVA is limiting—it works best with balanced designs, such as with the same sample size in each condition/facet; but in most real-world applications, data are not equally balanced in each condition. So, it is better to fit G-theory models in a mixed model framework, as exemplified in the source discussion.

Practical Challenges

G-theory is strong theoretically, but it has not been widely implemented. G-theory can be challenging because the researcher must specify, define, and assess the universe to which they would like to generalize their observations and to understand the reliability of the measure. Moreover, G-theory has had limited impact because of its omission of a latent variable framework (Dumenci, 2024) (such as in factor analysis, structural equation modeling, and item response theory), which is particularly useful in psychology because the most widely studied constructs in psychology tend to be latent and not directly observable. Nevertheless, there are extensions of G-theory to latent variables in structural equation modeling, as described in the source discussion.

Generate Data

For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Add Missing Data

Adding missing data to dataframes helps make examples more realistic to real-life data and helps you get in the habit of programming to account for missing data.

Combine Data into Dataframe

Below are examples implementing G-theory. The `pio_cross_dat` data file for these examples comes from the `gtheory` package (R-gtheory, n.d.). The examples are adapted from Huebner & Lucht, 2019.

Universe Score for Each Person

Universe scores for each person are generated using the following syntax and are presented in the source discussion.

Generalizability (G) Study

Generalizability studies can be conducted in an ANOVA or mixed model framework. Below, we fit a generalizability study model in each framework. In these models, the item, person, and their interaction appear to be the three facets that account for the most variance in scores. Thus, when designing future studies, it would be important to assess and evaluate these facets.

Mixed Model Framework

The mixed model framework for estimating generalizability is described by Jiang, 2018.

Decision (D) Study

The decision (D) study and generalizability (G) study, from which the generalizability and dependability coefficients can be estimated, were analyzed using the gtheory package (R-gtheory, n.d.).

Generalizability Coefficient

The generalizability coefficient is analogous to the reliability coefficient in CTT. It divides the estimated person variance component (the universe score variance) by the estimated observed-score variance (with some adjustment for the number of observations). In other words, variance in a reliable measure should mostly be due to person variance rather than variance as a function of items, occasion, raters, methods, or other factors. The generalizability coefficient uses relative error variance, so it characterizes the similarity in the relative standing of individuals, similar to CTT-based estimates of relative reliability, such as Cronbach's alpha. Thus, the generalizability coefficient is an index of relative reliability. The generalizability coefficient ranges from 0–1, and higher scores reflect better reliability.

Dependability Coefficient

The dependability coefficient is similar to the generalizability coefficient; however, it uses absolute error variance rather than relative error variance in the estimation. The dependability coefficient characterizes the absolute magnitude of differences across scores, not (just) the relative standing of individuals. Thus, the dependability coefficient is an index of absolute reliability. The dependability coefficient ranges from 0–1, and higher scores reflect better reliability.

Conclusion

G-theory provides an important reminder that reliability is not one thing. You cannot just say that a test "is reliable"; it is important to specify the facets across which the reliability and validity of a measure have been established (e.g., times, raters, items, groups, instruments). Generalizability theory can be a useful way of estimating multiple aspects of reliability and validity of measures in the same model.

Suggested Readings

Brennan, 2001

Questions

1. You want to see how generalizable the Antisocial Behavior subscale of the BPI is.

You conduct a generalizability study ("G study") to see how generalizable the scores are across participants (N = 3), two measurement occasions, and three raters. What is the universe score (estimate of true score) for each participant? What percent of variance is attributable to: (a) individual differences among participants, (b) different raters, (c) the measurement occasions, and (d) the interactive effect of raters and measurement occasions? Using a decision study ("D study"), what are the generalizability and dependability coefficients? Is the measure reliable across the universe (range of factors) we examined it in? Interpret the results from the G study and D study. The data from your study are in the source discussion below:

Answers

1. The universe score is [computed value omitted in this course adaptation], [computed value omitted in this course adaptation], and [computed value omitted in this course adaptation] for participants 1, 2, and 3, respectively.

The percent of variance attributable to each of those factors is: (a) participant: [computed value omitted in this course adaptation]\%, (b) rater: [computed value omitted in this course adaptation]\%, (c) the measurement occasion: [computed value omitted in this course adaptation]\%, and (d) the interactive effect of rater and measurement occasion: [computed value omitted in this course adaptation]\%. The generalizability coefficient is [computed value omitted in this course adaptation], and the dependability coefficient is [computed value omitted in this course adaptation]. The measure is fairly reliable across the universe examined, both in terms of relative differences (generalizability coefficient) and absolute differences (dependability coefficient). However, when using the measure in future work, it would be important to assess many participants (due to the strong presence of individual differences) across multiple occasions (due to the strong effect of occasion) and multiple raters (due to the moderated effect of rater as a function of participant). It will be important to account for the considerable sources of variance in scores on the measure in future work including: participant, occasion, participant × rater, and participant × occasion. By accounting for these sources of variance, we can get a purer estimate of each participant's true score and the population's true score.

Attribution and critical-reading note

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Read assessment claims critically. A score or method is a tool for inquiry, not a verdict about a child or family. Where source claims are dated, culture-bound, or illustrated with adult/general examples, the new lecture supplies child-adolescent cautions and contemporary framing.

Source module computerized-adaptive-testing.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Computer-Administered/Online Assessment

Computer-administered and online assessments have the potential to be both desirable and dangerous (Buchanan, 2002). The potential is to have a mental health screening instrument online that a person fills out, is automatically scored, and automatic feedback is provided with suggestions to follow a particular course of action. This would save the clinician time and could lead to a more productive therapy session. As an example, I have clients complete a brief measure of behavioral issues on a tablet when they arrive and are waiting in the waiting room. This is an example of measurement-based care, in which the participant's treatment response is assessed throughout treatment to help determine whether treatment is working or not, and whether to try something different. In addition, emerging techniques in machine learning may be useful for scoring of high-dimensional data in assessment in clinical practice (Galatzer-Levy & Onnela, 2023).

Advantages

There are many advantages of computer-administered or online assessment:

• Computer-administered and online assessment requires less time on the part of the clinician and administrators than traditional assessment methods.

They do not need the clinician to administer and score the assessment, and scores can be entered directly into the client's chart. This can be helpful for going back and finding data without losing it in a file cabinet; it is also helpful for compiling data to examine in research.

• Computer-administered and online assessments tend to be less costly than traditional assessment methods.

• Computer-administered and online assessments can increase access to services because some people can complete the assessment who otherwise would not be able to complete it.

For example, some people may be restricted by geographic or financial circumstances. You can provide the questionnaire in multiple languages and modalities, including written, spoken, and pictorial.

• People may disclose more information to a computer than to a person.

The degree of one's perceived anonymity matters. Perceived anonymity may be especially important for information on sensitive yet important issues, such as suicidality, homicidality, abuse, neglect, criminal behavior, etc.

• Computer-administered and online assessment allow precise timing of stimulus presentation, which is important for many domains including tests of memory.

• Computerized and online assessment has the potential to be more structured and the potential to do a more comprehensive and accurate assessment than an in-person interview, which tends to be less structured and more biased by clinical judgment (Garb, 2007).

• Computerized and online assessment tends to provide more comprehensive information than is usually collected in practice.

Structured approaches tend to ask more questions. Computers can increase the likelihood that the client answers all questions, because it can prevent clients from skipping questions. Computers can apply "branching logic" to adapt which questions are asked based on the

participant's responses (i.e., adaptive testing) to more efficiently gain greater coverage. Clinicians tend not to ask questions about important issues. Clinicians fail to examine disorders other than the hypothesized ones, which is known as diagnostic overshadowing due to the confirmation bias of looking for evidence-confirming information rather than disconfirming evidence. In general, computers give more diagnoses than clinicians. Clinicians often miss co-occurring disorders. Doing a more comprehensive assessment normalizes that people can and do experience the behavioral issues that they are asked about, and can increase self-disclosure. Greater structure leads to higher inter-rater reliability, less bias, and greater validity.

• Computerized and online assessment lends itself to continuous monitoring during treatment for measurement-based care.

Measurement-based care has been shown to be related to better treatment outcomes. Computerized or online assessment may be important in risk assessment that might otherwise go unrecognized. Treatment utility of computerized assessment has been demonstrated for suicide and substance use.

Validity Challenges

There are a number of challenges to the validity of computer-administered and online assessment:

• The impersonality of computers; however, this does not seem to be a strong threat.

• Lack of control in the testing situation.

• The possibility that extraneous factors can influence responses.

• Language and cultural differences; it is difficult to ask follow-up questions to assess understanding unless a clinician is present.

• Inability to observe the client's body language or nonverbals.

• Computers are not a great medium to assess constructs that could be affected by computer-related anxiety.

• Some constructs show different ratings online versus in-person.

• Computer-administered and online assessment may require different norms and cutoffs than paper-and-pencil assessment.

Different norms could be necessary, for example, due to increased self-disclosure and higher ratings of negative affect in computer-administered assessment compared to in-person assessment.

• Careless and fraudulent responding can be common when recruiting an online sample to complete a survey (Chandler et al., 2020).

Additional steps may need to be taken to ensure high-quality responses such as CAPTCHA (or reCAPTCHA), internet protocol (IP) verification of location, and attention or validity checks.

In general, online assessments of personality seem generally comparable to their paper-and-pencil versions, but their psychometric properties are not identical. In general, computer-administered and online assessments yield the same factor structure as in-person

assessments, but particular items may not function in the same way across computer versus in-person assessments.

Ethical Challenges

There are important ethical challenges of using computer-administered and online assessments:

• The internet and app stores provide a medium for the dissemination and proliferation of quackery.

Businesses sell access to assessment information despite no evidence of the reliability or validity of the measure.

• When safety issues arise, it can be a liability and ethical problem to have collected data regarding risk/safety (e.g., suicidality/homicidality/abuse/neglect) without promptly having reviewed the responses and taken the necessary steps to address.

• When giving potentially distressing or threatening feedback, it is important to provide opportunities for follow-up sessions or counseling.

When using online assessment in research, you can state that release of test scores is not possible, and that participants should consult a professional if they are worried or would like a professional assessment.

• An additional ethical challenge of computer-administered and online assessment deals with test security.

The validity of many tests is based on the assumption that the client is not knowledgeable about the test materials. Detailed information on many protected tests is available on the internet (Ruiz et al., 2002), and it could be used by malingerers to fake health or illness.

• Data security is another potential challenge of computer-administered and online assessments, including how to protect HIPAA compliance in the clinical context.

Best Practices

Below are best practices for computer-administered and online assessments:

• Only use measures with established reliability and validity, unless you are studying them, in which case you should evaluate their psychometrics.

• Have a trained clinician review the results of the assessment.

• Combine computer-administered assessment with clinical judgment.

Clients can be expected to make errors completing the assessment, so ask necessary follow-up questions after getting results from computer-administered assessments. Asking follow-up questions can help avoid false positive errors in diagnosis. Check whether a client mistakenly reported something; or whether they under-/over-reported psychopathology.

• Provide opportunities for follow-up sessions or counseling if distressing or threatening feedback is provided.

• It may be important to correct for a person's level of experience with computers when using computerized tasks for clinical decision-making (Lee Meeuw Kjoe et al., 2021).

Adaptive Testing

Adaptive testing involves having the respondent complete only those items that are needed to answer an assessment question. It involves changing which items are administered based on responses to previous items, and administering only a subset of the possible items. Adaptive testing is commonly used in intelligence testing, achievement testing, and aptitude testing. Adaptive testing is not yet commonly used for personality or psychopathology assessment. Most approaches to assessment of personality and psychopathology rely on conventional testing approaches using classical test theory. But many structured clinical interviews, use decision rules to skip a diagnosis, module, or section if a decision rule is not met. Adaptive testing has also been used with observational assessments (Granziol et al., 2022).

A goal of adaptive testing is to get the most accurate estimate of a person's level on a construct with the fewest items possible. Ideally, you would get similar results between adaptive testing and conventional testing that uses all items, when adaptive testing is done well.

There are multiple approaches to adaptive testing. Broadly, one class of approaches uses manual administration, whereas another class of approaches uses computerized adaptive testing (CAT) based on item response theory (IRT).

Manual Administration of Adaptive Testing (Adaptive Testing without IRT)

Manual administration of adaptive testing involves moving people up and down in the item difficulty according to their responses. Approaches to manual administration of adaptive testing include using skip rules, basal and ceiling criteria, and the countdown method.

Skip Rules

Many structured clinical interviews, such as the Structured Clinical Interview for DSM Disorders (SCID) and Mini-International Neuropsychiatric Interview (MINI), use decision rules to skip a diagnosis, module, or section if a decision rule is not met. The skip rules allow the interviews to be more efficient and therefore clinically practical.

Basal and Ceiling Approach

One approach to manual administration of adaptive testing involves establishing a respondent's basal and ceiling level. The basal level is the difficulty level at which the examinee answers almost all items correctly. The respondent's ceiling level is the difficulty level at which the examinee answers almost all items incorrectly. The basal and ceiling criteria set the rules for establishing the basal and ceiling level for each respondent and for when to terminate testing.

The goal of adaptive testing is to administer the fewest items necessary to get an accurate estimate of the person's ability, to save time and prevent the respondent from becoming bored from too-easy items or frustrated from too-difficult items. The basal and ceiling approach starts testing at the recommended starting point for a person's age or ability level. The examiner scores as items are administered. If the person gets too many items wrong in the beginning, the examiner moves to an earlier starting point, i.e., provides easier items, until the respondent gets most items correct, which establishes their basal level. Then, after establishing the respondent's basal level, the examiner proceeds forward to progressively

more difficult items until they reach their ceiling level, i.e., they get too many wrong in some set, which establishes their ceiling level.

Countdown Approach

The countdown approach of adaptive testing is a variant of the variable termination criterion approach to adaptive testing. It classifies the respondent into one of two groups—elevated or not elevated—based on whether they exceed the cutoff criterion on a given scale. The cutoff criterion is usually the raw score on the scale that corresponds to a clinical elevation.

Two countdown approaches include (1) the classification method and (2) the full scores on elevated scales (FSES) method. The countdown approach to adaptive testing is described by Forbey & Ben-Porath, 2007.

Classification Method Using the classification method to the countdown approach of adaptive testing, the examiner stops administering items once elevation is either ruled in or ruled out. The classification method only tells you whether a client produced an elevated score on the scale. It does not tell you their actual score on that scale.

Full Scores on Elevated Scales Method Using the full scores on elevated scales method to the countdown approach of adaptive testing, the examiner stops administering items only if elevation is ruled out. The approach generates a score on that scale only for people who produced an elevated score.

Summary

When doing manual administration with clinical scales, you see the greatest item savings when ordering items from least to most frequently endorsed (i.e., from most to least difficulty) so you can rule people out faster. Ordering items in this way results in a 20–30% item savings on the Minnesota Multiphasic Personality Inventory (MMPI), and corresponding time savings. Comparative studies using the MMPI tend to show comparable results (i.e., similar validity) between the countdown adaptive method and the conventional testing method. That is, the reduction in items does not impair the validity of the adaptive scales. However, you can get even greater item savings when using an IRT approach to adaptive testing. Using IRT, you can order the administration of items differently for each participant to administer the next item that will provide the most information (i.e., precision) for a respondent's ability given their responses on all previous items.

CAT with IRT

Computerized adaptive testing (CAT) based on IRT is widely used in educational testing, including the Graduate Record Examination (GRE) and the Graduate Management Admission Test (GMAT), but it is not widespread in mental health measurement for two reasons: (1) it works best with large item banks; large item banks are generally unavailable for mental health constructs; and (2) mental health constructs are multidimensional and CAT using IRT has primarily been restricted to unidimensional constructs, such as math achievement. However, that is changing; there are now multidimensional IRT approaches that allow simultaneously estimating people's scores on multiple dimensions, as described in the source discussion. For example, the higher-order construct of externalizing problems includes sub-dimensions including aggression and rule-breaking.

A CAT is designed to locate a person's level on the construct (theta) with as few items as possible. CAT has potential for greater item savings than the countdown approach, but it has

more assumptions. IRT can be used to create CATs using information gleaned from the IRT model's item parameters, including discrimination and difficulty (or severity). The item's discrimination indicates how strongly the item is related to the construct. For a strong measure, you want highly discriminating items that are strongly related to the construct. In terms of item difficulty/severity, you want items that span the full distribution of difficulty/severity so they are non-redundant.

An IRT-based CAT starts with a highly discriminating item at the 50th percentile of the distribution (severity) (or at whatever level is the best estimate of the person's ability/severity before the CAT is administered). Then, based on the participant's response, it generates a provisional estimate of the person's level on the construct and administers the item that will provide the most information. For example, if the respondent gets the first item correct, the CAT administers a highly discriminating item that might be at the 75th percentile of the distribution (severity). And so on, based on the participants' responses.

As described in the source discussion, item information is how much measurement precision for the construct is provided by a particular item. In other words, item information indicates how much the item reduces the standard error of measurement; i.e., how much the item reduces uncertainty of our estimate of the respondent's construct level. An IRT-based CAT continues to generate provisional estimates of the person's construct level and updates it based on new responses. The CAT tries to find the construct level where the respondent keeps getting items right and wrong (or endorsing versus not endorsing) 50% of the time. This process continues until the uncertainty in the person's estimated construct level is smaller than a pre-defined threshold—that is, when we are fairly confident (to some threshold) about a person's construct level.

CAT is a promising approach that saves time because it tailors which items are administered to which person and in which order based on their construct level to get the most reliable estimate in the shortest time possible. It administers items that are appropriate to the person's construct level, and it does not administer items that are way too easy or way too hard for the person, which saves time, boredom, and frustration. It shortens the assessment because not all items are administered, and it typically results in around 50% item savings or more. It also allows for increased measurement precision, because you are drilling down deeper (i.e., asking more items) near the person's construct level. You can control the measurement precision of the CAT because you can specify the threshold of measurement precision (i.e., standard error of measurement) at which testing is terminated. For example, you could have the CAT terminate when the standard error of measurement of a person's construct level becomes less than 0.2.

However, CATs assume that if you get a more difficult item correct, you would have gotten easier items correct, which might not be true in all contexts (especially for constructs that are not unidimensional). For example, just because a person does not endorse low-severity symptoms does not necessarily mean that they will not endorse higher-severity symptoms, especially when assessing a multidimensional construct.

Simpler CATs are built using unidimensional IRT. But many aspects of psychopathology are not unidimensional, and would violate assumptions of unidimensional IRT. For example, the externalizing spectrum includes sub-dimensions including aggression, disinhibition, and substance use. You can use multidimensional IRT to build CATs, though this is more complicated. For instance, you could estimate a bifactor model in which each item (e.g.,

"hits others") loads onto the general latent factor (e.g., externalizing problems) and its specific sub-dimension (e.g., aggression versus rule-breaking). Computerized adaptive testing of mental health disorders is reviewed by Gibbons et al., 2016.

Well-designed CATs show equivalent reliability and validity to their full-scale counterparts. By contrast, many short forms are not as accurate as their full-scale counterparts. Part of the reason that CATs tend to do better than short forms is that the CATs are adaptive; they determine which items to administer based on the participant's responses to prior items, unlike short forms. For guidelines on developing and evaluating short forms, see Smith et al., 2000.

Example of Unidimensional CAT

The computerized adaptive test (CAT) was fit using the mirtCAT package (R-mirtCAT, n.d.). The example of a CAT with unidimensional data is adapted from mirtCAT documentation: https://philchalmers.github.io/mirtCAT/html/unidim-exampleGUI.html (archived at https://perma.cc/3ZW4-DAUR)

Define Population IRT Parameters

For reproducibility, we set the seed below. Using the same seed will yield the same answer every time. There is nothing special about this particular seed.

Fit IRT Model

We can generate a mirt model object from known IRT parameters using the mirtCAT::generate.mirt_object function of the mirtCAT package (Chalmers, 2016).

Model Plots

A test characteristic curve of the measure is in the source discussion.

The test information and standard error of measurement as a function of the person's construct level (theta; theta) is in the source discussion. The test appears to measure theta (theta) with SE less than.4 within a range of −1.5 to 2.

Item Plots

Item characteristic curves are in the source discussion.

Item information curves are in the source discussion.

• Item 30 would be a good starting item; it has a difficulty closest to the 50th percentile (b = [computed value omitted in this course adaptation]) and a high discrimination (a = [computed value omitted in this course adaptation]).

• Item 70 would be a good starting item; it has a difficulty closest to the 50th percentile (b = [computed value omitted in this course adaptation]) and a high discrimination (a = [computed value omitted in this course adaptation]).

• Item 86 is an easy item (b = [computed value omitted in this course adaptation]).

• Item 16 is a difficult item (b = [computed value omitted in this course adaptation]).

• Item 53 has a low discrimination (a = [computed value omitted in this course adaptation]).

Item characteristic curves and information curves for these items are in Figures (the source discussion)–(the source discussion).

Run Computerized Adaptive Test (CAT)

Set the minimum standard error of measurement for the latent trait (theta; theta) that must be reached before stopping the CAT. You can lengthen the CAT by lowering the minimum standard error of measurement, or you can shorten the CAT by raising the minimum standard error of measurement.

Run the CAT using the mirtCAT::mirtCAT function of the mirtCAT package (Chalmers, 2016) and stop once the standard error of measurement for the latent trait (theta; theta) becomes [computed value omitted in this course adaptation] or lower.

CAT Standard Errors

Standard errors of measurement of a person's estimated construct level (theta; theta) as a function of the item (presented in the order that items were administered as part of the CAT) are in the source discussion.

Initially, the respondent got the first items correct, which raised the estimate of their construct level. However, as the respondent got more items incorrect, the CAT converged upon the estimate that the person has a construct level around theta = [computed value omitted in this course adaptation], which means that the person scored slightly above average (i.e., [computed value omitted in this course adaptation] standard deviations above the mean).

CAT 95% Confidence Interval

95% confidence intervals of a person's estimated construct level (theta; theta) as a function of the item (presented in the order that items were administered as part of the CAT) are in Figure the source discussion.

Creating a Computerized Adaptive Test From an Item Response Theory Model

You can create a computerized adaptive test from any item response theory model.

Create Items

For instance, below we create a matrix with the questions and response options from the graded response model that we fit in the source discussion in Chapter the source discussion on item response theory.

Run Computerized Adaptive Test

Then you can create and run the computerized adaptive model using the mirtCAT::mirtCAT function of the mirtCAT package (Chalmers, 2016):

Conclusion

Computer-administered and online assessments have the potential to be both desirable and dangerous. They have key advantages; at the same time, they have both validity and ethical challenges. Best practices for computer-administered and online assessments are provided. Adaptive testing involves having the respondent complete only those items that are needed to answer an assessment question, which can save immense time without sacrificing validity (if done well). There are many approaches to adaptive testing, including manual administration—such as skip rules, basal and ceiling criteria, and the countdown approach—and computerized adaptive testing (CAT) using item response theory. A CAT is designed to locate a person's level on the construct with as few items as possible. A CAT administers the items that will provide the most information based on participants' previous responses. CAT typically results in the greatest item savings—around 50% item savings or more.

Suggested Readings

Buchanan, 2002; Gibbons et al., 2016

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Source module psychophysiological-measures.qmd • Principles of Psychological Assessment: With Applied Examples in R • pinned GitHub snapshot a424bfcbad18. Full repository and license details appear in the attribution register.

Psychophysiological and Ambulatory Assessment

There are a number of alternative conceptualizations of psychopathology compared to the conceptualization of psychopathology provided in the Diagnostic and Statistical Manual of Mental Disorders (DSM). One alternative conceptualization of psychopathology is the p-factor (Caspi et al., 2014; Smith et al., 2020). The p-factor is a hierarchical view of psychopathology that accounts for covariation among various kinds of psychopathology. The p-factor is a general psychopathology factor, similar to the general intelligence factor (g), and has three sub-factors: internalizing problems, externalizing problems, and thought-disordered problems. Hierarchical conceptualizations tend to be advanced by lumpers, who group similar forms of psychopathology together, as opposed to splitters, who tend to split forms of psychopathology into separate categories.

Another hierarchical view of psychopathology is HiTOP, the Hierarchical Taxonomy of Psychopathology (Kotov et al., 2017; Kotov et al., 2021).

A third conceptualization of psychopathology is presented by the Research Domain Criteria, also called (RDoC), from the National Institute of Mental Health (NIMH). Recommendations for conducting assessment in a transdiagnostic way that is consistent with HiTOP and RDoC are provided by Stanton et al., 2020.

NIMH Research Domain Criteria (RDoC)

The National Institute of Mental Health (NIMH) has funded research to understand the etiology of mental illnesses and how best to intervene. Historically, NIMH funded considerable work on traditional DSM-based diagnostic categories. However, work studying DSM-defined diagnostic categories has not had as much impact as NIMH would have liked. So, the NIMH led an initiative known as the Research Domain Criteria, also called RDoC, to provide a template for research on psychopathology.

An overview of RDoC is provided by Kozak & Cuthbert, 2016. There are several ways in which RDoC is different from the DSM. Compared to the DSM, RDoC is dimensional, not categorical. RDoC views psychopathology as existing on continua, not as discrete categories. Another difference is that RDoC works from the ground up, starting with brain–behavior relations, and linking those to clinical symptoms. By contrast, the DSM is top-down, starting with diagnostic categories and determining what fits in those categories based on behavioral symptoms. However, as discussed in the source discussion, diagnoses are fictive categories that have poor diagnostic validity. The DSM-defined categories are based on behavioral presentations. That is, disorders are not things that people "have"; they are things that people "do".

The same behavior can occur for different underlying reasons, a principle known as equifinality. Different neurodevelopmental trajectories may underlie the same behavior for two different people. Behavioral presentations in the same disorder are heterogeneous. The pathophysiology in a given DSM category is not unitary. DSM categories also share symptoms and show lots of co-occurrence, so they are not entirely distinct. The DSM has resulted in the reification of these fictive diagnostic categories.

RDoC is a template for psychopathology research that consists of dimensional constructs integrating elements of psychology and biology. It incorporates a wider range of data, and multiple levels of analysis, including genetics, brain structure and function, and physiology. By contrast, the DSM only incorporates symptoms. In RDoC, there is a focus on narrower dimensions, such as cognition, emotion, reward seeking, fear, learning, memory, motivation, and perception, rather than disorders. The focus on narrow dimensions is based on the idea that there is greater potential to relate biological processes to simpler, lower-order, narrower dimensions of psychological constructs compared to disorders. These narrower dimensions of psychological constructs can be related to important clinical dimensions. Psychopathology is thought to represent extremes on the psychobiological distribution, e.g., high fearfulness in phobia versus low fearfulness in psychopathy. Thus, psychophysiological processes are a key emphasis in RDoC (Miller et al., 2016).

RDoC Matrix

The RDoC matrix is a matrix of two dimensions: six domains by eight units of analysis. In the RDoC matrix, domains are constructs that include elements, processes, mechanisms, and responses. The domains include: negative valence systems (i.e., negative affect: fear, anxiety, etc.), positive valence systems (i.e., positive affect: reward processing), cognitive systems (attention, perception, language, cognitive control, working memory), social processes (attachment, social communication, empathy), arousal and regulatory systems (sleep, circadian rhythms), sensorimotor systems (e.g., motor actions).

In the RDoC matrix, units of analysis are classes of measurement that are similar to levels of analysis. Units of analysis include: genes, molecules, cells, circuits, physiology, behaviors, self-reports, and paradigms.

The RDoC matrix is presented in the source discussion.

National Institute of Mental Health (NIMH) Research Domain Criteria (RDoC) Matrix.

In the RDoC framework, there is theoretical neutrality of the units of analysis. No unit of analysis is thought to be more important than any other; they are each thought to be important in their own right. That is why they are referred to as units of analysis rather than levels of analysis. "Levels" connotes order from basic to higher-order, and the developers of RDoC wanted to avoid suggesting that one level underlies another.

RDoC dimensional constructs are meant to be integrative rather than reductionistic. It involves moving away from the subjectivist tradition and moving toward a heterophenomenological approach. The subjectivist tradition is one in which subjective experience is deemed the primary measure of a phenomenon (e.g., fear). The heterophenomenological approach combines multiple units of analysis to identify phenomena.

The domains and constructs will need to be added and refined with further research. The RDoC matrix is just a tentative and incomplete starting template. The goal is to understand the bridge (i.e., mechanisms) that links the different units of analysis for the same construct, for example, how the biological and psychological bases of fear influence each other. Examples of mechanisms include endophenotypes, that are heritable, "unobservable", intermediary traits that signify disease liability, and that mediate the association between genotype and phenotypic expressions of psychopathology. An example of an endophenotype is depicted in the source discussion.

Example of an Endophenotype.

The idea is that genes influence endophenotypes and that endophenotypes lead to the phenotype. An endophenotype is similar to "intermediate phenotype", but an intermediate phenotype does not have the requirement of a genetic cause. An example of an intermediate phenotype is depicted in the source discussion. Endophenotypes and intermediate phenotypes are different from biomarkers, which are biological indicators (i.e., correlates) that are not necessarily causal. All endophenotypes are biomarkers, but not all biomarkers are endophenotypes.

Example of an Intermediate Phenotype.

One dimension that the RDoC does not include is development. It has been a challenge incorporating development into the RDoC matrix (Conradt et al., 2021; Durbin et al., 2022).

When considering the RDoC matrix, it is important to consider the RDoC dimensions—domains of functioning and units of analysis—within the context of environmental and developmental influences (Lupien et al., 2017; Woody & Gibb, 2015), as depicted in the source discussion.

Schematization Representation (From an Original idea from Woody & Gibb, 2015) of the Four-Dimensional Matrix of the RDoC Framework. Each row represents a sub-domain of functioning that could be studied in different studies or in a single study. Each column represents a unit of analysis. The different matrices are represented as a function of the environmental and developmental dimensions. For example, the domain of functioning in figure could represent the 'Negative Valence System' where '1' represents acute threat or fear, '2' represents potential threat of anxiety, '3' represents sustained threat, and '4' represents loss. The units of analysis could be represented by 'A' Genes; 'B' Stress Hormones; 'C' EEG; 'D' Brain imaging, and 'E': Metabolic markers. Each row of this schematized RDoC matrix represents a particular study (e.g., the gray boxes in Row #1 could represent a study measuring acute threat of fear as a function of stress hormones and brain imaging in an adult population) and the results of various studies are represented by the different filled boxes represented in rows and columns. Environmental and developmental factors could be added to all of these studies. (Figure reprinted from Lupien et al., 2017, Figure 1, p. 9. Lupien, S. J., Sasseville, M., François, N., Giguère, C. E., Boissonneault, J., Plusquellec, P., Godbout, R., Xiong, L., Potvin, S., Kouassi, E., & Lesage, A. (2017). The DSM5/RDoC debate on the future of mental health research: implication for studies on human stress and presentation of the signature bank. Stress, 20(1), 2-18. https://doi.org/10.1080/10253890.2017.1286324)

Psychophysiological Measures

Examples of psychophysiological measures include electroencephalography (EEG), event-related potential (ERP), (functional) magnetic resonance imaging (f)MRI, computerized axial tomography (CAT), magnetoencephalography (MEG), functional near infrared spectroscopy (fNIRS), electrocardiography (ECG or EKG), electromyography (EMG), electrooculography (EOG), eyetracking, and actigraphy. Miller et al., 2007 provide an overview of neuroimaging techniques for clinical assessment.

EEG caps can include 256 electrode sensors from millisecond to millisecond, with strong temporal resolution, yielding lots of data. Having more sensors allows better estimates of spatial localization. Psychophysiological measures are relevant because lots of behavior problems have important physiological facets. But it is not always clear whether biological processes cause psychopathology or the reverse, or whether there are third variable confounds that influence both psychophysiology and behavior problems (that explain why they are associated, even though they are non-causally related).

Psychophysiological measures are not invulnerable to basic measurement issues, including reliability and validity. Psychophysiological measures tend to be more expensive than questionnaire measures, so people traditionally were reluctant to consider issues of reliability and validity.

Galton and Cattell, nearly 200 years ago, were more interested in direct measures, like grip strength, than self-report. Hans Berger made the first EEG recording in 1924: EEG has been around a long time.

Reliability

Reliability matters, especially when you are examining individual differences (Dubois & Adolphs, 2016). From a generalizability theory point of view, we could examine the consistency or inconsistency of scores across facets and factors to determine which ones matter and which ones do not matter. Important considerations for evaluating reliability are discussed by Calamia, 2019.

Internal consistency involves examining the consistency of scores across trials to demonstrate that the measure assesses the same thing across trials. An example of internal consistency that could be applied to psychophysiological data is split-half reliability.

Another important form of reliability is test–retest reliability, which involves examining the consistency of scores across time. You could examine the consistency of individual differences across time, i.e., stability, and the consistency of scores within a person across time, i.e., repeatability. An example of repeatability would be examining a Bland-Altman plot.

Another important form of reliability is inter-rater reliability, which involves examining the consistency of scores across raters or processing methods. The goal is to generalize across instruments (machines, operators, etc.) to identify the reliability of the general method and not just the reliability of a particular operator or machine.

Parallel-forms reliability would involve examining the consistency of scores across slightly different stimuli thought to reflect the same cognitive processes. For instance, one could examine the reliability of scores across trials, time, raters, and stimuli. And if the reliability is low, it would be important to follow up to determine why the reliability is low.

It can be beneficial to conduct sensitivity analyses. Sensitivity analyses evaluate the robustness of findings by examining the extent to which findings differ based on changes in methods. For instance, you could examine whether findings differ when using different processing methods. If findings are consistent when using different processing methods, this provides further confidence in your findings because they do not appear to be due to the processing method chosen. However, if the findings differ across processing methods, it tells you that the processing method has an important consequence in relation to your outcome of interest. Sensitivity analysis can help you learn more about the phenomenon of interest.

Lots of Data

Psychophysiological measures yield lots of data. Some of the variance in the data reflects garbage (noise), some of the variance in the data reflects signal. So, when collecting lots of information, there's a possibility of finding an effect merely by chance. There are many researcher degrees of freedom. There are many brain voxels. There are often multiple conditions. And there are many ways to analyze the data. The researcher degrees of freedom can lead to p-hacking.

An example of the potential problem with researcher degrees of freedom is the "dead salmon study" (Bennett et al., 2009; Bennett et al., 2010). In the study, activation was detected in voxels of a dead salmon during fMRI scanning. The false positive activation detected was

due to a failure to account for many multiple comparisons. The point of the dead salmon study was not to invalidate fMRI—it was to show the importance of how rich data should be analyzed to adjust for multiple testing to lower the rate of Type I errors.

When dealing with lots of data, over-fitting is a common issue. As described in the source discussion, over-fitting involves explaining noise variance, a finding which would not generalize to a new sample. To avoid over-fitting when examining individual differences, such as with a correlation, it can be helpful to cross-validate the findings in an independent sample or in a hold-out sample. A hold-out sample is when you sub-divide a sample into a training and test data set. You develop the model on the training data set and see how the findings cross-validate on the hold-out (test) data set. Cross-validation requires larger sample (typically N greater than 100), which is difficult with psychophysiological measures. Most samples with psychophysiological data are relatively small, and as a result, have low power (Button et al., 2013; Button et al., 2013). With lots of data, researcher degrees of freedom, a small sample, and a statistical significance filter, reported findings have inflated estimates of effect sizes and poor replicability (Loken & Gelman, 2017).

One way to deal with the large data is to reduce the massive data down, and to use scoring and algorithms. For instance, you can examine the reliability of the scoring system, versus the reliability of the person conducting the scoring system. For data reduction, principal component analysis (PCA) is a useful technique. An emerging technique for integrating information from high-dimensional data is machine learning (Galatzer-Levy & Onnela, 2023).

Historically, psychophysiological data were averaged across subjects (group averaging) to improve the signal-to-noise ratio. For example, data were averaged for a clinical group versus controls. However, we think of psychopathology as dimensional and involving individual differences. There have been recent attempts to examine data at the individual differences level.

Issues such as reliability and validity are even more crucial when trying to make inferences about individual differences. It is important to remove as much noise as possible without sacrificing signal. Movement artifacts in psychophysiological measurements can lead to poor data quality. It is important to correct for artifacts because they do not occur at random. For example, patients with attention-deficit hyperactivity disorder (ADHD) and schizophrenia will likely show more movement artifacts than controls. One approach is to regress out the artifact (e.g., heart rate), if you have an independent measure of the artifact. If you do not have an independent measure of the artifact, you can use independent component analysis (ICA) to separate independent components of data and to remove artifacts.

Validity

Consideration of aspects of validity are crucial for any measure, including psychophysiological measures.

Consideration of construct validity, whether the measure actually assesses what it intends to assess, is crucial. For instance, performance on a measure of counting back by 7 is associated with many domains, including psychosis, anxiety, impulsivity, intelligence, etc. It is a measure of many constructs. It is important for our measures to have convergent validity in that they are associated with what they should be associated with, but also discriminant validity, in that they should not be associated with things that we do not expect them to be associated with.

If the research identifies activation in a specific part of the brain in association with the task, it is important to consider whether it is the same region across participants. For instance, the researcher should consider whether they are using a single atlas/head model for the whole sample or whether they are using a personalized atlas for each participant. Oftentimes, a given brain region is involved in lots of different cognitive processes and behaviors. This raises construct validity questions.

Ask yourself, "What is the evidence to support my interpretation of these effects?" Evidence is strengthened by showing evidence of convergent and discriminant validity. For example, differential deficits can provide stronger evidence of an effect. For instance, people with schizophrenia show deficits, abnormalities, or differences in many different brain processes—but an important question is whether they show relatively greater abnormalities in a given neural process than another neural process. This helps inform the specificity of the process (as opposed to other processes) for explaining differences in the condition of interest.

Another important question is: what is considered baseline? For heart rate reactivity and many psychophysiological measures, a comparison is made to a "baseline", which yields a contrast or difference score. But what should be considered "baseline"? There are many different possibilities for what could be considered "baseline," and each could yield different results. And, time during a participant's visit to the lab might not provide a good baseline because people may be anxious about being in a psychology lab. In addition, participants' brains are not resting during "resting-state".

Another important consideration is the ecological validity of the measures. Measures are often assessed in contexts and with stimuli that are not naturalistic. Psychophysiological measures may not reflect a participant's typical functioning. A goal would be to use more naturalistic contexts and task paradigms. Another challenge that biological measures have is in relating the biological criteria to actual behavior in the real world. It is not sufficient to show group differences on a biological measure; the researcher needs to show that the biological measure is associated with particular behaviors, that is, an independent measure in the same subjects.

When reliability and validity of psychophysiological measures are improved, a goal is to have norms for their use, so you know where a person stands compared to the population on the measure of interest.

Ambulatory Assessment

An ambulatory assessment is a measurement strategy designed to acquire minimally disruptive measures of a person engaging in normal activities in their natural environment. Ambulatory biosensors are assessment tools that measure physiological or motor activity (Haynes & Yoshioka, 2007). Ecological momentary assessment (EMA) is an example of an ambulatory assessment. EMA involves repeated sampling of people's current behaviors and experiences in real time, in their natural environments (Shiffman et al., 2008). EMA typically uses electronic diaries or mobile phones (Trull & Ebner-Priemer, 2013). Stone and colleagues (Stone et al., 2023) discuss important considerations for EMA assessment. EMA is a type of ambulatory assessment, but there are also other types of ambulatory assessment, such as physiological monitoring (e.g., sensors or actigraphs), global positioning system (GPS) tracking, and ecological momentary intervention (EMI) (Trull & Ebner-Priemer, 2013). Ambulatory assessment allows rich measurement including, for example, audio, pictures, video, geolocation (GPS), physiological functioning (e.g., heart rate), and physical activity (accelerometer). Ambulatory assessment may provide more precise measurement of phenotypes because they can assess contextual and environmental influences.

Ambulatory assessments can collect many measurements passively, without the respondent needing to do anything. Ambulatory assessment minimizes retrospective reporting biases because they can assess people in the moment. They also allow alerts and notifications for compliance reminders so that the participant completes the measurement. And they allow context-based triggers based on GPS. Queries can be random, event-based, time-based, or location-based. Some research suggests that event-based reporting may be preferable compared to signal-based reporting because participants may report that the target behavior occurs more frequently when using event-based reporting compared to signal-based reporting (Himmelstein et al., 2019).

Ambulatory assessments can also connect to external sensors via Bluetooth to record, for instance, blood pressure. They can provide data with high temporal resolution; for example, they can provide measurement samples from second to second or even millisecond to millisecond. When dealing with rich data, one important question is the level of granularity you want to analyze the data: seconds, days, weeks, etc. The optimal granularity depends on the research question.

Ambulatory assessment might provide opportunities for developing idiographic (personalized) models of the inter-relations among thoughts, feelings, behavior, physiology, and context (Wright & Zimmermann, 2019).

In addition, ambulatory assessment might have utility for intervention. For instance, it could help provide a just-in-time intervention when an event or context occurs. Ambulatory assessment can also be used for self-monitoring. And alerts can be used to increase homework compliance.

The three most popular versions of ambulatory biosensors are those that monitor cardiovascular activity (e.g., heart rate and blood pressure), physical activity (e.g., actigraph and pedometer), and cortisol levels. However, other ambulatory biosensors assess EEG, respiration, muscle tension, blood glucose, skin conductance, and other processes.

An applied example of an ambulatory biosensor is actigraphy. Actigraphy is the tracking of physical movement. Actigraphy uses an actigraph device. An actigraphy is a watch-like

device attached to the wrist or ankle that has an accelerometer. Smartphones can also serve as actigraphs because they have accelerometers. Actigraphy is a peripheral measure because it assesses gross motor movements. Actigraphy is also frequently used as a corresponding index of sleep because limb movements occur less frequently during sleep. Actigraphy is considered a better index of sleep than self-/informant-report, but not as good as polysomnography (PSG). However, actigraphy allows for greater ecological validity than a sleep study conducted in the lab, because a person can wear it while sleeping in their own bed at home.

In prior work, we took hundreds of sleep variables from actigraphy and using PCA to reduce the data down to key independent dimensions of sleep deficit: activity (sleep quality), (late) timing, (short) duration, and night-to-night variability (Staples et al., 2019).

Pros Pros of ambulatory assessment include:

• Ambulatory biosensors do not rely on participants to report their experience; they provide information about constructs that is derived from a different method—and they are useful in a multimethod assessment strategy.

Ambulatory biosensors lead to an increase in incremental validity and can improve specificity of decisions.

• Ambulatory biosensors are highly sensitive to change—they can track rapid changes in the variable of interest, with rapid timecourses (i.e., strong temporal resolution or precision).

• Ambulatory biosensors are portable, and have potential for greater ecological validity.

Biosensor variables collected in naturalistic settings are often only weakly associated with biosensor variables collected in lab settings, possibly due to reactivity in a lab setting.

• Ambulatory biosensors allow concurrent evaluation of contexts that influence psychophysiological responses.

Cons Cons of ambulatory assessment include:

• Ambulatory biosensors are expensive—they are potentially not cost-efficient depending on the use—i.e., they are potentially too expensive for clinical use.

• Some devices can be intrusive, cumbersome, and technical.

• There is limited psychometric evaluation conducted with many ambulatory methods.

Ambulatory biosensors are potentially unreliable. In addition, ambulatory biosensors are potentially not accurate for certain trait levels. For example, heart rate devices may be less accurate at high levels of physical activity. Ambulatory biosensors may not generalize well across time, devices, labs, and raters.

• The research investigator has less experimental control.

• There are practical challenges in conducting assessments outside of the research laboratory, as discussed by Holmlund et al., 2019 and Raugh et al., 2019.

Other Examples

Other examples of psychophysiological measures include eyetracking, which is used to assess attention, and wearables, which are smart "textiles" or clothing that has embedded biosensors.

Virtual Reality

Virtual reality (VR) is an emerging approach to assessment that allows structured assessment in simulated but life-like situations. It provides the investigator greater experimental control compared to ambulatory assessment approaches. Considerations in using virtual reality and other immersive technological approaches to assessment are described by Roberts et al., 2019.

Using Psychophysiological Measures

Ideally, we would be able to combine psychophysiological measures as objective measures with other measures, e.g., a performance-based assessment or self-/informant-report, in a multimethod measurement of constructs, as suggested by Patrick et al., 2019. Psychophysiological measures as part of a multimethod assessment approach helps to address the bias of participants and the bias in clinical judgment of clinicians. Ambulatory measures can provide incremental validity and increase the specificity of clinical judgment. Recommendations for using psychophysiological measures include using them in an ecologically valid way, including ambulatory approaches that allow ecological momentary assessment of contexts that influence psychophysiological responses, and conducting recordings over lengthy periods (Raugh et al., 2019). Recommendations for using ambulatory assessment in research on psychopathology are provided by Trull & Ebner-Priemer, 2020.

Challenges exist in selecting the granularity of analysis—how fine-grained to assess at each level of analysis—and how to combine measures across multiple levels of analysis because each may operate at different timescales. One option is to use multimodal imaging, e.g., simultaneous recordings of EEG and fMRI or EEG and fNIRS, to cancel out weaknesses of each. For instance, EEG has stronger temporal resolution than fMRI—it provides a faster timecourse—but EEG has weaker spatial resolution than fMRI—it provides less accurate estimates of where in the brain the neural activity is occurring.

It is worth considering combining multiple measures in a latent variable framework using, for example, factor analysis or structural equation modeling.

Given the costs, challenges, and time-consuming nature of psychophysiological data collection, science may be best positioned to address reliability and validity issues by sharing and combining data sets through consortia and open science. Consortia and data-sharing can help ensure large enough sample sizes for adequate statistical power, especially for individual difference analyses. Larger samples would also allow cross-validation of analyses in independent samples using out-of-sample prediction.

Conclusion

The National Institute of Mental Health (NIMH) provided the Research Domain Criteria (RDoC) as an alternative structure of psychopathology. Compared to the DSM, RDoC is dimensional and it works from the ground up, starting with brain–behavior relations, and linking those to clinical symptoms. RDoC spans multiple domains—such as positive and negative valience systems—and units of analysis, including genes, molecules, cells, circuits, physiology, behaviors, self-reports, and paradigms.

Psychophysiological measures include electroencephalography (EEG), event-related potential (ERP), (functional) magnetic resonance imaging (f)MRI, computerized axial tomography (CAT), magnetoencephalography (MEG), functional near infrared spectroscopy (fNIRS), electrocardiography (ECG or EKG), electromyography (EMG), electrooculography (EOG), eyetracking, and actigraphy. Psychophysiological measures are not invulnerable to basic measurement issues, including reliability and validity. Another important consideration of psychophysiological measures is how to handle the rich data without increasing the likelihood of Type I errors.

Another class of measurement includes ambulatory assessment. Ambulatory assessment is designed to acquire minimally disruptive measures of a person engaging in normal activities in their natural environment. Ambulatory assessment allows rich measurement including, for example, audio, pictures, video, geolocation (global positioning system [GPS]), and physical activity (accelerometer). Ambulatory biosensors are assessment tools that measure physiological or motor activity, such as those that monitor cardiovascular activity (e.g., heart rate and blood pressure), physical activity (e.g., actigraph and pedometer), cortisol levels, EEG, respiration, muscle tension, blood glucose, skin conductance, and other processes. Virtual reality is an emerging approach that allows structured assessment in simulated but life-like situations.

Suggested Readings

Miller et al., 2007; Haynes & Yoshioka, 2007

CHAPTER 19 / CAC.903

Report Writing, Feedback, and Communication

A report is an intervention. It shapes how a child is understood by parents, teachers, physicians, and sometimes courts. A good report is accurate, organized, readable, and useful. It distinguishes data from interpretation, states limitations, and offers actionable recommendations. A poor report can label a child for years. The counselor should write with care.

Structure

A typical report includes identifying information, referral question, methods, background, behavioral observations, results, formulation, diagnostic impressions if appropriate, recommendations, and limitations. The structure should serve the reader. A school report may emphasize classroom function. A medical report may emphasize symptoms and safety. A family feedback summary should be plain language.

Writing results

Results should be presented with metric, norm group, and uncertainty. Avoid jargon. Instead of "the examinee obtained a standard score of 85," write "Compared with children the same age, this score is in the low-average range. The confidence interval suggests the true score may be somewhat higher or lower." Then explain what it means for the referral question. Scores should not stand alone.

Formulation

The formulation links findings to a plan. It should include strengths, vulnerabilities, triggers, maintaining factors, and protective factors. It should be testable. A formulation that cannot guide intervention is incomplete. The counselor should avoid blaming language. "The child is manipulative" is not a formulation. "The child's crying appears to delay difficult tasks and is reinforced by escape" is more useful.

Recommendations

Recommendations should be specific, feasible, and prioritized. "Provide support" is vague. "Use a visual schedule, give one-step instructions, and check understanding before independent work" is actionable. Recommendations should match the setting and resources. The counselor should consider who will implement them and how progress will be reviewed.

Feedback session

Feedback should be a conversation. The counselor should explain findings, invite questions, and check understanding. With adolescents, feedback should respect autonomy. With caregivers, it should reduce shame and increase agency. The counselor should highlight strengths as well as needs. A family should leave knowing what to do next.

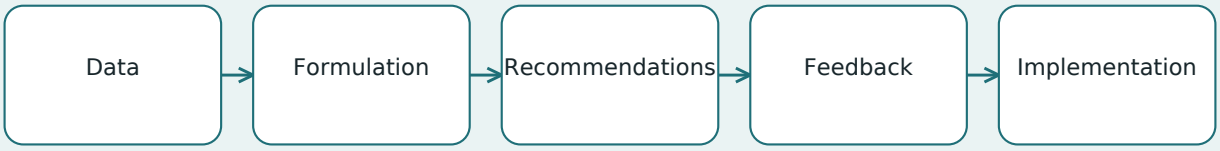
Real-life application

A parent receives a report that says "oppositional defiant disorder." The parent feels blamed. The counselor revises the feedback to explain the pattern, the role of escape, the child's anxiety, and the plan. The parent becomes an ally. The diagnosis may remain, but the communication changes the outcome.

Reflection

Write a one-paragraph feedback summary for a caregiver about a child's anxiety. Include strength, concern, uncertainty, and next step. A report is not finished when it is written. It is finished when it is understood.

Report to action



A report is useful when it leads to understood and feasible action.

CHAPTER 20 / CAC.903

Integrated Child and Adolescent Case Formulation

The final chapter brings the book together. Integrated assessment uses multiple methods, informants, and theories to answer a referral question and guide action. It is not a random collection of tests. It is a reasoned process that respects development, culture, ethics, and evidence. The doctoral counselor should be able to design, defend, and revise an assessment plan.

A decision framework

Begin with the referral question. Identify the decisions that will follow. Choose targets and methods that fit the child, context, and question. Consider evidence, feasibility, and ethics. Collect data. Organize findings into a formulation. Write recommendations. Provide feedback. Monitor outcomes. Revise as needed. This loop protects against rigid assessment.

Case example: school refusal

Amina, age nine, cries at the school door. The referral question is whether she has an anxiety disorder and what should be done. Methods include child interview, caregiver interview, teacher rating, observation at arrival, and review of recent events. Findings show separation fear, recent grief, bullying, and parental accommodation. The formulation links anxiety to avoidance and reinforcement. Recommendations include gradual exposure, caregiver coaching, school safety plan, and grief support. Monitoring tracks attendance and distress. The plan is reviewed after four weeks.

Case example: adolescent depression and risk

Daniel, age fifteen, has low mood, sleep disturbance, and self-harm thoughts. Assessment includes private interview, safety assessment, caregiver interview with consent limits, depression measure, and school functioning review. Findings show depression, family conflict, and protective connection to a coach. The formulation includes mood, isolation, and conflict. Recommendations include therapy, safety planning, family communication, and psychiatric consultation if needed. Monitoring is frequent. The assessment balances privacy and protection.

Case example: attention and learning

Leo, age eight, is referred for ADHD. Assessment includes developmental history, teacher and parent ratings, classroom observation, achievement testing, and sleep review. Findings show weak decoding, avoidance during reading, and adequate attention in preferred activities. The formulation emphasizes learning difficulty and secondary avoidance. Recommendations include reading intervention, classroom accommodations, and monitoring. The ADHD question is answered cautiously, not assumed.

Professional identity

The doctoral assessor is a scholar-practitioner. They read evidence, use supervision, document reasoning, and remain humble. They know that assessment can help or harm. They advocate for fair access and against misuse. They treat the child as a person with a future, not a score. The goal is not perfect certainty. The goal is responsible action under uncertainty.

Reflection

Write your own assessment philosophy in one page. Include your view of evidence, culture, ethics, and uncertainty. Then write three learning goals for supervision. Assessment is a craft, a science, and an ethical responsibility.



Integrated assessment holds technical skill inside ethical and cultural responsibility.

Glossary

Glossary

Adaptive functioning: Everyday skills needed for personal independence and social responsibility, including communication, self-care, and safety.

Assent: A child's affirmative agreement to participate in assessment, explained in developmentally appropriate language.

Base rate: The proportion of a population that has a condition or outcome. Base rates affect predictive values.

Behavioral coding: Systematic recording of behavior using defined categories, intervals, or events.

Case formulation: An organized hypothesis linking assessment findings to development, context, maintaining factors, and intervention.

Clinical utility: The degree to which assessment improves decisions, services, or client outcomes.

Confidence interval: A range around a score that reflects measurement uncertainty.

Construct: A concept or characteristic being measured, such as anxiety or working memory.

Content validity: Evidence that a measure adequately samples the domain it claims to assess.

Criterion validity: Evidence that scores relate to an external standard or outcome.

Cut score: A score threshold used to classify or make a decision.

Differential assessment: The process of distinguishing among possible explanations for a presenting concern.

Dynamic assessment: Assessment that examines learning potential through teaching, mediation, or supported performance.

Evidence-based assessment: Use of research and theory to select targets, methods, and interpretations.

False negative: A case that has the condition but is not identified by the assessment.

False positive: A case identified by the assessment that does not have the condition.

Formative construct: A construct formed by its indicators, which may not correlate strongly.

Functional behavioral assessment: Analysis of antecedents, behavior, and consequences to identify the function of behavior.

Generalizability theory: A framework for estimating how scores depend on persons, items, occasions, raters, and settings.

Incremental validity: Evidence that a method adds useful information beyond existing data.

Informant: A person who provides assessment information, such as a child, caregiver, or teacher.

Internal consistency: The degree to which items on a measure correlate as indicators of a reflective construct.

Inter-rater reliability: Agreement between different observers or raters.

Measurement error: Variation in observed scores that is not due to the construct being measured.

Measurement invariance: Evidence that a measure operates similarly across groups.

Multi-informant assessment: Use of reports from multiple people to capture behavior across contexts.

Negative predictive value: The probability that a negative screen is a true non-case.

Norm-referenced interpretation: Comparing a score to a reference group.

Percentile: A score indicating the percentage of the norm group scoring at or below a given point.

Positive predictive value: The probability that a positive screen is a true case.

Predictive bias: When a score predicts an outcome differently across groups.

Progress monitoring: Repeated brief assessment used to evaluate change and guide intervention.

Projective hypothesis: The idea that ambiguous stimuli may reveal inner experience; requires cautious interpretation.

Reflective construct: A construct that causes its indicators, which should correlate.

Reliability: Consistency or dependability of measurement.

Reliable change: Change that exceeds expected measurement error.

Screening: Brief assessment to identify possible concerns and need for further evaluation.

Sensitivity: The proportion of true cases correctly identified.

Specificity: The proportion of true non-cases correctly identified.

Standard error of measurement: An estimate of variability around an observed score due to measurement error.

Standardized test: A measure with uniform administration, scoring, and often norms.

Test security: Protection of test items, protocols, and data to preserve validity and privacy.

Validity: Evidence supporting the interpretation and use of scores for a specific purpose.

Applied Workbook

Applied Workbook: Assessment-to-Action Exercises

Use fictional or de-identified cases. Do not administer or interpret restricted tests without training, authorization, and supervision. These exercises practice reasoning, not unsupervised testing.

Exercise 1: Referral question rewrite

Take a vague referral such as "behavior problem." Rewrite it as an assessable question. List three methods and three informants. State one alternative hypothesis.

Exercise 2: Consent and assent script

Write a consent explanation for a caregiver and an assent explanation for an eight-year-old. Include purpose, privacy limits, and choice.

Exercise 3: Cultural formulation

Write five questions for a family from a different background. Include meaning, language, supports, power, and barriers. Write one assumption to avoid.

Exercise 4: Construct map

Choose a construct such as school refusal. List observable indicators. Decide whether the construct is reflective, formative, or mixed. Explain your reasoning.

Exercise 5: Score explanation

Explain a percentile, standard score, and confidence interval to a caregiver in plain language. Include what the score does not mean.

Exercise 6: Reliability audit

Choose a measure. Identify the reliability evidence most relevant to your decision. Write how much error is acceptable and why.

Exercise 7: Validity argument

Write the interpretation you want to make from a test. List three validity supports and two threats. State whether the assessment will help the child.

Exercise 8: Fairness review

Review a decision pathway in your setting. Identify who may be disadvantaged. Add two safeguards.

Exercise 9: Interview plan

Write a semi-structured interview outline for school refusal. Include safety, strengths, context, and developmental adaptation.

Exercise 10: ABC observation

Choose a behavior. Write an ABC hypothesis. Design an observation plan with definitions, setting, and coding method.

Exercise 11: Cognitive demand analysis

Choose a classroom behavior. List two cognitive demands that might explain it. Design a brief assessment plan.

Exercise 12: Intellectual and adaptive integration

Write why a full-scale score alone is insufficient. List three additional sources and how they would change interpretation.

Exercise 13: Multi-informant discrepancy

Write a case where child and parent reports disagree. Generate three explanations. Write how you would explore them.

Exercise 14: Expressive guardrails

Write three rules for using play or art responsibly. Include consent, interpretation limits, and documentation.

Exercise 15: Structured judgment

Write a prediction. List data and a simple decision rule. State how you would check accuracy.

Exercise 16: Screening pathway

Design a two-stage screening pathway. Identify sensitivity, specificity, base rate, and follow-up steps.

Exercise 17: Monitoring plan

Choose a target, measure, frequency, feedback method, and decision rule. Include a plan for missing data.

Exercise 18: Digital consent

Write a consent statement for a teen using a mood app. Include privacy, data use, and alternatives.

Exercise 19: Feedback summary

Write a one-paragraph caregiver feedback summary. Include strength, concern, uncertainty, and next step.

Exercise 20: Integrated case formulation

Write a full formulation for a fictional case. Include referral question, methods, findings, cultural context, ethics, recommendations, and monitoring.

Portfolio guidance

Collect your best exercises with feedback and revisions. A doctoral portfolio should show growth in reasoning, humility, and ethical decision-making. Keep all client information de-identified.

Clinical Casebook

Clinical Casebook: Assessment Reasoning in Practice

The following fictional cases are teaching examples. They are not diagnostic protocols and do not replace supervision. Each case asks you to practice the chapter's reasoning: clarify the question, choose methods, weigh evidence, and design a helpful next step.

Case 1: The referral that was too small

A teacher refers eight-year-old Musa for "ADHD testing." The teacher reports that Musa leaves his seat, interrupts, and fails to finish work. The parent reports that Musa is calm at home but forgets chores. The initial referral assumes one disorder. The assessment team rewrites the question: "What explains Musa's classroom inattention and activity, and what supports are needed?" The team collects teacher and parent ratings, classroom observation, sleep history, hearing screening, and academic review. Observation shows that Musa is most active during long written tasks and calmer during hands-on activities. Sleep history reveals late bedtimes. The formulation includes possible attention difficulties, sleep loss, and task mismatch. The team recommends sleep routine support, movement breaks, and further evaluation if impairment persists. The case teaches that a referral label is a starting point, not a conclusion.

Case 2: Permission without understanding

A clinic obtains a signed consent form from a mother before assessing ten-year-old Ada. The counselor begins testing without explaining the process to Ada. Ada becomes quiet and refuses items. The supervisor pauses the session and asks what Ada understood. Ada says she thought she was being punished. The counselor provides assent in simple language, uses a visual schedule, and offers a break. Ada participates more fully. The case teaches that legal permission is not enough. A child needs developmentally appropriate understanding and a sense of safety. Assent is an ongoing process, not a one-time signature.

Case 3: The interpreter who was not neutral

A family speaks a language the counselor does not know. The clinic asks the adolescent daughter to interpret for her mother. During the interview, the daughter softens the mother's concerns and omits family conflict. The counselor notices the mismatch and arranges a qualified interpreter. The mother then describes grief and financial stress. The case teaches that using a child as an interpreter can reverse roles and distort data. Language access is an ethical and validity issue. The counselor should protect the child from adult burdens and use qualified support.

Case 4: Laziness or reading failure?

A teacher describes nine-year-old Grace as lazy. Grace refuses reading tasks and says she hates school. The counselor avoids the label and maps the construct. Observation shows that Grace starts nonreading tasks but avoids decoding. Achievement assessment shows weak word reading. The formulation is reading difficulty with secondary avoidance. The plan includes structured reading intervention, accessible text, and praise for effort. The case teaches that behavior labels often hide skill deficits. Assessment should translate labels into observable processes.

Case 5: The score that frightened a parent

A parent hears that a child's standard score is 92 and concludes the child is "below normal." The counselor explains the average range, confidence interval, and functional meaning. The counselor also describes strengths in verbal reasoning and practical problem-solving. The parent's anxiety decreases. The case teaches that scores require translation. A number without context can harm a family's expectations. The counselor should explain what the score supports and what it does not support.

Case 6: Weekly scores that jumped around

A teen's depression scores move from 18 to 14 to 19 to 15 over four weeks. The counselor does not declare failure. The counselor reviews measurement error, life events, and functioning. The teen reports improved sleep but increased family conflict. The plan adds family communication work. The case teaches that fluctuation is not automatically change. Trends, context, and reliable change matter more than single points.

Case 7: A valid test used for the wrong question

A school uses a depression scale to decide which students need career counseling. The counselor objects that the tool was not designed for that decision. The team instead uses interest inventories, academic records, and student interviews. The case teaches that validity is purpose-specific. A measure can be strong for one question and weak for another. The counselor should match method to decision.

Case 8: Overrepresentation in discipline referrals

A district notices that one group of students is referred more often for emotional disturbance. The assessment team reviews language access, trauma exposure, instructional quality, and bias in referral language. They add bilingual interviews and classroom observation. Referrals become more accurate. The case teaches that fairness is a pathway issue. A test may not be the only source of bias. The counselor should examine the whole decision system.

Case 9: The adolescent who said everything was fine

Fourteen-year-old Tolu answers "fine" to every question. The counselor notices tension and avoids pressure. The counselor offers a drawing of a typical school day. Tolu draws a crowded hallway and labels it "too much." The interview reveals sensory overload and bullying. The case teaches that flexible methods can open communication. The counselor should not force disclosure but can create safer ways to share.

Case 10: Hitting that produced attention

A child hits peers during recess. Observation shows that hitting occurs after exclusion and ends when an adult intervenes. The team hypothesizes attention and escape from social confusion. The plan teaches help-seeking and peer-entry skills. Adults respond calmly and reinforce appropriate requests. The case teaches that behavior has function. Intervention should match the maintaining consequence, not only the topography.

Case 11: Not listening or not remembering?

A child fails to follow multi-step instructions. Cognitive assessment shows weak working memory. The classroom uses long verbal directions. The plan includes visual steps, chunked instructions, and check-backs. The case teaches that cognitive assessment should identify demands and supports. A behavior may reflect a mismatch between task and capacity.

Case 12: Low score after migration

A child recently migrated and scores low on a verbal test. Adaptive reports show strong daily skills in the home language. The counselor uses nonverbal tasks, dynamic assessment, and interpreter-supported interview. The report avoids a global deficit label. The case teaches that language exposure can depress scores. Intellectual assessment must be interpreted with opportunity and context.

Case 13: The teen who hid depression

A teen reports high depression; the parent reports low depression. The interview reveals private hopelessness and masking at home. The counselor takes the teen's report seriously and assesses safety. The case teaches that informant disagreement is meaningful. Inner experience may be visible only to the child. The counselor should not average away distress.

Case 14: The drawing that was not proof

A child draws a dark house after hospitalization. A trainee suggests the drawing proves trauma. The supervisor cautions against overinterpretation. The counselor asks the child about the drawing and follows play themes over time. The child introduces a helper and a key. The case teaches that expressive methods generate hypotheses. They do not replace corroborated evidence.

Case 15: The vivid prediction

A counselor predicts a teen will drop out because of a dramatic interview. A structured review shows moderate risk, not high risk. The counselor adjusts the plan to monitoring and support. The case teaches that vivid details can distort judgment. Structured rules and feedback improve calibration.

Case 16: The positive screen that was not a diagnosis

A school depression screen identifies many students. The counselor does not diagnose all. Follow-up interviews assess impairment, safety, and context. Some students need support; others had temporary stress. The case teaches that screening is a first step. Base rates and follow-up protect against overidentification.

Case 17: Monitoring that changed the plan

A teen's weekly scores improve, then rise after a family conflict. The counselor reviews the data with the teen and adds family communication work. The case teaches that monitoring is a conversation. Data should guide adjustment, not punish the client.

Case 18: The mood app that excluded a family

A clinic recommends a mood app, but a family lacks a smartphone. The counselor offers paper logs and phone check-ins. The case teaches that digital methods require equity planning. Technology should not become a barrier to care.

Case 19: The report that blamed the child

A report describes a child as manipulative. The supervisor asks for a functional formulation. The revised report describes crying that delays difficult tasks and is reinforced by escape. The plan becomes clearer. The case teaches that language shapes intervention. Blame labels close thinking; functional descriptions open options.

Case 20: The integrated formulation

A nine-year-old refuses school after grief and bullying. Assessment includes child and caregiver interviews, teacher ratings, arrival observation, and record review. The formulation links separation fear, avoidance, and accommodation. The plan includes gradual exposure, caregiver coaching, school safety, and grief support. Monitoring tracks attendance and distress. The case teaches that integrated assessment connects evidence to action. The child is not a score. The child is a person with a future.

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